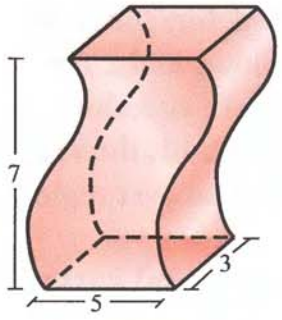



**Figure 5.1.11** Compute this volume.

**5.** A lumberjack cuts out a wedge-shaped piece $W$ of a cylindrical tree of radius $r$ obtained by making two saw cuts to the tree's center, one horizontally and one at an angle $\theta$. Compute the volume of the wedge $W$ using Cavalieri's principle. (See Figure 5.1.12.)

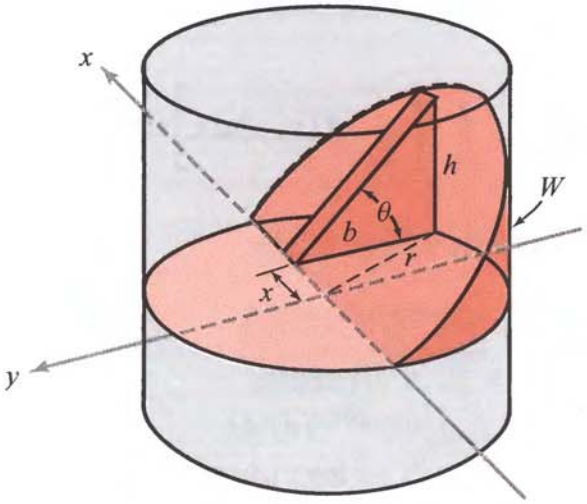
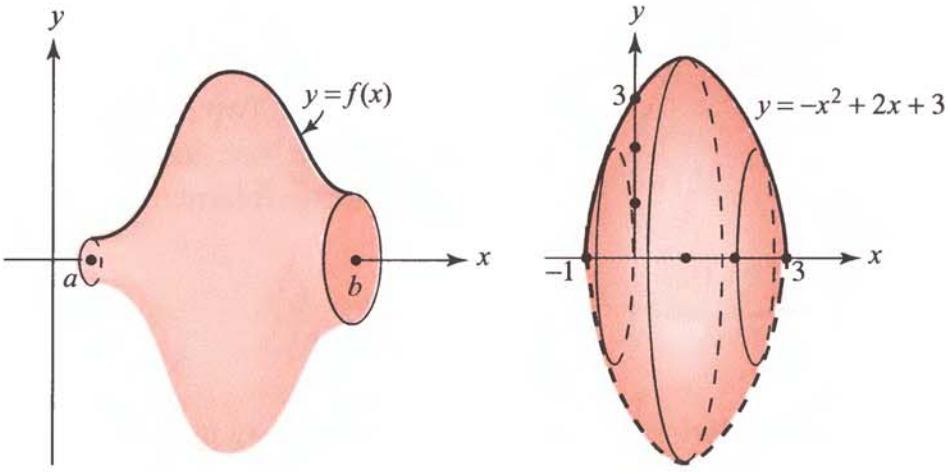



**Figure 5.1.12** Find the volume of $W$.

**6.** (a) Show that the volume of the solid of revolution shown in Figure 5.1.13(a) is

$$\pi \int_a^b [f(x)]^2 \, dx.$$

(b) Show that the volume of the region obtained by rotating the region under the graph of the parabola $y = -x^2 + 2x + 3$, $-1 \le x \le 3$, about the $x$ axis is $512\pi/15$ [see Figure 5.1.13(b)].




(a) (b)

**Figure 5.1.13** The solid of revolution (a) has volume $\pi \int_a^b [f(x)]^2 \, dx$. Part (b) shows the region between the graph of $y = -x^2 + 2x + 3$ and the $x$ axis rotated about the $x$ axis.

*Evaluate the double integrals in Exercises 7 to 9, where $R$ is the rectangle* $[0, 2] \times [-1, 0]$.

7. $\iint_R (x^2y^2 + x)\,dy\,dx$

8. $\iint_R \left(|y| \cos \frac{1}{4}\pi x\right) dy\,dx$

9. $\iint_R \left(-xe^x \sin \frac{1}{2}\pi y\right) dy\,dx$

10. Find the volume bounded by the graph of $f(x, y) = 1 + 2x + 3y$, the rectangle $[1, 2] \times [0, 1]$, and the four vertical sides of the rectangle $R$, as in Figure 5.1.1.

**11.** Repeat Exercise 10 for the function $f(x, y) = x^4 + y^2$ and the rectangle $[-1, 1] \times [-3, -2]$.

# 5.2 The Double Integral Over a Rectangle

We are ready to give a rigorous definition of the double integral as the limit of a sequence of sums. This will then be used to *define* the volume of the region under the graph of a function $f(x, y)$. We shall not require that $f(x, y) \geq 0$; but if $f(x, y)$ assumes negative values, we shall interpret the integral as a signed volume, just as for the area under the graph of a function of one variable. In addition, we shall discuss some of the fundamental algebraic properties of the double integral and prove Fubini's theorem, which states that the double integral can be calculated as an iterated integral. To begin, let us establish some notation for partitions and sums.

## Definition of the Integral

Consider a closed rectangle $R \subset \mathbb{R}^2$; that is, $R$ is a Cartesian product of two intervals: $R = [a, b] \times [c, d]$. By a ***regular partition*** of $R$ of order $n$ we mean the two ordered collections of $n + 1$ equally spaced points $\{x_j\}_{j=0}^n$ and $\{y_k\}_{k=0}^n$, that is, the points satisfying

$$a = x_0 < x_1 < \cdots < x_n = b, \qquad c = y_0 < y_1 < \cdots < y_n = d$$

and

$$x_{j+1} - x_j = \frac{b - a}{n}, \qquad y_{k+1} - y_k = \frac{d - c}{n}$$

(see Figure 5.2.1).

A function $f(x, y)$ is said to be ***bounded*** if there is a number $M > 0$ such that $-M \leq f(x, y) \leq M$ for all $(x, y)$ in the domain of $f$. A continuous function on a *closed* rectangle is always bounded, but, for example, $f(x, y) = 1/x$ on $(0, 1] \times [0, 1]$ is continuous but is not bounded, because $1/x$ becomes arbitrarily large for $x$ near 0. The rectangle $(0, 1] \times [0, 1]$ is not closed, because the endpoint 0 is missing in the first factor.

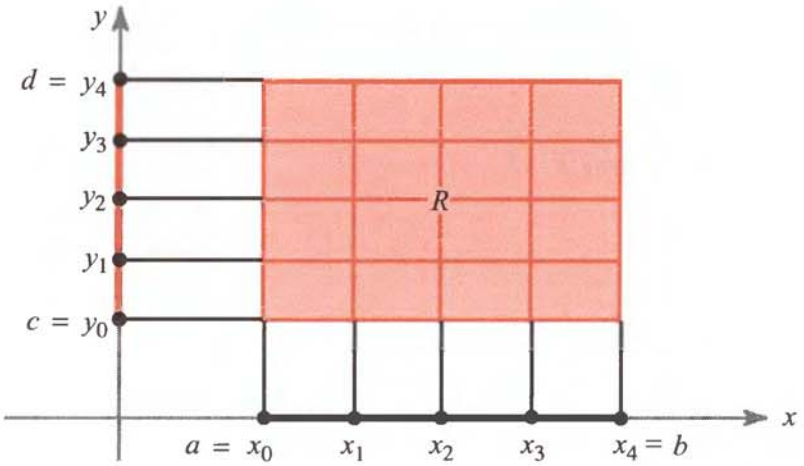



**Figure 5.2.1** A regular partition of a rectangle $R$, with $n = 4$.

Let $R_{jk}$ be the rectangle $[x_j, x_{j+1}] \times [y_k, y_{k+1}]$, and let $\mathbf{c}_{jk}$ be *any* point in $R_{jk}$. Suppose $f: R \to \mathbb{R}$ is a bounded real-valued function. Form the sum

$$S_n = \sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\, \Delta x\, \Delta y = \sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\, \Delta A, \tag{1}$$

where

$$\Delta x = x_{j+1} - x_j = \frac{b-a}{n}, \qquad \Delta y = y_{k+1} - y_k = \frac{d-c}{n},$$

and

$$\Delta A = \Delta x\, \Delta y.$$

This sum is taken over all $j$'s and $k$'s from 0 to $n-1$, and so there are $n^2$ terms. A sum of this type is called a ***Riemann sum*** for $f$.

**DEFINITION: Double Integral** If the sequence $\{S_n\}$ converges to a limit $S$ as $n \to \infty$ and if the limit $S$ is the same for any choice of points $\mathbf{c}_{jk}$ in the rectangles $R_{jk}$, then we say that $f$ is ***integrable*** over $R$ and we write

$$\iint_R f(x,y)\, dA, \qquad \iint_R f(x,y)\, dx\, dy, \qquad \text{or} \qquad \iint_R f\, dx\, dy$$

for the limit $S$.

Thus, we can rewrite integrability in the following way:

$$\lim_{n\to\infty} \sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\, \Delta x\, \Delta y = \iint_R f\, dx\, dy$$

for any choice of $\mathbf{c}_{jk} \in R_{jk}$.

## Properties of the Integral

The proof of the following basic theorem is presented in the Internet supplement for Chapter 5.

**THEOREM 1** Any continuous function defined on a closed rectangle $R$ is integrable.

If $f(x, y) \geq 0$, the existence of $\lim_{n\to\infty} S_n$ has a straightforward geometric meaning. Consider the graph of $z = f(x, y)$ as the top of a solid whose base is the rectangle $R$. If we take each $\mathbf{c}_{jk}$ to be a point where $f(x, y)$ has its minimum value[2] on $R_{jk}$, then $f(\mathbf{c}_{jk})\,\Delta x\,\Delta y$ represents the volume of a rectangular box with base $R_{jk}$. The sum $\sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\,\Delta x\,\Delta y$ equals the volume of an inscribed solid, part of which is shown in Figure 5.2.2.

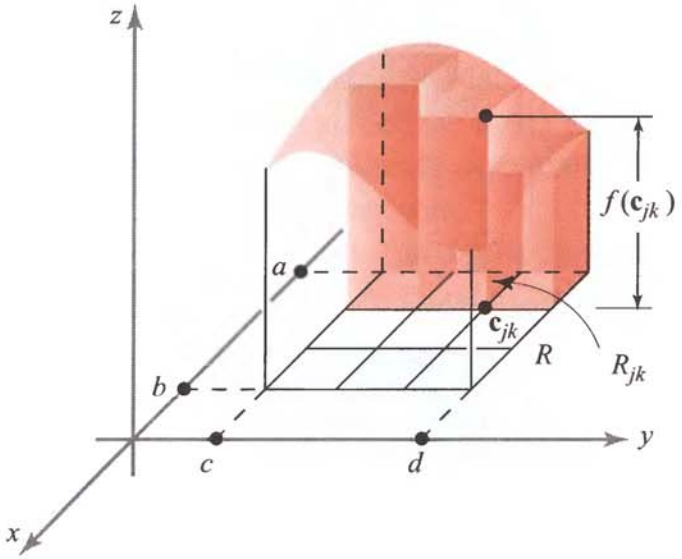



**Figure 5.2.2** The sum of inscribed boxes approximates the volume under the graph of $z = f(x, y)$.

Similarly, if $\mathbf{c}_{jk}$ is a point where $f(x, y)$ has its maximum on $R_{jk}$, then the sum $\sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\,\Delta x\,\Delta y$ is equal to the volume of a circumscribed solid (see Figure 5.2.3).

Therefore, if $\lim_{n\to\infty} S_n$ exists and is independent of $\mathbf{c}_{jk} \in R_{jk}$, it follows that the volumes of the inscribed and circumscribed solids approach the same limit as $n \to \infty$. It is therefore reasonable to call this limit the exact volume of the solid under the graph of $f$. Thus, the method of Riemann sums supports the concepts introduced on an intuitive basis in Section 5.1.

There is a theorem guaranteeing the existence of the integral of certain discontinuous functions as well. We shall need this result in the next section in order to discuss the integrals of functions over regions more general than rectangles. We shall be specifically interested in functions whose discontinuities lie on curves in the $xy$ plane. Figure 5.2.4 shows two functions defined on a rectangle $R$ whose discontinuities

[2]Such $\mathbf{c}_{jk}$ exist by virtue of the continuity of $f$ on $R$; see Theorem 7 in Section 3.3.

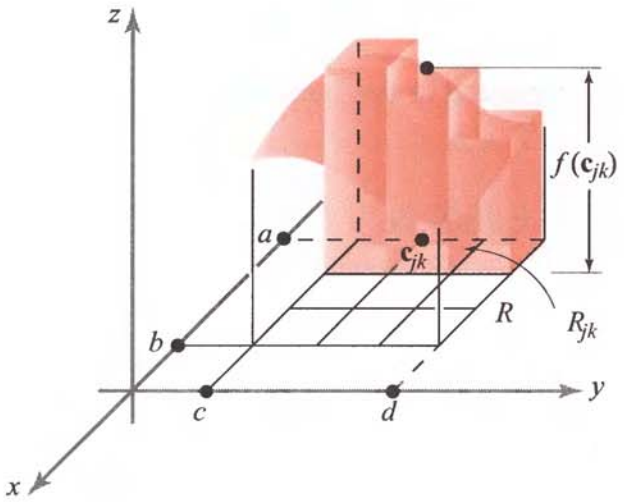



**Figure 5.2.3** The volume of circumscribed boxes also approximates the volume under $z = f(x, y)$.

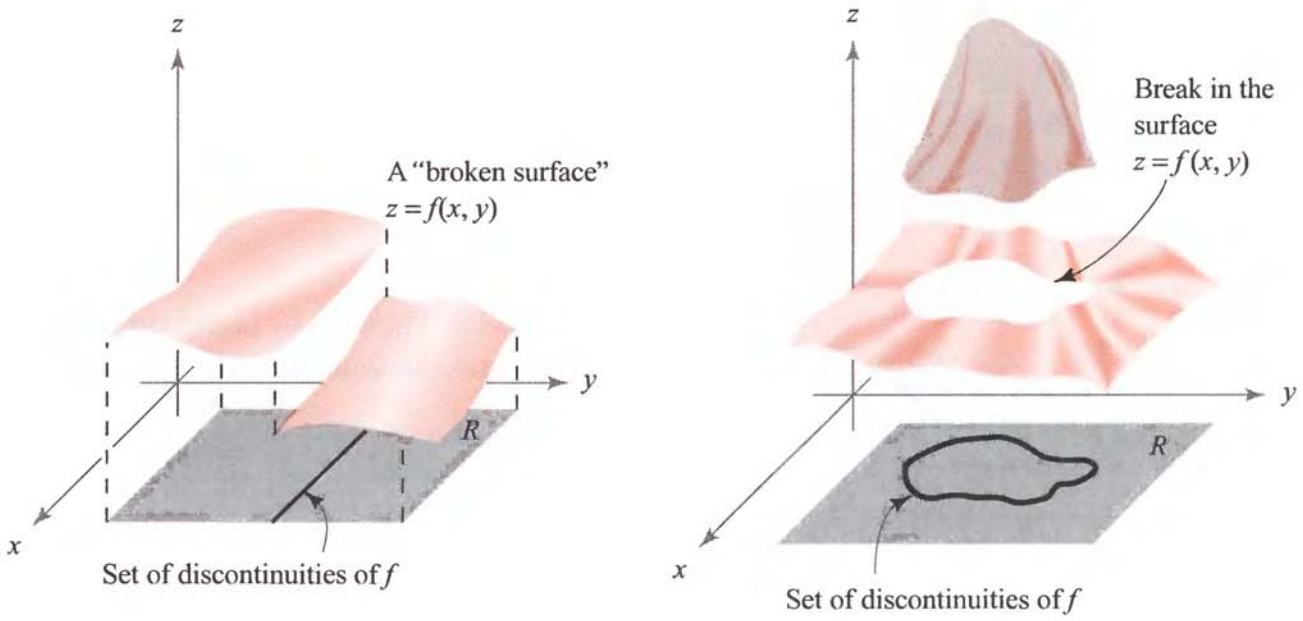



**Figure 5.2.4** What the graphs of discontinuous functions of two variables might look like.

lie along curves. In other words, $f$ is continuous at each point that is in $R$, but not necessarily on the curve.

Useful curves are graphs of functions such as $y = \phi(x), a \leq x \leq b$, or $x = \psi(y)$, $c \leq y \leq d$, or finite unions of such graphs. Some examples are shown in Figure 5.2.5.

The next theorem provides an important criterion for determining whether a function is integrable. The proof is discussed in the Internet supplement.

**THEOREM 2: Integrability of Bounded Functions** Let $f: R \to \mathbb{R}$ be a bounded real-valued function on the rectangle $R$, and suppose that the set of points where $f$ is discontinuous lies on a finite union of graphs of continuous functions. Then $f$ is integrable over $R$.

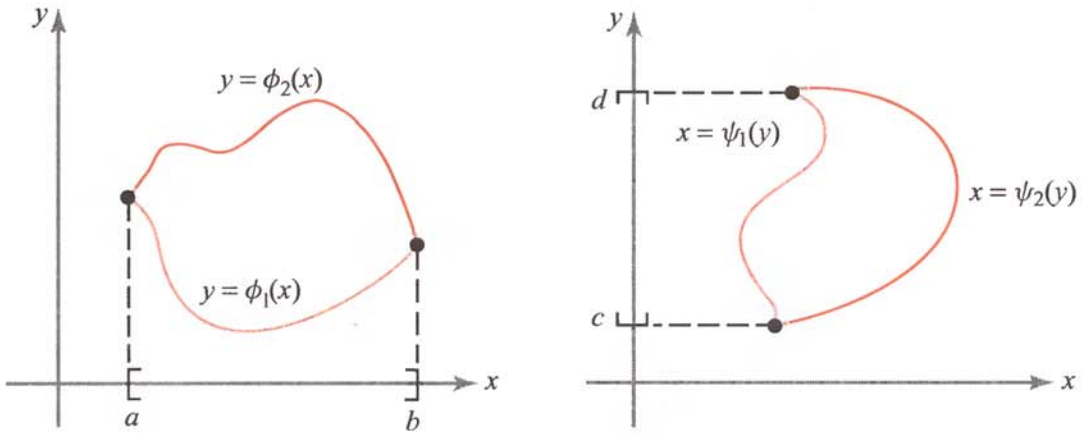



**Figure 5.2.5** Curves in the plane represented as graphs.

Using Theorem 2 and the remarks preceding it, we see that the functions sketched in Figure 5.2.4 are integrable over $R$, because these functions are bounded and continuous except on graphs of continuous functions.

From the definition of the integral as a limit of sums and the limit theorems, we can deduce some fundamental properties of the integral $\iint_R f(x, y)\, dA$; these properties are essentially the same as for the integral of a real-valued function of a single variable.

Let $f$ and $g$ be integrable functions on the rectangle $R$, and let $c$ be a constant. Then $f + g$ and $cf$ are integrable, and

(i) Linearity

$$\iint_R [f(x, y) + g(x, y)]\, dA = \iint_R f(x, y)\, dA + \iint_R g(x, y)\, dA.$$

(ii) Homogeneity

$$\iint_R cf(x, y)\, dA = c \iint_R f(x, y)\, dA.$$

(iii) Monotonicity If $f(x, y) \geq g(x, y)$, then

$$\iint_R f(x, y)\, dA \geq \iint_R g(x, y)\, dA.$$

(iv) Additivity If $R_i$, $i = 1, \ldots, m$, are pairwise disjoint rectangles such that $f$ is bounded and integrable over each $R_i$ and if $Q = R_1 \cup R_2 \cup \cdots \cup R_m$ is a rectangle, then $f: Q \to \mathbb{R}$ is integrable over $Q$ and

$$\iint_Q f(x, y)\, dA = \sum_{i=1}^{m} \iint_{R_i} f(x, y)\, dA.$$

Properties (i) and (ii) are a consequence of the definition of the integral as a limit of a sum and the following facts for convergent sequences $\{S_n\}$ and $\{T_n\}$, which are proved as with the limit theorems in Chapter 2:

$$\lim_{n\to\infty}(T_n + S_n) = \lim_{n\to\infty} T_n + \lim_{n\to\infty} S_n$$

$$\lim_{n\to\infty}(cS_n) = c\lim_{n\to\infty} S_n.$$

To demonstrate monotonicity, we first observe that if $h(x, y) \geq 0$ and $\{S_n\}$ is a sequence of Riemann sums that converges to $\iint_R h(x, y)\,dA$, then $S_n \geq 0$ for all $n$, so that $\iint_R h(x, y)\,dA = \lim_{n\to\infty} S_n \geq 0$. If $f(x, y) \geq g(x, y)$ for all $(x, y) \in R$, then $(f - g)(x, y) \geq 0$ for all $(x, y)$, and, using properties (i) and (ii), we have

$$\iint_R f(x, y)\,dA - \iint_R g(x, y)\,dA = \iint_R [f(x, y) - g(x, y)]\,dA \geq 0.$$

This proves property (iii). The proof of property (iv) is more technical and a special case is proved in the Internet supplement. It should be intuitively obvious.

Another important result is the inequality

$$\left|\iint_R f\,dA\right| \leq \iint_R |f|\,dA. \tag{2}$$

To see why formula (2) is true, note that, by the definition of absolute value,

$$-|f| \leq f \leq |f|;$$

therefore, from the monotonicity and homogeneity of integration (with $c = -1$),

$$-\iint_R |f|\,dA \leq \iint_R f\,dA \leq \iint_R |f|\,dA,$$

which is equivalent to formula (2).

## Fubini's Theorem

Although we have noted the integrability of a variety of functions, we have not yet established rigorously a general method of computing integrals. In the case of one variable, we avoid computing $\int_a^b f(x)\,dx$ from its definition as a limit of a sum by using the *fundamental theorem of integral calculus.* This important theorem tells us that *if $f$ is continuous, then*

$$\int_a^b f(x)\,dx = F(b) - F(a),$$

*where $F$ is an antiderivative of $f$; that is, $F' = f$.*

This technique will not work as stated for functions $f(x, y)$ of two variables. However, as we indicate in Section 5.1, we can often reduce a double integral over a rectangle to iterated single integrals; the fundamental theorem then applies to each of these single integrals. Fubini's theorem, which was mentioned in the last section, establishes this reduction to iterated integrals rigorously, by using Riemann sums. As we saw in Section 5.1, the reduction,

$$\iint_R f(x, y)\, dA = \int_a^b \left[ \int_c^d f(x, y)\, dy \right] dx = \int_c^d \left[ \int_a^b f(x, y)\, dx \right] dy,$$

is a consequence of Cavalieri's principle, at least if $f(x, y) \geq 0$. In terms of Riemann sums, it corresponds to the following equality:

$$\sum_{j,k=0}^{n-1} f(\mathbf{c}_{jk})\, \Delta x\, \Delta y = \sum_{j=0}^{n-1} \left( \sum_{k=0}^{n-1} f(\mathbf{c}_{jk})\, \Delta y \right) \Delta x = \sum_{k=0}^{n-1} \left( \sum_{j=0}^{n-1} f(\mathbf{c}_{jk})\, \Delta x \right) \Delta y,$$

which may be proved more generally as follows: *Let* $[a_{jk}]$ *be an* $n \times n$ *matrix, where* $0 \leq j \leq n-1$ *and* $0 \leq k \leq n-1$. *Let* $\sum_{j,k=0}^{n-1} a_{jk}$ *be the sum of the* $n^2$ *matrix entries. Then*

$$\sum_{j,k=0}^{n-1} a_{jk} = \sum_{j=0}^{n-1} \left( \sum_{k=0}^{n-1} a_{jk} \right) = \sum_{k=0}^{n-1} \left( \sum_{j=0}^{n-1} a_{jk} \right). \tag{3}$$

In the first equality, the right-hand side represents summing the matrix entries first by rows and then adding the results:

$$\begin{bmatrix} a_{00} & a_{01} & a_{02} & \cdots & \overrightarrow{a_{0k} \quad \cdots \quad a_{0(n-1)}} \\ \vdots & & & & \vdots \\ a_{j0} & a_{j1} & & \cdots & \overrightarrow{a_{jk} \quad \cdots \quad a_{j(n-1)}} \\ \vdots & & & & \vdots \\ a_{(n-1)0} & a_{(n-1)1} & & \cdots & \overrightarrow{a_{(n-1)k} \quad \cdots \quad a_{(n-1)(n-1)}} \end{bmatrix} \begin{matrix} \sum_{k=0}^{n-1} a_{0k} \\ \vdots \\ \sum_{k=0}^{n-1} a_{jk} \\ \vdots \\ \sum_{k=0}^{n-1} a_{(n-1)k} \downarrow \end{matrix}$$

$$\sum_{j=0}^{n-1} \left( \sum_{k=0}^{n-1} a_{jk} \right).$$

Clearly, this is equal to $\sum_{j,k=0}^{n-1} a_{jk}$, that is, the sum of all the $a_{jk}$. Similarly, the sum $\sum_{k=0}^{n-1} \left( \sum_{j=0}^{n-1} a_{jk} \right)$ represents a summing of the matrix entries by columns. This establishes equation (3) and makes the reduction to iterated integrals quite plausible

if we remember that integrals can be approximated by the corresponding Riemann sums. The actual proof of Fubini's theorem exploits this idea.

**THEOREM 3: Fubini's Theorem** Let $f$ be a continuous function with a rectangular domain $R = [a, b] \times [c, d]$. Then

$$\int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy = \iint_R f(x, y)\, dA. \quad (4)$$

*PROOF* We shall first show that

$$\int_a^b \int_c^d f(x, y)\, dy\, dx = \iint_R f(x, y)\, dA.$$

Let $c = y_0 < y_1 < \cdots < y_n = d$ be a partition of $[c, d]$ into $n$ equal parts. Define

$$F(x) = \int_c^d f(x, y)\, dy.$$

Then

$$F(x) = \sum_{k=0}^{n-1} \int_{y_k}^{y_{k+1}} f(x, y)\, dy.$$

Using the integral version of the mean-value theorem,[3] for each fixed $x$ and for each $k$ we have

$$\int_{y_k}^{y_{k+1}} f(x, y)\, dy = f(x, Y_k(x))(y_{k+1} - y_k)$$

(see Figure 5.2.6), where the point $Y_k(x)$ belongs to $[y_k, y_{k+1}]$ and may depend on $x$, $k$, and $n$.

We have thus shown that

$$F(x) = \sum_{k=0}^{n-1} f(x, Y_k(x))(y_{k+1} - y_k). \quad (5)$$

[3] This states that if $g(x)$ is continuous on $[a, b]$, then $\int_a^b g(x)\, dx = g(c)(b - a)$ for some point $c \in [a, b]$. The more general second mean-value theorem was proved in Section 3.2.

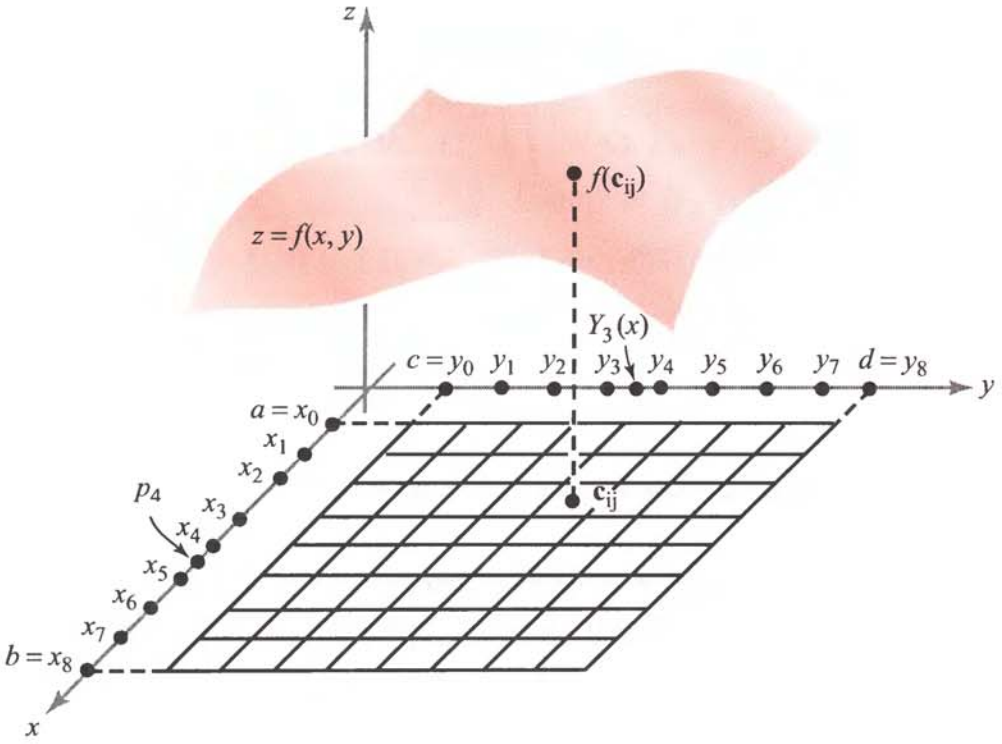



**Figure 5.2.6** The notation needed in the proof of Fubini's theorem; $n = 8$.

By the definition of the integral in one variable as a limit of Riemann sums,

$$\int_a^b F(x)\,dx = \int_a^b \left[\int_c^d f(x, y)\,dy\right] dx = \lim_{n\to\infty} \sum_{j=0}^{n-1} F(p_j)(x_{j+1} - x_j),$$

where $a = x_0 < x_1 < \cdots < x_n = b$ is a partition of the interval $[a, b]$ into $n$ equal parts and $p_j$ is any point in $[x_j, x_{j+1}]$. Setting $\mathbf{c}_{jk} = (p_j, Y_k(p_j)) \in R_{jk}$, we have [substituting $p_j$ for $x$ in equation (5)]

$$F(p_j) = \sum_{k=0}^{n-1} f(\mathbf{c}_{jk})(y_{k+1} - y_k).$$

Therefore,

$$\begin{aligned}\int_a^b \int_c^d f(x, y)\,dy\,dx = \int_a^b F(x)\,dx &= \lim_{n\to\infty} \sum_{j=0}^{n-1} F(p_j)(x_{j+1} - x_j) \\ &= \lim_{n\to\infty} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} f(\mathbf{c}_{jk})(y_{k+1} - y_k)(x_{j+1} - x_j) \\ &= \iint_R f(x, y)\,dA.\end{aligned}$$

Thus, we have proved that

$$\int_a^b \int_c^d f(x, y)\, dy\, dx = \iint_R f(x, y)\, dA.$$

By the same reasoning we can show that

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = \iint_R f(x, y)\, dA.$$

These two conclusions are exactly what we wanted to prove. ■

Fubini's theorem can be generalized to the case where $f$ is not necessarily continuous. Although we shall not present a proof, we state here this more general version.

**THEOREM 3′: Fubini's Theorem** Let $f$ be a bounded function with domain a rectangle $R = [a, b] \times [c, d]$, and suppose the discontinuities of $f$ lie on a finite union of graphs of continuous functions. If the integral $\int_c^d f(x, y)\, dy$ exists for each $x \in [a, b]$, then

$$\int_a^b \left[ \int_c^d f(x, y)\, dy \right] dx$$

exists and

$$\int_a^b \int_c^d f(x, y)\, dy\, dx = \iint_R f(x, y)\, dA.$$

Similarly, if $\int_a^b f(x, y)\, dx$ exists for each $y \in [c, d]$, then

$$\int_c^d \left[ \int_a^b f(x, y)\, dx \right] dy$$

exists and

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = \iint_R f(x, y)\, dA.$$

Thus, if all these conditions hold simultaneously,

$$\int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy = \iint_R f(x, y)\, dA.$$

The assumptions made for this version of Fubini's theorem are more complicated than those we made in Theorem 3. They are necessary because if $f$ is not continuous everywhere, for example, there is no guarantee that $\int_c^d f(x, y)\,dy$ will exist for each $x$.

**EXAMPLE 1** Compute $\iint_R (x^2 + y)\,dA$, where $R$ is the square $[0, 1] \times [0, 1]$.

SOLUTION By Fubini's theorem,

$$\iint_R (x^2 + y)\,dA = \int_0^1 \int_0^1 (x^2 + y)\,dx\,dy = \int_0^1 \left[ \int_0^1 (x^2 + y)\,dx \right] dy.$$

By the fundamental theorem of calculus, the $x$ integration may be performed:

$$\int_0^1 (x^2 + y)\,dx = \left[ \frac{x^3}{3} + yx \right]_{x=0}^1 = \frac{1}{3} + y.$$

Thus,

$$\iint_R (x^2 + y)\,dA = \int_0^1 \left[ \frac{1}{3} + y \right] dy = \left[ \frac{1}{3}y + \frac{y^2}{2} \right]_0^1 = \frac{5}{6}.$$

What we have done is hold $y$ fixed, integrate with respect to $x$, and then evaluate the result between the given limits for the $x$ variable. Next, we integrated the remaining function (of $y$ alone) with respect to $y$ to obtain the final answer. ▲

**EXAMPLE 2** A consequence of Fubini's theorem is that interchanging the order of integration in the iterated integrals does not change the answer. Verify this for Example 1.

SOLUTION We carry out the integration in the other order:

$$\int_0^1 \int_0^1 (x^2 + y)\,dy\,dx = \int_0^1 \left[ x^2 y + \frac{y^2}{2} \right]_{y=0}^1 dx = \int_0^1 \left[ x^2 + \frac{1}{2} \right] dx$$

$$= \left[ \frac{x^3}{3} + \frac{x}{2} \right]_0^1 = \frac{5}{6}. \quad ▲$$

We have seen that when $f(x, y) \geq 0$ on $R = [a, b] \times [c, d]$, the integral $\iint_R f(x, y)\,dA$ can be interpreted as a volume. If the function also takes on negative values, then the double integral can be thought of as the sum of all volumes lying between the surface $z = f(x, y)$ and the plane $z = 0$, bounded by the planes $x = a$, $x = b$, $y = c$, and $y = d$; here the volumes above $z = 0$ are counted as positive and those below as negative. However, Fubini's theorem as stated remains valid in the case where $f(x, y)$ is negative or changes sign on $R$; that is, there is no restriction on the sign of $f$ in the hypotheses of the theorem.

**EXAMPLE 3** Let $R$ be the rectangle $[-2, 1] \times [0, 1]$ and let $f$ be defined by $f(x, y) = y(x^3 - 12x)$; $f(x, y)$ takes on both positive and negative values on $R$. Evaluate the integral $\iint_R f(x, y)\,dx\,dy = \iint_R y(x^3 - 12x)\,dx\,dy$.

SOLUTION By Fubini's theorem, we may write

$$\iint_R y(x^3 - 12x)\,dx\,dy = \int_0^1 \left[ \int_{-2}^1 y(x^3 - 12x)\,dx \right] dy = \frac{57}{4} \int_0^1 y\,dy = \frac{57}{8}.$$

Alternatively, integrating first with respect to $y$, we find

$$\iint_R y(x^3 - 12x)\,dy\,dx = \int_{-2}^1 \left[ \int_0^1 (x^3 - 12x)y\,dy \right] dx$$

$$= \frac{1}{2} \int_{-2}^1 (x^3 - 12x)\,dx = \frac{1}{2}\left[ \frac{x^4}{4} - 6x^2 \right]_{-2}^1 = \frac{57}{8}. \quad \blacktriangle$$

## Historical Note

### *The Riemann Integral*

The first time most mathematics students encounter the name of Bernhard Riemann is in their calculus courses, where they read about the Riemann integral. Leibniz had thought of the integral of a function of one variable as an infinite sum (the $\int$ standing for a sum) of infinitesimal areas $f(x)\,dx$, where $dx$ is an "infinitesimal width" and $f(x)$ is the height of the corresponding "infinitesimally thin" rectangle. This intuitive approach sufficed for most purposes because the fundamental theorem

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

showed how to evaluate this (nebulously defined) integral when one knows the antiderivative $F$ of $f$.

However, Riemann was interested in applying integration to functions of one variable where the antiderivative was not known, and to functions in number theory or in general to those functions that "one need not find in nature."

Cauchy had already known that all continuous functions could be integrated and that the fundamental theorem was valid—that is, every continuous function had an antiderivative. However, his proofs were not entirely rigorous. For applications to number theory and to certain series

(called *Fourier series*), Riemann needed a clear, precise definition of the integral, which he presented in a paper in 1854. In this paper he defines his integral and gives necessary and sufficient conditions for a bounded function $f$ to be integrable over an interval $[a, b]$.

In 1876, the German mathematician Karl J. Thomae generalized Riemann's integral to apply to functions of several variables, as we do in this chapter. We further develop this approach in the Internet supplement.

In the first half of the nineteenth century, Cauchy had observed that for continuous function of two variables, Fubini's theorem was valid. But Cauchy also gave an example of an unbounded function of two variables for which the iterated integrals were not equal. In 1878, Thomae gave the first example of a bounded function of two variables where one iterated integral exists and the other does not. In these examples, the functions were not "Riemann integrable" in the sense described in this section. Cauchy and Thomae's examples demonstrated that one must apply caution and not necessarily assume that iterated integrals are always equal.

In 1902, the French mathematician Henri Lebesgue developed a truly sweeping generalization of the Riemann integral. Lebesgue's theory allowed integration of vastly more functions than did Riemann's approach. Perhaps, unforeseen by Lebesgue, his theory was to have a profound impact on the development of many areas of mathematics in the twentieth century—in particular the theory of partial differential equations. Mathematics students go into more depth about the Lebesgue integral in their first year of graduate study.

In 1907, the Italian mathematician Guido Fubini used the Lebesgue integral to state the most general form of the theorem on the equality of iterated integrals, the form that is studied today and used by working mathematicians and scientists in their research.

## EXERCISES

**1.** Evaluate each of the following integrals if $R = [0, 1] \times [0, 1]$.

(a) $\iint_R (x^3 + y^2)\, dA$

(b) $\iint_R ye^{xy}\, dA$

(c) $\iint_R (xy)^2 \cos x^3\, dA$

(d) $\iint_R \ln[(x + 1)(y + 1)]\, dA$

**2.** Evaluate each of the following integrals if $R = [0, 1] \times [0, 1]$.

(a) $\iint_R (x^m y^n)\, dx\, dy$, where $m, n > 0$

(b) $\iint_R (ax + by + c)\, dx\, dy$

(c) $\iint_R \sin(x + y)\, dx\, dy$

(d) $\iint_R (x^2 + 2xy + y\sqrt{x})\, dx\, dy$

**3.** Compute the volume of the region over the rectangle $[0, 1] \times [0, 1]$ and under the graph of $z = xy$.

**4.** Compute the volume of the solid bounded by the $xz$ plane, the $yz$ plane, the $xy$ plane, the planes $x = 1$ and $y = 1$, and the surface $z = x^2 + y^4$.

**5.** Let $f$ be continuous on $[a, b]$ and $g$ continuous on $[c, d]$. Show that

$$\iint_R [f(x)g(y)]\,dx\,dy = \left[\int_a^b f(x)\,dx\right]\left[\int_c^d g(y)\,dy\right],$$

where $R = [a, b] \times [c, d]$.

**6.** Compute the volume of the solid bounded by the surface $z = \sin y$, the planes $x = 1$, $x = 0$, $y = 0$, and $y = \pi/2$, and the $xy$ plane.

**7.** Compute the volume of the solid bounded by the graph $z = x^2 + y$, the rectangle $R = [0, 1] \times [1, 2]$, and the "vertical sides" of $R$.

**8.** Let $f$ be continuous on $R = [a, b] \times [c, d]$; for $a < x < b, c < y < d$, define

$$F(x, y) = \int_a^x \int_c^y f(u, v)\,dv\,du.$$

Show that $\partial^2 F/\partial x\,\partial y = \partial^2 F/\partial y\,\partial x = f(x, y)$. Use this example to discuss the relationship between Fubini's theorem and the equality of mixed partial derivatives.

**9.** Let $f$: $[0, 1] \times [0, 1] \to \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} 1 & x \text{ rational} \\ 2y & x \text{ irrational.} \end{cases}$$

Show that the iterated integral $\int_0^1 \left[\int_0^1 f(x, y)\,dy\right] dx$ exists but that $f$ is not integrable.

**10.** Express $\iint_R \cosh xy\,dx\,dy$ as a convergent sequence, where $R = [0, 1] \times [0, 1]$.

**11.** Although Fubini's theorem holds for most functions met in practice, one must still exercise some caution. This exercise gives a function for which it fails. By using a substitution involving the tangent function, show that

$$\int_0^1 \int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2}\,dy\,dx = \frac{\pi}{4}, \qquad \text{yet} \qquad \int_0^1 \int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2}\,dx\,dy = -\frac{\pi}{4}.$$

Why does this not contradict Theorem 3 or 3′?

**12.** Let $f$ be continuous, $f \geq 0$, on the rectangle $R$. If $\iint_R f\,dA = 0$, prove that $f = 0$ on $R$.

# 5.3 The Double Integral Over More General Regions

Our goal in this section is twofold: First, we wish to define the double integral of a function $f(x, y)$ over regions $D$ more general than rectangles; second, we want to develop a technique for evaluating this type of integral. To accomplish this, we shall define three special types of subsets of the $xy$ plane, and then extend the notion of the double integral to them.

## Elementary Regions

Suppose we are given two continuous real-valued functions $\phi_1\colon [a, b] \to \mathbb{R}$ and $\phi_2\colon [a, b] \to \mathbb{R}$ that satisfy $\phi_1(x) \leq \phi_2(x)$ for all $x \in [a, b]$. Let $D$ be the set of all points $(x, y)$ such that $x \in [a, b]$ and $\phi_1(x) \leq y \leq \phi_2(x)$. This region $D$ is said to be ***y-simple***. Figure 5.3.1 shows various examples of $y$-simple regions. The curves and straight-line segments that bound the region together constitute the ***boundary*** of $D$, denoted $\partial D$. We use the phrase $y$-simple because the region is described in a relatively simple way, using $y$ as a function of $x$.

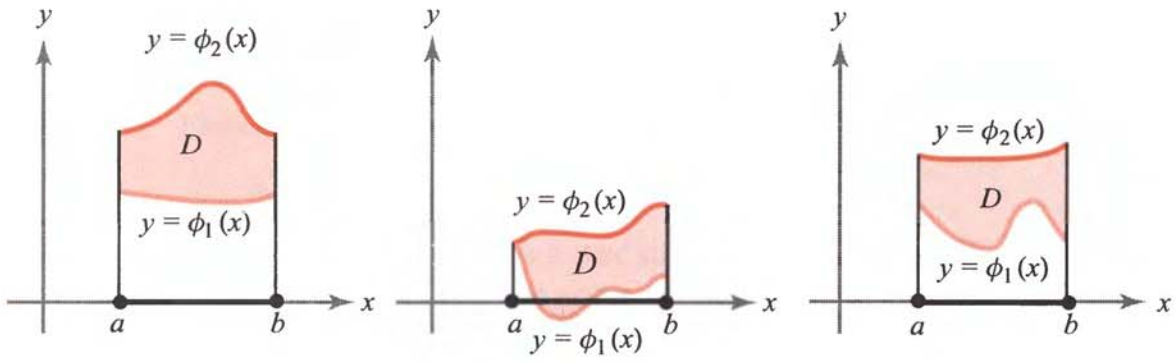



**Figure 5.3.1** Some $y$-simple regions.

We say that a region $D$ is ***x-simple*** if there are continuous functions $\psi_1$ and $\psi_2$ defined on $[c, d]$ such that $D$ is the set of points $(x, y)$ satisfying

$$y \in [c, d] \qquad \text{and} \qquad \psi_1(y) \leq x \leq \psi_2(y)$$

where $\psi_1(y) \leq \psi_2(y)$ for all $y \in [c, d]$. Again, the curves that bound the region $D$ constitute its boundary $\partial D$. Some examples of $x$-simple regions are shown in Figure 5.3.2. In this situation, $x$ is the distinguished variable, given as a function of $y$. Thus, the phrase $x$-simple is appropriate.

Finally, a ***simple*** region is one that is both $x$- and $y$-simple; that is, a simple region can be described as both an $x$-simple region and a $y$-simple region. An example of a simple region is a unit disk (see Figure 5.3.3).

Sometimes we will refer to any of the regions as ***elementary regions***. Note that the boundary $\partial D$ of an elementary region is the type of set of discontinuities of a function allowed in Theorem 2.

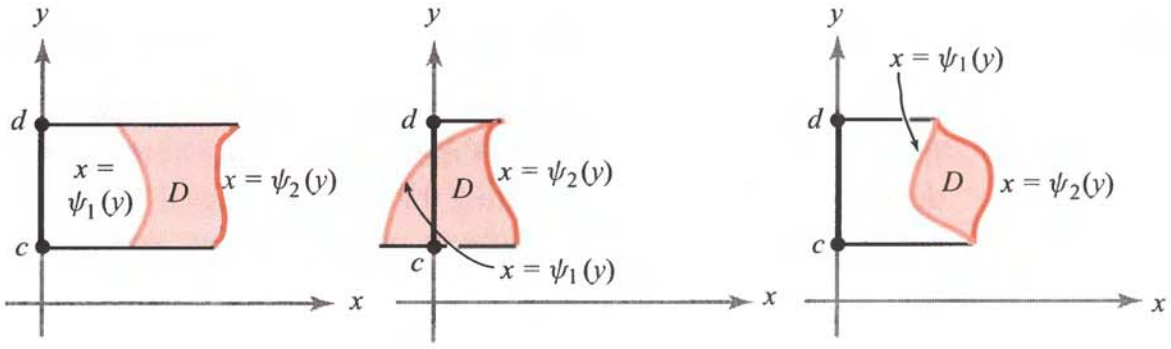



**Figure 5.3.2** Some $x$-simple regions.

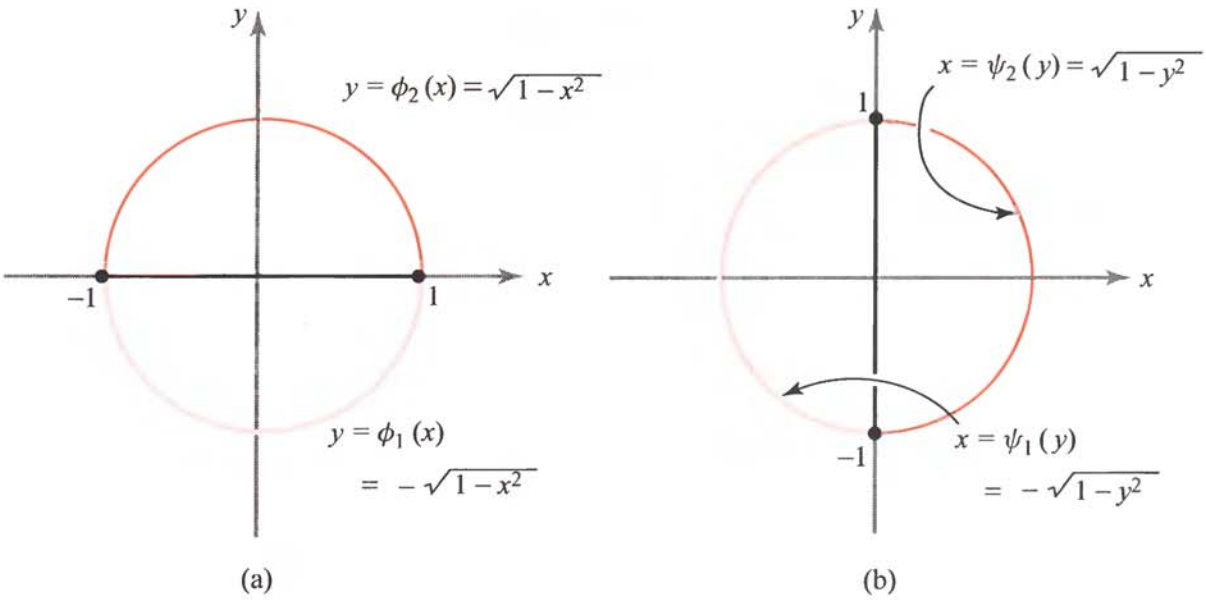



**Figure 5.3.3** The unit disk, a simple region: (a) as a $y$-simple region, and (b) as an $x$-simple region.

## The Integral over an Elementary Region

We can now use an interesting "trick" to extend the definition of the integral from rectangles to elementary regions.

**DEFINITION: Integral over an Elementary Region** If $D$ is an elementary region in the plane, choose a rectangle $R$ that contains $D$. Given $f: D \to \mathbb{R}$, where $f$ is continuous (and hence bounded), define $\iint_D f(x, y)\, dA$, the ***integral of $f$ over the set $D$*** as follows: Extend $f$ to a function $f^*$ defined on all of $R$ by

$$f^*(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \in D \\ 0 & \text{if } (x, y) \notin D \text{ and } (x, y) \in R. \end{cases}$$

Note that $f^*$ is bounded (because $f$ is) and continuous except possibly on the boundary of $D$ (see Figure 5.3.4). The boundary of $D$ consists of graphs of

continuous functions, and so $f^*$ is integrable over $R$ by Theorem 2, Section 5.2. Therefore, we can define

$$\iint_D f(x, y)\, dA = \iint_R f^*(x, y)\, dA.$$

When $f(x, y) \geq 0$ on $D$, we can interpret the integral $\iint_D f(x, y)\, dA$ as the volume of the three-dimensional region between the graph of $f$ and $D$, as is evident from Figure 5.3.4.

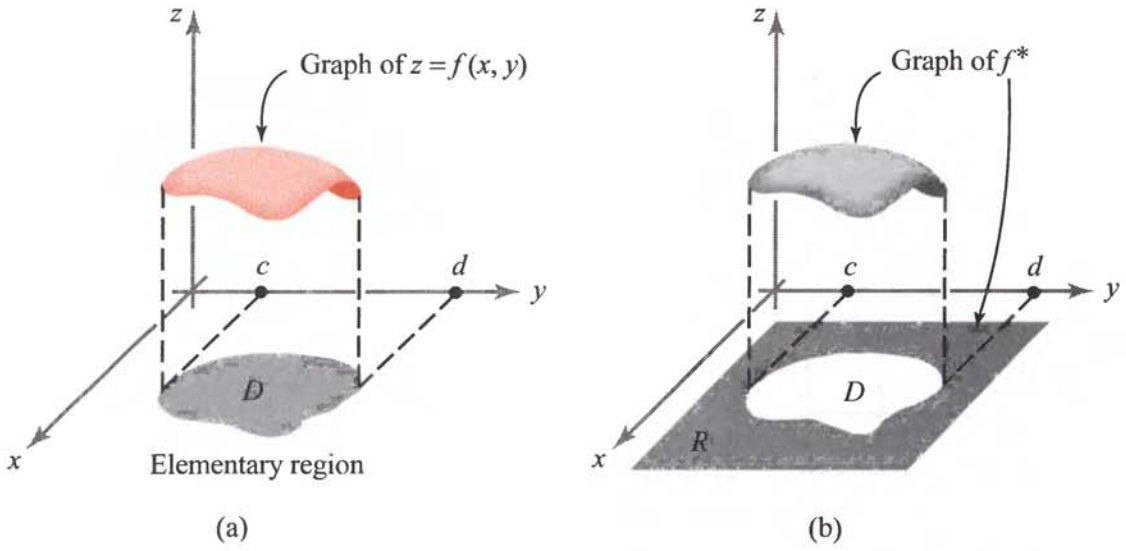



**Figure 5.3.4** (a) Graph of $z = f(x, y)$ over an elementary region $D$. (b) Shaded region shows graph of $z = f^*(x, y)$ on some rectangle $R$ containing $D$. From this picture we see that boundary points of $D$ may be points of discontinuity of $f^*$, because the graph of $z = f^*(x, y)$ can be broken at these points.

We have defined $\iint_D f(x, y)\, dx\, dy$ by choosing a rectangle $R$ that encloses $D$. It should be intuitively clear that the value of $\iint_D f(x, y)\, dx\, dy$ does not depend on the particular $R$ we select; we shall demonstrate this fact at the end of this section.

## Reduction to Iterated Integrals

If $R = [a, b] \times [c, d]$ is a rectangle containing $D$, we can use the results on iterated integrals in Section 5.2 to obtain

$$\iint_D f(x, y)\, dA = \iint_R f^*(x, y)\, dA = \int_a^b \int_c^d f^*(x, y)\, dy\, dx$$
$$= \int_c^d \int_a^b f^*(x, y)\, dx\, dy,$$

where $f^*$ equals $f$ in $D$ and zero outside $D$, as before. Assume that $D$ is a $y$-simple region determined by functions $\phi_1 : [a, b] \to \mathbb{R}$ and $\phi_2 : [a, b] \to \mathbb{R}$.

Consider the iterated integral

$$\int_a^b \int_c^d f^*(x, y)\, dy\, dx$$

and, in particular, the inner integral $\int_c^d f^*(x, y)\, dy$ for some fixed $x$ (Figure 5.3.5). By definition, $f^*(x, y) = 0$ if $y < \phi_1(x)$ or $y > \phi_2(x)$, so we obtain

$$\int_c^d f^*(x, y)\, dy = \int_{\phi_1(x)}^{\phi_2(x)} f^*(x, y)\, dy = \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy.$$

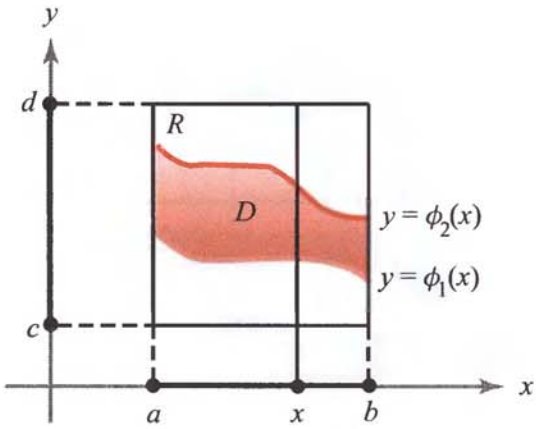



**Figure 5.3.5** The region between two graphs—a $y$-simple region.

We summarize what we have obtained in the following.

**THEOREM 4: Reduction to Iterated Integrals** If $D$ is a $y$-simple region, as shown in Figure 5.3.5, then

$$\iint_D f(x, y)\, dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy\, dx. \tag{1}$$

In the case $f(x, y) = 1$ for all $(x, y) \in D$, $\iint_D f(x, y)\, dA$ is the area of $D$. On the other hand, in this case, the right-hand side of formula (1) becomes:

$$\int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy\, dx = \int_a^b [\phi_2(x) - \phi_1(x)]\, dx = A(D),$$

which is the formula for the area of $D$ learned in one-variable calculus. Thus, formula (1) checks in this case.

**EXAMPLE 1** Find $\iint_T (x^3y + \cos x)\,dA$, where $T$ is the triangle consisting of all points $(x, y)$ such that $0 \le x \le \pi/2$, $0 \le y \le x$.

SOLUTION Referring to Figure 5.3.6 and formula (1), we have

$$\iint_T (x^3y + \cos x)\,dA = \int_0^{\pi/2} \int_0^x (x^3y + \cos x)\,dy\,dx$$

$$= \int_0^{\pi/2} \left[\frac{x^3y^2}{2} + y\cos x\right]_{y=0}^{x} dx = \int_0^{\pi/2} \left(\frac{x^5}{2} + x\cos x\right) dx$$

$$= \left[\frac{x^6}{12}\right]_0^{\pi/2} + \int_0^{\pi/2} (x\cos x)\,dx = \frac{\pi^6}{(12)(64)} + [x\sin x + \cos x]_0^{\pi/2}$$

$$= \frac{\pi^6}{768} + \frac{\pi}{2} - 1. \quad \blacktriangle$$

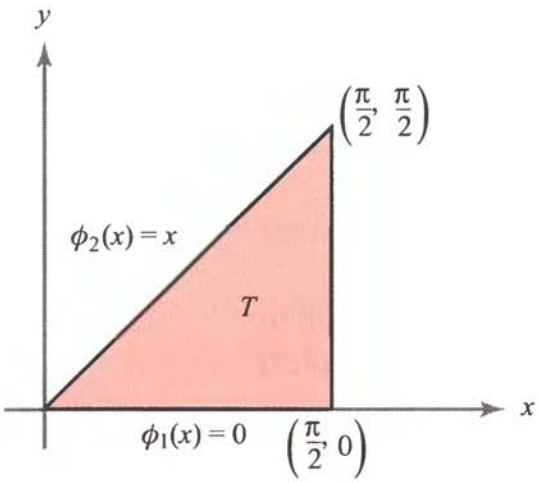



**Figure 5.3.6** A triangle $T$ represented as a $y$-simple region.

In the next example, we use formula (1) to find the volume of a solid whose base is a nonrectangular region $D$.

**EXAMPLE 2** Find the volume of the tetrahedron bounded by the planes $y = 0$, $z = 0$, $x = 0$, and $y - x + z = 1$ (Figure 5.3.7).

SOLUTION We first note that the given tetrahedron has a triangular base $D$ whose points $(x, y)$ satisfy $-1 \le x \le 0$ and $0 \le y \le 1 + x$; hence, $D$ is a $y$-simple region. In fact, $D$ is a simple region; see Figure 5.3.8.

For any point $(x, y)$ in $D$, the height of the surface $z$ above $(x, y)$ is $1 - y + x$. Thus, the volume we seek is given by the integral

$$\iint_D (1 - y + x)\,dA.$$

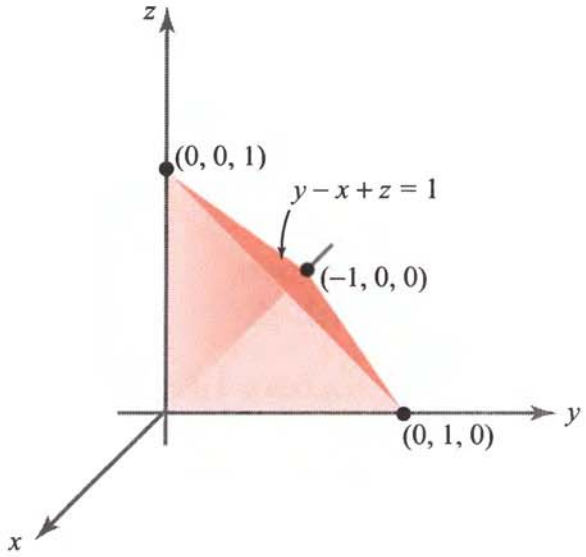



**Figure 5.3.7** A tetrahedron bounded by the planes $y = 0, z = 0, x = 0$, and $y - x + z = 1$.

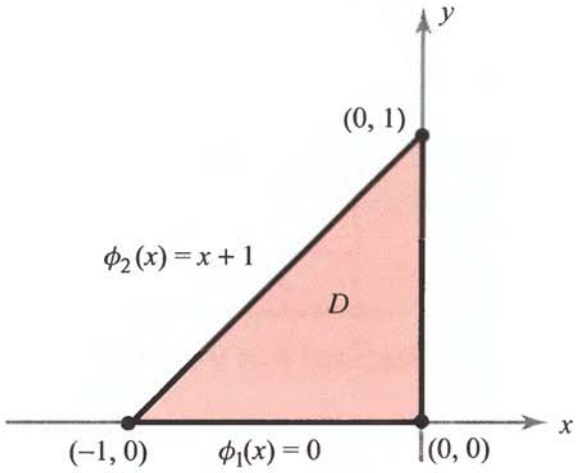



**Figure 5.3.8** The base of the tetrahedron in Figure 5.3.7 represented as a $y$-simple region.

Using formula (1) with $\phi_1(x) = 0$ and $\phi_2(x) = x + 1$, we have

$$\iint_D (1 - y + x)\,dA = \int_{-1}^{0} \int_{0}^{1+x} (1 - y + x)\,dy\,dx = \int_{-1}^{0} \left[(1 + x)y - \frac{y^2}{2}\right]_{y=0}^{1+x} dx$$

$$= \int_{-1}^{0} \left[\frac{(1 + x)^2}{2}\right] dx = \left[\frac{(1 + x)^3}{6}\right]_{-1}^{0} = \frac{1}{6}. \quad \blacktriangle$$

**EXAMPLE 3** Let $D$ be a $y$-simple region. Describe its area $A(D)$ as a limit of Riemann sums.

SOLUTION If we recall the definition, $A(D) = \iint_D dx\,dy$ is the integral over a containing rectangle $R$ of the function $f = 1$. A Riemann sum $S_n$ for this integral is obtained by dividing $R$ into subrectangles and forming the sum $S_n = \sum_{j,k=0}^{n-1} f^*(\mathbf{c}_{jk})\Delta x\,\Delta y$, as in formula (1) of Section 5.2. Now $f^*(\mathbf{c}_{jk})$ is 1 or 0, depending on whether or not $\mathbf{c}_{jk}$ is in $D$. Consider those subrectangles $R_{jk}$ that have nonvoid intersection with $D$, and choose $\mathbf{c}_{jk}$ in $D \cap R_{jk}$. Thus, $S_n$ is the sum of the areas of the subrectangles that meet $D$ and $A(D)$ is the limit of these as $n \to \infty$.

Thus, $A(D)$ is the limit of the areas of the rectangles "circumscribing" $D$. The reader should draw a figure to accompany this discussion. ▲

The methods for treating $x$-simple regions are entirely analogous. Specifically, we have the following.

**THEOREM 4′: Iterated Integrals for $x$-Simple Regions** Suppose that $D$ is the set of points $(x, y)$ such that $y \in [c, d]$ and $\psi_1(y) \le x \le \psi_2(y)$. If $f$ is continuous on $D$, then

$$\iint_D f(x, y)\, dA = \int_c^d \left[ \int_{\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx \right] dy. \tag{2}$$

To find the area of $D$, we substitute $f = 1$ in formula (2); this yields

$$\iint_D dA = \int_c^d (\psi_2(y) - \psi_1(y))\, dy.$$

Again, this result for area agrees with the results of single-variable calculus for the area of a region between two curves.

Either the method for $y$-simple or the method for $x$-simple regions can be used for integrals over simple regions.

It follows from formulas (1) and (2) that $\iint_D f\, dA$ is independent of the choice of the rectangle $R$ enclosing $D$ used in the definition of $\iint_D f\, dA$, because, if we had picked another rectangle enclosing $D$, we would have arrived at the same formula (1).

## EXERCISES

**1.** Evaluate the following iterated integrals and draw the regions $D$ determined by the limits. State whether the regions are $x$-simple, $y$-simple, or simple.

(a) $\displaystyle\int_0^1 \int_0^{x^2} dy\, dx$

(b) $\displaystyle\int_1^2 \int_{2x}^{3x+1} dy\, dx$

(c) $\displaystyle\int_0^1 \int_1^{e^x} (x + y)\, dy\, dx$

(d) $\displaystyle\int_0^1 \int_{x^3}^{x^2} y\, dy\, dx$

**2.** Evaluate the following integrals and sketch the corresponding regions.

(a) $\displaystyle\int_{-3}^2 \int_0^{y^2} (x^2 + y)\, dx\, dy$

(b) $\displaystyle\int_{-1}^1 \int_{-2|x|}^{|x|} e^{x+y}\, dy\, dx$

(c) $\displaystyle\int_0^1 \int_0^{(1-x^2)^{1/2}} dy\, dx$

(d) $\displaystyle\int_0^{\pi/2} \int_0^{\cos x} y \sin x\, dy\, dx$

(e) $\displaystyle\int_0^1 \int_{y^2}^{y} (x^n + y^m)\, dx\, dy, \qquad m, n > 0$

(f) $\displaystyle\int_{-1}^0 \int_0^{2(1-x^2)^{1/2}} x\, dy\, dx$

**3.** Use double integrals to compute the area of a circle of radius $r$.

**4.** Using double integrals, determine the area of an ellipse with semiaxes of length $a$ and $b$.

**5.** What is the volume of a barn that has a rectangular base 20 ft by 40 ft, vertical walls 30 ft high at the front (which we assume is on the 20-ft side of the barn), and 40 ft high at the rear? The barn has a flat roof. Use double integrals to compute the volume.

**6.** Let $D$ be the region bounded by the positive $x$ and $y$ axes and the line $3x + 4y = 10$. Compute

$$\iint_D (x^2 + y^2)\, dA.$$

**7.** Let $D$ be the region bounded by the $y$ axis and the parabola $x = -4y^2 + 3$. Compute

$$\iint_D x^3 y\, dx\, dy.$$

**8.** Evaluate $\int_0^1 \int_0^{x^2} (x^2 + xy - y^2)\, dy\, dx$. Describe this iterated integral as an integral over a certain region $D$ in the $xy$ plane.

**9.** Let $D$ be the region given as the set of $(x, y)$ where $1 \le x^2 + y^2 \le 2$ and $y \ge 0$. Is $D$ an elementary region? Evaluate $\iint_D f(x, y)\, dA$ where $f(x, y) = 1 + xy$.

**10.** Use the formula $A(D) = \iint_D dx\, dy$ to find the area enclosed by one period of the sine function $\sin x$, for $0 \le x \le 2\pi$, and the $x$ axis.

**11.** Find the volume of the region inside the surface $z = x^2 + y^2$ and between $z = 0$ and $z = 10$.

**12.** Set up the integral required to calculate the volume of a cone of base radius $r$ and height $h$.

**13.** Evaluate $\iint_D y\, dA$ where $D$ is the set of points $(x, y)$ such that $0 \le 2x/\pi \le y, y \le \sin x$.

**14.** From Exercise 5, Section 5.2, $\int_a^b \int_c^d f(x)g(y)\, dy\, dx = \left(\int_a^b f(x)\, dx\right)\left(\int_c^d g(y)\, dy\right)$. Is it true that $\iint_D f(x)g(y)\, dx\, dy = \left(\int_a^b f(x)\, dx\right)\left(\int_{\phi_1(a)}^{\phi_2(b)} g(y)\, dy\right)$ for $y$-simple regions?

**15.** Let $D$ be a region given as the set of $(x, y)$ with $-\phi(x) \le y \le \phi(x)$ and $a \le x \le b$, where $\phi$ is a nonnegative continuous function on the interval $[a, b]$. Let $f(x, y)$ be a function on $D$ such that $f(x, y) = -f(x, -y)$ for all $(x, y) \in D$. Argue that $\iint_D f(x, y)\, dA = 0$.

**16.** Use the methods of this section to show that the area of the parallelogram $D$ determined by two planar vectors $\mathbf{a}$ and $\mathbf{b}$ is $|a_1 b_2 - a_2 b_1|$, where $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j}$.

**17.** Describe the area $A(D)$ of a region as a limit of areas of inscribed rectangles, as in Example 3.

## 5.4 Changing the Order of Integration

Suppose that $D$ is a simple region—that is, it is both $x$-simple and $y$-simple. Thus, it can be given as the set of points $(x, y)$ such that

$$a \le x \le b, \qquad \phi_1(x) \le y \le \phi_2(x),$$

and also as the set of points $(x, y)$ such that

$$c \le y \le d, \qquad \psi_1(y) \le x \le \psi_2(y).$$

Hence, we have the formulas

$$\iint_D f(x, y)\, dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy\, dx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx\, dy.$$

If we are required to compute one of the preceding iterated integrals, we may do so by evaluating the other iterated integral; this technique is called *changing the order of integration*. It can be useful to make such a change when evaluating iterated integrals, because one of the iterated integrals may be more difficult to compute than the other.

**EXAMPLE 1** By changing the order of integration, evaluate

$$\int_0^a \int_0^{(a^2-x^2)^{1/2}} (a^2 - y^2)^{1/2}\, dy\, dx.$$

SOLUTION Note that $x$ varies between 0 and $a$, and for each such fixed $x$, we have $0 \le y \le (a^2 - x^2)^{1/2}$. Thus, the iterated integral is equivalent to the double integral

$$\iint_D (a^2 - y^2)^{1/2}\, dy\, dx,$$

where $D$ is the set of points $(x, y)$ such that $0 \le x \le a$ and $0 \le y \le (a^2 - x^2)^{1/2}$. But this is the representation of one quarter (the positive quadrant portion) of the disk of radius $a$; hence, $D$ can also be described as the set of points $(x, y)$ satisfying

$$0 \le y \le a, \qquad 0 \le x \le (a^2 - y^2)^{1/2}$$

(see Figure 5.4.1). Thus,

$$\begin{aligned}
\int_0^a \int_0^{(a^2-x^2)^{1/2}} (a^2 - y^2)^{1/2}\, dy\, dx &= \int_0^a \left[ \int_0^{(a^2-y^2)^{1/2}} (a^2 - y^2)^{1/2}\, dx \right] dy \\
&= \int_0^a [x(a^2 - y^2)^{1/2}]_{x=0}^{(a^2-y^2)^{1/2}}\, dy \\
&= \int_0^a (a^2 - y^2)\, dy = \left[ a^2 y - \frac{y^3}{3} \right]_0^a = \frac{2a^3}{3}. \quad \blacktriangle
\end{aligned}$$

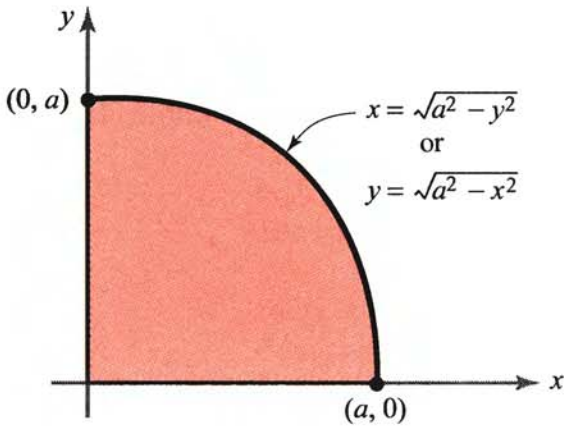



**Figure 5.4.1** The positive-quadrant portion of a disk of radius $a$.

We could have evaluated the initial iterated integral directly, but, as the reader can easily verify, changing the order of integration makes the problem simpler. The next example shows that it may not be obvious how to evaluate an iterated integral, and yet it may be relatively simple to evaluate the iterated integral obtained by changing the order of integration.

**EXAMPLE 2** Evaluate

$$\int_1^2 \int_0^{\log x} (x - 1)\sqrt{1 + e^{2y}}\, dy\, dx.$$

SOLUTION It will simplify matters if we first interchange the order of integration. First notice that the integral is equal to $\iint_D (x - 1)\sqrt{1 + e^{2y}}\, dA$, where $D$ is the set of $(x, y)$ such that

$$1 \le x \le 2 \qquad \text{and} \qquad 0 \le y \le \log x.$$

The region $D$ is simple (see Figure 5.4.2) and can also be described by

$$0 \le y \le \log 2 \qquad \text{and} \qquad e^y \le x \le 2.$$

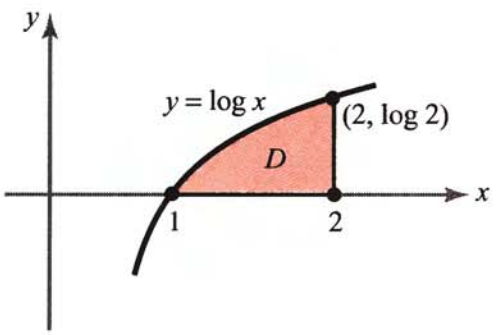



**Figure 5.4.2** $D$ is the region of integration for Example 2.

Thus, the given iterated integral is equal to

$$\int_0^{\log 2}\int_{e^y}^{2}(x-1)\sqrt{1+e^{2y}}\,dx\,dy = \int_0^{\log 2}\sqrt{1+e^{2y}}\left[\int_{e^y}^{2}(x-1)\,dx\right]dy$$

$$= \int_0^{\log 2}\sqrt{1+e^{2y}}\left[\frac{x^2}{2}-x\right]_{e^y}^{2}dy$$

$$= -\int_0^{\log 2}\left(\frac{e^{2y}}{2}-e^y\right)\sqrt{1+e^{2y}}\,dy$$

$$= -\frac{1}{2}\int_0^{\log 2}e^{2y}\sqrt{1+e^{2y}}\,dy + \int_0^{\log 2}e^y\sqrt{1+e^{2y}}\,dy. \tag{1}$$

In the first integral in expression (1), we substitute $u = e^{2y}$, and in the second, $v = e^y$. Hence, we obtain

$$-\frac{1}{4}\int_1^4\sqrt{1+u}\,du + \int_1^2\sqrt{1+v^2}\,dv. \tag{2}$$

Both integrals in expression (2) are easily found with techniques of one-variable calculus (or by consulting the table of integrals at the back of the book). For the first integral, we get

$$\frac{1}{4}\int_1^4\sqrt{1+u}\,du = \left[\frac{1}{6}(1+u)^{3/2}\right]_1^4 = \frac{1}{6}[(1+4)^{3/2}-2^{3/2}] = \frac{1}{6}[5^{3/2}-2^{3/2}]. \tag{3}$$

The second integral is

$$\int_1^2\sqrt{1+v^2}\,dv = \frac{1}{2}\left[v\sqrt{1+v^2}+\log\left(\sqrt{1+v^2}+v\right)\right]_1^2$$

$$= \frac{1}{2}\left[2\sqrt{5}+\log\left(\sqrt{5}+2\right)\right]-\frac{1}{2}\left[\sqrt{2}+\log\left(\sqrt{2}+1\right)\right] \tag{4}$$

(see formula 43 in the table of integrals at the back of the book). Finally, we subtract equation (3) from equation (4) to obtain the answer

$$\frac{1}{2}\left(2\sqrt{5}-\sqrt{2}+\log\frac{\sqrt{5}+2}{\sqrt{2}+1}\right)-\frac{1}{6}[5^{3/2}-2^{3/2}]. \quad \blacktriangle$$

## Mean Value Inequality

We conclude with an inequality that helps us estimate integrals. Suppose there are numbers $m$ and $M$ such that for all $(x, y) \in D$, and $m \le f(x, y) \le M$, then integrating over $D$, we get

$$m \cdot A(D) \le \iint_D f(x, y)\, dA \le M \cdot A(D), \tag{5}$$

where $A(D)$ is the area of the region $D$. Even though this inequality is obvious, it can help us *estimate* integrals that we cannot easily evaluate *exactly*.

**EXAMPLE 3** Consider the integral

$$\iint_D \frac{1}{\sqrt{1 + x^6 + y^8}}\, dx\, dy,$$

where $D$ is the unit square $[0, 1] \times [0, 1]$. Because the integrand satisfies, for $x$ and $y$ between 0 and 1,

$$\frac{1}{\sqrt{3}} \le \frac{1}{\sqrt{1 + x^6 + y^8}} \le 1,$$

and because the square has area 1, we get:

$$\frac{1}{\sqrt{3}} \le \iint_D \frac{1}{\sqrt{1 + x^6 + y^8}}\, dx\, dy \le 1. \quad \blacktriangle$$

## Mean Value Equality

The mean value inequality can be turned into an equality when $f$ is continuous. Here is the formal statement.

**THEOREM 5: Mean Value Theorem: Double Integrals** Suppose $f: D \to \mathbb{R}$ is continuous and $D$ is an elementary region. Then for some point $(x_0, y_0)$ in $D$ we have

$$\iint_D f(x, y)\, dA = f(x_0, y_0) A(D),$$

where $A(D)$ denotes the area of $D$.

*PROOF* We cannot prove this theorem with complete rigor, because it requires some concepts about continuous functions not proved in this course; but we can sketch the main ideas that underlie the proof.

Because $f$ is continuous on $D$, it has a maximum value $M$ and a minimum value $m$. Thus, $m \le f(x, y) \le M$ for all $(x, y) \in D$. Furthermore, $f(x_1, y_1) = m$ and $f(x_2, y_2) = M$ for some pairs $(x_1, y_1)$ and $(x_2, y_2)$ in $D$.

Dividing through inequality (5) by $A(D)$, we get

$$m \le \frac{1}{A(D)} \iint_D f(x, y)\, dA \le M. \tag{6}$$

Because a continuous function on $D$ takes on every value between its maximum and minimum values (this is the two-variable *intermediate value theorem* proved in advanced calculus; see also Review Exercise 32), and because the number $[1/A(D)] \iint_D f(x, y)\, dA$ is, by inequality (6), between these values, there must be a point $(x_0, y_0) \in D$ with

$$f(x_0, y_0) = \frac{1}{A(D)} \iint_D f(x, y)\, dA,$$

which is precisely the conclusion of Theorem 5. ■

## EXERCISES

**1.** In the following integrals, change the order of integration, sketch the corresponding regions, and evaluate the integral both ways.

(a) $\displaystyle\int_0^1 \int_x^1 xy\, dy\, dx$

(b) $\displaystyle\int_0^{\pi/2} \int_0^{\cos\theta} \cos\theta\, dr\, d\theta$

(c) $\displaystyle\int_0^1 \int_1^{2-y} (x + y)^2\, dx\, dy$

(d) $\displaystyle\int_a^b \int_a^y f(x, y)\, dx\, dy$ (express your answer in terms of antiderivatives).

**2.** Find

(a) $\displaystyle\int_{-1}^1 \int_{|y|}^1 (x + y)^2\, dx\, dy$

(b) $\displaystyle\int_{-3}^1 \int_{-\sqrt{(9-y^2)}}^{\sqrt{(9-y^2)}} x^2\, dx\, dy$

(c) $\displaystyle\int_0^4 \int_{y/2}^2 e^{x^2}\, dx\, dy$

(d) $\displaystyle\int_0^1 \int_{\tan^{-1} y}^{\pi/4} (\sec^5 x)\, dx\, dy$

**3.** If $f(x, y) = e^{\sin(x+y)}$ and $D = [-\pi, \pi] \times [-\pi, \pi]$, show that

$$\frac{1}{e} \le \frac{1}{4\pi^2} \iint_D f(x, y)\, dA \le e.$$

**4.** Show that

$$\frac{1}{2}(1 - \cos 1) \le \iint_{[0,1]\times[0,1]} \frac{\sin x}{1 + (xy)^4}\, dx\, dy \le 1.$$

**5.** If $D = [-1, 1] \times [-1, 2]$, show that

$$1 \le \iint_D \frac{dx\, dy}{x^2 + y^2 + 1} \le 6.$$

**6.** Using the mean value inequality, show that

$$\frac{1}{6} \leq \iint_D \frac{dA}{y - x + 3} \leq \frac{1}{4},$$

where $D$ is the triangle with vertices (0, 0), (1, 1), and (1, 0).

**7.** Compute the volume of an ellipsoid with semiaxes $a$, $b$, and $c$. (HINT: Use symmetry and first find the volume of one half of the ellipsoid.)

**8.** Compute $\iint_D f(x, y)\, dA$, where $f(x, y) = y^2\sqrt{x}$ and $D$ is the set of $(x, y)$ where $x > 0$, $y > x^2$, and $y < 10 - x^2$.

**9.** Find the volume of the region determined by $x^2 + y^2 + z^2 \leq 10$, $z \geq 2$. Use the disk method from one-variable calculus and state how the method is related to Cavalieri's principle.

**10.** Evaluate $\iint_D e^{x-y}\, dx\, dy$, where $D$ is the interior of the triangle with vertices (0, 0), (1, 3), and (2, 2).

**11.** Evaluate $\iint_D y^3(x^2 + y^2)^{-3/2}\, dx\, dy$, where $D$ is the region determined by the conditions $\frac{1}{2} \leq y \leq 1$ and $x^2 + y^2 \leq 1$.

**12.** Given that the double integral $\iint_D f(x, y)\, dx\, dy$ of a positive continuous function $f$ equals the iterated integral $\int_0^1 \left[\int_{x^2}^{x} f(x, y)\, dy\right] dx$, sketch the region $D$ and interchange the order of integration.

**13.** Given that the double integral $\iint_D f(x, y)\, dx\, dy$ of a positive continuous function $f$ equals the iterated integral $\int_0^1 \left[\int_{y}^{\sqrt{2-y^2}} f(x, y)\, dx\right] dy$, sketch the region $D$ and interchange the order of integration.

**14.** Prove that $2\int_a^b \int_x^b f(x)f(y)\, dy\, dx = \left(\int_a^b f(x)\, dx\right)^2$. [HINT: Notice that $\left(\int_a^b f(x)\, dx\right)^2 = \iint_{[a,b]\times[a,b]} f(x)f(y)\, dx\, dy$.]

**15.** Show that (see Exercise 27, Section 2.5)

$$\frac{d}{dx}\int_a^x \int_c^d f(x, y, z)\, dz\, dy = \int_c^d f(x, y, z)\, dz + \int_a^x \int_c^d f_x(x, y, z)\, dz\, dy.$$

## 5.5 The Triple Integral

Triple integrals are needed for many physical problems. For example, if the temperature inside an oven is not uniform, determining the average temperature involves "summing" the values of the temperature function at all points in the solid region enclosed by the oven walls and then dividing the answer by the total volume of the oven. Such a sum is expressed mathematically as a triple integral.

## Definition of the Triple Integral

Our objective now is to define the triple integral of a function $f(x, y, z)$ over a box (rectangular parallelepiped) $B = [a, b] \times [c, d] \times [p, q]$. Proceeding as in double integrals, we partition the three sides of $B$ into $n$ equal parts and form the sum

$$S_n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} f(\mathbf{c}_{ijk})\, \Delta V,$$

where $\mathbf{c}_{ijk}$ is a point in $B_{ijk}$, the $ijk$th rectangular parallelepiped (or box) in the partition of $B$, and $\Delta V$ is the volume of $B_{ijk}$ (see Figure 5.5.1).

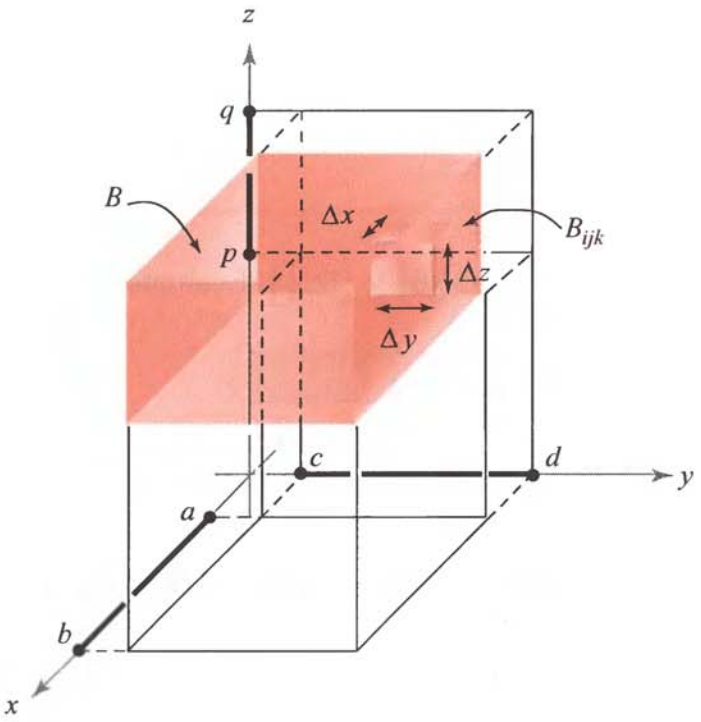



**Figure 5.5.1** A partition of a box $B$ into $n^3$ subboxes $B_{ijk}$.

**DEFINITION: Triple Integrals** Let $f$ be a bounded function of three variables defined on $B$. If $\lim_{n\to\infty} S_n = S$ exists and is independent of any choice of $\mathbf{c}_{ijk}$, we call $f$ ***integrable*** and call $S$ the ***triple integral*** (or simply the integral) of $f$ over $B$ and denote it by

$$\iiint_B f\, dV, \qquad \iiint_B f(x, y, z)\, dV \qquad \text{or} \qquad \iiint_B f(x, y, z)\, dx\, dy\, dz.$$

## Properties of Triple Integrals

As before, one can prove that continuous functions defined on $B$ are integrable. Moreover, bounded functions whose discontinuities are confined to graphs of continuous functions [such as $x = \alpha(y, z)$, $y = \beta(x, z)$, or $z = \gamma(x, y)$] are integrable. The other basic properties (such as the fact that the integral of a sum is the sum of the integrals)

for double integrals also hold for triple integrals. Especially important is the reduction to iterated integrals:

**Reduction to Iterated Integrals** Let $f(x, y, z)$ be integrable on the box $B = [a, b] \times [c, d] \times [p, q]$. Then any iterated integral that exists is equal to the triple integral; that is,

$$\begin{aligned}\iiint_B f(x, y, z)\,dx\,dy\,dz &= \int_p^q \int_c^d \int_a^b f(x, y, z)\,dx\,dy\,dz \\ &= \int_p^q \int_a^b \int_c^d f(x, y, z)\,dy\,dx\,dz \\ &= \int_a^b \int_p^q \int_c^d f(x, y, z)\,dy\,dz\,dx,\end{aligned}$$

and so on. (There are six possible orders altogether.)

**EXAMPLE 1** (a) Let $B$ be the box $[0, 1] \times [-\frac{1}{2}, 0] \times [0, \frac{1}{3}]$. Evaluate

$$\iiint_B (x + 2y + 3z)^2\,dx\,dy\,dz.$$

(b) Verify that we get the same answer if the integration is done in the order $y$ first, then $z$, and then $x$.

SOLUTION (a) According to the principle of reduction to iterated integrals, this integral may be evaluated as

$$\begin{aligned}&\int_0^{1/3} \int_{-1/2}^0 \int_0^1 (x + 2y + 3z)^2\,dx\,dy\,dz \\ &= \int_0^{1/3} \int_{-1/2}^0 \left[\frac{(x + 2y + 3z)^3}{3}\Big|_{x=0}^{1}\right] dy\,dz \\ &= \int_0^{1/3} \int_{-1/2}^0 \frac{1}{3}\left[(1 + 2y + 3z)^3 - (2y + 3z)^3\right] dy\,dz \\ &= \int_0^{1/3} \frac{1}{24}\left[(1 + 2y + 3z)^4 - (2y + 3z)^4\right]\Big|_{y=-1/2}^{0} dz \\ &= \int_0^{1/3} \frac{1}{24}\left[(3z + 1)^4 - 2(3z)^4 + (3z - 1)^4\right] dz \\ &= \frac{1}{24 \cdot 15}\left[(3z + 1)^5 - 2(3z)^5 + (3z - 1)^5\right]\Big|_{z=0}^{1/3} \\ &= \frac{1}{24 \cdot 15}(2^5 - 2) = \frac{1}{12}.\end{aligned}$$

(b)

$$\begin{aligned}
&\iiint_B (x+2y+3z)^2\,dy\,dz\,dx \\
&= \int_0^1 \int_0^{1/3} \int_{-1/2}^{0} (x+2y+3z)^2\,dy\,dz\,dx \\
&= \int_0^1 \int_0^{1/3} \left[ \frac{(x+2y+3z)^3}{6} \Big|_{y=-1/2}^{0} \right] dz\,dx \\
&= \int_0^1 \int_0^{1/3} \frac{1}{6}\left[(x+3z)^3 - (x+3z-1)^3\right] dz\,dx \\
&= \int_0^1 \frac{1}{6}\left\{ \left[ \frac{(x+3z)^4}{12} - \frac{(x+3z-1)^4}{12} \right] \Big|_{z=0}^{1/3} \right\} dx \\
&= \int_0^1 \frac{1}{72}\left[(x+1)^4 + (x-1)^4 - 2x^4\right] dx \\
&= \frac{1}{72}\frac{1}{5}\left[(x+1)^5 + (x-1)^5 - 2x^5\right]_{x=0}^{1} = \frac{1}{12}. \quad \blacktriangle
\end{aligned}$$

**EXAMPLE 2** Integrate $e^{x+y+z}$ over the box $[0, 1] \times [0, 1] \times [0, 1]$.

SOLUTION We perform the integrations in the standard order:

$$\begin{aligned}
&\int_0^1 \int_0^1 \int_0^1 e^{x+y+z}\,dx\,dy\,dz = \int_0^1 \int_0^1 (e^{x+y+z}|_{x=0}^{1})\,dy\,dz \\
&= \int_0^1 \int_0^1 (e^{1+y+z} - e^{y+z})\,dy\,dz = \int_0^1 \left[ e^{1+y+z} - e^{y+z} \right]_{y=0}^{1} dz \\
&= \int_0^1 \left[ e^{2+z} - 2e^{1+z} + e^{z} \right] dz = \left[ e^{2+z} - 2e^{1+z} + e^{z} \right]_0^1 \\
&= e^3 - 3e^2 + 3e - 1 = (e-1)^3. \quad \blacktriangle
\end{aligned}$$

As in the two-variable case, we define the integral of a function $f$ over a bounded region $W$ by defining a new function $f^*$, equal to $f$ on $W$ and zero outside $W$, and then setting

$$\iiint_W f(x, y, z)\,dx\,dy\,dz = \iiint_B f^*(x, y, z)\,dx\,dy\,dz,$$

where $B$ is any box containing the region $W$.

## Elementary Regions

As before, we restrict our attention to particularly simple regions. An ***elementary region*** in three-dimensional space is one defined by restricting one of the variables to be

between two functions of the remaining variables, the domains of these functions being an elementary (i.e., an $x$-simple or a $y$-simple) region in the plane. For example, if $D$ is an elementary region in the $xy$ plane and if $\gamma_1(x, y)$ and $\gamma_2(x, y)$ are two functions with $\gamma_2(x, y) \geq \gamma_1(x, y)$, an elementary region consists of all $(x, y, z)$ such that $(x, y)$ lies in $D$ and $\gamma_1(x, y) \leq z \leq \gamma_2(x, y)$. Figure 5.5.2 shows two elementary regions.

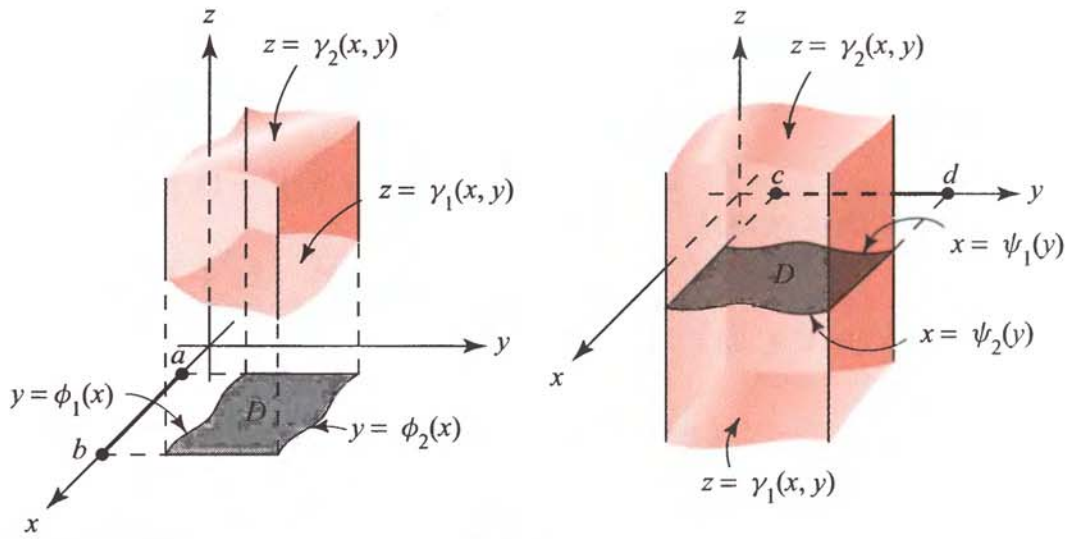



**Figure 5.5.2** Two elementary regions in space. The domain $D$ in the figure on the left is $y$-simple, while on the right it is $x$-simple.

**EXAMPLE 3** Describe the unit ball $x^2 + y^2 + z^2 \leq 1$ as an elementary region.

SOLUTION This can be done in several ways. One, in which $D$ is $y$-simple, is:

$$-1 \leq x \leq 1,$$
$$-\sqrt{1 - x^2} \leq y \leq \sqrt{1 - x^2},$$
$$-\sqrt{1 - x^2 - y^2} \leq z \leq \sqrt{1 - x^2 - y^2}.$$

In doing this, we first write the top and bottom hemispheres as $z = \sqrt{1 - x^2 - y^2}$ and $z = -\sqrt{1 - x^2 - y^2}$, respectively, where $x$ and $y$ vary over the unit disk (that is, $-\sqrt{1 - x^2} \leq y \leq \sqrt{1 - x^2}$ and $x$ varies between $-1$ and 1). (See Figure 5.5.3.) We can describe the region in other ways by interchanging the roles of $x$, $y$, and $z$ in the defining inequalities. ▲

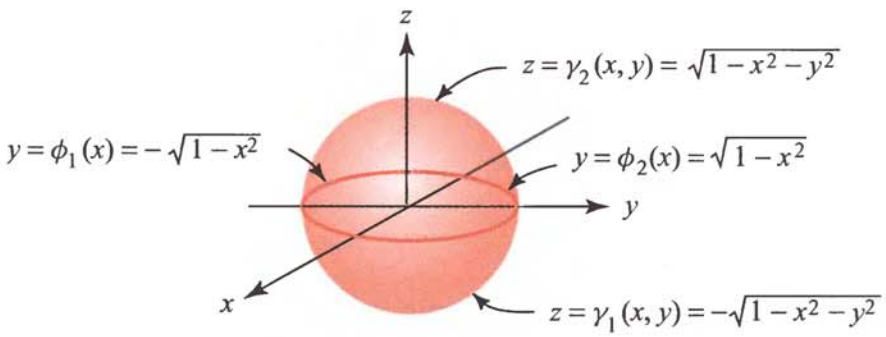



**Figure 5.5.3** The unit ball as an elementary region in space.

## Integrals over Elementary Regions

As with integrals in the plane, any function of three variables that is continuous over an elementary region is integrable on that region. An argument like that for double integrals shows that a triple integral over an elementary region can be rewritten as an iterated integral in which the limits of integration are functions. The formulas for such iterated integrals are given in the following box.

**Triple Integrals by Iterated Integration** Suppose that $W$ is an elementary region described by bounding $z$ between two functions of $x$ and $y$. Then either

$$\iiint_W f(x, y, z)\,dx\,dy\,dz = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \int_{\gamma_1(x,y)}^{\gamma_2(x,y)} f(x, y, z)\,dz\,dy\,dx$$

[see Figure 5.5.2 (left)] or

$$\iiint_W f(x, y, z)\,dx\,dy\,dz = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} \int_{\gamma_1(x,y)}^{\gamma_2(x,y)} f(x, y, z)\,dz\,dx\,dy$$

[see Figure 5.5.2 (right)].

If $f = 1$, we get the integral $\iiint_W dx\,dy\,dz$, which is the ***volume*** of the region $W$.

**EXAMPLE 4** Verify the volume formula for the ball of radius 1:

$$\iiint_W dx\,dy\,dz = \frac{4}{3}\pi,$$

where $W$ is the set of $(x, y, z)$ with $x^2 + y^2 + z^2 \le 1$.

SOLUTION We use the description of the unit ball from Example 3. From the first formula in the preceding box, the integral is

$$\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} dz\,dy\,dx.$$

Holding $y$ and $x$ fixed and integrating with respect to $z$ yields

$$\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \left[ z \Big|_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} \right] dy\,dx = 2 \int_{-1}^{1} \left[ \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (1 - x^2 - y^2)^{1/2}\,dy \right] dx.$$

Because $x$ is fixed in the $y$-integral, it can be expressed as $\int_{-a}^{a} (a^2 - y^2)^{1/2}\,dy$, where $a = (1 - x^2)^{1/2}$. This integral is the area of a semicircular region of radius

$a$, so that

$$\int_{-a}^{a} (a^2 - y^2)^{1/2}\, dy = \frac{a^2}{2}\pi.$$

(We could also have used a trigonometric substitution or a table of integrals.) Thus,

$$\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (1 - x^2 - y^2)^{1/2}\, dy = \frac{1 - x^2}{2}\pi,$$

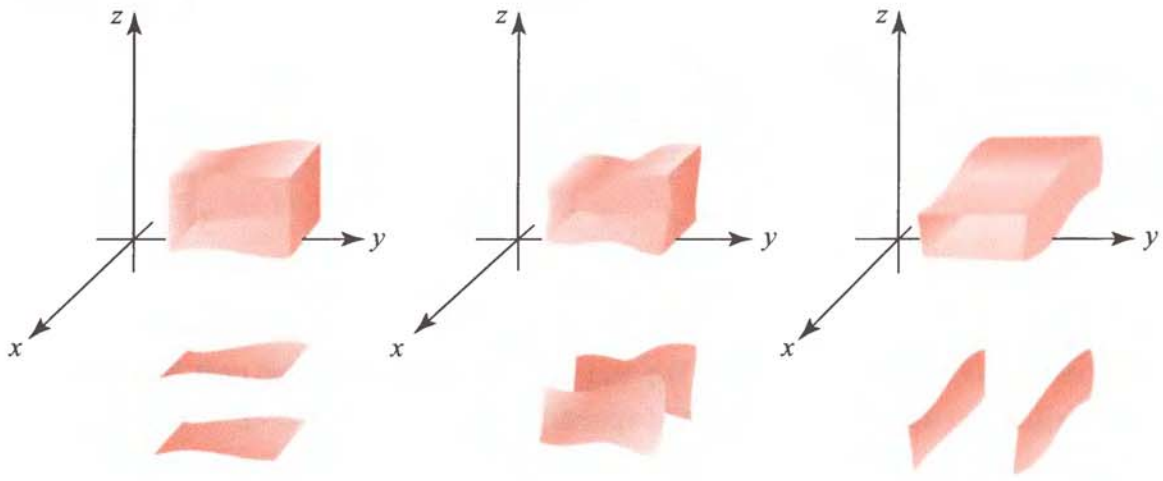
and so

$$2\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (1 - x^2 - y^2)^{1/2}\, dy\, dx = 2\int_{-1}^{1} \pi \frac{1 - x^2}{2}\, dx$$

$$= \pi \int_{-1}^{1} (1 - x^2)\, dx = \pi\left(x - \frac{x^3}{3}\right)\Big|_{x=-1}^{1} = \frac{4}{3}\pi. \quad \blacktriangle$$

Other types of elementary regions are shown in Figure 5.5.4. For instance, in the second region, $(y, z)$ lies in an elementary region in the $yz$ plane and $x$ lies between two graphs:

$$\rho_1(y, z) \le x \le \rho_2(y, z).$$




Top and bottom are surfaces $z = \gamma(x, y)$

Front and rear are surfaces $x = \rho(y, z)$

Left and right are surfaces $y = \delta(x, z)$

**Figure 5.5.4** Types of elementary regions in space.

As shown in Figure 5.5.5, some elementary regions can be simultaneously described in all three ways. We shall call these regions ***symmetric elementary regions***.

Corresponding to each description of a region as an elementary region is an integration formula. For instance, if $W$ is expressed as the set of all $(x, y, z)$ such that

$$c \le y \le d, \qquad \psi_1(y) \le z \le \psi_2(y), \qquad \rho_1(y, z) \le x \le \rho_2(y, z),$$

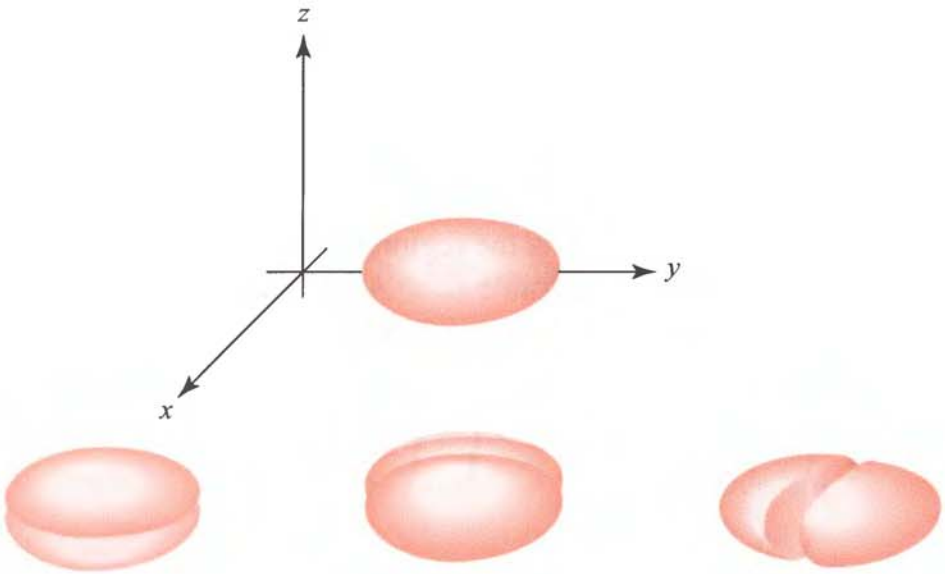



Figure 5.5.5 A symmetric elementary region can be described in three overall ways.

then

$$\iiint_W f(x,y,z)\,dx\,dy\,dz = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} \int_{\rho_1(y,z)}^{\rho_2(y,z)} f(x,y,z)\,dx\,dz\,dy.$$

**EXAMPLE 5** Let $W$ be the region bounded by the planes $x = 0$, $y = 0$, and $z = 2$, and the surface $z = x^2 + y^2$ and lying in the quadrant $x \geq 0$, $y \geq 0$. Compute $\iiint_W x\,dx\,dy\,dz$ and sketch the region.

SOLUTION *Method 1.* The region $W$ is sketched in Figure 5.5.6. As indicated in the figure, we may describe this region by the inequalities

$$0 \leq x \leq \sqrt{2}, \qquad 0 \leq y \leq \sqrt{2 - x^2}, \qquad x^2 + y^2 \leq z \leq 2.$$

Therefore,

$$\begin{aligned}
\iiint_W x\,dx\,dy\,dz &= \int_0^{\sqrt{2}} \left[ \int_0^{\sqrt{2-x^2}} \left( \int_{x^2+y^2}^{2} x\,dz \right) dy \right] dx \\
&= \int_0^{\sqrt{2}} \int_0^{\sqrt{2-x^2}} x(2 - x^2 - y^2)\,dy\,dx \\
&= \int_0^{\sqrt{2}} x\left[ (2-x^2)^{3/2} - \frac{(2-x^2)^{3/2}}{3} \right] dx \\
&= \int_0^{\sqrt{2}} \frac{2x}{3}(2-x^2)^{3/2}\,dx = \frac{-2(2-x^2)^{5/2}}{15}\Bigg|_0^{\sqrt{2}} \\
&= 2 \cdot \frac{2^{5/2}}{15} = \frac{8\sqrt{2}}{15}.
\end{aligned}$$

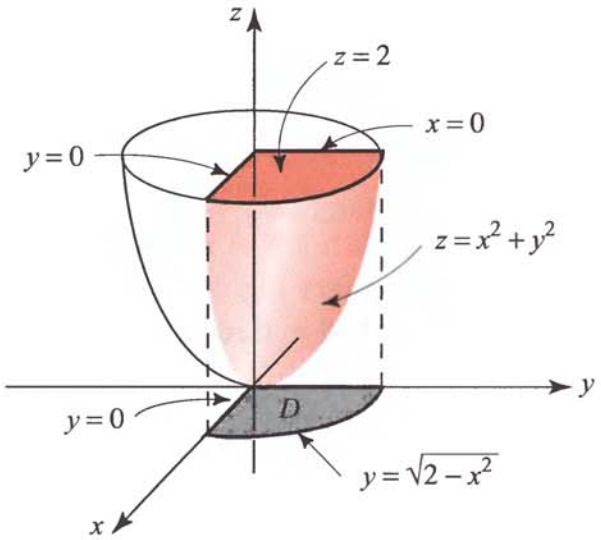



**Figure 5.5.6** $W$ is the region below the plane $z = 2$, above the paraboloid $z = x^2 + y^2$, and on the positive sides of the planes $x = 0$, $y = 0$.

*Method 2.* We can also place limits on $x$ first and describe $W$ by $0 \le x \le (z - y^2)^{1/2}$ and $(y, z)$ in $D$, where $D$ is the subset of the $yz$ plane with $0 \le z \le 2$ and $0 \le y \le z^{1/2}$ (see Figure 5.5.7).

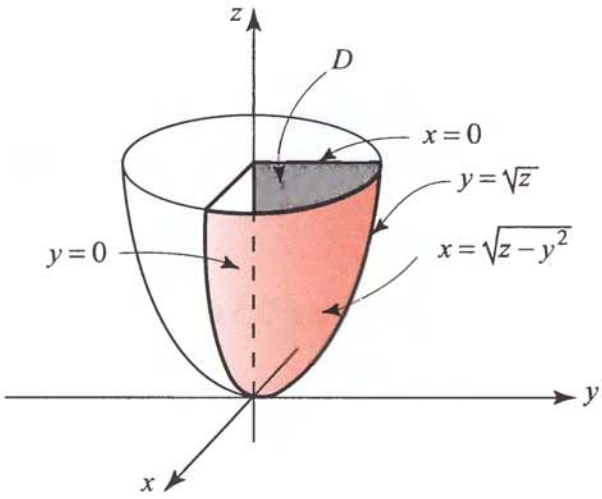



**Figure 5.5.7** A different description of the region in Example 5.

Therefore,

$$\begin{aligned}
\iiint_W x\,dx\,dy\,dz &= \iint_D \left( \int_0^{(z-y^2)^{1/2}} x\,dx \right) dy\,dz \\
&= \int_0^2 \left[ \int_0^{z^{1/2}} \left( \int_0^{(z-y^2)^{1/2}} x\,dx \right) dy \right] dz \\
&= \int_0^2 \int_0^{z^{1/2}} \left( \frac{z - y^2}{2} \right) dy\,dz \\
&= \frac{1}{2} \int_0^2 \left( z^{3/2} - \frac{z^{3/2}}{3} \right) dz = \frac{1}{2} \int_0^2 \frac{2}{3} z^{3/2}\,dz \\
&= \left[ \frac{2}{15} z^{5/2} \right]_0^2 = \frac{2}{15} 2^{5/2} = \frac{8\sqrt{2}}{15},
\end{aligned}$$

which agrees with our previous answer. ▲

**EXAMPLE 6** Evaluate

$$\int_0^1 \int_0^x \int_{x^2+y^2}^2 dz\, dy\, dx.$$

Sketch the region $W$ of integration and interpret.

SOLUTION

$$\int_0^1 \int_0^x \int_{x^2+y^2}^2 dz\, dy\, dx = \int_0^1 \int_0^x (2 - x^2 - y^2)\, dy\, dx$$

$$= \int_0^1 \left(2x - x^3 - \frac{x^3}{3}\right) dx = 1 - \frac{1}{4} - \frac{1}{12} = \frac{2}{3}.$$

This integral is the volume of the region sketched in Figure 5.5.8. ▲

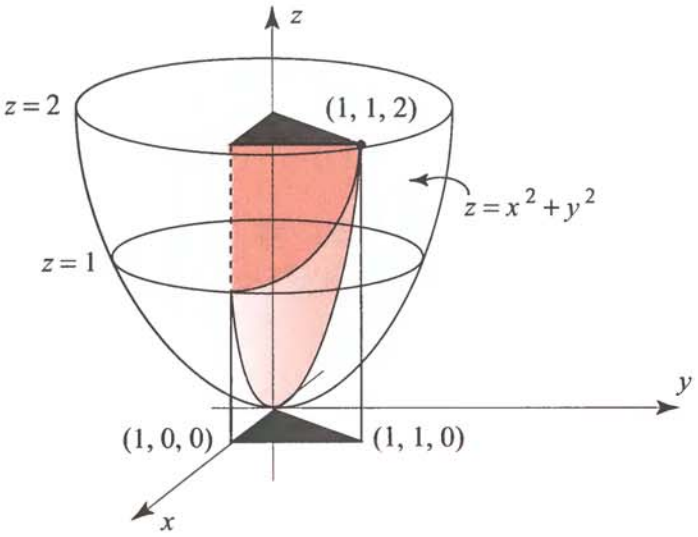



**Figure 5.5.8** The region $W$ lies between the paraboloid $z = x^2 + y^2$ and the plane $z = 2$, and above the region $D$.

## EXERCISES

*In Exercises 1 to 4, perform the indicated integration over the given box.*

1. $\iiint_B x^2\, dx\, dy\, dz$, $B = [0, 1] \times [0, 1] \times [0, 1]$

2. $\iiint_B e^{-xy}\, y\, dx\, dy\, dz$, $B = [0, 1] \times [0, 1] \times [0, 1]$

3. $\iiint_B (2x + 3y + z)\, dx\, dy\, dz$, $B = [0, 2] \times [-1, 1] \times [0, 1]$

4. $\iiint_B ze^{x+y}\, dx\, dy\, dz$, $B = [0, 1] \times [0, 1] \times [0, 1]$

*In Exercises 5 to 8, describe the given region as an elementary region.*

5. The region between the cone $z = \sqrt{x^2 + y^2}$ and the paraboloid $z = x^2 + y^2$.

**6.** The region cut out of the ball $x^2 + y^2 + z^2 \leq 4$ by the elliptic cylinder $2x^2 + z^2 = 1$, that is, the region inside the cylinder and the ball.

**7.** The region inside the sphere $x^2 + y^2 + z^2 = 1$ and above the plane $z = 0$.

**8.** The region bounded by the planes $x = 0$, $y = 0$, $z = 0$, $x + y = 4$, and $x = z - y - 1$.

*Find the volume of the region in Exercises 9 to 12.*

**9.** The region bounded by $z = x^2 + y^2$ and $z = 10 - x^2 - 2y^2$.

**10.** The solid bounded by $x^2 + 2y^2 = 2$, $z = 0$, and $x + y + 2z = 2$.

**11.** The solid bounded by $x = y$, $z = 0$, $y = 0$, $x = 1$, and $x + y + z = 0$.

**12.** The region common to the intersecting cylinders $x^2 + y^2 \leq a^2$ and $x^2 + z^2 \leq a^2$.

*Evaluate the integrals in Exercises 13 to 21.*

**13.** $\displaystyle\int_0^1 \int_1^2 \int_2^3 \cos[\pi(x + y + z)]\, dx\, dy\, dz$

**14.** $\displaystyle\int_0^1 \int_0^x \int_0^y (y + xz)\, dz\, dy\, dx$

**15.** $\displaystyle\iiint_W (x^2 + y^2 + z^2)\, dx\, dy\, dz$; $W$ is the region bounded by $x + y + z = a$ (where $a > 0$), $x = 0$, $y = 0$, and $z = 0$.

**16.** $\displaystyle\iiint_W z\, dx\, dy\, dz$; $W$ is the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, $z = 1$, and the cylinder $x^2 + y^2 = 1$, with $x \geq 0$, $y \geq 0$.

**17.** $\displaystyle\iiint_W x^2 \cos z\, dx\, dy\, dz$; $W$ is the region bounded by $z = 0$, $z = \pi$, $y = 0$, $y = 1$, $x = 0$, and $x + y = 1$.

**18.** $\displaystyle\int_0^2 \int_0^x \int_0^{x+y} dz\, dy\, dx$

**19.** $\displaystyle\iiint_W (1 - z^2)\, dx\, dy\, dz$; $W$ is the pyramid with top vertex at (0, 0, 1) and base vertices at (0, 0, 0), (1, 0, 0), (0, 1, 0), and (1, 1, 0).

**20.** $\displaystyle\iiint_W (x^2 + y^2)\, dx\, dy\, dz$; $W$ is the same pyramid as in Exercise 19.

**21.** $\displaystyle\int_0^1 \int_0^{2x} \int_{x^2+y^2}^{x+y} dz\, dy\, dx.$

**22.** (a) Sketch the region for the integral $\displaystyle\int_0^1 \int_0^x \int_0^y f(x, y, z)\, dz\, dy\, dx.$

(b) Write the integral with the integration order $dx\, dy\, dz$.

*For the regions in Exercises 23 to 26, find the appropriate limits $\phi_1(x)$, $\phi_2(x)$, $\gamma_1(x, y)$, and $\gamma_2(x, y)$, and write the triple integral over the region $W$ as an iterated integral in the form*

$$\iiint_W f\,dV = \int_a^b \left\{ \int_{\phi_1(x)}^{\phi_2(x)} \left[ \int_{\gamma_1(x,y)}^{\gamma_2(x,y)} f(x, y, z)\,dz \right] dy \right\} dx.$$

**23.** $W = \{(x, y, z) \mid \sqrt{x^2 + y^2} \leq z \leq 1\}$

**24.** $W = \{(x, y, z) \mid \frac{1}{2} \leq z \leq 1 \text{ and } x^2 + y^2 + z^2 \leq 1\}$

**25.** $W = \{(x, y, z) \mid x^2 + y^2 \leq 1, z \geq 0 \text{ and } x^2 + y^2 + z^2 \leq 4\}$

**26.** $W = \{(x, y, z) \mid |x| \leq 1, |y| \leq 1, z \geq 0 \text{ and } x^2 + y^2 + z^2 \leq 1\}$

**27.** Show that the formula using triple integrals for the volume under the graph of a positive function $f(x, y)$, on an elementary region $D$ in the plane, reduces to the double integral of $f$ over $D$.

**28.** Let $W$ be the region bounded by the planes $x = 0, y = 0, z = 0, x + y = 1$, and $z = x + y$.

(a) Find the volume of $W$.
(b) Evalute $\iiint_W x\,dx\,dy\,dz$.
(c) Evalute $\iiint_W y\,dx\,dy\,dz$.

**29.** Let $f$ be continuous and let $B_\varepsilon$ be the ball of radius $\varepsilon$ centered at the point $(x_0, y_0, z_0)$. Let vol $(B_\varepsilon)$ be the volume of $B_\varepsilon$. Prove that

$$\lim_{\varepsilon \to 0} \frac{1}{\text{vol}\,(B_\varepsilon)} \iiint_{B_\varepsilon} f(x, y, z)\,dV = f(x_0, y_0, z_0).$$

## REVIEW EXERCISES FOR CHAPTER 5

*Evaluate the integrals in Exercises 1 to 4.*

**1.** $\displaystyle\int_0^3 \int_{-x^2+1}^{x^2+1} xy\,dy\,dx$

**2.** $\displaystyle\int_0^1 \int_{\sqrt{x}}^1 (x + y)^2\,dy\,dx$

**3.** $\displaystyle\int_0^1 \int_{e^x}^{e^{2x}} x \ln y\,dy\,dx$

**4.** $\displaystyle\int_0^1 \int_1^2 \int_2^3 \cos[\pi(x + y + z)]\,dx\,dy\,dz.$

*Reverse the order of integration of the integrals in Exercises 5 to 8 and evaluate.*

**5.** The integral in Exercise 1.

**6.** The integral in Exercise 2.

**7.** The integral in Exercise 3.

**8.** The integral in Exercise 4.

9. Evaluate the integral $\int_0^1 \int_0^x \int_0^y (y + xz)\, dz\, dy\, dx$.

10. Evaluate $\int_0^1 \int_y^{y^2} e^{x/y}\, dx\, dy$.

11. Evaluate $\int_0^1 \int_0^{(\arcsin y)/y} y \cos xy\, dx\, dy$.

12. Change the order of integration and evaluate

$$\int_0^2 \int_{y/2}^1 (x + y)^2\, dx\, dy.$$

13. Show that evaluating $\iint_D dx\, dy$, where $D$ is a $y$-simple region, reproduces the formula from one-variable calculus for the area between two curves.

14. Change the order of integration and evaluate

$$\int_0^1 \int_{y^{1/2}}^1 (x^2 + y^3 x)\, dx\, dy.$$

15. Let $D$ be the region in the $xy$ plane inside the unit circle $x^2 + y^2 = 1$. Evaluate $\iint_D f(x, y)\, dx\, dy$ in each of the following cases:

(a) $f(x, y) = xy$ (b) $f(x, y) = x^2 y^2$ (c) $f(x, y) = x^3 y^3$

16. Find $\iint_D y[1 - \cos(\pi x/4)]\, dx\, dy$, where $D$ is the region in Figure 5.R.1.

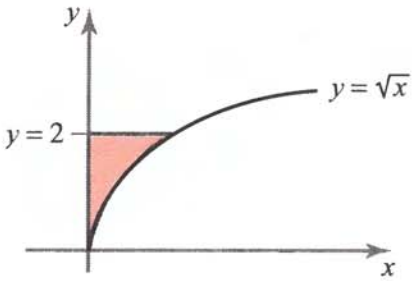



**Figure 5.R.1** The region of integration for Exercise 16.

*Evaluate the integrals in Exercises 17 to 24. Sketch and identify the type of the region (corresponding to the way the integral is written).*

17. $\int_0^{\pi} \int_{\sin x}^{3 \sin x} x(1 + y)\, dy\, dx$

18. $\int_0^1 \int_{x-1}^{x \cos(\pi x/2)} (x^2 + xy + 1)\, dy\, dx$

19. $\int_{-1}^1 \int_{y^{2/3}}^{(2-y)^2} \left(\frac{3}{2}\sqrt{x} - 2y\right) dx\, dy$

20. $\int_0^2 \int_{-3(\sqrt{4-x^2})/2}^{3(\sqrt{4-x^2})/2} \left(\frac{5}{\sqrt{2+x}} + y^3\right) dy\, dx$

21. $\int_0^1 \int_0^{x^2} (x^2 + xy - y^2)\, dy\, dx$

22. $\int_2^4 \int_{y^2-1}^{y^3} 3\, dx\, dy$

**23.** $\displaystyle\int_0^1 \int_{x^2}^{x} (x+y)^2\, dy\, dx$

**24.** $\displaystyle\int_0^1 \int_0^{3y} e^{x+y}\, dx\, dy$

*In Exercises 25 to 27, integrate the given function $f$ over the given region $D$.*

**25.** $f(x, y) = x - y$; $D$ is the triangle with vertices (0, 0), (1, 0), and (2, 1).

**26.** $f(x, y) = x^3 y + \cos x$; $D$ is the triangle defined by $0 \le x \le \pi/2$, $0 \le y \le x$.

**27.** $f(x, y) = x^2 + 2xy^2 + 2$; $D$ is the region bounded by the graph of $y = -x^2 + x$, the $x$ axis, and the lines $x = 0$ and $x = 2$.

*In Exercises 28 and 29, sketch the region of integration, interchange the order, and evaluate.*

**28.** $\displaystyle\int_1^4 \int_1^{\sqrt{x}} (x^2 + y^2)\, dy\, dx$

**29.** $\displaystyle\int_0^1 \int_{1-y}^{1} (x + y^2)\, dx\, dy$

**30.** Show that

$$4e^5 \le \iint_{[1,3]\times[2,4]} e^{x^2+y^2}\, dA \le 4e^{25}.$$

**31.** Show that

$$4\pi \le \iint_D (x^2 + y^2 + 1)\, dx\, dy \le 20\pi,$$

where $D$ is the disk of radius 2 centered at the origin.

**32.** Suppose $W$ is a ***path-connected region***, that is, given any two points of $W$ there is a continuous path joining them. If $f$ is a continuous function on $W$, use the intermediate-value theorem to show that there is at least one point in $W$ at which the value of $f$ is equal to the average of $f$ over $W$, that is, the integral of $f$ over $W$ divided by the volume of $W$. (Compare this with the mean-value theorem for double integrals.) What happens if $W$ is not connected?

**33.** Prove: $\int_0^x [\int_0^t F(u)\, du]\, dt = \int_0^x (x - u)F(u)\, du$.

*Evaluate the integrals in Exercises 34 to 36.*

**34.** $\displaystyle\int_0^1 \int_0^z \int_0^y xy^2 z^3\, dx\, dy\, dz$

**35.** $\displaystyle\int_0^1 \int_0^y \int_0^{x/\sqrt{3}} \frac{x}{x^2 + z^2}\, dz\, dx\, dy$

**36.** $\displaystyle\int_1^2 \int_1^z \int_{1/y}^2 yz^2\, dx\, dy\, dz$

**37.** Write the iterated integral $\int_0^1 \int_{1-x}^1 \int_x^1 f(x, y, z)\, dz\, dy\, dx$ as an integral over a region in $\mathbb{R}^3$ and then rewrite it in five other possible orders of integration.

# 6

# The Change of Variables Formula and Applications of Integration

*If you are stuck in a calculus problem and don't know what else to do, try integrating by parts or changing variables.*

*Jerry Kazdan*

*If that fails, go away, have a cup of coffee, and think!*

*Ute Müller*

The change of variables formula is one of the most powerful integration methods in single-variable calculus; it enables us to evaluate integrals such as

$$\int_0^1 x e^{x^2} dx$$

by using the substitution, or *change of variables* $u = x^2$, which reduces the problem to the easy task of integrating $e^u$ with respect to $u$. In this chapter, we develop the *multidimensional change of variables formula*, which is especially important and useful in evaluating multiple integrals in polar, cylindrical, and spherical coordinates.

One of the key ingredients in the change of variables formula is how to change variables in multidimensions. This involves the notion of mapping, which occurs in various interesting situations. For example, consider a deforming object, such as a swimming fish. As it changes its shape, one can imagine the instantaneous correspondence between points on the fish in its rest state and in its current shape. This type of correspondence is, in fact, the main idea behind a change of variables, in this case, of one three-dimensional region (the fish in its rest state) to another (the fish in its current shape).

The first section in this chapter describes the key concepts for mappings between regions of the plane. It goes on to develop the change of variables technique for double and then triple integrals. The chapter also includes some of the important physical applications of the integral.

# 6.1 The Geometry of Maps from $\mathbb{R}^2$ to $\mathbb{R}^2$

In this section, we shall be interested in maps from subsets of $\mathbb{R}^2$ to $\mathbb{R}^2$. The resulting geometric understanding will be useful in the next section, when we discuss the change of variables formula for multiple integrals.

## Maps of One Region to Another

Let $D^*$ be a subset of $\mathbb{R}^2$; suppose we consider a continuously differentiable map $T: D^* \to \mathbb{R}^2$, so $T$ takes points in $D^*$ to points in $\mathbb{R}^2$. We denote the set of image points by $D$ or by $T(D^*)$; hence, $D = T(D^*)$ is the set of all points $(x, y) \in \mathbb{R}^2$ such that

$$(x, y) = T(x^*, y^*) \qquad \text{for some} \qquad (x^*, y^*) \in D^*.$$

One way to understand the geometry of a map $T$ is to see how it *deforms* or changes $D^*$. For example, Figure 6.1.1 illustrates a map $T$ that takes a slightly twisted region into a disk.

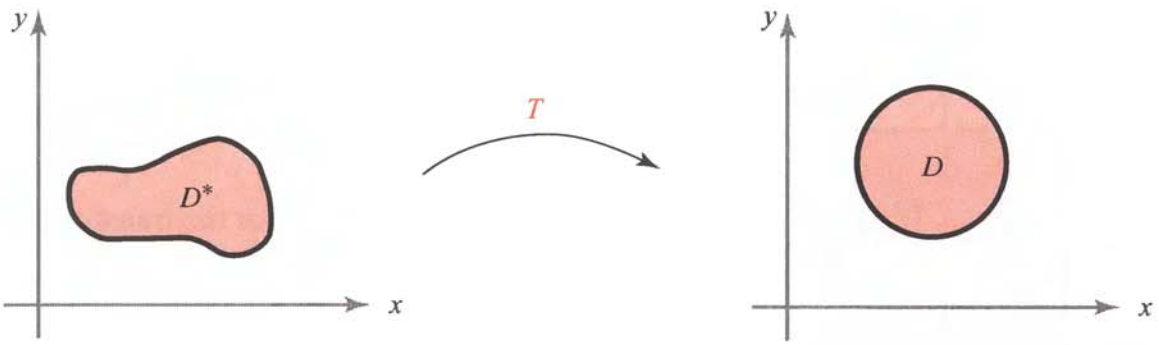



**Figure 6.1.1** A function $T$ from a region $D^*$ to a disk $D$.

**EXAMPLE 1** Let $D^* \subset \mathbb{R}^2$ be the rectangle $D^* = [0, 1] \times [0, 2\pi]$. Then all points in $D^*$ are of the form $(r, \theta)$, where $0 \le r \le 1, 0 \le \theta \le 2\pi$. Let $T$ be the polar coordinate "change of variables" defined by $T(r, \theta) = (r\cos\theta, r\sin\theta)$. Find the image set $D$.

SOLUTION Let $(x, y) = (r\cos\theta, r\sin\theta)$. Because of the identity $x^2 + y^2 = r^2\cos^2\theta + r^2\sin^2\theta = r^2 \le 1$, we see that the set of points $(x, y) \in \mathbb{R}^2$ such that $(x, y) \in D$ has the property that $x^2 + y^2 \le 1$, and so $D$ is contained in the unit disk. In addition, any point $(x, y)$ in the unit disk can be written as $(r\cos\theta, r\sin\theta)$ for some $0 \le r \le 1$ and $0 \le \theta \le 2\pi$. Thus, $D$ is the unit disk (see Figure 6.1.2). ▲

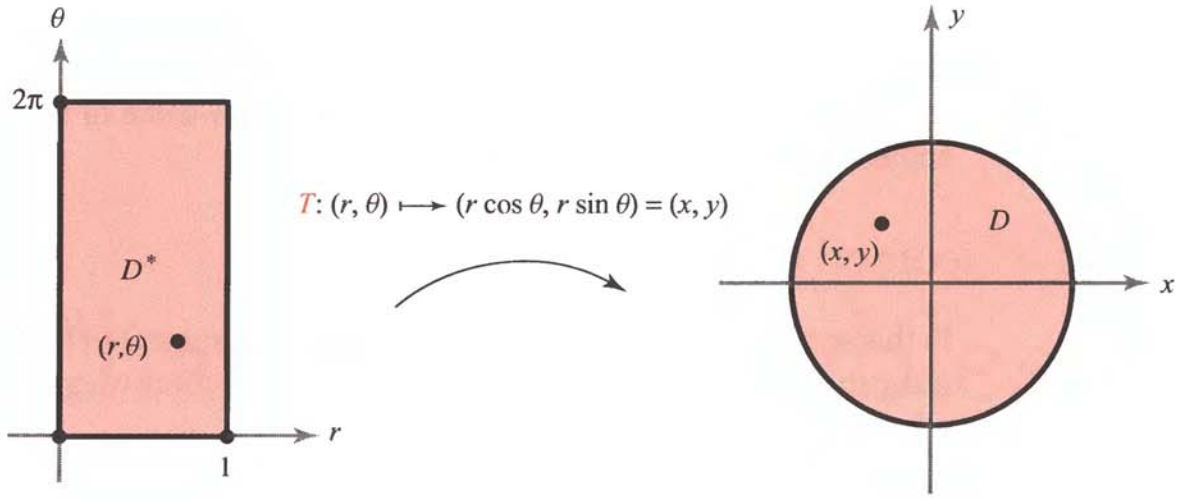



**Figure 6.1.2** $T$ gives a change of variables between Euclidean and polar coordinates. The unit circle is the image of a rectangle.

**EXAMPLE 2** Let $T$ be defined by $T(x, y) = ((x + y)/2, (x - y)/2)$ and let $D^* = [-1, 1] \times [-1, 1] \subset \mathbb{R}^2$ be a square with side of length 2 centered at the origin. Determine the image $D$ obtained by applying $T$ to $D^*$.

SOLUTION Let us first determine the effect of $T$ on the line $\mathbf{c}_1(t) = (t, 1)$, where $-1 \leq t \leq 1$ (see Figure 6.1.3). We have $T(\mathbf{c}_1(t)) = ((t + 1)/2, (t - 1)/2)$. The map $t \mapsto T(\mathbf{c}_1(t))$ is a parametrization of the line $y = x - 1$, $0 \leq x \leq 1$, because $(t - 1)/2 = (t + 1)/2 - 1$. This is the straight line segment joining $(1, 0)$ and $(0, -1)$.

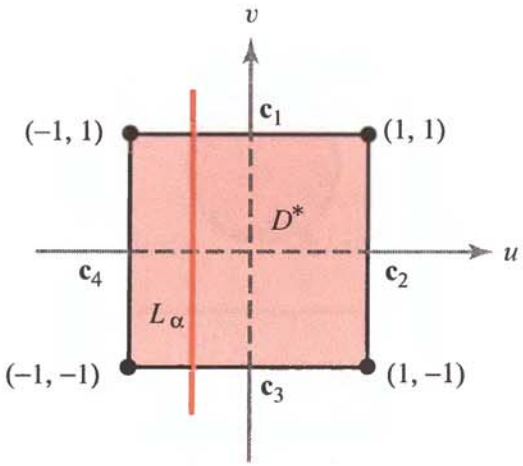



**Figure 6.1.3** Domain for the transformation $T$ of Example 2.

Let

$$\mathbf{c}_2(t) = (1, t), \qquad -1 \leq t \leq 1$$
$$\mathbf{c}_3(t) = (t, -1), \qquad -1 \leq t \leq 1$$
$$\mathbf{c}_4(t) = (-1, t), \qquad -1 \leq t \leq 1$$

be parametrizations of the other edges of the square $D^*$. Using the same argument as before, we see that $T \circ \mathbf{c}_2$ is a parametrization of the line $y = 1 - x$, $0 \leq x \leq 1$

[the straight line segment joining (0, 1) and (1, 0)]; $T \circ \mathbf{c}_3$ is the line $y = x + 1, -1 \leq x \leq 0$ joining (0, 1) and (−1, 0); and $T \circ \mathbf{c}_4$ is the line $y = -x - 1, -1 \leq x \leq 0$ joining (−1, 0) and (0, −1). By this time it seems reasonable to guess that $T$ "flips" the square $D^*$ over and takes it to the square $D$ whose vertices are (1, 0), (0, 1), (−1, 0), (0, −1) (Figure 6.1.4).

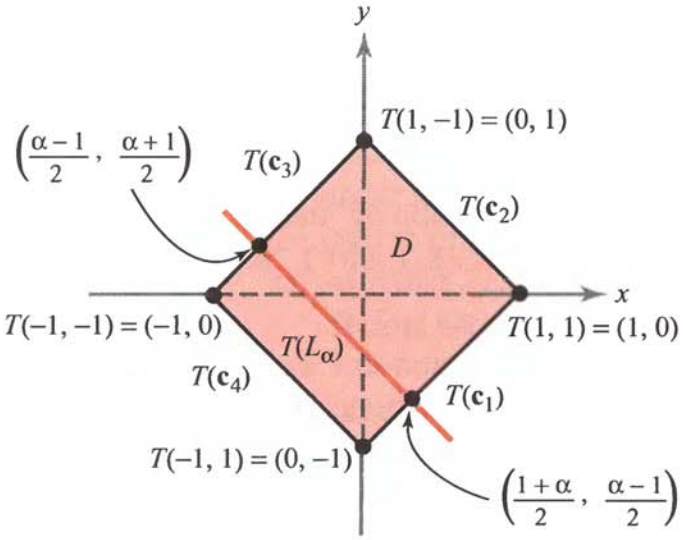



**Figure 6.1.4** The effect of $T$ on the region $D^*$.

To prove that this is indeed the case, let $-1 \leq \alpha \leq 1$ and let $L_\alpha$ (Figure 6.1.3) be a fixed line parametrized by $\mathbf{c}(t) = (\alpha, t), -1 \leq t \leq 1$; then $T(\mathbf{c}(t)) = ((\alpha + t)/2, (\alpha - t)/2)$ is a parametrization of the line $y = -x + \alpha, (\alpha - 1)/2 \leq x \leq (\alpha + 1)/2$. This line begins, for $t = -1$, at the point $((\alpha - 1)/2, (1 + \alpha)/2)$ and ends at the point $((1 + \alpha)/2, (\alpha - 1)/2)$; as is easily checked, these points lie on the lines $T \circ \mathbf{c}_3$ and $T \circ \mathbf{c}_1$, respectively. Thus, as $\alpha$ varies between −1 and 1, $L_\alpha$ sweeps out the square $D^*$ while $T(L_\alpha)$ sweeps out the square $D$ determined by the vertices (−1, 0), (0, 1), (1, 0), and (0, −1). ▲

## Images of Maps

The following theorem is a useful way to describe the image $T(D^*)$.

**THEOREM 1** Let $A$ be a $2 \times 2$ matrix with $\det A \neq 0$ and let $T$ be the linear mapping of $\mathbb{R}^2$ to $\mathbb{R}^2$ given by $T(\mathbf{x}) = A\mathbf{x}$ (matrix multiplication). Then $T$ transforms parallelograms into parallelograms and vertices into vertices. Moreover, if $T(D^*)$ is a parallelogram, $D^*$ must be a parallelogram.

The proof of Theorem 1 is left as Exercise 10 at the end of this section. This theorem simplifies the result of Example 2, because we need only find the vertices of $T(D^*)$ and then connect them by straight lines.

## One-to-One Maps

Although we cannot visualize the graph of a function $T: \mathbb{R}^2 \to \mathbb{R}^2$, it does help to consider how the function deforms subsets. However, simply looking at these

deformations does not give us a complete picture of the behavior of $T$. We may characterize $T$ further by using the notion of a one-to-one correspondence.

**DEFINITION** A mapping $T$ is ***one-to-one*** on $D^*$ if for $(u, v)$ and $(u', v') \in D^*$, $T(u, v) = T(u', v')$ implies that $u = u'$ and $v = v'$.

This statement means that *two different points of $D^*$ are not sent into the same point of $D$ by $T$*. For example, the function $T(x, y) = (x^2 + y^2, y^4)$ is not one-to-one, because $T(1, -1) = (2, 1) = T(1, 1)$ and yet $(1, -1) \neq (1, 1)$.

**EXAMPLE 3** Consider the polar coordinate mapping function $T: \mathbb{R}^2 \to \mathbb{R}^2$ described in Example 1, defined by $T(r, \theta) = (r\cos\theta, r\sin\theta)$. Show that $T$ is not one-to-one if its domain is all of $\mathbb{R}^2$.

SOLUTION If $\theta_1 \neq \theta_2$, then $T(0, \theta_1) = T(0, \theta_2)$, and so $T$ cannot be one-to-one. This observation implies that if $L$ is the side of the rectangle $D^* = [0, 1] \times [0, 2\pi]$ where $0 \leq \theta \leq 2\pi$ and $r = 0$ (Figure 6.1.5), then $T$ maps all of $L$ into a single point, the center of the unit disk $D$. However, if we consider the set $S^* = (0, 1] \times [0, 2\pi)$, then $T: S^* \to S$ is one-to-one (see Exercise 1). Evidently, in determining whether a function is one-to-one, the domain chosen must be carefully considered. ▲

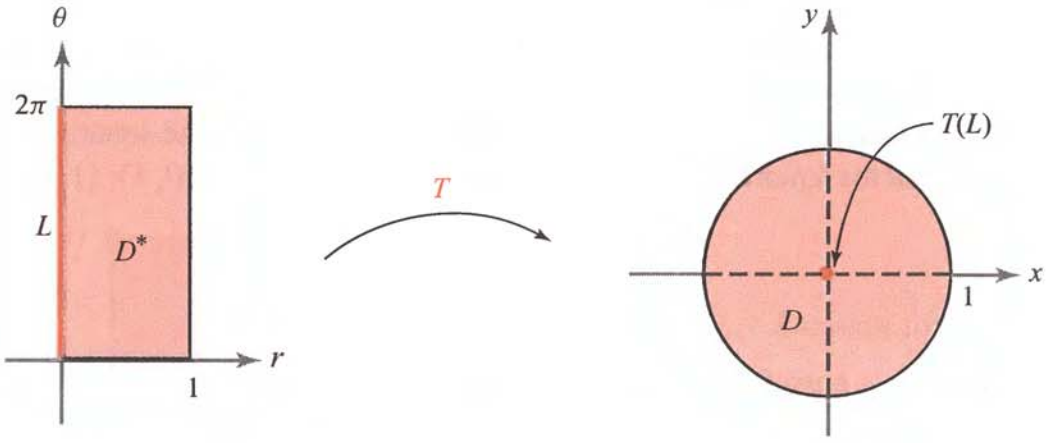



**Figure 6.1.5** The polar-coordinate transformation $T$ takes the line $L$ to the point $(0, 0)$.

**EXAMPLE 4** Show that the function $T: \mathbb{R}^2 \to \mathbb{R}^2$ of Example 2 is one-to-one.

SOLUTION Suppose $T(x, y) = T(x', y')$; then

$$\left(\frac{x+y}{2}, \frac{x-y}{2}\right) = \left(\frac{x'+y'}{2}, \frac{x'-y'}{2}\right)$$

and we have

$$x + y = x' + y',$$
$$x - y = x' - y'.$$

Adding, we have

$$2x = 2x'.$$

Thus, $x = x'$ and, similarly, subtracting gives $y = y'$, which shows that $T$ is one-to-one (with domain all of $\mathbb{R}^2$). Actually, because $T$ is linear and $T(\mathbf{x}) = A\mathbf{x}$, where $A$ is a $2 \times 2$ matrix, it would also suffice to show that $\det A \neq 0$ (see Exercise 8). ▲

## Onto Maps

In Examples 1 and 2, we have been determining the image $D = T(D^*)$ of a region $D^*$ under a mapping $T$. What will be of interest to us in the next section is, in part, the inverse problem: Namely, given $D$ and a one-to-one mapping $T$ of $\mathbb{R}^2$ to $\mathbb{R}^2$, find $D^*$ such that $T(D^*) = D$.

Before we examine this question in more detail, we introduce the notion of "onto."

DEFINITION The mapping $T$ is ***onto*** $D$ if for every point $(x, y) \in D$ there exists at least one point $(u, v)$ in the domain of $T$ such that $T(u, v) = (x, y)$.

Thus, if $T$ is onto, we *can solve* the equation $T(u, v) = (x, y)$ for $(u, v)$, given $(x, y) \in D$. If $T$ is, in addition, one-to-one, this *solution is unique*.

For *linear* mappings $T$ of $\mathbb{R}^2$ to $\mathbb{R}^2$ (or $\mathbb{R}^n$ to $\mathbb{R}^n$) it turns out that one-to-one and onto are equivalent notions (see Exercises 8 and 9).

If we are given a region $D$ and a mapping $T$, the determination of a region $D^*$ such that $T(D^*) = D$ will be possible only when for every $(x, y) \in D$ there is a $(u, v)$ in the domain of $T$ such that $T(u, v) = (x, y)$ (that is, $T$ must be onto $D$). The next example shows that this cannot always be done.

EXAMPLE 5 Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be given by $T(u, v) = (u, 0)$. Let $D$ be the square, $D = [0, 1] \times [0, 1]$. Because $T$ takes all of $\mathbb{R}^2$ to one axis, it is impossible to find a $D^*$ such that $T(D^*) = D$. ▲

Let us revisit Example 2 using these methods.

EXAMPLE 6 Let $T$ be defined as in Example 2 and let $D$ be the square whose vertices are $(1, 0)$, $(0, 1)$, $(-1, 0)$, $(0, -1)$. Find a $D^*$ with $T(D^*) = D$.

SOLUTION Because $T$ is linear and $T(\mathbf{x}) = A\mathbf{x}$, where $A$ is a $2 \times 2$ matrix satisfying $\det A \neq 0$, we know that $T: \mathbb{R}^2 \to \mathbb{R}^2$ is onto (see Exercises 8 and 9), and thus $D^*$ can be found. By Theorem 1, $D^*$ must be a parallelogram. In order to find $D^*$, it suffices to find the four points that are mapped onto vertices of $D$; then, by connecting these points, we will have found $D^*$. For the vertex $(1, 0)$ of $D$, we must solve $T(x, y) = (1, 0) = ((x + y)/2, (x - y)/2)$, so that $(x + y)/2 = 1$, $(x - y)/2 = 0$. Thus, $(x, y) = (1, 1)$ is a vertex of $D^*$. Solving for the other vertices, we find that $D^* = [-1, 1] \times [-1, 1]$. This is in agreement with what we found more laboriously in Example 2. ▲

**EXAMPLE 7** Let $D$ be the region in the first quadrant lying between the arcs of the circles $x^2 + y^2 = a^2$, $x^2 + y^2 = b^2$, $0 < a < b$ (see Figure 6.1.6). These circles have equations $r = a$ and $r = b$ in polar coordinates. Let $T$ be the polar-coordinate transformation given by $T(r, \theta) = (r\cos\theta, r\sin\theta) = (x, y)$. Find $D^*$ such that $T(D^*) = D$.

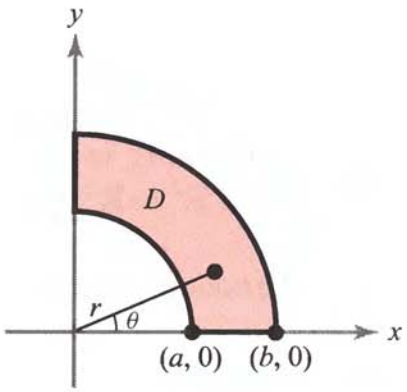



**Figure 6.1.6** We seek a region $D^*$ in the $\theta r$ plane whose image under the polar-coordinate mapping is $D$.

SOLUTION In the region $D$, $a^2 \le x^2 + y^2 \le b^2$; and because $r^2 = x^2 + y^2$, we see that $a \le r \le b$. Clearly, for this region $\theta$ varies between $0 \le \theta \le \pi/2$. Thus, if $D^* = [a, b] \times [0, \pi/2]$, we have $T(D^*) = D$ and $T$ is one-to-one. ▲

REMARK The inverse function theorem discussed in Section 3.5 is relevant to the material here. It states that if the determinant of $\mathbf{D}T(u_0, v_0)$ [which is the matrix of partial derivatives of $T$ evaluated at $(u_0, v_0)$] is not zero, then for $(u, v)$ near $(u_0, v_0)$ and $(x, y)$ near $(x_0, y_0) = T(u_0, v_0)$, the equation $T(u, v) = (x, y)$ can be *uniquely solved* for $(u, v)$ as functions of $(x, y)$. In particular, by uniqueness, $T$ is one-to-one near $(u_0, v_0)$; also, $T$ is onto a neighborhood of $(x_0, y_0)$, because $T(u, v) = (x, y)$ is solvable for $(u, v)$ if $(x, y)$ is near $(x_0, y_0)$.

However, even if $T$ is one-to-one near every point, and also onto, $T$ need not be *globally* one-to-one. Thus, one must exercise caution (see Exercise 12).

Surprisingly, if $D^*$ and $D$ are elementary regions and $T\colon D^* \to D$ has the property that the determinant of $\mathbf{D}T(u, v)$ is not zero for any $(u, v)$ in $D^*$ and if $T$ maps the boundary of $D^*$ in a one-to-one and onto manner to the boundary of $D$, then $T$ is one-to-one and onto from $D^*$ to $D$. (This proof is beyond the scope of this text.)

In summary, we have:

**One-to-One and Onto Mappings** A mapping $T\colon D^* \to D$ is ***one-to-one*** when it maps distinct points to distinct points. It is ***onto*** when the image of $D^*$ under $T$ is all of $D$.

A *linear* transformation of $\mathbb{R}^n$ to $\mathbb{R}^n$ given by multiplication by a matrix $A$ is one-to-one and onto when and only when $\det A \neq 0$.

## EXERCISES

**1.** Let $S^* = (0, 1] \times [0, 2\pi)$ and define $T(r, \theta) = (r\cos\theta, r\sin\theta)$. Determine the image set $S$. Show that $T$ is one-to-one on $S^*$.

**2.** Define

$$T(x^*, y^*) = \left(\frac{x^* - y^*}{\sqrt{2}}, \frac{x^* + y^*}{\sqrt{2}}\right).$$

Show that $T$ rotates the unit square, $D^* = [0, 1] \times [0, 1]$.

**3.** Let $D^* = [0, 1] \times [0, 1]$ and define $T$ on $D^*$ by $T(u, v) = (-u^2 + 4u, v)$. Find the image $D$. Is $T$ one-to-one?

**4.** Let $D^*$ be the parallelogram bounded by the lines $y = 3x - 4$, $y = 3x$, $y = \frac{1}{2}x$, and $y = \frac{1}{2}(x + 4)$. Let $D = [0, 1] \times [0, 1]$. Find a $T$ such that $D$ is the image of $D^*$ under $T$.

**5.** Let $D^* = [0, 1] \times [0, 1]$ and define $T$ on $D^*$ by $T(x^*, y^*) = (x^*y^*, x^*)$. Determine the image set $D$. Is $T$ one-to-one? If not, can we eliminate some subset of $D^*$ so that on the remainder $T$ is one-to-one?

**6.** Let $D^*$ be the parallelogram with vertices at $(-1, 3)$, $(0, 0)$, $(2, -1)$, and $(1, 2)$, and $D$ be the rectangle $D = [0, 1] \times [0, 1]$. Find a $T$ such that $D$ is the image set of $D^*$ under $T$.

**7.** Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be the spherical coordinate mapping defined by $(\rho, \phi, \theta) \mapsto (x, y, z)$, where

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

Let $D^*$ be the set of points $(\rho, \phi, \theta)$ such that $\phi \in [0, \pi]$, $\theta \in [0, 2\pi]$, $\rho \in [0, 1]$. Find $D = T(D^*)$. Is $T$ one-to-one? If not, can we eliminate some subset of $D^*$ so that, on the remainder, $T$ will be one-to-one?

*In Exercises 8 and 9, let $T(\mathbf{x}) = A\mathbf{x}$, where $A$ is a $2 \times 2$ matrix.*

**8.** Show that $T$ is one-to-one if and only if the determinant of $A$ is not zero.

**9.** Show that $\det A \neq 0$ if and only if $T$ is onto.

**10.** Suppose $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ is linear and is given by $T(\mathbf{x}) = A\mathbf{x}$, where $A$ is a $2 \times 2$ matrix. Show that if $\det A \neq 0$, then $T$ takes parallelograms onto parallelograms. [HINT: The general parallelogram in $\mathbb{R}^2$ can be described by the set of points $\mathbf{q} = \mathbf{p} + \lambda\mathbf{v} + \mu\mathbf{w}$ for $\lambda, \mu \in (0, 1)$ where $\mathbf{p}$, $\mathbf{v}$, $\mathbf{w}$ are vectors in $\mathbb{R}^2$ with $\mathbf{v}$ not a scalar multiple of $\mathbf{w}$.]

**11.** Suppose $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ is as in Exercise 10 and that $T(P^*) = P$ is a parallelogram. Show that $P^*$ is a parallelogram.

**12.** Consider the map $T\colon D \to D$, where $D$ is the unit disk in the plane, given by

$$T(r\cos\theta, r\sin\theta) = (r^2\cos 2\theta, r^2\sin 2\theta).$$

Using complex notation, $z = x + iy$, the map $T$ can be written as $T(z) = z^2$. Show that the Jacobian determinant of $T$ vanishes only at the origin. Thus, away from the origin, $T$ is locally one-to-one. However, show that $T$ is not globally one-to-one on $\mathbb{R}^2$ minus the origin.

## 6.2 The Change of Variables Theorem

Given two regions $D$ and $D^*$ in $\mathbb{R}^2$, a differentiable map $T$ on $D^*$ with image $D$, that is, $T(D^*) = D$, and any real-valued integrable function $f\colon D \to \mathbb{R}$, we would like to express $\iint_D f(x, y)\, dA$ as an integral over $D^*$ of the composite function $f \circ T$. In this section we shall see how to do this.

Assume that $D^*$ is a region in the $uv$ plane and that $D$ is a region in the $xy$ plane. The map $T$ is given by two coordinate functions:

$$T(u, v) = (x(u, v), y(u, v)) \qquad \text{for} \qquad (u, v) \in D^*.$$

At first, one might conjecture that

$$\iint_D f(x, y)\, dx\, dy \stackrel{?}{=} \iint_{D^*} f(x(u, v), y(u, v))\, du\, dv, \tag{1}$$

where $f \circ T(u, v) = f(x(u, v), y(u, v))$ is the composite function defined on $D^*$. However, if we consider the function $f\colon D \to \mathbb{R}^2$ where $f(x, y) = 1$, then equation (1) would imply

$$A(D) = \iint_D dx\, dy \stackrel{?}{=} \iint_{D^*} du\, dv = A(D^*). \tag{2}$$

But equation (2) will hold for only a few special cases and not for a general map $T$. For example, define $T$ by $T(u, v) = (-u^2 + 4u, v)$. Restrict $T$ to the unit square; that is, to the region $D^* = [0, 1] \times [0, 1]$ in the $uv$ plane (see Figure 6.2.1). Then, as in Exercise 3, Section 6.1, $T$ takes $D^*$ onto $D = [0, 3] \times [0, 1]$. Clearly, $A(D) \neq A(D^*)$, and so formula (2) is *not valid.*

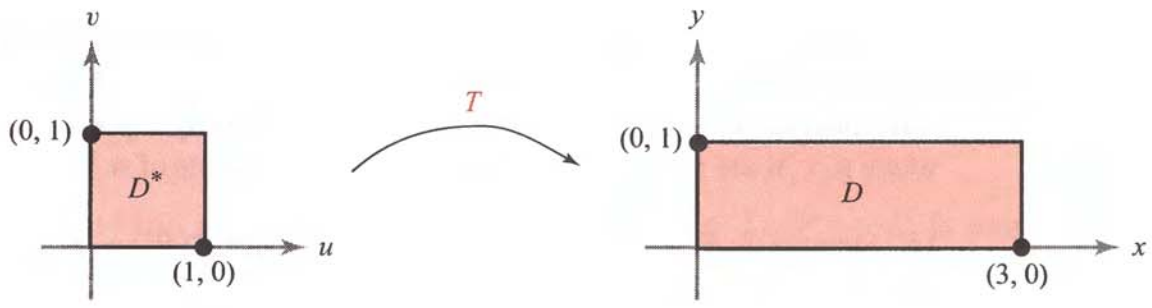



**Figure 6.2.1** The map $T\colon (u, v) \mapsto (-u^2 + 4u, v)$ takes the square $D^*$ onto the rectangle $D$.

## Jacobian Determinants

To rectify the incorrect formula (1), we need a measure of how a transformation $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ distorts the area of a region. This is given by the *Jacobian determinant*, which is defined as follows.

DEFINITION: Jacobian Determinant Let $T\colon D^* \subset \mathbb{R}^2 \to \mathbb{R}^2$ be a $C^1$ transformation given by $x = x(u, v)$ and $y = y(u, v)$. The ***Jacobian determinant*** of $T$, written $\partial(x, y)/\partial(u, v)$, is the determinant of the derivative matrix $\mathbf{D}T(u, v)$ of $T$:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix}.$$

EXAMPLE 1 The function from $\mathbb{R}^2$ to $\mathbb{R}^2$ that transforms polar coordinates into Cartesian coordinates is given by

$$x = r\cos\theta, \qquad y = r\sin\theta$$

and its Jacobian determinant is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r(\cos^2\theta + \sin^2\theta) = r. \quad \blacktriangle$$

Under suitable restrictions on the function $T$, we will argue below that the area of $D = T(D^*)$ is obtained by integrating the absolute value of the Jacobian $\partial(x, y)/\partial(u, v)$ over $D^*$; that is, we have the equation

$$A(D) = \iint_D dx\,dy = \iint_{D^*} \left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\,dv. \tag{3}$$

To illustrate: From Example 1 in Section 6.1, take $T\colon D^* \to D$, where $D = T(D^*)$ is the set of $(x, y)$ with $x^2 + y^2 \le 1$ and $D^* = [0, 1] \times [0, 2\pi]$, and $T(r, \theta) = (r\cos\theta, r\sin\theta)$. By formula (3),

$$A(D) = \iint_{D^*} \left|\frac{\partial(x, y)}{\partial(r, \theta)}\right| dr\,d\theta = \iint_{D^*} r\,dr\,d\theta \tag{4}$$

(here $r$ and $\theta$ play the role of $u$ and $v$). From the preceding computation it follows that

$$\iint_{D^*} r\,dr\,d\theta = \int_0^{2\pi}\int_0^1 r\,dr\,d\theta = \int_0^{2\pi}\left[\frac{r^2}{2}\right]_0^1 d\theta = \frac{1}{2}\int_0^{2\pi} d\theta = \pi$$

is the area of the unit disk $D$, confirming formula (3) in this case. In fact, we may recall from first-year calculus that equation (4) is the correct formula for the area of a region in polar coordinates.

It is not so easy to rigorously prove assertion (3). However, looked at in the proper way, it becomes quite plausible. Recall that $A(D) = \iint_D dx\,dy$ was obtained by dividing up $D$ into little rectangles, summing their areas, and then taking the limit of this sum as the size of the subrectangles tended to zero. The problem is that $T$ may map rectangles into regions whose area is not easy to compute. The solution is to approximate these images by simpler regions whose area we can compute. A useful tool for doing this is the derivative of $T$, which we know (from Chapter 2) gives the best linear approximation to $T$.

Consider a small rectangle $D^*$ in the $uv$ plane as shown in Figure 6.2.2. Let $T'$ denote the derivative of $T$ evaluated at $(u_0, v_0)$, so $T'$ is a $2 \times 2$ matrix. From our work in Chapter 2, we know that a good approximation to $T(u, v)$ is given by

$$T(u_0, v_0) + T'\begin{pmatrix}\Delta u\\ \Delta v\end{pmatrix},$$

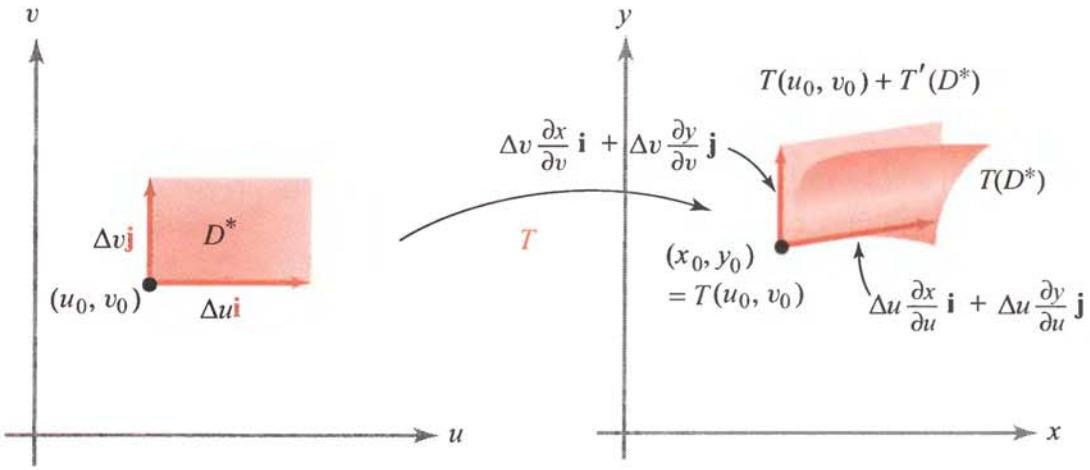



Figure 6.2.2 The effect of the transformation $T$ on a small rectangle $D^*$.

where $\Delta u = u - u_0$ and $\Delta v = v - v_0$. This mapping $T'$ takes $D^*$ into a parallelogram with vertex at $T(u_0, v_0)$ and with adjacent sides given by the vectors

$$T'(\Delta u\mathbf{i}) = \begin{bmatrix}\dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v}\\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v}\end{bmatrix}\begin{bmatrix}\Delta u\\ 0\end{bmatrix} = \Delta u\begin{bmatrix}\dfrac{\partial x}{\partial u}\\ \dfrac{\partial y}{\partial u}\end{bmatrix} = \Delta u\mathbf{T}_u$$

and

$$T'(\Delta v\mathbf{j}) = \begin{bmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{bmatrix} \begin{bmatrix} 0 \\ \Delta v \end{bmatrix} = \Delta v \begin{bmatrix} \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial v} \end{bmatrix} = \Delta v \mathbf{T}_v,$$

where

$$\mathbf{T}_u = \frac{\partial x}{\partial u}\mathbf{i} + \frac{\partial y}{\partial u}\mathbf{j} \quad \text{and} \quad \mathbf{T}_v = \frac{\partial x}{\partial v}\mathbf{i} + \frac{\partial y}{\partial v}\mathbf{j}$$

are evaluated at $(u_0, v_0)$.

Recall from Section 1.3 that the area of the parallelogram with sides equal to the vectors $a\mathbf{i} + b\mathbf{j}$ and $c\mathbf{i} + d\mathbf{j}$ is equal to the absolute value of the determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix}.$$

Thus, the area of $T(D^*)$ is approximately equal to the *absolute value* of

$$\begin{vmatrix} \dfrac{\partial x}{\partial u}\Delta u & \dfrac{\partial x}{\partial v}\Delta v \\ \dfrac{\partial y}{\partial u}\Delta u & \dfrac{\partial y}{\partial v}\Delta v \end{vmatrix} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} \Delta u\, \Delta v = \frac{\partial(x, y)}{\partial(u, v)} \Delta u\, \Delta v$$

evaluated at $(u_0, v_0)$.

This fact and a partitioning argument should make formula (3) plausible. Indeed, if we partition $D^*$ into small rectangles with sides of length $\Delta u$ and $\Delta v$, the images of these rectangles are approximated by parallelograms with sides $\mathbf{T}_u\, \Delta u$ and $\mathbf{T}_v\, \Delta v$, and hence with area $|\partial(x, y)/\partial(u, v)|\, \Delta u\, \Delta v$. Thus, the area of $D^*$ is approximately $\sum \Delta u\, \Delta v$, where the sum is taken over all the rectangles $R$ inside $D^*$ (see Figure 6.2.3). Hence, the area of $T(D^*)$ is approximately the sum

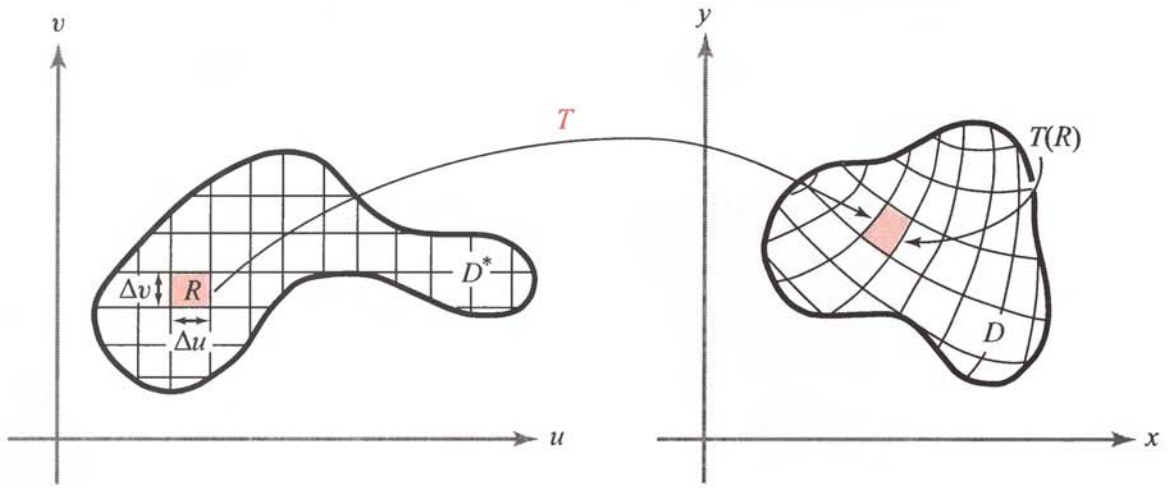



Figure 6.2.3 The area of the little rectangle $R$ is $\Delta u\, \Delta v$. The area of $T(R)$ is approximately $|\partial(x, y)/\partial(u, v)|\Delta u \Delta v$.

$\sum |\partial(x, y)/\partial(u, v)| \Delta u \, \Delta v$. In the limit, this sum becomes

$$\iint_{D^*} \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv.$$

Let us give another informal argument for the special case (4) of formula (3), that is, the case of polar coordinates. Consider a region $D$ in the $xy$ plane and a grid corresponding to a partition of the $r$ and $\theta$ variables (Figure 6.2.4). The area of the shaded region shown is approximately $(\Delta r)(r_{jk} \, \Delta\theta)$, because the arc length of a segment of a circle of radius $r$ subtending an angle $\phi$ is $r\phi$. The total area is then the limit of $\sum r_{jk} \, \Delta r \, \Delta\theta$; that is, $\iint_{D^*} r \, dr \, d\theta$. The key idea is thus that the $jk$th "polar rectangle" in the grid has area approximately equal to $r_{jk} \, \Delta r \, \Delta\theta$. (For $n$ large, the $jk$th polar rectangle will look like a rectangle with sides of lengths $r_{jk} \, \Delta\theta$ and $\Delta r$). This should provide some insight into why we say *the "area element $dx \, dy$" is transformed into the "area element $r \, dr \, d\theta$."*

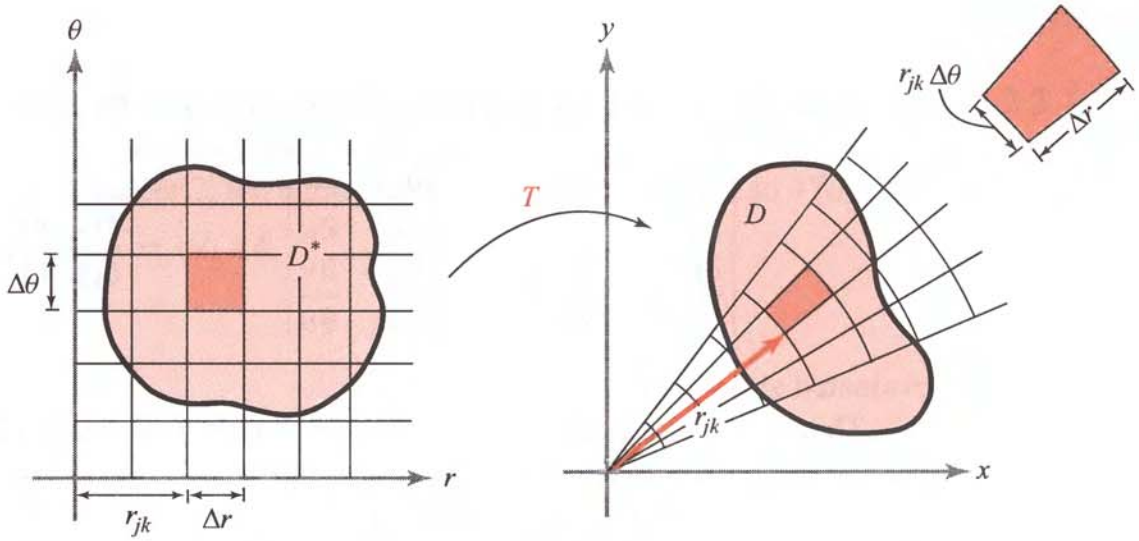



**Figure 6.2.4** $D^*$ is mapped to $D$ under the polar-coordinate mapping $T$.

**EXAMPLE 2** Let the elementary region $D$ in the $xy$ plane be bounded by the graph of a polar equation $r = f(\theta)$, where $\theta_0 \leq \theta \leq \theta_1$ and $f(\theta) \geq 0$ (see Figure 6.2.5). In the $r\theta$ plane we consider the $r$-simple region $D^*$ where $\theta_0 \leq \theta \leq \theta_1$

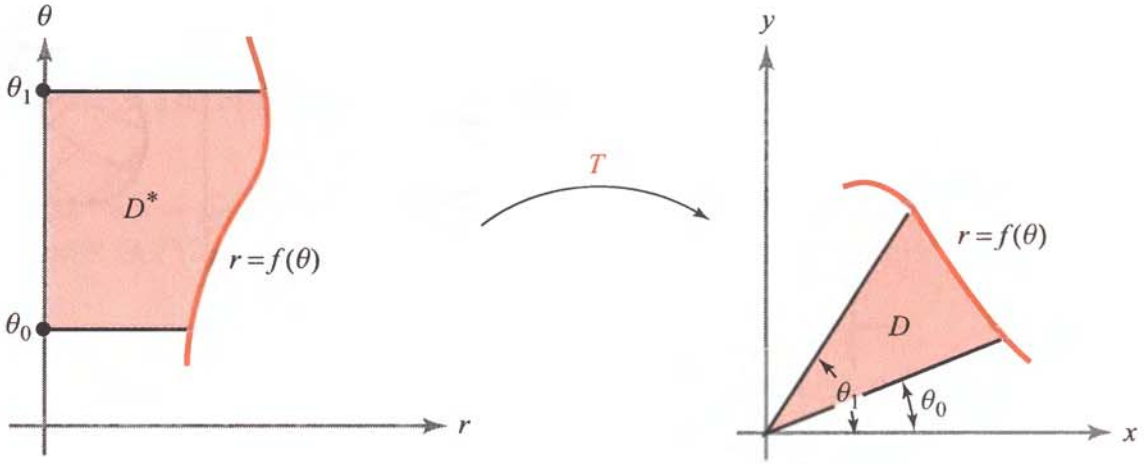



**Figure 6.2.5** The effect on the region $D^*$ of the polar-coordinate mapping.

and $0 \le r \le f(\theta)$. Under the transformation $x = r\cos\theta$, $y = r\sin\theta$, the region $D^*$ is carried onto the region $D$. Use equation (4) to calculate the area of $D$.

SOLUTION

$$A(D) = \iint_D dx\,dy = \iint_{D^*} \left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| dr\,d\theta$$

$$= \iint_{D^*} r\,dr\,d\theta = \int_{\theta_0}^{\theta_1} \left[\int_0^{f(\theta)} r\,dr\right] d\theta$$

$$= \int_{\theta_0}^{\theta_1} \left[\frac{r^2}{2}\right]_0^{f(\theta)} d\theta = \int_{\theta_0}^{\theta_1} \frac{[f(\theta)]^2}{2} d\theta$$

This formula for $A(D)$ should be familiar from one-variable calculus. ▲

## Change of Variables Formula

Before stating the two-variable change of variables formula, which is the culmination of this discussion, let us recall the corresponding theorem from one-variable calculus that goes under the name the *method of substitution*:

$$\int_a^b f(x(u))\frac{dx}{du}\,du = \int_{x(a)}^{x(b)} f(x)\,dx, \tag{5}$$

*where $f$ is continuous and $u \mapsto x(u)$ is continuously differentiable on $[a, b]$.*

*PROOF* Let $F$ be an antiderivative of $f$; that is, $F' = f$, whose existence is guaranteed by the fundamental theorem of calculus. The right-hand side of equation (5) becomes

$$\int_{x(a)}^{x(b)} f(x)\,dx = F(x(b)) - F(x(a)).$$

To evaluate the left-hand side of equation (5), let $G(u) = F(x(u))$. By the chain rule, $G'(u) = F'(x(u))x'(u) = f(x(u))x'(u)$. Hence, again by the fundamental theorem,

$$\int_a^b f(x(u))x'(u)\,du = \int_a^b G'(u)\,du = G(b) - G(a) = F(x(b)) - F(x(a)),$$

as required. ■

Suppose now that we have a $C^1$ *function* $u \mapsto x(u)$ that is one-to-one on $[a, b]$. Thus, we must have either $dx/du \ge 0$ on $[a, b]$ or $dx/du \le 0$ on $[a, b]$.[1] Let $I^*$ denote the interval $[a, b]$, and let $I$ denote the closed interval with endpoints $x(a)$

[1] If $dx/du$ is positive and then negative, the function $x = x(u)$ rises and then falls, and thus is not one-to-one; a similar statement applies if $dx/du$ is negative and then positive.

and $x(b)$. (Thus, $I = [x(a), x(b)]$ if $u \mapsto x(u)$ is increasing and $I = [x(b), x(a)]$ if $u \mapsto x(u)$ is decreasing.) With these conventions we can rewrite formula (5) as

$$\int_{I^*} f(x(u)) \left| \frac{dx}{du} \right| du = \int_I f(x)\, dx.$$

This formula generalizes to double integrals, as was already given informally in formula (3): $I^*$ becomes $D^*$, $I$ becomes $D$, and $|dx/du|$ is replaced by $|\partial(x, y)/\partial(u, v)|$. Let us state the result formally (the technical proof is omitted).

**THEOREM 2: Change of Variables: Double Integrals** Let $D$ and $D^*$ be elementary regions in the plane and let $T\colon D^* \to D$ be of class $C^1$; suppose that $T$ is one-to-one on $D^*$. Furthermore, suppose that $D = T(D^*)$. Then for any integrable function $f\colon D \to \mathbb{R}$, we have

$$\iint_D f(x, y)\, dx\, dy = \iint_{D^*} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv. \qquad (6)$$

One of the purposes of the change of variables theorem is to supply a method by which some double integrals can be simplified. One might encounter an integral $\iint_D f\, dA$ for which either the integrand $f$ or the region $D$ is complicated and for which direct evaluation is difficult. Therefore, a mapping $T$ is chosen so that the integral is easier to evaluate with the new integrand $f \circ T$ and with the new region $D^*$ [defined by $T(D^*) = D$]. Unfortunately, the problem may actually become more complicated if $T$ is not selected carefully.

**EXAMPLE 3** Let $P$ be the parallelogram bounded by $y = 2x$, $y = 2x - 2$, $y = x$, and $y = x + 1$ (see Figure 6.2.6). Evaluate $\iint_P xy\, dx\, dy$ by making the

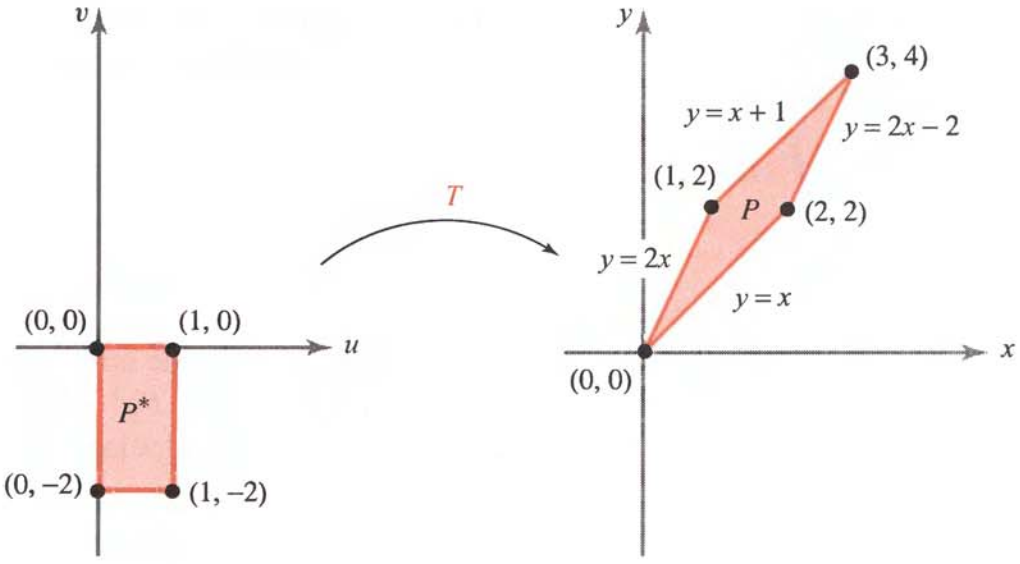



Figure 6.2.6 The effect of $T(u, v) = (u - v, 2u - v)$ on the rectangle $P^*$.

change of variables

$$x = u - v, \qquad y = 2u - v,$$

that is, $T(u, v) = (u - v, 2u - v)$.

SOLUTION The transformation $T$ has nonzero determinant and so is one-to-one (see Exercise 8, Section 6.1). It is designed so that it takes the *rectangle* $P^*$ bounded by $v = 0$, $v = -2$, $u = 0$, $u = 1$ onto $P$. The use of $T$ simplifies the region of integration from $P$ to $P^*$. Moreover,

$$\left|\frac{\partial(x, y)}{\partial(u, v)}\right| = \left|\det\begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}\right| = 1.$$

Therefore, by the change of variables formula,

$$\begin{aligned}
\iint_P xy\,dx\,dy &= \iint_{P^*} (u - v)(2u - v)\,du\,dv = \int_{-2}^{0}\int_{0}^{1} (2u^2 - 3vu + v^2)\,du\,dv \\
&= \int_{-2}^{0}\left[\frac{2}{3}u^3 - \frac{3u^2v}{2} + v^2u\right]_0^1 dv = \int_{-2}^{0}\left[\frac{2}{3} - \frac{3}{2}v + v^2\right] dv \\
&= \left[\frac{2}{3}v - \frac{3}{4}v^2 + \frac{v^3}{3}\right]_{-2}^{0} = -\left[\frac{2}{3}(-2) - 3 - \frac{8}{3}\right] \\
&= -\left[-\frac{12}{3} - 3\right] = 7. \quad \blacktriangle
\end{aligned}$$

## Integrals in Polar Coordinates

Suppose we consider the rectangle $D^*$ defined by $0 \le \theta \le 2\pi$, $0 \le r \le a$ in the $r\theta$ plane. The transformation $T$ given by $T(r, \theta) = (r\cos\theta, r\sin\theta)$ takes $D^*$ onto the disk $D$ with equation $x^2 + y^2 \le a^2$ in the $xy$ plane. This transformation represents the change from Cartesian coordinates to polar coordinates. However, $T$ does not satisfy the requirements of the change of variables theorem, because it is not one-to-one on $D^*$: In particular, $T$ sends all points with $r = 0$ to $(0, 0)$ (see Figure 6.2.7 and Example 3 of Section 6.1). Nevertheless, the change of variables theorem is valid in this case. Basically, the reason for this is that the set of points where $T$ is not one-to-one lies on an edge of $D^*$, which is the graph of a smooth curve and therefore, for the purpose of integration, can be neglected. In summary, the formula

**Change of Variables—Polar Coordinates**

$$\iint_D f(x, y)\,dx\,dy = \iint_{D^*} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta \qquad (7)$$

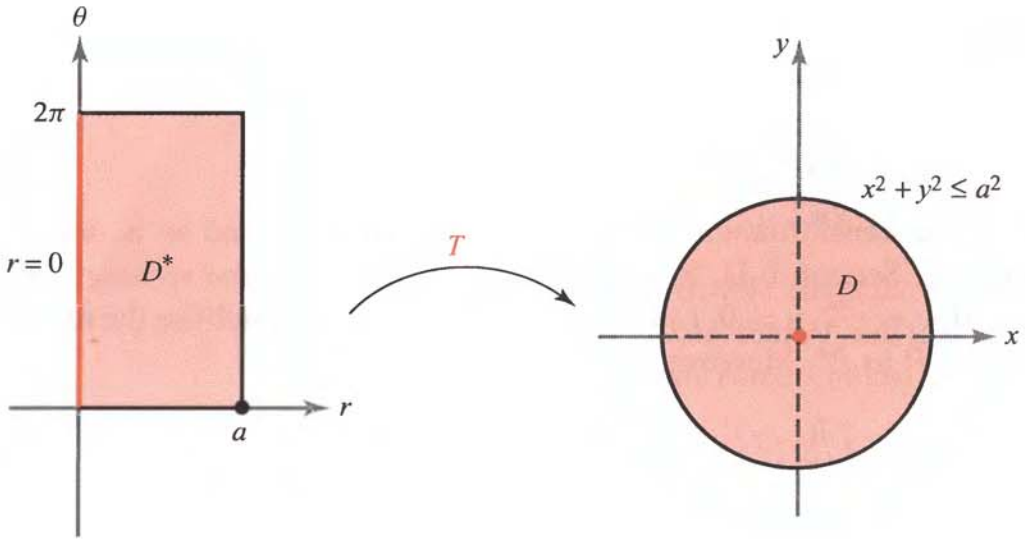



**Figure 6.2.7** The image of the rectangle $D^*$ under the polar-coordinate transformation is the disk $D$.

is valid when $T$ sends $D^*$ onto $D$ in a one-to-one fashion except possibly for points on the boundary of $D^*$.

**EXAMPLE 4** Evaluate $\iint_D \log(x^2 + y^2)\,dx\,dy$, where $D$ is the region in the first quadrant lying between the arcs of the circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$, where $0 < a < b$ (Figure 6.2.8).

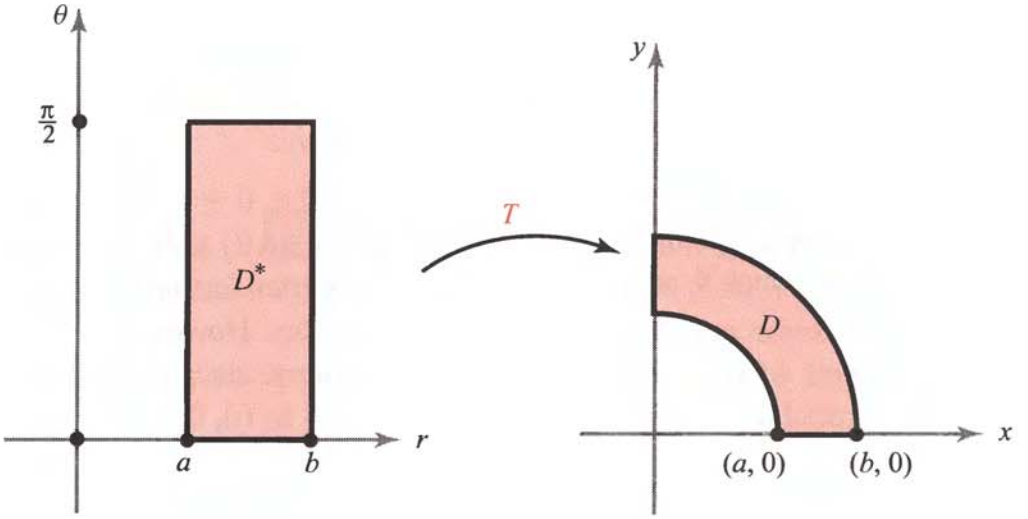



**Figure 6.2.8** The polar-coordinate mapping takes a rectangle $D^*$ onto part of an annulus $D$.

SOLUTION These circles have the simple equations $r = a$ and $r = b$ in polar coordinates. Moreover, $r^2 = x^2 + y^2$ appears in the integrand. Thus, a change to polar coordinates will simplify both the integrand and the region of integration. From Example 7, Section 6.1, the polar-coordinate transformation

$$x = r\cos\theta, \qquad y = r\sin\theta$$

sends the rectangle $D^*$ given by $a \le r \le b, 0 \le \theta \le \pi/2$ onto the region $D$. This transformation is one-to-one on $D^*$ and so, by formula (7), we have

$$\iint_D \log(x^2 + y^2)\, dx\, dy = \int_a^b \int_0^{\pi/2} r \log r^2\, d\theta\, dr$$
$$= \frac{\pi}{2} \int_a^b r \log r^2\, dr = \frac{\pi}{2} \int_a^b 2r \log r\, dr.$$

Applying integration by parts, or using the formula

$$\int x \log x\, dx = \frac{x^2}{2} \log x - \frac{x^2}{4}$$

from the table of integrals at the back of the book, we obtain the result

$$\frac{\pi}{2} \int_a^b 2r \log r\, dr = \frac{\pi}{2} \left[ b^2 \log b - a^2 \log a - \frac{1}{2}(b^2 - a^2) \right]. \quad \blacktriangle$$

**EXAMPLE 5** **The Gaussian Integral** One of the most beautiful applications of the change of variables formula, polar coordinates, and the reduction to iterated integrals is their application to the following formula, known as the ***Gaussian integral***:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Not only is this formula very attractive in its own right, but it is also useful in areas such as statistics. It also illustrates the unity of the transcendental numbers $e$ and $\pi$ nearly as well as does the classic formula $e^{i\pi} = -1$.

To carry out the integration of the Gaussian integral,[2] we first evaluate the double integral

$$\iint_{D_a} e^{-(x^2+y^2)}\, dx\, dy,$$

[2] The method that follows is admittedly not straightforward but requires a trick. The trick is to start with the desired formula and square both sides. You will then observe that the left-hand side resembles an iterated integral. There are several other ways to evaluate the Gaussian integral, but all of them require some nonobvious method. For the use of complex variables to evaluate it, see, for example, J. Marsden and M. Hoffman, *Basic Complex Analysis*, 3rd ed., W. H. Freeman, New York, 1998.

where $D_a$ is the disk $x^2 + y^2 \le a^2$. Because $r^2 = x^2 + y^2$, and $dx\,dy = r\,dr\,d\theta$, the change of variables formula gives

$$\iint_{D_a} e^{-(x^2+y^2)}\,dx\,dy = \int_0^{2\pi}\int_0^a e^{-r^2} r\,dr\,d\theta = \int_0^{2\pi} \left(-\frac{1}{2}e^{-r^2}\right)\Big|_0^a d\theta$$
$$= -\frac{1}{2}\int_0^{2\pi} (e^{-a^2} - 1)\,d\theta = \pi(1 - e^{-a^2}).$$

If we let $a \to \infty$ in this expression, we give meaning to the improper integral and get

$$\iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy = \pi.$$

Assuming (as shown in the Internet supplement) that we can also evaluate this improper integral as the limit of the integrals over the rectangles $R_a = [-a, a] \times [-a, a]$ as $a \to \infty$, we get

$$\lim_{a\to\infty} \iint_{R_a} e^{-(x^2+y^2)}\,dx\,dy = \pi.$$

By reduction to iterated integrals, we can write this as

$$\lim_{a\to\infty} \left[\int_{-a}^a e^{-x^2}\,dx \int_{-a}^a e^{-y^2}\,dy\right] = \left[\lim_{a\to\infty} \int_{-a}^a e^{-x^2}\,dx\right]^2 = \pi.$$

That is,

$$\left[\int_{-\infty}^{\infty} e^{-x^2}\,dx\right]^2 = \pi.$$

Thus, taking square roots, we arrive at the desired result.

Here is a variant of the Gaussian integral. Evaluate

$$\int_{-\infty}^{\infty} e^{-2x^2}\,dx.$$

To do this, use the change of variables formula $y = \sqrt{2}x$ to reduce the problem to the Gaussian integral just computed:

$$\int_{-\infty}^{\infty} e^{-2x^2}\,dx = \lim_{a\to\infty} \int_{-a}^a e^{-2x^2}\,dx = \lim_{a\to\infty} \int_{-\sqrt{2}a}^{\sqrt{2}a} e^{-y^2} \frac{dy}{\sqrt{2}}$$
$$= \frac{1}{\sqrt{2}} \int_{-\infty}^{\infty} e^{-y^2}\,dy = \frac{1}{\sqrt{2}}\sqrt{\pi} = \sqrt{\frac{\pi}{2}}. \quad \blacktriangle$$

## Change of Variables Formula for Triple Integrals

To state this formula, we first define the Jacobian of a transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$—it is a simple extension of the two-variable case.

DEFINITION Let $T\colon W \subset \mathbb{R}^3 \to \mathbb{R}^3$ be a $C^1$ function defined by the equations $x = x(u, v, w), y = y(u, v, w), z = z(u, v, w)$. Then the ***Jacobian*** of $T$, which is denoted $\partial(x, y, z)/\partial(u, v, w)$, is the determinant

$$\begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} & \dfrac{\partial x}{\partial w} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} & \dfrac{\partial y}{\partial w} \\ \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} & \dfrac{\partial z}{\partial w} \end{vmatrix}.$$

The absolute value of this determinant is equal to the volume of the parallelepiped determined by the three vectors

$$\mathbf{T}_u = \frac{\partial x}{\partial u}\mathbf{i} + \frac{\partial y}{\partial u}\mathbf{j} + \frac{\partial z}{\partial u}\mathbf{k},$$

$$\mathbf{T}_v = \frac{\partial x}{\partial v}\mathbf{i} + \frac{\partial y}{\partial v}\mathbf{j} + \frac{\partial z}{\partial v}\mathbf{k},$$

$$\mathbf{T}_w = \frac{\partial x}{\partial w}\mathbf{i} + \frac{\partial y}{\partial w}\mathbf{j} + \frac{\partial z}{\partial w}\mathbf{k}.$$

Just as in the two-variable case, the Jacobian measures how the transformation $T$ distorts the volume of its domain. Hence, for volume (triple) integrals, the change of variables formula takes the following form:

**Change of Variables Formula: Triple Integrals**

$$\begin{aligned} &\iiint_W f(x, y, z)\, dx\, dy\, dz \\ &= \iiint_{W^*} f(x(u, v, w), y(u, v, w), z(u, v, w)) \left|\frac{\partial(x, y, z)}{\partial(u, v, w)}\right| du\, dv\, dw, \end{aligned} \tag{8}$$

where $W^*$ is an elementary region in $uvw$ space corresponding to $W$ in $xyz$ space, under a mapping $T\colon (u, v, w) \mapsto (x(u, v, w), y(u, v, w), z(u, v, w))$, provided $T$ is of class $C^1$ and is one-to-one, except possibly on a set that is the union of graphs of functions of two variables.

## Cylindrical Coordinates

Let us apply formula (8) to cylindrical and then to spherical coordinates. First, we compute the Jacobian for the map defining the change to cylindrical coordinates. Because

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z,$$

we have

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r.$$

Thus, we obtain the formula

**Change of Variables—Cylindrical Coordinates**

$$\iiint_W f(x, y, z)\,dx\,dy\,dz = \iiint_{W^*} f(r\cos\theta, r\sin\theta, z)\,r\,dr\,d\theta\,dz. \quad (9)$$

## Spherical Coordinates

Next we consider the spherical coordinate system. Recall that it is given by

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

Therefore, we have

$$\frac{\partial(x, y, z)}{\partial(\rho, \theta, \phi)} = \begin{vmatrix} \sin\phi\cos\theta & -\rho\sin\phi\sin\theta & \rho\cos\phi\cos\theta \\ \sin\phi\sin\theta & \rho\sin\phi\cos\theta & \rho\cos\phi\sin\theta \\ \cos\phi & 0 & -\rho\sin\phi \end{vmatrix}.$$

Expanding along the last row, we get

$$\begin{aligned} \frac{\partial(x, y, z)}{\partial(\rho, \theta, \phi)} &= \cos\phi \begin{vmatrix} -\rho\sin\phi\sin\theta & \rho\cos\phi\cos\theta \\ \rho\sin\phi\cos\theta & \rho\cos\phi\sin\theta \end{vmatrix} \\ &\quad -\rho\sin\phi \begin{vmatrix} \sin\phi\cos\theta & -\rho\sin\phi\sin\theta \\ \sin\phi\sin\theta & \rho\sin\phi\cos\theta \end{vmatrix} \\ &= -\rho^2\cos^2\phi\sin\phi\sin^2\theta - \rho^2\cos^2\phi\sin\phi\cos^2\theta \\ &\quad -\rho^2\sin^3\phi\cos^2\theta - \rho^2\sin^3\phi\sin^2\theta \\ &= -\rho^2\cos^2\phi\sin\phi - \rho^2\sin^3\phi = -\rho^2\sin\phi. \end{aligned}$$

Thus, we arrive at the formula:

**Change of Variables—Spherical Coordinates**

$$\iiint_W f(x, y, z)\,dx\,dy\,dz = \iiint_{W^*} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2\sin\phi\,d\rho\,d\theta\,d\phi. \tag{10}$$

To prove formula (10), one must show that the transformation $S$ on the set $W^*$ is one-to-one except on a set that is the union of finitely many graphs of continuous functions. We shall leave this verification as Exercise 34.

**EXAMPLE 6** Evaluate

$$\iiint_W \exp(x^2 + y^2 + z^2)^{3/2}\,dV,$$

where $W$ is the unit ball in $\mathbb{R}^3$.

SOLUTION First note that we cannot *easily* integrate this function using iterated integrals (try it!). Hence (employing the strategy in the quote that opened this chapter), let us try a change of variables. The transformation into spherical coordinates seems appropriate, because then the entire quantity $x^2 + y^2 + z^2$ can be replaced by one variable, namely, $\rho^2$. If $W^*$ is the region such that

$$0 \le \rho \le 1, \qquad 0 \le \theta \le 2\pi, \qquad 0 \le \phi \le \pi,$$

we may apply formula (10) and write

$$\iiint_W \exp(x^2 + y^2 + z^2)^{3/2}\,dV = \iiint_{W^*} \rho^2 e^{\rho^3}\sin\phi\,d\rho\,d\theta\,d\phi.$$

This integral equals the iterated integral

$$\begin{aligned}\int_0^1\int_0^\pi\int_0^{2\pi} e^{\rho^3}\rho^2\sin\phi\,d\theta\,d\phi\,d\rho &= 2\pi\int_0^1\int_0^\pi e^{\rho^3}\rho^2\sin\phi\,d\phi\,d\rho \\ &= -2\pi\int_0^1 \rho^2 e^{\rho^3}[\cos\phi]_0^\pi\,d\rho \\ &= 4\pi\int_0^1 e^{\rho^3}\rho^2\,d\rho = \frac{4}{3}\pi\int_0^1 e^{\rho^3}(3\rho^2)\,d\rho \\ &= \left[\frac{4}{3}\pi e^{\rho^3}\right]_0^1 = \frac{4}{3}\pi(e-1). \quad \blacktriangle\end{aligned}$$

**EXAMPLE 7** Let $W$ be the ball of radius $R$ and center $(0, 0, 0)$ in $\mathbb{R}^3$. Find the volume of $W$.

SOLUTION The volume of $W$ is $\iiint_W dx\,dy\,dz$. This integral may be evaluated by reducing it to iterated integrals or by regarding $W$ as a volume of revolution, but let us evaluate it here by using spherical coordinates. We get

$$\iiint_W dx\,dy\,dz = \int_0^{\pi}\int_0^{2\pi}\int_0^{R} \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi = \frac{R^3}{3}\int_0^{\pi}\int_0^{2\pi} \sin\phi\, d\theta\, d\phi$$
$$= \frac{2\pi R^3}{3}\int_0^{\pi} \sin\phi\, d\phi = \frac{2\pi R^3}{3}\{-[\cos(\pi) - \cos(0)]\} = \frac{4\pi R^3}{3},$$

which is the standard formula for the volume of a solid sphere. ▲

## EXERCISES

**1.** Let $D$ be the unit disk: $x^2 + y^2 \leq 1$. Evaluate

$$\iint_D \exp(x^2 + y^2)\, dx\, dy$$

by making a change of variables to polar coordinates.

**2.** Let $D$ be the region $0 \leq y \leq x$ and $0 \leq x \leq 1$. Evaluate

$$\iint_D (x + y)\, dx\, dy$$

by making the change of variables $x = u + v$, $y = u - v$. Check your answer by evaluating the integral directly by using an iterated integral.

**3.** Let $T(u, v) = (x(u, v), y(u, v))$ be the mapping defined by $T(u, v) = (4u, 2u + 3v)$. Let $D^*$ be the rectangle $[0, 1] \times [1, 2]$. Find $D = T(D^*)$ and evaluate

(a) $\iint_D xy\, dx\, dy$ (b) $\iint_D (x - y)\, dx\, dy$

by making a change of variables to evaluate them as integrals over $D^*$.

**4.** Repeat Exercise 3 for $T(u, v) = (u, v(1 + u))$.

**5.** Evaluate

$$\iint_D \frac{dx\, dy}{\sqrt{1 + x + 2y}},$$

where $D = [0, 1] \times [0, 1]$, by setting $T(u, v) = (u, v/2)$ and evaluating an integral over $D^*$, where $T(D^*) = D$.

**6.** Define $T(u, v) = (u^2 - v^2, 2uv)$. Let $D^*$ be the set of $(u, v)$ with $u^2 + v^2 \leq 1$, $u \geq 0$, $v \geq 0$. Find $T(D^*) = D$. Evaluate $\iint_D dx\, dy$.

7. Let $T(u, v)$ be as in Exercise 6. By making a change of variables, "formally" evaluate the "improper" integral

$$\iint_D \frac{dx\,dy}{\sqrt{x^2+y^2}}.$$

[NOTE: This integral (and the one in the next exercise) is *improper*, because the integrand $1/\sqrt{x^2+y^2}$ is neither continuous nor bounded on the domain of integration. (The theory of improper integrals is discussed in Section 6.4.)]

8. Calculate $\iint_R \frac{1}{x+y}\,dy\,dx$, where $R$ is the region bounded by $x = 0$, $y = 0$, $x + y = 1$, $x + y = 4$, by using the mapping $T(u, v) = (u - uv, uv)$.

9. Evaluate $\iint_D (x^2+y^2)^{3/2}\,dx\,dy$ where $D$ is the disk $x^2 + y^2 \le 4$.

10. Let $D^*$ be a $v$-simple region in the $uv$ plane bounded by $v = g(u)$ and $v = h(u) \le g(u)$ for $a \le u \le b$. Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the transformation given by $x = u$ and $y = \psi(u, v)$, where $\psi$ is of class $C^1$ and $\partial\psi/\partial v$ is never zero. Assume that $T(D^*) = D$ is a $y$-simple region; show that if $f\colon D \to \mathbb{R}$ is continuous, then

$$\iint_D f(x, y)\,dx\,dy = \iint_{D^*} f(u, \psi(u, v))\left|\frac{\partial\psi}{\partial v}\right| du\,dv.$$

11. Use double integrals to find the area inside the curve $r = 1 + \sin\theta$.

12. (a) Express $\int_0^1 \int_0^{x^2} xy\,dy\,dx$ as an integral over the triangle $D^*$, which is the set of $(u, v)$ where $0 \le u \le 1$, $0 \le v \le u$. (HINT: Find a one-to-one mapping $T$ of $D^*$ onto the given region of integration.)

(b) Evaluate this integral directly and as an integral over $D^*$.

13. Integrate $ze^{x^2+y^2}$ over the cylinder $x^2 + y^2 \le 4$, $2 \le z \le 3$.

14. Let $D$ be the unit disk. Express $\iint_D (1 + x^2 + y^2)^{3/2}\,dx\,dy$ as an integral over $[0, 1] \times [0, 2\pi]$ and evaluate.

15. Using polar coordinates, find the area bounded by the *lemniscate* $(x^2 + y^2)^2 = 2a^2(x^2 - y^2)$.

16. Redo Exercise 11 of Section 5.3 using a change of variables and compare the effort involved in each method.

17. Calculate $\iint_R (x + y)^2 e^{x-y}\,dx\,dy$ where $R$ is the region bounded by $x + y = 1$, $x + y = 4$, $x - y = -1$, and $x - y = 1$.

18. Let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be defined by

$$T(u, v, w) = (u\cos v\cos w, u\sin v\cos w, u\sin w).$$

(a) Show that $T$ is onto the unit sphere; that is, every $(x, y, z)$ with $x^2 + y^2 + z^2 = 1$ can be written as $(x, y, z) = T(u, v, w)$ for some $(u, v, w)$.

(b) Show that $T$ is not one-to-one.

**19.** Integrate $x^2 + y^2 + z^2$ over the cylinder $x^2 + y^2 \leq 2$, $-2 \leq z \leq 3$.

**20.** Evaluate $\int_0^\infty e^{-4x^2}\,dx$.

**21.** Let $B$ be the unit ball. Evaluate

$$\iiint_B \frac{dx\,dy\,dz}{\sqrt{2 + x^2 + y^2 + z^2}}$$

by making the appropriate change of variables.

**22.** Evaluate $\iint_A [1/(x^2 + y^2)^2]\,dx\,dy$ where $A$ is determined by the conditions $x^2 + y^2 \leq 1$ and $x + y \geq 1$.

**23.** Evaluate $\iiint_W \frac{dx\,dy\,dz}{(x^2 + y^2 + z^2)^{3/2}}$, where $W$ is the solid bounded by the two spheres $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + z^2 = b^2$, where $0 < b < a$.

**24.** Evaluate $\iint_D x^2\,dx\,dy$ where $D$ is determined by the two conditions $0 \leq x \leq y$ and $x^2 + y^2 \leq 1$.

**25.** Integrate $\sqrt{x^2 + y^2 + z^2}\,e^{-(x^2+y^2+z^2)}$ over the region described in Exercise 23.

**26.** Evaluate the following by using cylindrical coordinates.

(a) $\iiint_B z\,dx\,dy\,dz$ where $B$ is the region within the cylinder $x^2 + y^2 = 1$ above the $xy$ plane and below the cone $z = (x^2 + y^2)^{1/2}$.

(b) $\iiint_W (x^2 + y^2 + z^2)^{-1/2}\,dx\,dy\,dz$ where $W$ is the region determined by the conditions $\frac{1}{2} \leq z \leq 1$ and $x^2 + y^2 + z^2 \leq 1$.

**27.** Evaluate $\iint_B (x + y)\,dx\,dy$ where $B$ is the rectangle in the $xy$ plane with vertices at $(0, 1)$, $(1, 0)$, $(3, 4)$, and $(4, 3)$.

**28.** Evaluate $\iint_D (x + y)\,dx\,dy$ where $D$ is the square with vertices at $(0, 0)$, $(1, 2)$, $(3, 1)$, and $(2, -1)$.

**29.** Let $E$ be the ellipsoid $(x^2/a^2) + (y^2/b^2) + (z^2/c^2) \leq 1$, where $a$, $b$, and $c$ are positive.

(a) Find the volume of $E$.

(b) Evaluate $\iiint_E [(x^2/a^2) + (y^2/b^2) + (z^2/c^2)]\,dx\,dy\,dz$. (HINT: Change variables and then use spherical coordinates.)

**30.** Using spherical coordinates, compute the integral of $f(\rho, \phi, \theta) = 1/\rho$ over the region in the first octant of $\mathbb{R}^3$, which is bounded by the cones $\phi = \pi/4$, $\phi = \arctan 2$ and the sphere $\rho = \sqrt{6}$.

**31.** The mapping $T(u, v) = (u^2 - v^2, 2uv)$ transforms the rectangle $1 \le u \le 2, 1 \le v \le 3$ of the $uv$ plane into a region $R$ of the $xy$ plane.

(a) Show that $T$ is one-to-one.
(b) Find the area of $R$ using the change of variables formula.

**32.** Let $R$ denote the region inside $x^2 + y^2 = 1$, but outside $x^2 + y^2 = 2y$ with $x \ge 0, y \ge 0$.

(a) Sketch this region.
(b) Let $u = x^2 + y^2, v = x^2 + y^2 - 2y$. Sketch the region $D$ in the $uv$ plane, which corresponds to $R$ under this change of coordinates.
(c) Compute $\iint_R xe^y \, dx \, dy$ using this change of coordinates.

**33.** Let $D$ be the region bounded by $x^{3/2} + y^{3/2} = a^{3/2}$, for $x \ge 0, y \ge 0$, and the coordinate axes $x = 0, y = 0$. Express $\iint_D f(x, y) \, dx \, dy$ as an integral over the triangle $D^*$, which is the set of points $0 \le u \le a, 0 \le v \le a - u$. (Do not attempt to evaluate.)

**34.** Show that $S(\rho, \theta, \phi) = (\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi)$, the spherical change-of-coordinate mapping, is one-to-one except on a set that is a union of finitely many graphs of continuous functions.

# 6.3 Applications

In this section, we shall discuss average values, centers of mass, moments of inertia, and the gravitational potential as applications.

## Averages

If $x_1, \ldots, x_n$ are $n$ numbers, their ***average*** is defined by

$$[x_i]_{\text{av}} = \frac{x_1 + \cdots + x_n}{n} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

Notice that if all the $x_i$ happen to have a common value $c$, then their average, of course, also equals $c$.

This concept leads one to define the average values of functions as follows.

**Average Values** The ***average value*** of a function of one variable on the interval $[a, b]$ is defined by

$$[f]_{\text{av}} = \frac{\int_a^b f(x) \, dx}{b - a}.$$

Likewise, for functions of two variables, the ratio of the integral to the area of $D$,

$$[f]_{\text{av}} = \frac{\iint_D f(x, y) \, dx \, dy}{\iint_D dx \, dy}, \tag{1}$$

is called the ***average value*** of $f$ over $D$. Similarly, the ***average value*** of a function $f$ on a region $W$ in three space is defined by

$$[f]_{\text{av}} = \frac{\iiint_W f(x, y, z)\, dx\, dy\, dz}{\iiint_W dx\, dy\, dz}.$$

Again, notice that the denominator is chosen so that if $f$ is a constant, say $c$, then $[f]_{\text{av}} = c$.

**EXAMPLE 1** Find the average value of $f(x, y) = x \sin^2(xy)$ on the region $D = [0, \pi] \times [0, \pi]$.

SOLUTION First, we compute

$$\begin{aligned}
\iint_D f(x, y)\, dx\, dy &= \int_0^\pi \int_0^\pi x \sin^2(xy)\, dx\, dy \\
&= \int_0^\pi \left[ \int_0^\pi \frac{1 - \cos(2xy)}{2} x\, dy \right] dx \\
&= \int_0^\pi \left[ \frac{y}{2} - \frac{\sin(2xy)}{4x} \right] x \Big|_{y=0}^{\pi} dx \\
&= \int_0^\pi \left[ \frac{\pi x}{2} - \frac{\sin(2\pi x)}{4} \right] dx = \left[ \frac{\pi x^2}{4} + \frac{\cos(2\pi x)}{8\pi} \right] \Big|_0^\pi \\
&= \frac{\pi^3}{4} + \frac{\cos(2\pi^2) - 1}{8\pi}.
\end{aligned}$$

Thus, the average value of $f$, by formula (1), is

$$\frac{\pi^3/4 + [\cos(2\pi^2) - 1]/8\pi}{\pi^2} = \frac{\pi}{4} + \frac{\cos(2\pi^2) - 1}{8\pi^3} \approx 0.7839. \quad \blacktriangle$$

**EXAMPLE 2** The temperature at points in the cube $W = [-1, 1] \times [-1, 1] \times [-1, 1]$ is proportional to the square of the distance from the origin.

(a) What is the average temperature?

(b) At which points of the cube is the temperature equal to the average temperature?

SOLUTION (a) Let $c$ be the constant of proportionality so $T = c(x^2 + y^2 + z^2)$ and the average temperature is $[T]_{\text{av}} = \frac{1}{8} \iiint_W T\, dx\, dy\, dz$,because the volume of the

cube is 8. Thus,

$$[T]_{\text{av}} = \frac{c}{8}\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1}(x^2+y^2+z^2)\,dx\,dy\,dz.$$

The triple integral is the sum of the integrals of $x^2$, $y^2$, and $z^2$. Because $x$, $y$, and $z$ enter symmetrically into the description of the cube, the three integrals will be equal, so that

$$[T]_{\text{av}} = \frac{3c}{8}\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1}z^2\,dx\,dy\,dz = \frac{3c}{8}\int_{-1}^{1}z^2\left(\int_{-1}^{1}\int_{-1}^{1}dx\,dy\right)dz.$$

The inner integral is equal to the area of the square $[-1, 1] \times [-1, 1]$. The area of that square is 4, and so

$$[T]_{\text{av}} = \frac{3c}{8}\int_{-1}^{1}4z^2\,dz = \frac{3c}{2}\left(\frac{z^3}{3}\right)\Big|_{-1}^{1} = c.$$

(b) The temperature is equal to the average temperature at all points satisfying $c(x^2 + y^2 + z^2) = c$, that is, which lie on the sphere $x^2 + y^2 + z^2 = 1$, which is inscribed in the cube $W$. ▲

## Centers of Mass

If masses $m_1, \ldots, m_n$ are placed at points $x_1, \ldots, x_n$ on the $x$ axis, their ***center of mass*** is defined to be

$$\bar{x} = \frac{\sum m_i x_i}{\sum m_i}. \tag{2}$$

This definition arises from the following observation: If one is balancing masses on a lever (Figure 6.3.1), the balance point $\bar{x}$ occurs where the total moment (mass times distance from the balance point) is zero, that is, where $\sum m_i(x_i - \bar{x}) = 0$. A physical principle, going back first to Archimedes and then in this generality to Newton, states that this condition means there is no tendency for the lever to rotate.

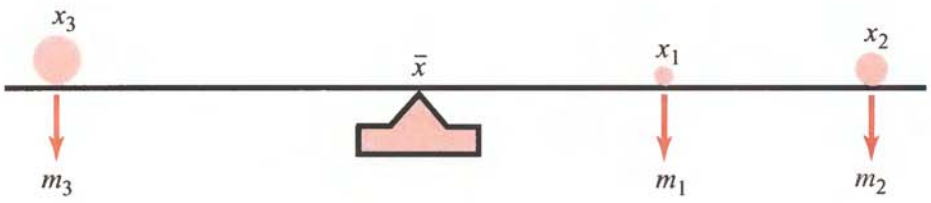



**Figure 6.3.1** The lever is balanced if $\Sigma(x_i - \bar{x})m_i = 0$.

For a continuous mass density $\delta(x)$ along the lever (measured in, say, grams/cm), the analog of formula (2) is

$$\bar{x} = \frac{\int x\delta(x)\,dx}{\int \delta(x)\,dx}. \tag{3}$$

For two-dimensional plates, this generalizes to:

**The Center of Mass of Two-Dimensional Plates**

$$\bar{x} = \frac{\iint_D x\delta(x, y)\,dx\,dy}{\iint_D \delta(x, y)\,dx\,dy} \quad \text{and} \quad \bar{y} = \frac{\iint_D y\delta(x, y)\,dx\,dy}{\iint_D \delta(x, y)\,dx\,dy}, \tag{4}$$

where again $\delta(x, y)$ is the mass density (see Figure 6.3.2).

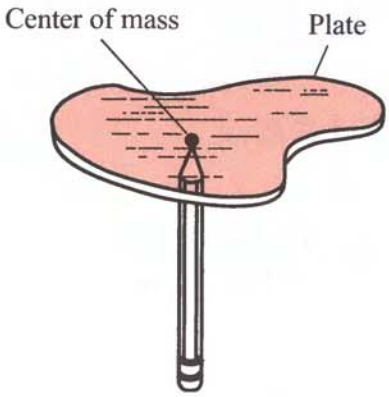



**Figure 6.3.2** The plate balances when supported at its center of mass.

**EXAMPLE 3** Find the center of mass of the rectangle $[0, 1] \times [0, 1]$ if the mass density is $e^{x+y}$.

SOLUTION First we compute the total mass:

$$\iint_D e^{x+y}\,dx\,dy = \int_0^1 \int_0^1 e^{x+y}\,dx\,dy = \int_0^1 \left(e^{x+y}|_{x=0}^1\right) dy = \int_0^1 (e^{1+y} - e^y)\,dy$$
$$= (e^{1+y} - e^y)|_{y=0}^1 = e^2 - e - (e - 1) = e^2 - 2e + 1.$$

The numerator in formula (4) for $\bar{x}$ is

$$\int_0^1 \int_0^1 xe^{x+y}\,dx\,dy = \int_0^1 (xe^{x+y} - e^{x+y})|_{x=0}^1\,dy = \int_0^1 [e^{1+y} - e^{1+y} - (0e^y - e^y)]\,dy$$
$$= \int_0^1 e^y\,dy = e^y|_{y=0}^1 = e - 1,$$

so that

$$\bar{x} = \frac{e - 1}{e^2 - 2e + 1} = \frac{e - 1}{(e - 1)^2} = \frac{1}{e - 1} \approx 0.582.$$

The roles of $x$ and $y$ may be interchanged in all these calculations, so that $\bar{y} = 1/(e - 1) \approx 0.582$ as well. ▲

For a region $W$ in space with mass density $\delta(x, y, z)$, we know that

$$\text{volume} = \iiint_W dx\, dy\, dz, \tag{5}$$

$$\text{mass} = \iiint_W \delta(x, y, z)\, dx\, dy\, dz. \tag{6}$$

If one denotes the coordinates of the center of mass by $(\bar{x}, \bar{y}, \bar{z})$, then the generalization of the formulas in the preceding box are as follows.

**Coordinates for the Center of Mass of Three-Dimensional Regions**

$$\bar{x} = \frac{\iiint_W x\delta(x, y, z)\, dx\, dy\, dz}{\text{mass}},$$

$$\bar{y} = \frac{\iiint_W y\delta(x, y, z)\, dx\, dy\, dz}{\text{mass}}, \tag{7}$$

$$\bar{z} = \frac{\iiint_W z\delta(x, y, z)\, dx\, dy\, dz}{\text{mass}}.$$

**EXAMPLE 4** The cube $[1, 2] \times [1, 2] \times [1, 2]$ has mass density given by $\delta(x, y, z) = (1 + x)e^z y$. Find the total mass of the box.

SOLUTION The mass of the box is, by formula (6),

$$\int_1^2 \int_1^2 \int_1^2 (1 + x)e^z y\, dx\, dy\, dz = \int_1^2 \int_1^2 \left[\left(x + \frac{x^2}{2}\right) e^z y\right]_{x=1}^{x=2} dy\, dz$$

$$= \int_1^2 \int_1^2 \frac{5}{2} e^z y\, dy\, dz = \int_1^2 \frac{15}{4} e^z\, dz = \left[\frac{15}{4} e^z\right]_{z=1}^{z=2} = \frac{15}{4}(e^2 - e). \quad \blacktriangle$$

If a region and its mass density are reflection-symmetric across a plane, then the center of mass lies on that plane. For example, in formula (7) for $\bar{x}$, if the region and mass density are symmetric in the $yz$ plane, then the integrand is odd in $x$, and so $\bar{x} = 0$. This kind of use of symmetry is illustrated in the next example.

**EXAMPLE 5** Find the center of mass of the hemispherical region $W$ defined by the inequalities $x^2 + y^2 + z^2 \leq 1, z \geq 0$. (Assume that the density is unity.)

SOLUTION By symmetry, the center of mass must lie on the $z$ axis, and so $\bar{x} = \bar{y} = 0$. To find $\bar{z}$, we must compute, by formula (7), the numerator $I = \iiint_W z\, dx\, dy\, dz$. The hemisphere is an elementary region, and thus the

integral becomes

$$I = \int_0^1 \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} \int_{-\sqrt{1-y^2-z^2}}^{\sqrt{1-y^2-z^2}} z \, dx \, dy \, dz.$$

Because $z$ is a constant for the $x$ and $y$ integrations, we can remove it from the first two integral signs, to obtain

$$I = \int_0^1 z \left( \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} \int_{-\sqrt{1-y^2-z^2}}^{\sqrt{1-y^2-z^2}} dx \, dy \right) dz.$$

Instead of calculating the inner two integrals explicitly, we observe that they equal the double integral $\iint_D dx\, dy$ over the disk $x^2 + y^2 \leq 1 - z^2$, considered as an $x$-simple region in the plane. The area of this disk is $\pi(1 - z^2)$, and so

$$I = \pi \int_0^1 z(1 - z^2)\, dz = \pi \int_0^1 (z - z^3)\, dz = \pi \left[ \frac{z^2}{2} - \frac{z^4}{4} \right]_0^1 = \frac{\pi}{4}.$$

The volume of the hemisphere is $\frac{2}{3}\pi$, and so $\bar{z} = (\pi/4)/(\frac{2}{3}\pi) = \frac{3}{8}$. ▲

## Historical Note

It is common knowledge that Archimedes observed the principle of the lever. Perhaps less known is that he was also responsible for discovering the concepts of center of mass and center of gravity. Only two of his works on mechanics have been handed down to us: *On Floating Bodies* and *On the Equilibrium and Centers of Mass of Plane Figures.* Both were translated into Latin by Niccolo Tartaglia, circa 1543.

In *Equilibrium...*, Archimedes began the field of applied mathematics, doing for mechanics what Euclid had accomplished for geometry. In this work he describes the principles behind all the machines of antiquity, including the lever, inclined plane, and pulley system.

Surprisingly, Archimedes never carefully defined the center of mass; the first proper definition was given by Pappus of Alexandria in 340 C.E. The concept of equilibrium was to have a profound effect on the development of mechanical engineering (through the introduction of gears), architecture,

and in art, permitting the construction of complex machines, large-scale buildings, and sculptures. Figure 6.3.3 shows sketches by Leonardo DaVinci, illustrating equilibrium positions of the human body.

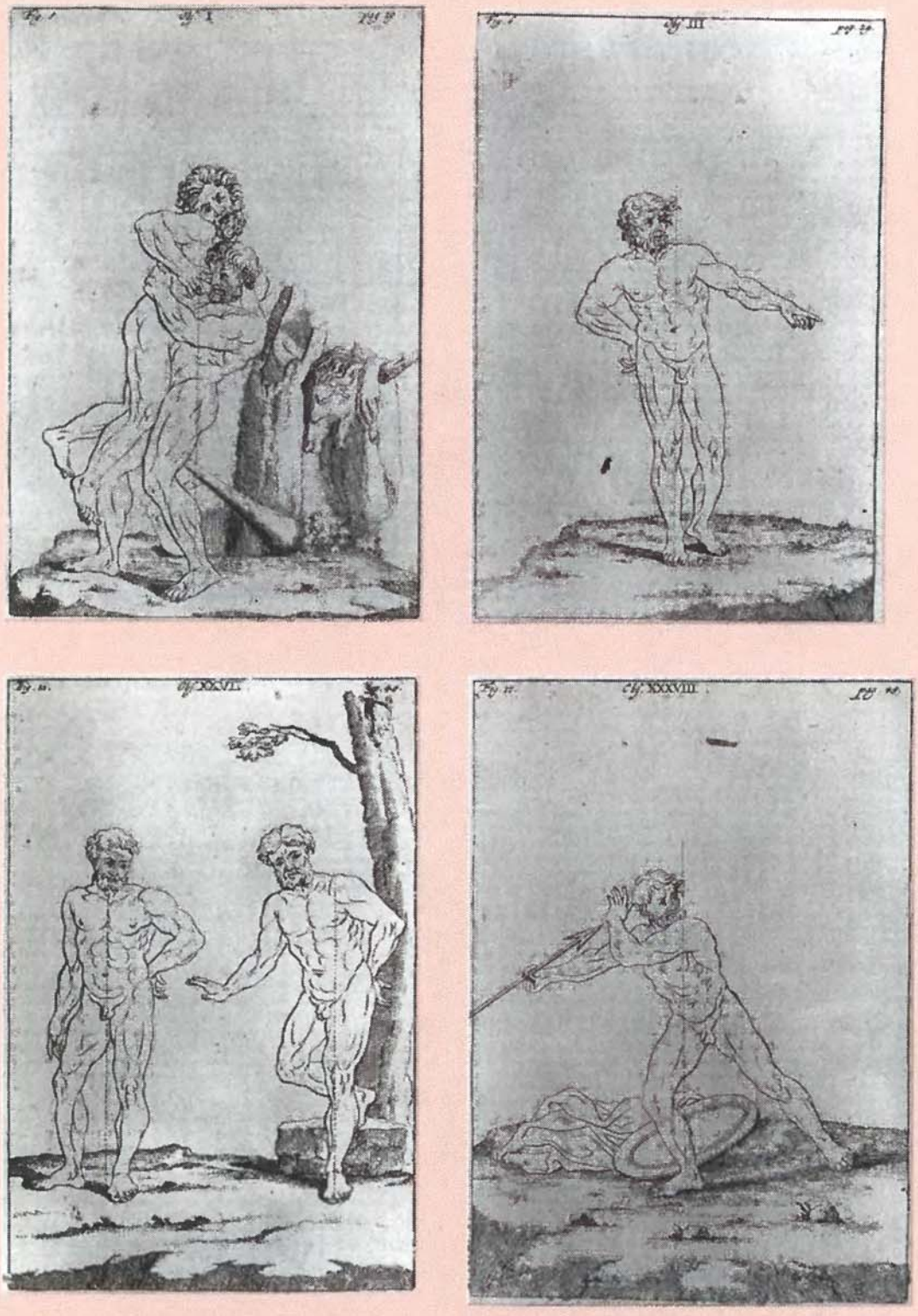


**Figure 6.3.3** Equilibrium positions of the human body, to be observed by the painter. The center of mass should be supported to maintain equilibrium.

## Moments of Inertia

Another important concept in mechanics, one that is needed in studying the dynamics of a rotating rigid body, is that of *moment of inertia*. If the solid $W$ has uniform density $\delta$, the ***moments of inertia*** $I_x$, $I_y$, and $I_z$ about the $x$, $y$, and $z$ axes, respectively, are defined by:

**Moments of Inertia About the Coordinate Axes**

$$I_x = \iiint_W (y^2 + z^2)\,\delta\,dx\,dy\,dz, \qquad I_y = \iiint_W (x^2 + z^2)\,\delta\,dx\,dy\,dz,$$
$$I_z = \iiint_W (x^2 + y^2)\,\delta\,dx\,dy\,dz. \tag{8}$$

The moment of inertia measures a body's response to efforts to rotate it; for example, as when one tries to rotate a merry-go-round. The moment of inertia is analogous to the mass of a body, which measures its response to efforts to translate it. In contrast to translational motion, however, the moments of inertia *depend on the shape and not just the total mass*. It is harder to spin up a large plate than a compact ball of the same mass.

For example, $I_x$ measures the body's response to forces attempting to rotate it about the $x$ axis. The factor $y^2 + z^2$, which is the square of the distance to the $x$ axis, weights masses farther away from the rotation axis more heavily. This is in agreement with the intuition just explained.

**EXAMPLE 6** Compute the moment of inertia $I_z$ for the solid above the $xy$ plane bounded by the paraboloid $z = x^2 + y^2$ and the cylinder $x^2 + y^2 = a^2$, assuming $a$ and the mass density to be constants.

SOLUTION The paraboloid and cylinder intersect at the plane $z = a^2$. Using cylindrical coordinates, we find from equation (8),

$$I_z = \int_0^a \int_0^{2\pi} \int_0^{r^2} \delta r^2 \cdot r\,dz\,d\theta\,dr = \delta \int_0^a \int_0^{2\pi} \int_0^{r^2} r^3\,dz\,d\theta\,dr = \frac{\pi\delta a^6}{3}. \quad \blacktriangle$$

## Gravitational Fields of Solid Objects

Another interesting physical application of triple integration is the determination of the gravitational fields of solid objects. Example 6, Section 2.6, showed that the gravitational force field $\mathbf{F}(x, y, z)$ of a particle is the negative of the gradient of a function $V(x, y, z)$ called the ***gravitational potential***. If there is a point mass $M$ at $(x, y, z)$, then the gravitational potential acting on a mass $m$ at $(x_1, y_1, z_1)$ due to this mass is $-GmM[(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2]^{-1/2}$, where $G$ is the universal gravitational constant.

If our attracting object occupies a domain $W$ with mass density $\delta(x, y, z)$, we may think of it as made of infinitesimal box-shaped regions with masses $dM = \delta(x, y, z)\,dx\,dy\,dz$ located at points $(x, y, z)$. The total gravitational potential $V$ for $W$ is then obtained by "summing" the potentials from the infinitesimal masses. Thus, we arrive at the triple integral (see Figure 6.3.4):

$$V(x_1, y_1, z_1) = -Gm \iiint_W \frac{\delta(x, y, z)\,dx\,dy\,dz}{\sqrt{(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2}}. \tag{9}$$

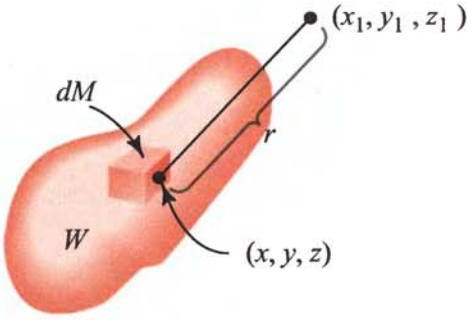



**Figure 6.3.4** The gravitational potential that produces a force acting on a mass $m$ at $(x_1, y_1, z_1)$ arising from the mass $dM = \delta(x, y, z)\,dx\,dy\,dz$ at $(x, y, z)$ is $-[Gm\delta(x, y, z)\,dx\,dy\,dz]/r$.

## Historical Note

The theory of gravitational force fields and gravitational potentials was developed by Sir Isaac Newton (1642–1727). Newton withheld publication of his gravitational theories for quite some time. The result that a spherical planet has the same gravitational field as it would have if its mass were all concentrated at the planet's center first appeared in his famous *Philosophiae Naturalis Principia Mathematica*, the first edition of which appeared in 1687. Using multiple integrals and spherical coordinates, we shall solve Newton's problem here; remarkably, Newton's published solution used only Euclidean geometry.

**EXAMPLE 7** Let $W$ be a region of constant density and total mass $M$. Show that the gravitational potential is given by

$$V(x_1, y_1, z_1) = \left[\frac{1}{r}\right]_{\text{av}} GMm,$$

where $[1/r]_{\text{av}}$ is the average over $W$ of

$$f(x, y, z) = \frac{1}{\sqrt{(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2}}.$$

SOLUTION According to formula (9),

$$-V(x_1, y_1, z_1) = Gm \iiint_W \frac{\delta \, dx\, dy\, dz}{\sqrt{(x-x_1)^2 + (y-y_1)^2 + (z-z_1)^2}}$$

$$= Gm\delta \iiint_W \frac{dx\, dy\, dz}{\sqrt{(x-x_1)^2 + (y-y_1)^2 + (z-z_1)^2}}$$

$$= Gm[\delta \text{ volume } (W)] \frac{\displaystyle\iiint_W \frac{dx\, dy\, dz}{\sqrt{(x-x_1)^2 + (y-y_1)^2 + (z-z_1)^2}}}{\text{volume } (W)}$$

$$= GmM \left[\frac{1}{r}\right]_{\text{av}}$$

are required. ▲

Let us now use formula (9) and spherical coordinates to find the gravitational potential $V(x_1, y_1, z_1)$ for a region $W$ with constant density between the concentric spheres $\rho = \rho_1$ and $\rho = \rho_2$, assuming the density is constant. Before evaluating the integral in formula (9), we make some observations that will simplify the computation. Because $G$, $m$, and the density are constants, we may ignore them at first. Because the attracting body, $W$, is symmetric with respect to rotations about the origin, the potential $V(x_1, y_1, z_1)$ should itself be symmetric—thus, $V(x_1, y_1, z_1)$ depends only on the distance $R = \sqrt{x_1^2 + y_1^2 + z_1^2}$ from the origin. Our computation will be simplest if we look at the point $(0, 0, R)$ on the $z$ axis (see Figure 6.3.5). Thus, we need to evaluate the integral

$$V(0, 0, R) = -\iiint_W \frac{dx\, dy\, dz}{\sqrt{x^2 + y^2 + (z-R)^2}}.$$

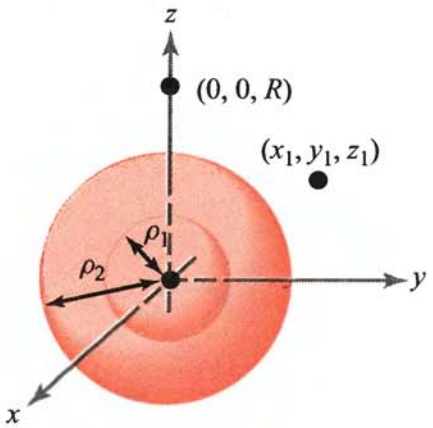



**Figure 6.3.5** The gravitational potential at $(x_1, y_1, z_1)$ is the same as at $(0, 0, R)$, where $R = \sqrt{x_1^2 + y_1^2 + z_1^2}$.

In spherical coordinates, $W$ is described by the inequalities $\rho_1 \le \rho \le \rho_2$, $0 \le \theta \le 2\pi$, and $0 \le \phi \le \pi$, and so

$$-V(0,0,R) = \int_{\rho_1}^{\rho_2} \int_0^{\pi} \int_0^{2\pi} \frac{\rho^2 \sin\phi \, d\theta \, d\phi \, d\rho}{\sqrt{\rho^2 \sin^2\phi(\cos^2\theta + \sin^2\theta) + (\rho\cos\phi - R)^2}}.$$

Replacing $\cos^2\theta + \sin^2\theta$ by 1, so that the integrand no longer involves $\theta$, we may integrate over $\theta$ to get

$$\begin{aligned} -V(0,0,R) &= 2\pi \int_{\rho_1}^{\rho_2} \int_0^{\pi} \frac{\rho^2 \sin\phi \, d\phi \, d\rho}{\sqrt{\rho^2 \sin^2\phi + (\rho\cos\phi - R)^2}} \\ &= 2\pi \int_{\rho_1}^{\rho_2} \rho^2 \left( \int_0^{\pi} \frac{\sin\phi \, d\phi}{\sqrt{\rho^2 - 2R\rho\cos\phi + R^2}} \right) d\rho. \end{aligned}$$

The inner integral over $\phi$ may be evaluated using the substitution $u = -2R\rho\cos\phi$. We get

$$\begin{aligned} \frac{1}{2R\rho} \int_{-2R\rho}^{2R\rho} (\rho^2 + u + R^2)^{-1/2} \, du &= \frac{2}{2R\rho} (\rho^2 + u + R^2)^{1/2} \Big|_{-2R\rho}^{2R\rho} \\ &= \frac{1}{R\rho} [(\rho^2 + 2R\rho + R^2)^{1/2} - (\rho^2 - 2R\rho + R^2)^{1/2} \\ &= \frac{1}{R\rho} \{[(\rho + R)^2]^{1/2} - [(\rho - R)^2]^{1/2}\} \\ &= \frac{1}{R\rho} (\rho + R - |\rho - R|). \end{aligned}$$

The expression $\rho + R$ is always positive, but $\rho - R$ may not be, so we must keep the absolute value sign. Substituting into the formula for $V$, we get

$$-V(0,0,R) = 2\pi \int_{\rho_1}^{\rho_2} \frac{\rho^2}{R\rho} (\rho + R - |\rho - R|) \, d\rho = \frac{2\pi}{R} \int_{\rho_1}^{\rho_2} \rho(\rho + R - |\rho - R|) \, d\rho.$$

We consider two possibilities for $R$, corresponding to the gravitational potential for objects *outside* and *inside* the hollow ball $W$.

**Case 1.** If $R \ge \rho_2$ [that is, if $(x_1, y_1, z_1)$ is outside $W$], then $|\rho - R| = R - \rho$ for all $\rho$ in the interval $[\rho_1, \rho_2]$, so that

$$-V(0,0,R) = \frac{2\pi}{R} \int_{\rho_1}^{\rho_2} \rho[\rho + R - (R - \rho)] \, d\rho = \frac{4\pi}{R} \int_{\rho_1}^{\rho_2} \rho^2 \, d\rho = \frac{1}{R} \frac{4\pi}{3} (\rho_2^3 - \rho_1^3).$$

The factor $(4\pi/3)(\rho_2^3 - \rho_1^3)$ equals the volume of $W$. Putting back the constants $G$, $m$, and the mass density, we find that *the gravitational potential* is $-GmM/R$, *where* $M$ *is the mass of* $W$. *Thus,* $V$ *is just as it would be if all the mass of* $W$ *were concentrated at the central point.*

**Case 2.** If $R \le \rho_1$ [that is, if $(x_1, y_1, z_1)$ is inside the hole], then $|\rho - R| = \rho - R$ for $\rho$ in $[\rho_1, \rho_2]$, and so

$$-V(0, 0, R) = (Gm)\frac{2\pi}{R}\int_{\rho_1}^{\rho_2} \rho[\rho + R - (\rho - R)]\,d\rho = (Gm)4\pi\int_{\rho_1}^{\rho_2} \rho\,d\rho$$
$$= (Gm)2\pi(\rho_2^2 - \rho_1^2).$$

The result is independent of $R$, and so the potential $V$ is *constant* inside the hole. Because the gravitational force is minus the gradient of $V$, we conclude that *there is no gravitational force inside a uniform hollow planet*!

We leave it to the reader to compute $V(0, 0, R)$ for the case $\rho_1 < R < \rho_2$.

A similar argument shows that the gravitational potential outside any *spherically symmetric* body of mass $M$ (even if the density is variable) is $V = GMm/R$, where $R$ is the distance to its center (which is its center of mass).

**EXAMPLE 8** Find the gravitational potential acting on a unit mass of a spherical star with a mass $M = 3.02 \times 10^{30}$ kg at a distance of $2.25 \times 10^{11}$ m from its center ($G = 6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2$).

SOLUTION The negative potential is

$$-V = \frac{GM}{R} = \frac{6.67 \times 10^{-11} \times 3.02 \times 10^{30}}{2.25 \times 10^{11}} = 8.95 \times 10^8\ \text{m}^2/\text{s}^2. \quad \blacktriangle$$

## EXERCISES

1. Find the average of $f(x, y) = y \sin xy$ over $D = [0, \pi] \times [0, \pi]$.

2. Find the average of $f(x, y) = e^{x+y}$ over the triangle with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$.

3. Find the center of mass of the region between $y = x^2$ and $y = x$ if the density is $x + y$.

4. Find the center of mass of the region between $y = 0$ and $y = x^2$, where $0 \le x \le \frac{1}{2}$.

5. A sculptured gold plate $D$ is defined by $0 \le x \le 2\pi$ and $0 \le y \le \pi$ (centimeters) and has mass density $\delta(x, y) = y^2 \sin^2 4x + 2$ (grams per square centimeter). If gold sells for \$7 per gram, how much is the gold in the plate worth?

6. In Exercise 5, what is the average mass density in grams per square centimeter?

7. (a) Find the mass of the box $[0, \frac{1}{2}] \times [0, 1] \times [0, 2]$, assuming the density to be uniform.
(b) Same as part (a), but with a mass density $\delta(x, y, z) = x^2 + 3y^2 + z + 1$.

**8.** Find the mass of the solid bounded by the cylinder $x^2 + y^2 = 2x$ and the cone $z^2 = x^2 + y^2$ if the density is $\delta = \sqrt{x^2 + y^2}$.

**9.** Find the center of mass of the region bounded by $x + y + z = 2$, $x = 0$, $y = 0$, and $z = 0$, assuming the density to be uniform.

**10.** Find the center of mass of the cylinder $x^2 + y^2 \leq 1$, $1 \leq z \leq 2$ if the density is $\delta = (x^2 + y^2)z^2$.

11. Find the average value of $\sin^2 \pi z \cos^2 \pi x$ over the cube $[0, 2] \times [0, 4] \times [0, 6]$.

**12.** Find the average value of $e^{-z}$ over the ball $x^2 + y^2 + z^2 \leq 1$.

**13.** A solid with constant density is bounded above by the plane $z = a$ and below by the cone described in spherical coordinates by $\phi = k$, where $k$ is a constant $0 < k < \pi/2$. Set up an integral for its moment of inertia about the $z$ axis.

14. Find the moment of inertia around the $y$ axis for the ball $x^2 + y^2 + z^2 \leq R^2$ if the mass density is a constant $\delta$.

**15.** Find the gravitational potential on a mass $m$ of a spherical planet with mass $M = 3 \times 10^{26}$ kg, at a distance of $2 \times 10^8$ m from its center.

**16.** Find the gravitational force exerted on a 70-kg object at the position in Exercise 15.

**17.** A body $W$ in $xyz$ coordinates is called *symmetric with respect to a given plane* if for every particle on one side of the plane there is a particle of equal mass located at its mirror image through the plane.

(a) Discuss the planes of symmetry for an automobile shell.

(b) Let the plane of symmetry be the $xy$ plane, and denote by $W^+$ and $W^-$ the portions of $W$ above and below the plane, respectively. By our assumption, the mass density $\delta(x, y, z)$ satisfies $\delta(x, y, -z) = \delta(x, y, z)$. Justify the following steps:

$$\begin{aligned}
\iiint_W \delta(x, y, z)\, dx\, dy\, dz &= \iiint_W z\delta(x, y, z)\, dx\, dy\, dz \\
&= \iiint_{W+} z\delta(x, y, z)\, dx\, dy\, dz + \iiint_{W-} z\delta(x, y, z)\, dx\, dy\, dz \\
&= \iiint_{W+} z\delta(x, y, z)\, dx\, dy\, dz + \iiint_{W+} -w\delta(u, v, -w)\, du\, dv\, dw \\
&= 0.
\end{aligned}$$

(c) Explain why part (b) proves that if a body is symmetrical with respect to a plane, then its center of mass lies in that plane.

(d) Derive this law of mechanics: *If a body is symmetric with respect to two planes, then its center of mass lies on their line of intersection.*

**18.** A uniform rectangular steel plate of sides $a$ and $b$ rotates about its center of mass with constant angular velocity $\omega$.

(a) The kinetic energy equals $\frac{1}{2}$(mass)(velocity)$^2$. Argue that the kinetic energy of any element of mass $\delta\,dx\,dy$ ($\delta$ = constant) is given by $\delta(\omega^2/2)(x^2+y^2)\,dx\,dy$, provided the origin (0, 0) is placed at the center of mass of the plate.

(b) Justify the formula for kinetic energy:

$$\text{K.E.} = \iint_{\text{plate}} \delta\frac{\omega^2}{2}(x^2+y^2)\,dx\,dy.$$

(c) Evaluate the integral, assuming that the plate is described by the inequalities $-a/2 \le x \le a/2$, $-b/2 \le y \le b/2$.

**19.** As is well known, the density of a typical planet is not constant throughout the planet. Assume that planet C.M.W. has a radius of $5 \times 10^8$ cm and a mass density (in grams per cubic centimeter)

$$\rho(x,y,z) = \begin{cases} \dfrac{3\times 10^4}{r}, & r \ge 10^4 \text{ cm}, \\ 3, & r \le 10^4 \text{ cm}, \end{cases}$$

where $r = \sqrt{x^2+y^2+z^2}$. Find a formula for the gravitational potential outside C.M.W.

## 6.4 Improper Integrals

In this section, we study improper integrals—that is, integrals in which the function may be unbounded or the region of integration is unbounded. We shall first recall the situation for functions of one variable.

### One-Variable Improper Integrals

In the study of integrals of functions of one variable, one encounters various types of "improper" integrals; that is, integrals of unbounded functions defined on intervals or integrals of functions over unbounded intervals. For example,

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx \qquad \text{and} \qquad \int_1^\infty \frac{dx}{x^2}$$

are improper integrals. They are evaluated using a limiting process; for instance,

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{a\to 0}\int_a^1 \frac{1}{\sqrt{x}}\,dx = \lim_{a\to 0}\left(2\sqrt{x}\Big|_a^1\right) = \lim_{a\to 0}(2-2\sqrt{a}) = 2$$

and

$$\int_1^\infty \frac{dx}{x^2} = \lim_{b\to\infty} \int_1^b \frac{dx}{x^2} = \lim_{b\to\infty} \left( -\frac{1}{x}\Big|_1^b \right) = \lim_{b\to\infty} \left(1 - \frac{1}{b}\right) = 1.$$

If, in such a limiting process, the limit does not exist (or is infinite), we say that the integral does not exist (or that the integral diverges).

## Improper Integrals in the Plane

Next, we describe three types of improper integrals of two variables over a region $D$. The first two types are described in the text below, and the third type (integrals over unbounded regions) is left to the exercises. We will evaluate all integrals using a limiting process, as in the one-variable case.

For simplicity of exposition, we first restrict ourselves to nonnegative functions $f$—that is, $f(x, y) \geq 0$ for all points $(x, y) \in D$—and to $y$-simple regions described as the set of $(x, y)$ such that

$$a \leq x \leq b, \qquad \phi_1(x) \leq y \leq \phi_2(x),$$

as in Figure 6.4.1.

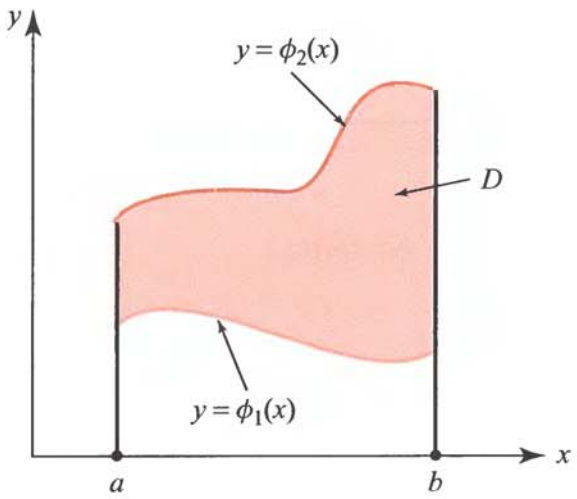



Figure 6.4.1 A $y$-simple domain.

In the first case we wish to consider, let's assume that $f: D \to \mathbb{R}$ is continuous except for points on the boundary of $D$. Consider, for example,

$$f(x, y) = \frac{1}{\sqrt{1 - x^2 - y^2}},$$

where $D$ is the unit disk $D = \{(x, y) | x^2 + y^2 \leq 1\}$. Clearly, $f$ is not defined on the boundary of $D$, where $x^2 + y^2 = 1$; yet it will be of practical interest to be able to evaluate $\iint_D f(x, y)\, dA$, because this integral represents the area of the upper hemisphere of the unit sphere in three space.

## Exhausting Regions

Our basic idea will be to integrate such an $f$ over a smaller region $D'$, where we know the integral exists, and then let $D'$ "tend" to $D$; that is, "exhaust" $D$ and see if $\iint_D f\,dA$ tends to some limit. With this in mind, we pick a special kind of $D'$, as follows.

Let $\eta > 0$ be small enough so that $a + \eta < b - \eta$. Let $\delta > 0$ be small enough so that $\phi_1(x) + \delta < \phi_2(x) - \delta$ for all $x$, $a \le x \le b$ (see Figure 6.4.2). If $\phi_2(x) = \phi_1(x)$ for some $x$, no such $\delta$ will exist, but we shall worry about this minor issue when it arises in our later examples. Then the region

$$D_{\eta,\delta} = \{(x, y) | a + \eta \le x \le b - \eta \quad \text{and} \quad \phi_1(x) + \delta \le y \le \phi_2(x) - \delta\}$$

is a subset of $D$, and as $(\eta, \delta) \to 0$, $D_{\eta,\delta}$ tends to $D$.

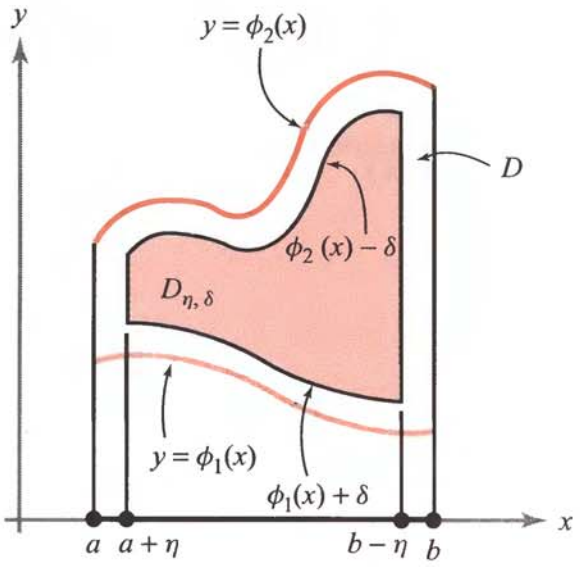



**Figure 6.4.2** A shrunken domain $D_{\eta,\delta}$ for improper integrals.

## Improper Integrals as Limits

Because $f$ is continuous and bounded on $D_{\eta,\delta}$, the integral $\iint D_{\eta,\delta} f\,dA$ exists. We can now ask what happens as the region $D_{\eta,\delta}$ expands to fill the region, $D$—that is, as $(\eta, \delta) \to (0, 0)$. Provided that

$$\lim_{(\eta,\delta)\to(0,0)} \iint_{D_{\eta,\delta}} f\,dA$$

exists, we say that the integral of $f$ over $D$ is ***convergent*** or that $f$ is ***integrable*** over $D$, and we define $\iint_D f\,dx\,dy$ to be equal to this limit.

**EXAMPLE 1** Evaluate

$$\iint_D \frac{1}{\sqrt[3]{xy}}\,dA,$$

where $D$ is the unit square $[0, 1] \times [0, 1]$.

SOLUTION $D$ is clearly a $y$-simple region. Choose $\eta > 0$ and $\delta > 0$ so that $D_{\eta,\delta} \subset D$, as in Figure 6.4.3. Then, by Fubini's theorem:

$$\begin{aligned}\iint_{D_{\eta,\delta}} \frac{1}{\sqrt[3]{xy}}\, dA &= \int_{\eta}^{1-\eta} \int_{\delta}^{1-\delta} \frac{1}{\sqrt[3]{xy}}\, dy\, dx \\ &= \int_{\eta}^{1-\eta} \frac{1}{\sqrt[3]{x}}\, dx \int_{\delta}^{1-\delta} \frac{1}{\sqrt[3]{y}}\, dy \\ &= \frac{3}{2}\left((1-\eta)^{2/3} - \eta^{2/3}\right) \cdot \frac{3}{2}\left((1-\delta)^{2/3} - \delta^{2/3}\right).\end{aligned}$$

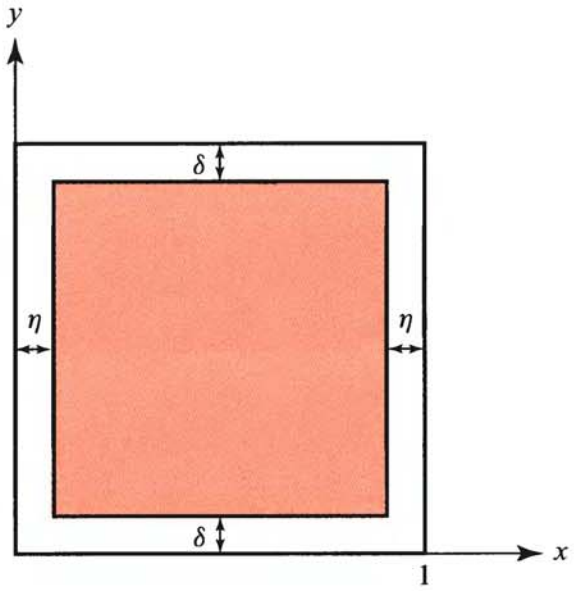



**Figure 6.4.3** The slightly shrunken unit square.

Letting $(\eta, \delta) \to (0, 0)$, we see that

$$\lim_{(\eta,\delta)\to(0,0)} \iint_{D_{\eta,\delta}} \frac{1}{\sqrt[3]{xy}}\, dy\, dx = \frac{3}{2}\frac{3}{2} = \frac{9}{4}. \quad \blacktriangle$$

Unfortunately, it may not always be possible to evaluate such limits so directly and simply. This is often the case in the most interesting examples, as with the surface area of the hemisphere, mentioned earlier. It's as if the "real world" always presents the greatest challenges to the mathematician! So let us expand a bit on our theoretical discussion.

## Improper Integrals as Limits of Iterated Integrals

Suppose $f$ is integrable over $D_{\eta,\delta}$. We can then apply Fubini's theorem to obtain

$$\iint_{D_{\eta,\delta}} f\, dA = \int_{a+\eta}^{b-\eta} \int_{\phi_1(x)+\delta}^{\phi_2(x)-\delta} f(x, y)\, dy\, dx.$$

Hence, if $f$ is integrable over $D$,

$$\iint_D f\,dA = \lim_{(\eta,\delta)\to(0,0)} \int_{a+\eta}^{b-\eta} \int_{\phi_1(x)+\delta}^{\phi_2(x)-\delta} f(x,y)\,dy\,dx. \tag{1}$$

Now $F(\eta,\delta) = \iint_{D_{\eta,\delta}} f\,dA$ is a function of two variables, $\eta$ and $\delta$, because as we change $\eta$ and $\delta$, we get another number. Now if $f$ is integrable, then

$$\lim_{(\eta,\delta)\to 0} F(\eta,\delta) = L$$

exists. It follows that the iterated limits

$$\lim_{\eta\to 0}\lim_{\delta\to 0} F(\eta,\delta) \qquad \text{and} \qquad \lim_{\delta\to 0}\lim_{\eta\to 0} F(\eta,\delta)$$

also exist and are both equal to $L$, which in our case is $\iint_D f\,dA$. Thus, the iterated limit

$$\lim_{\eta\to 0}\lim_{\delta\to 0} \int_{a+\eta}^{b-\eta} \int_{\phi_1(x)+\delta}^{\phi_1(x)-\delta} f(x,y)\,dy\,dx$$

also exists. Conversely, if the iterated limits exist, it does not generally follow that the limit $\lim_{(\eta,\delta)\to(0,0)} F(\eta,\delta)$ exists.

For example, if it were to turn out in some way that $F(\eta,\delta) = \eta\delta/(\eta^2+\delta^2)$, then $\lim_{\eta\to 0}\lim_{\delta\to 0} F(\eta,\delta) = \lim_{\delta\to 0}\lim_{\eta\to 0} F(\eta,\delta) = 0$; yet $\lim_{(\eta,\delta)\to 0} F(\eta,\delta)$ does not exist, because $F(\eta,\eta) = 1/2$ (see Section 2.2).

In view of this, consider expression (1) again. If $f$ is integrable, then

$$\begin{aligned}\iint_D f(x,y)\,dA &= \lim_{(\eta,\delta)\to(0,0)} \int_{a+\eta}^{b-\eta} \int_{\phi_1(x)+\delta}^{\phi_2(x)-\delta} f(x,y)\,dy\,dx \\ &= \lim_{\eta\to(0)}\lim_{\delta\to 0} \int_{a+\eta}^{b-\eta} \int_{\phi_1(x)+\delta}^{\phi_2(x)-\delta} f(x,y)\,dy\,dx.\end{aligned}$$

Now suppose that for each $x$,

$$\lim_{\delta\to 0} \int_{\phi_1(x)+\delta}^{\phi_2(x)-\delta} f(x,y)\,dy$$

exists. Denote this by $\int_{\phi_1(x)}^{\phi_2(x)} f(x,y)\,dy$. Suppose further that

$$\lim_{\eta\to 0} \int_{a+\eta}^{b-\eta} \int_{\phi_1(x)}^{\phi_2(x)} f(x,y)\,dy$$

also exists. We denote this limit by $\int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x,y)\,dy\,dx$. Then if all limits exist, *all limits must be equal*. Thus, $f$ is integrable and the iterated improper integral exists,

then necessarily

$$\iint_D f(x, y)\, dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy\, dx.$$

However, is it possible that the existence of *just* the iterated integrals *implies* the integrability of $f$? We turn to this important question next.

## Fubini's Theorem for Improper Integrals

For *integrals*, something truly *remarkable* happens. Unlike the case for iterated limits (as in the counterexample considered earlier), the existence of the iterated limits *does imply* the integrability of $f$ as long as $f \geq 0$. Thus, if $f \geq 0$ and if $\int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy\, dx$ exists as an iterated limit, then $f$ is integrable and

$$\iint_D f(x, y)\, dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy\, dx.$$

If $D$ is an $x$-simple region with the $x$ coordinate lying between two functions $\psi_1$ and $\psi_2$, and if

$$\int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx\, dy$$

exists as an improper integral, it again follows that $f$ is integrable and

$$\iint_D f(x, y)\, dA = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx\, dy.$$

All these results, which are the *improper* analogues of Theorems 4 and 4′ in Section 5.3, are known as *Fubini's theorem* for improper integrals, which we formally state.

**THEOREM 3: Fubini's Theorem** Let $D$ be an elementary region in the plane and $f \geq 0$ a function continuous except for points possibly on the boundary of $D$. If either of the integrals

$$\iint_D f(x, y)\, dA,$$

$$\int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\, dy\, dx, \qquad \text{for } y\text{-simple regions}$$

$$\int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx\, dy \qquad \text{for } x\text{-simple regions}$$

exist as improper integrals, $f$ is integrable and they are all equal.

The proof of this involves advanced concepts of analysis, so we omit it here. This result can be quite useful in calculation, as the next example shows.

**EXAMPLE 2** Let $f(x, y) = 1/\sqrt{1 - x^2 - y^2}$. Show that $f$ is integrable and that $\iint_D f(x, y)\, dA = 2\pi$, half the surface area of the unit sphere.

**SOLUTION** For $-1 < x < 1$, we have

$$\begin{aligned}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{dy}{\sqrt{1-x^2-y^2}} &= \lim_{\delta\to 0} \int_{-\sqrt{1-x^2}+\delta}^{\sqrt{1+x^2}-\delta} \frac{dy}{\sqrt{1-x^2-y^2}} \\ &= \lim_{\delta\to 0} \sin^{-1}\left(\frac{y}{\sqrt{1-x^2}}\right)\Bigg|_{-\sqrt{1-x^2}+\delta}^{\sqrt{1-x^2}-\delta} \\ &= \lim_{\delta\to 0} \left\{\sin^{-1}\left(1 - \frac{\delta}{\sqrt{1-x^2}}\right) - \sin^{-1}\left(-1 + \frac{\delta}{\sqrt{1-x^2}}\right)\right\} \\ &= \sin^{-1}(1) - \sin^{-1}(-1) = \frac{\pi}{2} - \frac{(-\pi)}{2} = \pi.\end{aligned}$$

Clearly,

$$\lim_{\eta\to 0} \int_{-1+\eta}^{1-\eta} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{dy\, dx}{\sqrt{1-x^2-y^2}} = \lim_{\eta\to 0} \int_{-1+\eta}^{1-\eta} \pi\, dx = \lim_{\eta\to 0} \pi(2 - 2\eta) = 2\pi.$$

Thus, $f$ is integrable. To see why this theorem is so useful, try to show directly from the definition that $f$ is integrable. It is not easy to do so! ▲

**EXAMPLE 3** Let $f(x, y) = 1/(x - y)$ and let $D$ be the set of $(x, y)$ satisfying $0 \le x \le 1$ and $0 \le y \le x$. Show that $f$ is *not* integrable over $D$.

**SOLUTION** Because the denominator of $f$ is zero on the line $y = x$, $f$ is unbounded on part of the boundary of $D$. Let $0 < \eta < 1$ and $0 < \delta < \eta$, and let $D_{\eta,\delta}$ be the set of $(x, y)$ with $\eta \le x \le 1 - \eta$ and $\delta \le y \le x - \delta$ (Figure 6.4.4).

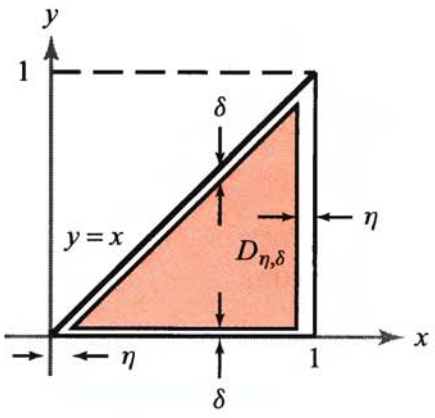



**Figure 6.4.4** The shrunken domain $D_{\eta,\delta}$ for a triangular domain $D$.

Here the region $D$ is $y$-simple with $\phi_1(x) = 0$, $\phi_2(x) = x$, and $\phi_1(0) = \phi_2(0)$. To ensure that $D_{\eta,\delta} \subset D$ and is depicted in the figure, we must choose $\delta$ a bit more

carefully. A little geometry shows that we should choose $2\delta \leq \eta$. Consider

$$\begin{aligned}\iint_{D_{\eta,\delta}} f\,dA &= \int_{\eta}^{1-\eta}\int_{\delta}^{x-\delta} \frac{1}{x-y}\,dy\,dx \\ &= \int_{\eta}^{1-\eta} [-\log(x-y)]|_{y=\delta}^{x-\delta}\,dx \\ &= \int_{\eta}^{1-\eta} [-\log(\delta)+\log(x-\delta)]\,dx \\ &= [-\log\delta]\int_{\eta}^{1-\eta} dx + \int_{\eta}^{1-\eta} \log(x-\delta)\,dx \\ &= -(1-2\eta)\log\delta + [(x-\delta)\log(x-\delta)-(x-\delta)]|_{\eta}^{1-\eta}.\end{aligned}$$

In the last step, we used the fact that $\int \log u\,du = u\log u - u$. Continuing the preceding set of qualities, we have

$$\begin{aligned}\iint_{D_{\eta,\delta}} f\,dA = &-(1-2\eta)\log\delta + (1-\eta-\delta)\log(1-\eta-\delta) \\ &-1(1-\eta-\delta) - (\eta-\delta)\log(\eta-\delta) + (\eta-\delta).\end{aligned}$$

As $(\eta, \delta) \to (0, 0)$, the second term converges to $1 \log 1 = 0$, and the third and fifth terms converge to $-1$ and 0, respectively. Let $v = \eta - \delta$. Because $v \log v \to 0$ as $v \to 0$ (a limit established by using L'Hôpital's rule from calculus[3]), we see that the fourth term goes to zero as $(\eta, \delta) \to (0, 0)$. It is the first term that will give us trouble. Now:

$$-(1-2\eta)\log\delta = -\log\delta + 2\eta\log\delta, \tag{2}$$

and it is not hard to see that this does not converge as $(\eta, \delta) \to (0, 0)$. For example, let $\eta = 2\delta$; then expression (2) becomes $-\log\delta + 4\delta\log\delta$. As before, $4\delta\log\delta \to 0$ as $\delta \to 0$, but $-\log\delta \to +\infty$ as $\delta \to 0$, which shows that expression (2) does not converge. Hence, $\lim_{(\eta,\delta)\to(0,0)} \iint_{D_{\eta,\delta}} f\,dA$ does not exist and so $f$ is not integrable. ▲

## Functions Unbounded at Isolated Points

We now consider nonnegative functions $f$ that become "infinite" or are undefined at isolated points in an $x$-simple or $y$-simple region $D$. For example, consider the function $f(x, y) = 1/\sqrt{x^2+y^2}$ on the unit disk $D = \{(x, y)|x^2 + y^2 \leq 1\}$. Again, $f \geq 0$, but $f$ is unbounded and is not defined at the origin.

[3] L' Hôpital's rule was discovered by Bernoulli and was reported in L'Hôpital's textbook.

Let $(x_0, y_0)$ be a point of a general region $D$ where a nonnegative function $f$ is undefined. Further, let $D_\delta = D_\delta(x_0, y_0)$ be the disk of radius $\delta$ centered at $(x_0, y_0)$ and let $D \backslash D_\delta$ denote the region $D$ with $D_\delta$ removed. Assume that $f$ is continuous at every point of $D$ except $(x_0, y_0)$. Then $\iint_{D\backslash D_\delta} f\, dA$ is defined. We say that $\iint_D f\, dA$ is ***convergent***, or that $f$ is ***integrable*** over $D$ if

$$\lim_{\delta \to 0} \iint_{D\backslash D_\delta} f\, dA$$

exists.

**EXAMPLE 4** Show that $f(x, y) = 1/\sqrt{x^2 + y^2}$ is integrable over the unit disk $D$ and evaluate $\iint_D f\, dA$.

SOLUTION Let $D_\delta$ be the disk of radius $\delta$ centered at the origin. Then $f$ is continuous everywhere on $D$ except at $(0, 0)$. Thus, $\iint_{D\backslash D_\delta} f\, dA$ exists. To evaluate this integral, we change variables to polar coordinates, $x = r\cos\theta$, $y = r\sin\theta$. Then $f(r\cos\theta, r\sin\theta) = 1/r$, and

$$\iint_{D\backslash D_\delta} f\, dA = \int_\delta^1 \int_0^{2\pi} \frac{1}{r} f\, d\theta\, dr = \int_\delta^1 \int_0^{2\pi} d\theta\, dr = 2\pi(1 - \delta).$$

Thus,

$$\iint_D f\, dA = \lim_{\delta \to 0} \iint_{D\backslash D_\delta} f\, dA = 2\pi. \quad \blacktriangle$$

More generally, one can, in an analogous manner, define the integral of nonnegative functions $f$ that are continuous except at a finite number of points in $D$. One can also combine both types of improper integrals; that is, one may consider functions that are continuous except at a finite number of points on $D$ or at points on the boundary of $D$, and define $\iint_D f\, dA$ appropriately.

If $f$ takes both positive and negative values, one can use a more advanced integration theory, called the *Lebesgue integral*, to generalize the notion of convergent integral $\iint_D f\, dA$. Using this theory, it is possible to show that if $\iint_D f\, dA$ exists, it can then be evaluated as an iterated integral. This latter fact is also known as Fubini's theorem.

## Unbounded Regions

As was mentioned previously, we will leave consideration of unbounded regions to the exercise section. However, we must point out that we have already addressed the main idea in Example 5 of Section 6.2 on the Gaussian integral. In that example, we integrated $\exp(-x^2 - y^2)$ over all of $\mathbb{R}^2$ by integrating first over a disk of radius $a$ and then letting $a \to \infty$.

## EXERCISES

*In Exercises 1 to 4, evaluate the following integrals if they exist (discuss how you define the integral if it was not given in the text).*

1. $\iint_D \frac{1}{\sqrt{xy}}\,dA$, where $D = [0, 1] \times [0, 1]$

2. $\iint_D \frac{1}{\sqrt{|x - y|}}\,dx\,dy$, where $D = \{(x, y) \mid 0 \le x \le 1, 0 \le y \le 1, y \le x\}$

3. $\iint_D (y/x)\,dx\,dy$, where $D$ is bounded by $x = 1, x = y$, and $x = 2y$

4. $\int_0^1 \int_0^{e^y} \log x\,dx\,dy$

5. (a) Evaluate

$$\iint_D \frac{dA}{(x^2 + y^2)^{2/3}},$$

where $D$ is the unit disk in $\mathbb{R}^2$

(b) Determine the real numbers $\lambda$ for which the integral

$$\iint_D \frac{dA}{(x^2 + y^2)^{\lambda}}$$

is convergent, where again $D$ is the unit disk.

6. (a) Discuss how you would define $\iint_D f\,dA$ if $D$ is an unbounded region, for example, the set of $(x, y)$ such that $a \le x < \infty$ and $\phi_1(x) \le y \le \phi_2(x)$, where $\phi_1 \le \phi_2$ are given (Figure 6.4.5).

(b) Evaluate $\iint_D xye^{-(x^2+y^2)}\,dx\,dy$ if $x \ge 0, 0 \le y \le 1$.

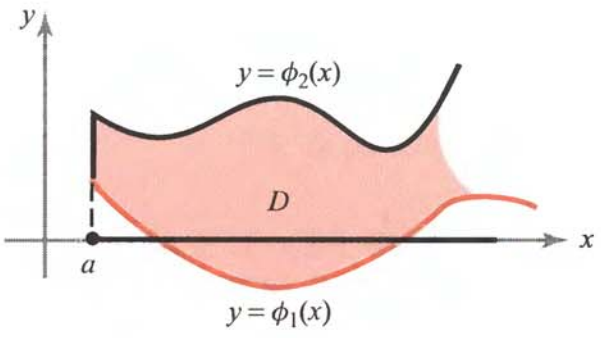



**Figure 6.4.5** An unbounded region $D$.

7. Using Exercise 6, integrate $e^{-xy}$ for $x \ge 0, 1 \le y \le 2$ in two ways. Assuming Fubini's theorem can be used, show that

$$\int_0^{\infty} \frac{e^{-x} - e^{-2x}}{x}\,dx = \log 2.$$

**8.** Show that the integral

$$\int_0^1 \int_0^a (x/\sqrt{a^2 - y^2})\, dy\, dx$$

exists, and compute its value.

**9.** Discuss whether the integral

$$\iint_D \frac{x + y}{x^2 + 2xy + y^2}\, dx\, dy$$

exists where $D = [0, 1] \times [0, 1]$. If it exists, compute its value.

**10.** One can also consider improper integrals of functions that fail to be continuous on entire curves lying in some region $D$. For example, by breaking $D = [0, 1] \times [0, 1]$ into two regions, define and then discuss the convergence of the integral

$$\iint_D \frac{1}{\sqrt{|x - y|}}\, dx\, dy.$$

**11.** Let $W$ be the first octant of the ball $x^2 + y^2 + z^2 \leq a^2$, where $x \geq 0, y \geq 0, z \geq 0$. Evaluate the improper integral

$$\iiint_W \frac{(x^2 + y^2 + z^2)^{1/4}}{\sqrt{z + (x^2 + y^2 + z^2)^2}}\, dx\, dy\, dz$$

by changing variables.

**12.** Let $f$ be a nonnegative function that may be unbounded and discontinuous on the boundary of an elementary region $D$. Let $g$ be a similar function such that $f(x, y) \leq g(x, y)$ whenever both are defined. Suppose $\iint_D g(x, y)\, dA$ exists. Argue informally that this implies the existence of $\iint_D f(x, y)\, dA$.

**13.** Use Exercise 12 to show that

$$\iint_D \frac{\sin^2(x - y)}{\sqrt{1 - x^2 - y^2}}\, dy\, dx$$

exists where $D$ is the unit disk $x^2 + y^2 \leq 1$.

**14.** Let $f$ be as in Exercise 12 and let $g$ be a function such that $0 \leq g(x, y) \leq f(x, y)$ whenever both are defined. Suppose that $\iint_D g(x, y)\, dA$ does not exist. Argue informally that $\iint_D f(x, y)\, dA$ cannot exist.

**15.** Use Exercise 14 to show that

$$\iint_D \frac{e^{x^2+y^2}}{x - y}\, dy\, dx$$

does not exist, where $D$ is the set of $(x, y)$ with $0 \leq x \leq 1$ and $0 \leq y \leq x$.

**16.** Let $D$ be the unbounded region defined as the set of $(x, y, z)$ with $x^2 + y^2 + z^2 \geq 1$. By making a change of variables, evaluate the improper integral

$$\iiint_D \frac{dx\,dy\,dz}{(x^2 + y^2 + z^2)^2}.$$

**17.** Evaluate

$$\int_0^1 \int_0^y \frac{x}{y}\,dx\,dy \qquad \text{and} \qquad \int_0^1 \int_x^1 \frac{x}{y}\,dy\,dx$$

Does Fubini's theorem apply?

**18.** In Exercise 11 of Section 5.2 we showed that

$$\int_0^1 \int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2}\,dy\,dx \neq \int_0^1 \int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2}\,dx\,dy.$$

Thus, Fubini's theorem does not hold here, even thought the iterated improper integrals both exist. What went wrong?

## REVIEW EXERCISES FOR CHAPTER 6

**1.** (a) Find a linear transformation taking the square $S = [0, 1] \times [0, 1]$ to the parallelogram $P$ with vertices $(0, 0)$, $(2, 0)$, $(1, 2)$, $(3, 2)$.

(b) Write down a change of variables formula appropriate to the transformation you found in part (a).

**2.** (a) Find the image of the square $[0, 1] \times [0, 1]$ under the transformation $T(x, y) = (2x, x + 3y)$.

(b) Write down a change of variables formula appropriate to the transformation and the region you found in part (a).

**3.** Let $B$ be the region in the first quadrant bounded by the curves $xy = 1$, $xy = 3$, $x^2 - y^2 = 1$, and $x^2 - y^2 = 4$. Evaluate $\iint_B (x^2 + y^2)\,dx\,dy$ using the change of variables $u = x^2 - y^2$, $v = xy$.

**4.** In parts (a) to (d), make the indicated change of variables. (Do not evaluate.)

(a) $\displaystyle\int_0^1 \int_{-1}^1 \int_{-\sqrt{(1-y^2)}}^{\sqrt{(1-y^2)}} (x^2 + y^2)^{1/2}\,dx\,dy\,dz$, cylindrical coordinates

(b) $\displaystyle\int_{-1}^1 \int_{-\sqrt{(1-y^2)}}^{\sqrt{(1-y^2)}} \int_{-\sqrt{(4-x^2-y^2)}}^{\sqrt{(4-x^2-y^2)}} xyz\,dz\,dx\,dy$, cylindrical coordinates

(c) $\displaystyle\int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{(2-y^2)}}^{\sqrt{(2-y^2)}} \int_{\sqrt{(x^2+y^2)}}^{\sqrt{(4-x^2-y^2)}} z^2\,dz\,dx\,dy$, spherical coordinates

(d) $\displaystyle\int_0^1 \int_0^{\pi/4} \int_0^{2\pi} \rho^3 \sin 2\phi\,d\theta\,d\phi\,d\rho$, rectangular coordinates

5. Find the volume inside the surfaces $x^2 + y^2 = z$ and $x^2 + y^2 + z^2 = 2$.

6. Find the volume enclosed by the cone $x^2 + y^2 = z^2$ and the plane $2z - y - 2 = 0$.

7. A cylindrical hole of diameter 1 is bored through a sphere of radius 2. Assuming that the axis of the cylinder passes through the center of the sphere, find the volume of the solid that remains.

8. Let $C_1$ and $C_2$ be two cylinders of infinite extent, of diameter 2, and with axes on the $x$ and $y$ axes, respectively. Find the volume of their intersection, $C_1 \cap C_2$.

9. Find the volume bounded by $x/a + y/b + z/c = 1$ and the coordinate planes.

10. Find the volume determined by $z \leq 6 - x^2 - y^2$ and $z \geq \sqrt{x^2 + y^2}$.

11. The *tetrahedron* defined by $x \geq 0$, $y \geq 0$, $z \geq 0$, $x + y + z \leq 1$ is to be sliced into $n$ segments of equal volume by planes parallel to the plane $x + y + z = 1$. Where should the slices be made?

12. Let $E$ be the solid ellipsoid $E = \{(x, y, z) \mid (x^2/a^2) + (y^2/b^2) + (z^2/c^2) \leq 1\}$ where $a > 0$, $b > 0$, and $c > 0$. Evaluate

$$\iiint xyz\, dx\, dy\, dz$$

(a) over the whole ellipsoid; and
(b) over that part of it in the first quadrant:

$$x \geq 0, \qquad y \geq 0, \qquad \text{and} \qquad z \geq 0, \qquad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1.$$

13. Find the volume of the "ice cream cone" defined by the inequalities $x^2 + y^2 \leq \frac{1}{5}z^2$, and $0 \leq z \leq 5 + \sqrt{5 - x^2 - y^2}$.

14. Let $\rho, \theta, \phi$ be spherical coordinates in $\mathbb{R}^3$ and suppose that a surface surrounding the origin is described by a continuous positive function $\rho = f(\theta, \phi)$. Show that the volume enclosed by the surface is

$$V = \frac{1}{3}\int_0^{2\pi}\int_0^{\pi} [f(\theta, \phi)]^3 \sin\phi\, d\phi\, d\theta.$$

15. Using an appropriate change of variables, evaluate

$$\iint_B \exp[(y - x)/(y + x)]\, dx\, dy$$

where $B$ is the interior of the triangle with vertices at $(0, 0)$, $(0, 1)$, and $(1, 0)$.

16. Suppose the density of a solid of radius $R$ is given by $(1 + d^3)^{-1}$ where $d$ is the distance to the center of the sphere. Find the total mass of the sphere.

**17.** The density of the material of a spherical shell whose inner radius is 1 m and whose outer radius is 2 m is $0.4d^2$ g/cm$^3$, where $d$ is the distance to the center of the sphere in meters. Find the total mass of the shell.

**18.** If the shell in Exercise 17 were dropped into a large tank of pure water, would it float? What if the shell leaked? (Assume that the density of water is exactly 1 g/cm$^3$.)

**19.** The temperature at points in the cube $C = \{(x, y, z) \mid -1 \le x \le 1, -1 \le y \le 1, \text{ and } -1 \le z \le 1\}$ is $32d^2$, where $d$ is the distance to the origin.

(a) What is the average temperature?
(b) At what points of the cube is the temperature equal to the average temperature?

**20.** Use cylindrical coordinates to find the center of mass of the region defined by

$$y^2 + z^2 \le \frac{1}{4}, \qquad (x-1)^2 + y^2 + z^2 \le 1, \qquad x \ge 1.$$

**21.** Find the center of mass of the solid hemisphere

$$V = \{(x, y, z) \mid x^2 + y^2 + z^2 \le a^2 \text{ and } z \ge 0\}$$

if the density is constant.

**22.** Evaluate $\iint_B e^{-x^2-y^2}\,dx\,dy$ where $B$ consists of those $(x, y)$ satisfying $x^2 + y^2 \le 1$ and $y \le 0$.

**23.** Evaluate

$$\iiint_S \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^{3/2}},$$

where $S$ is the solid bounded by the spheres $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + z^2 = b^2$, where $a > b > 0$.

**24.** Evaluate $\iiint_D (x^2 + y^2 + z^2)xyz\,dx\,dy\,dz$ over each of the following regions.

(a) The sphere $D = \{(x, y, z) \mid x^2 + y^2 + z^2 \le R^2\}$
(b) The hemisphere $D = \{(x, y, z) \mid x^2 + y^2 + z^2 \le R^2 \text{ and } z \ge 0\}$
(c) The octant $D = \{(x, y, z) \mid x \ge 0, y \ge 0, z \ge 0, \text{ and } z^2 + y^2 + z^2 \le R^2\}$

**25.** Let $C$ be the cone-shaped region $\{(x, y, z) \mid \sqrt{x^2 + y^2} \le z \le 1\}$ in $\mathbb{R}^3$ and evaluate the integral $\iiint_C (1 + \sqrt{x^2 + y^2})\,dx\,dy\,dz$.

**26.** Find $\iiint_{\mathbb{R}^3} f(x, y, z)\,dx\,dy\,dz$ where $f(x, y, z) = \exp\,[-(x^2 + y^2 + z^2)^{3/2}]$.

**27.** The *flexural rigidity* $EI$ of a uniform beam is the product of its Young's modulus of elasticity $E$ and the moment of inertia $I$ of the cross section of the beam with respect to a

horizontal line $l$ passing through the center of gravity of this cross section. Here

$$I = \iint_R [d(x, y)]^2 \, dx \, dy,$$

where $d(x, y) =$ the distance from $(x, y)$ to $l$ and $R =$ the cross section of the beam being considered.

(a) Assume that the cross section $R$ is the rectangle $-1 \leq x \leq 1, -1 \leq y \leq 2$, and $l$ is the line $y = 1/2$. Find $I$.

(b) Assume the cross section $R$ is a circle of radius 4 and $l$ is the $x$ axis. Find $I$, using polar coordinates.

**28.** Find, $\iiint_{\mathbb{R}^3} f(x, y, z) \, dx \, dy \, dz$ where

$$f(x, y, z) = \frac{1}{[1 + (x^2 + y^2 + z^2)^{3/2}]^{3/2}}.$$

**29.** Suppose $D$ is the unbounded region of $\mathbb{R}^2$ given by the set of $(x, y)$ with $0 \leq x < \infty$, $0 \leq y \leq x$. Let $f(x, y) = x^{-3/2} e^{y-x}$. Does the improper integral $\iint_D f(x, y) \, dx \, dy$ exist?

**30.** If the world were two-dimensional, the laws of physics would predict that the gravitational potential of a mass point is proportional to the logarithm of the distance from the point. Using polar coordinates, write an integral giving the gravitational potential of a disk of constant density.

**31.** (a) Evaluate the improper integral

$$\int_0^\infty \int_0^y x e^{-y^3} \, dx \, dy.$$

(b) Evaluate

$$\iint_B (x^4 + 2x^2y^2 + y^4) \, dx \, dy,$$

where $B$ is the portion of the disk of radius 2 [centered at (0, 0) in the first quadrant].

**32.** Let $f$ be a nonnegative function on an $x$-simple or a $y$-simple region $D \subset \mathbb{R}^2$, and that is continuous except for points on the boundary of $D$ and at most finitely many points interior to $D$. Give a suitable definition of $\iint_D f \, dA$.

**33.** Evaluate $\iint_{\mathbb{R}^2} f(x, y) \, dx \, dy$ where $f(x, y) = 1/(1 + x^2 + y^2)^{3/2}$. (HINT: You may assume that changing variables and Fubini's theorem are valid for improper integrals.)

# 7

# Integrals Over Paths and Surfaces

*I hold in fact: (1) That small portions of space are of a nature analogous to little hills on a surface which is on the average flat. (2) That this property of being curved or distorted is continually passed on from one portion of space to another after the manner of a wave. (3) That this variation of curvature of space is really what happens in that phenomenon which we call the motion of matter whether ponderable or ethereal. (4) That in this physical world nothing else takes place but this variation, subject, possibly, to the law of continuity.*

*W. K. Clifford (1870)*

In Chapter 5, we studied integration over regions in $\mathbb{R}^2$ and $\mathbb{R}^3$. In this chapter, we study integration over paths and surfaces. This is basic to an understanding of Chapter 8, in which we discuss the basic relation between vector differential calculus (Chapter 4) and vector integral calculus (this chapter), a relation that generalizes the fundamental theorem of calculus to several variables. This generalization is summarized in the theorems of Green, Gauss, and Stokes.

## 7.1 The Path Integral

This section introduces the concept of a path integral; this is one of the several ways in which integrals of functions of one variable can be generalized to functions of several variables. Besides those in Chapter 5, there are other generalizations, to be discussed in later sections.

Suppose we are given a scalar function $f\colon \mathbb{R}^3 \to \mathbb{R}$, so that $f$ sends points in $\mathbb{R}^3$ to real numbers. It will be useful to define the integral of such a function $f$ along a path

$\mathbf{c}: I = [a, b] \to \mathbb{R}^3$, where $\mathbf{c}(t) = (x(t), y(t), z(t))$. To relate this notion to something tangible, suppose that the image of $\mathbf{c}$ represents a wire. We can let $f(x, y, z)$ denote the mass density at $(x, y, z)$ and the integral of $f$ will be the total mass of the wire. By letting $f(x, y, z)$ indicate temperature, we can also use the integral to determine the average temperature along the wire. We first give the formal definition of the path integral and then, after the following example, further motivate it.

DEFINITION: Path Integrals The ***path integral***, or the ***integral of*** $f(x, y, z)$ ***along the path*** $\mathbf{c}$, is defined when $\mathbf{c}: I = [a, b] \to \mathbb{R}^3$ is of class $C^1$ and when the composite function $t \mapsto f(x(t), y(t), z(t))$ is continuous on $I$. We define this integral by the equation

$$\int_{\mathbf{c}} f\, ds = \int_a^b f(x(t), y(t), z(t)) \|\mathbf{c}'(t)\|\, dt.$$

Sometimes $\int_{\mathbf{c}} f\, ds$ is denoted

$$\int_{\mathbf{c}} f(x, y, z)\, ds$$

or

$$\int_a^b f(\mathbf{c}(t)) \|\mathbf{c}'(t)\|\, dt.$$

If $\mathbf{c}(t)$ is only piecewise $C^1$ or $f(\mathbf{c}(t))$ is piecewise continuous, we define $\int_{\mathbf{c}} f\, ds$ by breaking $[a, b]$ into pieces over which $f(\mathbf{c}(t))\|\mathbf{c}'(t)\|$ is continuous, and summing the integrals over the pieces.

When $f = 1$, we recover the definition of the arc length of $\mathbf{c}$. Also note that $f$ need only be defined on the image curve $C$ of $\mathbf{c}$ and not necessarily on the whole space in order for the preceding definition to make sense.

**EXAMPLE 1** Let $\mathbf{c}$ be the helix $\mathbf{c}: [0, 2\pi] \to \mathbb{R}^3$, $t \mapsto (\cos t, \sin t, t)$ (see Figure 2.4.9), and let $f(x, y, z) = x^2 + y^2 + z^2$. Evaluate the integral $\int_{\mathbf{c}} f(x, y, z)\, ds$.

SOLUTION First we compute $\|\mathbf{c}'(t)\|$:

$$\|\mathbf{c}'(t)\| = \sqrt{\left[\frac{d(\cos t)}{dt}\right]^2 + \left[\frac{d(\sin t)}{dt}\right]^2 + \left[\frac{dt}{dt}\right]^2} = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}.$$

Next, we substitute for $x$, $y$, and $z$ in terms of $t$ to obtain

$$f(x, y, z) = x^2 + y^2 + z^2 = \cos^2 t + \sin^2 t + t^2 = 1 + t^2$$

along $\mathbf{c}$. Inserting this information into the definition of the path integral yields

$$\int_{\mathbf{c}} f(x, y, z)\, ds = \int_0^{2\pi} (1 + t^2)\sqrt{2}\, dt = \sqrt{2}\left[t + \frac{t^3}{3}\right]_0^{2\pi} = \frac{2\sqrt{2}\pi}{3}(3 + 4\pi^2). \quad \blacktriangle$$

To motivate the definition of the path integral, we shall consider "Riemann-like" sums $S_N$ in the same general way we did to define arc length in Section 4.2. For simplicity, let $\mathbf{c}$ be of class $C^1$ on $I$. Subdivide the interval $I = [a, b]$ by means of a partition

$$a = t_0 < t_1 < \cdots < t_N = b.$$

This leads to a decomposition of $\mathbf{c}$ into paths $\mathbf{c}_i$ (Figure 7.1.1) defined on $[t_i, t_{i+1}]$ for $0 \le i \le N - 1$. Denote the arc length of $\mathbf{c}_i$ by $\Delta s_i$; thus,

$$\Delta s_i = \int_{t_i}^{t_{i+1}} \|\mathbf{c}'(t)\|\, dt.$$

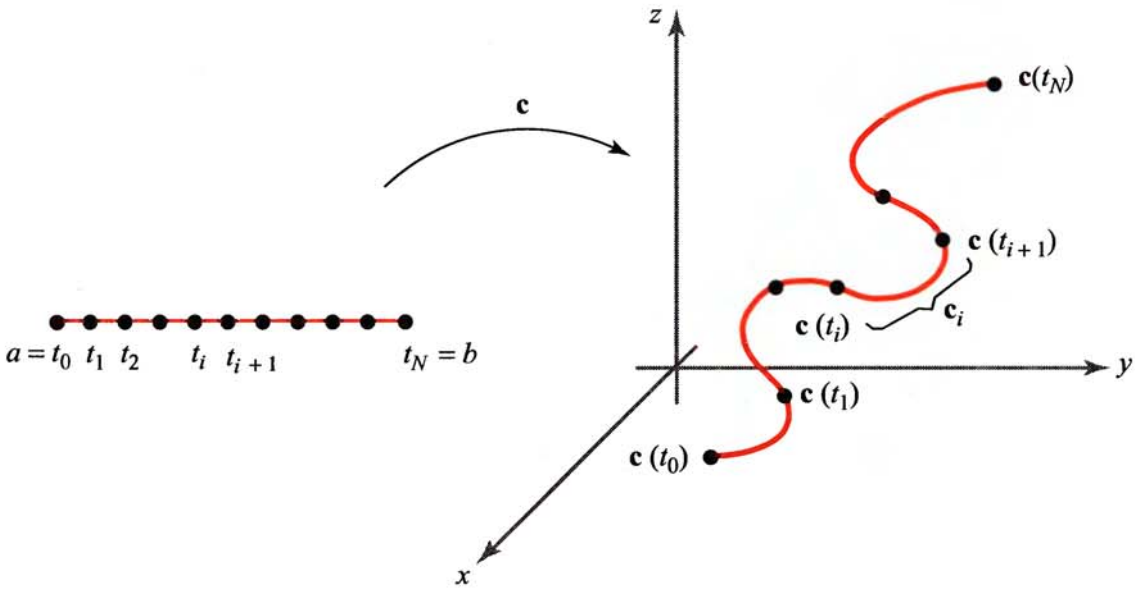



**Figure 7.1.1** Breaking $\mathbf{c}$ into smaller $\mathbf{c}_i$.

When $N$ is large, the arc length $\Delta s_i$ is small and $f(x, y, z)$ is approximately constant for points on $\mathbf{c}_i$. We consider the sums

$$S_N = \sum_{i=0}^{N-1} f(x_i, y_i, z_i)\, \Delta s_i,$$

where $(x_i, y_i, z_i) = \mathbf{c}(t)$ for some $t \in [t_i, t_{i+1}]$. By the mean-value theorem we know that $\Delta s_i = \|\mathbf{c}'(t_i^*)\| \Delta t_i$, where $t_i \le t_i^* \le t_{i+1}$ and $\Delta t_i = t_{i+1} - t_i$. From the theory of

Riemann sums, it can be shown that

$$\lim_{N\to\infty} S_N = \lim_{N\to\infty} \sum_{i=0}^{N-1} f(x_i, y_i, z_i)\|\mathbf{c}'(t_i^*)\|\Delta t_i = \int_I f(x(t), y(t), z(t))\|\mathbf{c}'(t)\|\,dt$$

$$= \int_{\mathbf{c}} f(x, y, z)\,ds.$$

## The Path Integral for Planar Curves

An important special case of the path integral occurs when the path $\mathbf{c}$ describes a plane curve. Suppose that all points $\mathbf{c}(t)$ lie in the $xy$ plane and $f$ is a real-valued function of two variables. The path integral of $f$ along $\mathbf{c}$ is

$$\int_{\mathbf{c}} f(x, y)\,ds = \int_a^b f(x(t), y(t))\sqrt{x'(t)^2 + y'(t)^2}\,dt.$$

When $f(x, y) \geq 0$, this integral has a geometric interpretation as the "area of a fence." We can construct a "fence" with base the image of $\mathbf{c}$ and with height $f(x, y)$ at $(x, y)$ (Figure 7.1.2). If $\mathbf{c}$ moves only once along the image of $\mathbf{c}$, the integral $\int_{\mathbf{c}} f(x, y)\,ds$ represents the area of a side of this fence. Readers should try to justify this interpretation for themselves, using an argument like the one used to justify the arc-length formula.

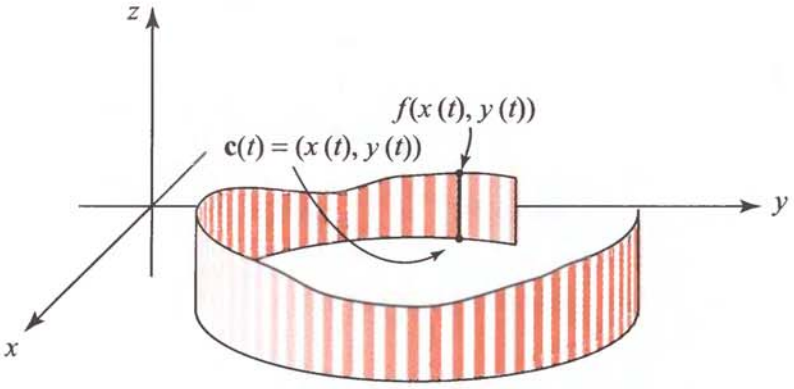



Figure 7.1.2 The path integral as the area of a fence.

**EXAMPLE 2** Tom Sawyer's aunt has asked him to whitewash both sides of the old fence shown in Figure 7.1.3. Tom estimates that for each 25 ft$^2$ of whitewashing he lets someone do for him, the willing victim will pay 5 cents. How much can Tom hope to earn, assuming his aunt will provide whitewash free of charge?

SOLUTION From Figure 7.1.3, the base of the fence in the first quadrant is the path $\mathbf{c}$: $[0, \pi/2] \to \mathbb{R}^2$, $t \mapsto (30\cos^3 t, 30\sin^3 t)$, and the height of the fence at $(x, y)$ is $f(x, y) = 1 + y/3$. The area of one side of the half of the fence is equal to the *integral* $\int_{\mathbf{c}} f(x, y)\,ds = \int_{\mathbf{c}}(1 + y/3)\,ds$. Because $\mathbf{c}'(t) = (-90\cos^2 t \sin t, 90\sin^2 t \cos t)$, we

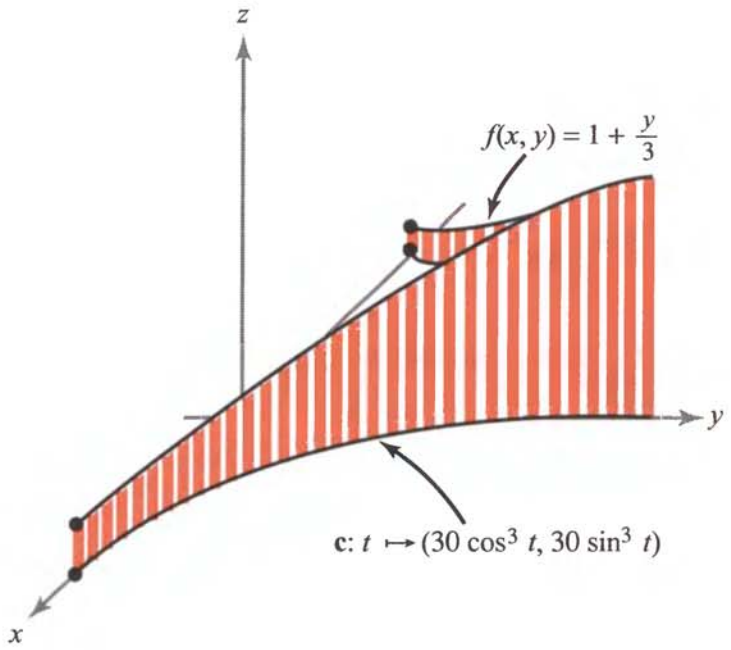



**Figure 7.1.3** Tom Sawyer's fence.

have $\|\mathbf{c}'(t)\| = 90 \sin t \cos t$. Thus, the integral is

$$\begin{aligned}\int_{\mathbf{c}} \left(1 + \frac{y}{3}\right) ds &= \int_0^{\pi/2} \left(1 + \frac{30 \sin^3 t}{3}\right) 90 \sin t \cos t \, dt \\ &= 90 \int_0^{\pi/2} (\sin t + 10 \sin^4 t) \cos t \, dt \\ &= 90 \left[\frac{\sin^2 t}{2} + 2 \sin^5 t\right]_0^{\pi/2} = 90\left(\frac{1}{2} + 2\right) = 225,\end{aligned}$$

which is the area in the first quadrant. Hence, the area of one side of the fence is 450 ft$^2$. Because both sides are to be whitewashed, we must multiply by 2 to find the total area, which is 900 ft$^2$. Dividing by 25 and then multiplying by 5, we find that Tom could realize as much as \$1.80 for the job. ▲

This concludes our study of integration of *scalar* functions over paths. In the next section we shall turn our attention to the integration of *vector fields* over paths, and we shall see many further applications of the path integral in Chapter 8, when we study vector analysis.

## Supplement to Section 7.1: The Total Curvature of a Curve

Exercises 12 to 17 of Section 4.2 described the notions of curvature $\kappa$ and torsion $\tau$ of a smooth curve $C$ in space. If $\mathbf{c}: [a, b] \to C \subset \mathbb{R}^3$ is a unit-speed parametrization of $C$, so that $\|\mathbf{c}'(t)\| = 1$, then the ***curvature*** $\kappa(p)$ at $p \in C$ is defined by $\kappa(p) = \|\mathbf{c}''(t)\|$, where $p = \mathbf{c}(t)$. A result of differential geometry is that two unit-speed curves with the same curvature and torsion can be obtained from one another by a rigid rotation, translation, or reflection.

The curvature $\kappa: C \to \mathbb{R}$ is a real-valued function on the set $C$, so we define the ***total curvature*** as its path integral over $C$: $\int_C \kappa \, ds$. There are some surprising facts that mathematicians have been able to prove about the total curvature. For one thing, if $C$ is a closed [that is, $\mathbf{c}(a) = \mathbf{c}(b)$] planar curve, then

$$\int_C \kappa \, ds \geq 2\pi,$$

and equals $2\pi$ only when $C$ is a circle. If $C$ is a closed space curve with

$$\int_C \kappa \, ds \leq 4\pi,$$

then $C$ is "unknotted"; that is, $C$ can be continuously deformed (without ever intersecting itself) into a planar circle. Therefore, for knotted curves,

$$\int_C \kappa \, ds > 4\pi.$$

See Figure 7.1.4.

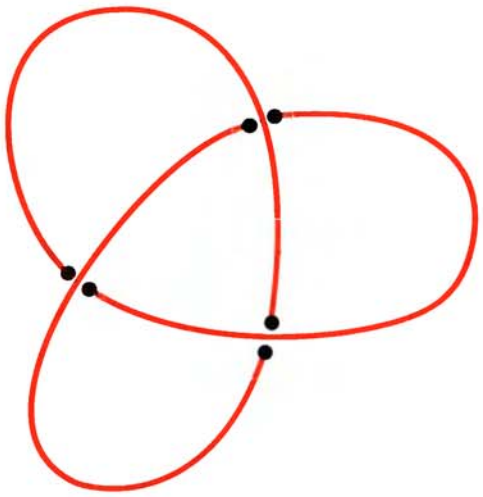


**Figure 7.1.4** A knotted curve in $\mathbb{R}^3$.

The formal statement of this fact is known as the *Fary–Milnor* theorem. Legend has it that John Milnor, a contemporary of John Nash's[1] at Princeton University, was asleep in a math class as the professor wrote three *unsolved* knot theory problems on the blackboard. At the end of the class, Milnor (still an undergraduate) woke up and, thinking the blackboard problems were assigned as homework, quickly wrote

[1] John Nash is the subject of Sylvia Nasar's best-selling biography, *A Beautiful Mind*, a fictionalized version of which was made into a movie in 2001.

them down. The following week he turned in the solution to all three problems—one of which was a proof of the Fary–Milnor theorem! Some years later, he was appointed a professor at Princeton, and in 1962 he was awarded (albeit for other work) a Fields medal, mathematics' highest honor, generally regarded as the mathematical Nobel Prize.

## EXERCISES

**1.** Let $f(x, y, z) = y$ and $\mathbf{c}(t) = (0, 0, t)$, $0 \le t \le 1$. Prove that $\int_{\mathbf{c}} f\, ds = 0$.

**2.** Evaluate the following path integrals $\int_{\mathbf{c}} f(x, y, z)\, ds$, where

(a) $f(x, y, z) = x + y + z$ and $\mathbf{c}$: $t \mapsto (\sin t, \cos t, t)$, $t \in [0, 2\pi]$
(b) $f(x, y, z) = \cos z$, $\mathbf{c}$ as in part (a)

**3.** Evaluate the following path integrals $\int_{\mathbf{c}} f(x, y, z)\, ds$, where

(a) $f(x, y, z) = \exp\sqrt{z}$, and $\mathbf{c}$: $t \mapsto (1, 2, t^2)$, $t \in [0, 1]$
(b) $f(x, y, z) = yz$, and $\mathbf{c}$: $t \mapsto (t, 3t, 2t)$, $t \in [1, 3]$

**4.** Evaluate the integral of $f(x, y, z)$ along the path $\mathbf{c}$, where

(a) $f(x, y, z) = x \cos z$, $\mathbf{c}$: $t \mapsto t\mathbf{i} + t^2\mathbf{j}$, $t \in [0, 1]$
(b) $f(x, y, z) = (x + y)/(y + z)$, and $\mathbf{c}$: $t \mapsto \left(t, \frac{2}{3}t^{3/2}, t\right)$, $t \in [1, 2]$

**5.** Let $f$: $\mathbb{R}^3 \backslash \{xz \text{ plane}\} \to \mathbb{R}$ be defined by $f(x, y, z) = 1/y^3$. Evaluate $\int_{\mathbf{c}} f(x, y, z)\, ds$, where $\mathbf{c}$: $[1, e] \to \mathbb{R}^3$ is given by $\mathbf{c}(t) = (\log t)\mathbf{i} + t\mathbf{j} + 2\mathbf{k}$.

**6.** (a) Show that the path integral of $f(x, y)$ along a path given in polar coordinates by $r = r(\theta)$, $\theta_1 \le \theta \le \theta_2$, is

$$\int_{\theta_1}^{\theta_2} f(r\cos\theta, r\sin\theta)\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta.$$

(b) Compute the arc length of the path $r = 1 + \cos\theta$, $0 \le \theta \le 2\pi$.

**7.** Let $f(x, y) = 2x - y$, and consider the path $x = t^4$, $y = t^4$, $-1 \le t \le 1$.

(a) Compute the integral of $f$ along this path and interpret the answer geometrically.
(b) Evaluate the arc-length function $s(t)$ and redo part (a) in terms of $s$ (you may wish to consult Exercise 2, Section 4.2).

*Exercises 8 to 11 are concerned with the application of the path integral to the problem of defining the average value of a scalar function along a path. Define the number*

$$\frac{\int_{\mathbf{c}} f(x, y, z)\, ds}{l(\mathbf{c})}$$

*to be the **average value** of $f$ along $\mathbf{c}$. Here $l(\mathbf{c})$ is the length of the path:*

$$l(\mathbf{c}) = \int_{\mathbf{c}} \|\mathbf{c}'(t)\|\, dt.$$

*(This is analogous to the average of a function over a region defined in Section 6.3.)*

**8.** (a) Justify the formula $[\int_{\mathbf{c}} f(x, y, z)\, ds]/l(\mathbf{c})$ for the average value of $f$ along $\mathbf{c}$ using Riemann sums.

(b) Show that the average value of $f$ along $\mathbf{c}$ in Example 1 is $(1 + \frac{4}{3}\pi^2)$.

(c) In Exercise 2(a) and (b) above, find the average value of $f$ over the given curves.

**9.** Find the average $y$ coordinate of the points on the semicircle parametrized by $\mathbf{c}$: $[0, \pi] \to \mathbb{R}^3, \theta \mapsto (0, a \sin\theta, a \cos\theta); a > 0$.

**10.** Suppose the semicircle in Exercise 9 is made of a wire with a uniform density of 2 grams per unit length.

(a) What is the total mass of the wire?

(b) Where is the center of mass of this configuration of wire? (Consult Section 6.3.)

**11.** Let $\mathbf{c}$ be the path given by $\mathbf{c}(t) = (t^2, t, 3)$ for $t \in [0, 1]$.

(a) Find $l(\mathbf{c})$, the length of the path.

(b) Find the average $y$ coordinate along the path $\mathbf{c}$.

**12.** If $f$: $[a, b] \to \mathbb{R}$ is piecewise continuously differentiable, let the *length of the graph* of $f$ on $[a, b]$ be defined as the length of the path $t \mapsto (t, f(t))$ for $t \in [a, b]$.

(a) Show that the length of the graph of $f$ on $[a, b]$ is

$$\int_a^b \sqrt{1 + [f'(x)]^2}\, dx.$$

(b) Find the length of the graph of $y = \log x$ from $x = 1$ to $x = 2$.

**13.** Find the mass of a wire formed by the intersection of the sphere $x^2 + y^2 + z^2 = 1$ and the plane $x + y + z = 0$ if the density at $(x, y, z)$ is given by $\rho(x, y, z) = x^2$ grams per unit length of wire.

**14.** Evaluate $\int_{\mathbf{c}} f\, ds$ where $f(x, y, z) = z$ and $\mathbf{c}(t) = (t \cos t, t \sin t, t)$ for $0 \le t \le t_0$.

**15.** Write the following limit as a path integral of $f(x, y, z) = xy$ over some path $\mathbf{c}$ on $[0, 1]$ and evaluate:

$$\lim_{N\to\infty} \sum_{i=1}^{N-1} t_i^2 \left(t_{i+1}^2 - t_i^2\right),$$

where $t_1, \ldots, t_N$ is a partition of $[0, 1]$.

**16.** Consider paths that connect the points $A = (0, 1)$ and $B = (1, 0)$ in the $xy$ plane, as in Figure 7.1.5.[2]

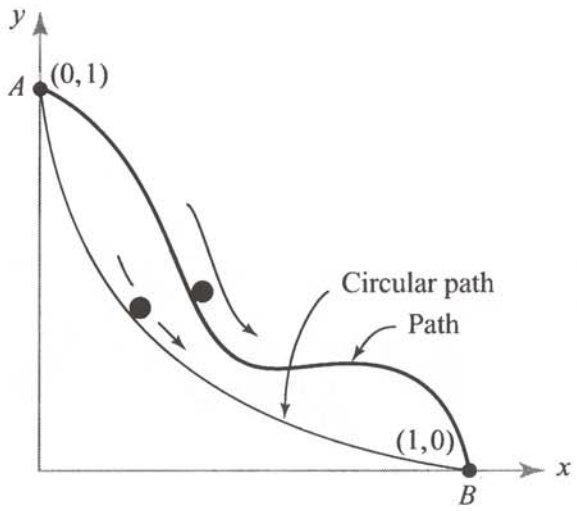



**Figure 7.1.5** A curve joining the points $A$ and $B$.

Galileo contemplated the following question: does a bead falling under the influence of gravity from a point $A$ to a point $B$ along a curve do so in *the least possible time* if that curve is a circular arc? For any given path, the time of transit $T$ is a path integral

$$T = \int \frac{dt}{v},$$

where the bead's velocity is $v = \sqrt{2gy}$, where $g$ is the gravitational constant. In 1697, Johann Bernoulli challenged the mathematical world to find the path in which the bead would roll from $A$ to $B$ in the least time. This solution would determine whether Galileo's considerations had been correct.

(a) Calculate $T$ for the straight-line path $y = 1 - x$.
(b) Write a formula for $T$ for Galileo's circular path, given by $(x - 1)^2 + (y - 1)^2 = 1$.

Incidentally, Newton was the first to send his solution [which turned out to be a cycloid—the same curve (inverted) that we studied in Example 2.4.4], but he did so anonymously. Bernoulli was not fooled, however. When he received the solution, he immediately knew its author, exclaiming, "I know the Lion from his paw." While the solution of this problem is a cycloid, it is known in the literature as the *brachistrochrone*. This was the beginning of the important field called the *calculus of variations*.

# 7.2 Line Integrals

We now consider the problem of integrating a *vector field* along a path. We will begin by considering the notion of *work* to motivate the general definition.

## Work Done by Force Fields

If **F** is a force field in space, then a test particle (for example, a small unit charge in an electric force field or a unit mass in a gravitational field) will experience the force **F**.

[2] We thank Tanya Leise for suggesting this exercise.

Suppose the particle moves along the image of a path $\mathbf{c}$ while being acted upon by $\mathbf{F}$. A fundamental concept is the *work done* by $\mathbf{F}$ on the particle as it traces out the path $\mathbf{c}$. If $\mathbf{c}$ is a straight-line displacement given by the vector $\mathbf{d}$ and if $\mathbf{F}$ is a constant force, then the work done by $\mathbf{F}$ in moving the particle along the path is the dot product $\mathbf{F} \cdot \mathbf{d}$:

$$\mathbf{F} \cdot \mathbf{d} = (\text{magnitude of force}) \times (\text{displacement in direction of force}).$$

If the path is curved, we can imagine that it is made up of a succession of infinitesimal straight-line displacements or that it is *approximated* by a finite number of straight-line displacements. Then (as in our derivation of the formulas for the path integral in the preceding section) we are led to the following formula for the work done by the force field $\mathbf{F}$ on a particle moving along a path $\mathbf{c}$: $[a, b] \to \mathbb{R}^3$:

$$\text{work done by } \mathbf{F} = \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\, dt.$$

We can further justify this derivation as follows. As $t$ ranges over a small interval $t$ to $t + \Delta t$, the particle moves from $\mathbf{c}(t)$ to $\mathbf{c}(t + \Delta t)$, a vector displacement of $\Delta \mathbf{s} = \mathbf{c}(t + \Delta t) - \mathbf{c}(t)$ (see Figure 7.2.1).

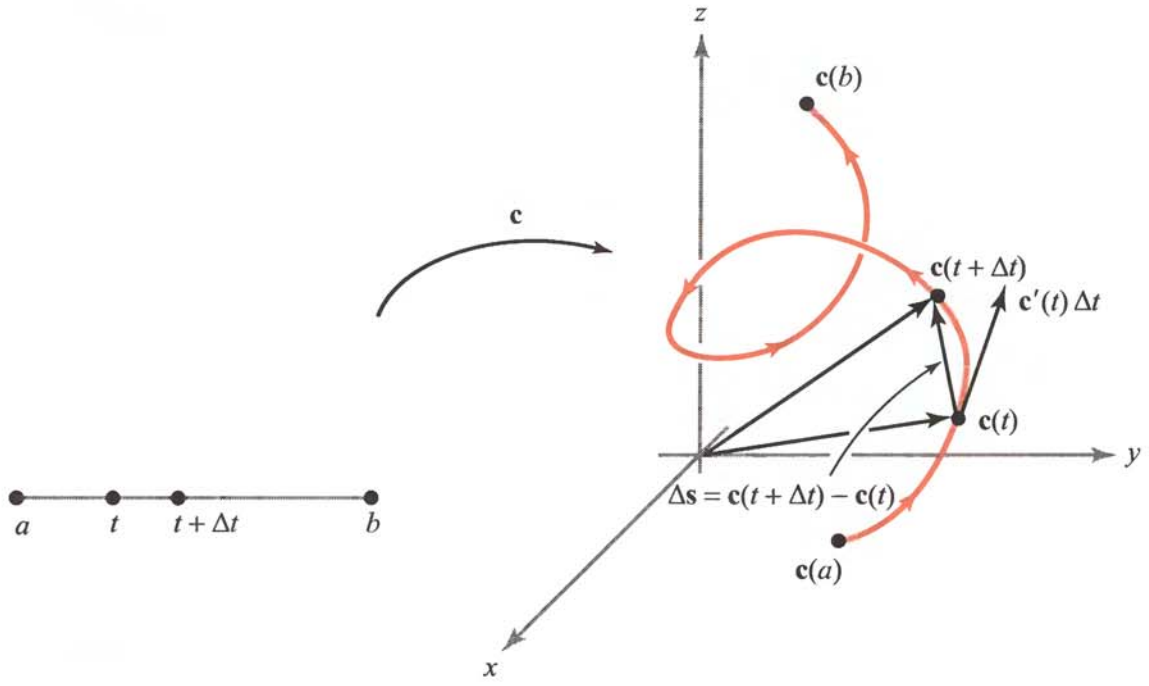



Figure 7.2.1 For small $\Delta t$, $\Delta \mathbf{s} = \mathbf{c}(t + \Delta t) - \mathbf{c}(t) \approx \mathbf{c}'(t)\Delta t$.

From the definition of the derivative, we get the approximation $\Delta \mathbf{s} \approx \mathbf{c}'(t)\Delta t$. The work done in going from $\mathbf{c}(t)$ to $\mathbf{c}(t + \Delta t)$ is therefore approximately

$$\mathbf{F}(\mathbf{c}(t)) \cdot \Delta \mathbf{s} \approx \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\Delta t.$$

If we subdivide the interval $[a, b]$ into $n$ equal parts $a = t_0 < t_1 < \cdots < t_n = b$, with $\Delta t = t_{i+1} - t_i$, then the work done by $\mathbf{F}$ is approximately

$$\sum_{i=0}^{n-1} \mathbf{F}(\mathbf{c}(t_i)) \cdot \Delta \mathbf{s} \approx \sum_{i=0}^{n-1} \mathbf{F}(\mathbf{c}(t_i)) \cdot \mathbf{c}'(t_i) \Delta t.$$

As $n \to \infty$, this approximation becomes better and better, and so it is reasonable to take as our definition of work to be the limit of the sum just given as $n \to \infty$. This limit is given by the integral

$$\int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\, dt.$$

## Definition of the Line Integral

The previous discussion of work motivates the following definition.

DEFINITION: Line Integrals Let $\mathbf{F}$ be a vector field on $\mathbb{R}^3$ that is continuous on the $C^1$ path $\mathbf{c}$: $[a, b] \to \mathbb{R}^3$. We define $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$, the ***line integral*** of $\mathbf{F}$ along $\mathbf{c}$, by the formula

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\, dt;$$

that is, we integrate the dot product of $\mathbf{F}$ with $\mathbf{c}'$ over the interval $[a, b]$.

As is the case with scalar functions, we can also define $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ if $\mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)$ is only piecewise continuous.

For paths $\mathbf{c}$ that satisfy $\mathbf{c}'(t) \neq \mathbf{0}$, there is another useful formula for the line integral: Namely, if $\mathbf{T}(t) = \mathbf{c}'(t)/\|\mathbf{c}'(t)\|$ denotes the unit tangent vector, we have

$$\begin{aligned} \int \mathbf{F} \cdot d\mathbf{s} &= \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\, dt && \text{(by definition)} \\ &= \int_a^b \left[ \mathbf{F}(\mathbf{c}(t)) \cdot \frac{\mathbf{c}'(t)}{\|\mathbf{c}'(t)\|} \right] \|\mathbf{c}'(t)\|\, dt && \text{(canceling } \|\mathbf{c}'(t)\|\text{)} \quad (1) \\ &= \int_a^b [\mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{T}(t)] \|\mathbf{c}'(t)\|\, dt. \end{aligned}$$

This formula says that $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ is equal to something that looks like the path integral of the tangential component $\mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{T}(t)$ of $\mathbf{F}$ along $\mathbf{c}$. In fact, the last part of formula (1) is analogous to the path integral of a scalar function $f$ along $\mathbf{c}$.[3]

To compute a line integral in any particular case, one can either use the original definition or integrate the tangential component of $\mathbf{F}$ along $\mathbf{c}$, as prescribed by formula (1), whichever is easier or more appropriate.

**EXAMPLE 1** Let $\mathbf{c}(t) = (\sin t, \cos t, t)$ with $0 \leq t \leq 2\pi$. Let the vector field $\mathbf{F}$ be defined by $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Compute $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$.

SOLUTION Here, $\mathbf{F}(\mathbf{c}(t)) = \mathbf{F}(\sin t, \cos t, t) = (\sin t)\mathbf{i} + (\cos t)\mathbf{j} + t\mathbf{k}$, and $\mathbf{c}'(t) = (\cos t)\mathbf{i} - (\sin t)\mathbf{j} + \mathbf{k}$. Therefore,

$$\mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t) = \sin t \cos t - \cos t \sin t + t = t,$$

and so

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} t\, dt = 2\pi^2. \quad \blacktriangle$$

Another common way of writing line integrals is

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} F_1\, dx + F_2\, dy + F_3\, dz,$$

where $F_1$, $F_2$, and $F_3$ are the components of the vector field $\mathbf{F}$. We call the expression $F_1\, dx + F_2\, dy + F_3\, dz$ a ***differential form***.[4] By *definition*, the integral of a differential form along a path $\mathbf{c}$, where $\mathbf{c}(t) = (x(t), y(t), z(t))$, is

$$\int_{\mathbf{c}} F_1\, dx + F_2\, dy + F_3\, dz = \int_a^b \left( F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}.$$

Note that we may think of $d\mathbf{s}$ as the differential form $d\mathbf{s} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$. Thus, the differential form $F_1\, dx + F_2\, dy + F_3\, dz$ may be written as the dot product $\mathbf{F} \cdot d\mathbf{s}$.

**EXAMPLE 2** Evaluate the line integral

$$\int_{\mathbf{c}} x^2\, dx + xy\, dy + dz,$$

where $\mathbf{c}$: $[0, 1] \to \mathbb{R}^3$ is given by $\mathbf{c}(t) = (t, t^2, 1) = (x(t), y(t), z(t))$.

---

[3] If $\mathbf{c}$ does not intersect itself (that is, if $\mathbf{c}(t_1) = \mathbf{c}(t_2)$ implies $t_1 = t_2$), then each point P on $C$ (the image curve of $\mathbf{c}$) can be written uniquely as $\mathbf{c}(t)$ for some $t$. If we define $f(\mathrm{P}) = f(\mathbf{c}(t)) = \mathbf{F}(\mathbf{c}) \cdot \mathbf{T}(t)$, $f$ is a function on $C$; by definition, its path integral along $\mathbf{c}$ is given by formula (1) and there is no difficulty in literally interpreting $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ as a path integral. If $\mathbf{c}$ intersects itself, we cannot define $f$ as a function on $C$ as before (why?); however, in this case it is still useful to think of the right side of formula (1) as a path integral.

[4] See Section 8.6 for a brief discussion of the general theory of differential forms.

SOLUTION We compute $dx/dt = 1$, $dy/dt = 2t$, $dz/dt = 0$; therefore,

$$\int_{\mathbf{c}} x^2\,dx + xy\,dy + dz = \int_0^1 \left([x(t)]^2 \frac{dx}{dt} + [x(t)y(t)] \frac{dy}{dt}\right) dt$$

$$= \int_0^1 (t^2 + 2t^4)\,dt = \left[\frac{1}{3}t^3 + \frac{2}{5}t^5\right]_0^1 = \frac{11}{15}. \quad \blacktriangle$$

**EXAMPLE 3** Evaluate the line integral

$$\int_{\mathbf{c}} \cos z\,dx + e^x\,dy + e^y\,dz,$$

where the path $\mathbf{c}$ is defined by $\mathbf{c}(t) = (1, t, e^t)$ and $0 \le t \le 2$.

SOLUTION We compute $dx/dt = 0$, $dy/dt = 1$, $dz/dt = e^t$, and so

$$\int_{\mathbf{c}} \cos z\,dx + e^x\,dy + e^y\,dz = \int_0^2 (0 + e + e^{2t})\,dt$$

$$= \left[et + \frac{1}{2}e^{2t}\right]_0^2 = 2e + \frac{1}{2}e^4 - \frac{1}{2}. \quad \blacktriangle$$

**EXAMPLE 4** Let $\mathbf{c}$ be the path

$$x = \cos^3\theta, \qquad y = \sin^3\theta, \qquad z = \theta, \qquad 0 \le \theta \le \frac{7\pi}{2}$$

(see Figure 7.2.2). Evaluate the integral $\int_{\mathbf{c}} (\sin z\,dx + \cos z\,dy - (xy)^{1/3}\,dz)$.

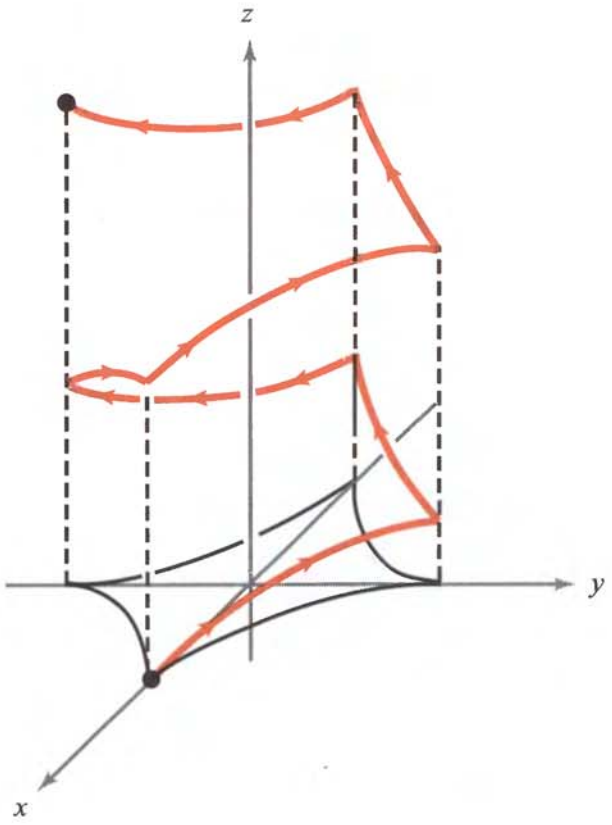



**Figure 7.2.2** The image of the path $x = \cos^3\theta, y = \sin^3\theta, z = \theta; 0 \le \theta \le 7\pi/2$.

SOLUTION In this case, we have

$$\frac{dx}{d\theta} = -3\cos^2\theta\sin\theta, \qquad \frac{dy}{d\theta} = 3\sin^2\theta\cos\theta, \qquad \frac{dz}{d\theta} = 1,$$

so the integral is

$$\int_{\mathbf{c}} \sin z\,dx + \cos z\,dy - (xy)^{1/3}\,dz$$
$$= \int_0^{7\pi/2} (-3\cos^2\theta\sin^2\theta + 3\sin^2\theta\cos^2\theta - \cos\theta\sin\theta)\,d\theta.$$

The first two terms cancel, and so we get

$$-\int_0^{7\pi/2} \cos\theta\sin\theta\,d\theta = -\left[\frac{1}{2}\sin^2\theta\right]_0^{7\pi/2} = -\frac{1}{2}. \quad \blacktriangle$$

**EXAMPLE 5** Suppose $\mathbf{F}$ is the force vector field $\mathbf{F}(x, y, z) = x^3\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Parametrize the circle of radius $a$ in the $yz$ plane by letting $\mathbf{c}(\theta)$ have components

$$x = 0, \qquad y = a\cos\theta, \qquad z = a\sin\theta, \qquad 0 \le \theta \le 2\pi.$$

Because $\mathbf{F}(\mathbf{c}(\theta)) \cdot \mathbf{c}'(\theta) = 0$, the force field $\mathbf{F}$ is normal to the circle at every point on the circle, so $\mathbf{F}$ will not do any work on a particle moving along the circle (Figure 7.2.3).

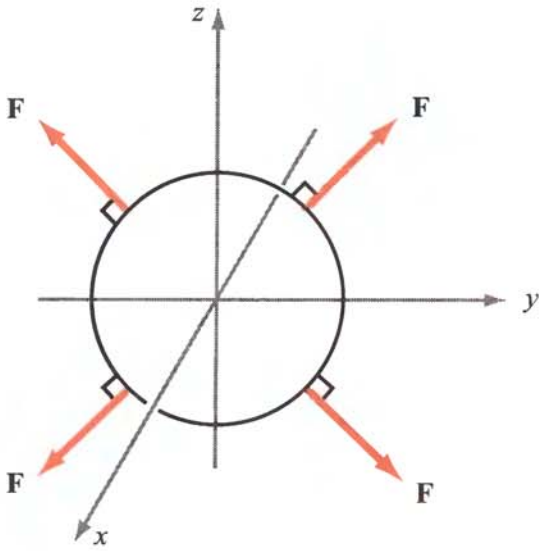



**Figure 7.2.3** A vector field $\mathbf{F}$ normal to a circle in the $yz$ plane.

We can verify by direct computation that the work done by $\mathbf{F}$ is zero:

$$W = \int_{\mathbf{c}} \mathbf{F}\cdot d\mathbf{s} = \int_{\mathbf{c}} x^3\,dx + y\,dy + z\,dz$$
$$= \int_0^{2\pi} (0 - a^2\cos\theta\sin\theta + a^2\cos\theta\sin\theta)\,d\theta = 0. \quad \blacktriangle$$

**EXAMPLE 6** If we consider the field and curve of Example 4, we see that the work done by the field is $-\frac{1}{2}$, a negative quantity. This means that the field impedes movement along the path. ▲

## Reparametrizations

The line integral $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ depends not only on the field $\mathbf{F}$ but also on the path $\mathbf{c}$: $[a, b] \to \mathbb{R}^3$. In general, if $\mathbf{c}_1$ and $\mathbf{c}_2$ are two different paths in $\mathbb{R}^3$, $\int_{\mathbf{c}_1} \mathbf{F} \cdot d\mathbf{s} \neq \int_{\mathbf{c}_2} \mathbf{F} \cdot d\mathbf{s}$. On the other hand, we shall see that it is true that $\int_{\mathbf{c}_1} \mathbf{F} \cdot d\mathbf{s} = \pm \int_{\mathbf{c}_2} \mathbf{F} \cdot d\mathbf{s}$ for every vector field $\mathbf{F}$ if $\mathbf{c}_1$ is what we call a ***reparametrization*** of $\mathbf{c}_2$; roughly speaking, this means that $\mathbf{c}_1$ and $\mathbf{c}_2$ are different descriptions of the same geometric curve.

DEFINITION Let $h$: $I \to I_1$ be a $C^1$ real-valued function that is a one-to-one map of an interval $I = [a, b]$ onto another interval $I_1 = [a_1, b_1]$. Let $\mathbf{c}$: $I_1 \to \mathbb{R}^3$ be a piecewise $C^1$ path. Then we call the composition

$$\mathbf{p} = \mathbf{c} \circ h\colon I \to \mathbb{R}^3$$

a ***reparametrization*** of $\mathbf{c}$.

This means that $\mathbf{p}(t) = \mathbf{c}(h(t))$, and so $h$ changes the variable; alternatively, one can think of $h$ as changing the speed at which a point moves along the path. Indeed, observe that $\mathbf{p}'(t) = \mathbf{c}'(h(t))h'(t)$, so that the velocity vector for $\mathbf{p}$ equals that for $\mathbf{c}$ but is multiplied by the scalar factor $h'(t)$.

It is implicit in the definition that $h$ must carry endpoints to endpoints; that is, either $h(a) = a_1$ and $h(b) = b_1$, or $h(a) = b_1$ and $h(b) = a_1$. We thus distinguish two types of reparametrizations. If $\mathbf{c} \circ h$ is a reparametrization of $\mathbf{c}$, then either

$$(\mathbf{c} \circ h)(a) = \mathbf{c}(a_1) \qquad \text{and} \qquad (\mathbf{c} \circ h)(b) = \mathbf{c}(b_1)$$

or

$$(\mathbf{c} \circ h)(a) = \mathbf{c}(b_1) \qquad \text{and} \qquad (\mathbf{c} \circ h)(b) = \mathbf{c}(a_1).$$

In the first case, the reparametrization is said to be ***orientation-preserving***, and a particle tracing the path $\mathbf{c} \circ h$ moves in the *same direction* as a particle tracing $\mathbf{c}$. In the second case, the reparametrization is described as ***orientation-reversing***, and a particle tracing the path $\mathbf{c} \circ h$ moves in the *opposite direction* to that of a particle tracing $\mathbf{c}$ (Figure 7.2.4).

For example, if $C$ is the image of a path $\mathbf{c}$, as shown in Figure 7.2.5, that is, $C = \mathbf{c}([a_1, b_1])$, and if $h$ is orientation-preserving, then $\mathbf{c} \circ h(t)$ will go from $\mathbf{c}(a_1)$ to $\mathbf{c}(b_1)$ as $t$ goes from $a$ to $b$; and if $h$ is orientation-reversing, $\mathbf{c} \circ h(t)$ will go from $\mathbf{c}(b_1)$ to $\mathbf{c}(a_1)$ as $t$ goes from $a$ to $b$.

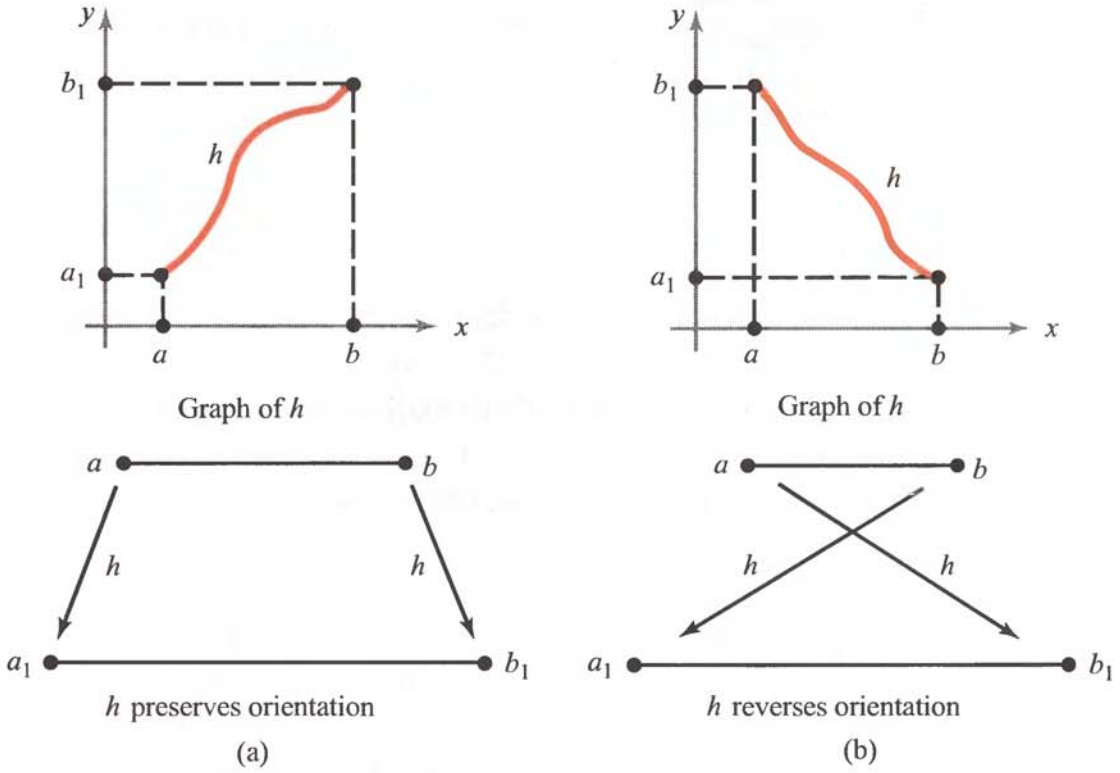



**Figure 7.2.4** Illustrating (a) an orientation-preserving reparametrization, and (b) an orientation-reversing reparametrization.

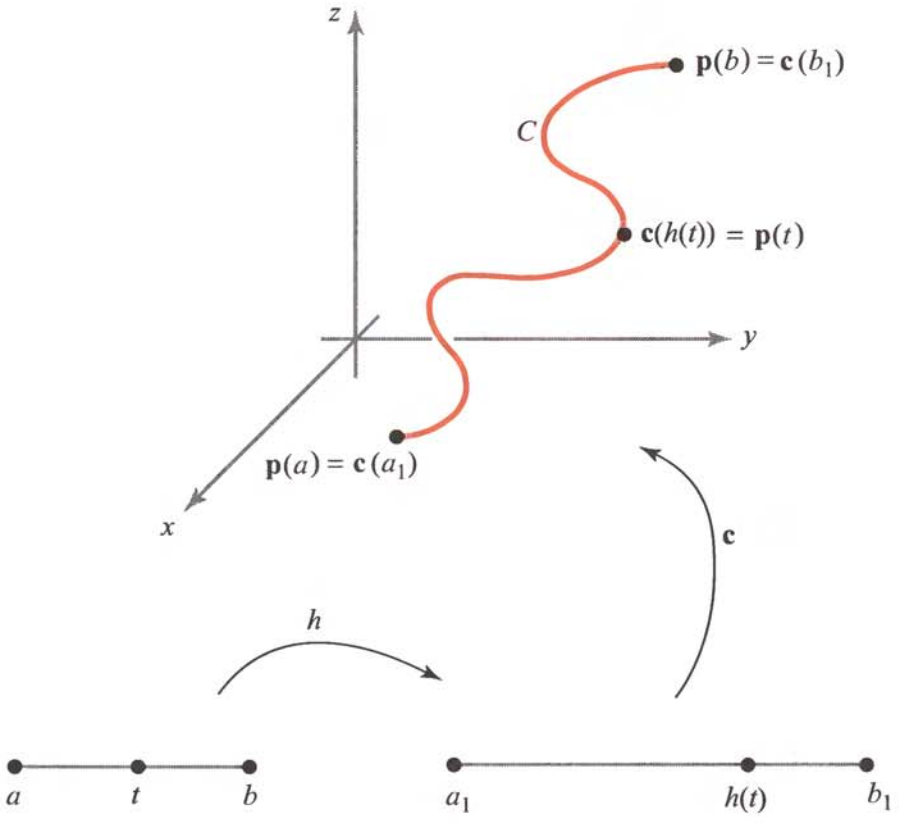



**Figure 7.2.5** The path $\mathbf{p} = \mathbf{c} \circ h$ is a reparametrization of $\mathbf{c}$.

**EXAMPLE 7** Let $\mathbf{c}\colon [a, b] \to \mathbb{R}^3$ be a piecewise $C^1$ path. Then:

(a) The path $\mathbf{c}_{\text{op}}\colon [a, b] \to \mathbb{R}^3, t \mapsto \mathbf{c}(a + b - t)$, is reparametrization of $\mathbf{c}$ corresponding to the map $h\colon [a, b] \to [a, b], t \mapsto a + b - t$; we call $\mathbf{c}_{\text{op}}$ the ***opposite path*** to $\mathbf{c}$. This reparametrization is orientation-reversing.

(b) The path $\mathbf{p}\colon [0, 1] \to \mathbb{R}^3, t \mapsto \mathbf{c}(a + (b - a)t)$, is an orientation-preserving reparametrization of $\mathbf{c}$ corresponding to a change of coordinates $h\colon [0, 1] \to [a, b], t \mapsto a + (b - a)t$. ▲

**THEOREM 1: Change of Parametrization for Line Integrals**
Let $\mathbf{F}$ be a vector field continuous on the $C^1$ path $\mathbf{c}\colon [a_1, b_1] \to \mathbb{R}^3$, and let $\mathbf{p}\colon [a, b] \to \mathbb{R}^3$ be a reparametrization of $\mathbf{c}$. If $\mathbf{p}$ is orientation-preserving, then

$$\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s},$$

and if $\mathbf{p}$ is orientation-reversing, then

$$\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = -\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}.$$

*PROOF* By hypothesis, we have a map $h$ such that $\mathbf{p} = \mathbf{c} \circ h$. By the chain rule,

$$\mathbf{p}'(t) = \mathbf{c}'(h(t))h'(t),$$

and so

$$\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = \int_a^b [\mathbf{F}(\mathbf{c}(h(t))) \cdot \mathbf{c}'(h(t))]h'(t)\, dt.$$

Changing variables with $s = h(t)$, this becomes

$$\int_{h(a)}^{h(b)} \mathbf{F}(\mathbf{c}(s)) \cdot \mathbf{c}'(s)\, ds$$

$$= \begin{cases} \displaystyle\int_{a_1}^{b_1} \mathbf{F}(\mathbf{c}(s)) \cdot \mathbf{c}'(s)\, ds = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} & \text{if } \mathbf{p} \text{ is orientation-preserving} \\ \displaystyle\int_{b_1}^{a_1} \mathbf{F}(\mathbf{c}(s)) \cdot \mathbf{c}'(s)\, ds = -\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} & \text{if } \mathbf{p} \text{ is orientation-reversing.} \end{cases}$$

■

Theorem 1 also holds for piecewise $C^1$ paths, as may be seen by breaking up the intervals into segments on which the paths are of class $C^1$ and summing the integrals over the separate intervals.

Thus, if it is convenient to reparametrize a path when evaluating an integral, Theorem 1 assures us that the value of the integral will not be affected, except possibly for the sign, depending on the orientation.

**EXAMPLE 8** Let $\mathbf{F}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$ and $\mathbf{c}$: $[-5, 10] \to \mathbb{R}^3$ be defined by $t \mapsto (t, t^2, t^3)$. Evaluate $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ and $\int_{\mathbf{c}_{op}} \mathbf{F} \cdot d\mathbf{s}$.

SOLUTION For the path $\mathbf{c}$, we have $dx/dt = 1$, $dy/dt = 2t$, $dz/dt = 3t^2$, and $\mathbf{F}(\mathbf{c}(t)) = t^5\mathbf{i} + t^4\mathbf{j} + t^3\mathbf{k}$. Therefore,

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_{-5}^{10} \left( F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt = \int_{-5}^{10} (t^5 + 2t^5 + 3t^5)\, dt = [t^6]_{-5}^{10} = 984,375.$$

On the other hand, for

$$\mathbf{c}_{op}\colon [-5, 10] \to \mathbb{R}^3, t \mapsto \mathbf{c}(5 - t) = (5 - t, (5 - t)^2, (5 - t)^3),$$

we have $dx/dt = -1$, $dy/dt = -10 + 2t = -2(5 - t)$, $dz/dt = -75 + 30t - 3t^2 = -3(5 - t)^2$, and $\mathbf{F}(\mathbf{c}_{op}(t)) = (5 - t)^5\mathbf{i} + (5 - t)^4\mathbf{j} + (5 - t)^3\mathbf{k}$. Therefore,

$$\int_{\mathbf{c}_{op}} \mathbf{F} \cdot d\mathbf{s} = \int_{-5}^{10} [-(5 - t)^5 - 2(5 - t)^5 - 3(5 - t)^5]\, dt = [(5 - t)^6]_{-5}^{10} = -984,375. \quad \blacktriangle$$

We are interested in reparametrizations, because if the image of a particular $\mathbf{c}$ can be represented in many ways, we want to be sure that path and line integrals depend only on the image curve and not on the particular parametrization. For example, for some problems the unit circle may be conveniently represented by the map $\mathbf{p}$ given by

$$x(t) = \cos 2t, \qquad y(t) = \sin 2t, \qquad 0 \le t \le \pi.$$

Theorem 1 guarantees that any integral computed for this representation will be the same as when we represent the circle by the map $\mathbf{c}$ given by

$$x(t) = \cos t, \qquad y(t) = \sin t, \qquad 0 \le t \le 2\pi,$$

because $\mathbf{p} = \mathbf{c} \circ h$, where $h(t) = 2t$, and thus $\mathbf{p}$ is an orientation-preserving reparametrization of $\mathbf{c}$. However, notice that the map $\gamma$ given by

$$\gamma(t) = (\cos t, \sin t), \qquad 0 \leq t \leq 4\pi$$

is *not* a reparametrization of $\mathbf{c}$. Although it traces out the same image (the circle), it does so twice. (Why does this imply that $\gamma$ is not a reparametrization of $\mathbf{c}$?)

The line integral is an *oriented integral*, in that a change of sign occurs (as we have seen in Theorem 1) if the orientation of the curve is reversed. The *path integral* does not have this property. This follows from the fact that changing $t$ to $-t$ (reversing orientation) just changes the sign of $\mathbf{c}'(t)$, not its length. This is one of the differences between line and path integrals. The following theorem, which is proved by the same method as Theorem 1, shows that path integrals are unchanged under reparametrizations—even orientation-reversing ones.

**THEOREM 2: Change of Parametrization for Path Integrals**
Let $\mathbf{c}$ be piecewise $C^1$, let $f$ be a continuous (real-valued) function on the image of $\mathbf{c}$, and let $\mathbf{p}$ be any reparametrization of $\mathbf{c}$. Then

$$\int_{\mathbf{c}} f(x, y, z)\, ds = \int_{\mathbf{p}} f(x, y, z)\, ds. \tag{2}$$

## Line Integrals of Gradient Fields

We next consider a useful technique for evaluating certain types of line integrals. Recall that a vector field $\mathbf{F}$ is a *gradient vector field* if $\mathbf{F} = \nabla f$ for some real-valued function $f$. Thus,

$$\mathbf{F} = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}.$$

Suppose $g$ and $G$ are real-valued continuous functions defined on a closed interval $[a, b]$, that $G$ is differentiable on $(a, b)$, and that $G' = g$. Then by the fundamental theorem of calculus

$$\int_a^b g(x)\, dx = G(b) - G(a).$$

Thus, the value of the integral of $g$ depends only on the value of $G$ at the endpoints of the interval $[a, b]$. Because $\nabla f$ represents the derivative of $f$, one can ask whether $\int_{\mathbf{c}} \nabla f \cdot d\mathbf{s}$ is completely determined by the value of $f$ at the endpoints $\mathbf{c}(a)$ and $\mathbf{c}(b)$. The answer is contained in the following *generalization of the fundamental theorem of calculus.*

**THEOREM 3: Line Integrals of Gradient Vector Fields** Suppose that $f\colon \mathbb{R}^3 \to \mathbb{R}$ is of class $C^1$ and that $\mathbf{c}\colon [a, b] \to \mathbb{R}^3$ is a piecewise $C^1$ path. Then

$$\int_{\mathbf{c}} \nabla f \cdot d\mathbf{s} = f(\mathbf{c}(b)) - f(\mathbf{c}(a)).$$

*PROOF* Apply the chain rule to the composite function

$$F\colon t \mapsto f(\mathbf{c}(t))$$

to obtain

$$F'(t) = (f \circ \mathbf{c})'(t) = \nabla f(\mathbf{c}(t)) \cdot \mathbf{c}'(t).$$

The function $F$ is a real-valued function of the variable $t$, and so, by the fundamental theorem of single-variable calculus,

$$\int_a^b F'(t)\,dt = F(b) - F(a) = f(\mathbf{c}(b)) - f(\mathbf{c}(a)).$$

Therefore,

$$\int_{\mathbf{c}} \nabla f \cdot d\mathbf{s} = \int_a^b \nabla f(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\,dt = \int_a^b F'(t)\,dt = F(b) - F(a)$$
$$= f(\mathbf{c}(b)) - f(\mathbf{c}(a)). \quad \blacksquare$$

**EXAMPLE 9** Let $\mathbf{c}$ be the path $\mathbf{c}(t) = (t^4/4, \sin^3(t\pi/2), 0)$, $t \in [0, 1]$. Evaluate

$$\int_{\mathbf{c}} y\,dx + x\,dy$$

(which means $\int_{\mathbf{c}} y\,dx + x\,dy + 0\,dz$).

SOLUTION We recognize $y\,dx + x\,dy$, or equivalently, the vector field $y\mathbf{i} + x\mathbf{j} + 0\mathbf{k}$, as the gradient of the function $f(x, y, z) = xy$. Thus,

$$\int_{\mathbf{c}} y\,dx + x\,dy = f(\mathbf{c}(1)) - f(\mathbf{c}(0)) = \frac{1}{4} \cdot 1 - 0 = \frac{1}{4}. \quad \blacktriangle$$

Obviously, if one can recognize the integrand as a gradient, then evaluation of the integral becomes much easier. For example, the reader should try to work out the integral in Example 9 directly. In one-variable calculus, every integral is, in principle,

obtainable by finding an antiderivative. For vector fields, however, this is not always true, because a given vector field need not be a gradient. This point will be examined in detail in Section 8.3, where we derive a test to determine when a vector field $\mathbf{F}$ is a gradient; that is, when $\mathbf{F} = \nabla f$ for some $f$.

## Line Integrals Over Geometric Curves

We have seen how to define path integrals (integrals of scalar functions) and line integrals (integrals of vector functions) over parametrized curves. We have also seen that our work is simplified if we make a judicious choice of parametrization. Because these integrals are independent of the parametrization (except possibly for the sign), it seems natural to express the theory in a way that is independent of the parametrization, and that is thereby more "geometric." We do this briefly and somewhat informally in the following discussion.

DEFINITION We define a ***simple curve*** $C$ to be the image of a piecewise $C^1$ map $\mathbf{c}\colon I \to \mathbb{R}^3$ that is one-to-one on an interval $I$; $\mathbf{c}$ is called a ***parametrization*** of $C$. Thus, a simple curve is one that does not intersect itself (Figure 7.2.6). If $I = [a, b]$, we call $\mathbf{c}(a)$ and $\mathbf{c}(b)$ ***endpoints*** of the curve.

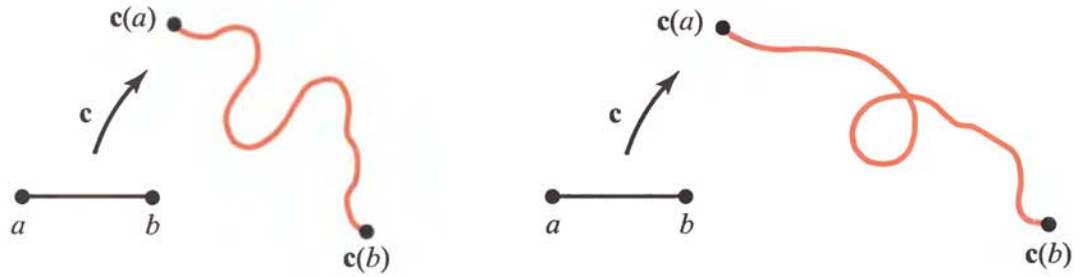



Figure 7.2.6 A simple curve that has no self-intersections is shown on the left. On the right is a curve with a self-intersection, so it is not simple.

Each simple curve $C$ has two orientations or directions associated with it. If P and Q are the endpoints of the curve, then we can consider $C$ as directed either from P to Q or from Q to P. The simple curve $C$ together with a sense of direction is called an ***oriented simple curve*** or ***directed simple curve*** (Figure 7.2.7).

Figure 7.2.7 There are two possible senses of direction on a curve joining P and Q.

**DEFINITION: Simple Closed Curves** By a ***simple closed curve*** we mean the image of a piecewise $C^1$ map $\mathbf{c}$: $[a, b] \to \mathbb{R}^3$ that is one-to-one on $[a, b)$ and satisfies $\mathbf{c}(a) = \mathbf{c}(b)$ (Figure 7.2.8). If $\mathbf{c}$ satisfies the condition $\mathbf{c}(a) = \mathbf{c}(b)$, but is not necessarily one-to-one on $[a, b)$, we call its image a ***closed curve***. Simple closed curves have two orientations, corresponding to the two possible directions of motion along the curve (Figure 7.2.9).

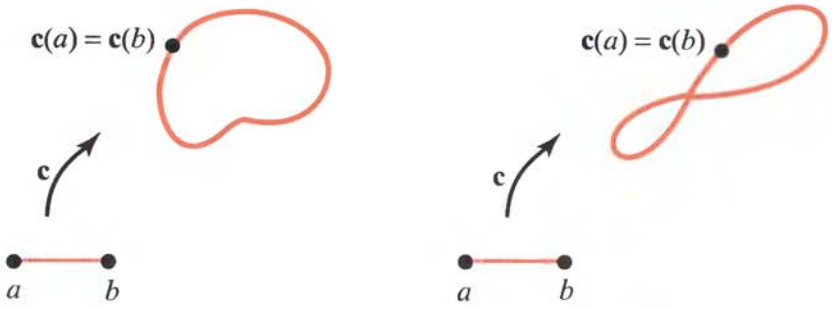



**Figure 7.2.8** A simple closed curve (left) and a closed curve that is not simple (right).

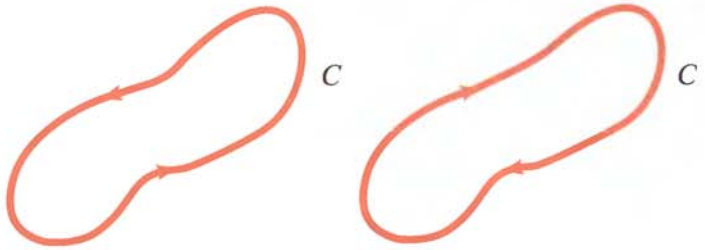



**Figure 7.2.9** Two possible orientations for a simple closed curve $C$.

If $C$ is an oriented simple curve or an oriented simple closed curve, we may unambiguously define line integrals along them.

**Line Integrals and Path Integrals Over Oriented Simple Curves and Simple Closed Curves** $C$:

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} \qquad \text{and} \qquad \int_C f\,ds = \int_{\mathbf{c}} f\,ds, \tag{3}$$

where $\mathbf{c}$ is any *orientation-preserving* parametrization of $C$.

These integrals do not depend on the choice of $\mathbf{c}$ as long as $\mathbf{c}$ is one-to-one (except possibly at the endpoints) by virtue of Theorems 1 and 2.[5] The point we want to make here is that *although a curve must be parametrized to make integration along it tractable, it is not necessary to include the parametrization in our notation for the integral.*

[5] We have not proved that any two one-to-one paths $\mathbf{c}$ and $\mathbf{p}$ with the same image must be reparametrizations of each other, but this technical point will be omitted.

**EXAMPLE 10** If $I = [a, b]$ is a closed interval on the $x$ axis, then $I$, as a curve, has two orientations: one corresponding to motion from $a$ to $b$ (left to right) and the other corresponding to motion from $b$ to $a$ (right to left). If $f$ is a real-valued function continuous on $I$, then denoting $I$ with the first orientation by $I^+$ and $I$ with the second orientation by $I^-$, we have

$$\int_{I^+} f(x)\,dx = \int_a^b f(x)\,dx = -\int_b^a f(x)\,dx = -\int_{I^-} f(x)\,dx. \quad \blacktriangle$$

A given simple closed curve can be parametrized in many different ways. Figure 7.2.10 shows $C$ represented as the image of a map $\mathbf{p}$, with $\mathbf{p}(t)$ progressing in a prescribed direction around an oriented curve $C$ as $t$ ranges from $a$ to $b$. Note that $\mathbf{p}'(t)$ points in this direction also. The speed with which we traverse $C$ may vary from parametrization to parametrization, but as long as the orientation is preserved, the integral will not, according to Theorems 1 and 2.

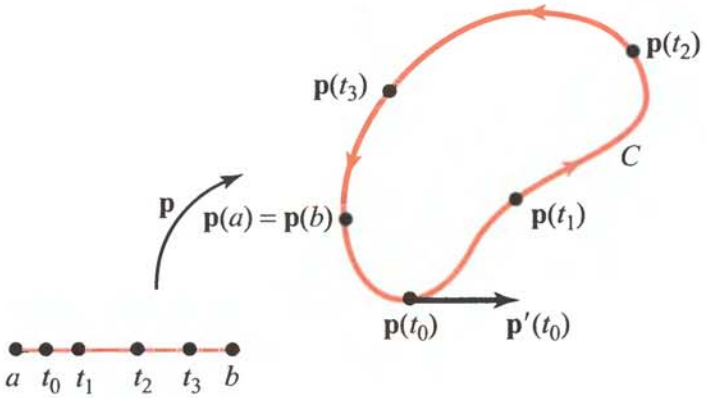



**Figure 7.2.10** As $t$ goes from $a$ to $b$, $\mathbf{p}(t)$ moves around the curve $C$ in some fixed direction.

The following precaution should be noted in regard to these remarks. It is possible to have two mappings $\mathbf{c}$ and $\mathbf{p}$ with the same image, and inducing the same orientation on the image, such that

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} \neq \int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s}.$$

For an example, let $\mathbf{c}(t) = (\cos t, \sin t, 0)$ and $\mathbf{p}(t) = (\cos 2t, \sin 2t, 0)$, $0 \leq t \leq 2\pi$, with $\mathbf{F}(x, y, z) = (y, 0, 0)$. Then $F_1(x, y, z) = y$, $F_2(x, y, z) = 0$, and $F_3(x, y, z) = 0$, so

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} F_1(\mathbf{c}(t)) \frac{dx}{dt}\,dt = -\int_0^{2\pi} \sin^2 t\,dt = -\pi.$$

But $\int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = -2\int_0^{2\pi} \sin^2 2t\,dt = -2\pi$. Clearly, $\mathbf{c}$ and $\mathbf{p}$ have the same image, namely, the unit circle in the $xy$ plane. Moreover, they traverse the unit circle in the same direction; yet $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} \neq \int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s}$. The reason for this is that $\mathbf{c}$ is one-to-one, but $\mathbf{p}$ is not ($\mathbf{p}$ traverses the unit circle *twice* in a counterclockwise direction); therefore, $\mathbf{p}$ is not a parametrization of the unit circle as a simple closed curve.

As a consequence of Theorem 1 and generalizing the notation in Example 10, we introduce the following convention:

**Line Integrals Over Curves with Opposite Orientations** Let $C^-$ be the same curve as $C$, but with the opposite orientation. Then

$$\int_C \mathbf{F} \cdot d\mathbf{s} = -\int_{C^-} \mathbf{F} \cdot d\mathbf{s}.$$

We also have:

**Line Integrals Over Curves Consisting of Several Components** Let $C$ be an oriented curve that is made up of several oriented component curves $C_i$, $i = 1, \ldots, k$, as in Figure 7.2.11. Then we shall write $C = C_1 + C_2 + \cdots + C_k$. Because we can parametrize $C$ by parametrizing the pieces $C_1, \ldots, C_k$ separately, one can prove that

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{C_1} \mathbf{F} \cdot d\mathbf{s} + \int_{C_2} \mathbf{F} \cdot d\mathbf{s} + \cdots + \int_{C_k} \mathbf{F} \cdot d\mathbf{s}. \tag{4}$$

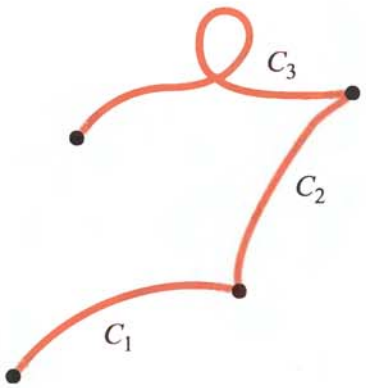



**Figure 7.2.11** A curve can be made up of several components.

One reason for writing a curve as a sum of components is that it may be easier to parametrize the components $C_i$ individually than it is to parametrize $C$ as a whole. If that is the case, formula (4) provides a convenient way of evaluating $\int_C \mathbf{F} \cdot d\mathbf{s}$.

## The $d\mathbf{r}$ Notation for Line Integrals

Sometimes one writes, as we occasionally do later, the line integral using the notation

$$\int_C \mathbf{F} \cdot d\mathbf{r}.$$

The reason is that we think of describing a $C^1$ path $\mathbf{c}$ in terms of a moving *position vector* based at the origin and ending at the point $\mathbf{c}(t)$ at time $t$. Position vectors are

often denoted by $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, and so the curve is described using the notation $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ in place of $\mathbf{c}(t)$. By definition, the line integral is given by

$$\int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt}\, dt.$$

Formally canceling the $dt$'s, and using the parametrization independence to replace the limits of integration with the geometric curve $C$, we arrive at the notation $\int_C \mathbf{F} \cdot d\mathbf{r}$.

**EXAMPLE 11** Consider $C$, the perimeter of the unit square in $\mathbb{R}^2$, oriented in the counterclockwise sense (see Figure 7.2.12). Evaluate the line integral

$$\int_C x^2\, dx + xy\, dy.$$

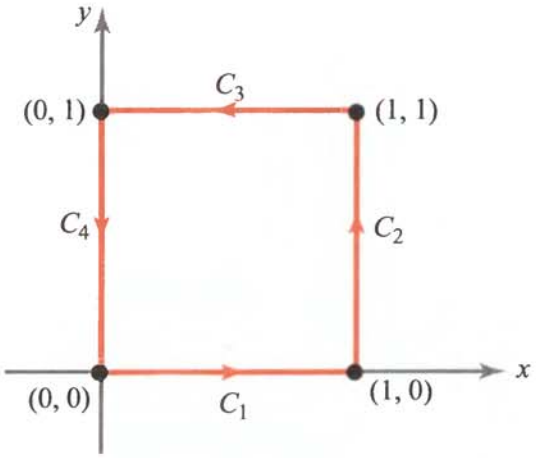



**Figure 7.2.12** The perimeter of the unit square, parametrized in four pieces.

SOLUTION We evaluate the integral using a convenient parametrization of $C$ that induces the given orientation. For example:

$$\mathbf{c}\colon [0, 4] \to \mathbb{R}^2, \qquad t \mapsto \begin{cases} (t, 0) & 0 \le t \le 1 \\ (1, t-1) & 1 \le t \le 2 \\ (3-t, 1) & 2 \le t \le 3 \\ (0, 4-t) & 3 \le t \le 4. \end{cases}$$

Then

$$\begin{aligned} \int_C x^2\, dx + xy\, dy &= \int_0^1 (t^2 + 0)\, dt + \int_1^2 [0 + (t-1)]\, dt \\ &\quad + \int_2^3 [-(3-t)^2 + 0]\, dt + \int_3^4 (0+0)\, dt \\ &= \frac{1}{3} + \frac{1}{2} + \left(-\frac{1}{3}\right) + 0 = \frac{1}{2}. \end{aligned}$$

Now let us reevaluate this line integral, using formula (4) and parametrizing the $C_i$ separately. Notice that $C = C_1 + C_2 + C_3 + C_4$, where $C_i$ are the oriented curves pictured in Figure 7.2.12. These can be parametrized as follows:

$$C_1\colon\ \mathbf{c}_1(t) = (t, 0),\ 0 \le t \le 1$$
$$C_2\colon\ \mathbf{c}_2(t) = (1, t),\ 0 \le t \le 1$$
$$C_3\colon\ \mathbf{c}_3(t) = (1 - t, 1),\ 0 \le t \le 1$$
$$C_4\colon\ \mathbf{c}_4(t) = (0, 1 - t),\ 0 \le t \le 1,$$

and so

$$\int_{C_1} x^2\,dx + xy\,dy = \int_0^1 t^2\,dt = \frac{1}{3}$$
$$\int_{C_2} x^2\,dx + xy\,dy = \int_0^1 t\,dt = \frac{1}{2}$$
$$\int_{C_3} x^2\,dx + xy\,dy = \int_0^1 -(1-t)^2\,dt = -\frac{1}{3}$$
$$\int_{C_4} x^2\,dx + xy\,dy = \int_0^1 0\,dt = 0.$$

Thus, again,

$$\int_C x^2\,dx + xy\,dy = \frac{1}{3} + \frac{1}{2} - \frac{1}{3} + 0 = \frac{1}{2}. \quad \blacktriangle$$

**EXAMPLE 12** An interesting application of the line integral is the mathematical formulation of Ampère's law, which relates electric currents to their magnetic effects.[6] Suppose $\mathbf{H}$ denotes a magnetic field in $\mathbb{R}^3$, and let $C$ be a closed oriented curve in $\mathbb{R}^3$. In appropriate physical units, Ampère's law states that

$$\int_C \mathbf{H} \cdot d\mathbf{s} = I,$$

where $I$ is the net current that passes through any surface bounded by $C$ (see Figure 7.2.13). $\blacktriangle$

Finally, let us mention that the line integral has another important physical meaning, specifically, the interpretation of $\int_C \mathbf{V} \cdot d\mathbf{s}$ as *circulation*, where $\mathbf{V}$ is the velocity

[6]The discovery that electric currents produce magnetic effects was made by Haas Christian Oersted circa 1820. See any elementary physics text for discussions of the physical basis of these ideas.

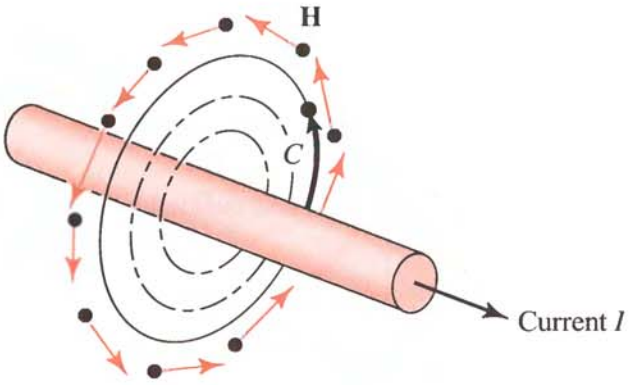



**Figure 7.2.13** The magnetic field **H** surrounding a wire carrying a current $I$ satisfies Ampère's law: $\int_C \mathbf{H} \cdot d\mathbf{s} = I$.

field of a fluid, as we shall discuss in Section 8.2. Thus, a wide variety of physical concepts, from the notion of work to electromagnetic fields and the motions of fluids, can be analyzed with the help of line integrals.

## EXERCISES

**1.** Let $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Evaluate the integral of $\mathbf{F}$ along each of the following paths:

(a) $\mathbf{c}(t) = (t, t, t)$, $0 \le t \le 1$
(b) $\mathbf{c}(t) = (\cos t, \sin t, 0)$, $0 \le t \le 2\pi$
(c) $\mathbf{c}(t) = (\sin t, 0, \cos t)$, $0 \le t \le 2\pi$
(d) $\mathbf{c}(t) = (t^2, 3t, 2t^3)$, $-1 \le t \le 2$

**2.** Evaluate each of the following line integrals:

(a) $\int_{\mathbf{c}} x\,dy - y\,dx$, $\mathbf{c}(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$
(b) $\int_{\mathbf{c}} x\,dx + y\,dy$, $\mathbf{c}(t) = (\cos \pi t, \sin \pi t)$, $0 \le t \le 2$
(c) $\int_{\mathbf{c}} yz\,dx + xz\,dy + xy\,dz$, where $\mathbf{c}$ consists of straight-line segments joining $(1, 0, 0)$ to $(0, 1, 0)$ to $(0, 0, 1)$
(d) $\int_{\mathbf{c}} x^2\,dx - xy\,dy + dz$, where $\mathbf{c}$ is the parabola $z = x^2$, $y = 0$ from $(-1, 0, 1)$ to $(1, 0, 1)$.

**3.** Consider the force field $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Compute the work done in moving a particle along the parabola $y = x^2$, $z = 0$, from $x = -1$ to $x = 2$.

**4.** Let $\mathbf{c}$ be a smooth path.

(a) Suppose $\mathbf{F}$ is perpendicular to $\mathbf{c}'(t)$ at the point $\mathbf{c}(t)$. Show that

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = 0.$$

(b) If $\mathbf{F}$ is parallel to $\mathbf{c}'(t)$ at $\mathbf{c}(t)$, show that

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \|\mathbf{F}\|\,ds.$$

[By parallel to $\mathbf{c}'(t)$ we mean that $\mathbf{F}(\mathbf{c}(t)) = \lambda(t)\mathbf{c}'(t)$, where $\lambda(t) > 0$.]

**5.** Suppose the path $\mathbf{c}$ has length $l$, and $\|\mathbf{F}\| \leq M$. Prove that

$$\left| \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} \right| \leq Ml.$$

**6.** Evaluate $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ where $\mathbf{F}(x, y, z) = y\mathbf{i} + 2x\mathbf{j} + y\mathbf{k}$ and the path $\mathbf{c}$ is defined by $\mathbf{c}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$, $0 \leq t \leq 1$.

**7.** Evaluate

$$\int_{\mathbf{c}} y\,dx + (3y^3 - x)\,dy + z\,dz$$

for each of the paths $\mathbf{c}(t) = (t, t^n, 0)$, $0 \leq t \leq 1$, where $n = 1, 2, 3, \ldots$.

**8.** This exercise refers to Example 12. Let $L$ be a very long wire, a planar section of which (with the plane perpendicular to the wire) is shown in Figure 7.2.14. Suppose this plane is the $xy$ plane. Experiments show that $\mathbf{H}$ is tangent to every circle in the $xy$ plane whose center is the axis of $L$, and that the magnitude of $\mathbf{H}$ is constant on every such circle $C$. Thus, $\mathbf{H} = H\mathbf{T}$, where $\mathbf{T}$ is a unit tangent vector to $C$ and $H$ is some scalar. Using this information, show that $H = I/2\pi r$, where $r$ is the radius of circle $C$ and $I$ is the current flowing in the wire.

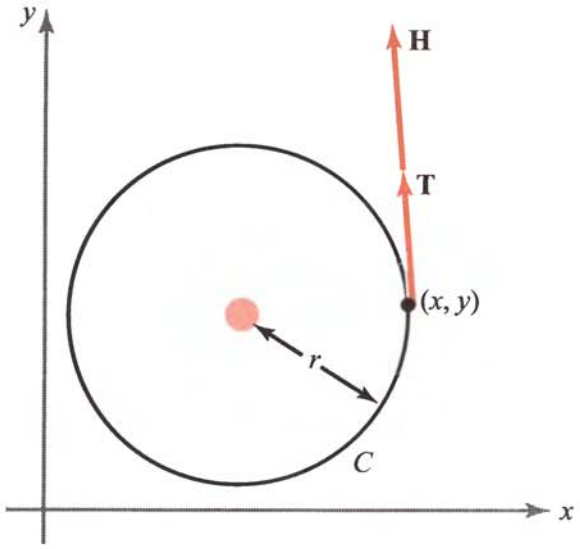



**Figure 7.2.14** A planar section of a long wire and a curve $C$ about the wire.

**9.** The image of the path $t \mapsto (\cos^3 t, \sin^3 t)$, $0 \leq t \leq 2\pi$ in the plane is shown in Figure 7.2.15. Evaluate the integral of the vector field $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$ around this curve.

**10.** Suppose $\mathbf{c}_1$ and $\mathbf{c}_2$ are two paths with the same endpoints and $\mathbf{F}$ is a vector field. Show that $\int_{\mathbf{c}_1} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}_2} \mathbf{F} \cdot d\mathbf{s}$ is equivalent to $\int_{c} \mathbf{F} \cdot d\mathbf{s} = 0$, where $C$ is the closed curve obtained by first moving along $\mathbf{c}_1$ and then moving along $\mathbf{c}_2$ in the opposite direction.

**11.** Let $\mathbf{c}(t)$ be a path and $\mathbf{T}$ the unit tangent vector. What is $\int_{\mathbf{c}} \mathbf{T} \cdot d\mathbf{s}$?

**12.** Let $\mathbf{F} = (z^3 + 2xy)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$. Show that the integral of $\mathbf{F}$ around the circumference of the unit square with vertices $(\pm 1, \pm 1)$ is zero.

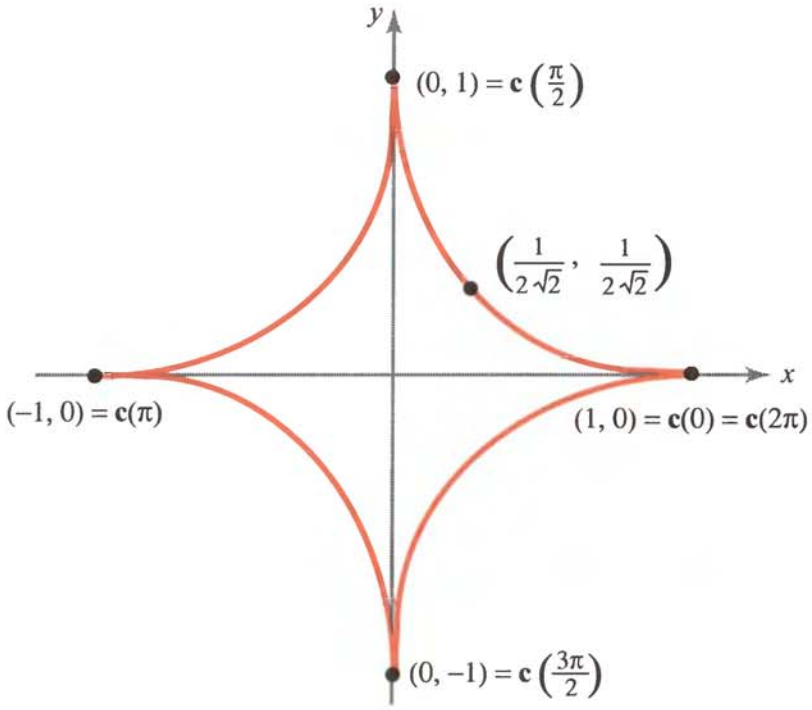



**Figure 7.2.15** The hypocycloid $\mathbf{c}(t) = (\cos^3 t, \sin^3 t)$ (Exercise 9).

**13.** Using the path in Exercise 9, observe that a $C^1$ map $\mathbf{c}$: $[a, b] \to \mathbb{R}^3$ can have an image that does not "look smooth." Do you think this could happen if $\mathbf{c}'(t)$ were always nonzero?

**14.** What is the value of the integral of a gradient field around a closed curve $C$?

**15.** Evaluate the line integral

$$\int_C 2xyz\,dx + x^2z\,dy + x^2y\,dz,$$

where $C$ is an oriented simple curve connecting $(1, 1, 1)$ to $(1, 2, 4)$.

**16.** Suppose $\nabla f(x, y, z) = 2xyze^{x^2}\mathbf{i} + ze^{x^2}\mathbf{j} + ye^{x^2}\mathbf{k}$. If $f(0, 0, 0) = 5$, find $f(1, 1, 2)$.

**17.** Consider the gravitational force field (with $G = m = M = 1$) defined [for $(x, y, z) \neq (0, 0, 0)$] by

$$\mathbf{F}(x, y, z) = -\frac{1}{(x^2 + y^2 + z^2)^{3/2}}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}).$$

Show that the work done by the gravitational force as a particle moves from $(x_1, y_1, z_1)$ to $(x_2, y_2, z_2)$ along any path depends only on the radii $R_1 = \sqrt{x_1^2 + y_1^2 + z_1^2}$ and $R_2 = \sqrt{x_2^2 + y_2^2 + z_2^2}$.

**18.** A cyclist rides up a mountain along the path shown in Figure 7.2.16. She makes one complete revolution around the mountain in reaching the top, while her vertical rate of climb is constant. Throughout the trip she exerts a force described by the vector field

$$\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j} + \mathbf{k}.$$

What is the work done by the cyclist in traveling from A to B? What is unrealistic about this model of a cyclist?

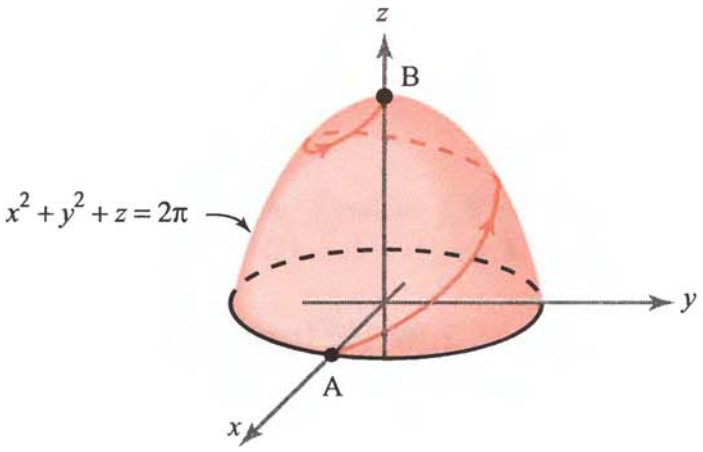



**Figure 7.2.16** How much work is done in cycling up this mountain?

**19.** Let $\mathbf{c}$: $[a, b] \to \mathbb{R}^3$ be a path such that $\mathbf{c}'(t) \neq \mathbf{0}$. Recall from Section 4.1 that when this condition holds, $\mathbf{c}$ is said to be ***regular***. Let the function $f$ be defined by the formula $f(x) = \int_a^x \|\mathbf{c}'(t)\|\, dt$.

(a) What is $df/dx$?

(b) Using the answer to part (a), prove that $f$: $[a, b] \to [0, L]$, where $L$ is the length of $\mathbf{c}$, has a differentiable inverse $g$: $[0, L] \to [a, b]$ satisfying $f \circ g(s) = s$, $g \circ f(x) = x$. (You may use the one-variable inverse function theorem stated at the beginning of Section 3.5.)

(c) Compute $dg/ds$.

(d) Recall that a path $s \mapsto \mathbf{b}(s)$ is said to be of unit speed, or parametrized by arc length, if $\|\mathbf{b}'(s)\| = 1$. Show that the reparametrization of $\mathbf{c}$ given by $\mathbf{b}(s) = \mathbf{c} \circ g(s)$ is of unit speed. Conclude that any regular path can be reparametrized by arc length. (Thus, for example, the Frenet formulas in Exercise 17 of Section 4.2 can be applied to the reparametrization $\mathbf{b}$.)

**20.** Along a "thermodynamic path" $C$ in $(V, T, P)$ space,

(i) The heat gained is $\int_C \Lambda_V\, dV + K_V\, dT$, where $\Lambda_V$, $K_V$ are functions of $(V, T, P)$, depending on the particular physical system.

(ii) The work done is $\int_C P\, dV$.

For a van der Waals gas, we have

$$P(V, T) = \frac{RT}{V - b} - \frac{a}{V^2}, \qquad J\Lambda_V = \frac{RT}{V - b}, \qquad \text{and} \qquad K_V = \text{constant},$$

where $R$, $b$, $a$, and $J$ are known constants. Initially the gas is at a temperature $T_0$ and volume $V_0$.

(a) An ***adiabatic*** process is a thermodynamic motion $(V(t), T(t), P(t))$ for which

$$\frac{dT}{dV} = \frac{dT/dt}{dV/dt} = -\frac{\Lambda_V}{K_V}.$$

If the van der Waals gas undergoes an adiabatic process in which the volume doubles to $2V_0$, compute

(1) the heat gained;
(2) the work done; and
(3) the final volume, temperature, and pressure.

(b) After the process indicated in part (a), the gas is cooled (or heated) at constant volume until the original temperature $T_0$ is achieved. Compute

(1) the heat gained;
(2) the work done; and
(3) the final volume, temperature, and pressure.

(c) After the process indicated in part (b), the gas is compressed until the gas returns to its original volume $V_0$. The temperature is held constant throughout the process. Compute

(1) the heat gained;
(2) the work done; and
(3) the final volume, temperature, and pressure.

(d) For the cyclic process described in parts (a), (b), (c), compute

(1) the total heat gained; and
(2) the total work done.

## 7.3 Parametrized Surfaces

In Sections 7.1 and 7.2, we studied integrals of scalar and vector functions along curves. Now we turn to integrals over surfaces and begin by studying the geometry of surfaces themselves.

### Graphs Are Too Restrictive

We are already used to one kind of surface, namely, the graph of a function $f(x, y)$. Graphs were extensively studied in Chapter 2, and we know how to compute their tangent planes. However, it would be unduly limiting to restrict ourselves to this case. For example, many surfaces arise as level surfaces of functions. Suppose our surface $S$ is the set of points $(x, y, z)$ where $x - z + z^3 = 0$. Here $S$ is a sheet that (relative to the $xy$ plane) doubles back on itself (see Figure 7.3.1). Obviously, we want to

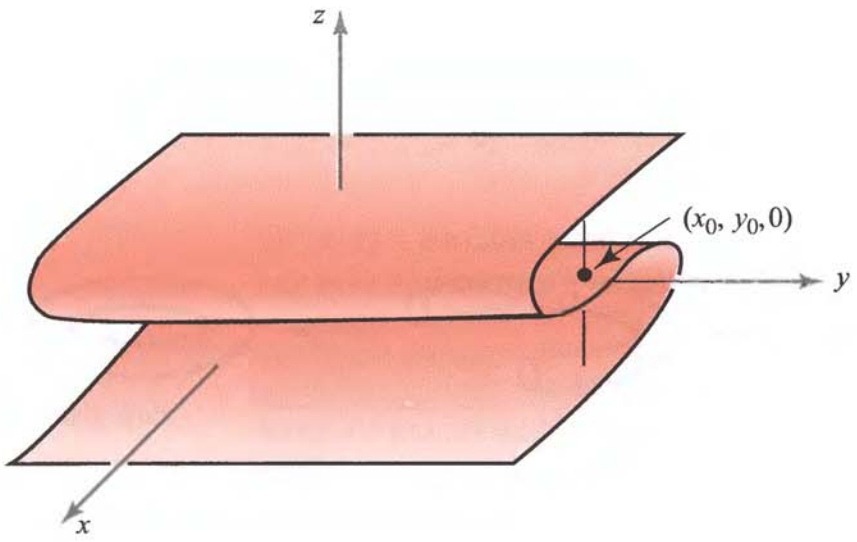



Figure 7.3.1 A surface that is not the graph of a function $z = f(x, y)$.

call $S$ a surface, because it is just a plane with a wrinkle. However, $S$ is *not* the graph of some function $z = f(x, y)$, because this means that for each $(x_0, y_0) \in \mathbb{R}^2$ there must be *one* $z_0$ with $(x_0, y_0, z_0) \in S$. As Figure 7.3.1 illustrates, this condition is violated.

Another example is the torus, or surface of a doughnut, which is depicted in Figure 7.3.2. Anyone would call a torus a surface; yet, by the same reasoning as before, a torus cannot be the graph of a differentiable function of two variables. These observations encourage us to extend our definition of a surface.

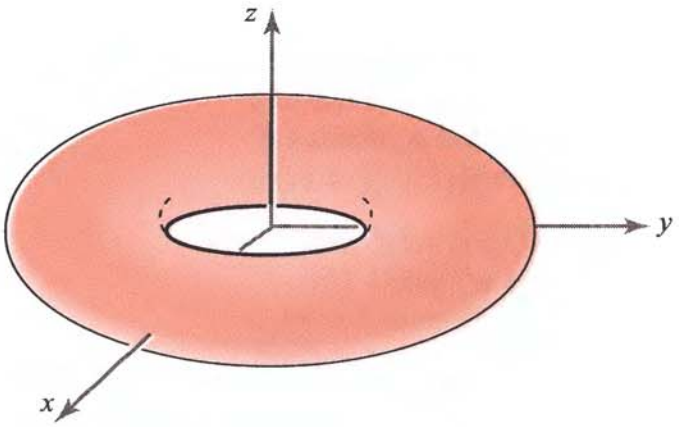



**Figure 7.3.2** The torus is not the graph of a function of the form $z = f(x, y)$.

The motivation for the extended definition that follows is partly that a surface can be thought of as being obtained from the plane by "rolling," "bending," and "pushing." For example, to get a torus, we take a portion of the plane and roll it (see Figure 7.3.3), then take the two "ends" and bring them together until they meet (Figure 7.3.4).

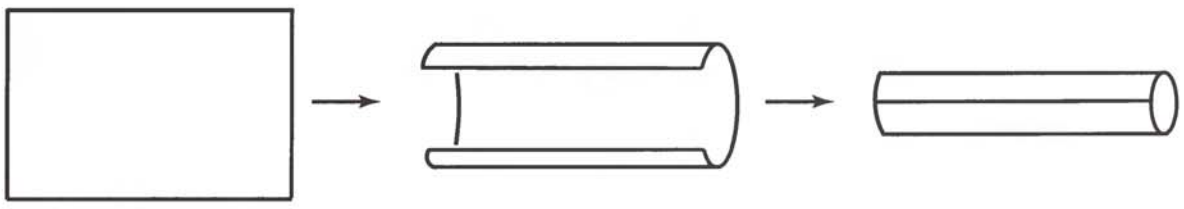


**Figure 7.3.3** The first step in obtaining a torus from a rectangle is to make a cylinder.

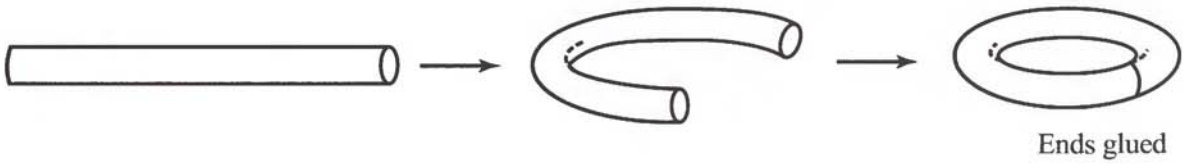



**Figure 7.3.4** Bend the cylinder and glue the ends to get a torus.

## Parametrized Surfaces as Mappings

In our study of differential calculus we dealt with mappings $f: A \subset \mathbb{R}^n \to \mathbb{R}^m$. Taking $n = 2$ and $m = 3$ corresponds to the case of a two-dimensional surface in 3-space. With surfaces, just as with curves, we want to distinguish a map (a parametrization) from its image (a geometric object). This leads us to the following definition.

**DEFINITION: Parametrized Surfaces** A ***parametrization of a surface*** is a function $\Phi$: $D \subset \mathbb{R}^2 \to \mathbb{R}^3$, where $D$ is some domain in $\mathbb{R}^2$. The ***surface*** $S$ corresponding to the function $\Phi$ is its image: $S = \Phi(D)$. We can write

$$\Phi(u, v) = (x(u, v), y(u, v), z(u, v)).$$

If $\Phi$ is differentiable or is of class $C^1$ [which is the same as saying that $x(u, v)$, $y(u, v)$, and $z(u, v)$ are differentiable or $C^1$ functions of $(u, v)$], we call $S$ a ***differentiable or a $C^1$ surface.***

We can think of $\Phi$ as twisting or bending the region $D$ in the plane to yield the surface $S$ (see Figure 7.3.5). Thus, each point $(u, v)$ in $D$ becomes a label for a point $(x(u, v), y(u, v), z(u, v))$ on $S$.

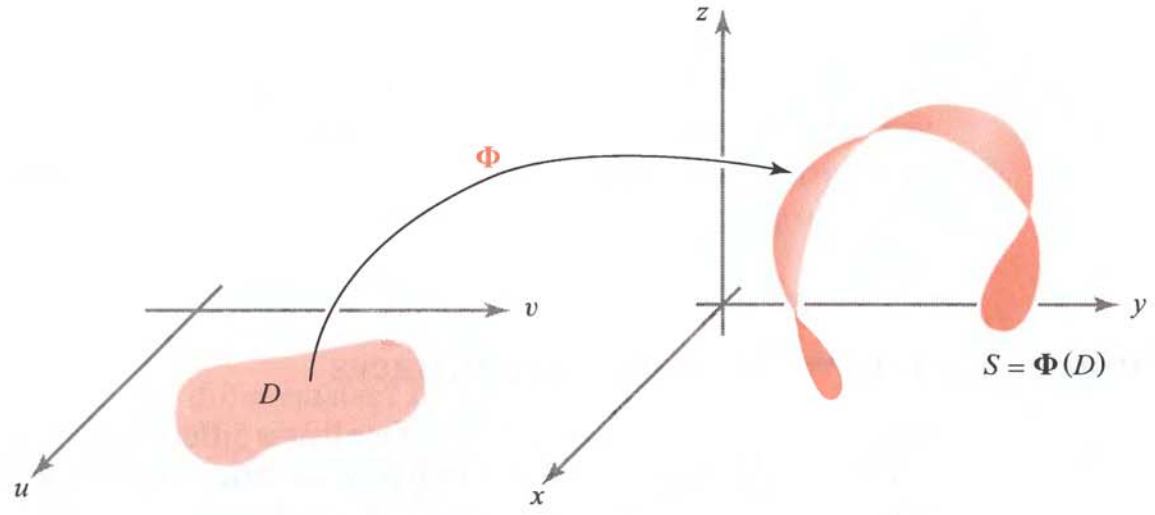



**Figure 7.3.5** $\Phi$ "twists" and "bends" $D$ onto the surface $S = \Phi(D)$.

Of course, surfaces need not bend or twist at all. In fact, planes are flat, as shown in our first, and simplest, example.

**EXAMPLE 1** In Section 1.3 we studied the equation of a plane $P$. We did so in terms of graphs and level sets. Now we examine the same notion using a parametrization.

Let $P$ be a plane that is parallel to two vectors $\alpha$ and $\beta$ and that passes through the tip of another vector $\gamma$, as in Figure 7.3.6.

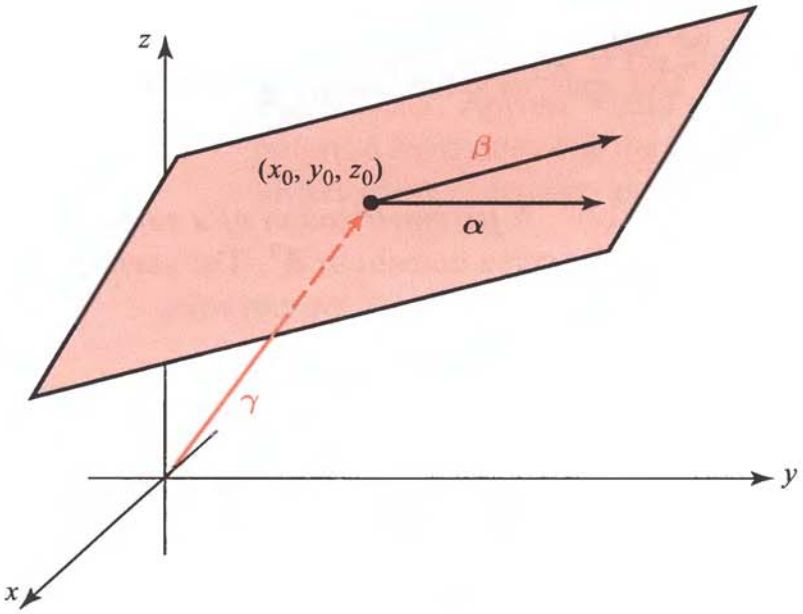



**Figure 7.3.6** Describing a plane parametrically.

Our goal in this example is to find a parametrization of this plane. Notice that the vector $\alpha \times \beta = \mathbf{N}$, which we also write as $A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$, is normal to $P$. If the tip of $\gamma$ is the point $(x_0, y_0, z_0)$, then the equation of $P$ as a level set (as discussed in Section 1.3) is given by:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

However, the set of all points on the plane $P$ can also be described by the set of all vectors that are $\gamma$ plus a linear combination of $\alpha$ and $\beta$. Using our preferred choice of real parameters $u$ and $v$, we arrive at the ***parametric equation of the plane*** $P$:

$$\Phi(u, v) = \alpha u + \beta v + \gamma. \quad \blacktriangle$$

## Tangent Vectors to Parametrized Surfaces

Suppose that $\Phi$ is a parametrized surface that is differentiable at $(u_0, v_0) \in \mathbb{R}^2$. Fixing $u$ at $u_0$, we get a map $\mathbb{R} \to \mathbb{R}^3$ given by $t \mapsto \Phi(u_0, t)$, whose image is a curve on the surface (Figure 7.3.7). From Chapters 2 and 4 we know that the vector tangent to this curve at the point $\Phi(u_0, v_0)$, which we denote by $\mathbf{T}_v$, is given by

$$\mathbf{T}_v = \frac{\partial \Phi}{\partial v} = \frac{\partial x}{\partial v}(u_0, v_0)\mathbf{i} + \frac{\partial y}{\partial v}(u_0, v_0)\mathbf{j} + \frac{\partial z}{\partial v}(u_0, v_0)\mathbf{k}.$$

Similarly, if we fix $v$ and consider the curve $t \mapsto \Phi(t, v_0)$, we obtain the tangent vector to this curve at $\Phi(u_0, v_0)$, given by

$$\mathbf{T}_u = \frac{\partial \Phi}{\partial u} = \frac{\partial x}{\partial u}(u_0, v_0)\mathbf{i} + \frac{\partial y}{\partial u}(u_0, v_0)\mathbf{j} + \frac{\partial z}{\partial u}(u_0, v_0)\mathbf{k}.$$

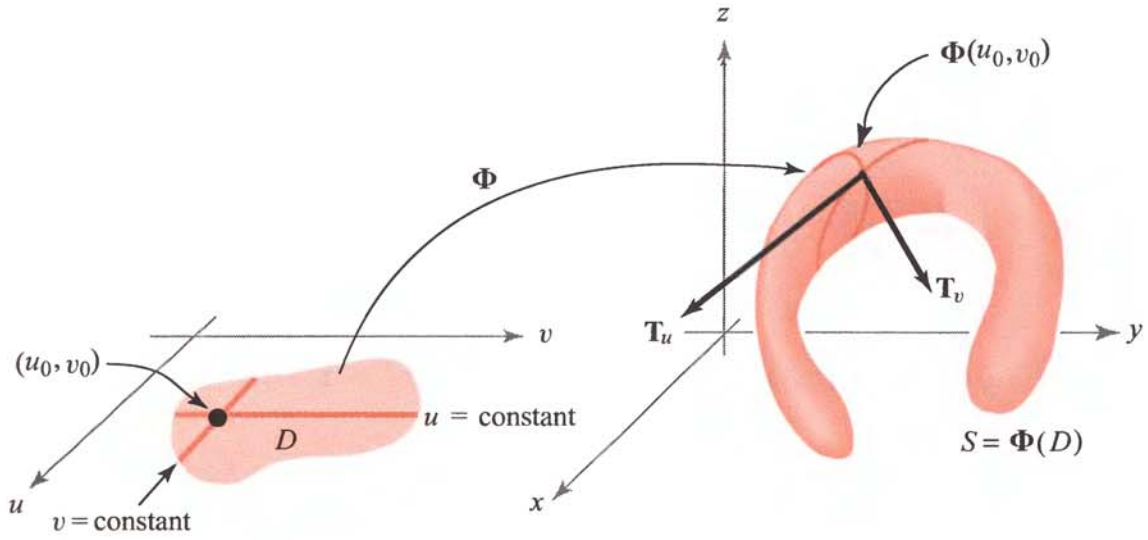



Figure 7.3.7 The tangent vectors $\mathbf{T}_u$ and $\mathbf{T}_v$ that are tangent to the curve on a surface $S$, and hence tangent to $S$.

## Regular Surfaces

Because the vectors $\mathbf{T}_u$ and $\mathbf{T}_v$ are tangent to two curves on the surface at a given point, the vector $\mathbf{T}_u \times \mathbf{T}_v$ ought to be normal to the surface at the same point.

We say that the surface $S$ is ***regular*** or ***smooth***[7] at $\Phi(u_0, v_0)$, provided that $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$ at $(u_0, v_0)$. The surface is called ***regular*** if it is regular at all points $\Phi(u_0, v_0) \in S$. The nonzero vector $\mathbf{T}_u \times \mathbf{T}_v$ is *normal* to $S$ (recall that the vector product of $\mathbf{T}_u$ and $\mathbf{T}_v$ is perpendicular to the plane spanned by $\mathbf{T}_u$ and $\mathbf{T}_v$); the fact that it is nonzero ensures that there will be a tangent plane. Intuitively, a smooth surface has no "corners."[8]

**EXAMPLE 2** Consider the surface given by the equations

$$x = u\cos v, \qquad y = u\sin v, \qquad z = u, \qquad u \geq 0.$$

Is this surface differentiable? Is it regular?

SOLUTION These equations describe the surface $z = \sqrt{x^2 + y^2}$ (square the equations for $x$, $y$, and $z$ to check this), which is shown in Figure 7.3.8. This surface is a cone with a "point" at (0, 0, 0); it is a differentiable surface because each component function is differentiable as a function of $u$ and $v$. However, *the surface*

[7]Strictly speaking, regularity depends on the parametrization $\Phi$ and not just on its image $S$. Therefore, this terminology is somewhat imprecise; however, it is descriptive and should not cause confusion. (See Exercise 15.)

[8]In Section 3.5, we showed that level surfaces $f(x, y, z) = 0$ were in fact graphs of functions of two variables in some neighborhood of a point $(x_0, y_0, z_0)$ satisfying $\nabla f(x_0, y_0, z_0) \neq 0$. This united two concepts of a surface—graphs and level sets. Again, using the implicit function theorem, it is likewise possible to show that the image of a parametrized surface $\Phi$ in the neighborhood of a point $(u_0, v_0)$ where $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$ is also the graph of a function of two variables. Thus, all definitions of a surface are consistent. (See Exercise 16.)

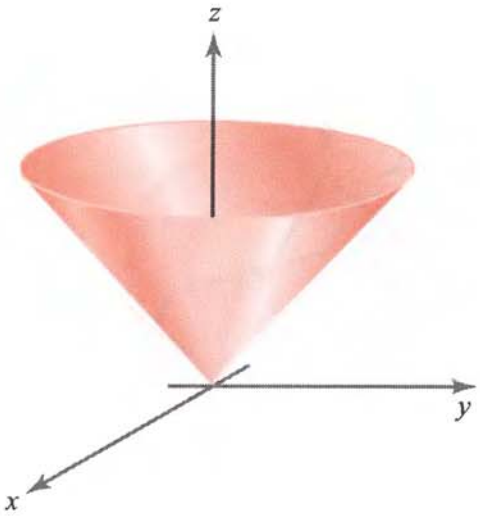



**Figure 7.3.8** The surface $z = \sqrt{x^2 + y^2}$ is a cone. It is not regular at its tip.

*is not regular at* $(0, 0, 0)$. To see this, compute $\mathbf{T}_u$ and $\mathbf{T}_v$ at $(0, 0) \in \mathbb{R}^2$:

$$\mathbf{T}_u = \frac{\partial \Phi}{\partial u} = \frac{\partial x}{\partial u}(0,0)\mathbf{i} + \frac{\partial y}{\partial u}(0,0)\mathbf{j} + \frac{\partial z}{\partial u}(0,0)\mathbf{k} = (\cos 0)\mathbf{i} + (\sin 0)\mathbf{j} + \mathbf{k} = \mathbf{i} + \mathbf{k},$$

and similarly,

$$\mathbf{T}_v = \frac{\partial \Phi}{\partial v} = 0(-\sin 0)\mathbf{i} + 0(\cos 0)\mathbf{j} + 0\mathbf{k} = \mathbf{0}.$$

Thus, $\mathbf{T}_u \times \mathbf{T}_v = \mathbf{0}$, and so, by definition, the surface is not regular at $(0, 0, 0)$. ▲

## Tangent Plane to a Parametrized Surface

We can use the fact that $\mathbf{n} = \mathbf{T}_u \times \mathbf{T}_v$ is normal to the surface to both formally define the tangent plane and to compute it.

**DEFINITION: The Tangent Plane to a Surface** If a parametrized surface $\Phi$: $D \subset \mathbb{R}^2 \to \mathbb{R}^3$ is *regular* at $\Phi(u_0, v_0)$, that is, if $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$ at $(u_0, v_0)$, we define the ***tangent plane*** of the surface at $\Phi(u_0, v_0)$ to be the plane determined by the vectors $\mathbf{T}_u$ and $\mathbf{T}_v$. Thus, $\mathbf{n} = \mathbf{T}_u \times \mathbf{T}_v$ is a normal vector, and an equation of the tangent plane at $(x_0, y_0, z_0)$ on the surface is given by

$$(x - x_0, y - y_0, z - z_0) \cdot \mathbf{n} = 0, \tag{1}$$

where $\mathbf{n}$ is evaluated at $(u_0, v_0)$; that is, the tangent plane is the set of $(x, y, z)$ satisfying (1). If $\mathbf{n} = (n_1, n_2, n_3) = n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k}$, then formula (1) becomes

$$n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0. \tag{1'}$$

**EXAMPLE 3** Let $\Phi\colon \mathbb{R}^2 \to \mathbb{R}^3$ be given by

$$x = u\cos v, \qquad y = u\sin v, \qquad z = u^2 + v^2.$$

Where does a tangent plane exist? Find the tangent plane at $\Phi(1, 0)$.

SOLUTION We compute

$$\mathbf{T}_u = (\cos v)\mathbf{i} + (\sin v)\mathbf{j} + 2u\mathbf{k} \qquad \text{and} \qquad \mathbf{T}_v = -u(\sin v)\mathbf{i} + u(\cos v)\mathbf{j} + 2v\mathbf{k},$$

so the tangent plane at the point $\Phi(u_0, v_0)$ is the set of vectors through $\Phi(u_0, v_0)$ perpendicular to

$$(\mathbf{T}_u \times \mathbf{T}_v)(u_0, v_0) = (-2u_0^2 \cos v_0 + 2v_0 \sin v_0, -2u_0^2 \sin v_0 - 2v_0 \cos v_0, u_0)$$

if this vector is nonzero. Because $\mathbf{T}_u \times \mathbf{T}_v$ is equal to $\mathbf{0}$ at $(u_0, v_0) = (0, 0)$, we cannot find a tangent plane at $\Phi(0, 0) = (0, 0, 0)$. However, we can find an equation of the tangent plane at all the other points, where $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$. At the point $\Phi(1, 0) = (1, 0, 1)$,

$$\mathbf{n} = (\mathbf{T}_u \times \mathbf{T}_v)(1, 0) = (-2, 0, 1) = -2\mathbf{i} + \mathbf{k}.$$

Because we have the vector $\mathbf{n}$ normal to the surface and a point $(1, 0, 1)$ on the surface, we can use formula $(1')$ to obtain an equation of the tangent plane:

$$-2(x - 1) + (z - 1) = 0; \text{ that is, } z = 2x - 1. \quad \blacktriangle$$

**EXAMPLE 4** Suppose a surface $S$ is the graph of a differentiable function $g\colon \mathbb{R}^2 \to \mathbb{R}$. Write $S$ in parametric form and show that the surface is smooth at all points $(u_0, v_0, g(u_0, v_0)) \in \mathbb{R}^3$.

SOLUTION Write $S$ in parametric form as follows:

$$x = u, \qquad y = v, \qquad z = g(u, v),$$

which is the same as $z = g(x, y)$. Then at the point $(u_0, v_0)$,

$$\mathbf{T}_u = \mathbf{i} + \frac{\partial g}{\partial u}(u_0, v_0)\mathbf{k} \qquad \text{and} \qquad \mathbf{T}_v = \mathbf{j} + \frac{\partial g}{\partial v}(u_0, v_0)\mathbf{k},$$

and for $(u_0, v_0) \in \mathbb{R}^2$,

$$\mathbf{n} = \mathbf{T}_u \times \mathbf{T}_v = -\frac{\partial g}{\partial u}(u_0, v_0)\mathbf{i} - \frac{\partial g}{\partial v}(u_0, v_0)\mathbf{j} + \mathbf{k} \neq \mathbf{0}. \tag{2}$$

This is nonzero because the coefficient of $\mathbf{k}$ is 1; consequently, the parametrization $(u, v) \mapsto (u, v, g(u, v))$ is regular at all points. Moreover, the tangent plane at the point $(x_0, y_0, z_0) = (u_0, v_0, g(u_0, v_0))$ is given, by formula (1), as

$$(x - x_0, y - y_0, z - z_0) \cdot \left( -\frac{\partial g}{\partial u}, -\frac{\partial g}{\partial v}, 1 \right) = 0,$$

where the partial derivatives are evaluated at $(u_0, v_0)$. Remembering that $x = u$ and $y = v$, we can write this as

$$z - z_0 = \left( \frac{\partial g}{\partial x} \right)(x - x_0) + \left( \frac{\partial g}{\partial y} \right)(y - y_0), \tag{3}$$

where $\partial g/\partial x$ and $\partial g/\partial y$ are evaluated at $(x_0, y_0)$. ▲

This example also shows that the definition of the tangent plane for parametrized surfaces agrees with the one for surfaces obtained as graphs, because equation (3) is the same formula we derived (in Chapter 2) for the plane tangent to $S$ at the point $(x_0, y_0, z_0) \in S$.

It is also useful to consider piecewise smooth surfaces, that is, surfaces composed of a certain number of images of smooth parametrized surfaces. For example, the surface of a cube in $\mathbb{R}^3$ is such a surface. These surfaces are considered in Section 7.4.

**EXAMPLE 5** Find a parametrization for the hyperboloid of one sheet:

$$x^2 + y^2 - z^2 = 1.$$

SOLUTION Because $x$ and $y$ appear in the combination $x^2 + y^2$, the surface is invariant under rotation about the $z$ axis, and so it is natural to write

$$x = r \cos\theta, \qquad y = r \sin\theta.$$

Then $x^2 + y^2 - z^2 = 1$ becomes $r^2 - z^2 = 1$. This we can conveniently parametrize by[9]

$$r = \cosh u, \qquad z = \sinh u.$$

Thus, a parametrization is

$$x = (\cosh u)(\cos\theta), \qquad y = (\cosh u)(\sin\theta), \qquad z = \sinh u,$$

where $0 \leq \theta < 2\pi$, $-\infty < u < \infty$. ▲

[9]Recall from one-variable calculus that $\cosh u = (e^u + e^{-u})/2$ and $\sinh u = (e^u - e^{-u})/2$. One easily verifies from these definitions that $\cosh^2 u - \sinh^2 u = 1$.

## EXERCISES

*In Exercises 1 to 3, find an equation for the plane tangent to the given surface at the specified point.*

**1.** $x = 2u, \quad y = u^2 + v, \quad z = v^2, \quad at\ (0, 1, 1)$

**2.** $x = u^2 - v^2, \quad y = u + v, \quad z = u^2 + 4v, \quad at\ (-\frac{1}{4}, \frac{1}{2}, 2)$

**3.** $x = u^2, \quad y = u \sin e^v, \quad z = \frac{1}{3} u \cos e^v, \quad at\ (13, -2, 1)$

**4.** At what points are the surfaces in Exercises 1 and 2 regular?

**5.** Find an expression for a unit vector normal to the surface

$$x = \cos v \sin u, \quad y = \sin v \sin u, \quad z = \cos u$$

at the image of a point $(u, v)$ for $u$ in $[0, \pi]$ and $v$ in $[0, 2\pi]$. Identify this surface.

**6.** Repeat Exercise 5 for the surface

$$x = 3 \cos\theta \sin\phi, \quad y = 2 \sin\theta \sin\phi, \quad z = \cos\phi$$

for $\theta$ in $[0, 2\pi]$ and $\phi$ in $[0, \pi]$.

**7.** Repeat Exercise 5 for the surface

$$x = \sin v, \quad y = u, \quad z = \cos v$$

for $0 \leq v \leq 2\pi$ and $-1 \leq u \leq 3$.

**8.** Repeat Exercise 5 for the surface

$$x = (2 - \cos v) \cos u, \quad y = (2 - \cos v) \sin u, \quad z = \sin v$$

for $-\pi \leq u \leq \pi$, $-\pi \leq v \leq \pi$. Is this surface regular?

**9.** (a) Develop a formula for the plane tangent to the surface $x = h(y, z)$.
(b) Obtain a similar formula for $y = k(x, z)$.

**10.** Find the equation of the plane tangent to the surface $x = u^2, y = v^2, z = u^2 + v^2$ at the point $u = 1, v = 1$.

**11.** Find a parametrization of the surface $z = 3x^2 + 8xy$ and use it to find the tangent plane at $x = 1, y = 0, z = 3$. Compare your answer with that using graphs.

**12.** Find a parametrization of the surface $x^3 + 3xy + z^2 = 2, z > 0$, and use it to find the tangent plane at the point $x = 1, y = 1/3, z = 0$. Compare your answer with that using level sets.

**13.** Consider the surface in $\mathbb{R}^3$ parametrized by

$$\Phi(r, \theta) = (r \cos\theta, r \sin\theta, \theta), \quad 0 \leq r \leq 1 \quad \text{and} \quad 0 \leq \theta \leq 4\pi.$$

(a) Sketch and describe the surface.
(b) Find an expression for a unit normal to the surface.
(c) Find an equation for the plane tangent to the surface at the point $(x_0, y_0, z_0)$.
(d) If $(x_0, y_0, z_0)$ is a point on the surface, show that the horizontal line segment of unit length from the $z$ axis through $(x_0, y_0, z_0)$ is contained in the surface and in the plane tangent to the surface at $(x_0, y_0, z_0)$.

**14.** Given a sphere of radius 2 centered at the origin, find the equation for the plane that is tangent to it at the point $(1, 1, \sqrt{2})$ by considering the sphere as:

(a) a surface parametrized by $\Phi(\theta, \phi) = (2\cos\theta\sin\phi, 2\sin\theta\sin\phi, 2\cos\phi)$;
(b) a level surface of $f(x, y, z) = x^2 + y^2 + z^2$; and
(c) the graph of $g(x, y) = \sqrt{4 - x^2 - y^2}$.

**15.** (a) Find a parametrization for the hyperboloid $x^2 + y^2 - z^2 = 25$.
(b) Find an expression for a unit normal to this surface.
(c) Find an equation for the plane tangent to the surface at $(x_0, y_0, 0)$, where $x_0^2 + y_0^2 = 25$.
(d) Show that the lines $(x_0, y_0, 0) + t(-y_0, x_0, 5)$ and $(x_0, y_0, 0) + t(y_0, -x_0, 5)$ lie in the surface *and* in the tangent plane found in part (c).

**16.** A parametrized surface is described by a differentiable function $\Phi\colon \mathbb{R}^2 \to \mathbb{R}^3$. According to Chapter 2, the derivative should give a linear approximation that yields a representation of the tangent plane. This exercise demonstrates that this is indeed the case.

(a) Assuming $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$, show that the range of the linear transformation $\mathbf{D}\Phi(u_0, v_0)$ is the plane spanned by $\mathbf{T}_u$ and $\mathbf{T}_v$. [Here $\mathbf{T}_u$ and $\mathbf{T}_v$ are evaluated at $(u_0, v_0)$.]
(b) Show that $\mathbf{w} \perp (\mathbf{T}_u \times \mathbf{T}_v)$ if and only if $\mathbf{w}$ is in the range of $\mathbf{D}\Phi(u_0, v_0)$.
(c) Show that the tangent plane as defined in this section is the same as the "parametrized plane"

$$(u, v) \mapsto \Phi(u_0, v_0) + \mathbf{D}\Phi(u_0, v_0)\begin{bmatrix} u - u_0 \\ v - v_0 \end{bmatrix}.$$

**17.** Consider the surfaces $\Phi_1(u, v) = (u, v, 0)$ and $\Phi_2(u, v) = (u^3, v^3, 0)$.

(a) Show that the image of $\Phi_1$ and of $\Phi_2$ is the $xy$ plane.
(b) Show that $\Phi_1$ describes a regular surface, yet $\Phi_2$ does not. Conclude that the notion of regularity of a surface $S$ depends on the existence of at least one regular parametrization for $S$.
(c) Prove that the tangent plane of $S$ is well defined independently of the regular (one-to-one) parametrization (you will need to use the inverse function theorem from Section 3.5).
(d) After these remarks, do you think you can find a regular parametrization of the cone of Figure 7.3.7?

**18.** Let $\Phi$ be a regular surface at $(u_0, v_0)$; that is, $\Phi$ is of class $C^1$ and $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$ at $(u_0, v_0)$.

(a) Use the implicit function theorem (Section 3.5) to show that the image of $\Phi$ *near* $(u_0, v_0)$ is the graph of a $C^1$ function of two variables. If the $z$ component of $\mathbf{T}_u \times \mathbf{T}_v$ is nonzero, we can write it as $z = f(x, y)$.
(b) Show that the tangent plane at $\Phi(u_0, v_0)$ defined by the plane spanned by $\mathbf{T}_u$ and $\mathbf{T}_v$ coincides with the tangent plane of the graph of $z = f(x, y)$ at this point.

# 7.4 Area of a Surface

Before proceeding to general surface integrals, let us first consider the problem of computing the area of a surface, just as we considered the problem of finding the arc length of a curve before discussing path integrals.

## Definition of Surface Area

In Section 7.3, we defined a parametrized surface $S$ to be the *image* of a function $\Phi$: $D \subset \mathbb{R}^2 \to \mathbb{R}^3$, written as $\Phi(u, v) = (x(u, v), y(u, v), z(u, v))$. The map $\Phi$ was called the parametrization of $S$ and $S$ was said to be regular at $\Phi(u, v) \in S$ provided that $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$, where

$$\mathbf{T}_u = \frac{\partial x}{\partial u}(u, v)\mathbf{i} + \frac{\partial y}{\partial u}(u, v)\mathbf{j} + \frac{\partial z}{\partial u}(u, v)\mathbf{k}$$

and

$$\mathbf{T}_v = \frac{\partial x}{\partial v}(u, v)\mathbf{i} + \frac{\partial y}{\partial v}(u, v)\mathbf{j} + \frac{\partial z}{\partial v}(u, v)\mathbf{k}.$$

Recall that a regular surface (loosely speaking) is one that has no corners or breaks.

In the rest of this chapter and in the next one, we shall consider only piecewise regular surfaces that are unions of images of parametrized surfaces $\Phi_i$: $D_i \to \mathbb{R}^3$ for which:

(i) $D_i$ is an elementary region in the plane;

(ii) $\Phi_i$ is of class $C^1$ and one-to-one, except possibly on the boundary of $D_i$; and

(iii) $S_i$, the image of $\Phi_i$ is regular, except possibly at a finite number of points.

**DEFINITION: Area of a Parametrized Surface** We define the ***surface area***[10] $A(S)$ of a parametrized surface by

$$A(S) = \iint_D \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv \tag{1}$$

where $\|\mathbf{T}_u \times \mathbf{T}_v\|$ is the norm of $\mathbf{T}_u \times \mathbf{T}_v$. If $S$ is a union of surfaces $S_i$, its area is the sum of the areas of the $S_i$.

[10] As we have not yet discussed the independence of parametrization, it may seem that $A(S)$ depends on the parametrization $\Phi$. We shall discuss independence of parametrization in Section 7.6; the use of this notation here should not cause confusion.

As the reader can easily verify, we have

$$\|\mathbf{T}_u \times \mathbf{T}_v\| = \sqrt{\left[\frac{\partial(x, y)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(y, z)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)}\right]^2}, \tag{2}$$

where

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix},$$

and so on. Thus, formula (1) becomes

$$A(S) = \iint_D \sqrt{\left[\frac{\partial(x, y)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(y, z)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)}\right]^2}\, du\, dv. \tag{3}$$

## Justification of the Area Formula

We can justify the definition of surface area by analyzing the integral $\iint_D \|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv$ in terms of Riemann sums. For simplicity, suppose $D$ is a rectangle; consider the $n$th regular partition of $D$, and let $R_{ij}$ be the $ij$th rectangle in the partition, with vertices $(u_i, v_j)$, $(u_{i+1}, v_j)$, $(u_i, v_{j+1})$, and $(u_{i+1}, v_{j+1})$, $0 \le i \le n-1$, $0 \le j \le n-1$. Denote the values of $\mathbf{T}_u$ and $\mathbf{T}_v$ at $(u_i, v_j)$ by $\mathbf{T}_{u_i}$ and $\mathbf{T}_{v_j}$. We can think of the vectors $\Delta u \mathbf{T}_{u_i}$ and $\Delta v \mathbf{T}_{v_j}$ as tangent to the surface at $\Phi(u_i, v_j) = (x_{ij}, y_{ij}, z_{ij})$, where $\Delta u = u_{i+1} - u_i$, $\Delta v = v_{j+1} - v_j$. Then these vectors form a parallelogram $P_{ij}$ that lies in the plane tangent to the surface at $(x_{ij}, y_{ij}, z_{ij})$ (see Figure 7.4.1).

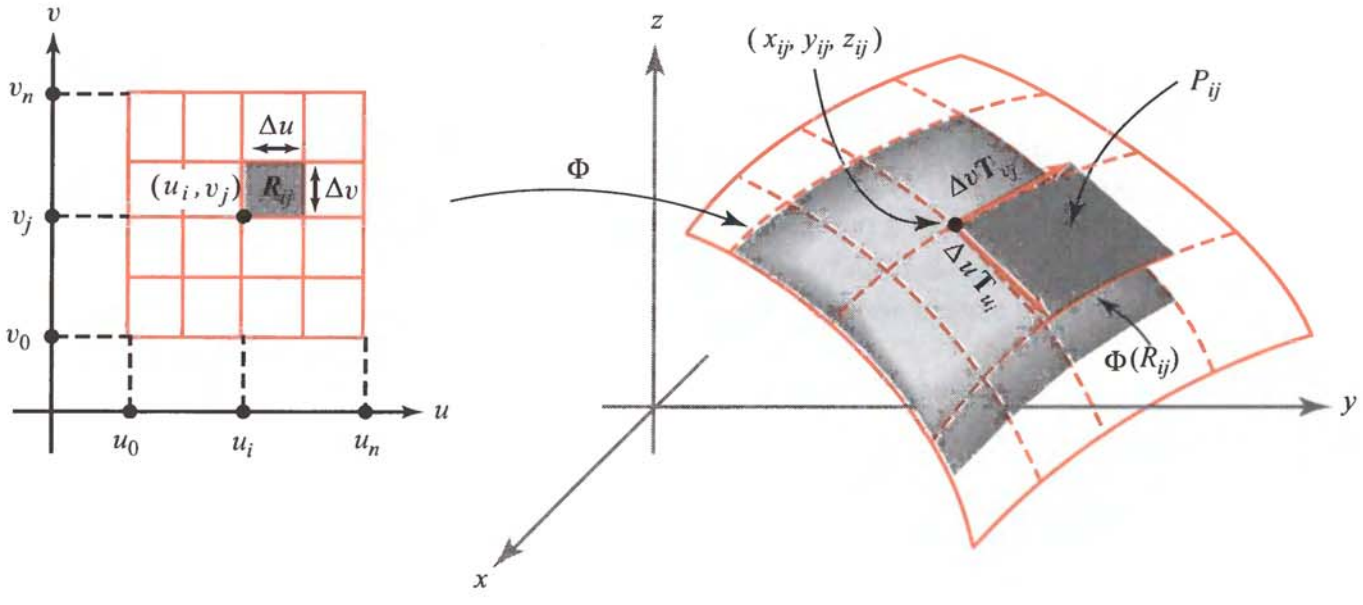



**Figure 7.4.1** $\|\mathbf{T}_{u_i} \times \mathbf{T}_{v_j}\|\, \Delta u\, \Delta v$ is equal to the area of a parallelogram that approximates the area of a patch on a surface $S = \Phi(D)$.

We thus have a "patchwork cover" of the surface by the $P_{ij}$. For $n$ large, the area of $P_{ij}$ is a good approximation to the area of $\Phi(R_{ij})$. Because the area of the parallelogram spanned by two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is $\|\mathbf{v}_1 \times \mathbf{v}_2\|$ (see Chapter 1), we see that

$$A(P_{ij}) = \|\Delta u \mathbf{T}_{u_i} \times \Delta v \mathbf{T}_{v_j}\| = \|\mathbf{T}_{u_i} \times \mathbf{T}_{v_j}\| \Delta u \, \Delta v.$$

Therefore, the area of the patchwork cover is

$$A_n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} A(P_{ij}) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \|\mathbf{T}_{u_i} \times \mathbf{T}_{v_j}\| \, \Delta u \, \Delta v.$$

As $n \to \infty$, the sums $A_n$ converge to the integral

$$\iint_D \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv.$$

Because $A_n$ should approximate the surface area better and better as $n \to \infty$, we are led to formula (1) as a reasonable definition of $A(S)$.

**EXAMPLE 1** Let $D$ be the region determined by $0 \le \theta \le 2\pi$, $0 \le r \le 1$ and let the function $\Phi$: $D \to \mathbb{R}^3$, defined by

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = r$$

be a parametrization of a cone $S$ (see Figure 7.3.8). Find its surface area.

SOLUTION In formula (3),

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r,$$

$$\frac{\partial(y, z)}{\partial(r, \theta)} = \begin{vmatrix} \sin\theta & r\cos\theta \\ 1 & 0 \end{vmatrix} = -r\cos\theta,$$

and

$$\frac{\partial(x, z)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ 1 & 0 \end{vmatrix} = r\sin\theta,$$

so the area integrand is

$$\|\mathbf{T}_r \times \mathbf{T}_\theta\| = \sqrt{r^2 + r^2\cos^2\theta + r^2\sin^2\theta} = r\sqrt{2}.$$

Clearly, $\|\mathbf{T}_r \times \mathbf{T}_\theta\|$ vanishes for $r = 0$, but $\Phi(0, \theta) = (0, 0, 0)$ for any $\theta$. Thus, $(0, 0, 0)$ is the only point where the surface is not regular. We have

$$\iint_D \|\mathbf{T}_r \times \mathbf{T}_\theta\| \, dr \, d\theta = \int_0^{2\pi} \int_0^1 \sqrt{2} r \, dr \, d\theta = \int_0^{2\pi} \frac{1}{2}\sqrt{2} \, d\theta = \sqrt{2}\pi.$$

To confirm that this is the area of $\Phi(D)$, we must verify that $\Phi$ is one-to-one (for points not on the boundary of $D$). Let $D^0$ be the set of $(r, \theta)$ with $0 < r < 1$ and $0 < \theta < 2\pi$. Hence, $D^0$ is $D$ without its boundary. To see that $\Phi$: $D^0 \to \mathbb{R}^3$ is one-to-one, assume that $\Phi(r, \theta) = \Phi(r', \theta')$ for $(r, \theta)$ and $(r', \theta') \in D^0$. Then

$$r \cos\theta = r' \cos\theta', \qquad r \sin\theta = r' \sin\theta', \qquad r = r'.$$

From these equations it follows that $\cos\theta = \cos\theta'$ and $\sin\theta = \sin\theta'$. Thus, either $\theta = \theta'$ or $\theta = \theta' + 2\pi n$. But the second case is impossible for $n$ a nonzero integer, because both $\theta$ and $\theta'$ belong to the open interval $(0, 2\pi)$, and thus cannot be more than $2\pi$ radians apart. This proves that off the boundary, $\Phi$ is one-to-one. (Is $\Phi$: $D \to \mathbb{R}^3$ one-to-one?) In future examples, we shall not usually verify that the parametrization is one-to-one when it is intuitively clear. ▲

**EXAMPLE 2** A ***helicoid*** is defined by $\Phi$: $D \to \mathbb{R}^3$, where

$$x = r \cos\theta, \qquad y = r \sin\theta, \qquad z = \theta$$

and $D$ is the region where $0 \le \theta \le 2\pi$ and $0 \le r \le 1$ (Figure 7.4.2). Find its area.

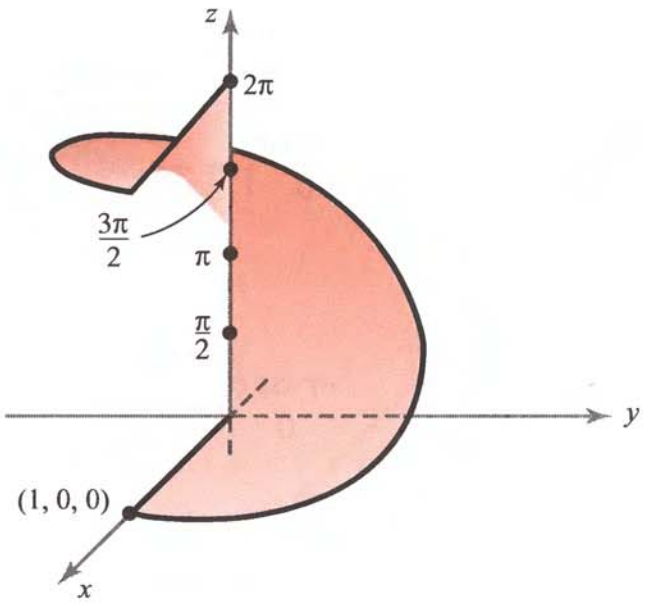



Figure 7.4.2 The helicoid $x = r\cos\theta, y = r\sin\theta, z = \theta$.

SOLUTION We compute $\partial(x, y)/\partial(r, \theta) = r$ as in Example 1, and

$$\frac{\partial(y, z)}{\partial(r, \theta)} = \begin{vmatrix} \sin\theta & r\cos\theta \\ 0 & 1 \end{vmatrix} = \sin\theta,$$

$$\frac{\partial(x, z)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ 0 & 1 \end{vmatrix} = \cos\theta.$$

The area integrand is therefore $\sqrt{r^2 + 1}$, which never vanishes, so the surface is regular. The area of the helicoid is

$$\iint_D \|\mathbf{T}_r \times \mathbf{T}_\theta\| \, dr \, d\theta = \int_0^{2\pi} \int_0^1 \sqrt{r^2 + 1} \, dr d\theta = 2\pi \int_0^1 \sqrt{r^2 + 1} \, dr.$$

After a little computation (using the table of integrals), we find that this integral is equal to

$$\pi[\sqrt{2} + \log(1 + \sqrt{2})]. \quad \blacktriangle$$

## Surface Area of a Graph

A surface $S$ given in the form $z = g(x, y)$, where $(x, y) \in D$, admits the parametrization

$$x = u, \qquad y = v, \qquad z = g(u, v)$$

for $(u, v) \in D$. When $g$ is of class $C^1$, this parametrization is smooth, and the formula for surface area reduces to

$$A(S) = \iint_D \left( \sqrt{\left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + 1} \right) dA, \tag{4}$$

after applying the formulas

$$\mathbf{T}_u = \mathbf{i} + \frac{\partial g}{\partial u}\mathbf{k}, \qquad \mathbf{T}_v = \mathbf{j} + \frac{\partial g}{\partial v}\mathbf{k},$$

and

$$\mathbf{T}_u \times \mathbf{T}_v = -\frac{\partial g}{\partial u}\mathbf{i} - \frac{\partial g}{\partial v}\mathbf{j} + \mathbf{k} = -\frac{\partial g}{\partial x}\mathbf{i} - \frac{\partial g}{\partial y}\mathbf{j} + \mathbf{k},$$

as noted in Example 4 of Section 7.3.

## Surfaces of Revolution

In most books on one-variable calculus, it is shown that the lateral surface area generated by revolving the graph of a function $y = f(x)$ about the $x$ axis is given by

$$A = 2\pi \int_a^b (|f(x)|\sqrt{1 + [f'(x)]^2})\, dx. \tag{5}$$

If the graph is revolved about the $y$ axis, the surface area is

$$A = 2\pi \int_a^b (|x|\sqrt{1 + [f'(x)]^2})\, dx. \tag{6}$$

We shall derive formula (5) by using the methods just developed; one can obtain formula (6) in a similar fashion (Exercise 10).

To derive formula (5) from formula (3), we must give a parametrization of $S$. Define the parametrization by

$$x = u, \qquad y = f(u)\cos v, \qquad z = f(u)\sin v$$

over the region $D$ given by

$$a \le u \le b, \qquad 0 \le v \le 2\pi.$$

This is indeed a parametrization of $S$, because for fixed $u$, the point

$$(u, f(u)\cos v, f(u)\sin v)$$

traces out a circle of radius $|f(u)|$ with the center $(u, 0, 0)$ (Figure 7.4.3).

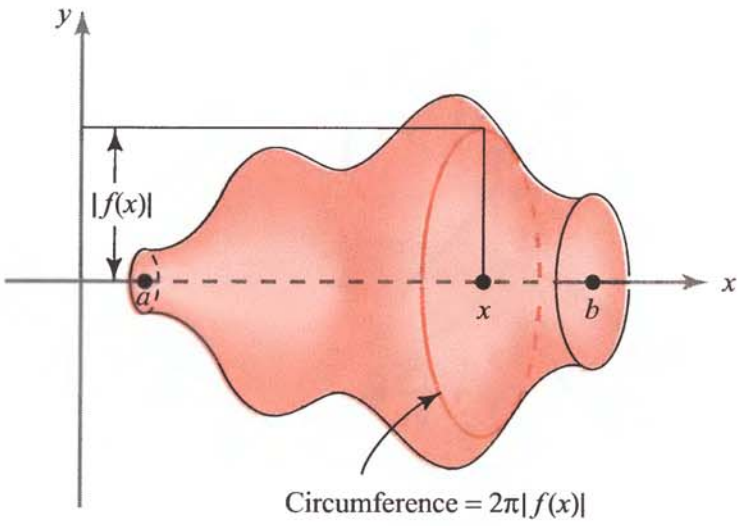



**Figure 7.4.3** The curve $y = f(x)$ rotated about the $x$ axis.

We calculate

$$\frac{\partial(x, y)}{\partial(u, v)} = -f(u)\sin v, \qquad \frac{\partial(y, z)}{\partial(u, v)} = f(u)f'(u), \qquad \frac{\partial(x, z)}{\partial(u, v)} = f(u)\cos v,$$

and so by formula (3)

$$
\begin{aligned}
A(S) &= \iint_D \sqrt{\left[\frac{\partial(x,y)}{\partial(u,v)}\right]^2 + \left[\frac{\partial(y,z)}{\partial(u,v)}\right]^2 + \left[\frac{\partial(x,z)}{\partial(u,v)}\right]^2}\, du\, dv \\
&= \iint_D \sqrt{[f(u)]^2 \sin^2 v + [f(u)]^2[f'(u)]^2 + [f(u)]^2 \cos^2 v}\, du\, dv \\
&= \iint_D |f(u)|\sqrt{1 + [f'(u)]^2}\, du\, dv \\
&= \int_a^b \int_0^{2\pi} |f(u)|\sqrt{1 + [f'(u)]^2}\, dv\, du \\
&= 2\pi \int_a^b |f(u)|\sqrt{1 + [f'(u)]^2}\, du,
\end{aligned}
$$

which is formula (5).

If $S$ is the surface of revolution, then $2\pi|f(x)|$ is the circumference of the vertical cross section to $S$ at the point $x$ (Figure 7.4.3). Observe that we can write

$$
2\pi \int_a^b |f(x)|\sqrt{1 + [f'(x)]^2}\, dx = \int_{\mathbf{c}} 2\pi |f(x)|\, ds,
$$

where the expression on the right is the path integral of $2\pi|f(x)|$ along the path given by $\mathbf{c}$: $[a, b] \to \mathbb{R}^2$, $t \mapsto (t, f(t))$. Therefore, *the lateral surface of a solid of revolution is obtained by integrating the cross-sectional circumference along the path that is the graph of the given function.*

## Historical Note

The most famous mathematician in ancient times was Archimedes. In addition to being an extraordinarily gifted mathematician, he was also an engineering genius on a scale never before seen and was greatly admired by his contemporaries and by later writers for his insights into mechanics. It was these talents that helped the people of the city of Syracuse in 214 B.C. to defend their city against the onslaught of the Roman legions under their commander Marcellus.

When the Romans besieged the city, they encountered an enemy whom Archimedes had supplied—totally unexpectedly—with powerful weapons, including artillery and burning mirrors, which, as legend has it, incinerated the Roman fleet.

The siege of Syracuse lasted two years, and the city finally fell as a result of acts of treason. In the aftermath of the assault, the old scientist was slain

by a Roman soldier, even though the commander had asked his men to spare Archimedes' life. As the story goes, Archimedes was sitting in front of his house studying some geometric figures he had drawn in the sand. When a Roman soldier approached, Archimedes shouted out, "Don't disturb my figures!" The ruffian, feeling insulted, slew Archimedes.

To honor this great man, Marcellus erected a tomb for Archimedes on which, according to Archimedes's own wishes, were depicted a cone, a sphere, and a cylinder (Figure 7.4.4).

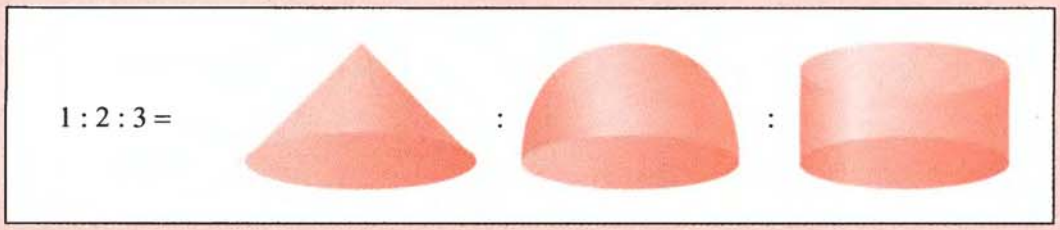



**Figure 7.4.4** Archimedes' theorem: The ratios of the volumes of a cone, a half ball, and a cylinder, all of the same height and radius, are 1:2:3.

Archimedes was incredibly proud of his calculation of the volume and surface area of the sphere, which justifiably were seen as truly outstanding accomplishments for their time. As in his works on centers of gravity, for which he provided no clear definition, Archimedes was able to compute the surface area of the sphere without having a clear definition of precisely what it was. However, as with many mathematical works, one knows the answer long before a proof or even the correct definition can be found.

The problem of properly defining surface areas is a difficult one. To Archimedes' credit, a careful theory of surface areas was not achieved until the twentieth century, after a long development that began in the seventeenth century with the discovery of calculus.

Christiaan Huygens (1629–1695) was the first person since Archimedes to give results on the areas of special surfaces beyond the sphere, and he obtained the areas of portions of surfaces of revolution, such as the paraboloid and hyperboloid.

The brilliant and prolific mathematician Leonhard Euler (1707–1783) presented the first fundamental work on the theory of surfaces in 1760 with *Recherches sur la courbure des surfaces*. However, it was in 1728, in a paper on shortest paths on surfaces, that Euler defined a surface as a graph $z = f(x,y)$. Euler was interested in studying the curvature of surfaces, and in 1771 he introduced the notion of the parametric surfaces that are described in this section.

After the rapid development of calculus in the early eighteenth century, formulas for the lengths of curves and areas of surfaces were developed. Although we do not know when all the area formulas presented in this section

first appeared, they were certainly common by the end of the eighteenth century. The underlying concepts of the length of a curve and the area of a surface were understood intuitively before this time, and the use of formulas from calculus to compute areas was considered a great achievement.

Augustin-Louis Cauchy (1789–1857) was the first to take the step of defining the quantities of length and surface areas by integrals as we have presented in this book. The question of defining surface area independent of integrals was taken up somewhat later, but this posed many difficult problems that were not properly resolved until this century.

We end this section by describing the fascinating classic area problem of Plateau, which has enjoyed a long history in mathematics. The Belgian physicist Joseph Plateau (1801–1883) carried out many experiments from 1830 to 1869 on surface tension and capillary phenomena, experiments that had enormous impact at the time and were repeated by notable nineteenth-century physicists, such as Michael Faraday (1791–1867). The corresponding collection of mathematical problems relating to soap films was named in 1904 after Plateau by the great French mathematician Henri Lebesgue (1875–1941).

If a wire is dipped into a soap or glycerine solution, then one usually withdraws a soap film spanning the wire. Some examples are given in Figure 7.4.5, although readers might like to perform the experiment for themselves. Plateau raised the mathematical question: For a given boundary (wire), how does one prove the existence of such a surface (soap film) and how many surfaces can there be? The underlying physical principle is that nature tends to minimize area; that is, the surface that forms should be a surface of least area among all possible surfaces that have the given curve as their boundary. This again is another example of the action principle of Maupertuis and Leibniz (c.f. Section 3.3).

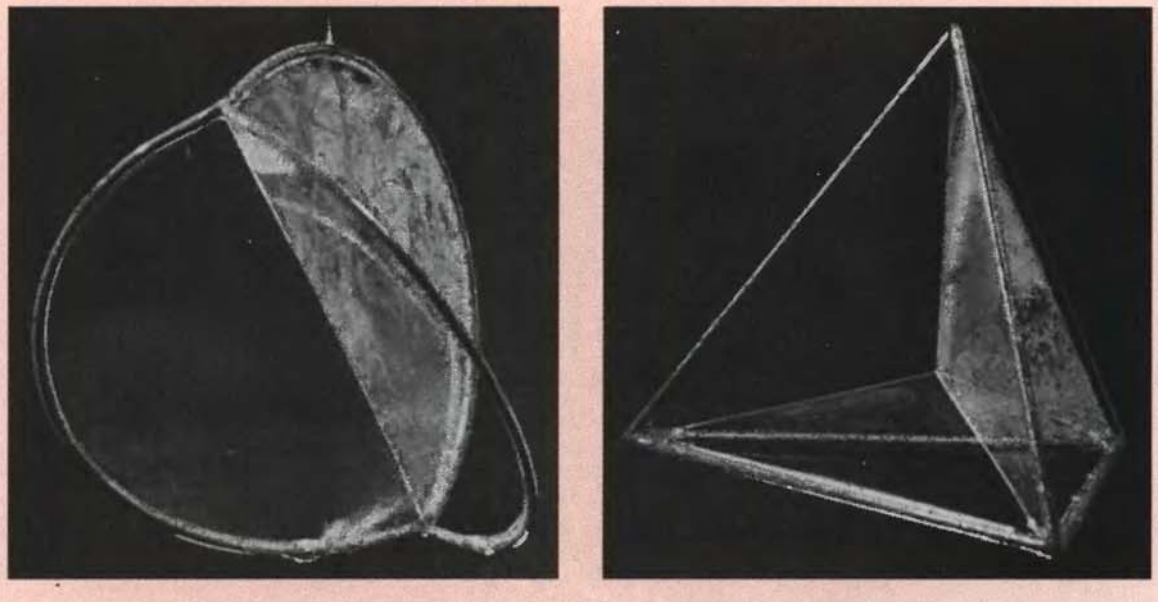


**Figure 7.4.5** Two soap films spanning wires.

For soap film surfaces that are disklike, the problem can be formulated in the following way. Let $D \subset \mathbb{R}^2$ be the unit disk defined to be the set $\{(x, y) \mid x^2 + y^2 \leq 1\}$ and let $\partial D$ be its boundary. Furthermore, suppose that the image $\Gamma$ of $\mathbf{c}: [0, 2\pi] \to \mathbb{R}^3$ is a simple closed curve, $\Gamma$ representing a wire in $\mathbb{R}^3$.

Let $\mathcal{S}$ be the set of all maps $\mathbf{\Phi}$: $D \to \mathbb{R}^3$ such that $\mathbf{\Phi}(\partial D) = \Gamma$, $\mathbf{\Phi}$ is of class $C^1$, and $\mathbf{\Phi}$ is one-to-one on $\partial D$. Each $\mathbf{\Phi} \in \mathcal{S}$ represents a parametric $C^1$ "disklike" surface "spanning" the wire $\Gamma$.

The soap films in Figure 7.4.5 are not disklike, but represent a system of multiple disklike surfaces. Figure 7.4.6 shows a contour that bounds two disklike surfaces and one nondisklike surface.

(a) (b) (c)

**Figure 7.4.6** Soap film surfaces; (b) and (c) are disklike surfaces, but (a) is not.

For each $\mathbf{\Phi} \in \mathcal{S}$, consider the area of the image surface, namely, $A(\mathbf{\Phi}) = \iint_D \|\mathbf{T}_u \times \mathbf{T}_v\| \, du \, dv$. This area is a function that assigns to each parametric surface its area. Plateau asked whether $A$ has a minimum on $\mathcal{S}$; that is, does there exist a $\mathbf{\Phi}_0$ such that $A(\mathbf{\Phi}_0) \leq A(\mathbf{\Phi})$ for all $\mathbf{\Phi} \in \mathcal{S}$? Unfortunately, the methods of this book are not adequate to solve this problem. We can tackle questions of finding minima of real-valued functions of several variables, but in no way can the set $\mathcal{S}$ be thought of as a region in $\mathbb{R}^n$ for *any* $n$!

In his own study of surfaces of least area, Weierstrass showed that if a minimum

$$\mathbf{\Phi}_0(u, v) = (x(u, v), y(u, v), z(u, v))$$

existed at all, it would have to satisfy (after suitable normalizations) the partial differential equations

(i) $\nabla^2 \mathbf{\Phi}_0 = 0$

(ii) $\dfrac{\partial \mathbf{\Phi}_0}{\partial u} \cdot \dfrac{\partial \mathbf{\Phi}_0}{\partial v} = 0$

(iii) $\left\| \dfrac{\partial \mathbf{\Phi}_0}{\partial u} \right\| = \left\| \dfrac{\partial \mathbf{\Phi}_0}{\partial v} \right\|$

where $\|\mathbf{w}\|$ denotes the "norm" or length of the vector $\mathbf{w}$. This example illustrates the intimate connections between problems of maxima and minima (the calculus of variations) and the subject of partial differential equations.

For well over 70 years, mathematicians such as Riemann, Weierstrass, H. A. Schwarz, Darboux, and Lebesgue puzzled over the challenge posed by Plateau. In 1931 the question was finally settled when Jesse Douglas showed that such a $\Phi_0$ existed. However, many questions about soap films remain unsolved, and this area of research is still active today.[11]

## EXERCISES

**1.** Find the surface area of the unit sphere $S$ represented parametrically by $\Phi\colon D \to S \subset \mathbb{R}^3$, where $D$ is the rectangle $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi$ and $\Phi$ is given by the equations

$$x = \cos\theta \sin\phi, \qquad y = \sin\theta \sin\phi, \qquad z = \cos\phi.$$

Note that we can represent the entire sphere parametrically, but we cannot represent it in the form $z = f(x, y)$.

**2.** In Exercise 1, what happens if we allow $\phi$ to vary from $-\pi/2$ to $\pi/2$? From 0 to $2\pi$? Why do we obtain different answers?

**3.** Find the area of the helicoid in Example 2 if the domain $D$ is $0 \le r \le 1$ and $0 \le \theta \le 3\pi$.

**4.** The torus $T$ can be represented parametrically by the function $\Phi\colon D \to \mathbb{R}^3$, where $\Phi$ is given by the coordinate functions $x = (R + \cos\phi)\cos\theta$, $y = (R + \cos\phi)\sin\theta$, $z = \sin\phi$; $D$ is the rectangle $[0, 2\pi] \times [0, 2\pi]$, that is, $0 \le \theta \le 2\pi$, $0 \le \phi \le 2\pi$; and $R > 1$ is fixed (see Figure 7.4.7). Show that $A(T) = (2\pi)^2 R$, first by using formula (3) and then by using formula (6).

**5.** Let $\Phi(u, v) = (u - v, u + v, uv)$ and let $D$ be the unit disk in the $uv$ plane. Find the area of $\Phi(D)$.

**6.** Find the area of the portion of the unit sphere that is cut out by the cone $z \ge \sqrt{x^2 + y^2}$ (see Exercise 1).

**7.** Show that the surface $x = 1/\sqrt{y^2 + z^2}$, where $1 \le x < \infty$, can be filled but not painted!

**8.** Find a parametrization of the surface $x^2 - y^2 = 1$, where $x > 0$, $-1 \le y \le 1$ and $0 \le z \le 1$. Use your answer to express the area of the surface as an integral.

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[11] For more information on this fascinating subject, the reader may consult *The Parsimonious Universe: Shape and Form in the Natural World*, by S. Hildebrandt and A. Tromba, Springer-Verlag, New York/Heidelberg, 1995.

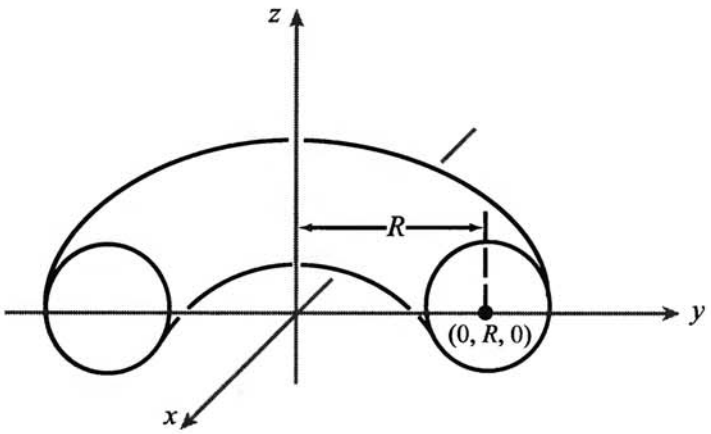


**Figure 7.4.7** A cross section of a torus.

**9.** Represent the ellipsoid $E$:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

parametrically and write out the integral for its surface area $A(E)$. (Do not evaluate the integral.)

**10.** Let the curve $y = f(x)$, $a \leq x \leq b$, be rotated about the $y$ axis. Show that the area of the surface swept out is given by equation (6); that is,

$$A = 2\pi \int_a^b |x| \sqrt{1 + [f'(x)]^2}\, dx.$$

Interpret the formula geometrically using arc length and slant height.

**11.** Find the area of the surface obtained by rotating the curve $y = x^2$, $0 \leq x \leq 1$, about the $y$ axis.

**12.** Use formula (4) to compute the surface area of the cone in Example 1.

**13.** Find the area of the surface defined by $x + y + z = 1$, $x^2 + 2y^2 \leq 1$.

**14.** Show that for the vectors $\mathbf{T}_u$ and $\mathbf{T}_v$, we have the formula

$$\|\mathbf{T}_u \times \mathbf{T}_v\| = \sqrt{\left[\frac{\partial(x, y)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(y, z)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)}\right]^2}.$$

**15.** Compute the area of the surface given by

$$x = r\cos\theta, \qquad y = 2r\cos\theta, \qquad z = \theta, \qquad 0 \leq r \leq 1, \qquad 0 \leq \theta \leq 2\pi.$$

Sketch.

**16.** Prove *Pappus' theorem*: Let $\mathbf{c}$: $[a, b] \to \mathbb{R}^2$ be a $C^1$ path whose image lies in the right half plane and is a simple closed curve. The area of the lateral surface generated by rotating

the image of **c** about the $y$ axis is equal to $2\pi\bar{x}l(\mathbf{c})$, where $\bar{x}$ is the average value of the $x$ coordinates of points on **c** and $l(\mathbf{c})$ is the length of **c**. (See Exercises 8 to 11, Section 7.1, for a discussion of average values.)

**17.** The cylinder $x^2 + y^2 = x$ divides the unit sphere $S$ into two regions $S_1$ and $S_2$, where $S_1$ is inside the cylinder and $S_2$ outside. Find the ratio of areas $A(S_2)/A(S_1)$.

**18.** Suppose a surface $S$ that is the graph of a function $z = f(x, y)$, where $(x, y) \in D \subset \mathbb{R}^2$ can also be described as the set of $(x, y, z) \in \mathbb{R}^3$ with $F(x, y, z) = 0$ (a level surface). Derive a formula for $A(S)$ that involves only $F$.

**19.** Calculate the area of the frustum shown in Figure 7.4.8 using (a) geometry alone and, second, (b) a surface area formula.

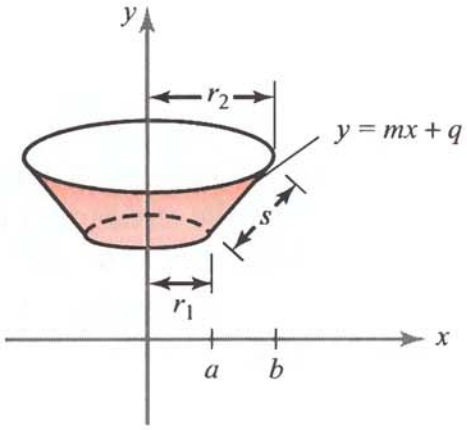


**Figure 7.4.8** A line segment revolved around the $y$ axis becomes a frustum of a cone.

**20.** A cylindrical hole of radius 1 is bored through a solid ball of radius 2 to form a ring coupler, as shown in Figure 7.4.9. Find the volume and outer surface area of this coupler.

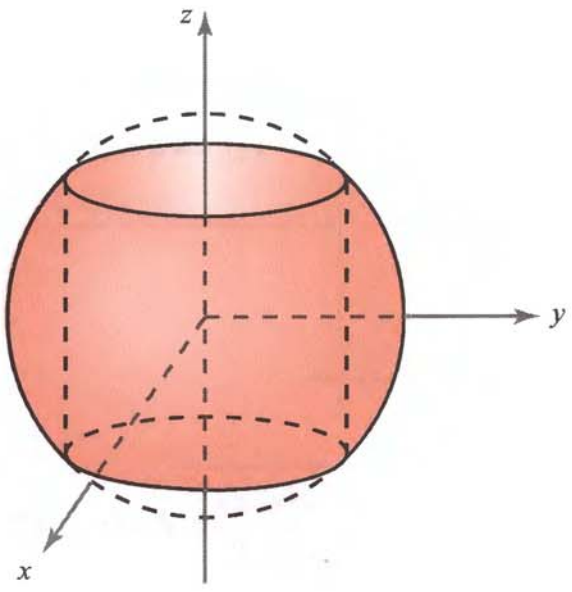


**Figure 7.4.9** Find the outer surface area and volume of the shaded region.

**21.** Find the area of the graph of the function $f(x, y) = \frac{2}{3}(x^{3/2} + y^{3/2})$ that lies over the domain $[0, 1] \times [0, 1]$.

**22.** Express the surface area of the following graphs over the indicated region $D$ as a double integral. Do not evaluate.

(a) $(x + 2y)^2$; $D = [-1, 2] \times [0, 2]$
(b) $xy + x/(y + 1)$; $D = [1, 4] \times [1, 2]$
(c) $xy^3 e^{x^2 y^2}$; $D =$ unit circle centered at the origin
(d) $y^3 \cos^2 x$; $D =$ triangle with vertices $(-1, 1)$, $(0, 2)$, and $(1, 1)$

**23.** Show that the surface area of the upper hemisphere of radius $R$, $z = \sqrt{R^2 - x^2 - y^2}$, can be computed by formula (4), evaluated as an improper integral.

## 7.5 Integrals of Scalar Functions Over Surfaces

Now we are ready to define the integral of a *scalar* function $f$ *over a surface* $S$. This concept is a natural generalization of the area of a surface, which corresponds to the integral over $S$ of the scalar function $f(x, y, z) = 1$. This is quite analogous to considering the path integral as a generalization of arc length. In the next section we shall deal with the integral of a *vector* function $\mathbf{F}$ over a surface. These concepts will play a crucial role in the vector analysis treated in the final chapter.

Let us start with a surface $S$ parametrized by a mapping $\Phi$: $D \to S \subset \mathbb{R}^3$, where $D$ is an elementary region, which we write as $\Phi(u, v) = (x(u, v), y(u, v), z(u, v))$.

**DEFINITION: The Integral of a Scalar Function Over a Surface** If $f(x, y, z)$ is a real-valued continuous function defined on a parametrized surface $S$, we define the ***integral of*** $f$ ***over*** $S$ to be

$$\iint_S f(x, y, z)\, dS = \iint_S f\, dS = \iint_D f(\Phi(u, v)) \|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv. \quad (1)$$

Written out, equation (1) becomes

$$\iint_S f\, dS = \iint_D f(x(u, v), y(u, v), z(u, v)) \sqrt{\left[\frac{\partial(x, y)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(y, z)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)}\right]^2}\, du\, dv. \quad (2)$$

Thus, if $f$ is identically 1, we recover the area formula (3) of Section 7.4. Like surface area, the surface integral is independent of the particular parametrization used. This will be discussed in Section 7.6.

We can gain some intuitive knowledge about this integral by considering it as a limit of sums. Let $D$ be a rectangle partitioned into $n^2$ rectangles $R_{ij}$ with areas $\Delta u\,\Delta v$. Let $S_{ij} = \Phi(R_{ij})$ *be the portion of the surface* $\Phi(D)$ corresponding to $R_{ij}$ (see Figure 7.5.1), and let $A(S_{ij})$ be the area of this portion of the surface. For large $n$, $f$ will be approximately constant on $S_{ij}$, and we form the sum

$$S_n = \sum_{i=0}^{n-1}\sum_{j=0}^{n-1} f(\Phi(u_i, v_j))A(S_{ij}), \tag{3}$$

where $(u_i, v_j) \in R_{ij}$. From Section 7.4 we have a formula for $A(S_{ij})$:

$$A(S_{ij}) = \iint_{R_{ij}} \|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv,$$

which, by the mean-value theorem for integrals, equals $\|\mathbf{T}_{u_i^*} \times \mathbf{T}_{v_j^*}\|\,\Delta u\,\Delta v$ for some point $(u_i^*, v_j^*)$ in $R_{ij}$. Hence, our sum becomes

$$S_n = \sum_{i=0}^{n-1}\sum_{j=0}^{n-1} f(\Phi(u_i, v_j))\|\mathbf{T}_{u_i^*} \times \mathbf{T}_{v_j^*}\|\Delta u\,\Delta v,$$

which is an approximating sum for the last integral in formula (1). Therefore,

$$\lim_{n\to\infty} S_n = \iint_S f\, dS.$$

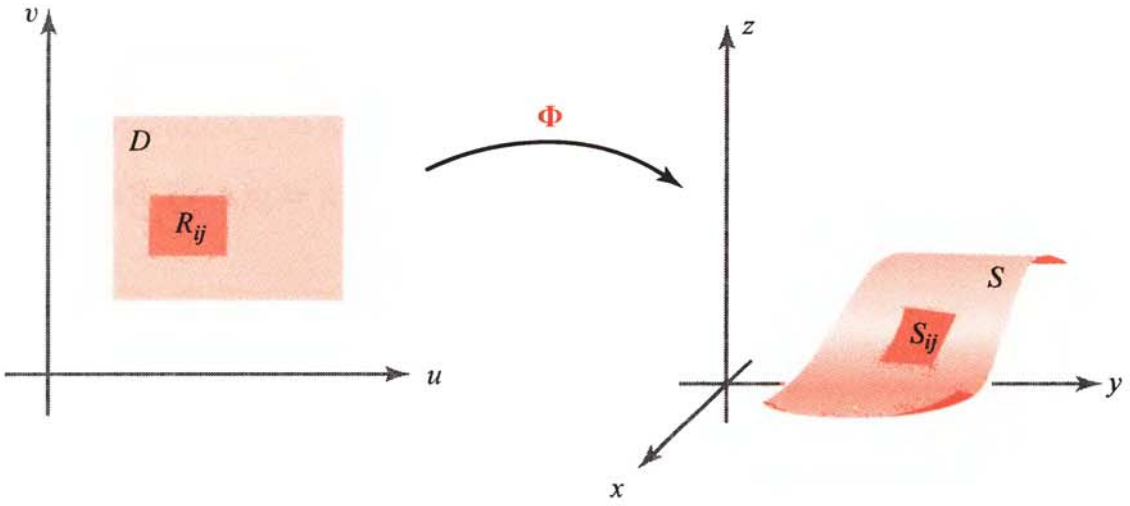


**Figure 7.5.1** $\Phi$ takes a portion $R_{ij}$ of $D$ to a portion of $S$.

Note that each term in the sum in formula (3) is the value of $f$ at some point $\Phi(u_i, v_j)$ times the area of $S_{ij}$. Compare this with the Riemann-sum interpretation of the path integral in Section 7.1.

If $S$ is a union of parametrized surfaces $S_i$, $i = 1, \ldots, N$, that do not intersect except possibly along curves defining their boundaries, then the integral of $f$ over $S$ is defined by

$$\iint_S f\,dS = \sum_{i=1}^{N} \iint_{S_i} f\,dS,$$

as we should expect. For example, the integral over the surface of a cube may be expressed as the sum of the integrals over the six sides.

**EXAMPLE 1** Suppose a helicoid is described as in Example 2, Section 7.4, and let $f$ be given by $f(x, y, z) = \sqrt{x^2 + y^2 + 1}$. Find $\iint_S f\,dS$.

SOLUTION As in Examples 1 and 2 of Section 7.4,

$$\frac{\partial(x, y)}{\partial(r, \theta)} = r, \qquad \frac{\partial(y, z)}{\partial(r, \theta)} = \sin\theta, \qquad \frac{\partial(x, z)}{\partial(r, \theta)} = \cos\theta.$$

Also, $f(r\cos\theta, r\sin\theta, \theta) = \sqrt{r^2 + 1}$. Therefore,

$$\iint_S f(x, y, z)\,dS = \iint_D f(\Phi(r, \theta))\|\mathbf{T}_r \times \mathbf{T}_\theta\|\,dr\,d\theta$$

$$= \int_0^{2\pi} \int_0^1 \sqrt{r^2 + 1}\sqrt{r^2 + 1}\,dr\,d\theta = \int_0^{2\pi} \frac{4}{3}\,d\theta = \frac{8}{3}\pi. \quad \blacktriangle$$

## Surface Integrals Over Graphs

Suppose $S$ is the graph of a $C^1$ function $z = g(x, y)$. Recall from Section 7.4 that we can parametrize $S$ by

$$x = u, \qquad y = v, \qquad z = g(u, v),$$

and that in this case

$$\|\mathbf{T}_u \times \mathbf{T}_v\| = \sqrt{1 + \left(\frac{\partial g}{\partial u}\right)^2 + \left(\frac{\partial g}{\partial v}\right)^2},$$

so

$$\iint_S f(x,y,z)\,dS = \iint_D f(x,y,g(x,y))\sqrt{1+\left(\frac{\partial g}{\partial x}\right)^2+\left(\frac{\partial g}{\partial y}\right)^2}\,dx\,dy. \quad (4)$$

**EXAMPLE 2** Let $S$ be the surface defined by $z = x^2 + y$, where $D$ is the region $0 \le x \le 1$, $-1 \le y \le 1$. Evaluate $\iint_S x\,dS$.

SOLUTION If we let $z = g(x,y) = x^2 + y$, formula (4) gives

$$\begin{aligned}\iint_S x\,dS &= \iint_D x\sqrt{1+\left(\frac{\partial g}{\partial x}\right)^2+\left(\frac{\partial g}{\partial y}\right)^2}\,dx\,dy = \int_{-1}^{1}\int_0^1 x\sqrt{1+4x^2+1}\,dx\,dy \\ &= \frac{1}{8}\int_{-1}^{1}\left[\int_0^1 (2+4x^2)^{1/2}(8x\,dx)\right]dy = \frac{2}{3}\cdot\frac{1}{8}\int_{-1}^{1}[(2+4x^2)^{3/2}]\Big|_0^1\,dy \\ &= \frac{1}{12}\int_{-1}^{1}(6^{3/2}-2^{3/2})\,dy = \frac{1}{6}(6^{3/2}-2^{3/2}) = \sqrt{6}-\frac{\sqrt{2}}{3} \\ &= \sqrt{2}\left(\sqrt{3}-\frac{1}{3}\right). \quad \blacktriangle\end{aligned}$$

**EXAMPLE 3** Evaluate $\iint_S z^2\,dS$, where $S$ is the unit sphere $x^2 + y^2 + z^2 = 1$.

SOLUTION For this problem, it is convenient to use spherical coordinates and to represent the sphere parametrically by the equations $x = \cos\theta\sin\phi$, $y = \sin\theta\sin\phi$, $z = \cos\phi$, over the region $D$ in the $\theta\phi$ plane given by the inequalities $0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$. From equation (1) we get

$$\iint_S z^2\,dS = \iint_D (\cos\phi)^2\|\mathbf{T}_\theta \times \mathbf{T}_\phi\|\,d\theta\,d\phi.$$

A little computation [use formula (2) of Section 7.4; see Exercise 6] shows that

$$\|\mathbf{T}_\theta \times \mathbf{T}_\phi\| = \sin\phi.$$

(Note that for $0 \le \phi \le \pi$, we have $\sin\phi \ge 0$). Thus,

$$\begin{aligned}\iint_S z^2\,dS &= \int_0^{2\pi}\int_0^{\pi}\cos^2\phi\sin\phi\,d\phi\,d\theta \\ &= \frac{1}{3}\int_0^{2\pi}[-\cos^3\phi]_0^{\pi}\,d\theta = \frac{2}{3}\int_0^{2\pi}d\theta = \frac{4\pi}{3}. \quad \blacktriangle\end{aligned}$$

This example also shows that on a sphere of radius $R$,

$$\iint_S f\, ds = \int_0^{2\pi} \int_0^{\pi} f(\phi, \theta) R^2 \sin\phi\, d\phi\, d\theta,$$

or, for short, the *area element on the sphere* is given by

$$dS = R^2 \sin\phi\, d\phi\, d\theta.$$

## Integrals Over Graphs

We now develop another formula for surface integrals when the surface can be represented as a graph. To do so, we let $S$ be the graph of $z = g(x, y)$ and consider formula (4). We claim that

$$\iint_S f(x, y, z)\, dS = \iint_D \frac{f(x, y, g(x, y))}{\cos\theta}\, dx\, dy, \tag{5}$$

where $\theta$ is the angle the normal to the surface makes with the unit vector $\mathbf{k}$ at the point $(x, y, g(x, y))$ (see Figure 7.5.2). Describing the surface by the equation $\phi(x, y, z) = z - g(x, y) = 0$, a normal vector $\mathbf{N}$ is $\nabla\phi$; that is,

$$\mathbf{N} = -\frac{\partial g}{\partial x}\mathbf{i} - \frac{\partial g}{\partial y}\mathbf{j} + \mathbf{k} \tag{6}$$

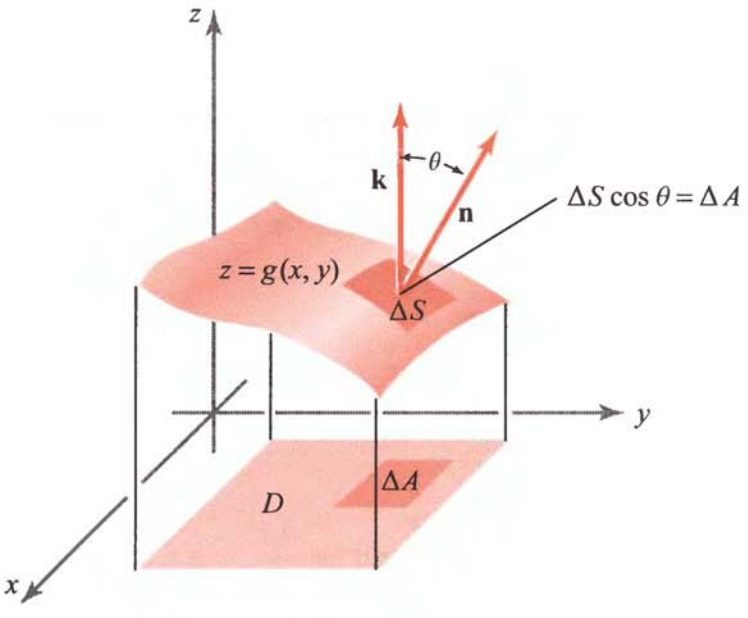


Figure 7.5.2 The area of a patch of area $\Delta S$ over a patch $\Delta A$ is $\Delta S = \Delta A/\cos\theta$, where $\theta$ is the angle the unit normal $\mathbf{n}$ makes with $\mathbf{k}$.

[see Example 4 of Section 7.3, or recall that the normal to a surface with equation $g(x, y, z) =$ constant is given by $\nabla g$]. Thus,

$$\cos\theta = \frac{\mathbf{N}\cdot\mathbf{k}}{\|\mathbf{N}\|} = \frac{1}{\sqrt{(\partial g/\partial x)^2 + (\partial g/\partial y)^2 + 1}}.$$

Substitution of this formula into equation (4) gives equation (5). Note that $\cos\theta = \mathbf{n}\cdot\mathbf{k}$, where $\mathbf{n} = \mathbf{N}/\|\mathbf{N}\|$ is the unit normal. Thus, we can write

$$d\mathbf{S} = \frac{dx\,dy}{\mathbf{n}\cdot\mathbf{k}}$$

The result is, in fact, obvious geometrically, for if a small rectangle in the $xy$ plane has area $\Delta A$, then the area of the portion above it on the surface is $\Delta S = \Delta A/\cos\theta$ (Figure 7.5.2). This intuitive approach can help us to remember formula (5) and to apply it in problems.

**EXAMPLE 4** Compute $\iint_S x\,dS$, where $S$ is the triangle with vertices (1, 0, 0), (0, 1, 0), (0, 0, 1) (see Figure 7.5.3).

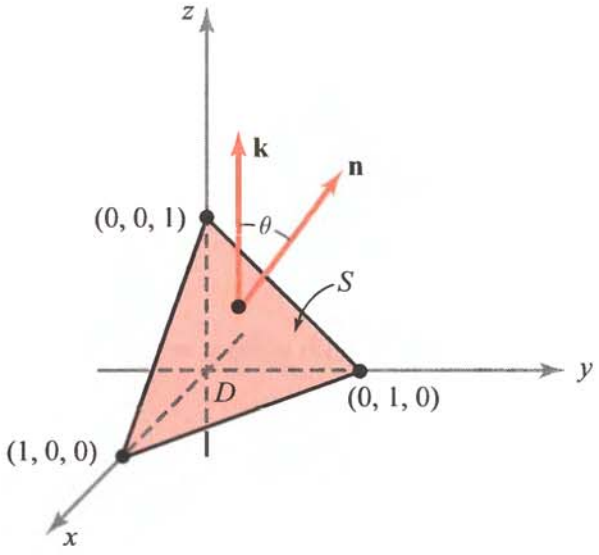


Figure 7.5.3 In computing a specific surface integral, one finds a formula for the unit normal **n** and computes the angle $\theta$ in preparation for formula (5).

SOLUTION This surface is the plane described by the equation $x + y + z = 1$. Because the surface is a plane, the angle $\theta$ is constant and a unit normal vector is $\mathbf{n} = (1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$. Thus, $\cos\theta = \mathbf{n}\cdot\mathbf{k} = 1/\sqrt{3}$, and by equation (5),

$$\iint_S x\,dS = \sqrt{3}\iint_D x\,dx\,dy,$$

where $D$ is the domain in the $xy$ plane. But

$$\sqrt{3}\iint_D x\,dx\,dy = \sqrt{3}\int_0^1\int_0^{1-x} x\,dy\,dx = \sqrt{3}\int_0^1 x(1-x)\,dx = \frac{\sqrt{3}}{6}. \quad \blacktriangle$$

Integrals of functions over surfaces are useful for computing the mass of a surface when the mass density function $m$ is known. The total mass of a surface with mass

density (per unit area) $m$ is given by

$$M(S) = \iint_S m(x, y, z)\, dS. \tag{7}$$

**EXAMPLE 5** Let $\Phi$: $D \to \mathbb{R}^3$ be the parametrization of the helicoid $S = \Phi(D)$ of Example 2 of Section 7.4. Recall that $\Phi(r, \theta) = (r\cos\theta, r\sin\theta, \theta)$, where $0 \le \theta \le 2\pi$, and $0 \le r \le 1$. Suppose $S$ has a mass density at $(x, y, z) \in S$ equal to twice the distance of $(x, y, z)$ from the central axis (see Figure 7.4.2), that is, $m(x, y, z) = 2\sqrt{x^2 + y^2} = 2r$, in the cylindrical coordinate system. Find the total mass of the surface.

SOLUTION Applying formula (7),

$$M(S) = \iint_S 2\sqrt{x^2 + y^2}\, dS = \iint_D 2r\, dS = \iint_D 2r \|\mathbf{T}_r \times \mathbf{T}_\theta\|\, dr\, d\theta.$$

From Example 2 of Section 7.4, we see that $\|\mathbf{T}_r \times \mathbf{T}_\theta\| = \sqrt{1 + r^2}$. Thus,

$$\begin{aligned} M(S) &= \iint_D 2r\sqrt{1 + r^2}\, dr\, d\theta = \int_0^{2\pi} \int_0^1 2r\sqrt{1 + r^2}\, dr\, d\theta \\ &= \int_0^{2\pi} \left[\frac{2}{3}(1 + r^2)^{3/2}\right]_0^1 d\theta = \int_0^{2\pi} \frac{2}{3}(2^{3/2} - 1)\, d\theta = \frac{4\pi}{3}(2^{3/2} - 1). \end{aligned}$$ ▲

## EXERCISES

**1.** Compute $\iint_S xy\, dS$, where $S$ is the surface of the tetrahedron with sides $z = 0$, $y = 0$, $x + z = 1$, and $x = y$.

**2.** Evaluate $\iint_S xyz\, dS$, where $S$ is the triangle with vertices (1, 0, 0), (0, 2, 0), and (0, 1, 1).

**3.** Evaluate $\iint_S z\, dS$, where $S$ is the upper hemisphere of radius $a$, that is, the set of $(x, y, z)$ with $z = \sqrt{a^2 - x^2 - y^2}$.

**4.** Evaluate $\iint_S (x + y + z)\, dS$, where $S$ is the boundary of the unit ball $B$; that is, $S$ is the set of $(x, y, z)$ with $x^2 + y^2 + z^2 = 1$. (HINT: Use the symmetry of the problem.)

**5.** (a) Compute the area of the portion of the cone $x^2 + y^2 = z^2$ with $z \ge 0$ that is inside the sphere $x^2 + y^2 + z^2 = 2Rz$, where $R$ is a positive constant.

(b) What is the area of that portion of the sphere that is inside the cone?

**6.** Verify that in spherical coordinates, on a sphere of radius $R$,

$$\|\mathbf{T}_\phi \times \mathbf{T}_\theta\|\, d\phi\, d\theta = R^2 \sin\phi\, d\phi\, d\theta.$$

**7.** Evaluate $\iint_S z\, dS$, where $S$ is the surface $z = x^2 + y^2$, $x^2 + y^2 \le 1$.

**8.** Evaluate the surface integral $\iint_S z^2\,dS$, where $S$ is the boundary of the cube $C = [-1, 1] \times [-1, 1] \times [-1, 1]$. (HINT: Do each face separately and add the results.)

**9.** Find the mass of a spherical surface $S$ of radius $R$ such that at each point $(x, y, z) \in S$ the mass density is equal to the distance of $(x, y, z)$ to some fixed point $(x_0, y_0, z_0) \in S$.

**10.** A metallic surface $S$ is in the shape of a hemisphere $z = \sqrt{R^2 - x^2 - y^2}$, where $(x, y)$ satisfies $0 \leq x^2 + y^2 \leq R^2$. The mass density at $(x, y, z) \in S$ is given by $m(x, y, z) = x^2 + y^2$. Find the total mass of $S$.

**11.** Let $S$ be the sphere of radius $R$.

(a) Argue by symmetry that

$$\iint_S x^2\,dS = \iint_S y^2\,dS = \iint_S z^2\,dS.$$

(b) Use this fact and some clever thinking to evaluate, with very little computation, the integral

$$\iint_S x^2\,dS.$$

(c) Does this help in Exercise 10?

**12.** (a) Use Riemann sums to justify the formula

$$\frac{1}{A(S)} \iint_S f(x, y, z)\,dS$$

for the *average value* of $f$ over the surface $S$.

(b) In Example 3 of this section, show that the average of $f(x, y, z) = z^2$ over the sphere is $1/3$.

(c) Define the ***center of gravity*** $(\bar{x}, \bar{y}, \bar{z})$ of a surface $S$ to be such that $\bar{x}$, $\bar{y}$, and $\bar{z}$ are the average values of the $x$, $y$, and $z$ coordinates on $S$. Show that the center of gravity of the triangle in Example 4 of this section is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

**13.** Find the $x$, $y$, and $z$ coordinates of the center of gravity of the octant of the solid sphere of radius $R$ and centered at the origin determined by $x \geq 0, y \geq 0, z \geq 0$. (HINT: Write this octant as a parametrized surface—see Example 3 of this section and Exercise 12.)

**14.** Find the $z$ coordinate of the center of gravity (the average $z$ coordinate) of the surface of a hemisphere ($z \leq 0$) with radius $r$ (see Exercise 12). Argue by symmetry that the average $x$ and $y$ coordinates are both zero.

**15.** Let $\Phi: D \subset \mathbb{R}^2 \to \mathbb{R}^3$ be a parametrization of a surface $S$ defined by

$$x = x(u, v), \qquad y = y(u, v), \qquad z = z(u, v).$$

(a) Let

$$\frac{\partial \Phi}{\partial u} = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right) \quad \text{and} \quad \frac{\partial \Phi}{\partial v} = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right),$$

that is, $\partial\Phi/\partial u = \mathbf{T}_u$ and $\partial\Phi/\partial v = \mathbf{T}_v$, and set

$$E = \left\|\frac{\partial \Phi}{\partial u}\right\|^2, \qquad F = \frac{\partial \Phi}{\partial u} \cdot \frac{\partial \Phi}{\partial v}, \qquad G = \left\|\frac{\partial \Phi}{\partial v}\right\|^2.$$

Show that

$$\sqrt{EG - F^2} = \|\mathbf{T}_u \times \mathbf{T}_v\|,$$

and that the surface area of $S$ is

$$A(S) = \iint_D \sqrt{EG - F^2}\, du\, dv.$$

In this notation, how can we express $\iint_S f\, dS$ for a general function of $f$?

(b) What does the formula for $A(S)$ become if the vectors $\partial\Phi/\partial u$ and $\partial\Phi/\partial v$ are orthogonal?

(c) Use parts (a) and (b) to compute the surface area of a sphere of radius $a$.

**16.** ***Dirichlet's functional*** for a parametrized surface $\Phi$: $D \to \mathbb{R}^3$ is defined by[12]

$$J(\Phi) = \frac{1}{2}\iint_D \left(\left\|\frac{\partial \Phi}{\partial u}\right\|^2 + \left\|\frac{\partial \Phi}{\partial v}\right\|^2\right) du\, dv.$$

Use Exercise 15 to argue that the area $A(\Phi) \le J(\Phi)$ and equality holds if

$$\text{(a)}\ \left\|\frac{\partial \Phi}{\partial u}\right\|^2 = \left\|\frac{\partial \Phi}{\partial v}\right\|^2 \quad \text{and} \quad \text{(b)}\ \frac{\partial \Phi}{\partial u} \cdot \frac{\partial \Phi}{\partial v} = 0.$$

Compare these equations with Exercise 15 and the remarks at the end of Section 7.4. A parametrization $\Phi$ that satisfies conditions (a) and (b) is said to be ***conformal***.

**17.** Let $D \subset \mathbb{R}^2$ and $\Phi$: $D \to \mathbb{R}^2$ be a smooth function $\Phi(u, v) = (x(u, v), y(u, v))$ satisfying conditions (a) and (b) of Exercise 16 and assume that

$$\det \begin{bmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{bmatrix} > 0.$$

---

[12] Dirichlet's functional played a major role in the mathematics of the nineteenth century. The mathematician Georg Friedrich Bernhard Riemann (1826–1866) used it to develop his complex function theory and to give a proof of the famous Riemann mapping theorem. Today it is still used extensively as a tool in the study of partial differential equations.

Show that $x$ and $y$ satisfy the ***Cauchy–Riemann equations*** $\partial x/\partial u = \partial y/\partial v$, $\partial x/\partial v = -\partial y/\partial u$. Conclude that $\nabla^2 \Phi = 0$ (i.e., each component of $\Phi$ is harmonic).

**18.** Let $S$ be a sphere of radius $r$ and $\mathbf{p}$ be a point inside or outside the sphere (but not on it). Show that

$$\iint_S \frac{1}{\|\mathbf{x} - \mathbf{p}\|}\, dS = \begin{cases} 4\pi r & \text{if } \mathbf{p} \text{ is inside } S \\ 4\pi r^2/d & \text{if } \mathbf{p} \text{ is outside } S, \end{cases}$$

where $d$ is the distance from $\mathbf{p}$ to the center of the sphere and the integration is over the sphere.

**19.** Find the surface area of that part of the cylinder $x^2 + z^2 = a^2$ that is inside the cylinder $x^2 + y^2 = 2ay$ and also in the positive octant ($x \geq 0, y \geq 0, z \geq 0$). Assume $a > 0$.

**20.** Let a surface $S$ be defined implicitly by $F(x, y, z) = 0$ for $(x, y)$ in a domain $D$ of $\mathbb{R}^2$. Show that

$$\iint_S \left|\frac{\partial F}{\partial z}\right| dS = \iint_D \sqrt{\left(\frac{\partial F}{\partial x}\right)^2 + \left(\frac{\partial F}{\partial y}\right)^2 + \left(\frac{\partial F}{\partial z}\right)^2}\, dx\, dy.$$

Compare with Exercise 18 of Section 7.4.

# 7.6 Surface Integrals of Vector Fields

The goal of this section is to develop the notion of the integral of a vector field over a surface. Recall that the definition of the line integral of a vector field was motivated by the fundamental physical notion of *work*. Similarly, there is a basic physical notion of *flux* that motivates the definition of the surface integral of a vector field.

For example, if the vector field is the velocity field of a fluid (perhaps the velocity field of a flowing river), and one puts an imagined mathematical surface into the fluid, one can ask: "What is the rate at which fluid is crossing the given surface (measured in, say, cubic meters per second)?" The answer is given by the surface integral of the fluid velocity vector field over the surface.

We shall come back to the physical interpretation shortly and reconcile it with the formal definition that we give first.

## Definition of the Surface Integral

We now define the integral of a vector field, denoted $\mathbf{F}$ over a surface $S$. We first give the definition and later in this section give its physical interpretation. This can also be used as a *motivation* for the definition if the reader so desires. Also, we shall start with a parametrized surface $\Phi$ and later study the question of independence of parametrization.

**DEFINITION: The Surface Integral of Vector Fields** Let $\mathbf{F}$ be a vector field defined on $S$, the image of a parametrized surface $\Phi$. The ***surface integral*** of $\mathbf{F}$ over $\Phi$, denoted by

$$\iint_{\Phi} \mathbf{F} \cdot d\mathbf{S},$$

is defined by (see Figure 7.6.1))

$$\iint_{\Phi} \mathbf{F} \cdot d\mathbf{S} = \iint_{D} \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)\, du\, dv.$$

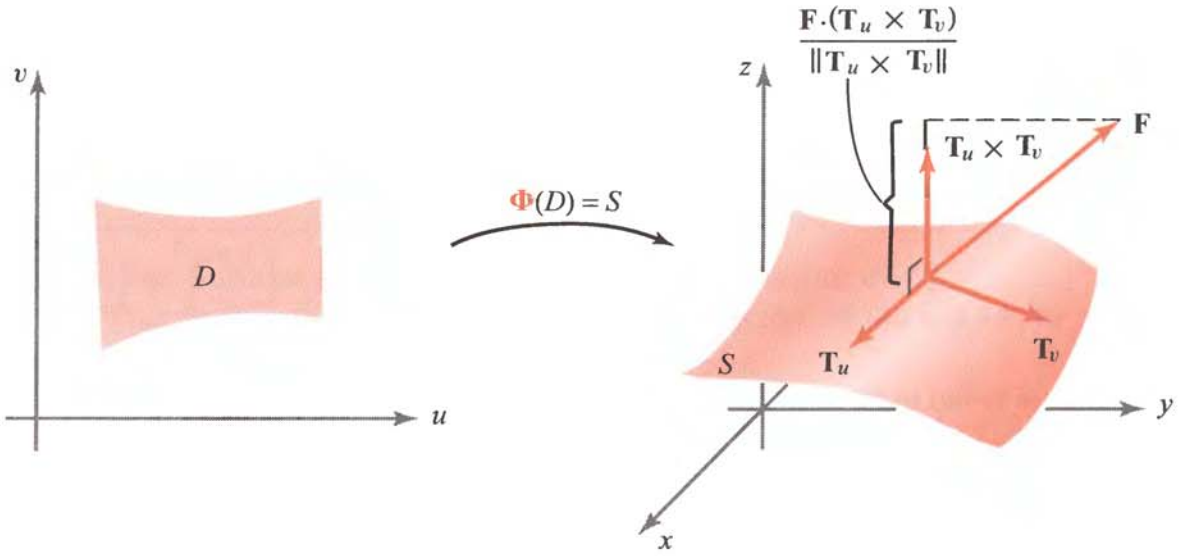


Figure 7.6.1 The geometric significance of $\mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)$.

**EXAMPLE 1** Let $D$ be the rectangle in the $\theta\phi$ plane defined by

$$0 \le \theta \le 2\pi, \qquad 0 \le \phi \le \pi,$$

and let the surface $S$ be defined by the parametrization $\Phi\colon D \to \mathbb{R}^3$ given by

$$x = \cos\theta \sin\phi, \qquad y = \sin\theta \sin\phi, \qquad z = \cos\phi.$$

(Thus, $\theta$ and $\phi$ are the angles of spherical coordinates, and $S$ is the unit sphere parametrized by $\Phi$.) Let $\mathbf{r}$ be the position vector $\mathbf{r}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Compute $\iint_{\Phi} \mathbf{r} \cdot d\mathbf{S}$.

SOLUTION First we find

$$\mathbf{T}_\theta = (-\sin\phi \sin\theta)\mathbf{i} + (\sin\phi \cos\theta)\mathbf{j}$$

$$\mathbf{T}_\phi = (\cos\theta \cos\phi)\mathbf{i} + (\sin\theta \cos\phi)\mathbf{j} - (\sin\phi)\mathbf{k},$$

and hence

$$\mathbf{T}_\theta \times \mathbf{T}_\phi = (-\sin^2\phi \cos\theta)\mathbf{i} - (\sin^2\phi \sin\theta)\mathbf{j} - (\sin\phi \cos\phi)\mathbf{k}.$$

Then we evaluate

$$\begin{aligned}\mathbf{r}\cdot(\mathbf{T}_\theta\times\mathbf{T}_\phi) &= (x\mathbf{i}+y\mathbf{j}+z\mathbf{k})\cdot(\mathbf{T}_\theta\times\mathbf{T}_\phi)\\ &= [(\cos\theta\sin\phi)\mathbf{i}+(\sin\theta\sin\phi)\mathbf{j}+(\cos\phi)\mathbf{k}]\\ &\quad\cdot(-\sin\phi)[(\sin\phi\cos\theta)\mathbf{i}+(\sin\phi\sin\theta)\mathbf{j}+(\cos\phi)\mathbf{k}]\\ &= (-\sin\phi)(\sin^2\phi\cos^2\theta+\sin^2\phi\sin^2\theta+\cos^2\phi) = -\sin\phi.\end{aligned}$$

Thus,

$$\iint_\Phi \mathbf{r}\cdot d\mathbf{S} = \iint_D -\sin\phi\,d\phi\,d\theta = \int_0^{2\pi}(-2)\,d\theta = -4\pi. \quad \blacktriangle$$

## Orientation

An analogy can be drawn between the surface integral $\iint_\Phi \mathbf{F}\cdot d\mathbf{S}$ and the line integral $\int_C \mathbf{F}\cdot d\mathbf{s}$. Recall that the line integral is an oriented integral. We needed the notion of orientation of a curve to extend the definition of $\int_C \mathbf{F}\cdot d\mathbf{s}$ to line integrals $\int_C \mathbf{F}\cdot d\mathbf{s}$ over oriented curves. We extend the definition of $\iint_\Phi \mathbf{F}\cdot d\mathbf{S}$ to *oriented surfaces* in a similar fashion; that is, given a surface $S$ parametrized by a mapping $\Phi$, we want to define $\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_\Phi \mathbf{F}\cdot d\mathbf{S}$ and show that it is independent of the parametrization, except possibly for the sign. To accomplish this, we need the notion of orientation of a surface.

DEFINITION: Oriented Surfaces An ***oriented surface*** is a two-sided surface with one side specified as the ***outside*** or ***positive side***; we call the other side the ***inside*** or ***negative side.***[13] At each point $(x, y, z) \in S$ there are two unit normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, where $\mathbf{n}_1 = -\mathbf{n}_2$ (see Figure 7.6.2). Each of these two normals can be associated with one side of the surface. Thus, to specify a side of a surface $S$, at each point we choose a unit normal vector $\mathbf{n}$ that points away from the positive side of $S$ at that point.

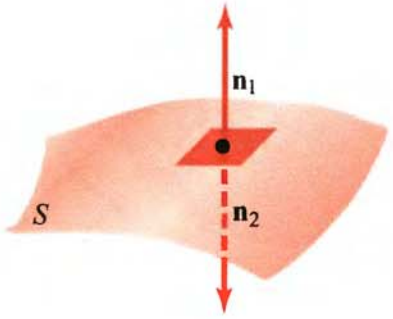


Figure 7.6.2 The two possible unit normals to a surface at a point.

[13] We use the term "side" in an intuitive sense. This concept can be developed rigorously, but this will not be done here. Also, the choice of the side to be named the "outside" is often dictated by the surface itself, as, for example, is the case with a sphere. In other cases, the naming is somewhat arbitrary (see the piece of surface depicted in Figure 7.6.2, for instance).

This definition assumes that our surface does have two sides. In fact, this is necessary, because there are examples of surfaces with only one side! The first known example of such a surface was the Möbius strip (named after the German mathematician and astronomer A. F. Möbius, who, along with the mathematician J. B. Listing, discovered it in 1858). Pictures of such a surface are given in Figures 7.6.3 and 7.6.4. At each point of $M$ there are two unit normals, $\mathbf{n}_1$ and $\mathbf{n}_2$. However, $\mathbf{n}_1$ does not determine a unique side of $M$, and neither does $\mathbf{n}_2$. To see this intuitively, we can slide $\mathbf{n}_2$ around the closed curve $C$ (Figure 7.6.3). When $\mathbf{n}_2$ returns to a fixed point $p$ on $C$ it will coincide with $\mathbf{n}_1$, showing that both $\mathbf{n}_1$ and $\mathbf{n}_2$ point away from the same side of $M$ and, consequently, that $M$ has only one side.

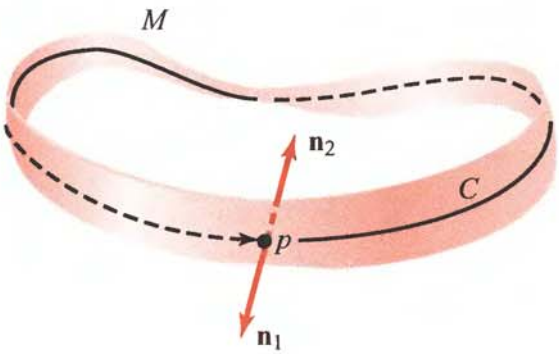


**Figure 7.6.3** The Möbius strip: Slide $\mathbf{n}_2$ around $C$ once; when $\mathbf{n}_2$ returns to its initial point, it will coincide with $\mathbf{n}_1 = -\mathbf{n}_2$.

Figure 7.6.4 is a Möbius strip as drawn by the well-known twentieth-century mathematician and artist M. C. Escher. It depicts ants crawling along the Möbius band. After one trip around the band (without crossing an edge) they end up on the "opposite side" of the surface.

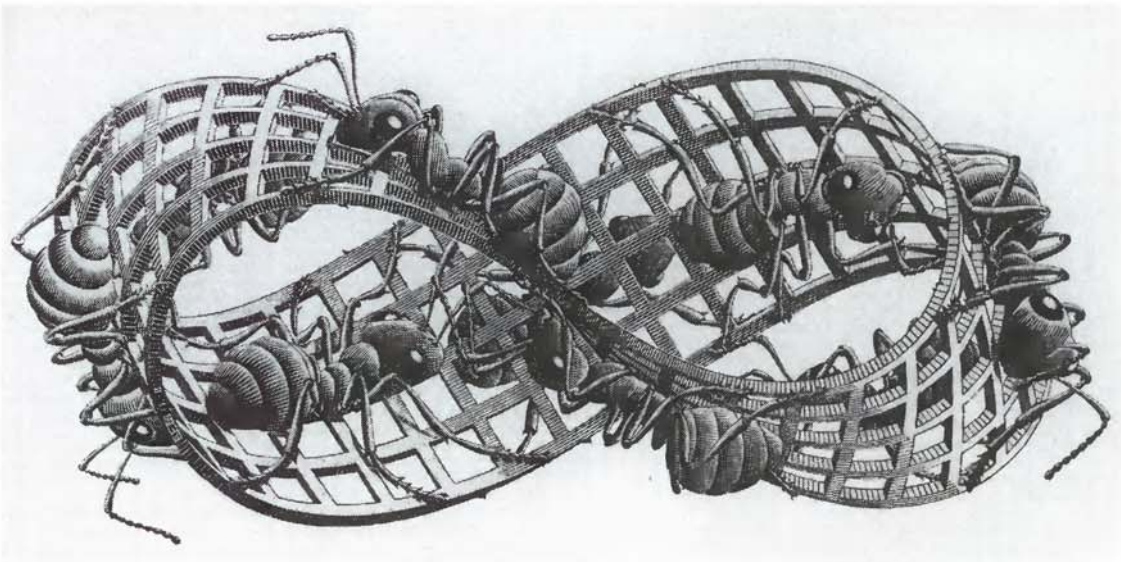

**Figure 7.6.4** Ants walking on a Möbius strip.

Let $\Phi\colon D \to \mathbb{R}^3$ be a parametrization of an oriented surface $S$ and suppose $S$ is regular at $\Phi(u_0, v_0)$, $(u_0, v_0) \in D$; thus, the vector $(\mathbf{T}_{u_0} \times \mathbf{T}_{v_0})/\|\mathbf{T}_{u_0} \times \mathbf{T}_{v_0}\|$ is defined. If $\mathbf{n}(\Phi(u_0, v_0))$ denotes the unit normal to $S$ at $\Phi(u_0, v_0)$, it follows

that

$$(\mathbf{T}_{u_0} \times \mathbf{T}_{v_0})/\|\mathbf{T}_{u_0} \times \mathbf{T}_{v_0}\| = \pm\mathbf{n}(\Phi(u_0, v_0)).$$

The parametrization $\Phi$ is said to be ***orientation-preserving*** if we have the $+$ sign; that is, if $(\mathbf{T}_u \times \mathbf{T}_v)/\|\mathbf{T}_u \times \mathbf{T}_v\| = \mathbf{n}(\Phi(u, v))$ at all $(u, v) \in D$ for which $S$ is smooth at $\Phi(u, v)$. In other words, $\Phi$ is orientation-preserving if the vector $\mathbf{T}_u \times \mathbf{T}_v$ points to the outside of the surface. If $\mathbf{T}_u \times \mathbf{T}_v$ points to the inside of the surface at all points $(u, v) \in D$ for which $S$ is regular at $\Phi(u, v)$, then $\Phi$ is said to be ***orientation-reversing***. Using the preceding notation, this condition corresponds to the choice $(\mathbf{T}_u \times \mathbf{T}_v)/\|\mathbf{T}_u \times \mathbf{T}_v\| = -\mathbf{n}(\Phi(u, v))$.

It follows from this discussion that the Möbius band $M$ cannot be parametrized by a single parametrization for which $\mathbf{n} = \mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$ and $\mathbf{n}$ is continuous over the whole surface[14] (if there were such a parameterization, then $M$ would indeed have two sides, one determined by $\mathbf{n}$ and one determined by $-\mathbf{n}$). The sphere in Example 1 can be parametrized by a single parametrization, but not by one that is everywhere one-to-one—see the discussion at the beginning of Section 7.4.

*Thus, any one-to-one parametrized surface for which $\mathbf{T}_u \times \mathbf{T}_v$ never vanishes can be considered as an oriented surface with a positive side determined by the direction of $\mathbf{T}_u \times \mathbf{T}_v$.*

**EXAMPLE 2** We can give the unit sphere $x^2 + y^2 + z^2 = 1$ in $\mathbb{R}^3$ (Figure 7.6.5) an orientation by selecting the unit vector $\mathbf{n}(x, y, z) = \mathbf{r}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, which points to the outside of the surface. This choice corresponds to our intuitive notion of outside for the sphere.

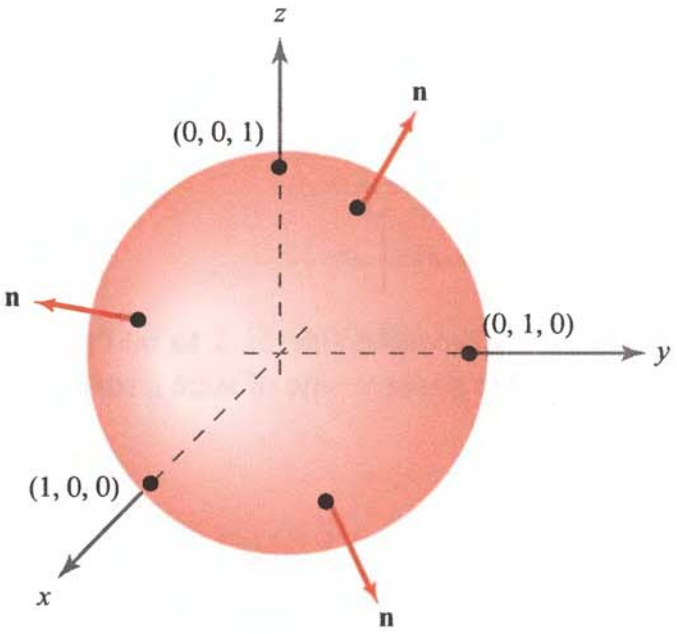


**Figure 7.6.5** The unit sphere oriented by its outward normal $\mathbf{n}$.

Now that the sphere $S$ is an oriented surface, consider the parametrization $\Phi$ of $S$ given in Example 1. The cross product of the tangent vectors $\mathbf{T}_\theta$ and $\mathbf{T}_\phi$—that is,

[14] There is a single parametrization obtained by cutting a strip of paper, twisting it, and gluing the ends, but it produces a discontinuous $\mathbf{n}$ on the surface.

a normal to $S$ is given by

$$(-\sin\phi)[(\cos\theta\sin\phi)\mathbf{i} + (\sin\theta\sin\phi)\mathbf{j} + (\cos\phi)\mathbf{k}] = -\mathbf{r}\sin\phi.$$

Because $-\sin\phi \leq 0$ for $0 \leq \phi \leq \pi$, this normal vector points inward from the sphere. Thus, the given parametrization $\Phi$ is *orientation-reversing*. By swapping the order of $\theta$ and $\phi$, we would get an orientation-preserving parametrization. ▲

## Orientation and the Vector Surface Element of a Sphere

Consider the sphere of radius $R$, namely, $x^2 + y^2 + z^2 = R^2$. It is standard practice to orient the sphere with the *outward unit normal.* In terms of the position vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, the outward unit normal is given by

$$\mathbf{n} = \frac{\mathbf{r}}{R}.$$

The order of spherical coordinates that goes along with this orientation, as is evident from Example 2, is given by the order $(\phi, \theta)$. The computation in Example 2 shows that the surface-area element is then given by

$$d\mathbf{S} = \mathbf{n} \cdot (\mathbf{T}_\phi \times \mathbf{T}_\theta)\, d\phi\, d\theta = \mathbf{r}R\sin\phi\, d\phi\, d\theta = \mathbf{n}R^2\sin\phi\, d\phi\, d\theta.$$

## The Orientation of Graphs

The next example discusses the orientation conventions for graphs. We shall compute the area element on graphs later in this section.

**EXAMPLE 3** Let $S$ be a surface described by $z = g(x, y)$. As in equation (6), Section 7.5, there are two unit normal vectors to $S$ at $(x_0, y_0, g(x_0, y_0))$, namely, $\pm\mathbf{n}$, where

$$\mathbf{n} = \frac{-\dfrac{\partial g}{\partial x}(x_0, y_0)\mathbf{i} - \dfrac{\partial g}{\partial y}(x_0, y_0)\mathbf{j} + \mathbf{k}}{\sqrt{\left[\dfrac{\partial g}{\partial x}(x_0, y_0)\right]^2 + \left[\dfrac{\partial g}{\partial y}(x_0, y_0)\right]^2 + 1}}.$$

We can orient all such surfaces by taking the positive side of $S$ to be the side away from which $\mathbf{n}$ points (Figure 7.6.6). Thus, the positive side of such a surface is

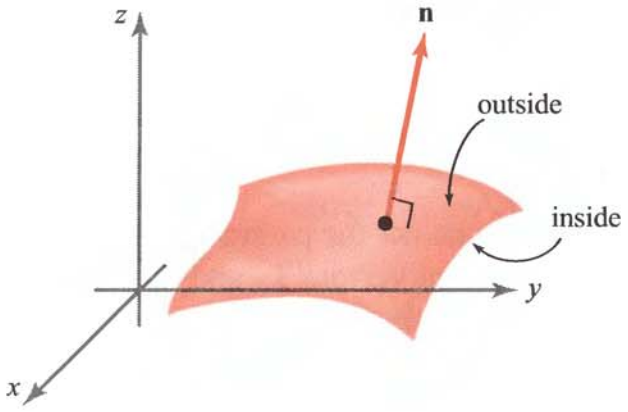


Figure 7.6.6 $\mathbf{n}$ points away from the outside of the surface.

determined by the unit normal $\mathbf{n}$ with positive $\mathbf{k}$ component—that is, it is *upward-pointing*. If we parametrize this surface by $\Phi(u, v) = (u, v, g(u, v))$, then $\Phi$ will be orientation-preserving. ▲

## Independence of Parametrization

We now state without proof a theorem showing that the integral over an oriented surface is independent of the parametrization. The proof of this theorem is analogous to that of Theorem 1 (Section 7.2); the heart of the proof is again the change of variables formula—this time applied to double integrals.

**THEOREM 4: Independence of Surface Integrals on Parametrizations** Let $S$ be an oriented surface and let $\Phi_1$ and $\Phi_2$ be two regular orientation-preserving parametrizations, with $\mathbf{F}$ a continuous vector field defined on $S$. Then

$$\iint_{\Phi_1} \mathbf{F} \cdot d\mathbf{S} = \iint_{\Phi_2} \mathbf{F} \cdot d\mathbf{S}.$$

If $\Phi_1$ is orientation-preserving and $\Phi_2$ orientation-reversing, then

$$\iint_{\Phi_1} \mathbf{F} \cdot d\mathbf{S} = -\iint_{\Phi_2} \mathbf{F} \cdot d\mathbf{S}.$$

If $f$ is a real-valued continuous function defined on $S$, and if $\Phi_1$ and $\Phi_2$ are parametrizations of $S$, then

$$\iint_{\Phi_1} f\,dS = \iint_{\Phi_2} f\,dS.$$

Note that if $f = 1$, we obtain

$$A(S) = \iint_{\Phi_1} dS = \iint_{\Phi_2} dS,$$

thus showing that area is independent of parametrization.

We can therefore unambiguously use the notation

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_{\Phi} \mathbf{F} \cdot d\mathbf{S}$$

(or a sum of such integrals, if $S$ is a union of parametrized surfaces that intersect only along their boundary curves) where $\Phi$ is an orientation-preserving parametrization.

Theorem 4 guarantees that the value of the integral does not depend on the selection of $\Phi$.

## Relation with Scalar Integrals

Recall from formula (1) of Section 7.2 that a line integral $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ can be thought of as the path integral of the tangential component of $\mathbf{F}$ along $\mathbf{c}$ (although for the case in which $\mathbf{c}$ intersects itself, the integral obtained is technically not a path integral). A similar situation holds for surface integrals, because we are assuming that the mappings $\Phi$ defining the surface $S$ are one-to-one except perhaps on the boundary of $D$, which can be ignored for the purposes of integration. Thus, in defining integrals over surfaces, we assume in this book that the surfaces are nonintersecting.

For an oriented smooth surface $S$ and an orientation-preserving parametrization $\Phi$ of $S$, we can express $\iint_S \mathbf{F} \cdot d\mathbf{S}$ as an integral of a real-valued function $f$ over the surface. Let $\mathbf{n} = (\mathbf{T}_u \times \mathbf{T}_v)/\|\mathbf{T}_u \times \mathbf{T}_v\|$ be the unit normal pointing to the outside of $S$. Then

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_\Phi \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)\, du\, dv \\ &= \iint_D \mathbf{F} \cdot \left( \frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} \right) \|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv \\ &= \iint_D (\mathbf{F} \cdot \mathbf{n}) \|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv = \iint_S (\mathbf{F} \cdot \mathbf{n})\, dS = \iint_S f\, dS,\end{aligned}$$

where $f = \mathbf{F} \cdot \mathbf{n}$. We have thus proved the following theorem.

**THEOREM 5** $\iint_S \mathbf{F} \cdot d\mathbf{S}$, the surface integral of $\mathbf{F}$ over $S$, is equal to the integral of the normal component of $\mathbf{F}$ over the surface. In short,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n}\, dS.$$

The observation in Theorem 5 can often save computational effort, as Example 4 demonstrates.

## The Physical Interpretation of Surface Integrals

The geometric and physical significance of the surface integral can be understood by expressing it as a limit of Riemann sums. For simplicity, we assume $D$ is a rectangle. Fix a parametrization $\Phi$ of $S$ that preserves orientation and partition the region $D$ into $n^2$ pieces $D_{ij}$, $0 \le i \le n-1$, $0 \le j \le n-1$. We let $\Delta u$ denote the length of the horizontal side of $D_{ij}$ and $\Delta v$ denote the length of the vertical side of $D_{ij}$. Let $(u, v)$ be a point in $D_{ij}$, and $(x, y, z) = \Phi(u, v)$ the corresponding point on the surface. We consider the parallelogram with sides $\Delta u\, \mathbf{T}_u$ and $\Delta v\, \mathbf{T}_v$ lying in the plane tangent to $S$ at $(x, y, z)$ and the parallelepiped formed by $\mathbf{F}$, $\Delta u\, \mathbf{T}_u$, and $\Delta v\, \mathbf{T}_v$. The volume of

the parallelepiped is the absolute value of the triple product

$$\mathbf{F} \cdot (\Delta u\, \mathbf{T}_u \times \Delta v\, \mathbf{T}_v) = \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)\, \Delta u\, \Delta v.$$

The vector $\mathbf{T}_u \times \mathbf{T}_v$ is normal to the surface at $(x, y, z)$ and points away from the outside of the surface. Thus, the number $\mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)$ is positive when the parallelepiped lies on the outside of the surface (Figure 7.6.7).

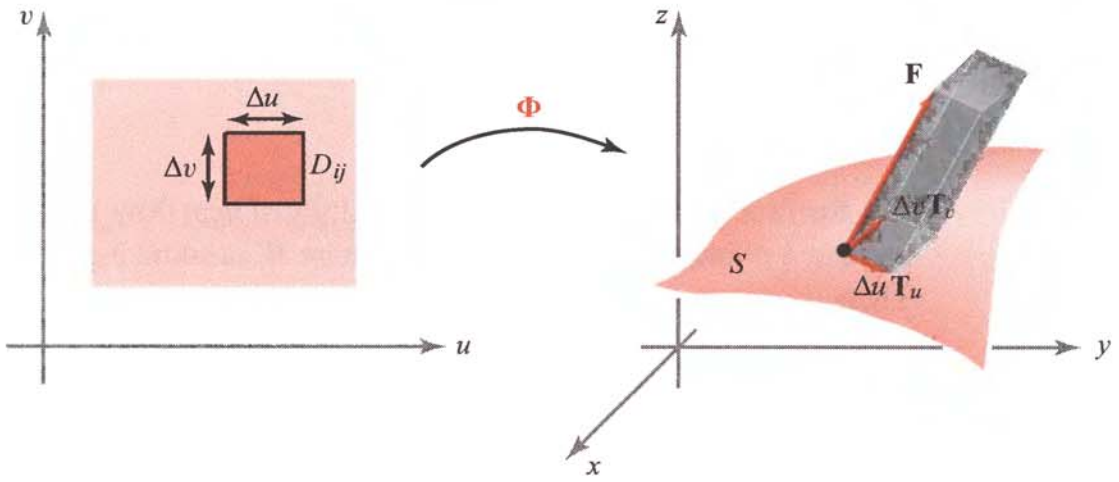


**Figure 7.6.7** $\mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) > 0$ when the parallelpiped formed by $\Delta v\, \mathbf{T}_v$, $\Delta u\, \mathbf{T}_u$, and $\mathbf{F}$ lies to the "outside" of the surface $S$.

In general, the parallelepiped lies on that side of the surface away from which $\mathbf{F}$ is pointing. If we think of $\mathbf{F}$ as the velocity field of a fluid, $\mathbf{F}(x, y, z)$ is pointing in the direction in which fluid is moving across the surface near $(x, y, z)$. Moreover, the number

$$|\mathbf{F} \cdot (\mathbf{T}_u\, \Delta u \times \mathbf{T}_v\, \Delta v)|$$

measures the amount of fluid that passes through the tangent parallelogram per unit time. Because the sign of $\mathbf{F} \cdot (\Delta u\, \mathbf{T}_u \times \Delta v\, \mathbf{T}_v)$ is positive if the vector $\mathbf{F}$ is pointing outward at $(x, y, z)$ and negative if $\mathbf{F}$ is pointing inward, $\sum_{i,j} \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)\, \Delta u\, \Delta v$ is an approximate measure of the net quantity of fluid to flow outward across the surface per unit time. (Remember that "outward" or "inward" depends on our choice of parametrization. Figure 7.6.8 illustrates $\mathbf{F}$ directed outward and inward, given $\mathbf{T}_u$ and $\mathbf{T}_v$.) Hence, *the integral* $\iint_S \mathbf{F} \cdot d\mathbf{S}$ *is the net quantity of fluid to flow across the surface per unit time, that is, the rate of fluid flow*. This integral is also called the ***flux*** of $\mathbf{F}$ across the surface.

In the case where $\mathbf{F}$ represents an electric or a magnetic field, $\iint_S \mathbf{F} \cdot d\mathbf{S}$ is also commonly known as the flux. The reader may be familiar with physical laws (such as *Faraday's law*) that relate flux of a vector field to a circulation (or current) in a bounding loop. This is the historical and physical basis of Stokes's theorem, which we will discuss in Section 8.2. The corresponding principle in fluid mechanics is called *Kelvin's circulation theorem*.

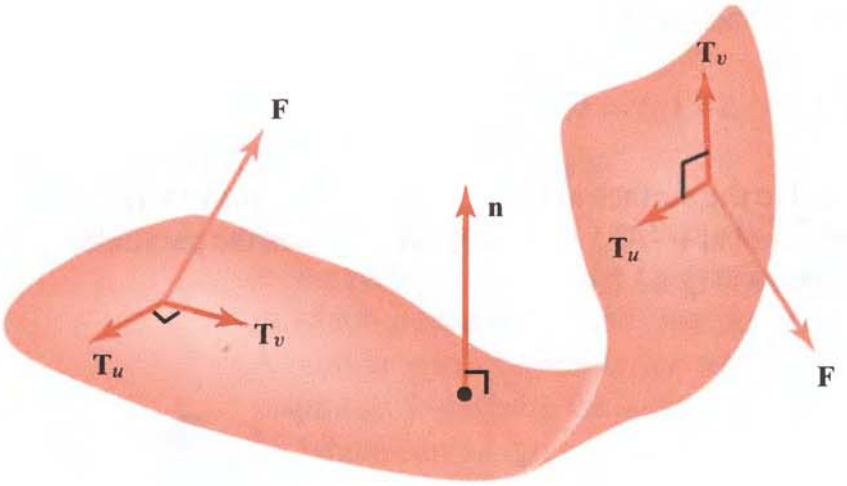


**Figure 7.6.8** When $\mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) > 0$ (left), $\mathbf{F}$ points outward; when $\mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) < 0$ (right), $\mathbf{F}$ points inward.

Surface integrals also apply to the study of heat flow. Let $T(x, y, z)$ be the temperature at a point $(x, y, z) \in W \subset \mathbb{R}^3$, where $W$ is some region and $T$ is a $C^1$ function. Then

$$\nabla T = \frac{\partial T}{\partial x}\mathbf{i} + \frac{\partial T}{\partial y}\mathbf{j} + \frac{\partial T}{\partial z}\mathbf{k}$$

represents the temperature gradient, and heat "flows" with the vector field $-k\,\nabla T = \mathbf{F}$, where $k$ is a positive constant (see Section 8.5). Therefore, $\iint_S \mathbf{F} \cdot d\mathbf{S}$ is the total rate of heat flow or flux across the surface $S$.

**EXAMPLE 4** Suppose a temperature function is given in $\mathbb{R}^3$ by the formula $T(x, y, z) = x^2 + y^2 + z^2$, and let $S$ be the unit sphere $x^2 + y^2 + z^2 = 1$ oriented with the outward normal (see Example 2). Find the heat flux across the surface $S$ if $k = 1$.

SOLUTION We have

$$\mathbf{F} = -\nabla T(x, y, z) = -2x\mathbf{i} - 2y\mathbf{j} - 2z\mathbf{k}.$$

On $S$, the vector $\mathbf{n}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the unit "outward" normal to $S$ at $(x, y, z)$, and $f(x, y, z) = \mathbf{F} \cdot \mathbf{n} = -2x^2 - 2y^2 - 2z^2 = -2$ is the normal component of $\mathbf{F}$. From Theorem 5 we can see that the surface integral of $\mathbf{F}$ is equal to the integral of its normal component $f = \mathbf{F} \cdot \mathbf{n}$ over $S$. Thus,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S f\,dS = -2 \iint_S dS = -2A(S) = -2(4\pi) = -8\pi.$$

The flux of heat is directed toward the center of the sphere (why toward?). Clearly, our observation that $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S f\,dS$ has saved us considerable computational time.

In this example, $\mathbf{F}(x, y, z) = -2x\mathbf{i} - 2y\mathbf{j} - 2z\mathbf{k}$ could also represent an electric field, in which case $\iint_S \mathbf{F} \cdot d\mathbf{S} = -8\pi$ would be the electric flux across $S$. ▲

**EXAMPLE 5** **Gauss' Law** There is an important physical law, due to the great mathematician and physicist K. F. Gauss, that relates the flux of an electric field $\mathbf{E}$ over a "closed" surface $S$ (for example, a sphere or an ellipsoid) to the net charge $Q$ enclosed by the surface, namely (in suitable units),

$$\iint_S \mathbf{E} \cdot d\mathbf{S} = Q \tag{1}$$

(see Figure 7.6.9). Gauss' law will be discussed in detail in Chapter 8. This law is analogous to Ampère's law (see Example 12, Section 7.2).

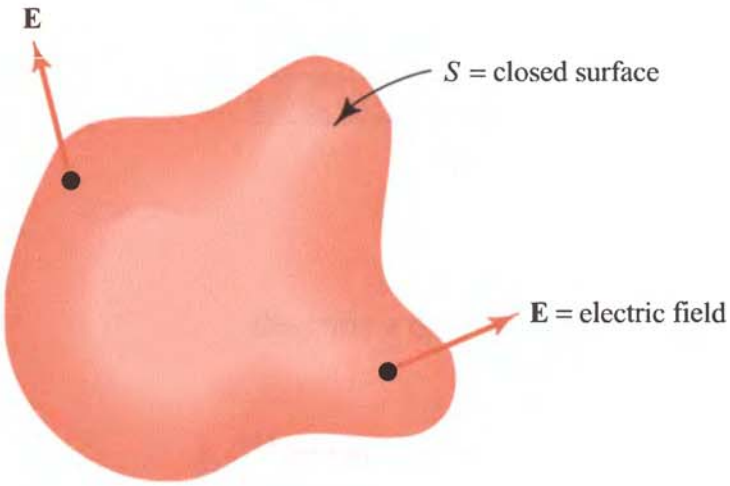


**Figure 7.6.9** Gauss' law: $\iint_S \mathbf{E} \cdot d\mathbf{S} = Q$, where $Q$ is the net charge inside $S$.

Suppose that $\mathbf{E} = E\mathbf{n}$; that is, $\mathbf{E}$ is a constant scalar multiple of the unit normal to $S$. Then Gauss' law, equation (1) in Example 5, becomes

$$\iint_S \mathbf{E} \cdot d\mathbf{S} = \iint_S E \, dS = E \iint_S dS = Q$$

because $E = \mathbf{E} \cdot \mathbf{n}$. Thus,

$$E = \frac{Q}{A(S)}. \tag{2}$$

In the case where $S$ is the sphere of radius $R$, equation (2) becomes

$$E = \frac{Q}{4\pi R^2} \tag{3}$$

(see Figure 7.6.10).

Now suppose that $\mathbf{E}$ arises from an isolated point charge, $Q$. From symmetry it is reasonable that $\mathbf{E} = E\mathbf{n}$, where $\mathbf{n}$ is the unit normal to any sphere centered at $Q$. Hence, equation (3) holds. Consider a second point charge, $Q_0$, located at a distance

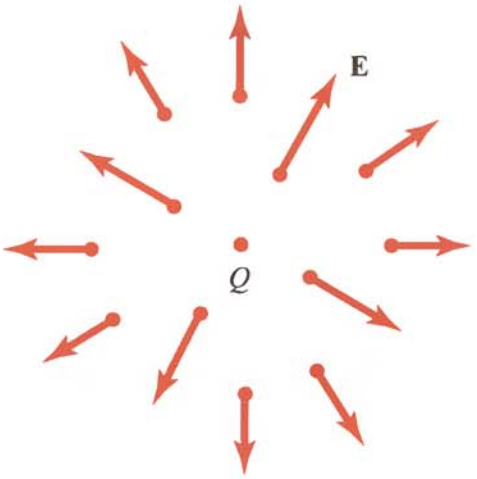


Figure 7.6.10 The field $\mathbf{E}$ due to a point charge $Q$ is $\mathbf{E} = Q\mathbf{n}/4\pi R^2$.

$R$ from $Q$. The force $\mathbf{F}$ that acts on this second charge, $Q_0$, is given by

$$\mathbf{F} = \mathbf{E}Q_0 = EQ_0\mathbf{n} = \frac{QQ_0}{4\pi R^2}\mathbf{n}.$$

If $F$ is the magnitude of $\mathbf{F}$, we have

$$F = \frac{QQ_0}{4\pi R^2},$$

which is ***Coulomb's law*** for the force between two point charges.[15] ▲

## Surface Integrals Over Graphs

Finally, let us derive the surface-integral formulas for vector fields $\mathbf{F}$ over surfaces $S$ that are graphs of functions. Consider the surface $S$ described by $z = g(x, y)$, where $(x, y) \in D$, where $S$ is oriented with the upward pointing unit normal:

$$\mathbf{n} = \frac{-\dfrac{\partial g}{\partial x}\mathbf{i} - \dfrac{\partial g}{\partial y}\mathbf{j} + \mathbf{k}}{\sqrt{\left(\dfrac{\partial g}{\partial x}\right)^2 + \left(\dfrac{\partial g}{\partial y}\right)^2 + 1}}.$$

We have seen that we can parametrize $S$ by $\Phi\colon D \to \mathbb{R}^3$ given by $\Phi(x, y) = (x, y, g(x, y))$. In this case, $\iint_S \mathbf{F} \cdot d\mathbf{S}$ can be written in a particularly simple form. We have

$$\mathbf{T}_x = \mathbf{i} + \frac{\partial g}{\partial x}\mathbf{k}, \qquad \mathbf{T}_y = \mathbf{j} + \frac{\partial g}{\partial y}\mathbf{k}.$$

[15] Sometimes one sees the formula $F = (1/4\pi\varepsilon_0)QQ_0/R^2$. The extra constant $\varepsilon_0$ appears when MKS units are used for measuring charge. We are using CGS, or Gaussian, units.

Thus, $\mathbf{T}_x \times \mathbf{T}_y = -(\partial g/\partial x)\mathbf{i} - (\partial g/\partial y)\mathbf{j} + \mathbf{k}$. If $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ is a continuous vector field, then we get

**The Surface Integral of a Vector Field Over a Graph $S$**

$$\begin{aligned}\iint_S \mathbf{F}\cdot d\mathbf{S} &= \iint_D \mathbf{F}\cdot(\mathbf{T}_x \times \mathbf{T}_y)\,dx\,dy \\ &= \iint_D \left[F_1\left(-\frac{\partial g}{\partial x}\right) + F_2\left(-\frac{\partial g}{\partial y}\right) + F_3\right]dx\,dy.\end{aligned} \tag{4}$$

**EXAMPLE 6** The equations

$$z = 12, \qquad x^2 + y^2 \le 25$$

describe a disk of radius 5 lying in the plane $z = 12$. Suppose $\mathbf{r}$ is the vector field

$$\mathbf{r}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$$

Compute $\iint_S \mathbf{r}\cdot d\mathbf{S}$.

SOLUTION We shall do this in three ways. First, we have $\partial z/\partial x = \partial z/\partial y = 0$, because $z = 12$ is constant on the disk, so

$$\mathbf{r}(x, y, z)\cdot(\mathbf{T}_x \times \mathbf{T}_y) = \mathbf{r}(x, y, z)\cdot(\mathbf{i}\times\mathbf{j}) = \mathbf{r}(x, y, z)\cdot\mathbf{k} = z.$$

Using the original definition at the beginning of this section, the integral becomes

$$\iint_S \mathbf{r}\cdot d\mathbf{S} = \iint_D z\,dx\,dy = \iint_D 12\,dx\,dy = 12(\text{area of } D) = 300\pi.$$

A second solution: Because the disk is parallel to the $xy$ plane, the outward unit normal is $\mathbf{k}$. Hence, $\mathbf{n}(x, y, z) = \mathbf{k}$ and $\mathbf{r}\cdot\mathbf{n} = z$. However, $\|\mathbf{T}_x \times \mathbf{T}_y\| = \|\mathbf{k}\| = 1$, and so we know from the discussion preceding Theorem 5 that

$$\iint_S \mathbf{r}\cdot d\mathbf{S} = \iint_S \mathbf{r}\cdot\mathbf{n}\,dS = \iint_S z\,dS = \iint_D 12\,dx\,dy = 300\pi.$$

Third, we may solve this problem by using formula (4) directly, with $g(x, y) = 12$ and $D$ the disk $x^2 + y^2 \leq 25$:

$$\iint_S \mathbf{r} \cdot d\mathbf{S} = \iint_D (x \cdot 0 + y \cdot 0 + 12)\, dx\, dy = 12(\text{area of } D) = 300\pi. \quad \blacktriangle$$

## Summary: Formulas for Surface Integrals

1. **Parametrized Surface: $\Phi(u, v)$**

   (a) Integral of a scalar function $f$:

   $$\iint_S f\, dS = \iint_D f(\Phi(u, v)) \|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv$$

   (b) Scalar surface element:

   $$dS = \|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv$$

   (c) Integral of a vector field $\mathbf{F}$:

   $$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)\, du\, dv$$

   (d) Vector surface element:

   $$d\mathbf{S} = (\mathbf{T}_u \times \mathbf{T}_v)\, du\, dv = \mathbf{n}\, dS$$

2. **Graph**: $z = g(x, y)$

   (a) Integral of a scalar function $f$:

   $$\iint_S f\, dS = \iint_D \frac{f(x, y, g(x, y))}{\cos\theta}\, dx\, dy$$

   (b) Scalar surface element:

   $$dS = \frac{dx\, dy}{\cos\theta} = \sqrt{\left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + 1}\, dx\, dy,$$

   where $\cos\theta = \mathbf{n} \cdot \mathbf{k}$, and $\mathbf{n}$ is a unit normal vector to the surface.

(c) Integral of a vector field $\mathbf{F}$:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left( -F_1 \frac{\partial g}{\partial x} - F_2 \frac{\partial g}{\partial y} + F_3 \right) dx\, dy$$

(d) Vector surface element:

$$d\mathbf{S} = \mathbf{n} \cdot dS = \left( -\frac{\partial g}{\partial x}\mathbf{i} - \frac{\partial g}{\partial y}\mathbf{j} + \mathbf{k} \right) dx\, dy$$

3. **Sphere:** $x^2 + y^2 + z^2 = R^2$

(a) Scalar surface element:

$$dS = R^2 \sin\phi\, d\phi\, d\theta$$

(b) Vector surface element:

$$d\mathbf{S} = (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})R \sin\phi\, d\phi\, d\theta = \mathbf{r}R \sin\phi\, d\phi\, d\theta = \mathbf{n}R^2 \sin\phi\, d\phi\, d\theta$$

## EXERCISES

**1.** Let the temperature of a point in $\mathbb{R}^3$ be given by $T(x, y, z) = 3x^2 + 3z^2$. Compute the heat flux across the surface $x^2 + z^2 = 2, 0 \le y \le 2$, if $k = 1$.

**2.** Compute the heat flux across the unit sphere $S$ if $T(x, y, z) = x$. Can you interpret your answer physically?

**3.** Let $S$ be the closed surface that consists of the hemisphere $x^2 + y^2 + z^2 = 1, z \ge 0$, and its base $x^2 + y^2 \le 1, z = 0$. Let $\mathbf{E}$ be the electric field defined by $\mathbf{E}(x, y, z) = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$. Find the electric flux across $S$. (HINT: Break $S$ into two pieces $S_1$ and $S_2$ and evaluate $\iint_{S_1} \mathbf{E} \cdot d\mathbf{S}$ and $\iint_{S_2} \mathbf{E} \cdot d\mathbf{S}$ separately.)

**4.** Let the velocity field of a fluid be described by $\mathbf{F} = \sqrt{y}\mathbf{i}$ (measured in meters per second). Compute how many cubic meters of fluid per second are crossing the surface $x^2 + z^2 = 1, 0 \le y \le 1, 0 \le x \le 1$.

**5.** Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where $S$ is the surface $x^2 + y^2 + 3z^2 = 1, z \le 0$, and $\mathbf{F}$ is the vector field $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + zx^3y^2\mathbf{k}$. (Let $\mathbf{n}$, the unit normal, be upward pointing.)

**6.** Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ where $\mathbf{F} = (x^2 + y - 4)\mathbf{i} + 3xy\mathbf{j} + (2xz + z^2)\mathbf{k}$ and $S$ is the surface $x^2 + y^2 + z^2 = 16, z \ge 0$. (Let $\mathbf{n}$, the unit normal, be upward pointing.)

**7.** Calculate the integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $S$ is the entire surface of the solid half ball $x^2 + y^2 + z^2 \leq 1, z \geq 0$, and $\mathbf{F} = (x + 3y^5)\mathbf{i} + (y + 10xz)\mathbf{j} + (z - xy)\mathbf{k}$. (Let $S$ be oriented by the outward pointing normal.)

**8.*** A restaurant is being built on the side of a mountain. The architect's plans are shown in Figure 7.6.11.

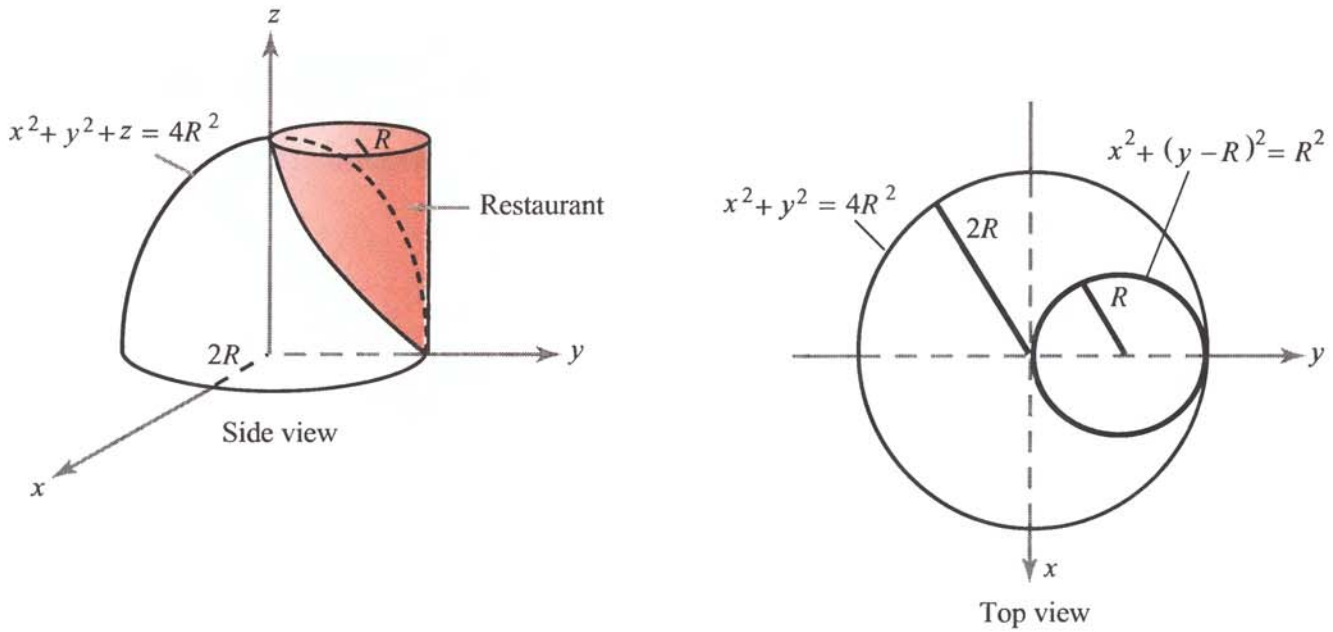


**Figure 7.6.11** Restaurant plans.

(a) The vertical curved wall of the restaurant is to be built of glass. What will be the surface area of this wall?

(b) To be large enough to be profitable, the consulting engineer informs the developer that the volume of the interior must exceed $\pi R^4/2$. For what $R$ does the proposed structure satisfy this requirement?

(c) During a typical summer day, the environs of the restaurant are subject to a temperature field given by

$$T(x, y, z) = 3x^2 + (y - R)^2 + 16z^2.$$

A heat flux density $\mathbf{V} = -k\,\nabla T$ ($k$ is a constant depending on the grade of insulation to be used) through all sides of the restaurant (including the top and the contact with the hill) produces a heat flux. What is this total heat flux? (Your answer will depend on $R$ and $k$.)

**9.** Find the flux of the vector field $\mathbf{V}(x, y, z) = 3xy^2\mathbf{i} + 3x^2y\mathbf{j} + z^3\mathbf{k}$ out of the unit sphere.

**10.** Evaluate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n}\, dA$, where $\mathbf{F}(x, y, z) = \mathbf{i} + \mathbf{j} + z(x^2 + y^2)^2\mathbf{k}$ and $S$ is the surface of the cylinder $x^2 + y^2 \leq 1, 0 \leq z \leq 1$.

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*The solution to this problem may be somewhat time-consuming.

**11.** Let $S$ be the surface of the unit sphere. Let $\mathbf{F}$ be a vector field and $F_r$ its radial component. Prove that

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi} F_r \sin\phi \, d\phi \, d\theta.$$

What is the corresponding formula for real-valued functions $f$?

**12.** Prove the following mean-value theorem for surface integrals: If $\mathbf{F}$ is a continuous vector field, then

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = [\mathbf{F}(\mathrm{Q}) \cdot \mathbf{n}(\mathrm{Q})] A(S)$$

for some point $\mathrm{Q} \in S$, where $A(S)$ is the area of $S$. [HINT: Prove it for real functions first, by reducing the problem to one of a double integral: Show that if $g \geq 0$, then

$$\iint_D fg \, dA = f(\mathrm{Q}) \iint_D g \, dA$$

for some $\mathrm{Q} \in D$ (do it by considering $(\iint_D fg \, dA)/(\iint_D g \, dA)$ and using the intermediate value theorem).]

**13.** Work out a formula like that in Exercise 11 for integration over the surface of a cylinder.

**14.** Let $S$ be a surface in $\mathbb{R}^3$ that is actually a subset $D$ of the $xy$ plane. Show that the integral of a scalar function $f(x, y, z)$ over $S$ reduces to the double integral of $f(x, y, z)$ over $D$. What does the surface integral of a vector field over $S$ become? (Make sure your answer is compatible with Example 6.)

**15.** Let the velocity field of a fluid be described by $\mathbf{F} = \mathbf{i} + x\mathbf{j} + z\mathbf{k}$ (measured in meters per second). Compute how many cubic meters of fluid per second are crossing the surface described by $x^2 + y^2 + z^2 = 1, z \geq 0$.

**16.** (a) A uniform fluid that flows vertically downward (heavy rain) is described by the vector field $\mathbf{F}(x, y, z) = (0, 0, -1)$. Find the total flux through the cone $z = (x^2 + y^2)^{1/2}$, $x^2 + y^2 \leq 1$.

(b) The rain is driven sideways by a strong wind so that it falls at a 45° angle, and it is described by $\mathbf{F}(x, y, z) = -(\sqrt{2}/2, 0, \sqrt{2}/2)$. Now what is the flux through the cone?

**17.** For $a > 0, b > 0, c > 0$, let $S$ be the upper half ellipsoid

$$S = \left\{ (x, y, z) \,\middle|\, \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, z \geq 0 \right\},$$

with orientation determined by the upward normal. Compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = (x^3, 0, 0)$.

**18.** If $S$ is the upper hemisphere $\{(x, y, z) \mid x^2 + y^2 + z^2 = 1, z \geq 0\}$ oriented by the normal pointing out of the sphere, compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$ for parts (a) and (b).

(a) $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j}$

(b) $\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j}$

(c) For each of these vector fields, compute $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ and $\int_C \mathbf{F} \cdot d\mathbf{s}$ where $C$ is the unit circle in the $xy$ plane traversed in the counterclockwise direction (as viewed from the positive $z$ axis). (Notice that $C$ is the boundary of $S$. The phenomenon illustrated here will be studied more thoroughly in the next chapter, using Stokes' theorem.)

# 7.7 Applications to Differential Geometry, Physics, and Forms of Life*

In the first half of the nineteenth century, the great German mathematician Karl Friedrich Gauss developed a theory of curved surfaces in $\mathbb{R}^3$. More than a century earlier, Isaac Newton had defined a measure of the curvature of a space curve, and Gauss was able to find extensions of this idea of curvature that would apply to surfaces. In so doing, Gauss made several remarkable discoveries.

## Curvature of Surfaces

For paths $\mathbf{c}: [a, b] \to \mathbb{R}^3$ that have unit speed—that is, $\|\mathbf{c}'(t)\| = 1$—the curvature $\kappa$ of the image curve $\kappa(\mathbf{c}(t))$ at the point $\mathbf{c}(t)$ is defined to be the length of the acceleration vector. That is, $\|\mathbf{c}''(t)\| = \kappa(\mathbf{c}(t))$. For paths $\mathbf{c}$ in space, the curvature is a true measure of the curvature of the geometric image curve $C$. As we saw at the end of Section 7.1, the "total curvature" $\int \kappa \, ds$ over $C$ has "topological" implications. The same, and even more, will hold for Gauss' definition of the total curvature of a surface. We begin with some definitions.

Let $\mathbf{\Phi}: D \to \mathbb{R}^3$ be a smooth parametrized surface. Then, as we know,

$$\mathbf{T}_u = \frac{\partial \mathbf{\Phi}}{\partial u} \qquad \text{and} \qquad \mathbf{T}_v = \frac{\partial \Phi}{\partial v}$$

are tangent vectors to the image surface $S = \mathbf{\Phi}(D)$ at the point $\mathbf{\Phi}(u, v)$. We will also assume that there is a well-defined normal vector; that is, we assume the surface is regular: $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$.

Let

$$E = \left\| \frac{\partial \mathbf{\Phi}}{\partial u} \right\|^2, \qquad F = \frac{\partial \mathbf{\Phi}}{\partial u} \cdot \frac{\partial \mathbf{\Phi}}{\partial v}, \qquad G = \left\| \frac{\partial \mathbf{\Phi}}{\partial v} \right\|^2.$$

In Exercise 15 of Section 7.5, we saw that

$$\|\mathbf{T}_u \times \mathbf{T}_v\|^2 = EG - F^2.$$

*This section can be skipped on a first reading without loss of continuity.

For notational reasons, we denote $EG - F^2$ by $W$. Furthermore, we let

$$\mathbf{N} = \frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} = \frac{\mathbf{T}_u \times \mathbf{T}_v}{\sqrt{W}}$$

denote the *unit* normal vector to the image surface at $p = \Phi(u, v)$. Next we will define two new measures of the curvature of a surface at $p$—the "Gauss curvature," $K(p)$, and the "mean curvature," $H(p)$. Both of these curvatures have deep connections to the curvature of space curves, which illuminate the meaning of their definitions, but we do not explore these here.

To define these two curvatures, we first define three new functions $\ell, m, n$ on $S$ as follows:

$$\begin{aligned} \ell(p) &= \mathbf{N}(u, v) \cdot \frac{\partial^2 \Phi}{\partial u^2} = \mathbf{N}(u, v) \cdot \Phi_{uu} \\ m(p) &= \mathbf{N}(u, v) \cdot \frac{\partial^2 \Phi}{\partial u \partial v} = \mathbf{N}(u, v) \cdot \Phi_{uv} \\ n(p) &= \mathbf{N}(u, v) \cdot \frac{\partial^2 \Phi}{\partial v^2} = \mathbf{N}(u, v) \cdot \Phi_{vv}. \end{aligned} \tag{1}$$

The ***Gauss curvature*** $K(p)$ of $S$ at $p$ is given by

$$K(p) = \frac{\ell n - m^2}{W}, \tag{2}$$

and the ***mean curvature*** $H(p)$ of $S$ at $p$ is defined by[16]

$$H(p) = \frac{G\ell + En - 2Fm}{2W}, \tag{3}$$

where the right-hand sides of both expressions are calculated at the point $p = \Phi(u, v)$.

**EXAMPLE 1** **Planes Have Zero Curvature** Let $\Phi(u, v) = \alpha u + \beta v + \gamma$, $(u, v) \in \mathbb{R}^2$, where $\alpha, \beta, \gamma$ are vectors in $\mathbb{R}^3$. According to Example 1 of Section 7.3, this determines a parametrized plane in $\mathbb{R}^3$. Show that at every point, both the Gauss and mean curvatures are zero, and hence $K$ and $H$ vanish identically.

SOLUTION Because $\Phi_{uu} = \Phi_{uv} = \Phi_{vv} \equiv 0$, the functions $\ell, m, n$ vanish everywhere, and so do $H$ and $K$. Thus, a plane has "zero" curvature. Hence, at least in this example, we ought to be convinced that $H$ and $K$ actually do measure the *flatness* of the plane. Conversely, one can show that if $H$ and $K$ vanish identically, then $S$ is part of a plane (see Exercise 10). ▲

[16]Technically speaking, $K(p)$ and $H(p)$ could, in principle, depend on the parametrization $\Phi$ of $S$, but one can show that they are, in fact, independent of $\Phi$.

**EXAMPLE 2** **Curvature of a Hemisphere** Let

$$\Phi(u, v) = (u, v, g(u, v)),$$

where $g(u, v) = \sqrt{R^2 - u^2 - v^2}$ is a parametrization of the "upper hemisphere" of radius $R$. Show that the Gauss curvature at every point is $1/R^2$ and the mean curvature is $1/R$.

SOLUTION We must first calculate the following quantities:

$$\mathbf{T}_u, \mathbf{T}_v, \mathbf{T}_u \times \mathbf{T}_v, \Phi_{uu}, \Phi_{vv}, \Phi_{uv}, E, G, F, \ell, m, n.$$

First of all, we have

$$\Phi_u = \mathbf{T}_u = \mathbf{i} - \frac{u}{\sqrt{R^2 - u^2 - v^2}}\mathbf{k}$$

$$\Phi_v = \mathbf{T}_v = \mathbf{j} - \frac{v}{\sqrt{R^2 - v^2 - v^2}}\mathbf{k}.$$

From formula (2) in Section 7.3, we have

$$\mathbf{T}_u \times \mathbf{T}_v = -\frac{\partial g}{\partial u}\mathbf{i} - \frac{\partial g}{\partial v}\mathbf{j} + \mathbf{k}$$

$$= \frac{u}{\sqrt{R^2 - u^2 - v^2}}\mathbf{i} + \frac{v}{\sqrt{R^2 - u^2 - v^2}}\mathbf{j} + \mathbf{k}.$$

Therefore,

$$E = \|\Phi_u\|^2 = 1 + \frac{u^2}{R^2 - u^2 - v^2} = \frac{R^2 - v^2}{R^2 - u^2 - v^2}$$

$$G = \|\Phi_v\|^2 = \frac{R^2 - u^2}{R^2 - u^2 - v^2}$$

$$F = \Phi_u \cdot \Phi_v = \frac{uv}{R^2 - u^2 - v^2}.$$

From Exercise 15 of Section 7.5, we know that

$$\|\mathbf{T}_u \times \mathbf{T}_v\|^2 = EG - F^2 = \frac{(R^2 - v^2)(R^2 - u^2) - u^2v^2}{(R^2 - u^2 - v^2)^2}$$

$$= \frac{R^4 - R^2u^2 - R^2v^2}{(R^2 - u^2 - v^2)^2} = \frac{R^2}{(R^2 - u^2 - v^2)} = W.$$

Now a direct calculation shows that

$$\Phi_{uu} = \frac{R^2 - v^2}{(R^2 - u^2 - v^2)^{3/2}}\mathbf{k}$$

$$\Phi_{vv} = \frac{R^2 - u^2}{(R^2 - u^2 - v^2)^{3/2}}\mathbf{k}$$

$$\Phi_{uv} = \frac{uv}{(R^2 - u^2 - v^2)^{3/2}}\mathbf{k}.$$

Furthermore,

$$\begin{aligned}\mathbf{N} &= \frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} = \frac{\mathbf{T}_u \times \mathbf{T}_v}{\sqrt{W}} \\ &= \frac{\sqrt{R^2 - u^2 - v^2}}{R} \cdot \left( \frac{u}{\sqrt{R^2 - u^2 - v^2}}\mathbf{i} + \frac{v}{\sqrt{R^2 - u^2 - v^2}}\mathbf{j} + \mathbf{k} \right) \\ &= \frac{1}{R}\left(u\mathbf{i} + v\mathbf{j} + \sqrt{R^2 - u^2 - v^2}\,\mathbf{k}\right).\end{aligned}$$

Thus,

$$\ell = \mathbf{N} \cdot \Phi_{uv} = \frac{1}{R}\left(\frac{R^2 - v^2}{R^2 - u^2 - v^2}\right)$$

$$n = \mathbf{N} \cdot \Phi_{vv} = \frac{1}{R}\left(\frac{R^2 - u^2}{R^2 - u^2 - v^2}\right)$$

$$m = \mathbf{N} \cdot \Phi_{uv} = \frac{1}{R}\left(\frac{uv}{R^2 - u^2 - v^2}\right).$$

Therefore,

$$\begin{aligned}\ell n - m^2 &= \frac{1}{R^2}\left(\frac{(R^2 - v^2)(R^2 - u^2) - u^2v^2}{(R^2 - u^2 - v^2)^2}\right) \\ &= \frac{1}{R^2 - u^2 - v^2}.\end{aligned}$$

Dividing this by $W$ yields $K = 1/R^2$. Thus, the Gauss curvature does not change from point to point on the hemisphere; that is, it is constant. This conforms to our intuition that the sphere is perfectly symmetrical and that its curvature is everywhere equal. Hence, the mean curvature should also be constant. This is verified by the

following calculation:

$$\begin{aligned} H &= \frac{G\ell + En - 2Fm}{2W} \\ &= \frac{1}{2W}\left\{\left(\frac{R^2-u^2}{R^2-u^2-v^2}\right)\frac{1}{R}\left(\frac{R^2-v^2}{R^2-u^2-v^2}\right)\right. \\ &\quad \left. + \left(\frac{R^2-v^2}{R^2-u^2-v^2}\right)\frac{1}{R}\left(\frac{R^2-u^2}{R^2-u^2-v^2}\right) - 2\frac{u^2v^2}{(R^2-u^2-v^2)^2}\right\} \\ &= \frac{1}{W}\left\{\frac{R}{R^2-u^2-v^2}\right\} = \frac{1}{R}. \end{aligned}$$ ▲

## Surfaces of Constant Curvature

Surfaces of constant Gauss and mean curvature are of great interest to mathematicians. It was known in the nineteenth century that the only closed and bounded smooth surfaces with "no boundary" and with constant Gauss curvature were spheres. In the twentieth century, the Russian mathematician Alexandrov showed that the only closed and bounded smooth surfaces without a boundary that do not intersect themselves and that have constant mean curvature must also be spheres. Mathematicians believed that Alexandrov's result held even if the surface *was* allowed to intersect itself, but no one could find a proof. In 1984, Professor Henry Wente (Toledo, Ohio) startled the world by finding a self-intersecting torus of constant mean curvature.

Surfaces of constant mean curvature are physically relevant and occur throughout nature. Soap bubble formations have constant nonzero mean curvature (see Figure 7.7.1), and soap film formations (containing no air) have constant mean curvature zero (see Figures 7.7.2 and 7.7.3).

Figure 7.7.1 Soap bubble formation; $H$ = constant.

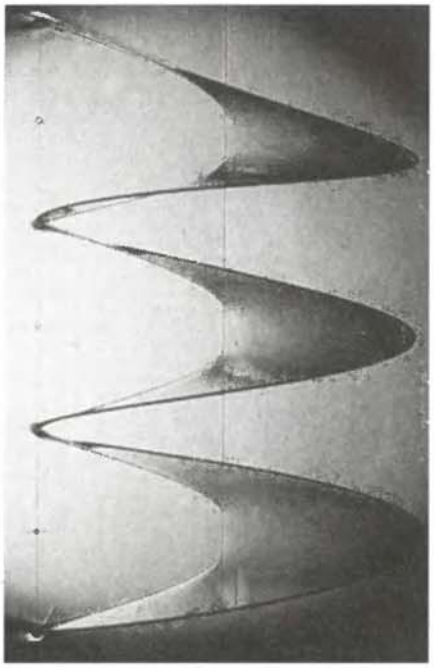


Figure 7.7.2 A helicoid, $H = 0$.

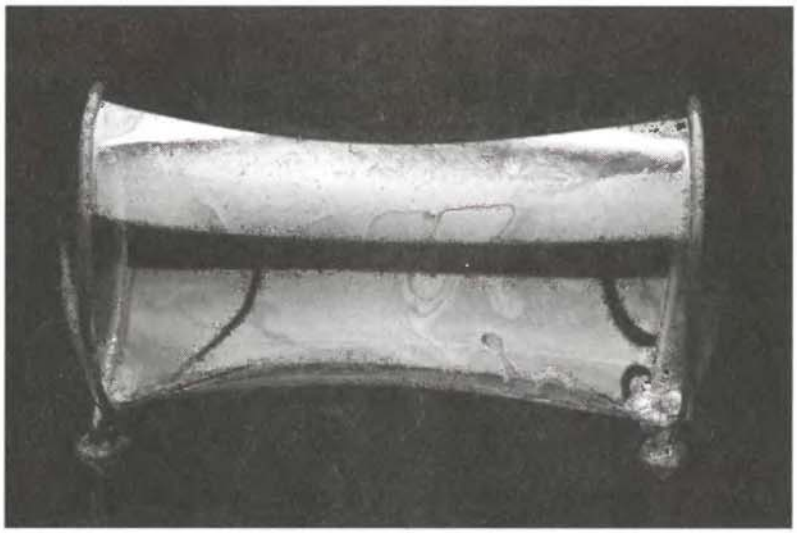


Figure 7.7.3 A soap film, $H = 0$, spanning two circular wires; this one is the catenoid.

In the early nineteenth century, the French mathematician Delaunay discovered all surfaces of revolution that have constant mean curvature. They are the cylinder, sphere, catenoid, unduloid, and nodoid. The catenoid exists as a soap film surface spanning two circular contours.

## Optimal Shapes in Nature

Throughout the ages, people have speculated on why things are shaped the way they are. Why are the earth and the stars "round" and not cubical? Why are life forms shaped the way they are?

In 1917, the British natural philosopher D'Arcy Thompson published a provocative work entitled *On Growth and Form*, in which he investigated the forces behind the creation of living forms in nature. He wrote:

> In an organism, great or small, it is not merely the nature of the motions of the living substance which we must interpret in terms of force (according to kinetics), but also the conformation of the organism itself, whose permanence or equilibrium is explained by the interaction or balance of forces, as described in statics.

Surprisingly, Thompson discovered *all* of Delaunay's surfaces in the form of unicellular organisms (see Figure 7.7.4). The constant mean curvature of these organisms

can be explained by minimum principles similar to those described in the Historical Note in Section 3.3. In 1952, Watson and Crick determined that the structure of DNA is that of a double helix, a discovery that set the stage for the genetic revolution. We know from soap films, as in Figure 7.2.2, that nature likes helicoid forms, and nature tends to repeat patterns. A better understanding of the scientific principles underlying life may ultimately help mathematics play a more prominent role in this area of theoretical biology.

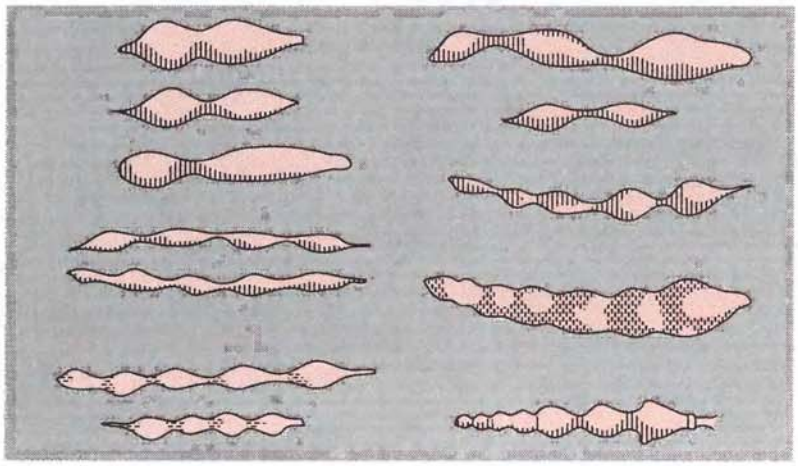


**Figure 7.7.4** Surfaces of revolution of constant mean curvature as unicellulars.

## Curvature and Physics

The theory of curved surfaces, initiated by Gauss, has had a profound effect on physics. Gauss realized that the Gauss curvature K of a surface depended only on the measure of distance *on the surface itself*; that is, curvature was *intrinsic to the surface.* This is not true of the mean curvature $H$. Thus, beings "living" on the surface would be able to tell that the surface was curving, without any reference to an "external" world. Gauss himself found this mathematical result to be so striking that he named it *theorema egregium*, or "remarkable theorem." Gauss' theory was generalized by his student Bernhard Riemann to $n$-dimensional surfaces for which one could describe a notion of curvature.

Recall that Newton created the idea of a gravitational force acting over vast galactic distances, pulling galaxies together as well as pushing them apart (see Figure 7.7.5). In the early 1900s, Albert Einstein used Riemann's ideas to develop the *general theory of relativity*, a theory of gravitation that eliminated the need to consider forces (as Newton did) acting over great distances. Einstein's theory explained the bending of light by the sun, black holes, the expansion of the universe, the formation of galaxies, and the Big Bang itself. For most applications, including the dynamics of our solar system, Newton's theory suffices and is commonly used today by NASA to plan space missions, as we saw in Section 4.1. But for

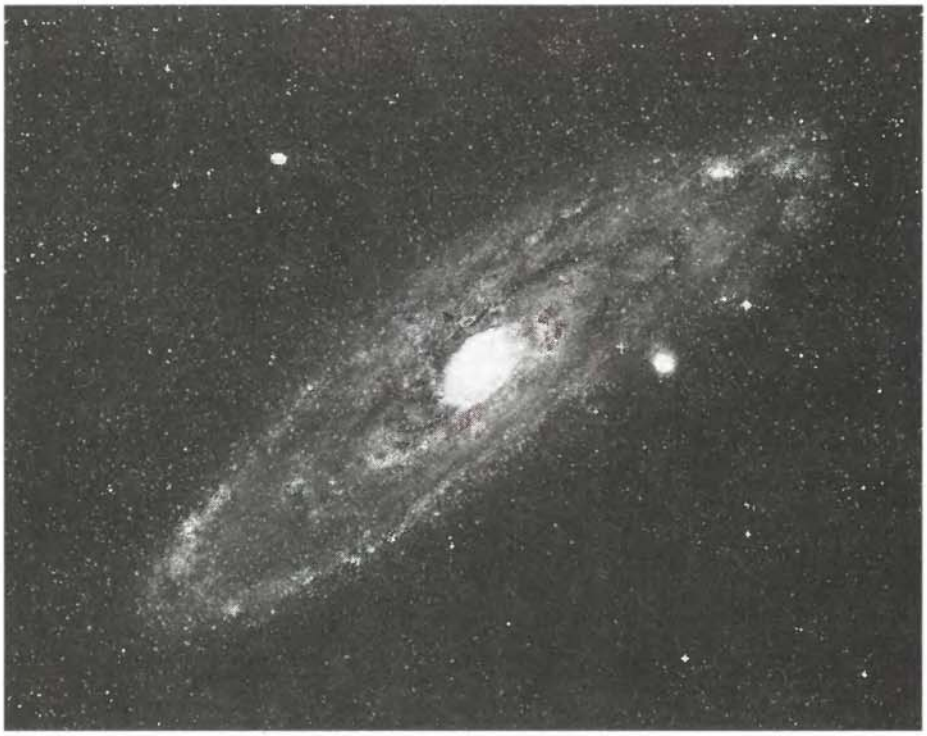


**Figure 7.7.5** The Andromeda Galaxy. It will collide with the Milky Way in roughly 2 billion years.

cosmological applications on the grand scale, Einstein's theory replaced that of Isaac Newton, published in his *Principia* in 1687.

As a testament to his genius, and despite the astounding success of this theory, Newton was nevertheless disturbed by questions about *how* this gravitational force acted. He could give no other explanation than to say, "I have not been able to deduce from phenomena the reason for these properties of gravitation, and I do not invent hypotheses; for anything which cannot be deduced from phenomena should be called an hypothesis." Moreover, in a letter to his friend, Richard Bentley, Newton wrote:

> That gravity should be innate, inherent and essential to matter, so that one body may act upon another at a distance, through which their action may be conveyed from one to another, is to me so great an absurdity that I believe no man, who has in philosophical matters a competent faculty of thinking, can ever fall into it.

Newton coined the term *action at a distance* (which means "force acting at a distance") to describe the mysterious effect of gravitation over large distances. This effect is as difficult to understand today as it was in Newton's time.

Johann Bernoulli found it difficult to believe in the concept of a force that acts through a vacuum of space over distances of even hundreds of millions of miles. He viewed this force as a concept revolting to minds unaccustomed to accepting

**Figure 7.7.6** Albert Einstein (1879–1955) at his desk in the Patent Office, Bern, 1905.

any principle in physics, save those that are incontestable and evident. Additionally, Leibniz considered gravitation to be an incorporeal and inexplicable power, philosophically void.

It was perhaps Albert Einstein's greatest inspiration (see Figure 7.7.6) to replace Newton's model of gravitation with a model that would have thrilled the early Greeks—*a geometric model of gravitation*. In Einstein's theory, the concept of a force acting through great distances has been replaced by the *curvature* of a space–time[17] world. As the quote at the beginning of the chapter illustrates, W. K. Clifford had a premonition of events to come! In order to elucidate Einstein's scheme, we shall present an oversimplified model that conveys some of his basic ideas.

We represent space by a surface that we imagine as an originally flat trampoline (the vacuum state), which is at some point strongly deformed by the weight of a gigantic steel ball (the sun). A tiny steel ball rolling on the trampoline is our planet Earth (see Figure 7.7.7)

If we roll the small steel ball across the flat trampoline, it will travel in a straight-line path. However, if we now place the gigantic steel ball in the center, it will cause the trampoline to bend, or "curve," even "far away" from the large ball. If we then give our little ball a push, it will no longer travel in a straight line but in a curved

[17] Space–time is locally like $\mathbb{R}^4$ with three space coordinates and one time coordinate.

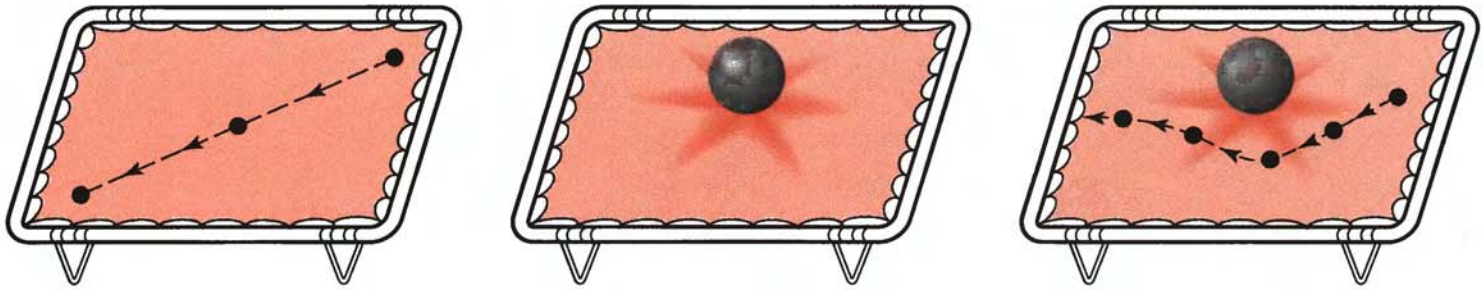



**Figure 7.7.7** (a) A particle on a taut trampoline moves in a straight line. (b) A heavy steel ball distorts the trampoline. (c) A particle on the distorted trampoline follows a curved path.

path. The big ball affects the trajectory of the little ball by curving the space around it. With just the right push, the little ball might even orbit the big one for a while. This trampoline model explains how a large body could, by curving space, influence a small one over great distance.

Einstein stated that space–time is curved by matter and energy. In this curved space–time, even light rays are *bent* as they pass near massive objects like our sun. Thanks to Gauss and Riemann, the curvature of space–time requires no external "universe" in which it curves.

The equations that tell one how much space and time are curved by matter and energy are known as *Einstein's field equations.* A description of them is beyond the scope of this book, but the mathematical kernel from which these equations arise is not; this kernel is based on another remarkable result of the research of Gauss and Bonnet.

## Gauss–Bonnet Theorem

In Example 2, we computed the Gauss curvature $K$ of the sphere $x^2 + y^2 + z^2 = R^2$ of radius $R$ and found it to be the constant $1/R^2$. The Gauss curvature $K$ is a scalar-valued function over the surface, and as such we can integrate it over the surface. We wish to consider a constant times this surface integral, namely,

$$\frac{1}{2\pi} \iint_S K \, dA.$$

For the sphere of radius $R$, this quantity becomes

$$\frac{1}{2\pi R^2} \iint_S dA = \frac{4\pi R^2}{2\pi R^2} = 2.$$

What Gauss and Bonnet discovered was that if $S$ is *any* "sphere-like" closed surface (closed and bounded, but with no boundary, as in Figure 7.7.8), then

$$\frac{1}{2\pi}\iint_S K\,dA = 2$$

still holds.[18]

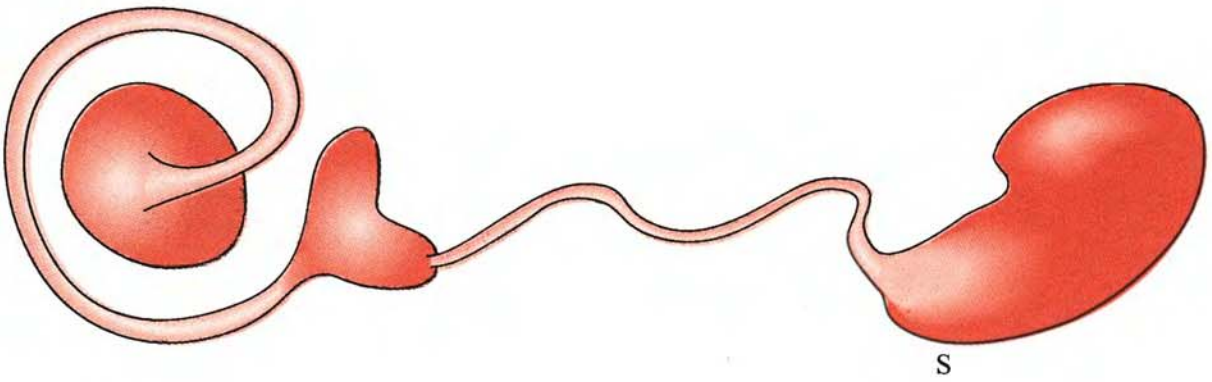



**Figure 7.7.8** A deformed sphere. $1/2\pi \iint_S K\,dA = 2$.

Thus, the integral

$$\frac{1}{2\pi}\iint_S K\,dA$$

always equals the integer 2, and is therefore a *topological invariant* of the surface. That the integral of curvature should be an interesting quantity should be already clear from the discussion at the end of Section 7.1.

Now consider a torus, or doughnut. The torus can be considered as coming from the sphere by cutting out two disks and gluing in a handle (see Figure 7.7.9).

Moreover, we can continue this process adding 1, 2, 3, ..., $g$ handles to the sphere. If $g$ handles are attached, we call the resulting surface a surface of genus $g$, as in Figure 7.7.10. Notice that the torus has genus 1.

If two surfaces have a different genus, they are topologically distinct, and thus cannot be obtained from one another by bending or stretching. Interestingly, even two surfaces with the same genus can sit in space in quite different and complex ways, as in Figure 7.7.11. Astonishingly, even though the integral (or total curvature) given by $(1/2\pi)\iint_S K\,dA$ depends on the genus, it does not depend on how the surface sits in space (and thus not on $K$).

[18]Roughly speaking, this means that $S$ can be obtained from the sphere by bending and stretching (like with a balloon) but not tearing (the balloon bursts!)

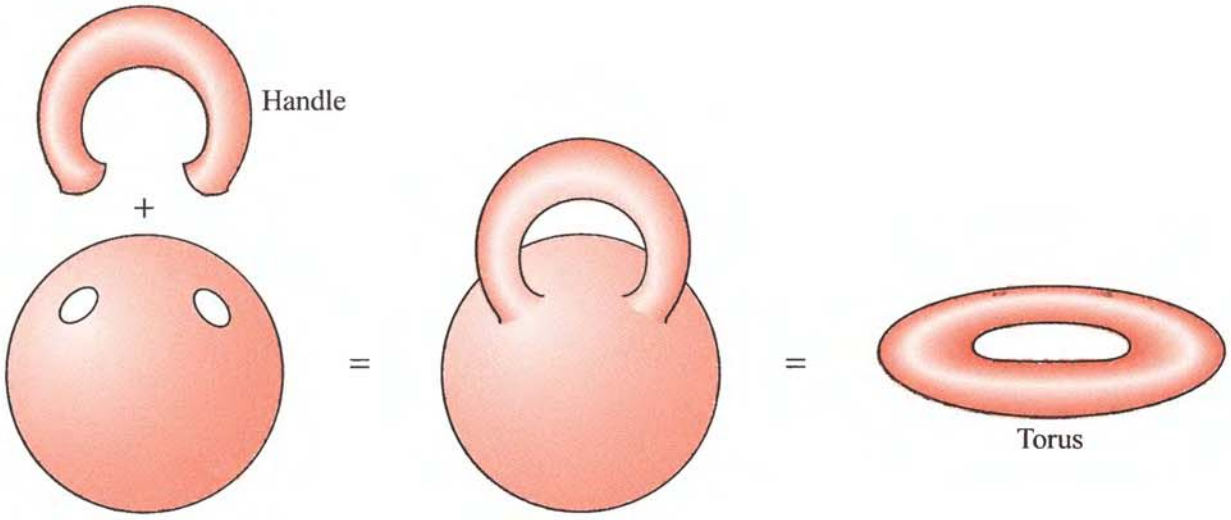



**Figure 7.7.9** Gluing a handle to a sphere to obtain a torus.

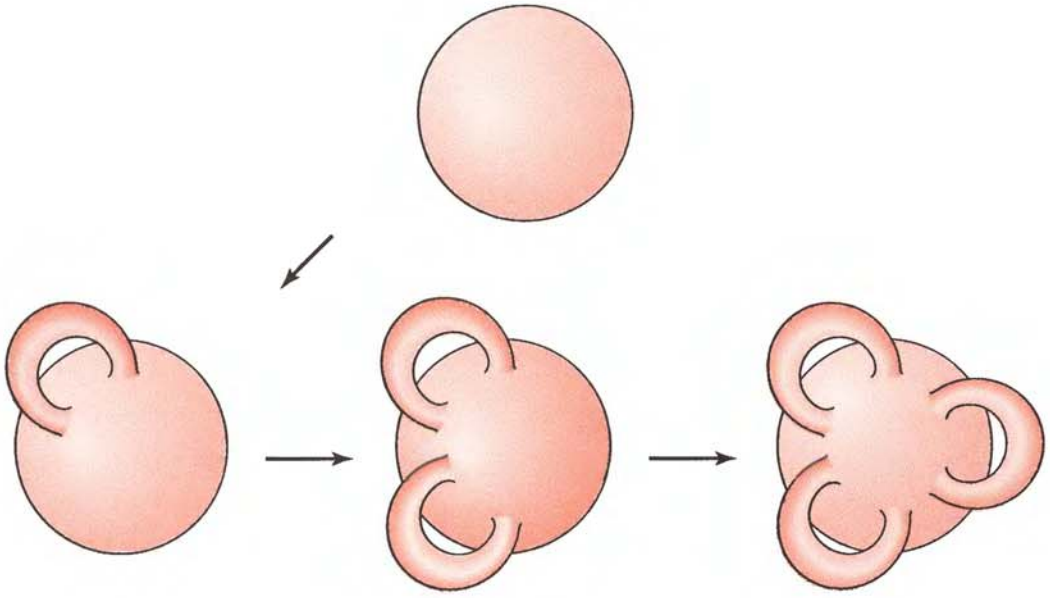


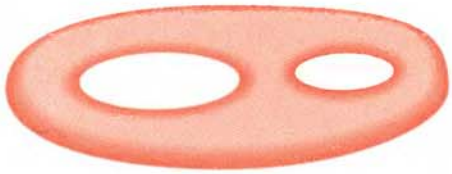
**Figure 7.7.10** A sphere with 0, 1, 2, 3 handles attached.

Gauss and Bonnet proved that

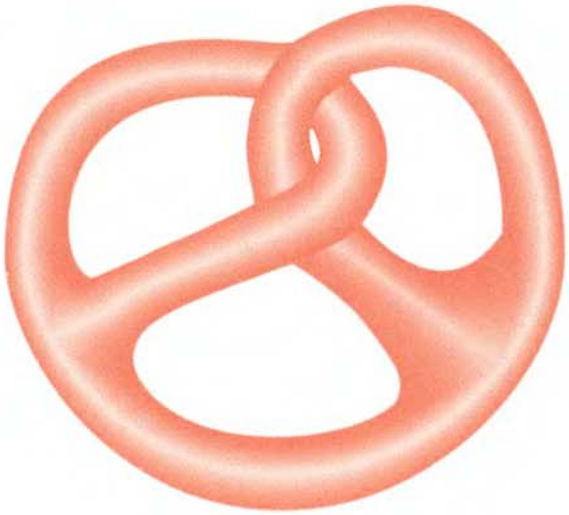
$$\frac{1}{2\pi}\iint_S K\, dA = 2 - 2g.$$

Thus, for the sphere ($g = 0$), it is always 2 (already verified); for the torus, it is always 0 (see Exercise 8).

There is something even more remarkable connected to the theorem of Gauss–Bonnet, observed by the great German mathematician David Hilbert (Figure 7.7.12).



Simple double doughnut



Baker's pretzel

**Figure 7.7.11** Two manifestations of a surface $S$ in $\mathbb{R}^3$ of genus 2.

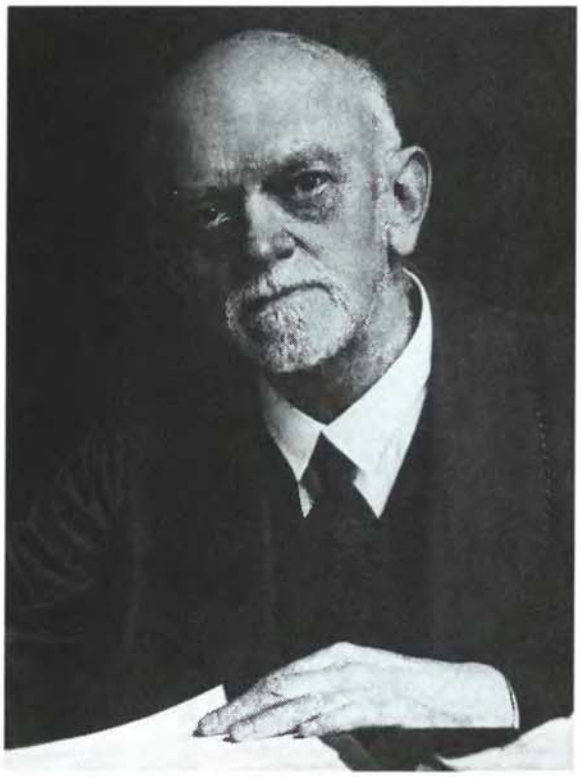


**Figure 7.7.12** David Hilbert (1862–1943) was a leading mathematician of his time.

Hilbert observed that the Gauss–Bonnet theorem is, in effect, a two-dimensional version of Einstein's field equations. In the physics literature, this fact is known as *Hilbert's action principle* in general relativity.[19] Not surprisingly, similar geometric ideas are being employed by contemporary researchers in an effort to unify gravity and quantum mechanics—to "quantize" gravity, so to speak.

[19] See C. Misner, K. Thorne, and A. Wheeler, *Gravitation*, Freeman, New York, 1972.

## EXERCISES

**1.** The helicoid can be described by

$$\Phi(u, v) = (u \cos v, u \sin v, bv), \text{ where } b \neq 0.$$

Show that $H = 0$ and that $K = -b^2/(b^2 + u^2)^2$. In Figures 7.7.1 and 7.7.5, we see that the helicoid is actually a soap film surface. Surfaces in which $H = 0$ are called ***minimal surfaces.***

**2.** Consider the saddle surface $z = xy$. Show that

$$K = \frac{-1}{(1 + x^2 + y^2)^2},$$

and that

$$H = \frac{-xy}{(1 + x^2 + y^2)^{3/2}}.$$

**3.** Show that $\Phi(u, v) = (u, v, \log \cos v - \log \cos u)$ has mean curvature zero (and is thus a minimal surface; see Exercise 1).

**4.** Find the Gauss curvature of the elliptic paraboloid

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}.$$

**5.** Find the Gauss curvature of the hyperbolic paraboloid

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}.$$

**6.** Compute the Gauss curvature of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

**7.** Show that Enneper's surface

$$\Phi(u, v) = \left(u - \frac{u^3}{3} + uv^2, v - \frac{v^3}{3} + u^2 v, u^2 - v^2\right)$$

is a minimal surface ($H = 0$).

**8.** Consider the torus $T$ given in Exercise 4, Section 7.4. Compute its Gauss curvature and verify the theorem of Gauss–Bonnet. [HINT: Show that $\|T_\theta \times T_\phi\|^2 = (R + \cos\phi)^2$ and $K = \cos\phi/(R + \cos\phi)$.]

**9.** Let $\Phi(u, v) = (u, h(u) \cos v, h(u) \sin v)$, $h > 0$, be a surface of revolution. Show that $K = -h''/h\{1 + (h')^2\}^2$.

**10.** A parametrization $\Phi$ of a surface $S$ is said to be ***conformal*** (see Section 7.4), provided that $E = G$, $F = 0$. Assume that $\Phi$ conformally parametrizes $S$.[20] Show that if $H$ and $K$ vanish identically, then $S$ must be part of a plane in $\mathbb{R}^3$.

## REVIEW EXERCISES FOR CHAPTER 7

**1.** Integrate $f(x, y, z) = xyz$ along the following paths:

(a) $\mathbf{c}(t) = (e^t \cos t, e^t \sin t, 3), 0 \le t \le 2\pi$
(b) $\mathbf{c}(t) = (\cos t, \sin t, t), 0 \le t \le 2\pi$
(c) $\mathbf{c}(t) = \frac{3}{2}t^2\mathbf{i} + 2t^2\mathbf{j} + t\mathbf{k}, 0 \le t \le 1$
(d) $\mathbf{c}(t) = t\mathbf{i} + (1/\sqrt{2})t^2\mathbf{j} + \frac{1}{3}t^3\mathbf{k}, 0 \le t \le 1$

**2.** Compute the integral of $f$ along the path $\mathbf{c}$ in each of the following cases:

(a) $f(x, y, z) = x + y + yz; \mathbf{c}(t) = (\sin t, \cos t, t), 0 \le t \le 2\pi$
(b) $f(x, y, z) = x + \cos^2 z; \mathbf{c}(t) = (\sin t, \cos t, t), 0 \le t \le 2\pi$
(c) $f(x, y, z) = x + y + z; \mathbf{c}(t) = (t, t^2, \frac{2}{3}t^3), 0 \le t \le 1$

**3.** Compute each of the following line integrals:

(a) $\int_C (\sin \pi x)\, dy - (\cos \pi y)\, dz$, where $C$ is the triangle whose vertices are (1, 0, 0), (0, 1, 0), and (0, 0, 1), in that order

(b) $\int_C (\sin z)\, dx + (\cos z)\, dy - (xy)^{1/3}\, dz$, where $C$ is the path $\mathbf{c}(\theta) = (\cos^3 \theta, \sin^3 \theta, \theta)$, $0 \le \theta \le 7\pi/2$

**4.** If $\mathbf{F}(\mathbf{x})$ is orthogonal to $\mathbf{c}'(t)$ at each point on the curve $\mathbf{x} = \mathbf{c}(t)$, what can you say about $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$?

**5.** Find the work done by the force $\mathbf{F}(x, y) = (x^2 - y^2)\mathbf{i} + 2xy\mathbf{j}$ in moving a particle counterclockwise around the square with corners (0, 0), (a, 0), (a, a), (0, a), $a > 0$.

**6.** A ring in the shape of the curve $x^2 + y^2 = a^2$ is formed of thin wire weighing $|x| + |y|$ grams per unit length at $(x, y)$. Find the mass of the ring.

**7.** Find a parametrization for each of the following surfaces;

(a) $x^2 + y^2 + z^2 - 4x - 6y = 12$
(b) $2x^2 + y^2 + z^2 - 8x = 1$
(c) $4x^2 + 9y^2 - 2z^2 = 8$

**8.** Find the area of the surface defined by $\Phi$: $(u, v) \mapsto (x, y, z)$, where

$$x = h(u, v) = u + v, \qquad y = g(u, v) = u, \qquad z = f(u, v) = v;$$

$0 \le u \le 1, 0 \le v \le 1$. Sketch.

---

[20] Gauss proved that conformal parametrization of a surface always exists. The result of this exercise remains valid even if $\Phi$ is not conformal, but the proof is more difficult.

**9.** Write a formula for the surface area of $\Phi$: $(r, \theta) \mapsto (x, y, z)$, where

$$x = r\cos\theta, \qquad y = 2r\sin\theta, \qquad z = r;$$

$0 \le r \le 1, 0 \le \theta \le 2\pi$. Describe the surface.

**10.** Suppose $z = f(x, y)$ and $(\partial f/\partial x)^2 + (\partial f/\partial y)^2 = c, c > 0$. Show that the area of the graph of $f$ lying over a region $D$ in the $xy$ plane is $\sqrt{1+c}$ times the area of $D$.

**11.** Compute the integral of $f(x, y, z) = x^2 + y^2 + z^2$ over the surface in Review Exercise 8.

**12.** Find $\iint_S f\, dS$ in each of the following cases:

(a) $f(x, y, z) = x$; $S$ is the part of the plane $x + y + z = 1$ in the positive octant defined by $x \ge 0, y \ge 0, z \ge 0$

(b) $f(x, y, z) = x^2$; $S$ is the part of the plane $x = z$ inside the cylinder $x^2 + y^2 = 1$

(c) $f(x, y, z) = x$; $S$ is the part of the cylinder $x^2 + y^2 = 2x$ with $0 \le z \le \sqrt{x^2 + y^2}$

**13.** Compute the integral of $f(x, y, z) = xyz$ over the rectangle with vertices (1, 0, 1), (2, 0, 0), (1, 1, 1), and (2, 1, 0).

**14.** Compute the integral of $x + y$ over the surface of the unit sphere.

**15.** Compute the surface integral of $x$ over the triangle with vertices (1, 1, 1), (2, 1, 1), and (2, 0, 3).

**16.** A paraboloid of revolution $S$ is parametrized by $\Phi(u, v) = (u\cos v, u\sin v, u^2)$, $0 \le u \le 2, 0 \le v \le 2\pi$.

(a) Find an equation in $x$, $y$, and $z$ describing the surface.

(b) What are the geometric meanings of the parameters $u$ and $v$?

(c) Find a unit vector orthogonal to the surface at $\Phi(u, v)$.

(d) Find the equation for the tangent plane at $\Phi(u_0, v_0) = (1, 1, 2)$ and express your answer in the following two ways:

(i) parametrized by $u$ and $v$; and

(ii) in terms of $x$, $y$, and $z$.

(e) Find the area of $S$.

**17.** Let $f(x, y, z) = xe^y \cos \pi z$.

(a) Compute $\mathbf{F} = \nabla f$.

(b) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{s}$ where $\mathbf{c}(t) = (3\cos^4 t, 5\sin^7 t, 0), 0 \le t \le \pi$.

**18.** Let $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $S$ is the upper hemisphere of the unit sphere $x^2 + y^2 + z^2 = 1$.

**19.** Let $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Evaluate $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$ where $\mathbf{c}(t) = (e^t, t, t^2), 0 \le t \le 1$.

**20.** Let $\mathbf{F} = \nabla f$ for a given scalar function. Let $\mathbf{c}(t)$ be a closed curve, that is, $\mathbf{c}(b) = \mathbf{c}(a)$. Show that $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = 0$.

**21.** Consider the surface $\Phi(u, v) = (u^2 \cos v, u^2 \sin v, u)$. Compute the unit normal at $u = 1, v = 0$. Compute the equation of the tangent plane at this point.

**22.** Let $S$ be the part of the cone $z^2 = x^2 + y^2$ with $z$ between 1 and 2 oriented by the normal pointing out of the cone. Compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = (x^2, y^2, z^2)$.

**23.** Let $\mathbf{F} = x\mathbf{i} + x^2\mathbf{j} + yz\mathbf{k}$ represent the velocity field of a fluid (velocity measured in meters per second). Compute how many cubic meters of fluid per second are crossing the $xy$ plane through the square $0 \le x \le 1, 0 \le y \le 1$.

**24.** Show that the surface area of the part of the sphere $x^2 + y^2 + z^2 = 1$ lying above the rectangle $[-a, a] \times [-a, a]$, where $2a^2 < 1$, in the $xy$ plane is

$$A = 2\int_{-a}^{a} \sin^{-1}\left(\frac{a}{\sqrt{1 - x^2}}\right) dx.$$

**25.** Let $S$ be a surface and $C$ a closed curve bounding $S$. Verify the equality

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{s}$$

if $\mathbf{F}$ is a gradient field (use Review Exercise 20).

**26.** Calculate $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = (x, y, -y)$ and $S$ is the cylindrical surface defined by $x^2 + y^2 = 1, 0 \le z \le 1$, with normal pointing out of the cylinder.

**27.** Let $S$ be the portion of the cylinder $x^2 + y^2 = 4$ between the planes $z = 0$ and $z = x + 3$. Compute the following:

(a) $\iint_S x^2\, dS$ (b) $\iint_S y^2\, dS$ (c) $\iint_S z^2\, dS$

**28.** Let $\Gamma$ be the curve of intersection of the plane $z = ax + by$ with the cylinder $x^2 + y^2 = 1$. Find all values of the real numbers $a$ and $b$ such that $a^2 + b^2 = 1$ and

$$\int_\Gamma y\, dx + (z - x)\, dy - y\, dz = 0.$$

**29.** A circular helix that lies on the cylinder $x^2 + y^2 = R^2$ with pitch $p$ may be described parametrically by

$$x = R\cos\theta, \qquad y = R\sin\theta, \qquad z = p\theta, \qquad \theta \ge 0.$$

A particle slides under the action of gravity (which acts parallel to the $z$ axis) without friction along the helix. If the particle starts out at the height $z_0 > 0$, then when it reaches the height $z$, $0 \leq z < z_0$, along the helix, its speed is given by

$$\frac{ds}{dt} = \sqrt{(z_0 - z)2g},$$

where $s$ is arc length along the helix, $g$ is the constant of gravity, and $t$ is time.

(a) Find the length of the part of the helix between the planes $z = z_0$ and $z = z_1$, $0 \leq z_1 < z_0$.

(b) Compute the time $T_0$ it takes the particle to reach the plane $z = 0$.

# 8

# The Integral Theorems of Vector Analysis

*All the theory of the motion of fluids has just been reduced to the solution of analytic formulas.*

*Leonhard Euler*

*Fluids are a lot easier to drink than they are to understand.*

*Alan Newell*

We are now prepared to tie together vector differential calculus and vector integral calculus. This will be done by means of the important theorems of Green, Gauss, and Stokes. We shall also point out some of the physical applications of these theorems to the study of electricity and magnetism, hydrodynamics, heat conduction, and differential equations.

The basic integral theorems in vector analysis had their origins in applications. For example, Green's theorem, discovered about 1828, arose in connection with potential theory (this includes gravitational and electrical potentials). Gauss' theorem—the divergence theorem—arose in connection with the study of capillarity (this theorem should be jointly credited to the Russian mathematician Ostrogradsky, who discovered the theorem around the same time as Gauss). Stokes' theorem was first suggested in a letter to Stokes from the physicist Lord Kelvin in 1850 and was used by Stokes on the examination for the Smith Prize in 1854.

## 8.1 Green's Theorem

Green's theorem relates a line integral along a closed curve $C$ in the plane $\mathbb{R}^2$ to a double integral over the region enclosed by $C$. This important result will be generalized in the following sections to curves and surfaces in $\mathbb{R}^3$. We shall be referring to line

integrals around curves that are the boundaries of elementary regions (see Section 5.3). To understand the ideas in this section, you may also need to refer to Section 7.2.

## Simple and Elementary Regions and their Boundaries

A simple closed curve $C$ that is the boundary of an elementary region has two orientations—counterclockwise (positive) and clockwise (negative). We denote $C$ with the counterclockwise orientation as $C^+$, and with the clockwise orientation as $C^-$ (Figure 8.1.1).

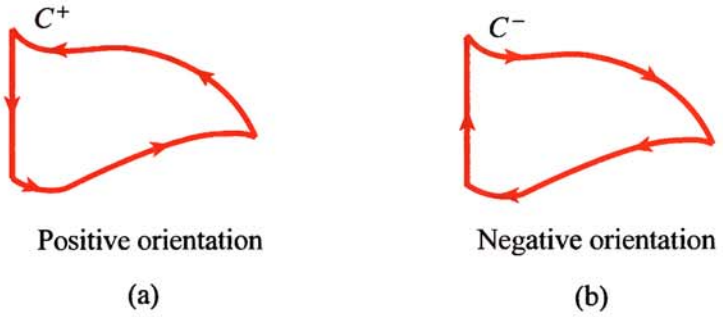



**Figure 8.1.1** (a) Positive orientation of $C$, and (b) negative orientation of $C$.

The boundary $C$ of a $y$-simple region can be decomposed into bottom and top portions, $C_1$ and $C_2$, and (if applicable) left and right vertical portions, $B_1$ and $B_2$. Following Figure 8.1.2, we write,

$$C^+ = C_1^+ + B_2^+ + C_2^- + B_1^-,$$

where the pluses denote the curves oriented in the direction of left to right or bottom to top, and the minuses denote the curves oriented from right to left or from top to bottom.

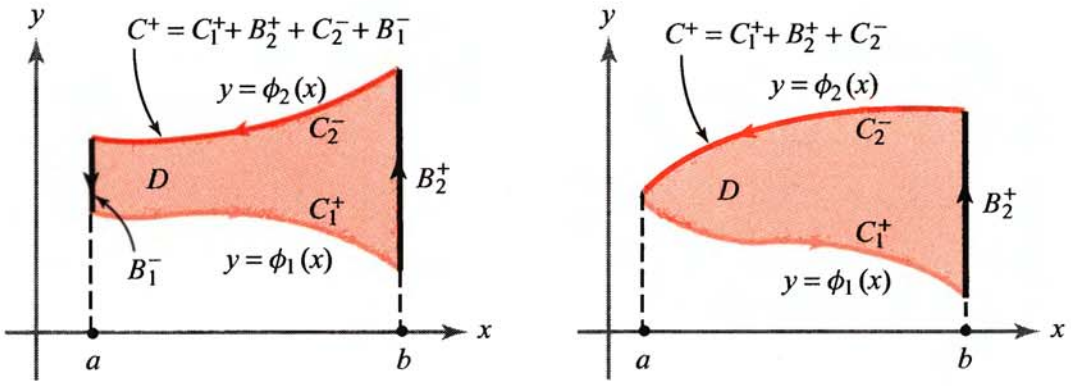



**Figure 8.1.2** Two examples showing how to break the positively oriented boundary of a $y$-simple region $D$ into oriented components.

We may make a similar decomposition of the boundary of an $x$-simple region into left and right portions, and upper and lower horizontal portions (if applicable) (Figure 8.1.3).

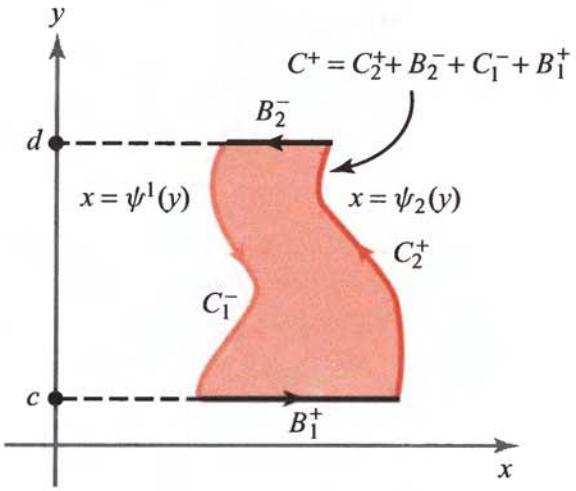



**Figure 8.1.3** An example showing how to break the positively oriented boundary of an $x$-simple region $D$ into oriented components.

Similarly, the boundary of a simple region has *two* decompositions—one into upper and lower halves, the other into left and right halves.

## Green's Theorem

We shall now prove two lemmas in preparation for Green's theorem.

LEMMA 1 Let $D$ be a $y$-simple region and let $C$ be its boundary. Suppose $P\colon D \to \mathbb{R}$ is of class $C^1$. Then

$$\int_{C^+} P\,dx = -\iint_D \frac{\partial P}{\partial y}\,dx\,dy.$$

(The left-hand side denotes the line integral $\int_{C^+} P\,dx + Q\,dy$ where $Q = 0$.)

*PROOF* Suppose $D$ is described by

$$a \le x \le b \qquad \phi_1(x) \le y \le \phi_2(x).$$

We decompose $C^+$ by writing $C^+ = C_1^+ + B_2^+ + C_2^- + B_1^-$ (see Figure 8.1.2). By Fubini's theorem, we may evaluate the double integral as an iterated integral and then use the fundamental theorem of calculus:

$$\begin{aligned}\iint_D \frac{\partial P}{\partial y}(x,y)\,dx\,dy &= \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \frac{\partial P}{\partial y}(x,y)\,dy\,dx \\ &= \int_a^b [P(x,\phi_2(x)) - P(x,\phi_1(x))]\,dx.\end{aligned}$$

However, because $C_1^+$ can be parametrized by $x \mapsto (x, \phi_1(x))$, $a \le x \le b$, and $C_2^+$ can be parametrized by $x \mapsto (x, \phi_2(x))$, $a \le x \le b$, we have

$$\int_a^b P(x, \phi_1(x))\, dx = \int_{C_1^+} P(x, y)\, dx$$

and

$$\int_a^b P(x, \phi_2(x))\, dx = \int_{C_2^+} P(x, y)\, dx.$$

Thus, by reversing orientations,

$$-\int_a^b P(x, \phi_2(x))\, dx = \int_{C_2^-} P(x, y)\, dx.$$

Hence,

$$\iint_D \frac{\partial P}{\partial y}\, dx\, dy = -\int_{C_1^+} P\, dx - \int_{C_2^-} P\, dx.$$

Because $x$ is constant on $B_2^+$ and $B_1^-$, we have

$$\int_{B_2^+} P\, dx = 0 = \int_{B_1^-} P\, dx,$$

so

$$\int_{C^+} P\, dx = \int_{C_1^+} P\, dx + \int_{B_2^+} P\, dx + \int_{C_2^-} P\, dx + \int_{B_1^-} P\, dx = \int_{C_1^+} P\, dx + \int_{C_2^-} P\, dx.$$

Thus,

$$\iint_D \frac{\partial P}{\partial y}\, dx\, dy = -\int_{C_1^+} P\, dx - \int_{C_2^-} P\, dx = -\int_{C^+} P\, dx. \quad \blacksquare$$

We now prove the analogous lemma with the roles of $x$ and $y$ interchanged.

LEMMA 2 Let $D$ be an $x$-simple region with boundary $C$. Then if $Q$: $D \to \mathbb{R}$ is $C^1$,

$$\int_{C^+} Q\, dy = \iint_D \frac{\partial Q}{\partial x}\, dx\, dy.$$

The negative sign does not occur here, because reversing the role of $x$ and $y$ corresponds to a change of orientation for the plane.

*PROOF* Suppose $D$ is given by

$$\psi_1(y) \le x \le \psi_2(y), \qquad c \le y \le d.$$

Using the notation of Figure 8.1.3, and noting that $y$ is constant on $B_1^+$ and $B_2^-$, we have

$$\int_{C^+} Q\,dy = \int_{C_1^- + B_1^+ + C_2^+ + B_2^-} Q\,dy = \int_{C_2^+} Q\,dy + \int_{C_1^-} Q\,dy,$$

where $C_2^+$ is the curve parametrized by $y \mapsto (\psi_2(y), y)$, $c \le y \le d$, and $C_1^+$ is the curve $y \mapsto (\psi_1(y), y)$, $c \le y \le d$. Applying Fubini's theorem and the fundamental theorem of calculus, we obtain

$$\begin{aligned}\iint_D \frac{\partial Q}{\partial x}\,dx\,dy &= \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} \frac{\partial Q}{\partial x}\,dx\,dy = \int_c^d [Q(\psi_2(y), y) - Q(\psi_1(y), y)]\,dy \\ &= \int_{C_2^+} Q\,dy - \int_{C_1^+} Q\,dy = \int_{C_2^+} Q\,dy + \int_{C_1^-} Q\,dy = \int_{C^+} Q\,dy. \quad \blacksquare\end{aligned}$$

Adding the results of Lemmas 1 and 2 proves the following important theorem.

**THEOREM 1: Green's Theorem** Let $D$ be a simple region and let $C$ be its boundary. Suppose $P\colon D \to \mathbb{R}$ and $Q\colon D \to \mathbb{R}$ are of class $C^1$. Then

$$\int_{C^+} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.$$

The correct (positive) orientation for the boundary curves of region $D$ can be remembered by the following device: *If you walk along the curve C with the correct orientation, the region D will be on your left* (see Figure 8.1.4).

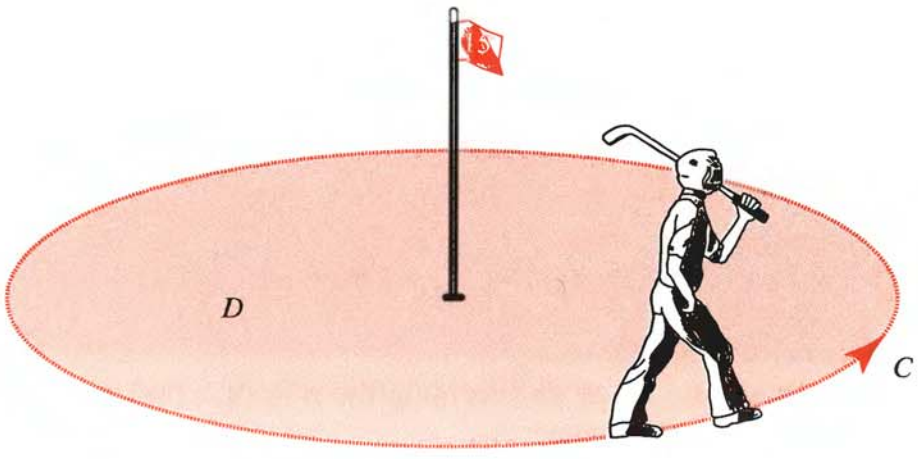



**Figure 8.1.4** The correct orientation for the boundary of a region $D$.

## Generalizing Green's Theorem

Green's theorem actually applies to any "decent" region in $\mathbb{R}^2$. For instance, Green's theorem applies to regions that are not simple, but that can be broken up into pieces, each of which is simple. An example is shown in Figure 8.1.5. The region $D$ is an annulus; its boundary consists of two curves $C = C_1 + C_2$ with the indicated orientations. (Note that for the inner region the correct orientation to ensure the validity of Green's theorem is *clockwise*; *the device in Figure 8.1.4 still works for remembering the orientation*!) If Theorem 1 is applied to each of the regions $D_1$, $D_2$, $D_3$, and $D_4$ and the results are summed, the equality of Green's theorem will be obtained for $D$ and its boundary curve $C$. This works because the integrals along the interior lines in opposite directions cancel. This trick, in fact, shows that Green's theorem holds for virtually all regions with reasonable boundaries that one is likely to encounter (see Exercise 8).

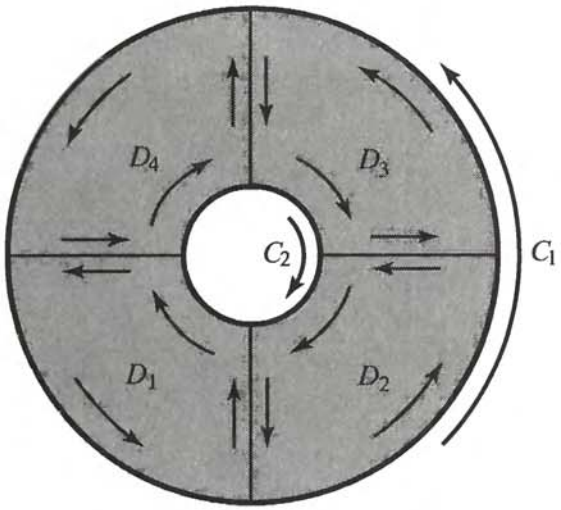



**Figure 8.1.5** Green's theorem applies to $D = D_1 \cup D_2 \cup D_3 \cup D_4$.

Let us use the notation $\partial D$ for the oriented curve $C^+$, that is, the boundary curve of $D$ oriented in the sense as described by the device in Figure 8.1.4. Then we may write Green's theorem as

$$\int_{\partial D} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.$$

Green's theorem is very useful because it relates a line integral around the boundary of a region to an area integral over the interior of the region, and in many cases it is easier to evaluate the line integral than the area integral or vice versa. For example, if we know that $P$ vanishes on the boundary, we can immediately conclude that $\iint_D (\partial P/\partial y)\,dx\,dy = 0$ even though $\partial P/\partial y$ need not vanish on the interior. (Can you construct such a $P$ on the unit square?)

**EXAMPLE 1** Verify Green's theorem for $P(x, y) = x$ and $Q(x, y) = xy$ where $D$ is the unit disk $x^2 + y^2 \leq 1$.

SOLUTION We do this by evaluating both sides in Green's theorem directly. The boundary of $D$ is the unit circle parametrized by $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$,

and so

$$\int_{\partial D} P\,dx + Q\,dy = \int_0^{2\pi} [(\cos t)(-\sin t) + \cos t \sin t \cos t]\,dt$$
$$= \left[\frac{\cos^2 t}{2}\right]_0^{2\pi} + \left[-\frac{\cos^3 t}{3}\right]_0^{2\pi} = 0.$$

On the other hand,

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \iint_D y\,dx\,dy,$$

which is also zero by symmetry. Thus, Green's theorem is verified in this case. ▲

## Areas

We can use Green's theorem to obtain a formula for the area of a region bounded by a simple closed curve.

**THEOREM 2: Area of a Region** If $C$ is a simple closed curve that bounds a region to which Green's theorem applies, then the area of the region $D$ bounded by $C = \partial D$ is

$$A = \frac{1}{2}\int_{\partial D} x\,dy - y\,dx.$$

*PROOF* Let $P(x, y) = -y$, $Q(x, y) = x$; then by Green's theorem we have

$$\frac{1}{2}\int_{\partial D} x\,dy - y\,dx = \frac{1}{2}\iint_D \left[\frac{\partial x}{\partial x} - \frac{\partial(-y)}{\partial y}\right] dx\,dy$$
$$= \frac{1}{2}\iint_D [1 + 1]\,dx\,dy = \iint_D dx\,dy = A. \quad ■$$

**EXAMPLE 2** Let $a > 0$. Compute the area (see Figure 8.1.6) of the region enclosed by the hypocycloid defined by $x^{2/3} + y^{2/3} = a^{2/3}$ using the parametrization

$$x = a\cos^3\theta, \qquad y = a\sin^3\theta, \qquad 0 \le \theta \le 2\pi.$$

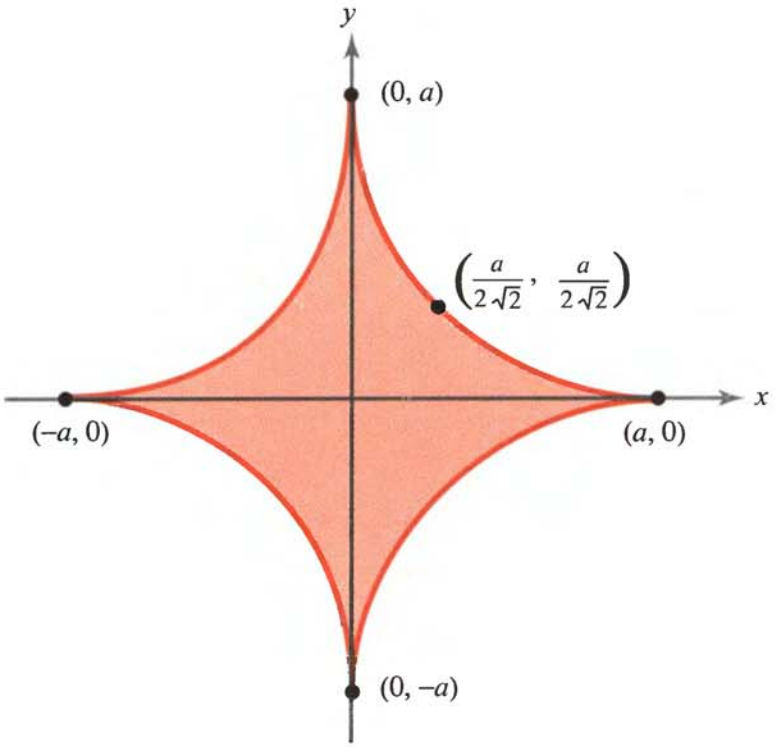



**Figure 8.1.6** The hypocycloid $x = a\cos^3\theta$, $y = a\sin^3\theta$, $0 \le \theta \le 2\pi$.

SOLUTION From the preceding box, and using the trigonometric identities $\cos^2\theta + \sin^2\theta = 1$, $\sin 2\theta = 2\sin\theta\cos\theta$, and $\sin^2\phi = (1 - \cos 2\phi)/2$, we get

$$\begin{aligned} A &= \frac{1}{2}\int_{\partial D} x\,dy - y\,dx \\ &= \frac{1}{2}\int_0^{2\pi} [(a\cos^3\theta)(3a\sin^2\theta\cos\theta) - (a\sin^3\theta)(-3a\cos^2\theta\sin\theta)]\,d\theta \\ &= \frac{3}{2}a^2\int_0^{2\pi} (\sin^2\theta\cos^4\theta + \cos^2\theta\sin^4\theta)\,d\theta = \frac{3}{2}a^2\int_0^{2\pi} \sin^2\theta\cos^2\theta\,d\theta \\ &= \frac{3}{8}a^2\int_0^{2\pi} \sin^2 2\theta\,d\theta = \frac{3}{8}a^2\int_0^{2\pi} \left(\frac{1-\cos 4\theta}{2}\right)d\theta \\ &= \frac{3}{16}a^2\int_0^{2\pi} d\theta - \frac{3}{16}a^2\int_0^{2\pi} \cos 4\theta\,d\theta = \frac{3}{8}\pi a^2. \quad \blacktriangle \end{aligned}$$

## Vector Form using the Curl

The statement of Green's theorem can be neatly rewritten in the language of vector fields. As we will see, this points the way to one possible generalization to $\mathbb{R}^3$.

**THEOREM 3: Vector Form of Green's Theorem** Let $D \subset \mathbb{R}^2$ be a region to which Green's theorem applies, let $\partial D$ be its (positively oriented) boundary, and let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ be a $C^1$ vector field on $D$. Then

$$\int_{\partial D} \mathbf{F}\cdot d\mathbf{s} = \iint_D (\operatorname{curl}\mathbf{F})\cdot\mathbf{k}\,dA = \iint_D (\nabla\times\mathbf{F})\cdot\mathbf{k}\,dA$$

(see Figure 8.1.7).

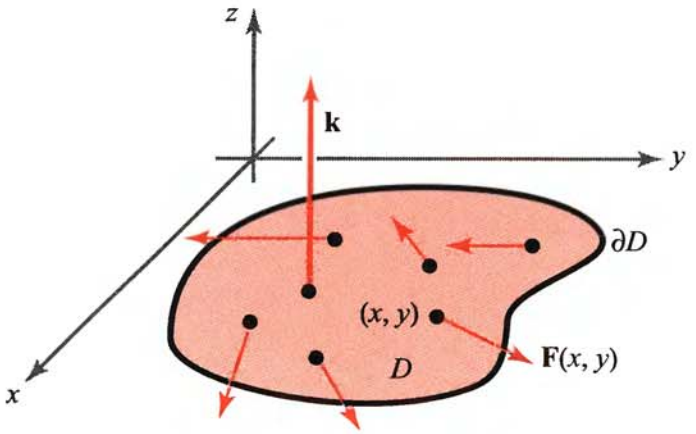



**Figure 8.1.7** The vector form of Green's theorem.

This result follows from Theorem 1 and the fact that $(\nabla \times \mathbf{F}) \cdot \mathbf{k} = \partial Q/\partial x - \partial P/\partial y$. We ask the reader to supply the details in Exercise 14.

**EXAMPLE 3** Let $\mathbf{F} = (xy^2, y + x)$. Integrate $(\nabla \times \mathbf{F}) \cdot \mathbf{k}$ over the region in the first quadrant bounded by the curves $y = x^2$ and $y = x$.

SOLUTION *Method 1.* We first compute the curl

$$\nabla \times \mathbf{F} = \left(0, 0, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) = (1 - 2xy)\mathbf{k}.$$

Thus, $(\nabla \times \mathbf{F}) \cdot \mathbf{k} = 1 - 2xy$. This can be integrated over the given region $D$ (see Figure 8.1.8) using an iterated integral as follows:

$$\iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dx\, dy = \int_0^1 \int_{x^2}^{x} (1 - 2xy)\, dy\, dx = \int_0^1 [y - xy^2]\Big|_{x^2}^{x} dx$$

$$= \int_0^1 [x - x^3 - x^2 + x^5]\, dx = \frac{1}{2} - \frac{1}{4} - \frac{1}{3} + \frac{1}{6} = \frac{1}{12}.$$

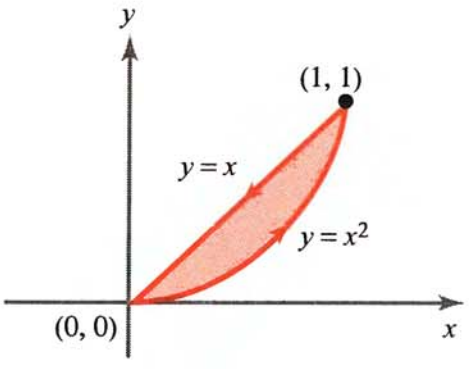



**Figure 8.1.8** The region bounded by the curves $y = x^2$ and $y = x$.

*Method 2.* Here we use Theorem 3 to obtain

$$\iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dx\, dy = \int_{\partial D} \mathbf{F} \cdot d\mathbf{s}.$$

The line integral of $\mathbf{F}$ along the curve $y = x$ from left to right is

$$\int_0^1 F_1\,dx + F_2\,dy = \int_0^1 (x^3 + 2x)\,dx = \frac{1}{4} + 1 = \frac{5}{4}.$$

Along the curve $y = x^2$ we get

$$\int_0^1 F_1\,dx + F_2\,dy = \int_0^1 x^5\,dx + (x + x^2)(2x\,dx) = \frac{1}{6} + \frac{2}{3} + \frac{1}{2} = \frac{4}{3}.$$

Thus, remembering that the integral along $y = x$ is to be taken from right to left, as in Figure 8.1.8,

$$\int_{\partial D} \mathbf{F} \cdot d\mathbf{s} = \frac{4}{3} - \frac{5}{4} = \frac{1}{12}. \quad \blacktriangle$$

## Vector Form Using the Divergence

There is another form of Green's theorem that can be generalized to $\mathbb{R}^3$.

**THEOREM 4: Divergence Theorem in the Plane** Let $D \subset \mathbb{R}^2$ be a region to which Green's theorem applies and let $\partial D$ be its boundary. Let $\mathbf{n}$ denote the outward unit normal to $\partial D$. If $\mathbf{c}\colon [a, b] \to \mathbb{R}^2$, $t \mapsto \mathbf{c}(t) = (x(t), y(t))$ is a positively oriented parametrization of $\partial D$, $\mathbf{n}$ is given by

$$\mathbf{n} = \frac{(y'(t), -x'(t))}{\sqrt{[x'(t)]^2 + [y'(t)]^2}}$$

(see Figure 8.1.9). Let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ be a $C^1$ vector field on $D$. Then

$$\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\,ds = \iint_D \operatorname{div} \mathbf{F}\,dA.$$

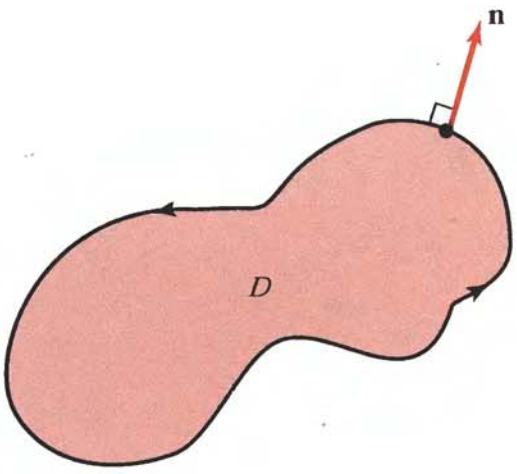



Figure 8.1.9 $\mathbf{n}$ is the outward unit normal to $\partial D$.

*PROOF* Recall that $\mathbf{c}'(t) = (x'(t), y'(t))$ is tangent to $\partial D$, and note that $\mathbf{n} \cdot \mathbf{c}' = 0$. Thus, $\mathbf{n}$ is normal to the boundary. The sign of $\mathbf{n}$ is chosen to make it correspond to the *outward* (rather than the inward) direction. By the definition of the line integral (see Section 7.2),

$$\begin{aligned}\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds &= \int_a^b \frac{P(x(t), y(t))y'(t) - Q(x(t), y(t))x'(t)}{\sqrt{[x'(t)]^2 + [y'(t)]^2}} \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt \\ &= \int_a^b [P(x(t), y(t))y'(t) - Q(x(t), y(t))x'(t)]\, dt \\ &= \int_{\partial D} P\, dy - Q\, dx.\end{aligned}$$

By Green's theorem, this equals

$$\iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dx\, dy = \iint_D \operatorname{div} \mathbf{F}\, dA. \quad \blacksquare$$

**EXAMPLE 4** Let $\mathbf{F} = y^3\mathbf{i} + x^5\mathbf{j}$. Compute the integral of the normal component of $\mathbf{F}$ around the unit square.

SOLUTION This can be done using the divergence theorem. Indeed,

$$\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = \iint_D \operatorname{div} \mathbf{F}\, dA.$$

But $\operatorname{div} \mathbf{F} = 0$, and so the integral is zero. ▲

## EXERCISES

**1.** Evaluate $\int_C y\, dx - x\, dy$ where $C$ is the boundary of the square $[-1, 1] \times [-1, 1]$ oriented in the counterclockwise direction, using Green's theorem.

**2.** Find the area of the disk $D$ of radius $R$ using Green's theorem.

**3.** Verify Green's theorem for the disk $D$ with center $(0, 0)$ and radius $R$ and the functions:

(a) $P(x, y) = xy^2$, $Q(x, y) = -yx^2$
(b) $P(x, y) = x + y$, $Q(x, y) = y$
(c) $P(x, y) = xy = Q(x, y)$
(d) $P(x, y) = 2y$, $Q(x, y) = x$

**4.** Using the divergence theorem, show that $\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = 0$ where $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$ and $D$ is the unit disk. Verify this directly.

**5.** Find the area bounded by one arc of the cycloid $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$, where $a > 0$, and $0 \le \theta \le 2\pi$, and the $x$ axis (use Green's theorem).

**6.** Under the conditions of Green's theorem, prove that

(a) $$\int_{\partial D} PQ\,dx + PQ\,dy = \iint_D \left[ Q\left(\frac{\partial P}{\partial x} - \frac{\partial P}{\partial y}\right) + P\left(\frac{\partial Q}{\partial x} - \frac{\partial Q}{\partial y}\right)\right] dx\,dy$$

(b) $$\int_{\partial D} \left(Q\frac{\partial P}{\partial x} - P\frac{\partial Q}{\partial x}\right) dx + \left(P\frac{\partial Q}{\partial y} - Q\frac{\partial P}{\partial y}\right) dy$$
$$= 2\iint_D \left(P\frac{\partial^2 Q}{\partial x\,\partial y} - Q\frac{\partial^2 P}{\partial x\,\partial y}\right) dx\,dy$$

**7.** Evaluate the line integral

$$\int_C (2x^3 - y^3)\,dx + (x^3 + y^3)\,dy,$$

where $C$ is the unit circle, and verify Green's theorem for this case.

**8.** Prove the following generalization of Green's theorem: Let $D$ be a region in the $xy$ plane with boundary a finite number of oriented simple closed curves. Suppose that by means of a finite number of line segments parallel to the coordinate axes, $D$ can be decomposed into a finite number of simple regions $D_i$ with the boundary of each $D_i$ oriented counterclockwise (see Figure 8.1.5). Then if $P$ and $Q$ are of class $C^1$ on $D$,

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \int_{\partial D} P\,dx + Q\,dy,$$

where $\partial D$ is the oriented boundary of $D$. (HINT: Apply Green's theorem to each $D_i$.)

**9.** Verify Green's theorem for the integrand of Exercise 7 (that is, with $P = 2x^3 - y^3$ and $Q = x^3 + y^3$) and the annular region $D$ described by $a \le x^2 + y^2 \le b$, with boundaries oriented as in Figure 8.1.5.

**10.** Let $D$ be a region for which Green's theorem holds. Suppose $f$ is harmonic; that is,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

on $D$. Prove that

$$\int_{\partial D} \frac{\partial f}{\partial y}\,dx - \frac{\partial f}{\partial x}\,dy = 0.$$

**11.** (a) Verify the divergence theorem for $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$ and $D$ the unit disk $x^2 + y^2 \le 1$.
(b) Evaluate the integral of the normal component of $2xy\mathbf{i} - y^2\mathbf{j}$ around the ellipse defined by $x^2/a^2 + y^2/b^2 = 1$.

**12.** Let $P(x, y) = -y/(x^2 + y^2)$ and $Q(x, y) = x/(x^2 + y^2)$. Assuming $D$ is the unit disk, investigate why Green's theorem fails for this $P$ and $Q$.

**13.** Use Green's theorem to evaluate $\int_{C^+} (y^2 + x^3)\,dx + x^4\,dy$, where $C^+$ is the perimeter of the square $[0, 1] \times [0, 1]$ in the counterclockwise direction.

**14.** Verify Theorem 3 by showing that $(\nabla \times \mathbf{F}) \cdot \mathbf{k} = \partial Q/\partial x - \partial P/\partial y$.

**15.** Use Theorem 2 to compute the area inside the ellipse $x^2/a^2 + y^2/b^2 = 1$.

**16.** Use Theorem 2 to recover the formula $A = \frac{1}{2}\int_a^b r^2\, d\theta$ for a region in polar coordinates.

**17.** Sketch the proof of Green's theorem for the region shown in Figure 8.1.10.

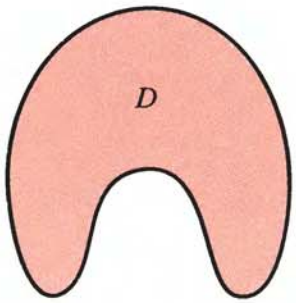



**Figure 8.1.10** Prove Green's theorem for this region.

**18.** Prove the identity

$$\int_{\partial D} \phi \nabla\phi \cdot \mathbf{n}\, ds = \iint_D (\phi \nabla^2\phi + \nabla\phi \cdot \nabla\phi)\, dA.$$

**19.** Use Green's theorem to find the area of one loop of the four-leafed rose $r = 3\sin 2\theta$. (HINT: $x\, dy - y\, dx = r^2 d\theta$).

**20.** Show that if $C$ is a simple closed curve that bounds a region to which Green's theorem applies, then the area of the region $D$ bounded by $C$ is

$$A = \int_{\partial D} x\, dy = -\int_{\partial D} y\, dx.$$

Show how this implies Theorem 2.

*Exercises 21 to 29 illustrate the application of Green's theorem to partial differential equations. (Further applications are given in the Internet supplement.) They are particularly concerned with solutions to Laplace's equation, that is, with harmonic functions. For these exercises, let $D$ be an open region in $\mathbb{R}^2$ with boundary $\partial D$. Let $u$: $D \cup \partial D \to \mathbb{R}$ be a continuous function that is of class $C^2$ on $D$. Suppose $\mathbf{p} \in D$ and the closed disks $B_\rho = B_\rho(\mathbf{p})$ of radius $\rho$ centered at $\mathbf{p}$ are contained in $D$ for $0 < \rho \leq R$. Define $I(\rho)$ by*

$$I(\rho) = \frac{1}{\rho}\int_{\partial B_\rho} u\, ds.$$

**21.** Show that $\lim_{\rho \to 0} I(\rho) = 2\pi u(\mathbf{p})$.

**22.** Let $\mathbf{n}$ denote the outward unit normal to $\partial B_\rho$ and $\partial u/\partial n = \nabla u \cdot \mathbf{n}$. Show that

$$\int_{\partial B_\rho} \frac{\partial u}{\partial n}\, ds = \iint_{B_\rho} \nabla^2 u\, dA.$$

**23.** Using Exercise 22, show that $I'(\rho) = (1/\rho) \iint_{B_\rho} \nabla^2 u \, dA$.

**24.** Suppose $u$ satisfies Laplace's equation: $\nabla^2 u = 0$ on $D$. Use the preceding exercises to show that

$$u(\mathbf{p}) = \frac{1}{2\pi R} \int_{\partial B_R} u \, ds.$$

(This expresses the fact that the value of a harmonic function at a point is the average of its values on the circumference of any disk centered about it.)

**25.** Use Exercise 24 to show that if $u$ is harmonic (i.e., if $\nabla^2 u = 0$), then $u(\mathbf{p})$ can be expressed as an area integral

$$u(\mathbf{p}) = \frac{1}{\pi R^2} \iint_{B_R} u \, dA.$$

**26.** Suppose $u$ is a harmonic function defined on $D$ (i.e., $\nabla^2 u = 0$ on $D$) and that $u$ has a local maximum (or minimum) at a point $\mathbf{p}$ in $D$.

(a) Show that $u$ must be constant on some disk centered at $\mathbf{p}$. (HINT: Use the results of Exercise 25.)

(b) Suppose that $D$ is path-connected [i.e., for any points $\mathbf{p}$ and $\mathbf{q}$ in $D$, there is a continuous path $\mathbf{c}$: $[0, 1] \to D$ such that $\mathbf{c}(0) = \mathbf{p}$ and $\mathbf{c}(1) = \mathbf{q}$] and that for some $\mathbf{p}$ the maximum or minimum at $\mathbf{p}$ is absolute; thus, $u(\mathbf{q}) \leq u(\mathbf{p})$ or $u(\mathbf{q}) \geq u(\mathbf{p})$ for every $\mathbf{q}$ in $D$. Show that $u$ must be constant on $D$.

(The result in this Exercise is called a ***strong maximum*** or ***minimum principle*** for harmonic functions. Compare this with Exercises 36 to 40 in Section 3.3.)

**27.** A function is said to be ***subharmonic*** on $D$ if $\nabla^2 u \geq 0$ everywhere in $D$. It is said to be ***superharmonic*** if $\nabla^2 u \leq 0$.

(a) Derive a strong maximum principle for subharmonic functions.
(b) Derive a strong minimum principle for superharmonic functions.

**28.** Suppose $D$ is the disk $\{(x, y) \mid x^2 + y^2 < 1\}$ and $C$ is the circle $\{(x, y) \mid x^2 + y^2 = 1\}$. In the Internet supplement, we shall show that if $f$ is a continuous real-valued function on $C$, then there is a continuous function $u$ on $D \cup C$ that agrees with $f$ on $C$ and is harmonic on $D$. That is, $f$ has a harmonic extension to the disk. Assuming this, show the following:

(a) If $q$ is a nonconstant continuous function on $D \cup C$ that is subharmonic (but not harmonic) on $D$, then there is a continuous function $u$ on $D \cup C$ that is harmonic on $D$ such that $u$ agrees with $q$ on $C$ and $q < u$ everywhere on $D$.

(b) The same assertion holds if "subharmonic" is replaced by "superharmonic" and "$q < u$" by "$q > u$."

**29.** Let $D$ be as in Exercise 28. Let $f$: $D \to \mathbb{R}$ be continuous. Show that a solution to the equation $\nabla^2 u = 0$ satisfying $u(\mathbf{x}) = f(\mathbf{x})$ for all $\mathbf{x} \in \partial D$ is unique.

**30.** Use Green's theorem to prove the change of variables formula in the following special case:

$$\iint_D dx\,dy = \iint_{D^*} \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

for a transformation $(u, v) \mapsto (x(u, v), y(u, v))$.

## 8.2 Stokes' Theorem

Stokes' theorem relates the line integral of a vector field around a simple closed curve $C$ in $\mathbb{R}^3$ to an integral over a surface $S$ for which $C$ is the boundary. In this regard it is very much like Green's theorem.

### Stokes' Theorem for Graphs

Let us begin by recalling a few facts from Chapter 7. Consider a surface $S$ that is the graph of a function $f(x, y)$, so that $S$ is parametrized by

$$\begin{cases} x = u \\ y = v \\ z = f(u, v) = f(x, y) \end{cases}$$

for $(u, v)$ in some domain $D$ in the plane. The integral of a vector function $\mathbf{F}$ over $S$ was developed in Section 7.6 as

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \left[F_1\left(-\frac{\partial z}{\partial x}\right) + F_2\left(-\frac{\partial z}{\partial y}\right) + F_3\right] dx\,dy, \tag{1}$$

where $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$.

In Section 8.1, we first assumed that the regions $D$ under consideration were simple; while this was used in our proof of Green's theorem, we noted there that the theorem is valid for a wider class of regions. In this section we assume that $D$ is a region whose boundary is a simple closed curve and to which Green's theorem applies. Green's theorem involves choosing an orientation on the boundary of $D$, as was explained in Section 8.1. The choice of orientation that validates Green's theorem will be called ***positive***. Recall that if $D$ is simple, then the positive orientation is the counterclockwise one.

Suppose that $\mathbf{c}$: $[a, b] \to \mathbb{R}^2$, $\mathbf{c}(t) = (x(t), y(t))$ is a parametrization of $\partial D$ in the positive direction. Then we define the ***boundary curve*** $\partial S$ to be the oriented simple closed curve that is the image of the mapping $\mathbf{p}$: $t \mapsto (x(t), y(t), f(x(t), y(t)))$ with the orientation induced by $\mathbf{p}$ (Figure 8.2.1).

To remember this orientation (that is, the positive direction) on $\partial S$, imagine that you are an "observer" walking along the boundary of the surface with the normal as your upright direction; you are moving in the positive direction if the surface is on your left. This orientation on $\partial S$ is often called the ***orientation induced by an upward normal*** $\mathbf{n}$.

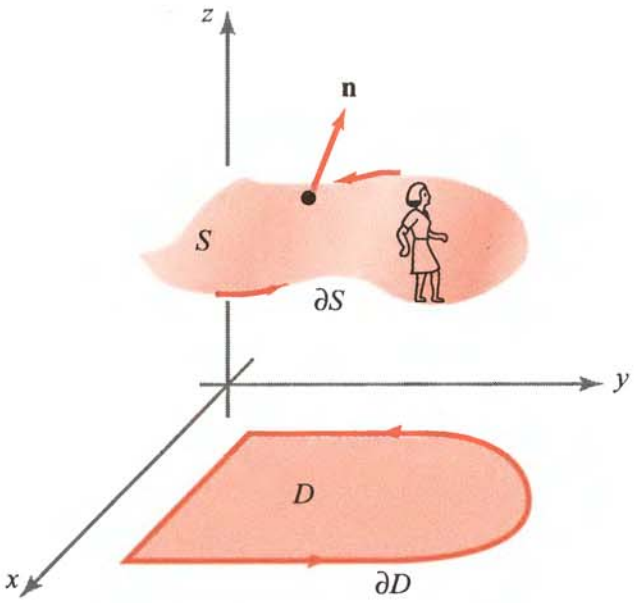



Figure 8.2.1 The induced orientation on $\partial S$: As you walk around the boundary, the surface should be on your left.

**THEOREM 5: Stokes' Theorem for Graphs** Let $S$ be the oriented surface defined by a $C^2$ function $z = f(x, y)$, where $(x, y) \in D$, a region to which Green's theorem applies, and let $\mathbf{F}$ be a $C^1$ vector field on $S$. Then if $\partial S$ denotes the oriented boundary curve of $S$ as just defined, we have

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$

Remember that $\int_{\partial S} \mathbf{F} \cdot d\mathbf{s}$ is the integral around $\partial S$ of the tangential component of $\mathbf{F}$, while $\iint_S \mathbf{G} \cdot d\mathbf{S}$ is the integral over $S$ of $\mathbf{G} \cdot \mathbf{n}$, the normal component of $\mathbf{G}$ (see Sections 7.2 and 7.6). Thus, Stokes' theorem says that the integral of *the normal component of the curl of a vector field* $\mathbf{F}$ *over a surface* $S$ *is equal to the integral of the tangential component of* $\mathbf{F}$ *around the boundary* $\partial S$.

*PROOF* If $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$, then

$$\operatorname{curl} \mathbf{F} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\mathbf{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\mathbf{k}.$$

Therefore, we use formula (1) to write

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_D \left[ \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\left(-\frac{\partial z}{\partial x}\right) + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\left(-\frac{\partial z}{\partial y}\right) + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) \right] dA. \quad (2)$$

On the other hand,

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{p}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{p}} F_1\, dx + F_2\, dy + F_3\, dz,$$

where $\mathbf{p}$: $[a, b] \to \mathbb{R}^3$, $\mathbf{p}(t) = (x(t), y(t), f(x(t), y(t)))$ is the orientation-preserving parametrization of the oriented simple closed curve $\partial S$ discussed earlier. Thus,

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_a^b \left( F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt. \tag{3}$$

By the chain rule

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

Substituting this expression into equation (3), we obtain

$$\begin{aligned} \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} &= \int_a^b \left[ \left( F_1 + F_3 \frac{\partial z}{\partial x} \right) \frac{dx}{dt} + \left( F_2 + F_3 \frac{\partial z}{\partial y} \right) \frac{dy}{dt} \right] dt \\ &= \int_{\mathbf{c}} \left( F_1 + F_3 \frac{\partial z}{\partial x} \right) dx + \left( F_2 + F_3 \frac{\partial z}{\partial y} \right) dy \\ &= \int_{\partial D} \left( F_1 + F_3 \frac{\partial z}{\partial x} \right) dx + \left( F_2 + F_3 \frac{\partial z}{\partial y} \right) dy. \end{aligned} \tag{4}$$

Applying Green's theorem to equation (4) yields (we are assuming that Green's theorem applies to $D$)

$$\iint_D \left[ \frac{\partial (F_2 + F_3\, \partial z/\partial y)}{\partial x} - \frac{\partial (F_1 + F_3\, \partial z/\partial x)}{\partial y} \right] dA.$$

Now we use the chain rule, remembering that $F_1$, $F_2$, and $F_3$ are functions of $x$, $y$, and $z$ and that $z$ is a function of $x$ and $y$, to obtain

$$\begin{aligned} \iint_D \Bigg[ &\left( \frac{\partial F_2}{\partial x} + \frac{\partial F_2}{\partial z}\frac{\partial z}{\partial x} + \frac{\partial F_3}{\partial x}\frac{\partial z}{\partial y} + \frac{\partial F_3}{\partial z}\frac{\partial z}{\partial x}\frac{\partial z}{\partial y} + F_3 \frac{\partial^2 z}{\partial x \partial y} \right) \\ &- \left( \frac{\partial F_1}{\partial y} + \frac{\partial F_1}{\partial z}\frac{\partial z}{\partial y} + \frac{\partial F_3}{\partial y}\frac{\partial z}{\partial x} + \frac{\partial F_3}{\partial z}\frac{\partial z}{\partial y}\frac{\partial z}{\partial x} + F_3 \frac{\partial^2 z}{\partial y \partial x} \right) \Bigg] dA. \end{aligned}$$

Because mixed partials are equal, the last two terms in each parenthesis cancel each other, and we can rearrange terms to obtain the integral of equation (2), which completes the proof. ■

**EXAMPLE 1** Let $\mathbf{F} = ye^z\mathbf{i} + xe^z\mathbf{j} + xye^z\mathbf{k}$. Show that the integral of $\mathbf{F}$ around an oriented simple closed curve $C$ that is the boundary of a surface $S$ is 0. (Assume $S$ is the graph of a function, as in Theorem 5.)

SOLUTION Indeed, $\int_C \mathbf{F} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, by Stokes' theorem. But we compute

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ ye^z & xe^z & xye^z \end{vmatrix} = \mathbf{0},$$

and so $\int_C \mathbf{F} \cdot d\mathbf{s} = 0$. Alternatively, we can observe that $\mathbf{F} = \nabla(xye^z)$, so its integral around a closed curve is zero. ▲

**EXAMPLE 2** Use Stokes' theorem to evaluate the line integral

$$\int_C -y^3\,dx + x^3\,dy - z^3\,dz,$$

where $C$ is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $x + y + z = 1$, and the orientation on $C$ corresponds to counterclockwise motion in the $xy$ plane.

SOLUTION The curve $C$ bounds the surface $S$ defined by the equation $z = 1 - x - y = f(x, y)$ for $(x, y)$ in the set $D = \{(x, y) \mid x^2 + y^2 \le 1\}$ (Figure 8.2.2). We set $\mathbf{F} = -y^3\mathbf{i} + x^3\mathbf{j} - z^3\mathbf{k}$, which has curl $\nabla \times \mathbf{F} = (3x^2 + 3y^2)\mathbf{k}$. Then, by

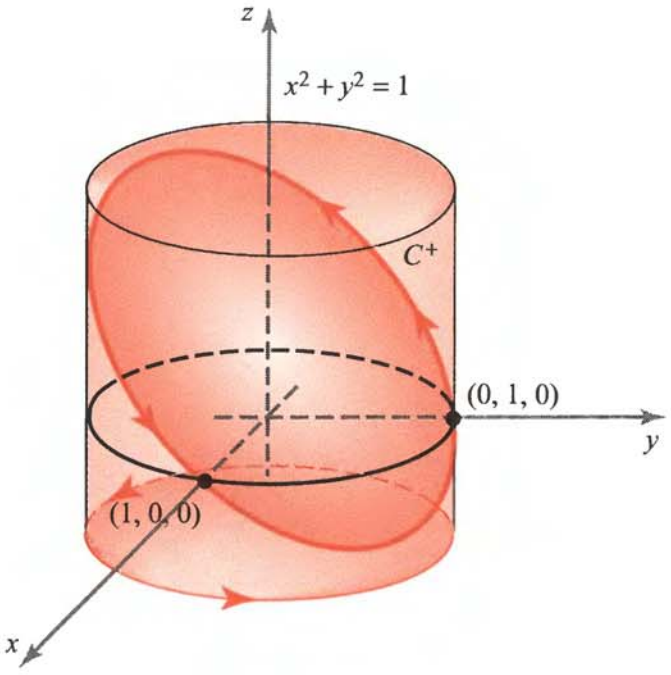



Figure 8.2.2 The curve $C$ is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $x + y + z = 1$.

Stokes' theorem, the line integral is equal to the surface integral

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

But $\nabla \times \mathbf{F}$ has only a $\mathbf{k}$ component. Thus, by formula (1) we have

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_D (3x^2 + 3y^2)\, dx\, dy.$$

This integral can be evaluated by changing to polar coordinates. Doing this, we get

$$3\iint_D (x^2 + y^2)\, dx\, dy = 3\int_0^1 \int_0^{2\pi} r^2 \cdot r\, d\theta\, dr = 6\pi \int_0^1 r^3\, dr = \frac{6\pi}{4} = \frac{3\pi}{2}.$$

Let us verify this result by *directly* evaluating the line integral

$$\int_C -y^3\, dx + x^3\, dy - z^3\, dz.$$

We can parametrize the curve $\partial D$ by the equations

$$x = \cos t, \qquad y = \sin t, \qquad z = 0, \qquad 0 \le t \le 2\pi.$$

The curve $C$ is therefore parametrized by the equations

$$x = \cos t, \qquad y = \sin t, \qquad z = 1 - \sin t - \cos t, \qquad 0 \le t \le 2\pi.$$

Thus,

$$\begin{aligned}
&\int_C -y^3\, dx + x^3\, dy - z^3\, dz \\
&= \int_0^{2\pi} [(-\sin^3 t)(-\sin t) + (\cos^3 t)(\cos t) \\
&\qquad - (1 - \sin t - \cos t)^3(-\cos t + \sin t)]\, dt \\
&= \int_0^{2\pi} (\cos^4 t + \sin^4 t)\, dt - \int_0^{2\pi} (1 - \sin t - \cos t)^3(-\cos t + \sin t)\, dt.
\end{aligned}$$

The second integrand is of the form $u^3\, du$, where $u = 1 - \sin t - \cos t$, and thus the integral is equal to

$$\frac{1}{4}[(1 - \sin t - \cos t)^4]_0^{2\pi} = 0.$$

Hence, we are left with

$$\int_0^{2\pi} (\cos^4 t + \sin^4 t)\, dt.$$

This integral can be evaluated using formulas (18) and (19) of the table of integrals. We can also proceed as follows. Using the trigonometric identities

$$\sin^2 t = \frac{1 - \cos 2t}{2}, \qquad \cos^2 t = \frac{1 + \cos 2t}{2},$$

substituting and squaring these expressions, we reduce the preceding integral to

$$\frac{1}{2}\int_0^{2\pi} (1 + \cos^2 2t)\, dt = \pi + \frac{1}{2}\int_0^{2\pi} \cos^2 2t\, dt.$$

Again using the identity $\cos^2 2t = (1 + \cos 4t)/2$, we find

$$\begin{aligned}\pi + \frac{1}{4}\int_0^{2\pi} (1 + \cos 4t)\, dt &= \pi + \frac{1}{4}\int_0^{2\pi} dt + \frac{1}{4}\int_0^{2\pi} \cos 4t\, dt \\ &= \pi + \frac{\pi}{2} + 0 = \frac{3\pi}{2}. \quad \blacktriangle\end{aligned}$$

## Stokes' Theorem for Parametrized Surfaces

To simplify the proof of Stokes' theorem given earlier, we assumed that the surface $S$ could be described as the graph of a function $z = f(x, y)$, $(x, y) \in D$, where $D$ is some region to which Green's theorem applies. However, without too much more effort we can obtain a *more general theorem* for oriented parametrized surfaces $S$. The main complication is in the definition of $\partial S$.

Suppose $\mathbf{\Phi}$: $D \to \mathbb{R}^3$ is a parametrization of a surface $S$ and $\mathbf{c}(t) = (u(t), v(t))$ is a parametrization of $\partial D$. We might be tempted to define $\partial S$ as the curve parametrized by $t \mapsto \mathbf{p}(t) = \mathbf{\Phi}(u(t), v(t))$. However, with this definition, $\partial S$ might not be the boundary of $S$ in any reasonable geometric sense.

For example, we would conclude that the boundary of the unit sphere $S$ parametrized by spherical coordinates in $\mathbb{R}^3$ is half of the great circle on $S$ lying in the $xz$ plane, but clearly in a geometric sense $S$ is a smooth surface (no points or cusps) with no boundary or edge at all (see Figure 8.2.3 and Exercise 20). Thus, this great circle is in some sense the "mistaken" boundary of $S$.

We can get around this difficulty by assuming that $\mathbf{\Phi}$ is one-to-one on all of $D$. Then the image of $\partial D$ under $\mathbf{\Phi}$, namely, $\mathbf{\Phi}(\partial D)$, will be the geometric boundary of $S = \mathbf{\Phi}(D)$. If $\mathbf{c}(t) = (u(t), v(t))$ is a parametrization of $\partial D$ in the positive direction,

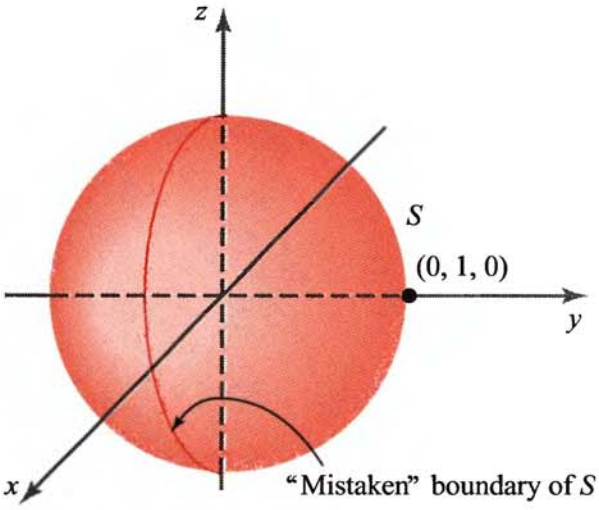



**Figure 8.2.3** The surface $S$ is a portion of a sphere.

we define $\partial S$ to be the oriented simple closed curve that is the image of the mapping $\mathbf{p}$: $t \mapsto \mathbf{\Phi}(u(t), v(t))$, with the orientation of $\partial S$ induced by $\mathbf{p}$ (see Figure 8.2.1).

**THEOREM 6: Stokes' Theorem: Parametrized Surfaces** Let $S$ be an oriented surface defined by a one-to-one parametrization $\mathbf{\Phi}$: $D \subset \mathbb{R}^2 \to S$, where $D$ is a region to which Green's theorem applies. Let $\partial S$ denote the oriented boundary of $S$ and let $\mathbf{F}$ be a $C^1$ vector field on $S$. Then

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$

If $S$ has no boundary, and this includes surfaces such as the sphere, then the integral on the left is zero (see Exercise 17).

This is proved in the same way as Theorem 5.

**EXAMPLE 3** Let $S$ be the surface shown in Figure 8.2.4, with the indicated orientation. Let $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + e^{xz}\mathbf{k}$. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$.

SOLUTION This surface could be parametrized using spherical coordinates based at the center of the sphere. However, we need not explicitly find $\mathbf{\Phi}$ in order to solve this problem. By Theorem 6, $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}$, and so if we parametrize $\partial S$ by $x(t) = \cos t$, $y(t) = \sin t$, $0 \le t \le 2\pi$, we determine

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} \left(y\frac{dx}{dt} - x\frac{dy}{dt}\right)dt = \int_0^{2\pi} (-\sin^2 t - \cos^2 t)\,dt = -\int_0^{2\pi} dt = -2\pi$$

and therefore $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = -2\pi$. ▲

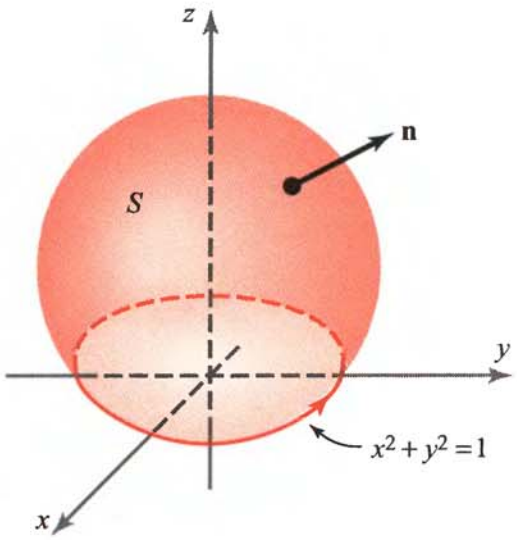



**Figure 8.2.4** This surface $S$ is a portion of a sphere sitting on top of the circle $x^2 + y^2 = 1$. It does *not* include the disk $x^2 + y^2 < 1$ in the $xy$ plane.

## The Curl as Circulation per Unit Area

Let us now use Stokes' theorem to justify the physical interpretation of $\nabla \times \mathbf{F}$ in terms of paddle wheels that was proposed in Chapter 4. Paraphrasing Theorem 6, we have

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n}\, dS = \iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_{\partial S} F_T\, ds,$$

where $F_T$ is the tangential component of $\mathbf{F}$. This says that the integral of the normal component of the curl of a vector field over an oriented surface $S$ is equal to the line integral of $\mathbf{F}$ along $\partial S$, which in turn is equal to the path integral of the tangential component of $\mathbf{F}$ over $\partial S$.

Suppose $\mathbf{V}$ represents the velocity vector field of a fluid. Consider a point P and a unit vector $\mathbf{n}$. Let $S_\rho$ denote the disk of radius $\rho$ and center P, which is perpendicular to $\mathbf{n}$. By Stokes' theorem,

$$\iint_{S_\rho} \text{curl } \mathbf{V} \cdot d\mathbf{S} = \iint_{S_\rho} \text{curl } \mathbf{V} \cdot \mathbf{n}\, dS = \int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s},$$

where $\partial S_\rho$ has the orientation induced by $\mathbf{n}$ (see Figure 8.2.5).

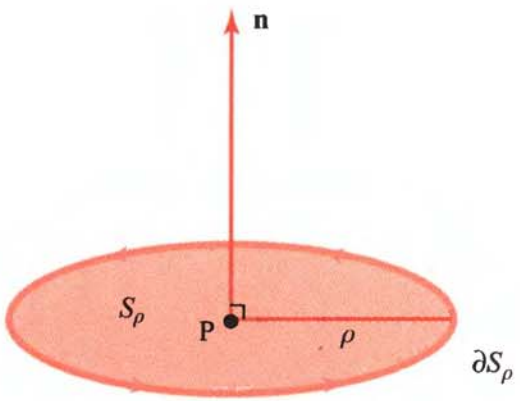



**Figure 8.2.5** A normal $\mathbf{n}$ induces an orientation on the boundary $\partial S_\rho$ of the disk $S_\rho$.

By the mean-value theorem for integrals (Exercise 12, Section 7.6), there is a point Q in $S_\rho$ such that

$$\iint_{S_\rho} \text{curl } \mathbf{V} \cdot \mathbf{n}\, dS = [\text{curl } \mathbf{V}(\text{Q}) \cdot \mathbf{n}] A(S_\rho)$$

where $A(S_\rho) = \pi\rho^2$ is the area of $S_\rho$, curl $\mathbf{V}$(Q) is the value of curl $\mathbf{V}$ at Q. Thus,

$$\begin{aligned}\lim_{\rho\to 0} \frac{1}{A(S_\rho)} \int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s} &= \lim_{\rho\to 0} \frac{1}{A(S_\rho)} \iint_{S_\rho} (\text{curl } \mathbf{V}) \cdot d\mathbf{S} \\ &= \lim_{\rho\to 0} \text{curl } \mathbf{V}(\text{Q}) \cdot \mathbf{n} = \text{curl } \mathbf{V}(\text{P}) \cdot \mathbf{n}.\end{aligned}$$

Thus,[1]

$$\text{curl } \mathbf{V}(\text{P}) \cdot \mathbf{n} = \lim_{\rho\to 0} \frac{1}{A(S_\rho)} \int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s}. \tag{5}$$

Let us pause to consider the physical meaning of $\int_C \mathbf{V} \cdot d\mathbf{s}$ when $\mathbf{V}$ is the velocity field of a fluid. Suppose, for example, that $\mathbf{V}$ points in the direction tangent to the oriented curve $C$ (Figure 8.2.6). Then clearly $\int_C \mathbf{V} \cdot d\mathbf{s} > 0$, and particles on $C$ tend to rotate counterclockwise. If $\mathbf{V}$ is pointing in the opposite direction, then $\int_C \mathbf{V} \cdot d\mathbf{s} < 0$ and particles tend to rotate clockwise. If $\mathbf{V}$ is perpendicular to $C$, then particles don't rotate on $C$ at all and $\int_C \mathbf{V} \cdot d\mathbf{s} = 0$. In general, $\int_C \mathbf{V} \cdot d\mathbf{s}$, being the integral of the tangential component of $\mathbf{V}$, represents the net amount of turning of the fluid in a counterclockwise direction around $C$. One therefore refers to $\int_C \mathbf{V} \cdot d\mathbf{s}$ as the ***circulation*** of $\mathbf{V}$ around $C$ (see Figure 8.2.7).

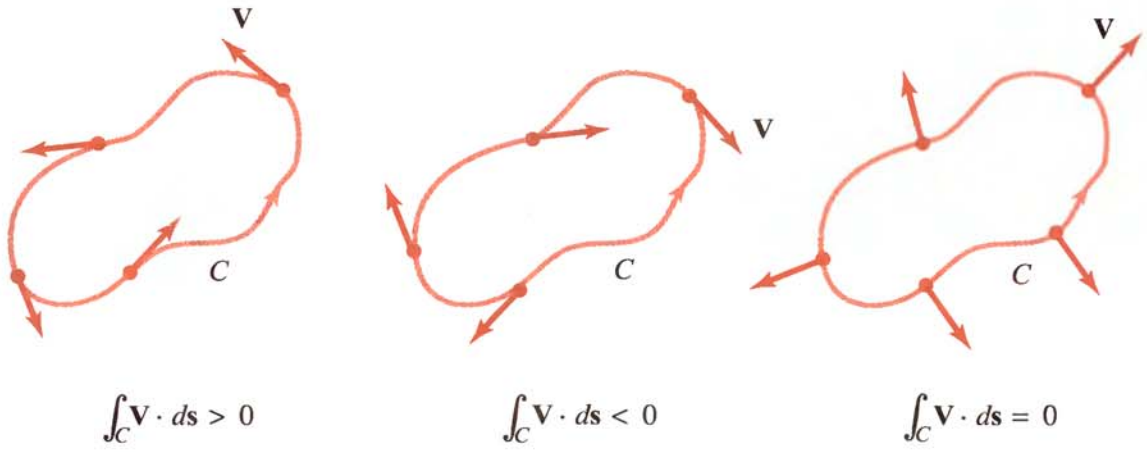



**Figure 8.2.6** The intuitive meaning of the possible signs of $\int_C \mathbf{V} \cdot d\mathbf{s}$.

[1] Some informal texts adopt equation (5) as the *definition* of the curl, and use it to "prove" Stokes' theorem. However, this raises the danger of circular reasoning, for to show that equation (5) really defines a vector "curl $\mathbf{V}$(P)" requires Stokes' theorem, or some similar argument.

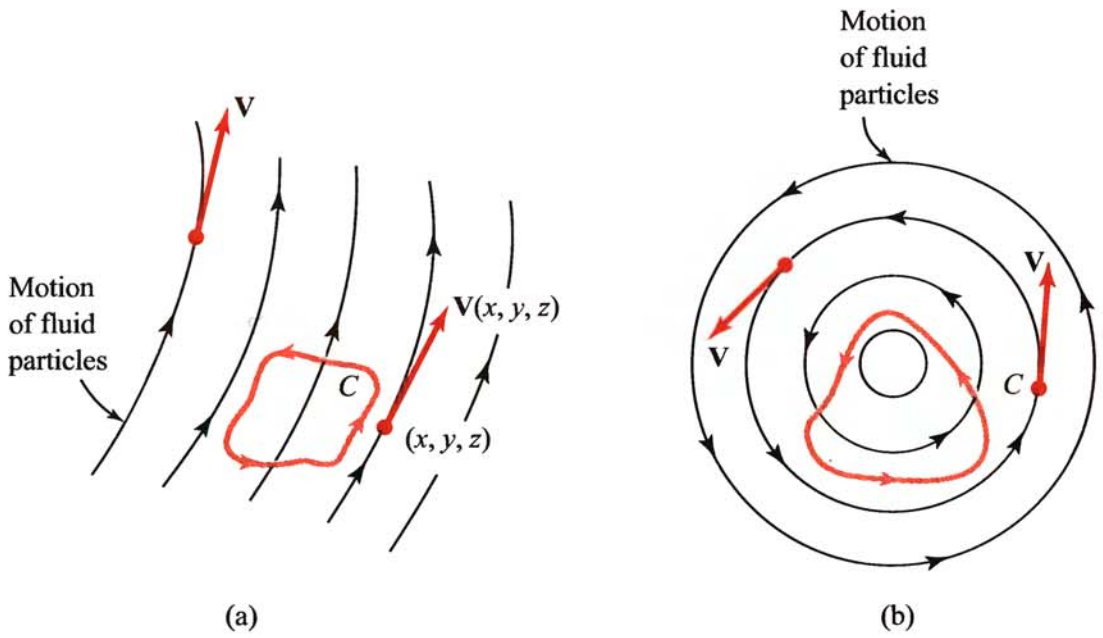



**Figure 8.2.7** Circulation of a vector field (velocity field of a fluid): (a) Circulation about $C$ is zero; (b) nonzero circulation about $C$ ("whirlpool").

These results allow us to see just what curl $\mathbf{V}$ means for the motion of a fluid. The circulation $\int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s}$ is the net velocity of the fluid around $\partial S_\rho$, so that (curl $\mathbf{V}$) $\cdot$ $\mathbf{n}$ represents the turning or rotating effect of the fluid around the axis $\mathbf{n}$.

**Circulation and Curl** The dot product of curl $\mathbf{V}(\mathrm{P})$ with a unit vector $\mathbf{n}$, namely, curl $\mathbf{V}(\mathrm{P}) \cdot \mathbf{n}$, equals the circulation of $\mathbf{V}$ per unit area at P on a surface perpendicular to $\mathbf{n}$.

Observe that the magnitude of curl $\mathbf{V}(\mathrm{P}) \cdot \mathbf{n}$ is maximized when $\mathbf{n} = \text{curl } \mathbf{V}/\|\text{curl } \mathbf{V}\|$ (evaluated at P). Therefore, the rotating effect at P is greatest about the axis that is parallel to curl $\mathbf{V}/\|\text{curl } \mathbf{V}\|$. Thus, curl $\mathbf{V}$ is aptly called the ***vorticity vector***.

We can use these ideas to compute the curl in cylindrical coordinates.

**EXAMPLE 4** Let the unit vectors $\mathbf{e}_r$, $\mathbf{e}_\theta$, $\mathbf{e}_z$ associated to cylindrical coordinates be as shown in Figure 8.2.8. Let $\mathbf{F} = F_r\mathbf{e}_r + F_\theta\mathbf{e}_\theta + F_z\mathbf{e}_z$. (The subscripts here denote components of $\mathbf{F}$, not partial derivatives.) Find a formula for the $\mathbf{e}_r$ component of $\nabla \times \mathbf{F}$ in cylindrical coordinates.

SOLUTION Let $S$ be the surface shown in Figure 8.2.9.

The area of $S$ is $r\,d\theta\,dz$ and the unit normal is $\mathbf{e}_r$. The integral of $\mathbf{F}$ around the edges of $S$ is approximately

$$[F_\theta(r,\theta,z) - F_\theta(r,\theta,z+dz)]r\,d\theta + [F_z(r,\theta+d\theta,z) - F_z(r,\theta,z)]\,dz$$
$$\approx -\frac{\partial F_\theta}{\partial z}\,dz\,r\,d\theta + \frac{\partial F_z}{\partial \theta}\,d\theta\,dz.$$

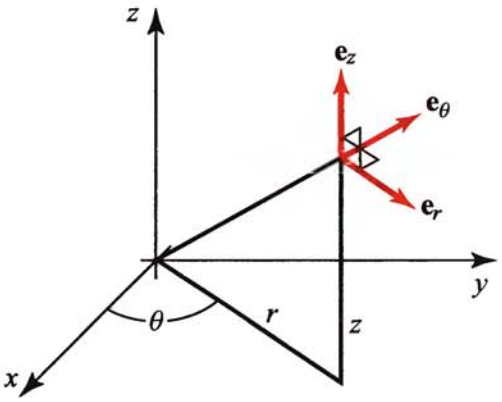



**Figure 8.2.8** Orthonormal vectors $\mathbf{e}_r$, $\mathbf{e}_\theta$, and $\mathbf{e}_z$ associated with cylindrical coordinates. The vector $\mathbf{e}_r$ is parallel to the line labeled $r$.

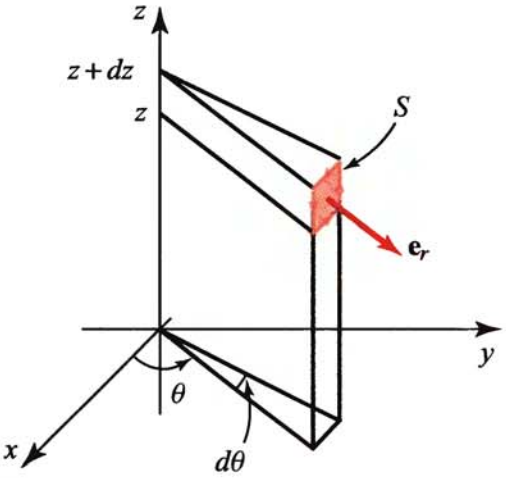



**Figure 8.2.9** A surface element in cylindrical coordinates.

Thus, the circulation per unit area is this expression divided by $r\,d\theta dz$, namely,

$$\frac{1}{r}\frac{\partial F_z}{\partial \theta} - \frac{\partial F_\theta}{\partial z}.$$

According to the previous box, this must be the $\mathbf{e}_r$ component of the curl. ▲

## Gradient, Divergence, and Curl in Cylindrical and Spherical Coordinates

By similar arguments to Example 4, one finds that the curl in cylindrical coordinates is given by

$$\nabla \times \mathbf{F} = \frac{1}{r}\begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\theta & \mathbf{e}_z \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial \theta} & \dfrac{\partial}{\partial z} \\ F_r & rF_\theta & F_z \end{vmatrix}.$$

We can find other important vector quantities expressed in different coordinate systems. For example, the chain rule shows that the gradient in cylindrical coordinates is

$$\nabla f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{\partial f}{\partial z}\mathbf{e}_z,$$

and in Section 8.4 we will establish related techniques that give the following formula for the divergence in cylindrical coordinates:

$$\nabla \cdot \mathbf{F} = \frac{1}{r}\left[\frac{\partial}{\partial r}(rF_r) + \frac{\partial F_\theta}{\partial \theta} + \frac{\partial}{\partial z}(rF_z)\right].$$

Corresponding formulas for gradient, divergence, and curl in spherical coordinates are

$$\nabla f = \frac{\partial f}{\partial \rho}\mathbf{e}_\rho + \frac{1}{\rho}\frac{\partial f}{\partial \phi}\mathbf{e}_\phi + \frac{1}{\rho \sin\phi}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta$$

$$\nabla \cdot \mathbf{F} = \frac{1}{\rho^2}\frac{\partial}{\partial \rho}(\rho^2 F_\rho) + \frac{1}{\rho \sin\phi}\frac{\partial}{\partial \phi}(\sin\phi F_\phi) + \frac{1}{\rho \sin\phi}\frac{\partial F_\theta}{\partial \theta}$$

and

$$\nabla \times \mathbf{F} = \left[\frac{1}{\rho \sin\phi}\frac{\partial}{\partial \phi}(\sin\phi F_\theta) - \frac{1}{\rho \sin\phi}\frac{\partial F_\phi}{\partial \theta}\right]\mathbf{e}_\rho + \left[\frac{1}{\rho \sin\phi}\frac{\partial F_\rho}{\partial \theta} - \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho F_\theta)\right]\mathbf{e}_\phi + \left[\frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho F_\phi) - \frac{1}{\rho}\frac{\partial F_\rho}{\partial \phi}\right]\mathbf{e}_\theta$$

where $\mathbf{e}_\rho$, $\mathbf{e}_\phi$, $\mathbf{e}_\theta$ are as shown in Figure 8.2.10 and where $\mathbf{F} = F_\rho \mathbf{e}_\rho + F_\phi \mathbf{e}_\phi + F_\theta \mathbf{e}_\theta$.

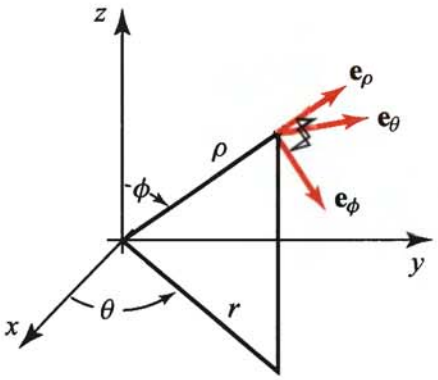



**Figure 8.2.10** Orthonormal vectors $\mathbf{e}_\rho$, $\mathbf{e}_\phi$, and $\mathbf{e}_\theta$ associated with spherical coordinates.

## Faraday's Law

Vector calculus plays an essential role in the theory of electromagnetism. The next example shows how Stokes' theorem applies.

**EXAMPLE 5** Let $\mathbf{E}$ and $\mathbf{H}$ be time-dependent electric and magnetic fields, respectively, in space. Let $S$ be a surface with boundary $C$. We define

$$\int_C \mathbf{E} \cdot d\mathbf{s} = \text{voltage around } C,$$

$$\iint_S \mathbf{H} \cdot d\mathbf{S} = \text{magnetic flux across } S.$$

***Faraday's law*** (see Figure 8.2.11) states that the *voltage around C equals the negative rate of change of magnetic flux through S.* Show that Faraday's law follows from the following differential equation (one of the Maxwell equations):

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{H}}{\partial t}.$$

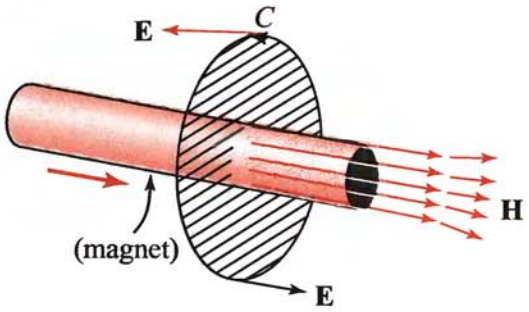



**Figure 8.2.11** Faraday's law.

SOLUTION Assume that $-\partial \mathbf{H}/\partial t = \nabla \times \mathbf{E}$ holds. By Stokes' theorem,

$$\int_C \mathbf{E} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}.$$

Assuming that we can move $\partial/\partial t$ under the integral sign, we get

$$-\frac{\partial}{\partial t} \iint_S \mathbf{H} \cdot d\mathbf{S} = \iint_S -\frac{\partial \mathbf{H}}{\partial t} \cdot d\mathbf{S} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = \int_C \mathbf{E} \cdot d\mathbf{s}$$

and so

$$\int_C \mathbf{E} \cdot d\mathbf{s} = -\frac{\partial}{\partial t} \iint_S \mathbf{H} \cdot d\mathbf{S},$$

which is Faraday's law. ▲

## Supplement to Section 8.2: Stokes' Theorem, Astronauts, and Falling Cats

### *Falling Cats*

Have you ever wondered how a falling cat can right itself? Released from a resting position with its feet above its head, the cat is able to execute a 180° reorientation and land safely on its feet. This well-known phenomenon has fascinated people for many years—especially in cities like New York, where cats have been known to survive falls of 8 to 30 stories!

There have been many incorrect explanations as to how cats are able to right themselves, including the idea that it has to do with how the cat twirls its tail. This cannot be right because Manx cats, which have no tails, can also perform this feat!

Figure 8.2.12
The falling cat rights itself by wriggling its body parts.

One observes, as in Figure 8.2.12, that the cat achieves this net change in orientation by wriggling, to create *changes in its internal shape or configuration.* On the surface, this provides a seeming contradiction; because the cat is dropped from a resting position, it has zero angular momentum at the beginning of the fall and hence, according to a basic law of physics called *conservation of angular momentum,* the cat has zero angular momentum throughout the duration of its fall.[2] Amazingly, the cat has effectively changed its angular position while maintaining zero angular momentum!

The exact process by which this occurs is subtle; intuitive reasoning can lead one astray and, as we have indicated, many false explanations have been offered throughout the history of trying to solve this mystery.[3] Recently, new and interesting insights have been discovered using geometric methods that, in fact, are related to curvature (see Section 7.7).[4]

The way that curvature and geometry are related to the falling cat phenomenon is not easy to explain in full detail, but we can explain a similar phenomenon that is easy to understand. One of the points to emphasize is that *Stokes' theorem is the key to proving all of the relevant theorems.*

### *Reorienting Astronauts*

Another example to help visualize this effect is to consider astronauts who wish to reorient themselves in a free-space environment. As with the falling cat, this motion can again be achieved using internal gyrations, or *shape changes.* For instance, consider astronauts moving their arms much like the motion of arms stirring liquid in a large kettle. The arms are held out forward, to lie in a horizontal plane that goes through the shoulders, parallel to the floor; the hands are clasped together and remain in this horizontal plane during the circular stirring motion. At the point of maximum extension of the arms, the inertia of the body about a vertical axis is also at a maximum. Conservation of angular momentum requires that the body rotate in an opposite and proportional manner to the motion of the arms. As the arms rotate around and are brought in, however, the inertia of the body is reduced. The motion of the body in reaction is therefore also reduced. Thus, in one complete cycle of arm movement, the body undergoes a net rotation opposite the direction of arm motion. When the desired

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[2] We saw an instance of the law of conservation of angular momentum in Section 4.1, Exercise 20.

[3] Another favorite fallacious argument, showing that a cat *cannot* turn itself over(!), is this: "Accept from physics that angular momentum is the moment of inertia times angular velocity [moments of inertia are discussed in Section 6.3]. But the angular momentum of the cat is zero, so the angular velocity must also be zero. Because angular velocity is the rate of change of the angular position, the angular position is constant. Thus, the cat cannot turn itself over." What is wrong? This argument ignores the fact that the cat changes its *shape,* and hence its moment of inertia, during the fall.

[4] See T. R. Kane and M. Scher, "A Dynamical Explanation of the Falling Cat Phenomenon," *Int. J. Solids Struct.,* 5 (1969): 663–670. See also R. Montgomery, "Isoholonomic Problems and Some Applications," *Commun. Math. Phys.,* 128 (1990): 565–592; R. Montgomery, "How Much Does a Rigid Body Rotate? A Berry's Phase from the 18th Century," *Am. J. Pys.,* 59 (1991b): 394–398. See also J. E. Marsden and J. Ostrowski, "Symmetries in Motion: Geometric Foundations of Motion Control," *Nonlinear Science Today* (1998), http://link.springer-ny.com; R. Batterman, "Falling Cats, Parallel Parking, and Polarized Light," *Philos. Soc. Arch.* (2002); http://philsci-archive.pitt.edu/documents/disk0/00/00/05/83, http://www.its.caltech.edu/~mleok/falling_cats.htm, and references therein.

orientation is achieved, the astronaut needs merely to stop the arm motion in order to come to rest. One often refers to the extra motion that is achieved as ***geometric phase***.

### *Link with Non-Euclidean Geometry*

The theory of geometric phases also shows up in an interesting way in non-Euclidean geometry—as in the geometry of triangles drawn on a sphere. A simple way to explain this link is as follows. Hold your hand at arm's length, but allow rotation in your shoulder joints. Move your hand along three great circles, forming a triangle on the sphere; during the motion along each arc, always keep your thumb *parallel*; that is, it should move in such a way that it forms a *fixed* angle with the direction of motion along each arc and does not rotate when switching arcs. After completing the circuit around the triangle, your thumb will return rotated through an angle relative to its starting position (see Figure 8.2.13). Can you see in Figure 8.2.13 that the angle of rotation is $90^\circ$ (or $\pi/2$ radians) and that this is what happens when you do the thumb experiment yourself?

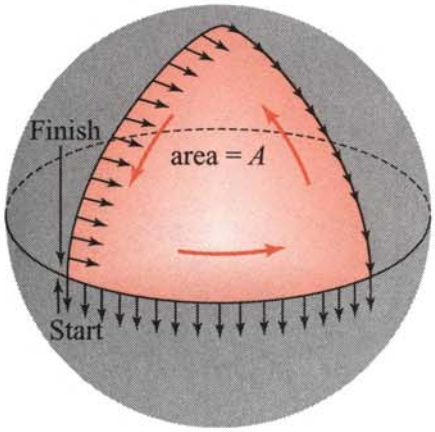



**Figure 8.2.13** A parallel movement of your thumb around a spherical triangle produces a phase shift.

For general spherical triangles, this angle (in radians) is given by $\Theta = \Delta - \pi$, where $\Delta$ is the sum of the angles of the triangle. The fact that $\Theta$ is strictly positive (!) is one of the basic truths of non-Euclidean geometry—the sum of the angles of a right triangle on a sphere is greater than $\pi$! This angle is also related to the *area* $A$ enclosed by the triangle through the relation $\Theta = A/r^2$, where $r$ is the radius of the sphere. The rotational shift of the thumb during the course of its cyclic journey around the spherical triangle is directly related to the curvature of the sphere and to the area enclosed by the path that is traced out. Notice first that for a spherical triangle that is $1/8$ of the sphere, $A = 4\pi r^2/8 = \pi r^2/2$. Thus, $A/r^2 = \pi/2$. Notice also that when $r \to \infty$, the sphere becomes flatter and thus approaches a Euclidean plane, in which case $\Theta = 0$.

The cyclic journey of the thumb around the closed path is analogous to the cyclic internal motions made by the cat during its fall; the $90^\circ$ shift in the direction of the thumb after one trip around is analogous to the $180^\circ$ reorientation of the cat. A deeper look at the underlying mathematics shows that, in fact, they are both instances of the same phenomenon (called *holonomy*)—and Stokes' theorem is the key to understanding it.

## EXERCISES

**1.** Redo Exercise 5 of Section 7.6 using Stokes' theorem.

**2.** Redo Exercise 6 of Section 7.6 using Stokes' theorem.

**3.** Verify Stokes' theorem for the upper hemisphere $z = \sqrt{1 - x^2 - y^2}$, $z \geq 0$, and the radial vector field $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

**4.** Let $S$ be a surface with boundary $\partial S$, and suppose $\mathbf{E}$ is an electric field that is perpendicular to $\partial S$. Show that the induced magnetic flux across $S$ is constant in time. (HINT: Use Faraday's law.)

**5.** Let $S$ be the capped cylindrical surface shown in Figure 8.2.14. $S$ is the union of two surfaces, $S_1$ and $S_2$, where $S_1$ is the set of $(x, y, z)$ with $x^2 + y^2 = 1$, $0 \leq z \leq 1$, and $S_2$ is the set of $(x, y, z)$ with $x^2 + y^2 + (z - 1)^2 = 1$, $z \geq 1$. Set $\mathbf{F}(x, y, z) = (zx + z^2y + x)\mathbf{i} + (z^3yx + y)\mathbf{j} + z^4x^2\mathbf{k}$. Compute $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$. (HINT: Stokes' theorem holds for this surface.)

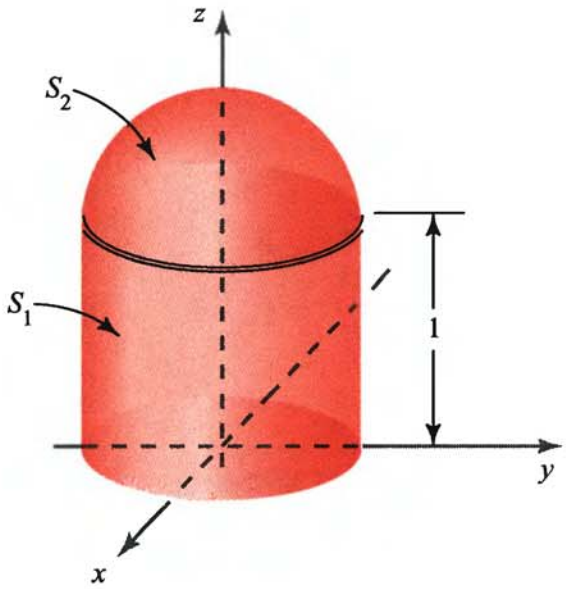



**Figure 8.2.14** The capped cylinder is the union of $S_1$ and $S_2$.

**6.** Let $\mathbf{c}$ consist of straight lines joining $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ and let $S$ be the triangle with these vertices. Verify Stokes' theorem directly with $\mathbf{F} = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$.

**7.** Evaluate the integral $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where $S$ is the portion of the surface of a sphere defined by $x^2 + y^2 + z^2 = 1$ and $x + y + z \geq 1$, and where $\mathbf{F} = \mathbf{r} \times (\mathbf{i} + \mathbf{j} + \mathbf{k})$, $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

**8.** Show that the calculation in Exercise 7 can be simplified by observing that $\int_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \int_{\partial \Sigma} \mathbf{F} \cdot d\mathbf{r}$ for any other surface $\Sigma$. By picking $\Sigma$ appropriately, $\iint_\Sigma (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ may be easy to compute. Show that this is the case if $\Sigma$ is taken to be the portion of the plane $x + y + z = 1$ inside the circle $\partial S$.

**9.** Calculate the surface integral $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ where $S$ is the hemisphere $x^2 + y^2 + z^2 = 1$, $x \geq 0$ and $\mathbf{F} = x^3\mathbf{i} - y^3\mathbf{j}$.

**10.** Find $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where $S$ is the ellipsoid $x^2 + y^2 + 2z^2 = 10$ and $\mathbf{F}$ is the vector field $\mathbf{F} = (\sin xy)\mathbf{i} + e^x\mathbf{j} - yz\mathbf{k}$.

**11.** Let $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + zx^3y^2\mathbf{k}$. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dA$, where $S$ is the surface defined by $x^2 + y^2 + z^2 = 1, z \leq 0$.

**12.** A hot-air balloon has the truncated spherical shape shown in Figure 8.2.15. The hot gases escape through the porous envelope with a velocity vector field

$$\mathbf{V}(x, y, z) = \nabla \times \Phi(x, y, z) \qquad \text{where} \qquad \Phi(x, y, z) = -y\mathbf{i} + x\mathbf{j}.$$

If $R = 5$, compute the volume flow rate of the gases through the surface.

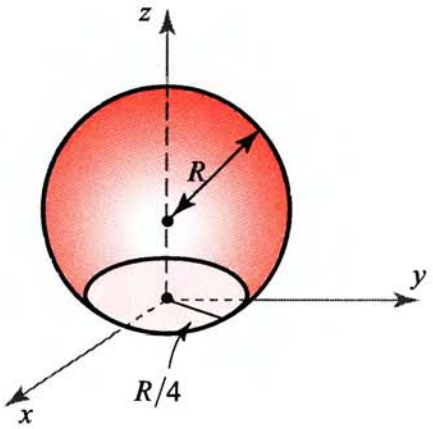



**Figure 8.2.15** A hot-air balloon.

**13.** Prove that Faraday's law implies $\nabla \times \mathbf{E} = -\partial \mathbf{H}/\partial t$.

**14.** Let $S$ be a surface and let $\mathbf{F}$ be perpendicular to the tangent to the boundary of $S$. Show that

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0.$$

What does this mean physically if $\mathbf{F}$ is an electric field?

**15.** Consider two surfaces $S_1$, $S_2$ with the same boundary $\partial S$. Describe with sketches how $S_1$ and $S_2$ must be oriented to ensure that

$$\iint_{S_1} (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_{S_2} (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

**16.** For a surface $S$ and a fixed vector $\mathbf{v}$, prove that

$$2 \iint_S \mathbf{v} \cdot \mathbf{n}\, dS = \int_{\partial S} (\mathbf{v} \times \mathbf{r}) \cdot d\mathbf{s},$$

where $\mathbf{r}(x, y, z) = (x, y, z)$.

**17.** Argue informally that if $S$ is a closed surface, then

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0$$

(see Exercise 15). (A *closed surface* is one that forms the boundary of a region in space; thus, for example, a sphere is a closed surface.)

**18.** If $C$ is a closed curve that is the boundary of a surface $S$, and $f$ and $g$ are $C^2$ functions, show that

(a) $\displaystyle\int_C f\nabla g \cdot d\mathbf{s} = \iint_S (\nabla f \times \nabla g) \cdot d\mathbf{S}$

(b) $\displaystyle\int_C (f\nabla g + g\nabla f) \cdot d\mathbf{s} = 0$

**19.** (a) If $C$ is a closed curve that is the boundary of a surface $S$, and $\mathbf{v}$ is a constant vector, show that

$$\int_C \mathbf{v} \cdot d\mathbf{s} = 0.$$

(b) Show that this is true even if $C$ is not the boundary of a surface $S$.

**20.** Show that $\Phi$: $D \to \mathbb{R}^3$, $D = [0, \pi] \times [0, 2\pi]$, $\Phi(\phi, \theta) = (\cos\theta \sin\phi, \sin\theta \sin\phi, \cos\phi)$, which parametrizes the unit sphere, takes the boundary of $D$ to half of a great circle on $S$.

**21.** Verify Theorem 6 for the helicoid $\Phi(r, \theta) = (r\cos\theta, r\sin\theta, \theta)$, $(r, \theta) \in [0, 1] \times [0, \pi/2]$, and the vector field $\mathbf{F}(x, y, z) = (z, x, y)$.

**22.** Prove Theorem 6.

**23.** Let $\mathbf{F} = x^2\mathbf{i} + (2xy + x)\mathbf{j} + z\mathbf{k}$. Let $C$ be the circle $x^2 + y^2 = 1$ and $S$ the disk $x^2 + y^2 \leq 1$ within the plane $z = 0$.

(a) Determine the flux of $\mathbf{F}$ out of $S$.
(b) Determine the circulation of $\mathbf{F}$ around $C$.
(c) Find the flux of $\nabla \times \mathbf{F}$. Verify Stokes' theorem directly in this case.

**24.** Let $S$ be a surface with boundary $\partial S$, and suppose that $\mathbf{E}$ is an electric field that is perpendicular to $\partial S$. Use Faraday's law to show that the induced magnetic flux across $S$ is constant in time.

**25.** Integrate $\nabla \times \mathbf{F}$, $\mathbf{F} = (3y, -xz, -yz^2)$ over the portion of the surface $2z = x^2 + y^2$ below the plane $z = 2$, both directly and by using Stokes' theorem.

**26.** ***Ampère's law*** states that if the electric current density is described by a vector field $\mathbf{J}$ and the induced magnetic field is $\mathbf{H}$, then the circulation of $\mathbf{H}$ around the boundary $C$ of a surface $S$ equals the integral of $\mathbf{J}$ over $S$ (i.e., the total current crossing $S$). See Figure 8.2.16. Show that this is implied by the steady-state *Maxwell equation* $\nabla \times \mathbf{H} = \mathbf{J}$.

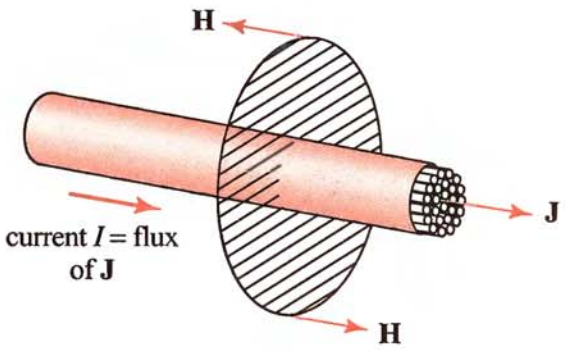



Figure 8.2.16 Ampère's law.

**27.** Faraday's law relates the line integral of the electric field around a loop $C$ to the surface integral of the rate of change of the magnetic field over a surface $S$ with boundary $C$. Regarding the equation $\nabla \times \mathbf{E} = -\partial \mathbf{H}/\partial t$ as the basic equation, Faraday's law is a consequence of Stokes' theorem, as we have seen in Example 4.

Suppose we are given electric and magnetic fields in space that satisfy $\nabla \times \mathbf{E} = -\partial \mathbf{H}/\partial t$. Suppose $C$ is the boundary of the Möbius band shown in Figures 7.6.3 and 7.6.4. Because the Möbius band cannot be oriented, Stokes' theorem does not apply. What becomes of Faraday's law? What do you guess $\int_C \mathbf{E} \cdot d\mathbf{s}$ equals?

**28.** (a) If in spherical coordinates, we write

$$\mathbf{e}_r = \alpha \mathbf{i} + \beta \mathbf{j} + \gamma \mathbf{k}, \text{ find } \alpha, \beta, \text{ and } \gamma.$$

(b) Find similar formulas for $\mathbf{e}_\phi$ and $\mathbf{e}_\theta$.

## 8.3 Conservative Fields

We saw in Section 7.2 that for a gradient force field $\mathbf{F} = \nabla f$, line integrals of $\mathbf{F}$ were evaluated as follows:

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = f(\mathbf{c}(b)) - f(\mathbf{c}(a)).$$

The value of the integral depends only on the endpoints $\mathbf{c}(b)$ and $\mathbf{c}(a)$ of the path. In other words, if we used another path with the same endpoints, we would still get the same answer. This leads us to say that the integral is *path-independent.*

Gradient fields are important in many physical problems. For example, if $V = -f$ represents a potential energy (gravitational, electrical, and so on), then $\mathbf{F}$ represents a force.[5] Consider the example of a particle of mass $m$ in the field of the earth; in this case, one takes $f$ to be $GmM/r$ or $V = -GmM/r$, where $G$ is the gravitational constant, $M$ is the mass of the earth, and $r$ is the distance from the center of the

[5] If the minus sign is used, then $V$ is *decreasing* in the direction $\mathbf{F}$.

earth. The corresponding force is $\mathbf{F} = -(GmM/r^3)\mathbf{r} = -(GmM/r^2)\mathbf{n}$, where $\mathbf{n}$ is the unit radial vector. Note that $\mathbf{F}$ fails to be defined at the point $r = 0$.

## When are Vector Fields Gradients?

We wish to characterize those vector fields that can be written as a gradient. Our task is simplified considerably by Stokes' theorem.

**THEOREM 7: Conservative Fields** Let $\mathbf{F}$ be a $C^1$ vector field defined on $\mathbb{R}^3$ except possibly for a finite number of points. The following conditions on $\mathbf{F}$ are all equivalent:

(i) For any oriented simple closed curve $C$, $\int_C \mathbf{F} \cdot d\mathbf{s} = 0$.

(ii) For any two oriented simple curves $C_1$ and $C_2$ that have the same endpoints,

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{s} = \int_{C_2} \mathbf{F} \cdot d\mathbf{s}.$$

(iii) $\mathbf{F}$ is the gradient of some function $f$; that is, $\mathbf{F} = \nabla f$ (and if $\mathbf{F}$ has one or more exceptional points where it fails to be defined, $f$ is also undefined there).

(iv) $\nabla \times \mathbf{F} = \mathbf{0}$.

A vector field satisfying one (and, hence, all) of the conditions (i)–(iv) is called a ***conservative vector field***.[6]

*PROOF* We shall establish the following chain of implications, which will prove the theorem:

$$\text{(i)} \Rightarrow \text{(ii)} \Rightarrow \text{(iii)} \Rightarrow \text{(iv)} \Rightarrow \text{(i)}.$$

First we show that condition (i) implies condition (ii). Suppose $\mathbf{c}_1$ and $\mathbf{c}_2$ are parametrizations representing $C_1$ and $C_2$, with the same endpoints. Construct the closed curve $\mathbf{c}$ obtained by first traversing $\mathbf{c}_1$ and then $-\mathbf{c}_2$ (Figure 8.3.1), or, symbolically, the curve $\mathbf{c} = \mathbf{c}_1 - \mathbf{c}_2$. Assuming $\mathbf{c}$ is simple, condition (i) gives

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}_1} \mathbf{F} \cdot d\mathbf{s} - \int_{\mathbf{c}_2} \mathbf{F} \cdot d\mathbf{s} = 0,$$

and so condition (ii) holds. (If $\mathbf{c}$ is not simple, an additional argument, omitted here, is needed.)

---

[6] In the plane $\mathbb{R}^2$, exceptional points are not allowed (see Exercise 12). Theorem 7 can be proved in the same way if $\mathbf{F}$ is defined and is of class $C^1$ only on an open convex set in $\mathbb{R}^2$ or $\mathbb{R}^3$. (A set $D$ is convex if $P, Q \in D$ implies the line joining P and Q also belongs to $D$.)

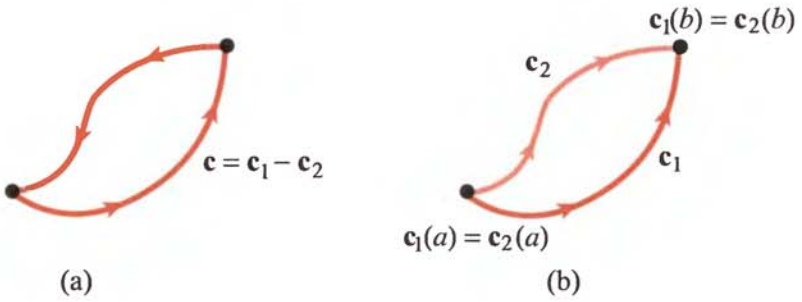



**Figure 8.3.1** Constructing (a) an oriented simple closed curve $\mathbf{c}_1 - \mathbf{c}_2$ from (b) two oriented simple curves.

Next we prove that condition (ii) implies condition (iii). Let $C$ be any oriented simple curve joining a point such as $(0, 0, 0)$ to $(x, y, z)$, and suppose $C$ is represented by the parametrization $\mathbf{c}$ [if $(0, 0, 0)$ is the exceptional point of $\mathbf{F}$, we can choose a different starting point for $\mathbf{c}$ without affecting the argument]. Define $f(x, y, z)$ to be $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$. By hypothesis (ii), $f(x, y, z)$ is independent of $C$. We shall show that $\mathbf{F} = \text{grad } f$. Indeed, choose $\mathbf{c}$ to be the path shown in Figure 8.3.2, so that

$$f(x, y, z) = \int_0^x F_1(t, 0, 0)\, dt + \int_0^y F_2(x, t, 0)\, dt + \int_0^z F_3(x, y, t)\, dt,$$

where $\mathbf{F} = (F_1, F_2, F_3)$.

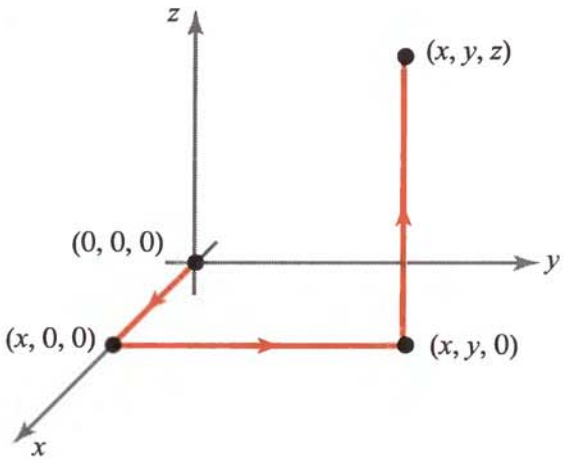



**Figure 8.3.2** A path joining $(0, 0, 0)$ to $(x, y, z)$.

It follows from the fundamental theorem of calculus that $\partial f/\partial z = F_3$. We can repeat this processing using two other paths from $(0, 0, 0)$ to $(x, y, z)$ [for example, by drawing the lines from $(0, 0, 0)$ to $(0, y, 0)$ to $(x, y, 0)$ to $(x, y, z)$], and we can similarly show that $\partial f/\partial x = F_1$ and $\partial f/\partial y = F_2$ (see Exercise 22). Thus, $\nabla f = \mathbf{F}$.

Third, condition (iii) implies condition (iv), because, as proved in Section 4.4,

$$\nabla \times \nabla f = \mathbf{0}.$$

Finally, let $\mathbf{c}$ represent a closed curve $C$ and let $S$ be any surface whose boundary is $\mathbf{c}$ (if $\mathbf{F}$ has exceptional points, choose $S$ to avoid them). Figure 8.3.3 indicates that we can probably always find such a surface; however, a formal proof of this would require the development of more sophisticated mathematical ideas than we can present here.

By Stokes' theorem,

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n}\, dS.$$

Because $\nabla \times \mathbf{F} = \mathbf{0}$, this integral vanishes, so that condition (iv) $\Rightarrow$ condition (i). ■

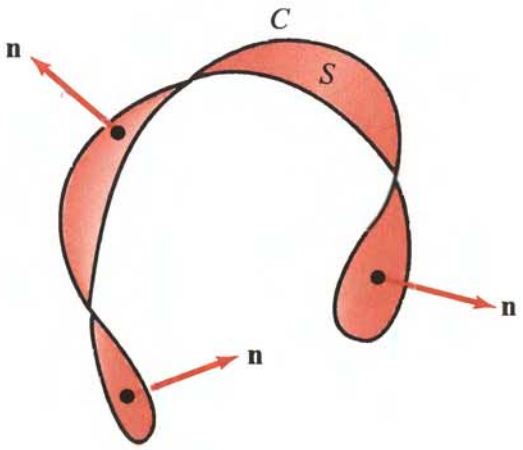



**Figure 8.3.3** A surface $S$ spanning a curve $C$.

## Physical Interpretations of $\int_C \mathbf{F} \cdot d\mathbf{s}$

We have already seen that one interpretation of the line integral is as the work done by $\mathbf{F}$ in moving a particle along $C$. A second interpretation is the notion of circulation, which we encountered at the end of the last section. Recall that in this case, we think of $\mathbf{F}$ as the velocity field of a fluid; that is, to each point P in space, $\mathbf{F}$ assigns the velocity vector of the fluid at P (in the last section, $\mathbf{F}$ was designated $\mathbf{V}$). Take $C$ to be a closed curve, and let $\Delta\mathbf{s}$ be a small directed chord of $C$. Then $\mathbf{F} \cdot \Delta\mathbf{s}$ is approximately the tangential component of $\mathbf{F}$ times $\|\Delta\mathbf{s}\|$. The circulation $\int_C \mathbf{F} \cdot d\mathbf{s}$ is the net tangential component around $C$. A small paddle wheel placed in the fluid would rotate if it is centered at a point where $\mathbf{F}$ vanishes and if the circulation of the fluid is nonzero, or $\int_C \mathbf{F} \cdot d\mathbf{s} \neq 0$ for small loops $C$ (see Figure 8.3.4).

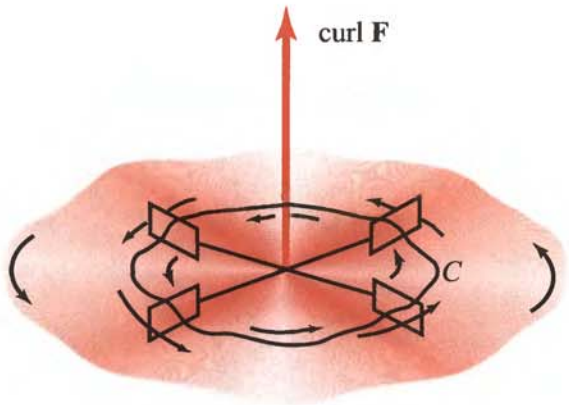



**Figure 8.3.4** $\int_C \mathbf{F} \cdot d\mathbf{s} \neq 0$ implies that a paddle wheel in a fluid with velocity field $\mathbf{F}$ will rotate around its axis.

There is a similar interpretation in electromagnetic theory: If $\mathbf{F}$ represents an electric field, then a current will flow around a loop $C$ if $\int_C \mathbf{F} \cdot d\mathbf{s} \neq 0$.

By Theorem 7, a field $\mathbf{F}$ has no circulation if and only if curl $\mathbf{F} = \nabla \times \mathbf{F} = \mathbf{0}$. Hence, a vector field $\mathbf{F}$ with curl $\mathbf{F} = \mathbf{0}$ is called ***irrotational***. We have therefore proved that *a vector field in $\mathbb{R}^3$ is irrotational if and only if it is a gradient field for some function, that is, if and only if* $\mathbf{F} = \nabla f$. The function $f$ is called a ***potential*** for $\mathbf{F}$.

**EXAMPLE 1** Consider the vector field $\mathbf{F}$ on $\mathbb{R}^3$ defined by

$$\mathbf{F}(x, y, z) = y\mathbf{i} + (z \cos yz + x)\mathbf{j} + (y \cos yz)\mathbf{k}.$$

Show that $\mathbf{F}$ is irrotational and find a scalar potential for $\mathbf{F}$.

SOLUTION We compute $\nabla \times \mathbf{F}$:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ y & x + z\cos yz & y\ \cos yz \end{vmatrix}$$

$$= (\cos yz - yz \sin yz - \cos yz + yz \sin yz)\mathbf{i} + (0 - 0)\mathbf{j} + (1 - 1)\mathbf{k}$$

$$= 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = \mathbf{0},$$

so $\mathbf{F}$ is irrotational. Thus, a potential exists by Theorem 7. We can find it in several ways.

*Method 1.* By the technique used to prove that condition (ii) implies condition (iii) in Theorem 7, we can set

$$f(x, y, z) = \int_0^x F_1(t, 0, 0)\, dt + \int_0^y F_2(x, t, 0)\, dt + \int_0^z F_3(x, y, t)\, dt$$

$$= \int_0^x 0\, dt + \int_0^y x\, dt + \int_0^z y \cos yt\, dt$$

$$= 0 + xy + \sin yz = xy + \sin yz.$$

One easily verifies that $\nabla f = \mathbf{F}$, as required:

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k} = y\mathbf{i} + (x + z\cos yz)\mathbf{j} + (y \cos yz)\mathbf{k}.$$

*Method 2.* Because we know that $f$ exists, we know that we can solve the system of equations

$$\frac{\partial f}{\partial x} = y, \qquad \frac{\partial f}{\partial y} = x + z\cos yz, \qquad \frac{\partial f}{\partial z} = y \cos yz,$$

for $f(x, y, z)$. These are equivalent to the simultaneous equations

(a) $f(x, y, z) = xy + h_1(y, z)$

(b) $f(x, y, z) = \sin yz + xy + h_2(x, z)$

(c) $f(x, y, z) = \sin yz + h_3(x, y)$

for functions $h_1, h_2, h_3$ independent of $x$, $y$, and $z$ (respectively). When $h_1(y, z) = \sin yz$, $h_2(x, z) = 0$, and $h_3(x, y) = xy$, the three equations agree and so yield a potential for $\mathbf{F}$. However, we have only guessed at the values of $h_1$, $h_2$, and $h_3$. To derive the formula for $f$ more systematically, we note that because $f(x, y, z) = xy + h_1(y, z)$ and $\partial f/\partial z = y \cos yz$, we find that

$$\frac{\partial h_1(y, z)}{\partial z} = y \cos yz$$

or

$$h_1(y, z) = \int y \cos yz \, dz + g(y) = \sin yz + g(y).$$

Therefore, substituting this back into equation (a), we get

$$f(x, y, z) = xy + \sin yz + g(y);$$

but by equation (b),

$$g(y) = h_2(x, z).$$

Because the right side of this equation is a function of $x$ and $z$ and the left side is a function of $y$ alone, we conclude that they must equal some constant $C$. Thus,

$$f(x, y, z) = xy + \sin yz + C$$

and we have determined $f$ up to a constant. ▲

**EXAMPLE 2** A mass $M$ at the origin in $\mathbb{R}^3$ exerts a force on a mass $m$ located at $\mathbf{r} = (x, y, z)$ with magnitude $GmM/r^2$ and directed toward the origin. Here, $G$ is the gravitational constant, which depends on the units of measurement, and $r = \|\mathbf{r}\| = \sqrt{x^2 + y^2 + z^2}$. If we remember that $-\mathbf{r}/r$ is a unit vector directed toward the origin, then we can write the force field as

$$\mathbf{F}(x, y, z) = -\frac{GmM\mathbf{r}}{r^3}.$$

Show that **F** is irrotational and find a scalar potential for **F**. (Notice that **F** is not defined at the origin, but Theorem 7 still applies, because it allows an exceptional point.)

SOLUTION First let us verify that $\nabla \times \mathbf{F} = \mathbf{0}$. Referring to formula 10 in the table of vector identities in Section 4.4, we get

$$\nabla \times \mathbf{F} = -GmM\left[\nabla\left(\frac{1}{r^3}\right) \times \mathbf{r} + \frac{1}{r^3}\nabla \times \mathbf{r}\right].$$

But $\nabla(1/r^3) = -3\mathbf{r}/r^5$ (see Exercise 30, Section 4.4), and so the first term vanishes, because $\mathbf{r} \times \mathbf{r} = \mathbf{0}$. The second term vanishes, because

$$\nabla \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z}\right)\mathbf{i} + \left(\frac{\partial x}{\partial z} - \frac{\partial z}{\partial x}\right)\mathbf{j} + \left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y}\right)\mathbf{k} = \mathbf{0}.$$

Hence, $\nabla \times \mathbf{F} = 0$ (for $\mathbf{r} \neq 0$).

If we recall the formula $\nabla(r^n) = nr^{n-2}\mathbf{r}$ (again, see Exercise 30, Section 4.4), then we can read off a scalar potential for **F** by inspection. We have $\mathbf{F} = -\nabla V$, where $V(x, y, z) = -GmM/r$ is called the ***gravitational potential energy***.

[We observe in passing that by Theorem 3 of Section 7.2, the work done by **F** in moving a particle of mass $m$ from a point $\mathrm{P}_1$ to a point $\mathrm{P}_2$ is given by

$$V(\mathrm{P}_1) - V(\mathrm{P}_2) = GmM\left(\frac{1}{r_2} - \frac{1}{r_1}\right),$$

where $r_1$ is the radial distance of $\mathrm{P}_1$ from the origin, with $r_2$ similarly defined.] ▲

## The Planar Case

By the same proof, Theorem 7 is also true for $C^1$ vector fields **F** on $\mathbb{R}^2$. In this case, we require that **F** has *no* exceptional points; that is, **F** is smooth everywhere (see Exercise 12). Notice, however, that the conclusion *might* still hold even if there are exceptional points, an example being $(x\mathbf{i} + y\mathbf{j})/(x^2 + y^2)^{3/2}$. An example where the conclusion does *not* hold is $(-y\mathbf{i} + x\mathbf{j})/(x^2 + y^2)$, as shown in Exercise 12.

If $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$, then

$$\nabla \times \mathbf{F} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}.$$

Sometimes $\partial Q/\partial x - \partial P/\partial y$ is called the ***scalar curl*** of **F**. Therefore, the condition $\nabla \times \mathbf{F} = 0$ reduces to

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Thus, we have:

COROLLARY 1 If $\mathbf{F}$ is a $C^1$ vector field on $\mathbb{R}^2$ of the form $P\mathbf{i} + Q\mathbf{j}$ that satisfies $\partial P/\partial y = \partial Q/\partial x$, then $\mathbf{F} = \nabla f$ for some $f$ on $\mathbb{R}^2$.

We emphasize again that this corollary can be false if $\mathbf{F}$ fails to be of class $C^1$ at even a single point (an example is given in Exercise 12). In $\mathbb{R}^3$, however, as already noted, exceptions at single points are allowed (see Theorem 7).

**EXAMPLE 3** (a) Determine whether the vector field

$$\mathbf{F} = e^{xy}\mathbf{i} + e^{x+y}\mathbf{j}$$

is a gradient field.

(b) Repeat part (a) for

$$\mathbf{F} = (2x\cos y)\mathbf{i} - (x^2 \sin y)\mathbf{j}.$$

SOLUTION (a) Here $P(x, y) = e^{xy}$ and $Q(x, y) = e^{x+y}$, and so we compute

$$\frac{\partial P}{\partial y} = xe^{xy}, \qquad \frac{\partial Q}{\partial x} = e^{x+y}.$$

These are not equal, and so $\mathbf{F}$ cannot have a potential function.

(b) In this case, we find

$$\frac{\partial P}{\partial y} = -2x \sin y = \frac{\partial Q}{\partial x},$$

and so $\mathbf{F}$ has a potential function $f$. To compute $f$ we solve the equations

$$\frac{\partial f}{\partial x} = 2x\cos y, \qquad \frac{\partial f}{\partial y} = -x^2 \sin y.$$

Thus, $f(x, y) = x^2 \cos y + h_1(y)$ and $f(x, y) = x^2 \cos y + h_2(x)$. If $h_1$ and $h_2$ are the same constant, then both equations are satisfied, and so $f(x, y) = x^2 \cos y$ is a potential for $\mathbf{F}$. ▲

**EXAMPLE 4** Let $\mathbf{c}$: $[1, 2] \to \mathbb{R}^2$ be given by $x = e^{t-1}$, $y = \sin(\pi/t)$. Compute the integral

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} 2x \cos y \, dx - x^2 \sin y \, dy,$$

where $\mathbf{F} = (2x \cos y)\mathbf{i} - (x^2 \sin y)\mathbf{j}$.

SOLUTION The endpoints are $\mathbf{c}(1) = (1, 0)$ and $\mathbf{c}(2) = (e, 1)$. Because $\partial(2x \cos y)/\partial y = \partial(-x^2 \sin y)/\partial x$, $\mathbf{F}$ is irrotational and hence a gradient vector field (as we saw in Example 3). Thus, by Theorem 7, we can replace $\mathbf{c}$ by any piecewise $C^1$ curve having the same endpoints, in particular, by the polygonal path from $(1, 0)$ to $(e, 0)$ to $(e, 1)$. Thus, the line integral must be equal to

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_1^e 2t \cos 0 \, dt + \int_0^1 -e^2 \sin t \, dt = (e^2 - 1) + e^2(\cos 1 - 1)$$
$$= e^2 \cos 1 - 1.$$

Alternatively, using Theorem 3 of Section 7.2, we have

$$\int_{\mathbf{c}} 2x \cos y \, dx - x^2 \sin y \, dy = \int_{\mathbf{c}} \nabla f \cdot d\mathbf{s} = f(\mathbf{c}(2)) - f(\mathbf{c}(1)) = e^2 \cos 1 - 1,$$

because $f(x, y) = x^2 \cos y$ is a potential function for $\mathbf{F}$. Evidently, this technique is simpler than computing the integral directly. ▲

We conclude this section with a theorem that is quite similar in spirit to Theorem 7. Theorem 7 was motivated partly as a converse to the result that curl $\nabla f = \mathbf{0}$ for any $C^1$ function $f\colon \mathbb{R}^3 \to \mathbb{R}$—or, if curl $\mathbf{F} = 0$, then $\mathbf{F} = \nabla f$. We also know [formula (9) in the table of vector identities in Section 4.4] that div (curl $\mathbf{G}$) $= 0$ for any $C^2$ vector field $\mathbf{G}$. We may ask about the converse statement: If div $\mathbf{F} = 0$, is $\mathbf{F}$ the curl of a vector field $\mathbf{G}$? The following theorem answers this in the affirmative.

**THEOREM 8** If $\mathbf{F}$ is a $C^1$ vector field on all of $\mathbb{R}^3$ with div $\mathbf{F} = 0$, then there exists a $C^1$ vector field $\mathbf{G}$ with $\mathbf{F} =$ curl $\mathbf{G}$.

The proof is outlined in Exercise 16. We should warn the reader at this point that, unlike the $\mathbf{F}$ in Theorem 7, the vector field $\mathbf{F}$ in Theorem 8 is not allowed to have an exceptional point. For example, the gravitational force field $\mathbf{F} = -(GmM\mathbf{r}/r^3)$ has the property that div $\mathbf{F} = 0$, and yet there is no $\mathbf{G}$ for which $\mathbf{F} =$ curl $\mathbf{G}$ (see Exercise 25). Theorem 8 does not apply, because the gravitational force field $\mathbf{F}$ is not defined at $\mathbf{0} \in \mathbb{R}^3$.

## EXERCISES

**1.** Show that any two potential functions for a vector field on $\mathbb{R}^3$ differ at most by a constant.

**2.** (a) Let $\mathbf{F}(x, y) = (xy, y^2)$ and let $\mathbf{c}$ be the path $y = 2x^2$ joining $(0, 0)$ to $(1, 2)$ in $\mathbb{R}^2$. Evaluate $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$.

(b) Does the integral in part (a) depend on the path joining $(0, 0)$ to $(1, 2)$?

**3.** Let $\mathbf{F}(x, y, z) = (2xyz + \sin x)\mathbf{i} + x^2 z\mathbf{j} + x^2 y\mathbf{k}$. Find a function $f$ such that $\mathbf{F} = \nabla f$.

**4.** Evaluate $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{c}(t) = (\cos^5 t, \sin^3 t, t^4)$, $0 \le t \le \pi$, and $\mathbf{F}$ is as in Exercise 3.

**5.** If $f(x)$ is a smooth function of one variable, must $\mathbf{F}(x, y) = f(x)\mathbf{i} + f(y)\mathbf{j}$ be a gradient?

**6.** (a) Show that $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$ is the gradient of $f(x, y, z) = 1/r$.
(b) What is the work done by the force $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$ in moving a particle from a point $\mathbf{r}_0 \in \mathbb{R}^3$ "to $\infty$," where $\mathbf{r}(x, y, z) = (x, y, z)$?

**7.** Let $\mathbf{F}(x, y, z) = xy\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Can there exist a function $f$ such that $\mathbf{F} = \nabla f$?

**8.** Let $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ and suppose each $F_k$ satisfies the homogeneity condition

$$F_k(tx, ty, tz) = tF_k(x, y, z), \qquad k = 1, 2, 3.$$

Suppose also $\nabla \times \mathbf{F} = \mathbf{0}$. Prove that $\mathbf{F} = \nabla f$, where

$$2f(x, y, z) = xF_1(x, y, z) + yF_2(x, y, z) + zF_3(x, y, z).$$

(HINT: Use Review Exercise 23, Chapter 2.)

**9.** Let $\mathbf{F}(x, y, z) = (e^x \sin y)\mathbf{i} + (e^x \cos y)\mathbf{j} + z^2\mathbf{k}$. Evaluate the integral $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{c}(t) = (\sqrt{t}, t^3, \exp\sqrt{t})$, $0 \le t \le 1$.

**10.** Let a fluid have the velocity field $\mathbf{F}(x, y, z) = xy\mathbf{i} + yz\mathbf{j} + xz\mathbf{k}$. What is the circulation around the unit circle in the $xy$ plane? Interpret your answer.

**11.** The mass of the earth is approximately $6 \times 10^{27}$ g and that of the sun is 330,000 times as much. The gravitational constant is $6.7 \times 10^{-8}$ cm$^3$/s$^2 \cdot$ g. The distance of the earth from the sun is about $1.5 \times 10^{12}$ cm. Compute, approximately, the work necessary to increase the distance of the earth from the sun by 1 cm.

**12.** (a) Show that $\int_C (x\,dy - y\,dx)/(x^2 + y^2) = 2\pi$, where $C$ is the unit circle.
(b) Conclude that the associated vector field $[-y/(x^2 + y^2)]\mathbf{i} + [x/(x^2 + y^2)]\mathbf{j}$ is not a conservative field.
(c) Show, however, that $\partial P/\partial y = \partial Q/\partial x$. Does this contradict the corollary to Theorem 7? If not, why not?

**13.** Determine which of the following vector fields $\mathbf{F}$ in the plane is the gradient of a scalar function $f$. If such an $f$ exists, find it.

(a) $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$
(b) $\mathbf{F}(x, y) = xy\mathbf{i} + xy\mathbf{j}$
(c) $\mathbf{F}(x, y) = (x^2 + y^2)\mathbf{i} + 2xy\mathbf{j}$

**14.** Repeat Exercise 13 for the following vector fields:

(a) $\mathbf{F}(x, y) = (\cos xy - xy \sin xy)\mathbf{i} - (x^2 \sin xy)\mathbf{j}$
(b) $\mathbf{F}(x, y) = (x\sqrt{x^2y^2 + 1})\mathbf{i} + (y\sqrt{x^2y^2 + 1})\mathbf{j}$
(c) $\mathbf{F}(x, y) = (2x \cos y + \cos y)\mathbf{i} - (x^2 \sin y + x \sin y)\mathbf{j}$

**15.** Show that the following vector fields are conservative. Calculate $\int_C \mathbf{F} \cdot d\mathbf{s}$ for the given curve.

(a) $\mathbf{F} = (xy^2 + 3x^2y)\mathbf{i} + (x + y)x^2\mathbf{j}$; $C$ is the curve consisting of line segments from $(1, 1)$ to $(0, 2)$ to $(3, 0)$.

(b) $\mathbf{F} = \dfrac{2x}{y^2 + 1}\mathbf{i} - \dfrac{2y(x^2 + 1)}{(y^2 + 1)^2}\mathbf{j}$; $C$ is parametrized by $x = t^3 - 1$, $y = t^6 - t$, $0 \le t \le 1$.

(c) $\mathbf{F} = [\cos(xy^2) - xy^2 \sin(xy^2)]\mathbf{i} - 2x^2y \sin(xy^2)\mathbf{j}$; $C$ is the curve $(e^t, e^{t+1})$, $-1 \le t \le 0$.

**16.** Prove Theorem 8. [HINT: Define $\mathbf{G} = G_1\mathbf{i} + G_2\mathbf{j} + G_3\mathbf{k}$ by

$$G_1(x, y, z) = \int_0^z F_2(x, y, t)\, dt - \int_0^y F_3(x, t, 0)\, dt$$

$$G_2(x, y, z) = -\int_0^z F_1(x, y, t)\, dt$$

and $G_3(x, y, z) = 0$.]

**17.** Is each of the following vector fields the curl of some other vector field? If so, find the vector field.

(a) $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

(b) $\mathbf{F} = (x^2 + 1)\mathbf{i} + (z - 2xy)\mathbf{j} + y\mathbf{k}$

**18.** Let $\mathbf{F} = xz\mathbf{i} - yz\mathbf{j} + y\mathbf{k}$. Verify that $\nabla \cdot \mathbf{F} = 0$. Find a $\mathbf{G}$ such that $\mathbf{F} = \nabla \times \mathbf{G}$.

**19.** Repeat Exercise 18 for $\mathbf{F} = y^2\mathbf{i} + z^2\mathbf{j} + x^2\mathbf{k}$.

**20.** Let $\mathbf{F} = xe^y\mathbf{i} - (x \cos z)\mathbf{j} - ze^y\mathbf{k}$. Find a $\mathbf{G}$ such that $\mathbf{F} = \nabla \times \mathbf{G}$.

**21.** Let $\mathbf{F} = (x \cos y)\mathbf{i} - (\sin y)\mathbf{j} + (\sin x)\mathbf{k}$. Find a $\mathbf{G}$ such that $\mathbf{F} = \nabla \times \mathbf{G}$.

**22.** By using different paths from $(0, 0, 0)$ to $(x, y, z)$, show that the function $f$ defined in the proof of Theorem 7 for "condition (ii) implies condition (iii)" satisfies $\partial f/\partial x = F_1$ and $\partial f/\partial y = F_2$.

**23.** Let $\mathbf{F}$ be the vector field on $\mathbb{R}^3$ given by $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$.

(a) Show that $\mathbf{F}$ is rotational, that is, $\mathbf{F}$ is not irrotational.

(b) Suppose $\mathbf{F}$ represents the velocity vector field of a fluid. Show that if we place a cork in this fluid it will revolve in a plane parallel to the $xy$ plane, in a circular trajectory about the $z$ axis.

(c) In what direction does the cork revolve?

**24.** Let $\mathbf{G}$ be the vector field on $\mathbb{R}^3 \setminus \{z \text{ axis}\}$ defined by

$$\mathbf{G} = \frac{-y}{x^2 + y^2}\mathbf{i} + \frac{x}{x^2 + y^2}\mathbf{j}.$$

(a) Show that **G** is irrotational.
(b) Show that the result of Exercise 23(b) holds for **G** also.
(c) How can we resolve the fact that the trajectories of **F** and **G** are both the same (circular about the $z$ axis) yet **F** is rotational and **G** is not? (HINT: The property of being rotational is a local condition, that is, a property of the fluid in the neighborhood of a point.)

**25.** Let $\mathbf{F} = -(GmM\mathbf{r}/r^3)$ be the gravitational force field defined on $\mathbb{R}^3\backslash\{\mathbf{0}\}$.

(a) Show that div $\mathbf{F} = 0$.
(b) Show that $\mathbf{F} \neq \text{curl } \mathbf{G}$ for any $C^1$ vector field **G** on $\mathbb{R}^3\backslash\{\mathbf{0}\}$.

# 8.4 Gauss' Theorem

Gauss' theorem states that the flux of a vector field out of a closed surface equals the integral of the divergence of that vector field over the volume enclosed by the surface. The result parallels Stokes' theorem and Green's theorem in that it relates an integral over a closed geometrical object (curve or surface) to an integral over a contained region (surface or volume).

## Elementary Regions and their Boundaries

We shall begin by asking the reader to review the various elementary regions in space that were introduced when we considered the volume integral; these regions are illustrated in Figures 5.5.2 and 5.5.4. As these figures indicate, the boundary of an elementary region in $\mathbb{R}^3$ is a surface made up of a finite number (at most six, at least two) of surfaces that can be described as graphs of functions from $\mathbb{R}^2$ to $\mathbb{R}$. This kind of surface is called a *closed surface.* The surfaces $S_1, S_2, \ldots, S_N$ composing such a closed surface are called its *faces.*

**EXAMPLE 1** The cube in Figure 8.4.1(a) is an elementary region, and in fact a symmetric elementary region, with six rectangles composing its boundary. The sphere in Figure 8.4.1(b) is the boundary of a solid ball, which is also a symmetric elementary region. ▲

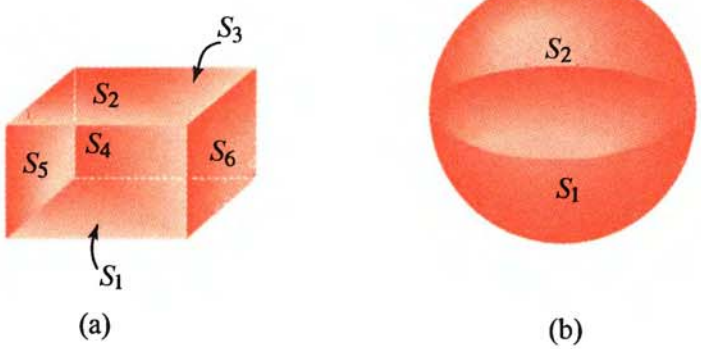



**Figure 8.4.1** (a) Symmetric elementary regions and (b) the surface $S_i$ composing their boundaries.

Closed surfaces can be oriented in two ways. The outward orientation makes the normal point outward into space, and the inward orientation makes the normal point into the bounded region (Figure 8.4.2).

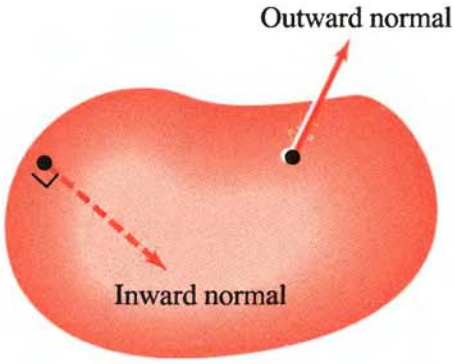



**Figure 8.4.2** Two possible orientations for a closed surface.

Suppose $S$ is a closed surface oriented in one of these two ways and $\mathbf{F}$ is a vector field on $S$. Then, as we defined it in Section 7.6,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \sum_i \iint_{S_i} \mathbf{F} \cdot d\mathbf{S}.$$

If $S$ is given the outward orientation, the integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ measures the total flux of $\mathbf{F}$ outward across $S$. That is, if we think of $\mathbf{F}$ as the velocity field of a fluid, $\iint_S \mathbf{F} \cdot d\mathbf{S}$ indicates the amount of fluid leaving the region bounded by $S$ per unit time. If $S$ is given the inward orientation, the integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ measures the total flux of $\mathbf{F}$ inward across $S$.

We recall another common way of writing these surface integrals, a way that explicitly specifies the orientation of $S$. Let the orientation of $S$ be given by a unit normal vector $\mathbf{n}(x, y, z)$ at each point of $S$. Then we have the oriented integral

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n})\, dS,$$

that is, the integral of the normal component of $\mathbf{F}$ over $S$. In the remainder of this section, if $S$ is a closed surface enclosing a region $W$, we adopt the default convention that $S = \partial W$ is given the outward orientation, with outward unit normal $\mathbf{n}(x, y, z)$ at each point $(x, y, z) \in S$. Furthermore, we denote the surface with the opposite (inward) orientation by $\partial W_{\text{op}}$. Then the associated unit normal direction for this orientation is $-\mathbf{n}$. Thus,

$$\iint_{\partial W} \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n})\, dS = -\iint_S [\mathbf{F} \cdot (-\mathbf{n})]\, dS = -\iint_{\partial W_{\text{op}}} \mathbf{F} \cdot d\mathbf{S}.$$

**EXAMPLE 2** The unit cube $W$ given by

$$0 \le x \le 1, \qquad 0 \le y \le 1, \qquad 0 \le z \le 1$$

is a symmetric elementary region in space (see Figures 8.4.3 and 5.5.5).

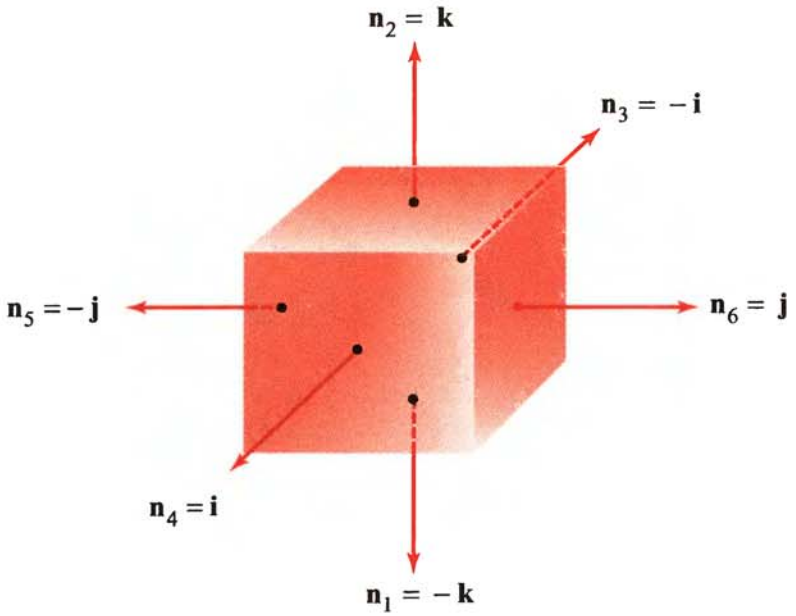



**Figure 8.4.3** The outward orientation on the cube.

We write the faces as

$$\begin{aligned}
S_1&: z = 0, \quad 0 \le x \le 1, \quad 0 \le y \le 1\\
S_2&: z = 1, \quad 0 \le x \le 1, \quad 0 \le y \le 1\\
S_3&: x = 0, \quad 0 \le y \le 1, \quad 0 \le z \le 1\\
S_4&: x = 1, \quad 0 \le y \le 1, \quad 0 \le z \le 1\\
S_5&: y = 0, \quad 0 \le x \le 1, \quad 0 \le z \le 1\\
S_6&: y = 1, \quad 0 \le x \le 1, \quad 0 \le z \le 1.
\end{aligned}$$

From Figure 8.4.3 we see that

$$\begin{aligned}
\mathbf{n}_2 &= \mathbf{k} = -\mathbf{n}_1,\\
\mathbf{n}_4 &= \mathbf{i} = -\mathbf{n}_3,\\
\mathbf{n}_6 &= \mathbf{j} = -\mathbf{n}_5,
\end{aligned}$$

and so for a continuous vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$,

$$\begin{aligned}
\iint_{\partial W} \mathbf{F}\cdot d\mathbf{S} = \iint_S \mathbf{F}\cdot\mathbf{n}\,dS &= -\iint_{S_1} F_3\,dS + \iint_{S_2} F_3\,dS - \iint_{S_3} F_1\,dS\\
&\quad + \iint_{S_4} F_1\,dS - \iint_{S_5} F_2\,dS + \iint_{S_6} F_2\,dS. \quad \blacktriangle
\end{aligned}$$

## Gauss' Theorem

We have now come to the last of the three central theorems of this chapter. This theorem relates surface integrals to volume integrals; in other words, the theorem states that if $W$ is a region in $\mathbb{R}^3$, then the flux of a vector field $\mathbf{F}$ outward across the closed surface $\partial W$ is equal to the integral of div $\mathbf{F}$ over $W$. We begin by assuming that $W$ is a symmetric elementary region (Figure 5.5.5).

**THEOREM 9: Gauss' Divergence Theorem** Let $W$ be a symmetric elementary region in space. Denote by $\partial W$ the oriented closed surface that bounds $W$. Let $\mathbf{F}$ be a smooth vector field defined on $W$. Then

$$\iiint_W (\nabla \cdot \mathbf{F})\, dV = \iint_{\partial W} \mathbf{F} \cdot d\mathbf{S}$$

or, alternatively,

$$\iiint_W (\text{div } \mathbf{F})\, dV = \iint_{\partial W} (\mathbf{F} \cdot \mathbf{n})\, dS.$$

*PROOF* If $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$, then by definition, the divergence of $\mathbf{F}$ is given by div $\mathbf{F} = \partial P/\partial x + \partial Q/\partial y + \partial R/\partial z$, so we can write (using additivity of the volume integral)

$$\iiint_W \text{div } \mathbf{F}\, dV = \iiint_W \frac{\partial P}{\partial x}\, dV + \iiint_W \frac{\partial Q}{\partial y}\, dV + \iiint_W \frac{\partial R}{\partial z}\, dV.$$

On the other hand, the surface integral in question is

$$\begin{aligned}\iint_{\partial W} \mathbf{F} \cdot \mathbf{n}\, dS &= \iint_{\partial W} (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) \cdot \mathbf{n}\, dS \\ &= \iint_{\partial W} P\mathbf{i} \cdot \mathbf{n}\, dS + \iint_{\partial W} Q\mathbf{j} \cdot \mathbf{n}\, dS + \iint_{\partial W} R\mathbf{k} \cdot \mathbf{n}\, dS.\end{aligned}$$

The theorem will follow if we establish the three equalities

$$\iint_{\partial W} P\mathbf{i} \cdot \mathbf{n}\, dS = \iiint_W \frac{\partial P}{\partial x}\, dV, \tag{1}$$

$$\iint_{\partial W} Q\mathbf{j} \cdot \mathbf{n}\, dS = \iiint_W \frac{\partial Q}{\partial y}\, dV, \tag{2}$$

and

$$\iint_{\partial W} R\mathbf{k}\cdot\mathbf{n}\,dS = \iiint_W \frac{\partial R}{\partial z}\,dV. \tag{3}$$

We shall prove equation (3); the other two equalities can be proved in an analogous fashion.

Because $W$ is a symmetric elementary region, there is a pair of functions

$$z = g_1(x, y), \qquad z = g_2(x, y),$$

with common domain an elementary region $D$ in the $xy$ plane, such that $W$ is the set of all points $(x, y, z)$ satisfying

$$g_1(x, y) \le z \le g_2(x, y), \qquad (x, y) \in D.$$

By reduction to iterated integrals, we have

$$\iiint_W \frac{\partial R}{\partial z}\,dV = \iint_D \left( \int_{z=g_1(x,y)}^{z=g_2(x,y)} \frac{\partial R}{\partial z}\,dz \right) dx\,dy,$$

and so, by the fundamental theorem of calculus,

$$\iiint_W \frac{\partial R}{\partial z}\,dV = \iint_D [R(x, y, g_2(x, y)) - R(x, y, g_1(x, y))]\,dx\,dy. \tag{4}$$

The boundary of $W$ is a closed surface whose top $S_2$ is the graph of $z = g_2(x, y)$, where $(x, y) \in D$, and whose bottom $S_1$ is the graph of $z = g_1(x, y)$, $(x, y) \in D$. The four other sides of $\partial W$ consist of surfaces $S_3$, $S_4$, $S_5$, and $S_6$, whose normals are always perpendicular to the $z$ axis. (See Figure 8.4.4. Note that some of the other four sides

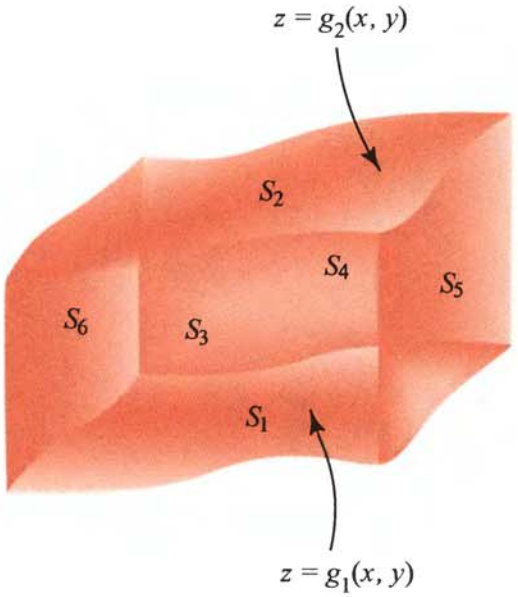



**Figure 8.4.4** A symmetric elementary region $W$ for which $\iint_{\partial W} R\mathbf{k}\cdot dS = \iiint_W (\partial R/\partial z)\,dV$. The four sides of $\partial W$, which are $S_3$, $S_4$, $S_5$, $S_6$ have normals perpendicular to the $z$ axis.

might be absent; for instance, if $W$ is a solid ball and $\partial W$ is a sphere.) By definition,

$$\iint_{\partial W} R\mathbf{k}\cdot\mathbf{n}\,dS = \iint_{S_1} R\mathbf{k}\cdot\mathbf{n}_1\,dS + \iint_{S_2} R\mathbf{k}\cdot\mathbf{n}_2\,dS + \sum_{i=3}^{6}\iint_{S_i} R\mathbf{k}\cdot\mathbf{n}_i\,dS.$$

Because the normal $\mathbf{n}_i$ is perpendicular to $\mathbf{k}$ on each of $S_3$, $S_4$, $S_5$, $S_6$, we have $\mathbf{k}\cdot\mathbf{n} = 0$ along these faces, and so the integral reduces to

$$\iint_{\partial W} R\mathbf{k}\cdot\mathbf{n}\,ds = \iint_{S_1} R\mathbf{k}\cdot d\mathbf{S}_1 + \iint_{S_2} R\mathbf{k}\cdot d\mathbf{S}_2. \tag{5}$$

The surface $S_1$ is defined by $z = g_1(x, y)$, and

$$d\mathbf{S}_1 = \left(\frac{\partial g_1}{\partial x}\mathbf{i} + \frac{\partial g_1}{\partial y}\mathbf{j} - \mathbf{k}\right)dx\,dy$$

(the negative of the general formula for $d\mathbf{S}$ for graphs from Section 7.6, because the normal is downward pointing). Therefore,

$$\iint_{S_1} R\mathbf{k}\cdot d\mathbf{S}_1 = -\iint_D R(x, y, g_1(x, y))\,dx\,dy. \tag{6}$$

Similarly, for the top face $S_2$,

$$d\mathbf{S}_2 = \left(-\frac{\partial g_2}{\partial x}\mathbf{i} - \frac{\partial g_2}{\partial y}\mathbf{j} + \mathbf{k}\right)dx\,dy.$$

Therefore,

$$\iint_{S_2} R\mathbf{k}\cdot d\mathbf{S}_2 = \iint_D R(x, y, g_2(x, y))\,dx\,dy. \tag{7}$$

Substituting equations (6) and (7) into equation (5) and then comparing with equation (4), we obtain

$$\iiint_W \frac{\partial R}{\partial z}\,dV = \iint_{\partial W} R(\mathbf{k}\cdot\mathbf{n})\,dS.$$

The remaining equalities, (1) and (2), can be established in the same way to complete the proof. ■

## Generalizing Gauss' Theorem

The reader should note that the proof of Gauss' theorem is similar to that of Green's theorem. By the procedure used in Exercise 8 of Section 8.1, we can extend Gauss' theorem to any region that can be broken up into symmetric elementary regions. This

includes all regions of interest to us. An example of a region to which Gauss' theorem applies is the region between two closed surfaces, one inside the other. The surface of this region consists of two pieces oriented as shown in Figure 8.4.5. We shall apply the divergence theorem to such a region when we prove Gauss' law in Theorem 10.

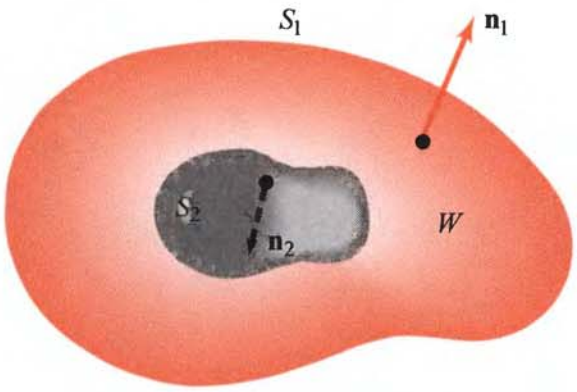



Figure 8.4.5 A more general region to which Gauss' theorem applies.

**EXAMPLE 3** Consider $\mathbf{F} = 2x\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$. Let $S$ be the unit sphere defined by $x^2 + y^2 + z^2 = 1$. Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$.

SOLUTION By Gauss' theorem,

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_W (\operatorname{div} \mathbf{F})\, dV,$$

where $W$ is the ball bounded by the sphere. The integral on the right is

$$2\iiint_W (1 + y + z)\, dV = 2\iiint_W dV + 2\iiint_W y\, dV + 2\iiint_W z\, dV.$$

By symmetry, we can argue that $\iiint_W y\, dV = \iiint_W z\, dV = 0$ (see Exercise 17, Section 6.3). Thus, because a sphere of radius $R$ has volume $4\pi R^3/3$,

$$2\iiint_W (1 + y + z)\, dV = 2\iiint_W dV = \frac{8\pi}{3}.$$

Readers can convince themselves that direct computation of $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$ proves unwieldy. ▲

**EXAMPLE 4** Use the divergence theorem to evaluate

$$\iint_{\partial W} (x^2 + y + z)\, dS,$$

where $W$ is the solid ball $x^2 + y^2 + z^2 \leq 1$.

SOLUTION To apply Gauss' divergence theorem, we find a vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ on $W$ with $\mathbf{F}\cdot\mathbf{n} = x^2 + y + z$. At any point $(x, y, z) \in \partial W$, the outward unit normal $\mathbf{n}$ to $\partial W$ is

$$\mathbf{n} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k},$$

because on $\partial W$, $x^2 + y^2 + z^2 = 1$ and the radius vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is normal to the sphere $\partial W$ (Figure 8.4.6).

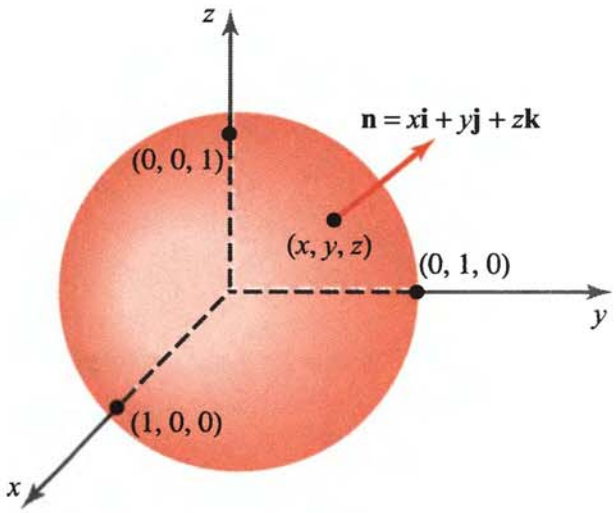



**Figure 8.4.6** $\mathbf{n}$ is the unit normal to $\partial W$, the boundary of the ball $W$.

Therefore, if $\mathbf{F}$ is the desired vector field, then

$$\mathbf{F}\cdot\mathbf{n} = F_1x + F_2y + F_2z.$$

We set

$$F_1x = x^2, \qquad F_2y = y, \qquad F_3z = z$$

and solve for $F_1$, $F_2$, and $F_3$ to find that $\mathbf{F} = x\mathbf{i} + \mathbf{j} + \mathbf{k}$. Computing div $\mathbf{F}$, we get

$$\operatorname{div}\mathbf{F} = 1 + 0 + 0 = 1.$$

Thus, by Gauss' divergence theorem,

$$\iint_{\partial W} (x^2 + y + z)\,dS = \iiint_W dV = \text{volume}\,(W) = \frac{4}{3}\pi. \quad \blacktriangle$$

## The Divergence as the Flux per Unit Volume

The physical meaning of divergence is that at a point P, div $\mathbf{F}$(P) is the rate of net outward flux at P per unit volume. This follows from Gauss' theorem and the mean-value theorem for integrals: If $W_\rho$ is a ball in $\mathbb{R}^3$ of radius $\rho$ centered at P, then there

is a point $Q \in W_\rho$ such that

$$\iint_{\partial W_\rho} \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_{W_\rho} \operatorname{div} \mathbf{F}\, dV = \operatorname{div} \mathbf{F}(Q) \cdot \text{volume}\,(W_\rho)$$

and so

$$\operatorname{div} \mathbf{F}(P) = \lim_{\rho \to 0} \operatorname{div} \mathbf{F}(Q) = \lim_{\rho \to 0} \frac{1}{V(W_\rho)} \iint_{\partial W_\rho} \mathbf{F} \cdot \mathbf{n}\, dS.$$

This is analogous to the limit formulation of the curl given at the end of Section 8.2. Thus, if $\operatorname{div} \mathbf{F}(P) > 0$, we consider P to be a ***source***, for there is a net outward flow near P. If $\operatorname{div} \mathbf{F}(P) < 0$, P is called a ***sink*** for $\mathbf{F}$.

A $C^1$ vector field $\mathbf{F}$ defined on $\mathbb{R}^3$ is said to be ***divergence free*** if $\operatorname{div} \mathbf{F} = 0$. If $\mathbf{F}$ is divergence-free, we have $\iint_S \mathbf{F} \cdot d\mathbf{S} = 0$ for all closed surfaces $S$. The converse can also be demonstrated readily using Gauss' theorem: If $\iint_S \mathbf{F} \cdot d\mathbf{S} = 0$ for all closed surfaces $S$, then $\mathbf{F}$ is divergence-free. If $\mathbf{F}$ is divergence-free, we thus see that the flux of $\mathbf{F}$ across any closed surface $S$ is 0, so that if $\mathbf{F}$ is the velocity field of a fluid, the net amount of fluid that flows out of any region will be 0. Thus, exactly as much fluid must flow into the region as flows out (in unit time). A fluid with this property is therefore described as ***incompressible***.

**EXAMPLE 5** Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = xy^2\mathbf{i} + x^2y\mathbf{j} + y\mathbf{k}$ and $S$ is the surface of the cylinder $x^2 + y^2 = 1$, bounded by the planes $z = 1$ and $z = -1$, and including the portions $x^2 + y^2 \leq 1$ when $z = \pm 1$.

SOLUTION One can compute this integral directly, but it is easier to use the divergence theorem.

Now $S$ is the boundary of the region $W$ given by $x^2 + y^2 \leq 1, -1 \leq z \leq 1$. Thus, $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_W (\operatorname{div} \mathbf{F})\, dV$. Moreover,

$$\begin{aligned}\iiint_W (\operatorname{div} \mathbf{F})\, dV = \iiint_W (x^2 + y^2)\, dx\, dy\, dz &= \int_{-1}^{1} \left( \iint_{x^2+y^2 \leq 1} (x^2 + y^2)\, dx\, dy \right) dz \\ &= 2 \iint_{x^2+y^2 \leq 1} (x^2 + y^2)\, dx\, dy.\end{aligned}$$

Before evaluating the double integral, we note that the surface integral satisfies

$$\iint_{\partial W} \mathbf{F} \cdot \mathbf{n}\, dS = 2 \iint_{x^2+y^2 \leq 1} (x^2 + y^2)\, dx\, dy > 0.$$

This means that $\iint_{\partial W} \mathbf{F} \cdot d\mathbf{S}$, the net flux of $\mathbf{F}$ out of the cylinder, is positive.

We change variables to polar coordinates to evaluate the double integral:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad 0 \le r \le 1, \qquad 0 \le \theta \le 2\pi.$$

Hence, we have $\partial(x, y)/\partial(r, \theta) = r$ and $x^2 + y^2 = r^2$. Thus,

$$\iint_{x^2+y^2\le 1} (x^2 + y^2)\,dx\,dy = \int_0^{2\pi} \left( \int_0^1 r^3\,dr \right) d\theta = \frac{1}{2}\pi.$$

Therefore, $\iiint_W \operatorname{div} \mathbf{F}\,dV = \pi$. ▲

## Gauss' Law

As we remarked earlier, Gauss' divergence theorem can be applied to regions in space more general than symmetric elementary regions. To conclude this section, we shall use this observation *in the proof* of the following important results.

**THEOREM 10: Gauss' Law** Let $M$ be a symmetric elementary region in $\mathbb{R}^3$. Then if $(0, 0, 0) \notin \partial M$, we have

$$\iint_{\partial M} \frac{\mathbf{r}\cdot\mathbf{n}}{r^3}\,dS = \begin{cases} 4\pi & \text{if } (0,0,0) \in M \\ 0 & \text{if } (0,0,0) \notin M, \end{cases}$$

where

$$\mathbf{r}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

and

$$r(x, y, z) = \|\mathbf{r}(x, y, z)\| = \sqrt{x^2 + y^2 + z^2}.$$

*PROOF OF GAUSS' LAW* First suppose $(0, 0, 0) \notin M$. Then $\mathbf{r}/r^3$ is a $C^1$ vector field on $M$ and $\partial M$, and so by the divergence theorem,

$$\iint_{\partial M} \frac{\mathbf{r}\cdot\mathbf{n}}{r^3}\,dS = \iiint_M \nabla\cdot\left(\frac{\mathbf{r}}{r^3}\right) dV.$$

But $\nabla\cdot(\mathbf{r}/r^3) = 0$ for $r \neq 0$, as the reader can easily verify (see Exercise 30, Section 4.4). Thus,

$$\iint_{\partial M} \frac{\mathbf{r}\cdot\mathbf{n}}{r^3}\,dS = 0.$$

Now let us suppose $(0, 0, 0) \in M$. We can no longer use the preceding method, because $\mathbf{r}/r^3$ is not smooth on $M$, on account of the zero denominator at $\mathbf{r} = (0, 0, 0)$. Because $(0, 0, 0) \in M$ and $(0, 0, 0) \notin \partial M$, there is an $\varepsilon > 0$ such that the ball $N$ of radius $\varepsilon$ centered at $(0, 0, 0)$ is contained completely inside $M$. Let $W$ be the region between $M$ and $N$. Then $W$ has boundary $\partial N \cup \partial M = S$. But the orientation on $\partial N$ induced by the *outward* normal on $W$ is the opposite of that obtained from $N$ (see Figure 8.4.7).

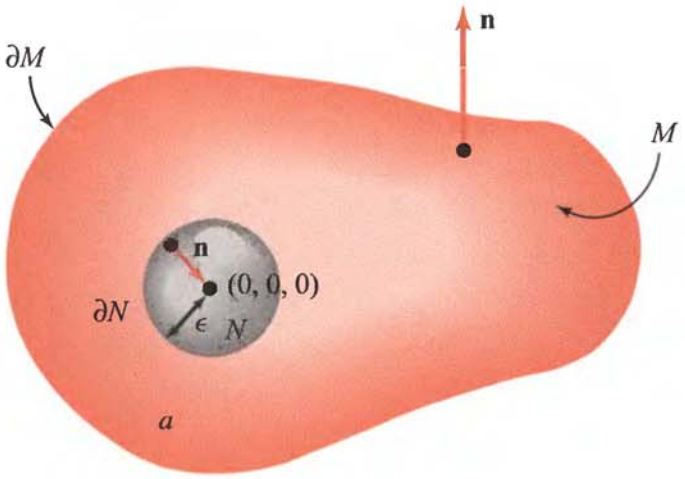



**Figure 8.4.7** Induced outward orientation on $S$; $W$ is $M$ minus the ball $N$.

Now $\nabla \cdot (\mathbf{r}/r^3) = 0$ on $W$, and so, by the divergence theorem applied to the (nonelementary) region $W$,

$$\iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS = \iiint_W \nabla \cdot \left(\frac{\mathbf{r}}{r^3}\right) dV = 0.$$

Because

$$\iint_S \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS = \iint_{\partial M} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS + \iint_{\partial N} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS,$$

where $\mathbf{n}$ is the outward normal to $S$, we have

$$\iint_{\partial M} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS = -\iint_{\partial N} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS.$$

However, on $\partial N$, $\mathbf{n} = -\mathbf{r}/r$ and $r = \varepsilon$, because $\partial N$ is a sphere of radius $\varepsilon$, so that

$$-\iint_{\partial N} \frac{\mathbf{r} \cdot \mathbf{n}}{r^3}\, dS = \iint_{\partial N} \frac{\varepsilon^2}{\varepsilon^4}\, dS = \frac{1}{\varepsilon^2} \iint_{\partial N} dS.$$

But $\iint_{\partial N} dS = 4\pi\varepsilon^2$, the surface area of the sphere of radius $\varepsilon$. This proves the result. ■

**EXAMPLE 6** Gauss' law has the following physical interpretation. The potential due to a point charge $Q$ at (0, 0, 0) is given by

$$\phi(x, y, z) = \frac{Q}{4\pi r} = \frac{Q}{4\pi\sqrt{x^2 + y^2 + z^2}},$$

and the corresponding electric field is

$$\mathbf{E} = -\nabla\phi = \frac{Q}{4\pi}\left(\frac{\mathbf{r}}{r^3}\right).$$

Thus, Theorem 10 states that the total electric flux $\iint_{\partial M} \mathbf{E} \cdot d\mathbf{S}$ (that is, the flux of $\mathbf{E}$ out of a closed surface $\partial M$) equals $Q$ if the charge lies inside $M$ and zero otherwise. Note that even if $(0, 0, 0) \notin M$, $\mathbf{E}$ will still be *nonzero* on $M$.

For a continuous charge distribution described by a charge density $\rho$ in a region $W$, the field $\mathbf{E}$ is related to the density $\rho$ by

$$\operatorname{div} \mathbf{E} = \nabla \cdot \mathbf{E} = \rho.$$

Thus, by Gauss' theorem,

$$\iint_{\partial W} \mathbf{E} \cdot d\mathbf{S} = \iiint_W \rho \, dV = Q;$$

that is, the flux out of a surface is equal to the total charge inside. ▲

## Divergence in Spherical Coordinates

We next use Gauss' theorem to derive the formula

$$\operatorname{div} \mathbf{F} = \frac{1}{\rho^2}\frac{\partial}{\partial\rho}(\rho^2 F_\rho) + \frac{1}{\rho \sin\phi}\frac{\partial}{\partial\phi}(\sin\phi \, F_\phi) + \frac{1}{\rho \sin\phi}\frac{\partial F_\theta}{\partial\theta} \tag{8}$$

for the divergence of a vector field $\mathbf{F}$ in spherical coordinates, which was stated in Section 8.2. (Again, the subscripts here denote *components*, *not* partial derivatives.) The method is to use the formula

$$\operatorname{div} \mathbf{F}(\mathrm{P}) = \lim_{W \to P} \frac{1}{V(W)} \iint_{\partial W} \mathbf{F} \cdot \mathbf{n} \, dS, \tag{9}$$

where $W$ is a region with volume $V(W)$, which shrinks down to a point $\mathbf{P}$ (in the main text we took a ball, but one can use regions of any shape). Let $W$ be the shaded region in Figure 8.4.8.

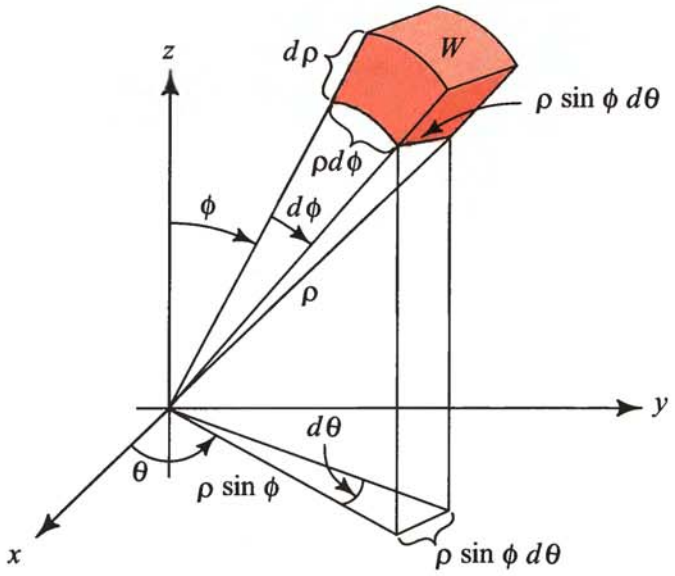



**Figure 8.4.8** Infinitesimal volume determined by $d\rho, d\theta, d\phi$ at $(\rho, \theta, \phi)$.

For the two faces orthogonal to the radial direction, the surface integral in equation (9) is, approximately,

$$\begin{aligned} &F_\rho(\rho + d\rho, \phi, \theta) \times \text{ (area of outer face)} - F_\rho(\rho, \phi, \theta) \times \text{ (area of inner face)} \\ &\quad \approx F_\rho(\rho + d\rho, \phi, \theta)(\rho + d\rho)^2 \sin\phi \, d\phi \, d\theta - F_\rho(\rho, \phi, \theta)\rho^2 \sin\phi \, d\phi \, d\theta \\ &\quad \approx \frac{\partial}{\partial \rho}(F_\rho \rho^2 \sin\phi) \, d\rho \, d\phi \, d\theta \end{aligned} \tag{10}$$

by the one-variable mean-value theorem. Dividing by the volume of the region $W$, namely, $\rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$, we see that the contribution to the right-hand side of equation (9) is

$$\frac{1}{\rho^2} \frac{\partial}{\partial \rho}(\rho^2 F_\rho) \tag{11}$$

for these faces. Likewise, the contribution from the faces orthogonal to the $\phi$ direction is

$$\frac{1}{\rho \sin\phi} \frac{\partial}{\partial \phi}(\sin\phi F_\phi), \qquad \text{and for the } \theta \text{ direction,} \qquad \frac{1}{\rho \sin\phi} \frac{\partial F_\theta}{\partial \theta}.$$

Substituting (11) and these expressions in equation (9) and taking the limit gives equation (8).

## EXERCISES

**1.** Use the divergence theorem to calculate the flux of $\mathbf{F} = (x - y)\mathbf{i} + (y - z)\mathbf{j} + (z - x)\mathbf{k}$ out of the unit sphere.

**2.** Let $\mathbf{F} = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$. Evaluate the surface integral of $\mathbf{F}$ over the unit sphere.

**3.** Evaluate $\iint_{\partial W} \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $W$ is the unit cube (in the first octant). Perform the calculation directly and check by using the divergence theorem.

**4.** Repeat Exercise 3 for

(a) $\mathbf{F} = \mathbf{i} + \mathbf{j} + \mathbf{k}$
(b) $\mathbf{F} = x^2\mathbf{i} + x^2\mathbf{j} + z^2\mathbf{k}$

**5.** Let $\mathbf{F} = y\mathbf{i} + z\mathbf{j} + xz\mathbf{k}$. Evaluate $\iint_{\partial W} \mathbf{F} \cdot d\mathbf{S}$ for each of the following regions $W$:

(a) $x^2 + y^2 \leq z \leq 1$
(b) $x^2 + y^2 \leq z \leq 1$ and $x \geq 0$
(c) $x^2 + y^2 \leq z \leq 1$ and $x \leq 0$

**6.** Repeat Exercise 5 for $\mathbf{F} = (x - y)\mathbf{i} + (y - z)\mathbf{j} + (z - x)\mathbf{k}$. [The solution to part (b) only is in the Study Guide to this text.]

**7.** Find the flux of the vector field $\mathbf{F} = (x - y^2)\mathbf{i} + y\mathbf{j} + x^3\mathbf{k}$ out of the rectangular solid $[0, 1] \times [1, 2] \times [1, 4]$.

**8.** Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F} = 3xy^2\mathbf{i} + 3x^2y\mathbf{j} + z^3\mathbf{k}$ and $S$ is the surface of the unit sphere.

**9.** Evaluate $\iint_{\partial W} \mathbf{F} \cdot \mathbf{n}\, dA$, where $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} - z\mathbf{k}$ and $W$ is the unit cube in the first octant. Perform the calculation directly and check by using the divergence theorem.

**10.** Evaluate the surface integral $\iint_{\partial S} \mathbf{F} \cdot \mathbf{n}\, dA$, where $\mathbf{F}(x, y, z) = \mathbf{i} + \mathbf{j} + z(x^2 + y^2)^2\mathbf{k}$ and $\partial S$ is the surface of the cylinder $x^2 + y^2 \leq 1, 0 \leq z \leq 1$.

**11.** Prove that

$$\iiint_W (\nabla f) \cdot \mathbf{F}\, dx\, dy\, dz = \iint_{\partial W} f\, \mathbf{F} \cdot \mathbf{n}\, dS - \iiint_W f \nabla \cdot \mathbf{F}\, dx\, dy\, dz.$$

**12.** Prove the identity

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G}).$$

**13.** Show that $\iiint_W (1/r^2)\, dx\, dy\, dz = \iint_{\partial W} (\mathbf{r} \cdot \mathbf{n}/r^2)\, dS$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

**14.** Fix vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathbb{R}^3$ and numbers ("charges") $q_1, \ldots, q_k$. Define the function $\phi$ by $\phi(x, y, z) = \sum_{i=1}^{k} q_i/(4\pi \|\mathbf{r} - \mathbf{v}_i\|)$, where $\mathbf{r} = (x, y, z)$. Show that for a closed surface $S$ and $\mathbf{E} = -\nabla\phi$,

$$\iint_S \mathbf{E} \cdot d\mathbf{S} = Q,$$

where $Q$ is the total charge inside $S$. (Assume that Gauss' law from Theorem 10 applies and that none of the charges are on $S$.)

**15.** Prove ***Green's identities***

$$\iint_{\partial W} f\nabla g \cdot \mathbf{n}\, dS = \iiint_W (f\nabla^2 g + \nabla f \cdot \nabla g)\, dV$$

and

$$\iint_{\partial W} (f\nabla g - g\nabla f) \cdot \mathbf{n}\, dS = \iiint_W (f\nabla^2 g - g\nabla^2 f)\, dV.$$

**16.** Suppose $\mathbf{F}$ satisfies div $\mathbf{F} = 0$ and curl $\mathbf{F} = \mathbf{0}$ on all of $\mathbb{R}^3$. Show that we can write $\mathbf{F} = \nabla f$, where $\nabla^2 f = 0$.

**17.** Let $\rho$ be a continuous function on $\mathbb{R}^3$ such that $\rho(\mathbf{q}) = 0$ except for $\mathbf{q}$ in some region $W$. Let $\mathbf{q} \in W$ be denoted by $\mathbf{q} = (x, y, z)$. The ***potential*** of $\rho$ is defined to be the function

$$\phi(\mathbf{p}) = \iiint_W \frac{\rho(\mathbf{q})}{4\pi \|\mathbf{p} - \mathbf{q}\|}\, dV(\mathbf{q}),$$

where $\|\mathbf{p} - \mathbf{q}\|$ is the distance between $\mathbf{p}$ and $\mathbf{q}$.

(a) Using the method of Theorem 10, show that

$$\iint_{\partial W} \nabla\phi \cdot \mathbf{n}\, dS = -\iiint_W \rho\, dV$$

for those regions $W$ that can be partitioned into a finite union of symmetric elementary regions.

(b) Show that $\phi$ satisfies ***Poisson's equation***

$$\nabla^2\phi = -\rho.$$

[HINT: Use part (a).] (Notice that if $\rho$ is a charge density, then the integral defining $\phi$ may be thought of as the sum of the potential at $\mathbf{p}$ caused by point charges distributed over $W$ according to the density $\rho$.)

**18.** Suppose $\mathbf{F}$ is tangent to the closed surface $S = \partial W$ of a region $W$. Prove that

$$\iiint_W (\text{div } \mathbf{F})\, dV = 0.$$

**19.** Use Gauss' law and symmetry to prove that the electric field due to a charge $Q$ evenly spread over the surface of a sphere is the same outside the surface as the field from a point charge $Q$ located at the center of the sphere. What is the field inside the sphere?

**20.** Reformulate Exercise 19 in terms of gravitational fields.

**21.** Show how Gauss' law can be used to solve part (b) of Exercise 25 in Section 8.3.

**22.** Let $S$ be a closed surface. Use Gauss' theorem to show that if $\mathbf{F}$ is a $C^2$ vector field, then we have $\iint_s (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0$.

**23.** Let $S$ be the surface of region $W$. Show that

$$\iint_S \mathbf{r} \cdot \mathbf{n}\, dS = 3 \text{ volume } (W).$$

Explain this geometrically.

## 8.5 Some Differential Equations of Mechanics and Technology

Isaac Newton reputedly said, "All in nature reduces to differential equations." This point of view was paraphrased by Max Planck (see the Historical Note in Section 3.3): "... Present day physics, as far as it is theoretically organized, is completely governed by a system of space–time differential equations."

In this section, we apply the central theorems of vector analysis to the derivation of the differential equations governing heat transfer, electromagnetism, and the motion of some fluids.

Keep in mind the importance of these problems in modern technology. For example, a good understanding of fluids and the ability to do computations to solve their governing equations is at the heart of how one builds a modern airplane or designs a submarine. For instance, the flow of air (the fluid in this case) over the wings of an aircraft is very subtle, even though the governing equations are relatively simple. We shall derive a slightly idealized form of these equations in this section. Likewise, the equations of electromagnetism, as we will discuss in the following paragraphs, is central to the communications industry; wireless, television, and much of the operation of modern electronic devices, including computers, depends on these and related fundamental equations.

### Conservation Laws

As preparation for deriving the equations of a fluid, let us first discuss an important equation that is referred to as a ***conservation*** equation. For fluids, it expresses the conservation of mass; for electromagnetic theory, it expresses the conservation of charge. We shall apply these ideas to the equation for heat conduction and to electromagnetism.

Let $\mathbf{V}(t, x, y, z)$ be a $C^1$ vector field on $\mathbb{R}^3$ for each $t$ and let $\rho(t, x, y, z)$ be a $C^1$ real-valued function. By the ***law of conservation of mass*** for $\mathbf{V}$ and $\rho$, we mean that the condition

$$\frac{d}{dt} \iiint_W \rho\, dV = -\iint_{\partial W} \mathbf{J} \cdot \mathbf{n}\, dS$$

holds for all regions $W$ in $\mathbb{R}^3$, where $\mathbf{J} = \rho\mathbf{V}$ (see Figure 8.5.1).

If we think of $\rho$ as a mass density ($\rho$ could also be a charge density)—that is, the mass per unit volume—and of $\mathbf{V}$ as the velocity field of a fluid, the condition simply says that the rate of change of total mass in $W$ equals the rate at which mass flows *into* $W$. Recall that $\iint_{\partial W} \mathbf{J} \cdot \mathbf{n}\, dS$ is called the *flux* of $\mathbf{J}$. We need the following result.

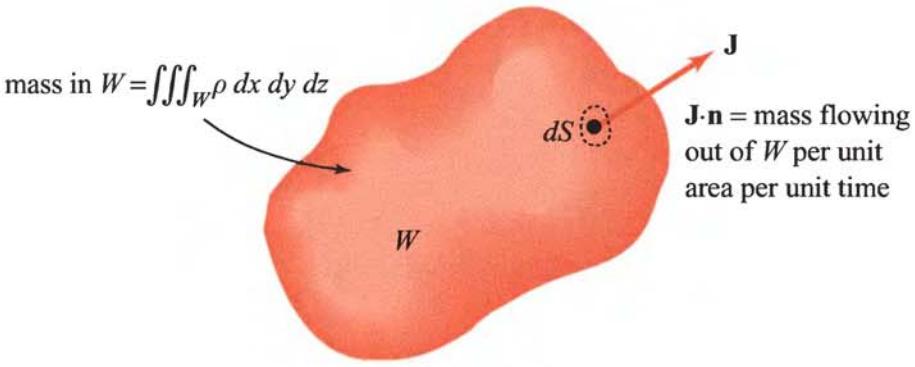



**Figure 8.5.1** The rate of change of mass in $W$ equals the rate at which mass crosses $\partial W$.

**THEOREM 11** For $\mathbf{V}$ and $\rho$ (a smooth vector field and a scalar field on $\mathbb{R}^3$), the law of conservation of mass for $\mathbf{V}$ and $\rho$ is equivalent to the condition

$$\operatorname{div} \mathbf{J} + \frac{\partial \rho}{\partial t} = 0. \tag{1}$$

That is,

$$\rho \operatorname{div} \mathbf{V} + \mathbf{V} \cdot \nabla \rho + \frac{\partial \rho}{\partial t} = 0. \tag{1'}$$

Here, div $\mathbf{J}$ means that we compute div $\mathbf{J}$ for $t$ held fixed, and $\partial\rho/\partial t$ means we differentiate $\rho$ with respect to $t$ for $x, y, z$ fixed.

*PROOF* First, observe that by differentiating under the integral, we get

$$\frac{d}{dt} \iiint_W \rho\, dx\, dy\, dz = \iiint_W \frac{\partial \rho}{\partial t}\, dx\, dy\, dz$$

and also

$$\iint_{\partial W} \mathbf{J} \cdot \mathbf{n}\, dS = \iiint_W \operatorname{div} \mathbf{J}\, d\mathbf{V}$$

by the divergence theorem. Thus, conservation of mass is equivalent to the condition

$$\iiint_W \left( \operatorname{div} \mathbf{J} + \frac{\partial \rho}{\partial t} \right) dx\, dy\, dz = 0.$$

Because this is to hold for all regions $W$, it is equivalent to div $\mathbf{J} + \partial\rho/\partial t = 0$. ■

The equation div $\mathbf{J} + \partial\rho/\partial t = 0$ is called the ***equation of continuity***. An interesting remark is that using the change of variables formula, the law of conservation

of mass may be shown to be equivalent to the condition

$$\frac{d}{dt}\iiint_{W_t} \rho \, dV = 0,$$

where $W_t$ is the image of $W$ obtained by moving each point in $W$ along flow lines of $\mathbf{V}$ for time $t$. This result is a special case of the *transport theorem* that we discuss next.

## The Transport Theorem

The transport theorem is an interesting application of the divergence theorem that will be needed in our derivation of the equations of a fluid.

THEOREM 12 Let $\mathbf{F}$ be a vector field on $\mathbb{R}^3$ and denote the flow line of $\mathbf{F}$ starting at $\mathbf{x}$ after time $t$ by $\phi(\mathbf{x}, t)$. (See the Internet supplement to Section 4.4 for more information.) Let $J(\mathbf{x}, t)$ be the Jacobian of the map $\phi_t$: $\mathbf{x} \mapsto \phi(\mathbf{x}, t)$ for $t$ fixed. Then

$$\frac{\partial}{\partial t} J(\mathbf{x}, t) = [\operatorname{div} \mathbf{F}(\phi(\mathbf{x}, t))] J(\mathbf{x}, t).$$

For a given function $f(x, y, z, t)$ and a region $W \subset \mathbb{R}^3$, the ***transport equation*** holds:

$$\frac{d}{dt}\iiint_{W_t} f(x, y, z, t)\, dx\, dy\, dz = \iiint_{W_t} \left( \frac{Df}{Dt} + f \operatorname{div} \mathbf{F} \right) dx\, dy\, dz,$$

where $W_t = \phi_t(W)$, which is the region moving with the flow, and where

$$\frac{Df}{Dt} = \partial f/\partial t + \nabla f \cdot \mathbf{F}$$

is the ***material derivative***.

Taking $f = 1$, Theorem 12 implies that the following assertions are equivalent (which justifies the use of the term *incompressible*):

1. $\operatorname{div} \mathbf{F} = 0$
2. volume $(W_t)$ = volume $(W)$
3. $J(\mathbf{x}, t) = 1$

Let $\phi$, $J$, $\mathbf{F}$, $f$ be as just defined. There is also a vector form of the transport theorem, namely,

$$\frac{d}{dt}\iiint_{W_t} (f\mathbf{F})\, dx\, dy\, dz$$

$$= \iiint_{W_t} \left[ \frac{\partial}{\partial t}(f\mathbf{F}) + \mathbf{F} \cdot \nabla(f\mathbf{F}) + (f\mathbf{F}) \operatorname{div} \mathbf{F} \right] dx\, dy\, dz,$$

where $\mathbf{F} \cdot \nabla(f\mathbf{F})$ denotes the $3 \times 3$ derivative matrix $\mathbf{D}(f\mathbf{F})$ operating on the column vector $\mathbf{F}$; in Cartesian coordinates, $\mathbf{F} \cdot \nabla\mathbf{G}$ is the vector whose $i$th component is

$$\sum_{j=1}^{3} F_j \frac{\partial G^i}{\partial x_j} = F_1 \frac{\partial G^i}{\partial x} + F_2 \frac{\partial G^i}{\partial y} + F_3 \frac{\partial G^i}{\partial z}.$$

We shall leave the proofs of these results, which are extensions of the arguments used to prove Theorem 11, to the reader (see the exercises).

## Derivation of Euler's Equation of a Perfect Fluid

The continuity equation is not sufficient to completely determine the motion of a fluid—we need other conditions.

The fluids that the continuity equation governs can be compressible. If $\operatorname{div} \mathbf{V} = 0$ (incompressible case) and $\rho$ is constant, equation (1′) follows automatically. But in general, even for incompressible fluids, the equation is not automatic, because $\rho$ can depend on $(x, y, z)$ and $t$. Thus, even if the equation $\operatorname{div} \mathbf{V} = 0$ holds, $\operatorname{div}(\rho\mathbf{V}) \neq 0$ may still be true.

Here we discuss Euler's equation for a perfect fluid. Consider a nonviscous fluid moving in space with a velocity field $\mathbf{V}$. When we say that the fluid is *perfect*, we mean that if $W$ is any portion of the fluid, forces of pressure act on the boundary of $W$ along its normal. We assume that the force per unit area acting on $\partial W$ is $-p\mathbf{n}$, where $p(x, y, z, t)$ is some function called the ***pressure*** (see Figure 8.5.2). Thus, the total pressure force acting on $W$ is

$$\mathbf{F}_{\partial W} = \text{ force } = -\iint_{\partial W} p\mathbf{n}\, dS.$$

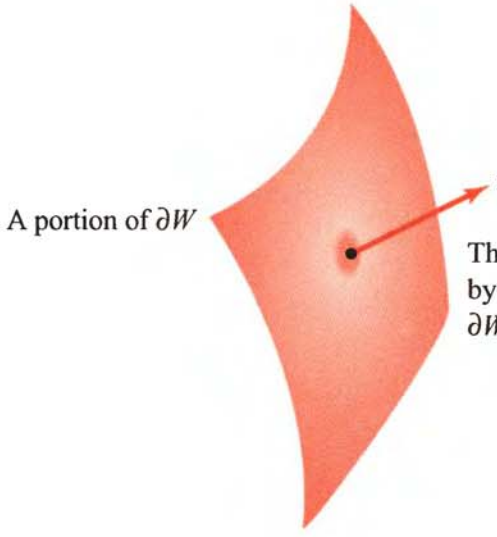



Figure 8.5.2 The force acting on $\partial W$ per unit area is $-p\mathbf{n}$.

This is a *vector* quantity; the $i$th component of $\mathbf{F}_{\partial W}$ is the integral of the $i$th component of $p\mathbf{n}$ over the surface $\partial W$ (this is therefore the surface integral of a

real-valued function). If $\mathbf{e}$ is any fixed vector in space, we have

$$\mathbf{F}_{\partial W} \cdot \mathbf{e} = -\iint_{\partial W} p\mathbf{e} \cdot \mathbf{n}\, dS,$$

which is the integral of a scalar over $\partial W$. By the divergence theorem and identity (7) in the table of vector identities (Section 4.4), we get

$$\mathbf{E} \cdot \mathbf{F}_{\partial W} = -\iiint_W \operatorname{div}(p\mathbf{E})\, dx\, dy\, dz = -\iiint_W (\operatorname{grad} p) \cdot \mathbf{E}\, dx\, dy\, dz,$$

so that

$$\mathbf{F}_{\partial W} = -\iiint_W \nabla p\, dx\, dy\, dz.$$

Now we apply *Newton's second law* to a moving region $W_t$. As in the transport theorem, $W_t = \phi_t(W)$, where $\phi_t(\mathbf{x}) = \phi(\mathbf{x}, t)$ denotes the flow of $\mathbf{V}$. The rate of change of momentum of the fluid in $W_t$ equals the force acting on it:

$$\frac{d}{dt}\iiint_{W_t} p\mathbf{V}\, dx\, dy\, dz = \mathbf{F}_{\partial W_t} = \iiint_{W_t} \nabla p\, dx\, dy\, dz.$$

We apply the vector form of the transport theorem to the left-hand side to get

$$\iiint_{W_t} \left[\frac{\partial}{\partial t}(\rho\mathbf{V}) + \mathbf{V} \cdot \nabla(\rho\mathbf{V}) + p\mathbf{V} \operatorname{div} \mathbf{V}\right] dx\, dy\, dz = -\iiint_{W_t} \nabla p\, dx\, dy\, dz.$$

Because $W_t$ is arbitrary, this is equivalent to

$$\frac{\partial}{\partial t}(\rho\mathbf{V}) + \mathbf{V} \cdot \nabla(\rho\mathbf{V}) + \rho\mathbf{V} \operatorname{div} \mathbf{V} = -\nabla p.$$

Simplification using the equation of continuity, namely, formula (1′), gives

$$\rho\left(\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V} \cdot \nabla\mathbf{V}\right) = -\nabla p. \tag{2}$$

This is ***Euler's equation for a perfect fluid***. For compressible fluids, $p$ is a given function of $\rho$ (for instance, for many gases, $p = A\rho^{\gamma}$ for constants $A$ and $\gamma$). On the other hand, if the fluid is incompressible, $\rho$ is to be determined from the condition $\operatorname{div} \mathbf{V} = 0$. Equations (1) and (2) then govern the motion of the fluid.

## Historical Note

The equations describing the motion of a fluid were first derived by Leonhard Euler in 1755, in a paper entitled "General Principles of the Motion of Fluids." Euler did basic work in mechanics as well as voluminous work in pure mathematics, a small part of which has already been discussed in this book; he essentially began the subject of analytical mechanics (as opposed to the Euclidean geometric methods used by Newton). He is responsible for the equations of a rigid body (equations that apply, for example, to a tumbling satellite) and the formulation of many basic equations of mechanics in terms of variational principles; that is, by the methods of maxima and minima of real-valued functions. Euler wrote the first comprehensive textbook on calculus and contributed to virtually all branches of mathematics. He wrote several books and hundreds of research papers even after he became totally blind, and he was working on a new treatise on fluid mechanics at the time of his death in 1783. Euler's equations for a fluid were eventually modified by Navier and Stokes to include viscous effects; the resulting Navier–Stokes equations are described in virtually every textbook on fluid mechanics.[7] Stokes is, of course, also responsible for developing Stokes' theorem, one of the main results discussed in this text!

## Conservation of Energy and the Derivation of the Heat Equation

If $T(t, x, y, z)$ (a $C^2$ function) denotes the temperature in a body at time $t$, then $\nabla T$ represents the temperature gradient: Heat "flows" with the vector field $-\nabla T = \mathbf{F}$. Note that $\nabla T$ points in the direction of *increasing* $T$. Because heat flows from hot to cold, we have inserted a minus sign to reflect this physically observable fact. The energy density, that is, the energy per unit volume, is $c\rho_0 T$, where $c$ is a constant (called the specific heat) and $\rho_0$ is the mass density, assumed constant. (We accept these assertions from elementary physics.) The ***energy flux vector*** is defined to be $\mathbf{J} = k\mathbf{F}$, where $k$ is a constant called the ***conductivity***.

One now makes the hypothesis that energy is conserved. This means that $\mathbf{J}$ and $\rho = c\rho_0 T$ should obey the law of conservation of mass, with $\rho$ playing the role of "mass" (note that it is *energy density*, not mass); that is,

$$\frac{d}{dt}\iiint_W \rho\, dV = -\iint_{\partial W} \mathbf{J}\cdot\mathbf{n}\, dS.$$

By Theorem 11, this assertion is equivalent to

$$\operatorname{div}\mathbf{J} + \frac{\partial \rho}{\partial t} = 0.$$

[7] The Clay Foundation has offered a prize of $1 million to anyone who shows that for the incompressible Navier–Stokes equations, smooth data at $t = 0$ lead to smooth solutions for all $t > 0$.

But

$$\operatorname{div} \mathbf{J} = \operatorname{div}(-k\nabla T) = -k\nabla^2 T.$$

(Recall that $\nabla^2 T = \partial^2 T/\partial x^2 + \partial^2 T/\partial y^2 + \partial^2 T/\partial z^2$ and $\nabla^2$ is the Laplace operator.) Continuing, we have

$$\frac{\partial \rho}{\partial t} = \frac{\partial (c\rho_0 T)}{\partial t} = c\rho_0 \frac{\partial T}{\partial t}.$$

Thus, the equation $\operatorname{div} \mathbf{J} + \partial \rho/\partial t = 0$ becomes

$$\frac{\partial T}{\partial t} = \frac{k}{c\rho_0} \nabla^2 T = k\nabla^2 T, \tag{3}$$

where $\kappa = k/c\rho_0$ is called the ***diffusivity***. Equation (3) is the important ***heat equation***.

Just as equations (1) and (2) govern the flow of an ideal fluid, equation (3) governs the conduction of heat in the following sense. If $T(0, x, y, z)$ is a given initial temperature distribution, then a unique $T(t, x, y, z)$ is determined that satisfies equation (3). In other words, the initial condition at $t = 0$ gives the result for $t > 0$. Notice that if $T$ does not change with time (the steady-state case), then we must have $\nabla^2 T = 0$ (Laplace's equation).

## Maxwell's Equations and the Prediction of Radio Waves: The Communication Revolution Begins

We now return to *Maxwell's equations*, which govern the propagation of electromagnetic fields. The form of these equations depends on the physical units one is employing, and changing units introduces factors like $4\pi$ and the velocity of light. We shall choose the system in which Maxwell's equations are simplest.

Let $\mathbf{E}$ and $\mathbf{H}$ be $C^1$ functions of $(t, x, y, z)$ that are vector fields for each $t$. They satisfy (by definition) ***Maxwell's equation with charge density*** $\rho(t, x, y, z)$ and ***current density*** $\mathbf{J}(t, x, y, z)$ when the following conditions hold:

$$\nabla \cdot \mathbf{E} = \rho \text{ (Gauss' law)}, \tag{4}$$

$$\nabla \cdot \mathbf{H} = 0 \text{ (no negative sources)}, \tag{5}$$

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{H}}{\partial t} = \mathbf{0} \text{ (Faraday's law)}, \tag{6}$$

and

$$\nabla \times \mathbf{H} - \frac{\partial \mathbf{E}}{\partial t} = J \text{ (Ampère's law)}, \tag{7}$$

Of these laws, equations (4) and (6) were described in integral form in Sections 8.2 and 8.4; historically, they arose in these forms as physically observed laws. Ampère's law was mentioned for a special case in Section 7.2, Example 12.

Physically, one interprets $\mathbf{E}$ as the ***electric field*** and $\mathbf{H}$ as the ***magnetic field***. According to the preceding equations, as time $t$ progresses, these fields interact with each other, and with any charges and currents that are present. For example, the propagation of electromagnetic waves (TV signals, radio waves, light from the sun, etc.) in a vacuum is governed by these equations with $\mathbf{J} = 0$ and $\rho = 0$.

Because $\nabla \cdot \mathbf{H} = 0$, we can apply Theorem 8 (from Section 8.3) to conclude that $\mathbf{H} = \nabla \times \mathbf{A}$ for some vector field $\mathbf{A}$. (We are assuming that $\mathbf{H}$ is defined on all of $\mathbb{R}^3$ for each time $t$.) The vector field $\mathbf{A}$ is not unique, and we can use $\mathbf{A}' = \mathbf{A} + \nabla f$ equally well for any function $f(t, x, y, z)$, because $\nabla \times \nabla f = 0$. (This freedom in the choice of $\mathbf{A}$ is called *gauge freedom.*) For any such choice of $\mathbf{A}$, we have, by equation (6),

$$\begin{aligned}\mathbf{0} = \nabla \times \mathbf{E} + \frac{\partial \mathbf{H}}{\partial t} &= \nabla \times \mathbf{E} + \frac{\partial}{\partial t} \nabla \times \mathbf{A} \\ &= \nabla \times \mathbf{E} \times \nabla \times \frac{\partial \mathbf{A}}{\partial t} \\ &= \nabla \times \left( \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right).\end{aligned}$$

Applying Theorem 7 (from Section 8.3), there is a real-valued function $\phi$ on $\mathbb{R}^3$ such that

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla \phi.$$

Substituting this equation and $\mathbf{H} = \nabla \times \mathbf{A}$ into equation (7), and using the vector identity (whose proof we leave as an exercise)

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A},$$

we get

$$\begin{aligned}\mathbf{J} = \nabla \times \mathbf{H} - \frac{\partial \mathbf{E}}{\partial t} &= \nabla \times (\nabla \times \mathbf{A}) - \frac{\partial}{\partial t} \left( -\frac{\partial \mathbf{A}}{\partial t} - \nabla \phi \right) \\ &= \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} + \frac{\partial^2 \mathbf{A}}{\partial t^2} + \frac{\partial}{\partial t}(\nabla \phi).\end{aligned}$$

Thus,

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mathbf{J} + \nabla(\nabla \cdot \mathbf{A}) + \frac{\partial}{\partial t}(\nabla \phi).$$

That is,

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mathbf{J} + \nabla\left(\nabla \cdot \mathbf{A} + \frac{\partial \phi}{\partial t}\right). \tag{8}$$

Again using the equation $\mathbf{E} + \partial \mathbf{A}/\partial t = -\nabla\phi$ and the equation $\nabla \cdot \mathbf{E} = \rho$, we obtain

$$\rho = \nabla \cdot \mathbf{E} = \nabla \cdot \left(-\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}\right) = -\nabla^2\phi - \frac{\partial(\nabla \cdot \mathbf{A})}{\partial t}.$$

That is,

$$\nabla^2 \phi = -\rho - \frac{\partial(\nabla \cdot \mathbf{A})}{\partial t}. \tag{9}$$

Now let us exploit the freedom in our choice of $\mathbf{A}$. We impose the "condition"

$$\nabla \cdot \mathbf{A} + \frac{\partial \phi}{\partial t} = 0. \tag{10}$$

We must be sure we can do this. Supposing we have a given $\mathbf{A}_0$ and a corresponding $\phi_0$, can we choose a new $\mathbf{A} = \mathbf{A}_0 + \nabla f$ and then a new $\phi$ such that $\nabla \cdot \mathbf{A} + \partial\phi/\partial t = 0$? With this new $\mathbf{A}$, the new $\phi$ is $\phi_0 - \partial f/\partial t$; we leave verification as an exercise for the reader. Condition (10) on $f$ then becomes

$$0 = \nabla \cdot (\mathbf{A}_0 + \nabla f) = \frac{\partial(\phi_0 - \partial f/\partial t)}{\partial t} = \nabla \cdot \mathbf{A}_0 + \nabla^2 f + \frac{\partial \phi_0}{\partial t} - \frac{\partial^2 f}{\partial t^2}$$

or

$$\nabla^2 f - \frac{\partial^2 f}{\partial t^2} = -\left(\nabla \cdot \mathbf{A}_0 + \frac{\partial \phi_0}{\partial t}\right). \tag{11}$$

Thus, to be able to choose $\mathbf{A}$ and $\phi$ satisfying $\nabla \cdot \mathbf{A} + \partial\phi/\partial t = 0$, we must be able to solve equation (11) for $f$. One can indeed do this under general conditions, although we do not prove it here. Equation (11) is called the ***inhomogeneous wave equation***.

If we accept that $\mathbf{A}$ and $\phi$ can be chosen to satisfy $\nabla \cdot \mathbf{A} + \partial\phi/\partial t = 0$, then equations (8) and (9) for $\mathbf{A}$ and $\phi$ become

$$\nabla^2 \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mathbf{J}; \tag{8'}$$

$$\nabla^2 \phi - \frac{\partial^2 \phi}{\partial t^2} = -\rho. \tag{9'}$$

Conversely, if $\mathbf{A}$ and $\phi$ satisfy the equations $\nabla \cdot \mathbf{A} + \partial\phi/\partial t = 0$, $\nabla^2\phi - \partial^2\phi/\partial t^2 = -\rho$, and $\nabla^2\mathbf{A} - \partial^2\mathbf{A}/\partial t^2 = -\mathbf{J}$, then $\mathbf{E} = -\nabla\phi - \partial\mathbf{A}/\partial t$ and $\mathbf{H} = \nabla \times \mathbf{A}$ satisfy Maxwell's equations. *This procedure then "reduces" Maxwell's equations to a study of the wave equation.*[8]

Since the eighteenth century, solutions to the wave equation have been well studied (one learns these in most courses on differential equations). To indicate the wavelike nature of the solutions, for example, observe that for any function $f$,

$$\phi(t, x, y, z) = f(x - t)$$

solves the wave equation $\nabla^2\phi - (\partial^2\phi/\partial t^2) = 0$. This solution just propagates the graph of $f$ like a wave; thus, one might conjecture that solutions of Maxwell's equations are wavelike in nature. Historically, all of this was Maxwell's great achievement, and it soon led to Hertz's discovery of radio waves. To quote from the Feynman *Lectures on Physics* (Vol. II):

> From a long view of the history of mankind—seen from, say, ten thousand years from now—there can be little doubt that the most significant event of the nineteenth century will be judged as Maxwell's discovery of the laws of electrodynamics. The American Civil War will pale into provincial insignificance in comparison with this important scientific event of the same decade.

Mathematics again shows its uncanny ability not only to *describe* but to *predict* natural phenomena.

There are other techniques (called Green's function methods) for dealing with the basic equations of mechanics and mathematical physics that also rely on vector calculus. Some of these methods are discussed in the Internet supplement for this book.

## EXERCISES

**1.** Use a direct argument (or the proof of Theorem 1 in the Internet supplement to Section 4.4) to show that

$$\frac{\partial}{\partial t} J(\mathbf{x}, t) = [\operatorname{div} \mathbf{F}(\phi(\mathbf{x}, t))] J(\mathbf{x}, t).$$

---

[8]There are variations on this procedure. For further details, see, for example, *Differential Equations of Applied Mathematics*, by G. F. D. Duff and D. Naylor, Wiley, New York, 1966, or books on electromagnetic theory, such as *Classical Electrodynamics*, by J. D. Jackson, Wiley, New York, 1962.

**2.** Using the change of variables theorem and Exercise 1, show that if $f(x, y, z, t)$ is a given function and $W \subset \mathbb{R}^3$ is any region, then the ***transport equation*** holds:

$$\frac{d}{dt}\iiint_{W_t} f(x, y, z, t)\, dx\, dy\, dz = \iiint_{W_t} \left(\frac{Df}{Dt} + f \operatorname{div} \mathbf{F}\right) dx\, dy\, dz$$

where $W_t = \phi_t(W)$, which is the region moving with the flow, and where $Df/Dt = \partial f/\partial t + \nabla f \cdot \mathbf{F}$ is the material derivative.

**3.** Use the transport equation to show that

$$\frac{d}{dt}\iiint_{W_t} \rho\, dx\, dy\, dz = 0$$

is equivalent to the law of conservation of mass.

**4.** Using Exercise 3 and the change of variables theorem, show that $\rho(\mathbf{x}, t)$ can be expressed in terms of the Jacobian $J(\mathbf{x}, t)$ of the flow map $\phi(\mathbf{x}, t)$ and $\rho(\mathbf{x}, 0)$ by the equation

$$\rho(\mathbf{x}, t)J(\mathbf{x}, t) = \rho(\mathbf{x}, 0).$$

What can you conclude from this for incompressible flow?

**5.** Prove the vector form of the transport theorem, namely,

$$\frac{d}{dt}\iiint_{W_t} (f\mathbf{F})\, dx\, dy\, dz = \iiint_{W_t} \left[\frac{\partial}{\partial t}(f\mathbf{F}) + \mathbf{F} \cdot \nabla(f\mathbf{F}) + (f\mathbf{F}) \operatorname{div} \mathbf{F}\right] dx\, dy\, dz,$$

where $\mathbf{F} \cdot \nabla(f\mathbf{F})$ denotes the $3 \times 3$ derivative matrix $\mathbf{D}\,(f\mathbf{F})$ operating on the column vector $\mathbf{F}$; in Cartesian coordinates, $\mathbf{F} \cdot \nabla\mathbf{G}$ is the vector whose $i$th component is

$$\sum_{j=1}^{3} F_j \frac{\partial G^i}{\partial x_j} = F_1 \frac{\partial G^i}{\partial x} + F_2 \frac{\partial G^i}{\partial y} + F_3 \frac{\partial G^i}{\partial z}.$$

**6.** Let $\mathbf{V}$ be a vector field with flow $\phi(\mathbf{x}, t)$ and let $\mathbf{V}$ and $\rho$ satisfy the law of conservation of mass. Let $W_t$ be the region transported with the flow. Prove the following version of the transport theorem:

$$\frac{d}{dt}\iiint_{W_t} \rho f\, dx\, dy\, dz = \iiint_{W_t} \rho \frac{Df}{Dt}\, dx\, dy\, dz.$$

**7.** (*Bernoulli's law*) (a) Let $\mathbf{V}$, $\rho$ satisfy the law of conservation of mass and equation (2) (Euler's equation for a perfect fluid). Suppose $\mathbf{V}$ is irrotational and hence that $\mathbf{V} = \nabla\phi$ for a function $\phi$. Show that if $C$ is a path connecting two points $P_1$ and $P_2$, then

$$\left(\frac{\partial \phi}{\partial t} + \frac{1}{2}\|\mathbf{V}\|^2\right)\Bigg|_{P_1}^{P_2} + \int_C \frac{dp}{\rho} = 0.$$

[HINT: You may need the vector identity, $(\mathbf{V}\cdot\nabla)\mathbf{V} = \frac{1}{2}\nabla(\|\mathbf{V}\|^2) + (\nabla\times\mathbf{V})\times\mathbf{V}$.]

(b) If in part (a), $\mathbf{V}$ is stationary—that is, $\partial\mathbf{V}/\partial t = 0$—and $\rho$ is constant, show that

$$\frac{1}{2}\|\mathbf{V}\|^2 + \frac{p}{\rho}$$

is constant in space. Deduce that, in this situation, *higher pressure is associated with lower fluid speed.*

**8.** Using Exercise 7, show that if $\phi$ satisfies Laplace's equation $\nabla^2\phi = 0$, then $\mathbf{V} = \nabla\phi$ is a stationary solution to Euler's equation for a perfect *incompressible* fluid with constant density.

**9.** Verify that Maxwell's equations imply the equation of continuity for $\mathbf{J}$ and $\rho$.

**10.** For a steady-state charge distribution and divergence-free current distribution, the electric and magnetic fields $\mathbf{E}(x, y, z)$ and $\mathbf{H}(x, y, z)$ satisfy

$$\nabla\times\mathbf{E} = \mathbf{0}, \qquad \nabla\cdot\mathbf{H} = 0, \qquad \nabla\cdot\mathbf{J} = 0, \qquad \nabla\cdot\mathbf{E} = \rho, \qquad \text{and} \qquad \nabla\times\mathbf{H} = \mathbf{J}.$$

Here $\rho = \rho(x, y, z)$ and $\mathbf{J}(x, y, z)$ are assumed to be known. The radiation that the fields produce through a surface $S$ is determined by a radiation flux density vector field, called the ***Poynting*** vector field,

$$\mathbf{P} = \mathbf{E}\times\mathbf{H}.$$

(a) If $S$ is a *closed* surface, show that the radiation flux—that is, the flux of $\mathbf{P}$ through $S$—is given by

$$\iint_S \mathbf{P}\cdot d\mathbf{S} = -\iiint_V \mathbf{E}\cdot\mathbf{J}\,dV,$$

where $V$ is the region enclosed by $S$.

(b) Examples of such fields are

$$\mathbf{E}(x, y, z) = z\mathbf{j} + y\mathbf{k} \qquad \text{and} \qquad \mathbf{H}(x, y, z) = -xy\mathbf{i} + x\mathbf{j} + yz\mathbf{k}.$$

In this case, find the flux of the Poynting vector through the hemispherical shell shown in Figure 8.5.3. (Notice that it is an *open* surface.)

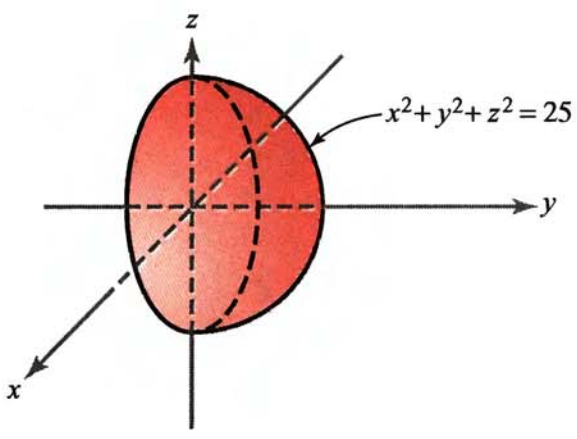



**Figure 8.5.3** The surface for Exercise 10.

(c) The fields of part (b) produce a Poynting vector field passing through the toroidal surface shown in Figure 8.5.4. What is the flux through this torus?

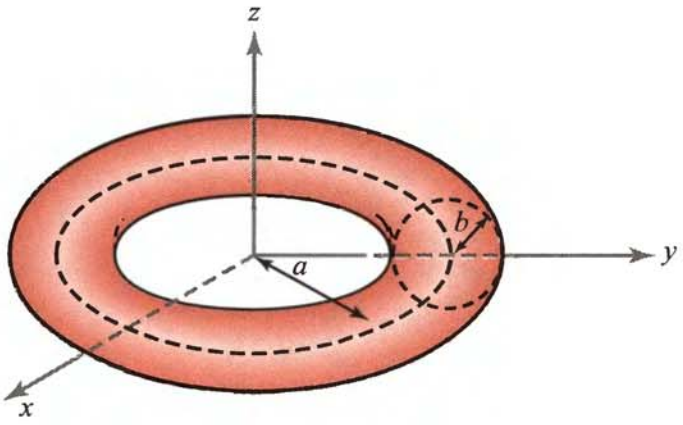



**Figure 8.5.4** The surface for Exercise 10(c).

## 8.6 Differential Forms

The theory of differential forms provides an elegant way of formulating Green's, Stokes', and Gauss' theorems as one statement, the ***fundamental theorem of calculus***. The birth of the concept of a differential form is another dramatic example of how mathematics speaks to mathematicians and drives its own development. These three theorems are, in reality, generalizations of the fundamental theorem of calculus of Newton and Leibniz for functions of one variable,

$$\int_a^b f'(x)\,dx = f(b) - f(a)$$

to two and three dimensions.

Recall that Bernhard Riemann created the concept of $n$-dimensional spaces. If the fundamental theorem of calculus was truly *fundamental,* then it should generalize to arbitrary dimensions. But wait! The cross product, and therefore the curl, does not generalize to higher dimensions, as we remarked in footnote 3, in Section 1.3. Thus, some new idea is needed.

Recall that Hamilton searched for almost 15 years for his quaternions, which ultimately led to the discovery of the cross product. What is the nonexistence of a cross product in higher dimensions telling us? If the fundamental theorem of calculus is the core concept, this suggests the existence of a mathematical language in which it can be formulated in $n$-dimensions. In order to achieve this, mathematicians realized that they were forced to move away from vectors and on to the discovery of *dual* vectors and an entirely new mathematical object, a *differential form.* In this new language, all of the theorems of Green, Stokes, and Gauss have the same elegant and extraordinarily simple form.

Simply and very briefly stated, an expression of the type $P\,dx + Q\,dy$ is a 1-form, or a *differential one-form* on a region in the $xy$ plane, and $F\,dx\,dy$ is a 2-form. Analogously, one can define the notion of an $n$-form. There is an operation $d$, which

takes $n$-forms to $n+1$-forms. It is like a generalized curl and has the property that for $\omega = P\,dx + Q\,dy$, we have

$$d\omega = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

and so in this notation, Green's theorem becomes

$$\int_{\partial D} \omega = \int_D d\omega,$$

which, interestingly, just switches the boundary operator $\partial$ with the $d$ operator. However, differential forms are more than just notation. They create a beautiful theory that generalizes to $n$-dimensions.

In general, if $M$ is an oriented surface of dimension $n$ with an $(n-1)$-dimensional boundary $\partial M$ and if $\omega$ is an $(n-1)$-form on $M$, then the fundamental theorem of calculus (also called the *generalized Stokes' theorem*) reads

$$\boxed{\int_{\partial M} \omega = \int_M d\omega.}$$

A useful thing for the reader to contemplate at this stage is the sense in which the fundamental theorem of calculus becomes a special instance of this result.

In this section, we shall give a very elementary exposition of the theory of forms. Because our primary goal is to show that the theorems of Green, Stokes, and Gauss can be unified under a single theorem, we shall be satisfied with less than the strongest possible version of these theorems. Moreover, we shall introduce forms in a purely axiomatic and nonconstructive manner, thereby avoiding the tremendous number of formal algebraic preliminaries that are usually required for their construction. To the purist our approach will be far from complete, but to the student it may be comprehensible. We hope that this will motivate some students to delve further into the theory of differential forms.

We shall begin by introducing the notion of a 0-form.

## 0-Forms

Let $K$ be an open set in $\mathbb{R}^3$. A 0-***form*** on $K$ is a real-valued function $f: K \to \mathbb{R}$. When we differentiate $f$ once, it is assumed to be of class $C^1$, and $C^2$ when we differentiate twice.

Given two 0-forms $f_1$ and $f_2$ on $K$, we can add them in the usual way to get a new 0-form $f_1 + f_2$ or multiply them to get a 0-form $f_1 f_2$.

**EXAMPLE 1** $f_1(x, y, z) = xy + yz$ and $f_2(x, y, z) = y \sin xz$ are 0-forms on $\mathbb{R}^3$:

$$(f_1 + f_2)(x, y, z) = xy + yz + y \sin xz$$

and

$$(f_1 f_2)(x, y, z) = y^2 x \sin xz + y^2 z \sin xz. \quad \blacktriangle$$

## 1-Forms

The ***basic* 1-*forms*** are the expressions $dx$, $dy$, and $dz$. At present we consider these to be only formal symbols. A 1-***form*** $\omega$ on an open set $K$ is a formal linear combination

$$\omega = P(x, y, z)\,dx + Q(x, y, z)\,dy + R(x, y, z)\,dz,$$

or simply

$$\omega = P\,dx + Q\,dy + R\,dz,$$

where $P$, $Q$, and $R$ are real-valued functions on $K$. By the expression $P\,dx$ we mean the 1-form $P\,dx + 0 \cdot dy + 0 \cdot dz$, and similarly for $Q\,dy$ and $R\,dz$. Also, the order of $P\,dx$, $Q\,dy$, and $R\,dz$ is immaterial, and so

$$P\,dx + Q\,dy + R\,dz = R\,dz + P\,dx + Q\,dy, \text{ etc.}$$

Given two 1-forms $\omega_1 = P_1\,dx + Q_1\,dy + R_1\,dz$ and $\omega_2 = P_2\,dx + Q_2\,dy + R_2\,dz$, we can add them to get a new 1-form $\omega_1 + \omega_2$ defined by

$$\omega_1 + \omega_2 = (P_1 + P_2)\,dx + (Q_1 + Q_2)\,dy + (R_1 + R_2)\,dz,$$

and given a 0-form $f$, we can form the 1-form $f\omega_1$ defined by

$$f\omega_1 = (fP_1)\,dx + (fQ_1)\,dy + (fR_1)\,dz.$$

**EXAMPLE 2** Let $\omega_1 = (x + y^2)\,dx + (zy)\,dy + (e^{xyz})\,dz$ and $\omega_2 = \sin y\,dx + \sin x\,dy$ be 1-forms. Then

$$\omega_1 + \omega_2 = (x + y^2 + \sin y)\,dx + (zy + \sin x)\,dy + (e^{xyz})\,dz.$$

If $f(x, y, z) = x$, then

$$f\omega_2 = x \sin y\,dx + x \sin x\,dy. \quad \blacktriangle$$

## 2-Forms

The ***basic* 2-*forms*** are the formal expressions $dx\,dy$, $dy\,dz$, and $dz\,dx$. These expressions should be thought of as products of $dx$ and $dy$, $dy$ and $dz$, and $dz$ and $dx$.

A 2-***form*** $\eta$ on $K$ is a formal expression

$$\eta = F\,dx\,dy + G\,dy\,dz + H\,dz\,dx,$$

where $F$, $G$, and $H$ are real-valued functions on $K$. The order of $F\,dx\,dy$, $G\,dy\,dz$, and $H\,dz\,dx$ is immaterial; for example,

$$F\,dx\,dy + G\,dy\,dz + H\,dz\,dx = H\,dz\,dx + F\,dx\,dy + G\,dy\,dz, \text{ etc.}$$

At this point it is useful to note that in a 2-form the basic 1-forms $dx$, $dy$, and $dz$ always appear in cyclic pairs (see Figure 8.6.1), that is, $dx\,dy$, $dy\,dz$, and $dz\,dx$.

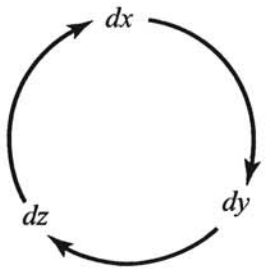



**Figure 8.6.1** The cyclic order of *dx, dy,* and *dz.*

By analogy with 0-forms and 1-forms, we can add two 2-forms

$$\eta_i = F_i\,dx\,dy + G_i\,dy\,dz + H_i\,dz\,dx,$$

$i = 1$ and 2, to obtain a new 2-form,

$$\eta_1 + \eta_2 = (F_1 + F_2)\,dx\,dy + (G_1 + G_2)\,dy\,dz + (H_1 + H_2)\,dz\,dx.$$

Similarly, if $f$ is a 0-form and $\eta$ is a 2-form, we can take the product

$$f\eta = (fF)\,dx\,dy + (fG)\,dy\,dz + (fH)\,dz\,dx.$$

Finally, by the expression $F\,dx\,dy$ we mean the 2-form $F\,dx\,dy + 0 \cdot dy\,dz + 0 \cdot dz\,dx$.

**EXAMPLE 3** The expressions

$$\eta_1 = x^2\,dx\,dy + y^3 x\,dy\,dz + \sin zy\,dz\,dx$$

and

$$\eta_2 = y\,dy\,dz$$

are 2-forms. Their sum is

$$\eta_1 + \eta_2 = x^2\,dx\,dy + (y^3x + y)\,dy\,dz + \sin zy\,dz\,dx.$$

If $f(x, y, z) = xy$, then

$$f\eta_2 = xy^2\,dy\,dz. \quad \blacktriangle$$

## 3-Forms

A ***basic* 3-*form*** is a formal expression $dx\,dy\,dz$ (in this specific cyclic order, as in Figure 8.6.1). A ***3-form*** $\nu$ on an open set $K \subset \mathbb{R}^3$ is an expression of the form $\nu = f(x, y, z)\,dx\,dy\,dz$, where $f$ is a real-valued function on $K$.

We can add two 3-forms and we can multiply them by 0-forms in the obvious way. There seems to be little difference between a 0-form and a 3-form, because both involve a single real-valued function. But we distinguish them for a purpose that will become clear when we multiply and differentiate forms.

**EXAMPLE 4** Let $\nu_1 = y\,dx\,dy\,dz$, $\nu_2 = e^{x^2}\,dx\,dy\,dz$, and $f(x, y, z) = xyz$. Then $\nu_1 + \nu_2 = (y + e^{x^2})\,dx\,dy\,dz$ and $f\nu_1 = y^2xz\,dx\,dy\,dz$. $\blacktriangle$

Although we can add two 0-forms, two 1-forms, two 2-forms, or two 3-forms, we do not need to add a $k$-form and a $j$-form if $k \neq j$. For example, we shall not need to write

$$f(x, y, z)\,dx\,dy + g(x, y, z)\,dz.$$

Now that we have defined these formal objects (forms), one can legitimately ask what they are good for, how they are used, and, perhaps most important, what they mean. The answer to the first question will become clear as we proceed, but we can immediately describe how to use and interpret them.

A real-valued function on a domain $K$ in $\mathbb{R}^3$ is a rule that assigns a real number to each point in $K$. Differential forms are, in some sense, generalizations of the real-valued functions we have studied in calculus. In fact, 0-forms on an open set $K$ are just functions on $K$. Thus, a 0-form $f$ takes points in $K$ to real numbers.

We should like to interpret differential $k$-forms (for $k \geq 1$) not as functions on points in $K$, but as functions on geometric objects such as curves and surfaces. Many of the early Greek geometers viewed lines and curves as being made up of infinitely many points, and planes and surfaces as being made up of infinitely many curves. Consequently, there is at least some historical justification for applying this geometric hierarchy to the interpretation of differential forms.

Given an open subset $K \subset \mathbb{R}^3$, we shall distinguish four types of subsets of $K$ (see Figure 8.6.2):

(i) points in $K$,

(ii) oriented simple curves and oriented simple closed curves, $C$, in $K$,

(iii) oriented surfaces, $S \subset K$,

(iv) elementary subregions, $R \subset K$.

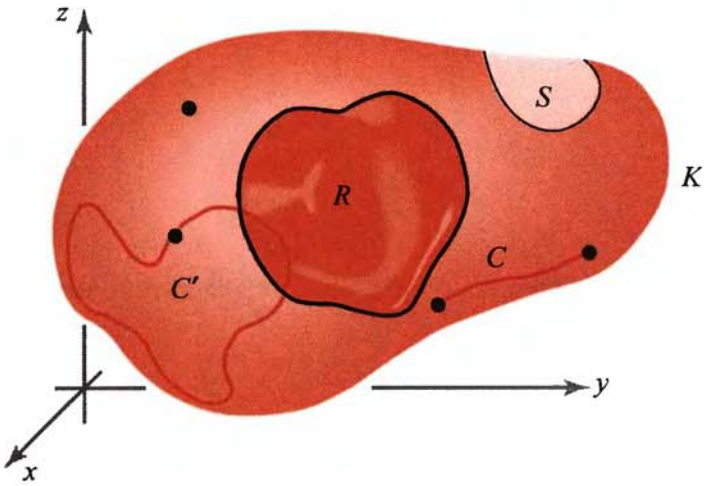



**Figure 8.6.2** The four geometric types of subsets of an open set $K \subset \mathbb{R}^3$ to which the theory of forms applies.

## The Integral of 1-Forms Over Curves

We shall begin with 1-forms. Let

$$\omega = P(x, y, z)\,dx + Q(x, y, z)\,dy + R(x, y, z)\,dz$$

be a 1-form on $K$ and let $C$ be an oriented simple curve as in Figure 8.6.2. The real number that $\omega$ assigns to $C$ is given by the formula

$$\int_C \omega = \int_C P(x, y, z)\,dx + Q(x, y, z)\,dy + R(x, y, z)\,dz. \tag{1}$$

Recall (see Section 7.2) that this integral is evaluated as follows. Suppose that $\mathbf{c}: [a, b] \to K$, $\mathbf{c}(t) = (x(t), y(t), z(t))$ is an orientation-preserving parametrization of $C$. Then

$$\int_C \omega = \int_{\mathbf{c}} \omega = \int_a^b \left[ P(x(t), y(t), z(t)) \cdot \frac{dx}{dt} + Q(x(t), y(t), z(t)) \cdot \frac{dy}{dt} + R(x(t), y(t), z(t)) \cdot \frac{dz}{dt} \right] dt.$$

Theorem 1 of Section 7.2 guarantees that $\int_C \omega$ does not depend on the choice of the parametrization $\mathbf{c}$.

We can thus interpret a 1-form $\omega$ on $K$ as a rule assigning a real number to each oriented curve $C \subset K$; a 2-form $\eta$ will similarly be seen to be a rule assigning a real number to each oriented surface $S \subset K$; and a 3-form $\nu$ will be a rule assigning a real number to each elementary subregion of $K$. The rules for associating real numbers with curves, surfaces, and regions are completely contained in the formal expressions we have defined.

**EXAMPLE 5** Let $\omega = xy\,dx + y^2\,dy + dz$ be a 1-form on $\mathbb{R}^3$ and let $C$ be the oriented simple curve in $\mathbb{R}^3$ described by the parametrization $\mathbf{c}(t) = (t^2, t^3, 1)$, $0 \leq t \leq 1$. $C$ is oriented by choosing the positive direction of $C$ to be the direction in which $\mathbf{c}(t)$ traverses $C$ as $t$ goes from 0 to 1. Then, by formula (1),

$$\int_C \omega = \int_0^1 [t^5(2t) + t^6(3t^2) + 0]\,dt = \int_0^1 (2t^6 + 3t^8)dt = \frac{13}{21}.$$

Thus, this 1-form $\omega$ assigns to each oriented simple curve and each oriented simple closed curve $C$ in $\mathbb{R}^3$ the number $\int_C \omega$. ▲

## The Integral of 2-Forms over Surfaces

A 2-form $\eta$ on an open set $K \subset \mathbb{R}^3$ can similarly be interpreted as a function that associates with each oriented surface $S \subset K$ a real number. This is accomplished by means of the notion of integration of 2-forms over surfaces. Let

$$\eta = F(x, y, z)\,dx\,dy + G(x, y, z)\,dy\,dz + H(x, y, z)\,dz\,dx$$

be a 2-form on $K$, and let $S \subset K$ be an oriented surface parametrized by a function $\mathbf{\Phi}: D \to \mathbb{R}^3$, $D \subset \mathbb{R}^2$, $\mathbf{\Phi}(u, v) = (x(u, v), y(u, v), z(u, v))$ (see Section 7.3).

**DEFINITION** If $S$ is such a surface and $\eta$ is a 2-form on $K$, we define $\iint_S \eta$ by the formula

$$\begin{aligned}\iint_S \eta &= \iint_S F\,dx\,dy + G\,dy\,dz + H\,dz\,dx \\ &= \iint_D \Bigg[ F(x(u, v), y(u, v), z(u, v)) \cdot \frac{\partial(x, y)}{\partial(u, v)} \\ &\quad + G(x(u, v), y(u, v), z(u, v)) \cdot \frac{\partial(y, z)}{\partial(u, v)} \\ &\quad + H(x(u, v), y(u, v), z(u, v)) \cdot \frac{\partial(z, x)}{\partial(u, v)} \Bigg] du\,dv,\end{aligned} \tag{2}$$

where

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix}, \quad \frac{\partial(y,z)}{\partial(u,v)} = \begin{vmatrix} \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \\ \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} \end{vmatrix}, \quad \frac{\partial(z,x)}{\partial(u,v)} = \begin{vmatrix} \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} \\ \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \end{vmatrix}.$$

If $S$ is composed of several pieces $S_i$, $i = 1, \ldots, k$, as in Figure 8.4.4, each with its own parametrization $\mathbf{\Phi}_i$, we define

$$\iint_S \eta = \sum_{i=1}^{k} \iint_{S_i} \eta.$$

One must verify that $\iint_S \eta$ does not depend on the choice of parametrization $\mathbf{\Phi}$. This result is essentially (but not obviously) contained in Theorem 4, Section 7.6.

**EXAMPLE 6** Let $\eta = z^2\,dx\,dy$ be a 2-form on $\mathbb{R}^3$, and let $S$ be the upper unit hemisphere in $\mathbb{R}^3$. Find $\iint_S \eta$.

SOLUTION Let us parametrize $S$ by

$$\mathbf{\Phi}(u,v) = (\sin u \cos v, \sin u \sin v, \cos u),$$

where $(u,v) \in D = [0, \pi/2] \times [0, 2\pi]$. By formula (2),

$$\iint_S \eta = \iint_D \cos^2 u \left[\frac{\partial(x,y)}{\partial(u,v)}\right] du\,dv,$$

where

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \cos u \cos v & -\sin u \sin v \\ \cos u \sin v & \sin u \cos v \end{vmatrix}$$
$$= \sin u \cos u \cos^2 v + \cos u \sin u \sin^2 v = \sin u \cos u.$$

Therefore,

$$\iint_S \eta = \iint_D \cos^2 u \cos u \sin u\,du\,dv$$
$$= \int_0^{2\pi} \int_0^{\pi/2} \cos^3 u \sin u\,du\,dv = \int_0^{2\pi} \left[-\frac{\cos^4 u}{4}\right]_0^{\pi/2} dv = \frac{\pi}{2}. \quad \blacktriangle$$

**EXAMPLE 7** Evaluate $\iint_S x\,dy\,dz + y\,dx\,dy$, where $S$ is the oriented surface described by the parametrization $x = u + v$, $y = u^2 - v^2$, $z = uv$, where $(u,v) \in D = [0,1] \times [0,1]$.

SOLUTION By definition, we have

$$\frac{\partial(y,z)}{\partial(u,v)} = \begin{vmatrix} 2u & -2v \\ v & u \end{vmatrix} = 2(u^2+v^2);$$

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 1 & 1 \\ 2u & -2v \end{vmatrix} = -2(u+v).$$

Consequently,

$$\begin{aligned}\iint_S x\,dy\,dz + y\,dx\,dy &= \iint_D [(u+v)(2)(u^2+v^2) + (u^2-v^2)(-2)(u+v)]\,du\,dv \\ &= 4\iint_D (v^3+uv^2)\,du\,dv = 4\int_0^1\int_0^1 (v^3+uv^2)\,du\,dv \\ &= 4\int_0^1 \left[uv^3 + \frac{u^2v^2}{2}\right]_0^1 dv = 4\int_0^1 \left(v^3+\frac{v^2}{2}\right) dv \\ &= \left[v^4 + \frac{2v^3}{3}\right]_0^1 = 1+\frac{2}{3} = \frac{5}{3}. \quad \blacktriangle\end{aligned}$$

## The Integral of 3-Forms over Regions

Finally, we must interpret 3-forms as functions on the elementary subregions of $K$. Let $\nu = f(x,y,z)\,dx\,dy\,dz$ be a 3-form and let $R \subset K$ be an elementary subregion of $K$. Then to each such $R \subset K$ we assign the number

$$\iiint_R \nu = \iiint_R f(x,y,z)\,dx\,dy\,dz, \tag{3}$$

which is just the ordinary triple integral of $f$ over $R$, as described in Section 5.5.

**EXAMPLE 8** Suppose $\nu = (x+z)\,dx\,dy\,dz$ and $R = [0,1]\times[0,1]\times[0,1]$. Evaluate $\iiint_R \nu$.

SOLUTION We compute:

$$\begin{aligned}\iiint_R \nu &= \iiint_R (x+z)\,dx\,dy\,dz = \int_0^1\int_0^1\int_0^1 (x+z)\,dx\,dy\,dz \\ &= \int_0^1\int_0^1 \left[\frac{x^2}{2}+zx\right]_0^1 dy\,dz = \int_0^1\int_0^1 \left(\frac{1}{2}+z\right) dy\,dz = \int_0^1 \left(\frac{1}{2}+z\right) dz \\ &= \left[\frac{z}{2}+\frac{z^2}{2}\right]_0^1 = 1. \quad \blacktriangle\end{aligned}$$

## The Algebra of Forms

We now discuss the algebra (or rules of multiplication) of forms that, together with differentiation of forms, will enable us to state Green's, Stokes', and Gauss' theorems in terms of differential forms.

If $\omega$ is a $k$-form and $\eta$ is an $l$-form on $K$, $0 \leq k+l \leq 3$, there is a product called the ***wedge product*** $\omega \wedge \eta$ of $\omega$ and $\eta$ that is a $k+l$ form on $K$. The wedge product satisfies the following laws:

(i) For each $k$ there is a zero $k$-form 0 with the property that $0+\omega=\omega$ for all $k$-forms $\omega$ and $0 \wedge \eta = 0$ for all $l$-forms $\eta$ if $0 \leq k+l \leq 3$.

(ii) (*Distributivity*) If $f$ is a 0-form, then

$$(f\omega_1 + \omega_2) \wedge \eta = f(\omega_1 \wedge \eta) + (\omega_2 \wedge \eta).$$

(iii) (*Anticommutativity*) $\omega \wedge \eta = (-1)^{kl}(\eta \wedge \omega)$.

(iv) (*Associativity*) If $\omega_1, \omega_2, \omega_3$ are $k_1, k_2, k_3$ forms, respectively, with $k_1 + k_2 + k_3 \leq 3$, then

$$\omega_1 \wedge (\omega_2 \wedge \omega_3) = (\omega_1 \wedge \omega_2) \wedge \omega_3.$$

(v) (*Homogeneity with respect to functions*) If $f$ is a 0-form, then

$$\omega \wedge (f\eta) = (f\omega) \wedge \eta = f(\omega \wedge \eta).$$

Notice that rules (ii) and (iii) actually imply rule (v).

(vi) The following multiplication rules for 1-forms hold:

$$\begin{aligned}
dx \wedge dy &= dx\,dy \\
dy \wedge dx &= -dx\,dy = (-1)(dx \wedge dy) \\
dy \wedge dz &= dy\,dz = (-1)(dz \wedge dy) \\
dz \wedge dx &= dz\,dx = (-1)(dx \wedge dz) \\
dx \wedge dx &= 0,\ dy \wedge dy = 0,\ dz \wedge dz = 0 \\
dx \wedge (dy \wedge dz) &= (dx \wedge dy) \wedge dz = dx\,dy\,dz.
\end{aligned}$$

(vii) If $f$ is a 0-form and $\omega$ is any $k$-form, then $f \wedge \omega = f\omega$.

Using laws (i) to (vii), we can now find a unique product of any $l$-form $\eta$ and any $k$-form $\omega$, if $0 \leq k+l \leq 3$.

**EXAMPLE 9** Show that $dx \wedge dy\,dz = dx\,dy\,dz$.

SOLUTION By rule (vi), $dy\,dz = dy \wedge dz$. Therefore,

$$dx \wedge dy\,dz = dx \wedge (dy \wedge dz) = dx\,dy\,dz. \quad \blacktriangle$$

**EXAMPLE 10** If $\omega = x\,dx + y\,dy$ and $\eta = zy\,dx + xz\,dy + xy\,dz$, find $\omega \wedge \eta$.

SOLUTION Computing $\omega \wedge \eta$, to get

$$\begin{aligned}
\omega \wedge \eta &= (x\,dx + y\,dy) \wedge (zy\,dx + xz\,dy + xy\,dz) \\
&= [(x\,dx + y\,dy) \wedge (zy\,dx)] + [(x\,dx + y\,dy) \wedge (xz\,dy)] \\
&\quad + [(x\,dx + y\,dy) \wedge (xy\,dz)] \\
&= xyz(dx \wedge dx) + zy^2(dy \wedge dx) + x^2z(dx \wedge dy) + xyz(dy \wedge dy) \\
&\quad + x^2y(dx \wedge dz) + xy^2(dy \wedge dz) \\
&= -zy^2\,dx\,dy + x^2z\,dx\,dy - x^2y\,dz\,dx + xy^2\,dy\,dz \\
&= (x^2z - y^2z)\,dx\,dy - x^2y\,dz\,dx + xy^2\,dy\,dz. \quad \blacktriangle
\end{aligned}$$

**EXAMPLE 11** If $\omega = x\,dx - y\,dy$ and $\eta = x\,dy\,dz + z\,dx\,dy$, find $\omega \wedge \eta$.

SOLUTION

$$\begin{aligned}
\omega \wedge \eta &= (x\,dx - y\,dy) \wedge (x\,dy\,dz + z\,dx\,dy) \\
&= [(x\,dx - y\,dy) \wedge (x\,dy\,dz)] + [(x\,dx - y\,dy) \wedge (z\,dx\,dy)] \\
&= (x^2\,dx \wedge dy\,dz) - (xy\,dy \wedge dy\,dz) + (xz\,dx \wedge dx\,dy) \\
&\quad - (yz\,dy \wedge dx\,dy) \\
&= [x^2\,dx \wedge (dy \wedge dz)] - [xy\,dy \wedge (dy \wedge dz)] + [xz\,dx \wedge (dx \wedge dy)] \\
&\quad - [yz\,dy \wedge (dx \wedge dy)] \\
&= x^2\,dx\,dy\,dz - [xy(dy \wedge dy) \wedge dz] + [xz(dx \wedge dx) \wedge dy] \\
&\quad - [yz(dy \wedge dx) \wedge dy] \\
&= x^2\,dx\,dy\,dz - xy(0 \wedge dz) + xz(0 \wedge dy) + [yz(dy \wedge dy) \wedge dx] \\
&= x^2\,dx\,dy\,dz. \quad \blacktriangle
\end{aligned}$$

The last major step in the development of this theory is to show how to differentiate forms. The derivative of a $k$-form is a $(k+1)$-form if $k < 3$, and the derivative of a

3-form is always zero. If $\omega$ is a $k$-form, we shall denote the derivative of $\omega$ by $d\omega$. The operation $d$ has the following properties:

(1) If $f: K \to \mathbb{R}$ is a 0-form, then

$$df = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy + \frac{\partial f}{\partial z}\,dz.$$

(2) *(Linearity)* If $\omega_1$ and $\omega_2$ are $k$-forms, then

$$d(\omega_1 + \omega_2) = d\omega_1 + d\omega_2.$$

(3) If $\omega$ is a $k$-form and $\eta$ is an $l$-form,

$$d(\omega \wedge \eta) = (d\omega \wedge \eta) + (-1)^k(\omega \wedge d\eta).$$

(4) $d(d\omega) = 0$ and $d(dx) = d(dy) = d(dz) = 0$ or, simply, $d^2 = 0$.

Properties (1) to (4) provide enough information to allow us to uniquely differentiate any form.

**EXAMPLE 12** Let $\omega = P(x, y, z)\,dx + Q(x, y, z)\,dy$ be a 1-form on some open set $K \subset \mathbb{R}^3$. Find $d\omega$.

SOLUTION

$$\begin{aligned}
&d[P(x, y, z)\,dx + Q(x, y, z)\,dy] \\
&\quad = d[P(x, y, z) \wedge dx] + d[Q(x, y, z) \wedge dy] && \text{(using 2)} \\
&\quad = (dP \wedge dx) + [P \wedge d(dx)] + (dQ \wedge dy) + [Q \wedge d(dy)] && \text{(using 3)} \\
&\quad = (dP \wedge dx) + (dQ \wedge dy) && \text{(using 4)} \\
&\quad = \left(\frac{\partial P}{\partial x}\,dx + \frac{\partial P}{\partial y}\,dy + \frac{\partial P}{\partial z}\,dz\right) \wedge dx \\
&\qquad + \left(\frac{\partial Q}{\partial x}\,dx + \frac{\partial Q}{\partial y}\,dy + \frac{\partial Q}{\partial z}\,dz\right) \wedge dy && \text{(using 1)} \\
&\quad = \left(\frac{\partial P}{\partial x}\,dx \wedge dx\right) + \left(\frac{\partial P}{\partial y}\,dy \wedge dx\right) + \left(\frac{\partial P}{\partial z}\,dz \wedge dx\right) \\
&\qquad + \left(\frac{\partial Q}{\partial x}\,dx \wedge dy\right) + \left(\frac{\partial Q}{\partial y}\,dy \wedge dy\right) + \left(\frac{\partial Q}{\partial z}\,dz \wedge dy\right) \\
&\quad = -\frac{\partial P}{\partial y}\,dx\,dy + \frac{\partial P}{\partial z}\,dz\,dx + \frac{\partial Q}{\partial x}\,dx\,dy - \frac{\partial Q}{\partial z}\,dy\,dz \\
&\quad = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy + \frac{\partial P}{\partial z}\,dz\,dx - \frac{\partial Q}{\partial z}\,dy\,dz. \quad \blacktriangle
\end{aligned}$$

**EXAMPLE 13** Let $f$ be a 0-form. Using only differentiation rules (1) to (3) and the fact that $d(dx) = d(dy) = d(dz) = 0$, show that $d(df) = 0$.

SOLUTION By rule (1),

$$df = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy + \frac{\partial f}{\partial z}\,dz,$$

and so

$$d(df) = d\left(\frac{\partial f}{\partial x}\,dx\right) + d\left(\frac{\partial f}{\partial y}\,dy\right) + d\left(\frac{\partial f}{\partial z}\,dz\right).$$

Working only with the first term, using rule (3), we get

$$\begin{aligned} d\left(\frac{\partial f}{\partial x}\,dx\right) = d\left(\frac{\partial f}{\partial x} \wedge dx\right) &= d\left(\frac{\partial f}{\partial x}\right) \wedge dx + \frac{\partial f}{\partial x} \wedge d(dx) \\ &= \left(\frac{\partial^2 f}{\partial x^2}\,dx + \frac{\partial^2 f}{\partial y \partial x}\,dy + \frac{\partial^2 f}{\partial z \partial x}\,dz\right) \wedge dx + 0 \\ &= \frac{\partial^2 f}{\partial y \partial x}\,dy \wedge dx + \frac{\partial^2 f}{\partial z \partial x}\,dz \wedge dx \\ &= -\frac{\partial^2 f}{\partial y \partial x}\,dx\,dy + \frac{\partial^2 f}{\partial z \partial x}\,dz\,dx. \end{aligned}$$

Similarly, we find that

$$d\left(\frac{\partial f}{\partial y}\,dy\right) = \frac{\partial^2 f}{\partial x \partial y}\,dx\,dy - \frac{\partial^2 f}{\partial z \partial y}\,dy\,dz$$

and

$$d\left(\frac{\partial f}{\partial z}\,dz\right) = -\frac{\partial^2 f}{\partial x \partial z}\,dz\,dx + \frac{\partial^2 f}{\partial y \partial z}\,dy\,dz.$$

Adding these up, we get $d(df) = 0$ by the equality of mixed partial derivatives. ▲

**EXAMPLE 14** Show that $d(dx\,dy)$, $d(dy\,dz)$, and $d(dz\,dx)$ are all zero.

SOLUTION To prove the first case, we use property (3):

$$d(dx\,dy) = d(dx \wedge dy) = [d(dx) \wedge dy - dx \wedge d(dy)] = 0.$$

The other cases are similar. ▲

**EXAMPLE 15** If $\eta = F(x, y, z)\,dx\,dy + G(x, y, z)\,dy\,dz + H(x, y, z)\,dz\,dx$, find $d\eta$.

SOLUTION By property (2),

$$d\eta = d(F\,dx\,dy) + d(G\,dy\,dz) + d(H\,dz\,dx).$$

We shall compute $d(F\,dx\,dy)$. Using property (3) again, we get

$$d(F\,dx\,dy) = d(F \wedge dx\,dy) = dF \wedge (dx\,dy) + F \wedge d(dx\,dy).$$

By Example 14, $d(dx\,dy) = 0$, so we are left with

$$\begin{aligned} dF \wedge (dx\,dy) &= \left(\frac{\partial F}{\partial x}\,dx + \frac{\partial F}{\partial y}\,dy + \frac{\partial F}{\partial z}\,dz\right) \wedge (dx \wedge dy) \\ &= \left[\frac{\partial F}{\partial x}dx \wedge (dx \wedge dy)\right] + \left[\frac{\partial F}{\partial y}dy \wedge (dx \wedge dy)\right] \\ &\quad + \left[\frac{\partial F}{\partial z}dz \wedge (dx \wedge dy)\right]. \end{aligned}$$

Now

$$\begin{aligned} dx \wedge (dx \wedge dy) &= (dx \wedge dx) \wedge dy = 0 \wedge dy = 0, \\ dy \wedge (dx \wedge dy) &= -dy \wedge (dy \wedge dx) \\ &= -(dy \wedge dy) \wedge dx = 0 \wedge dx = 0, \end{aligned}$$

and

$$dz \wedge (dx \wedge dy) = (-1)^2(dx \wedge dy) \wedge dz = dx\,dy\,dz.$$

Consequently,

$$d(F\,dx\,dy) = \frac{\partial F}{\partial z}\,dx\,dy\,dz.$$

Analogously, we find that

$$d(G\,dy\,dz) = \frac{\partial G}{\partial x}\,dx\,dy\,dz \qquad \text{and} \qquad d(H\,dz\,dx) = \frac{\partial H}{\partial y}\,dx\,dy\,dz.$$

Therefore,

$$d\eta = \left(\frac{\partial F}{\partial z} + \frac{\partial G}{\partial x} + \frac{\partial H}{\partial y}\right) dx\, dy\, dz. \quad \blacktriangle$$

We have now developed all the concepts needed to reformulate Green's, Stokes', and Gauss' theorems in the language of forms.

**THEOREM 13: Green's Theorem** Let $D$ be an elementary region in the $xy$ plane, with $\partial D$ given the counterclockwise orientation. Suppose $\omega = P(x, y)\,dx + Q(x, y)\,dy$ is a 1-form on some open set $K$ in $\mathbb{R}^3$ that contains $D$. Then

$$\int_{\partial D} \omega = \iint_D d\omega.$$

Here $d\omega$ is a 2-form on $K$ and $D$ is in fact a surface in $\mathbb{R}^3$ parametrized by $\mathbf{\Phi}\colon D \to \mathbb{R}^3$, $\mathbf{\Phi}(x, y) = (x, y, 0)$. Because $P$ and $Q$ are explicitly *not* functions of $z$, then $\partial P/\partial z$ and $\partial Q/\partial z = 0$, and by Example 12, $d\omega = (\partial Q/\partial x - \partial P/\partial y)\,dx\,dy$. Consequently, Theorem 13 means nothing more than that

$$\int_{\partial D} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy,$$

which is precisely Green's theorem. Hence, Theorem 13 holds. Likewise, we have the following theorems.

**THEOREM 14: Stokes' Theorem** Let $S$ be an oriented surface in $\mathbb{R}^3$ with a boundary consisting of a simple closed curve $\partial S$ (Figure 8.6.3) oriented as the boundary of $S$ (see Figure 8.2.1). Suppose that $\omega$ is a 1-form on some open set $K$ that contains $S$. Then

$$\int_{\partial S} \omega = \iint_S d\omega.$$

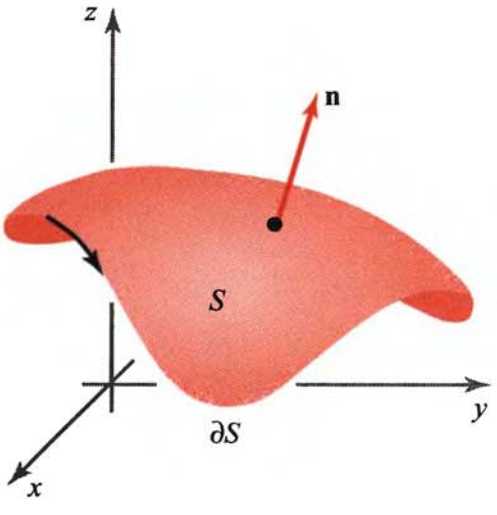



**Figure 8.6.3** An oriented surface to which Stokes' theorem applies.

**THEOREM 15: Gauss' Theorem** Let $W \subset \mathbb{R}^3$ be an elementary region with $\partial W$ given the outward orientation (see Section 8.4). If $\eta$ is a 2-form on some region $K$ containing $W$, then

$$\iint_{\partial W} \eta = \iiint_W d\eta.$$

The reader has probably noticed the strong similarity in the statements of these theorems. In the vector-field formulations, we used divergence for regions in $\mathbb{R}^3$ (Gauss' theorem) and the curl for surfaces in $\mathbb{R}^3$ (Stokes' theorem) and regions in $\mathbb{R}^2$ (Green's theorem). Here we just use the unified notion of the derivative of a differential form for all three theorems; and, in fact, we can state all theorems as one by introducing a little more terminology.

By an ***oriented 2-manifold with boundary*** in $\mathbb{R}^3$ we mean a surface in $\mathbb{R}^3$ whose boundary is a simple closed curve with orientation as described in Section 8.2. By an ***oriented 3-manifold*** in $\mathbb{R}^3$ we mean an elementary region in $\mathbb{R}^3$ (we assume its boundary, which is a surface, is given the outward orientation discussed in Section 8.4). We call the following unified theorem "Stokes' theorem," according to the current convention.

**THEOREM 16: General Stokes' Theorem** Let $M$ be an oriented $k$-manifold in $\mathbb{R}^3$ ($k = 2$ or 3) contained in some open set $K$. Suppose $\omega$ is $(k-1)$-form on $K$. Then

$$\int_{\partial M} \omega = \int_M d\omega.$$

Here the integral is interpreted as a single, double, or triple integral, as is appropriate. In fact, it is this form of Stokes' theorem that generalizes to spaces of arbitrary dimension.

## EXERCISES

**1.** Evaluate $\omega \wedge \eta$ if

(a) $\omega = 2x\,dx + y\,dy$
$\eta = x^3\,dx + y^2\,dy$

(b) $\omega = x\,dx - y\,dy$
$\eta = y\,dx + x\,dy$

(c) $\omega = x\,dx + y\,dy + z\,dz$
$\eta = z\,dx\,dy + x\,dy\,dz + y\,dz\,dx$

(d) $\omega = xy\,dy\,dz + x^2\,dx\,dy$
$\eta = dx + dz$

(e) $\omega = e^{xyz}\,dx\,dy$
$\eta = e^{-xyz}\,dz$

**2.** Prove that

$$(a_1\,dx + a_2\,dy + a_3\,dz) \wedge (b_1\,dy\,dz + b_2\,dz\,dx + b_3\,dx\,dy) = \left(\sum_{i=1}^{3} a_i b_i\right) dx\,dy\,dz.$$

**3.** Find $d\omega$ in the following examples:

(a) $\omega = x^2 y + y^3$
(b) $\omega = y^2 \cos x\,dy + xy\,dx + dz$
(c) $\omega = xy\,dy + (x+y)^2\,dx$
(d) $\omega = x\,dx\,dy + z\,dy\,dz + y\,dz\,dx$
(e) $\omega = (x^2 + y^2)\,dy\,dz$
(f) $\omega = (x^2 + y^2 + z^2)\,dz$
(g) $\omega = \dfrac{-x}{x^2+y^2}\,dx + \dfrac{y}{x^2+y^2}\,dy$
(h) $\omega = x^2 y\,dy\,dz$

**4.** Let $\mathbf{V}\colon K \to \mathbb{R}^3$ be a vector field defined by $\mathbf{V}(x, y, z) = G(x, y, z)\mathbf{i} + H(x, y, z)\mathbf{j} + F(x, y, z)\mathbf{k}$, and let $\eta$ be the 2-form on $K$ given by

$$\eta = F\,dx\,dy + G\,dy\,dz + H\,dz\,dx.$$

Show that $d\eta = (\operatorname{div} \mathbf{V})\,dx\,dy\,dz$.

**5.** If $\mathbf{V} = A(x, y, z)\mathbf{i} + B(x, y, z)\mathbf{j} + C(x, y, z)\mathbf{k}$ is a vector field on $K \subset \mathbb{R}^3$, define the operation $\text{Form}_2$: Vector Fields $\to$ 2-forms by

$$\text{Form}_2(\mathbf{V}) = A\,dy\,dz + B\,dz\,dx + C\,dx\,dy.$$

(a) Show that $\text{Form}_2(\alpha\mathbf{V}_1 + \mathbf{V}_2) = \alpha\ \text{Form}_2(\mathbf{V}_1) + \text{Form}_2(\mathbf{V}_2)$, where $\alpha$ is a real number.
(b) Show that $\text{Form}_2(\text{curl } \mathbf{V}) = d\omega$, where $\omega = A\,dx + B\,dy + C\,dz$.

**6.** Using the differential form version of Stokes' theorem, prove the vector field version in Section 8.2. Repeat for Gauss' theorem.

**7.** Interpret Theorem 16 in the case $k = 1$.

**8.** Let $\omega = (x+y)\,dz + (y+z)\,dx + (x+z)\,dy$, and let $S$ be the upper part of the unit sphere; that is, $S$ is the set of $(x, y, z)$ with $x^2 + y^2 + z^2 = 1$ and $z \geq 0$. $\partial S$ is the unit circle in the $xy$ plane. Evaluate $\int_{\partial S} \omega$ both directly and by Stokes' theorem.

**9.** Let $T$ be the triangular solid bounded by the $xy$ plane, the $xz$ plane, the $yz$ plane, and the plane $2x + 3y + 6z = 12$. Compute

$$\iint_{\partial T} F_1\,dx\,dy + F_2\,dy\,dz + F_3\,dz\,dx$$

directly and by Gauss' theorem, if

(a) $F_1 = 3y,\ F_2 = 18z,\ F_3 = -12$; and
(b) $F_1 = z,\ F_2 = x^2,\ F_3 = y$.

**10.** Evaluate $\iint_S \omega$ where $\omega = z\,dx\,dy + x\,dy\,dz + y\,dz\,dx$ and $S$ is the unit sphere, directly and by Gauss' theorem.

**11.** Let $R$ be an elementary region in $\mathbb{R}^3$. Show that the volume of $R$ is given by the formula

$$v(R) = \frac{1}{3}\iint_{\partial R} x\,dy\,dz + y\,dz\,dx + z\,dx\,dy.$$

**12.** In Section 4.2, we saw that the length $l(\mathbf{c})$ of a curve $\mathbf{c}(t) = (x(t), y(t), z(t))$, $a \le t \le b$, was given by the formula

$$l(\mathbf{c}) = \int d\mathbf{s} = \int_a^b \left(\frac{ds}{dt}\right) dt$$

where, loosely speaking, $(ds)^2 = (dx)^2 + (dy)^2 + (dz)^2$, that is,

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}.$$

Now suppose a surface $S$ is given in parametrized form by $\Phi(u, v) = (x(u, v), y(u, v), z(u, v))$, where $(u, v) \in D$. Show that the area of $S$ can be expressed as

$$A(S) = \iint_D dS$$

where formally $(dS)^2 = (dx \wedge dy)^2 + (dy \wedge dz)^2 + (dz \wedge dx)^2$, a formula requiring interpretation. (HINT:

$$dx = \frac{\partial x}{\partial u}\,du + \frac{\partial x}{\partial v}\,dv,$$

and similarly for $dy$ and $dz$. Use the law of forms for the basic 1-forms $du$ and $dv$. Then $dS$ turns out to be a function times the basic 2-form $du\,dv$, which we can integrate over $D$.)

## REVIEW EXERCISES FOR CHAPTER 8

**1.** Let $\mathbf{F} = 2yz\mathbf{i} + (-x + 3y + 2)\mathbf{j} + (x^2 + z)\mathbf{k}$. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where $S$ is the cylinder $x^2 + y^2 = a^2$, $0 \le z \le 1$ (without the top and bottom). What if the top and bottom are included?

**2.** Let $W$ be a region in $\mathbb{R}^3$ with boundary $\partial W$. Prove the identity

$$\iint_{\partial W} [\mathbf{F} \times (\nabla \times \mathbf{G})] \cdot dS = \iiint_W (\nabla \times \mathbf{F}) \cdot (\nabla \times \mathbf{G})\,dV - \iiint_W \mathbf{F} \cdot (\nabla \times \nabla \times \mathbf{G})\,dV.$$

**3.** Let $\mathbf{F} = x^2y\mathbf{i} + z^8\mathbf{j} - 2xyz\mathbf{k}$. Evaluate the integral of $\mathbf{F}$ over the surface of the unit cube.

4. Verify Green's theorem for the line integral

$$\int_C x^2 y\,dx + y\,dy,$$

when $C$ is the boundary of the region between the curves $y = x$ and $y = x^3$, $0 \le x \le 1$.

5. (a) Show that $\mathbf{F} = (x^3 - 2xy^3)\mathbf{i} - 3x^2y^2\mathbf{j}$ is a gradient vector field.
(b) Evaluate the integral of $\mathbf{F}$ along the path $x = \cos^3\theta$, $y = \sin^3\theta$, $0 \le \theta \le \pi/2$.

6. Can you derive Green's theorem in the plane from Gauss' theorem?

7. (a) Show that $\mathbf{F} = 6xy(\cos z)\mathbf{i} + 3x^2(\cos z)\mathbf{j} - 3x^2y(\sin z)\mathbf{k}$ is conservative (see Section 8.3).
(b) Find $f$ such that $\mathbf{F} = \nabla f$.
(c) Evaluate the integral of $\mathbf{F}$ along the curve $x = \cos^3\theta$, $y = \sin^3\theta$, $z = 0$, $0 \le \theta \le \pi/2$.

8. Let $\mathbf{r}(x, y, z) = (x, y, z)$, $r = \|\mathbf{r}\|$. Show that $\nabla^2(\log r) = 1/r^2$ and $\nabla^2(r^n) = n(n+1)r^{n-2}$.

9. Let the velocity of a fluid be described by $\mathbf{F} = 6xz\mathbf{i} + x^2y\mathbf{j} + yz\mathbf{k}$. Compute the rate at which fluid is leaving the unit cube.

10. Let $\mathbf{F} = x^2\mathbf{i} + (x^2y - 2xy)\mathbf{j} - x^2z\mathbf{k}$. Does there exist a $\mathbf{G}$ such that $\mathbf{F} = \nabla \times \mathbf{G}$?

11. Let $\mathbf{a}$ be a constant vector and $\mathbf{F} = \mathbf{a} \times \mathbf{r}$ [as usual, $\mathbf{r}(x, y, z) = (x, y, z)$]. Is $\mathbf{F}$ conservative? If so, find a potential for it.

12. (For students who have read Section 8.5.) Consider the case of incompressible fluid flow with velocity field $\mathbf{F}$ and density $\rho$.

(a) If $\rho$ is constant for each fixed $t$, then show that $\rho$ is constant in $t$ as well.
(b) If $\rho$ is constant in $t$, then show that $\mathbf{F} \cdot \nabla\rho = 0$.

13. (a) Let $f(x, y, z) = 3xye^{z^2}$. Compute $\nabla f$.
(b) Let $\mathbf{c}(t) = (3\cos^3 t, \sin^2 t, e^t)$, $0 \le t \le \pi$. Evaluate

$$\int_{\mathbf{c}} \nabla f \cdot d\mathbf{s}.$$

(c) Verify directly Stokes' theorem for gradient vector fields $\mathbf{F} = \nabla f$.

14. Using Green's theorem, or otherwise, evaluate $\int_C x^3\,dy - y^3\,dx$, where $C$ is the unit circle $(x^2 + y^2 = 1)$.

15. Evaluate the integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + 3\mathbf{k}$ and where $S$ is the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.

16. (a) State Stokes' theorem for surfaces in $\mathbb{R}^3$.

(b) Let $\mathbf{F}$ be a vector field on $\mathbb{R}^3$ satisfying $\nabla \times \mathbf{F} = \mathbf{0}$. Use Stokes' theorem to show that $\int_C \mathbf{F} \cdot d\mathbf{s} = 0$ where $C$ is a closed curve.

17. Use Green's theorem to find the area of the loop of the curve $x = a \sin\theta \cos\theta$, $y = a \sin^2\theta$, for $a > 0$ and $0 \le \theta \le \pi$.

**18.** Evaluate $\int_C yz\,dx + xz\,dy + xy\,dz$ where $C$ is the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the surface $z = y^2$.

**19.** Evaluate $\int_C (x + y)\,dx + (2x - z)\,dy + (y + z)\,dz$ where $C$ is the perimeter of the triangle connecting $(2, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 6)$ in that order.

**20.** Which of the following are conservative fields on $\mathbb{R}^3$? For those that are, find a function $f$ such that $\mathbf{F} = \nabla f$.

(a) $\mathbf{F}(x, y, z) = 3x^2y\mathbf{i} + x^3\mathbf{j} + 5\mathbf{k}$
(b) $\mathbf{F}(x, y, z) = (x + z)\mathbf{i} - (y + z)\mathbf{j} + (x - y)\mathbf{k}$
(c) $\mathbf{F}(x, y, z) = 2xy^3\mathbf{i} + x^2z^3\mathbf{j} + 3x^2yz^2\mathbf{k}$

**21.** Consider the following two vector fields in $\mathbb{R}^3$:

(i) $\mathbf{F}(x, y, z) = y^2\mathbf{i} - z^2\mathbf{j} + x^2\mathbf{k}$
(ii) $\mathbf{G}(x, y, z) = (x^3 - 3xy^2)\mathbf{i} + (y^3 - 3x^2y)\mathbf{j} + z\mathbf{k}$

(a) Which of these fields (if any) are conservative on $\mathbb{R}^3$? (That is, which are gradient fields?) Give reasons for your answer.
(b) Find potential for the fields that are conservative.
(c) Let $\alpha$ be the path that goes from $(0, 0, 0)$ to $(1, 1, 1)$ by following edges of the cube $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$ from $(0, 0, 0)$ to $(0, 0, 1)$ to $(0, 1, 1)$ to $(1, 1, 1)$. Let $\beta$ be the path from $(0, 0, 0)$ to $(1, 1, 1)$ directly along the diagonal of the cube. Find the values of the line integrals

$$\int_\alpha \mathbf{F} \cdot d\mathbf{s}, \quad \int_\alpha \mathbf{G} \cdot d\mathbf{s}, \quad \int_\beta \mathbf{F} \cdot d\mathbf{s}, \quad \int_\beta \mathbf{G} \cdot d\mathbf{s}.$$

**22.** Consider the *constant* vector field $\mathbf{F}(x, y, z) = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$ in $\mathbb{R}^3$.

(a) Find a scalar field $\phi(x, y, z)$ in $\mathbb{R}^3$ such that $\nabla\phi = \mathbf{F}$ in $\mathbb{R}^3$ and $\phi(0, 0, 0) = 0$.
(b) On the sphere $\Sigma$ of radius 2 about the origin find all the points at which
  (i) $\phi$ is a maximum, and
  (ii) $\phi$ is a minimum.
(c) Compute the maximum and minimum values of $\phi$ on $\Sigma$.

**23.** Let $\mathbf{F}$ be a $C^1$ vector field and suppose $\nabla \cdot \mathbf{F}(x_0, y_0, z_0) > 0$. Show that for a sufficiently small sphere $S$ centered at $(x_0, y_0, z_0)$, the flux of $\mathbf{F}$ out of $S$ is positive.

**24.** Let $B \subset \mathbb{R}^3$ be a planar region, and let $\mathrm{O} \in \mathbb{R}^3$ be a point. If we connect all points in $B$ to O we get a cone, say $C$, with vertex O and base $B$. Show that

$$\text{volume } (C) = \frac{1}{3} \text{ area } (B)\, h,$$

where $h$ is the distance of O from the plane of $B$, using the following steps.

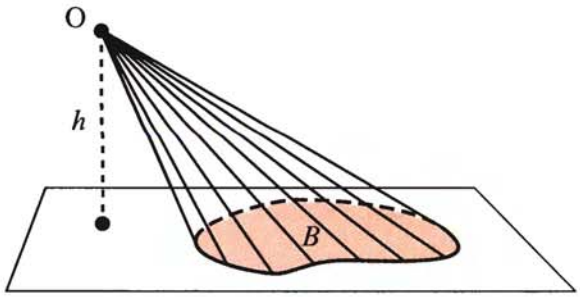



Figure 8.R.1

1. Let O be the origin of the coordinate system. Define $\mathbf{r}(x, y, z) := (x, y, z)$. Evaluate the flux of $\mathbf{r}$ through the boundary of $C$, that is, $\iint_{\partial C} \mathbf{r} \cdot \mathbf{n}\, dA$, where $\mathbf{n}$ is the outward unit normal to $\partial C$.
2. Evaluate the divergence of $\mathbf{r}$ in $C$, that is, $\iiint_C \nabla \cdot \mathbf{r}\, dV$.
3. Use Gauss' theorem, which states that the total divergence of a vector field within a region enclosed by a surface is equal to the flux of that vector field through the surface:

$$\iiint_C \nabla \cdot \mathbf{r}\, dV = \iint_{\partial C} \mathbf{r} \cdot \mathbf{n}\, dA.$$

# Answers to Odd-Numbered Exercises

*Some solutions requiring proofs may be incomplete or be omitted.*

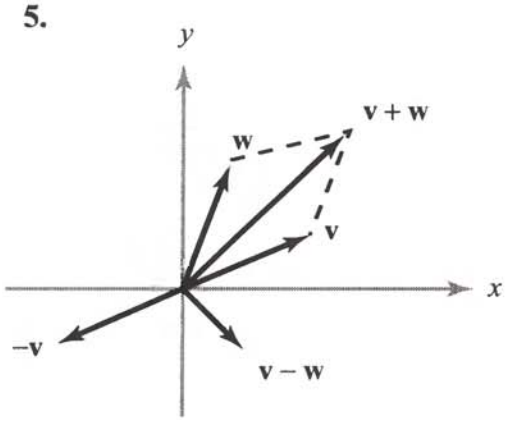
## Chapter 1

### Section 1.1

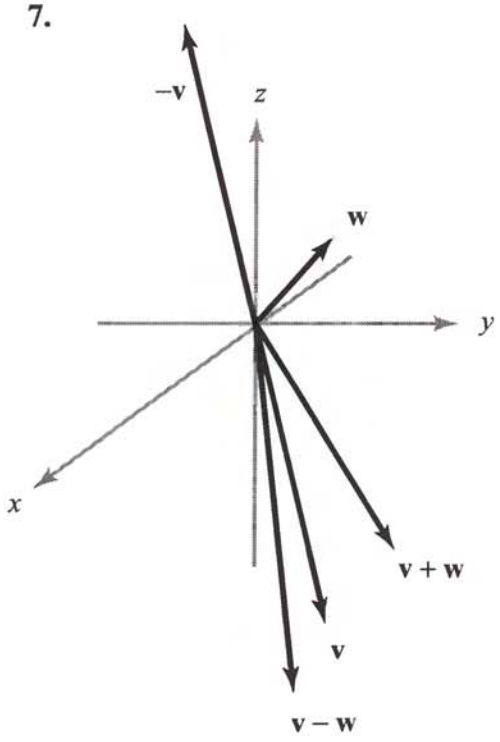
**1.** 4; 17

**3.** $(-104 + 16a, -24 - 4b, -22 + 26c)$

**5.**




**7.**




**9.** $x = 0, z = 0, y \in \mathbb{R}; x = 0, y = 0, z \in \mathbb{R}; y = 0, x, z \in \mathbb{R}; x = 0, y, z \in \mathbb{R}$

**11.** $\{(2s, 7s + 2t, 7t) \mid s \in \mathbb{R}, t \in \mathbb{R}\}$

**13.** $\mathbf{l}(t) = -\mathbf{i} + (t - 1)\mathbf{j} - \mathbf{k}$

**15.** $\mathbf{l}(t) = (2t - 1)\mathbf{i} - \mathbf{j} + (3t - 1)\mathbf{k}$

**17.** $\{s\mathbf{i} + 3s\mathbf{k} - 2t\mathbf{j} \mid 0 \leq s \leq 1, 0 \leq t \leq 1\}$

**19.** If $(x, y, z)$ lies on the line, then $x = 2 + t$, $y = -2 + t$, and $z = -1 + t$. Therefore, $2x - 3y + z - 2 = 4 + 2t + 6 - 3t - 1 + t - 2 = 7$, which is not zero. Hence, no $(x, y, z)$ satisfies both conditions.

**21.** Yes.

**23.** The set of vectors of the form

$$\mathbf{v} = p\mathbf{a} + q\mathbf{b} + r\mathbf{c}$$

where $0 \le p \le 1$, $0 \le q \le 1$, and $0 \le r \le 1$.

**25.** All points of the form

$$(x_0 + t(x_1 - x_0) + s(x_2 - x_0),\ y_0 + t(y_1 - y_0) + s(y_2 - y_0),\ z_0 + t(z_1 - z_0) + s(z_2 - z_0))$$

for real numbers $t$ and $s$.

**27.** If one vertex is placed at the origin and the two adjacent sides are $\mathbf{u}$ and $\mathbf{v}$, the new triangle has sides $b\mathbf{u}$, $b\mathbf{v}$, and $b(\mathbf{u} - \mathbf{v})$.

**29.** $(1, 0, 1) + (0, 2, 1) = (0, 2, 0) + (1, 0, 2)$

**31.** Two such lines (there are many others) are $x = 1, y = t, z = t$ and $x = 1, y = t, z = -t$.

## Section 1.2

**1.** 6

**3.** 99°

**5.** No, it is 75.7; it would be zero only if the vectors were parallel.

**7.** $\|\mathbf{u}\| = \sqrt{5}, \|\mathbf{v}\| = \sqrt{2}, \mathbf{u} \cdot \mathbf{v} = -3$

**9.** $\|\mathbf{u}\| = \sqrt{11}, \|\mathbf{v}\| = \sqrt{62}, \mathbf{u} \cdot \mathbf{v} = -14$

**11.** $\|\mathbf{u}\| = \sqrt{14}, \|\mathbf{v}\| = \sqrt{26}, \mathbf{u} \cdot \mathbf{v} = -17$

**13.** In Exercise 9, $\cos^{-1}(-14/\sqrt{11}\sqrt{62})$; in Exercise 10, $\pi/2$; and in 11, $\cos^{-1}(-17/\sqrt{14}\sqrt{26})$.

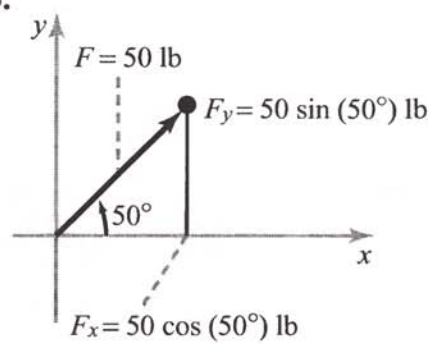
**15.** $-4(-\mathbf{i} + \mathbf{j} + \mathbf{k})/3$

**17.** Any $(x, y, z)$ with $x + y + z = 0$; for example, $(1, -1, 0)$ and $(0, 1, -1)$.

**19.** $\mathbf{i} + 4\mathbf{j}$, $\theta \approx 0.24$ radian east of north

**21.** (a) 12:03 P.M. (b) 4.95 km

**23.**




**25.** (4.9, 4.9, 4.9) and (−4.9, −4.9, 4.9) N.

**27.** (a) $\mathbf{F} = (3\sqrt{2}\mathbf{i} + 3\sqrt{2}\mathbf{j})$ (b) ≈0.322 radian or 18.4° (c) $18\sqrt{2}$

## Section 1.3

**1.** $\begin{vmatrix} 1 & 2 & 1 \\ 3 & 0 & 1 \\ 2 & 0 & 2 \end{vmatrix} = -8, \quad \begin{vmatrix} 3 & 0 & 1 \\ 1 & 2 & 1 \\ 2 & 0 & 2 \end{vmatrix} = 8.$

**3.** $-3\mathbf{i} + \mathbf{j} + 5\mathbf{k}$

**5.** $\sqrt{35}$

**7.** 10

**9.** $\pm\mathbf{k}$

**11.** $\pm(113\mathbf{i} + 17\mathbf{j} - 103\mathbf{k})/\sqrt{23,667}$

**13.** $\mathbf{u} + \mathbf{v} = 3\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}; \mathbf{u} \cdot \mathbf{v} = 6; \|\mathbf{u}\| = \sqrt{6}; \|\mathbf{v}\| = 3; \mathbf{u} \times \mathbf{v} = -3\mathbf{i} + 3\mathbf{k}$

**15.** (a) $x + y + z - 1 = 0$
(b) $x + 2y + 3z - 6 = 0$
(c) $5x + 2z = 25$
(d) $x + 2y - 3z = 13$

**17.** (a) The parallel planes $Ax + By + Cz + D = 0$ and $\sigma Ax + \sigma By + \sigma Cz + D' = 0$ are identical when $D' = \sigma D$ and otherwise never intersect.
(b) In a line.

**19.** The line $x = t, y = 2t, z = -5t$.

**21.** (a) Do the first by working out each side in coordinates, and then use that and $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = -(\mathbf{b} \times \mathbf{c}) \times \mathbf{a}$ to get the second.
(b) Use the identities in part (a) to write the quantity in terms of inner products.
(c) Use the identities in part (a) and collect terms.

**23.** Compute the results of Cramer's rule and check that they satisfy the equation.

**25.** $x - 2y + 3z + 12 = 0$

**27.** $4x - 6y - 10z = 14$

**29.** $10x - 17y + z + 25 = 0$

**31.** For Exercise 19, note that $(2, -3, 1) \cdot (1, 1, 1) = 0$, and so the line and plane are parallel and $(2, -2, -1)$ does not lie in the plane. For Exercise 20, the line and plane are parallel and $(1, -1, 2)$ *does* lie in the plane.

**33.** $\sqrt{2}/13$

**35.** (a) Show that $\mathbf{M}$ satisfies the geometric properties of $\mathbf{R} \times \mathbf{F}$.
(b) $2\sqrt{3}$

**37.** Show that $n_1(\mathbf{N} \times \mathbf{a})$ and $n_2(\mathbf{N} \times \mathbf{b})$ have the same magnitude and direction.

**39.** One method is to write out all terms in the left-hand side and see that the terms involving $\lambda$ all cancel. Another method is to first observe that the determinant is linear in each row or column and that if any row or column is repeated, the answer is zero. Then

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 + \lambda a_1 & b_2 + \lambda b_1 & c_2 + \lambda c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \lambda \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

## Section 1.4

**1.** (a)

| Cylindrical | | | Rectangular | | | Spherical | | |
|---|---|---|---|---|---|---|---|---|
| $r$ | $\theta$ | $z$ | $x$ | $y$ | $z$ | $\rho$ | $\theta$ | $\phi$ |
| 1 | 45° | 1 | $\sqrt{2}/2$ | $\sqrt{2}/2$ | 1 | $\sqrt{2}$ | 45° | 45° |
| 2 | $\pi/2$ | −4 | 0 | 2 | −4 | $2\sqrt{5}$ | $\pi/2$ | $\pi - \arccos(2\sqrt{5}/5)$ |
| 0 | 45° | 10 | 0 | 0 | 10 | 10 | 45° | 0 |
| 3 | $\pi/6$ | 4 | $3\sqrt{3}/2$ | 3/2 | 4 | 5 | $\pi/6$ | $\arccos\frac{4}{5}$ |
| 1 | $\pi/6$ | 0 | $\sqrt{3}/2$ | $\frac{1}{2}$ | 0 | 1 | $\pi/6$ | $\pi/2$ |
| 2 | $3\pi/4$ | −2 | $-\sqrt{2}$ | $\sqrt{2}$ | −2 | $2\sqrt{2}$ | $3\pi/4$ | $3\pi/4$ |

(b)

| Rectangular | | | Spherical | | | Cylindrical | | |
|---|---|---|---|---|---|---|---|---|
| $x$ | $y$ | $z$ | $\rho$ | $\theta$ | $\phi$ | $r$ | $\theta$ | $z$ |
| 2 | 1 | −2 | 3 | $\arctan\frac{1}{2}$ | $\pi - \arccos(2/3)$ | $\sqrt{5}$ | $\arctan\frac{1}{2}$ | −2 |
| 0 | 3 | 4 | 5 | $\pi/2$ | $\arccos(4/5)$ | 3 | $\pi/2$ | 4 |
| $\sqrt{2}$ | 1 | 1 | 2 | $\arctan(\sqrt{2}/2)$ | $\pi/3$ | $\sqrt{3}$ | $\arctan(\sqrt{2}/2)$ | 1 |
| $-2\sqrt{3}$ | −2 | 3 | 5 | $7\pi/6$ | $\arccos\frac{3}{5}$ | 4 | $7\pi/6$ | 3 |

**3.** (a) Rotation by $\pi$ around the $z$ axis
(b) Reflection across the $xy$ plane
(c) Rotation by $\pi/2$ about the $z$ axis together with a radial expansion by a factor of 2

**5.** No; $(\rho, \theta, \phi)$ and $(-\rho, \theta + \pi, \pi - \phi)$ represent the same point.

**7.** (a) $\mathbf{e}_\rho = (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})/\sqrt{x^2 + y^2 + z^2} = (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})/\rho$
$\mathbf{e}_\theta = (-y\mathbf{i} + x\mathbf{j})/\sqrt{x^2 + y^2} = (-y\mathbf{i} + x\mathbf{j})/r$
$\mathbf{e}_\phi = (xz\mathbf{i} + yz\mathbf{j} - (x^2 + y^2)\mathbf{k})/r\rho$
(b) $\mathbf{e}_\theta \times \mathbf{j} = -y\mathbf{k}/\sqrt{x^2 + y^2}$, $\mathbf{e}_\phi \times \mathbf{j} = (xz/r\rho)\mathbf{k} + (r/\rho)\mathbf{i}$

**9.** (a) The length of $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is $(x^2 + y^2 + z^2)^{1/2} = \rho$
(b) $\cos\phi = z/(x^2 + y^2 + z^2)^{1/2}$ (c) $\cos\theta = x/(x^2 + y^2)^{1/2}$

**11.** $0 \le r \le a, 0 \le \theta \le 2\pi$ means that $(r, \theta, z)$ is inside the cylinder with radius $a$ centered on the $z$ axis, and $|z| \le b$ means that it is no more than a distance $b$ from the $xy$ plane.

**13.** $-d/(6\cos\phi) \le \rho \le d/2, 0 \le \theta \le 2\pi$, and $\pi - \cos^{-1}(\frac{1}{3}) \le \phi \le \pi$

**15.** This is a surface whose cross section with each surface $z = c$ is four-petaled rose. The petals shrink to zero as $|c|$ changes from 0 to 1.

## Section 1.5

**1.** 7

**3.** $|\mathbf{x}\cdot\mathbf{y}| = 10 = \sqrt{5}\sqrt{20} = \|\mathbf{x}\|\|\mathbf{y}\|$, so $|\mathbf{x}\cdot\mathbf{y}| \le \|\mathbf{x}\|\|\mathbf{y}\|$ is true.
$\|\mathbf{x}+\mathbf{y}\| = 3\sqrt{5} = \|\mathbf{x}\| + \|\mathbf{y}\|$, so $\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$ is true.

**5.** $|\mathbf{x}\cdot\mathbf{y}| = 5 < \sqrt{65} = \|\mathbf{x}\|\|\mathbf{y}\|$, so $|\mathbf{x}\cdot\mathbf{y}| \le \|\mathbf{x}\|\|\mathbf{y}\|$ is true.
$\|\mathbf{x}+\mathbf{y}\| = \sqrt{28} < \sqrt{5} + \sqrt{13} = \|\mathbf{x}\| + \|\mathbf{y}\|$, so $\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$ is true.

**7.** $AB = \begin{bmatrix} -1 & -1 & 3 \\ -1 & 11 & 3 \\ -6 & 5 & 8 \end{bmatrix}$, $\det A = -5$, $\det B = -24$,
$\det AB = 120 (= \det A \det B)$, $\det(A+B) = -61 (\neq \det A + \det B)$

**9.** HINT: For $k = 2$ use the triangle inequality to show that $\|\mathbf{x}_1 + \mathbf{x}_2\| \le \|\mathbf{x}_1\| + \|\mathbf{x}_2\|$; then for $k = i+1$ note that $\|\mathbf{x}_1 + \mathbf{x}_2 + \cdots + \mathbf{x}_{i+1}\| \le \|\mathbf{x}_1 + \mathbf{x}_2 + \cdots + \mathbf{x}_i\| + \|\mathbf{x}_{i+1}\|$.

**11.** (a) Check $n = 1$ and $n = 2$ directly. Then reduce an $n \times n$ determinant to a sum of $(n-1) \times (n-1)$ determinants and use induction.

(b) The argument is similar to that for part (a). Suppose the first row is multiplied by $\lambda$. The first term of the sum will be $\lambda a_{11}$ times an $(n-1) \times (n-1)$ determinant with no factors of $\lambda$. The other terms obtained (by expanding across the first row) are similar.

**13.** Not necessarily. Try $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

**15.** (a) The sum of two continuous functions and a scalar multiple of a continuous function are continuous.

(b) (i) $(\alpha f + \beta g)\cdot h = \int_0^1 (\alpha f + \beta g)(x)h(x)\,dx$
$= \int_0^1 f(x)h(x)\,dx + \beta \int_0^1 g(x)h(x)\,dx$
$= \alpha f\cdot h + \beta g\cdot h.$

(ii) $f\cdot g = \int_0^1 f(x)g(x)\,dx = \int_0^1 g(x)f(x)\,dx = g\cdot f.$

In conditions (iii) and (iv), the integrand is a perfect square. Therefore, the integral in nonnegative and can be 0 only if the integrand is 0 everywhere. If $f(x) \neq 0$ for some $x$, then it would be positive in a neighborhood of $x$ by continuity, and the integral would be positive.

**17.** Compute the matrix product in both orders and check that you get the identity.

**19.** $(\det A)(\det A^{-1}) = \det(AA^{-1}) = \det(I) = 1$

## Review Exercises for Chapter 1

**1.** $\mathbf{v}+\mathbf{w} = 4\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$; $3\mathbf{v} = 9\mathbf{i} + 12\mathbf{j} + 15\mathbf{k}$; $6\mathbf{v} + 8\mathbf{w} = 26\mathbf{i} + 16\mathbf{j} + 38\mathbf{k}$; $-2\mathbf{v} = -6\mathbf{i} - 8\mathbf{j} - 10\mathbf{k}$; $\mathbf{v}\cdot\mathbf{w} = 4$; $\mathbf{v}\times\mathbf{w} = 9\mathbf{i} + 2\mathbf{j} - 7\mathbf{k}$. Your sketch should display $\mathbf{v}$, $\mathbf{w}$, $3\mathbf{v}$, $6\mathbf{v}$, $8\mathbf{w}$, $6\mathbf{v}+8\mathbf{w}$, $\mathbf{v}\cdot\mathbf{w}$ as the projection of $\mathbf{v}$ along $\mathbf{w}$ and $\mathbf{v}\times\mathbf{w}$ as a vector perpendicular to both $\mathbf{v}$ and $\mathbf{w}$.

**3.** (a) $\mathbf{l}(t) = -\mathbf{i} + (2+t)\mathbf{j} - \mathbf{k}$ (c) $-2x + y + 2z = 9$
(b) $\mathbf{l}(t) = (3t-3)\mathbf{i} + (t+1)\mathbf{j} - t\mathbf{k}$

**5.** (a) 0 (b) 5 (c) $-10$

**7.** (a) $\pi/2$ (b) $5/(2\sqrt{15})$ (c) $-10/(\sqrt{6}\sqrt{17})$

**9.** $\{st\mathbf{a} + s(1-t)\mathbf{b} \mid 0 \le t \le 1 \text{ and } 0 \le s \le 1\}$

**11.** Let $\mathbf{v} = (a_1, a_2, a_3)$, $\mathbf{w} = (b_1, b_2, b_3)$, and apply the CBS inequality.

**13.** The area is the absolute value of

$$\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 \\ b_1 + \lambda a_1 & b_2 + \lambda a_2 \end{vmatrix}.$$

(A multiple of one row of a determinant can be added to another row without changing its value.) Your sketch should show two parallelograms with the same base and height.

**15.** The cosines of the two parts of the angle are equal, because
$\mathbf{a}\cdot\mathbf{v}/\|\mathbf{a}\|\|\mathbf{v}\| = (\mathbf{a}\cdot\mathbf{b} + \|\mathbf{a}\|\|\mathbf{b}\|)/\|\mathbf{v}\| = \mathbf{b}\cdot\mathbf{v}/\|\mathbf{b}\|\|\mathbf{v}\|$.

**17.** $\mathbf{i}\times\mathbf{j} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \mathbf{k}$; etc.

**19.** (a) HINT: The length of the projection of the vector connecting any pair of points, one on each line, onto $(\mathbf{a}_1 \times \mathbf{a}_2)/\|\mathbf{a}_1 \times \mathbf{a}_2\|$ is $d$.
(b) $\sqrt{2}$

**21.** (a) Note that

$$\frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 1 & 0 & 0 \\ x_1 & x_2 - x_1 & x_3 - x_1 \\ y_1 & y_2 - y_1 & y_3 - y_1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{vmatrix}.$$

(b) $\frac{1}{2}$

**23.**

| Rectangular | Spherical | (plot omitted) |
|---|---|---|
| (a) $(\sqrt{2}/2, \sqrt{2}/2, 1)$ | (a) $(\sqrt{2}, \pi/4, \pi/4)$ | |
| (b) $(3\sqrt{3}/2, 3/2, -4)$ | (b) $(5, \pi/6, \arccos(-4/5))$ | |
| (c) $(0, 0, 1)$ | (c) $(1, \pi/4, 0)$ | |
| (d) $(0, -2, 1)$ | (d) $(\sqrt{5}, 3\pi/2, \arccos(\sqrt{5}/5))$ | |
| (e) $(0, 2, 1)$ | (e) $(\sqrt{5}, \pi/2, \arccos(\sqrt{5}/5))$ | |

**25.** $z = r^2\cos 2\theta$; $\cos\phi = \rho\sin^2\phi\cos 2\theta$

**27.** $|\mathbf{x}\cdot\mathbf{y}| = 6 < \sqrt{98} = \|\mathbf{x}\|\|\mathbf{y}\|$; $\|\mathbf{x}+\mathbf{y}\| = \sqrt{33} < \sqrt{14}+\sqrt{7} = \|\mathbf{x}\| + \|\mathbf{y}\|$

**29.** (a) The associative law for matrix multiplication may be checked as follows:

$$[(AB)C]_{ij} = \sum_{k=1}^{n}(AB)_{ik}C_{kj} = \sum_{k=1}^{n}\sum_{l=1}^{n} A_{il}B_{lk}C_{kj}$$

$$= \sum_{l=1}^{n} A_i(BC)_{lj} = [A(BC)]_{ij}.$$

Use this with $C$ taken to be a column vector.

(b) The matrix for the composition is the product matrix.

**31.** $\mathbb{R}^n$ is spanned by the vectors $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n$. If $\mathbf{v} \in \mathbb{R}^n$, then

$$T\mathbf{v} = T\left[\sum_{i=1}^{n}(\mathbf{v}\cdot\mathbf{e}_i)\mathbf{e}_i\right] = \sum_{i=1}^{n}(\mathbf{v}\cdot\mathbf{e}_i)T\mathbf{e}_i.$$

Let $a_{ij} = (T\mathbf{e}_j \cdot \mathbf{e}_i)$, so that

$$T\mathbf{e}_j = \sum_{j=1}^{n} a_{ij}\mathbf{e}_i.$$

Then

$$T\mathbf{v}\cdot\mathbf{e}_k = \sum_{i=1}^{n}(\mathbf{v}\cdot\mathbf{e}_i)a_{ki}.$$

That is, if

$$\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}, \qquad \text{then} \qquad T\mathbf{v} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix},$$

as desired.

**33.** (a) $70\cos\theta + 20\sin\theta$ (b) $(21\sqrt{3}+6)$ ft · lb

**35.** Each side equals $2xy - 7yz + 5z^2 - 48x + 54y - 5z - 96$. (Or switch the first two columns and then subtract the first row from the second.)

**37.** Add the last row to the first and subtract it from the second.

**39.** (a) $\frac{1}{6}\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ (b) 1/3

**41.** Use the fact that $\|\mathbf{a}\|^2 = \mathbf{a} \cdot \mathbf{a}$, expand both sides, and use the definition of $\mathbf{c}$.

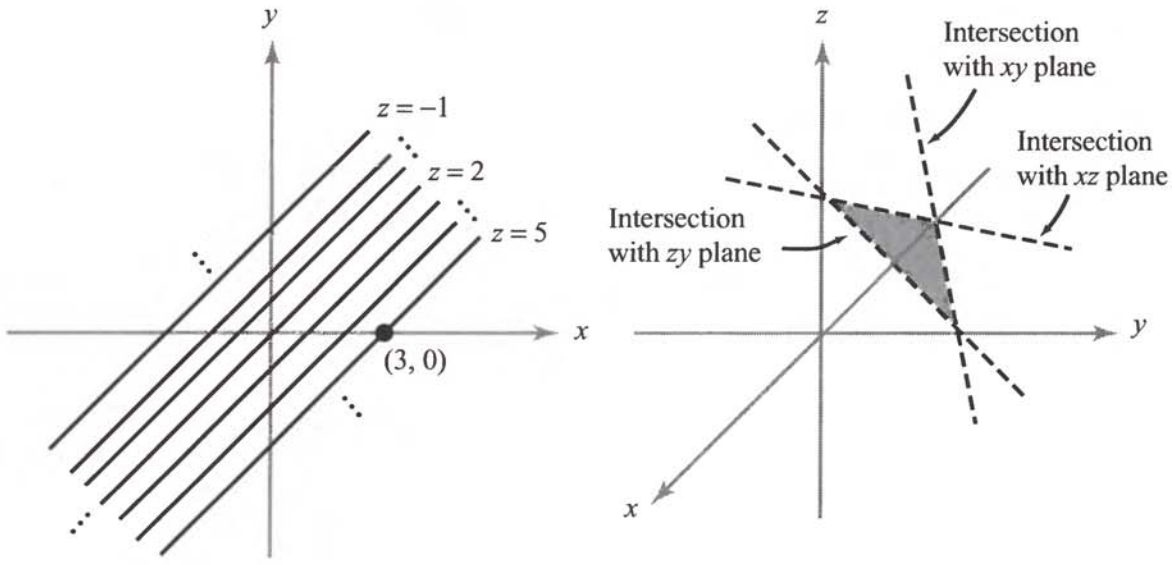
**43.** $(1/\sqrt{38})\mathbf{i} - (6/\sqrt{38})\mathbf{j} + (1/\sqrt{38})\mathbf{k}$ **45.** $(2/\sqrt{5})\mathbf{i} - (1/\sqrt{5})\mathbf{j}$

**47.** $(\sqrt{3}/2)\mathbf{i} + (1/2\sqrt{2})\mathbf{j} + (1/2\sqrt{2})\mathbf{k}$

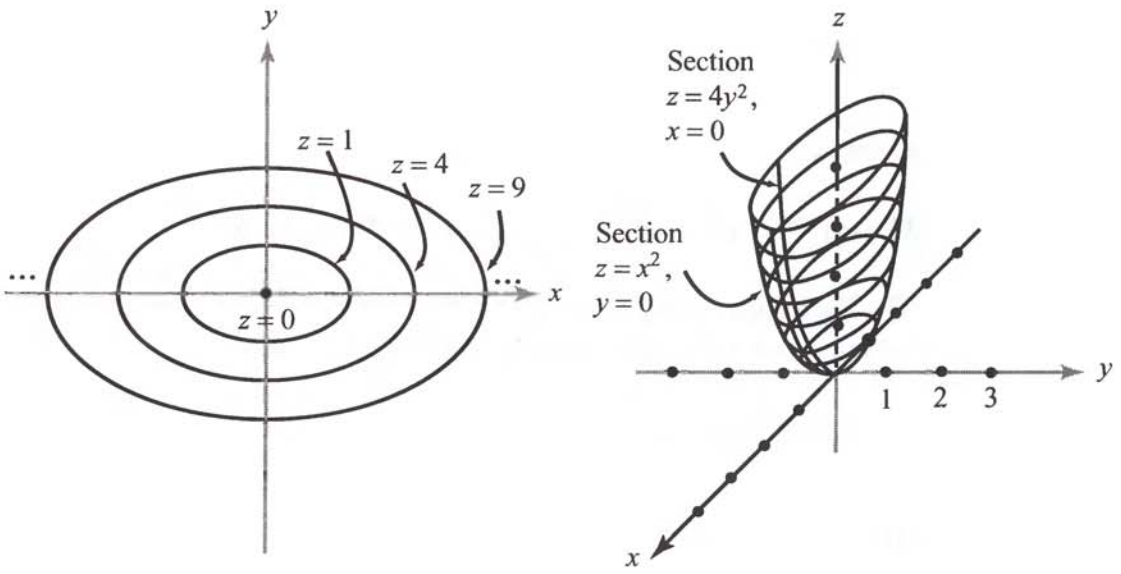
# Chapter 2

## Section 2.1

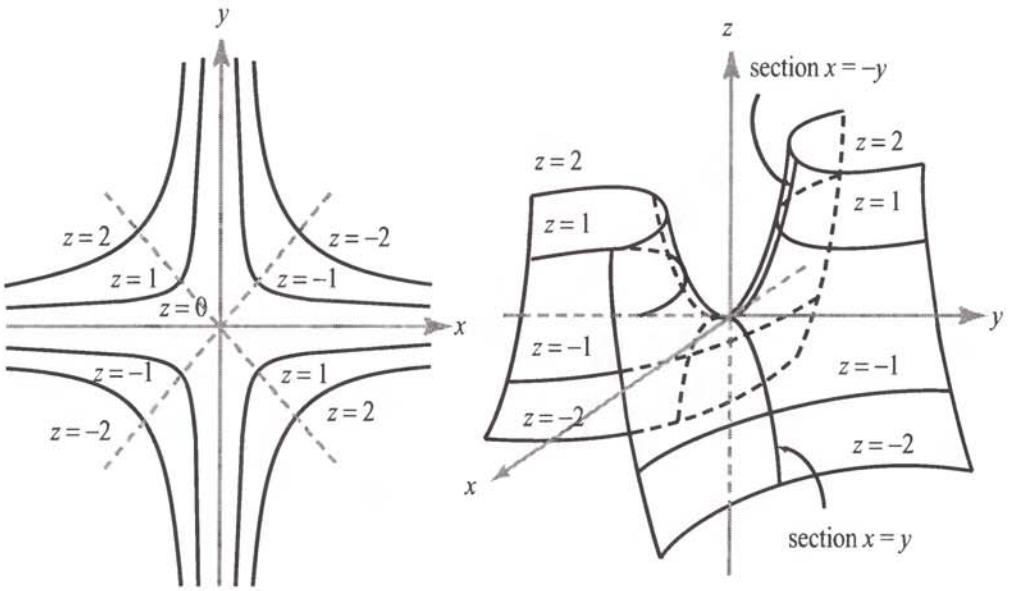
**1.** The level curves and graphs are sketched below. The graph in part (c) is a hyperbolic paraboloid like that of Example 4, but rotated 45° and vertically compressed by a factor of 1/4. To see this, use the variables $u = x + y$ and $v = x - y$. Then $z = (v^2 - u^2)/4$.




(a)



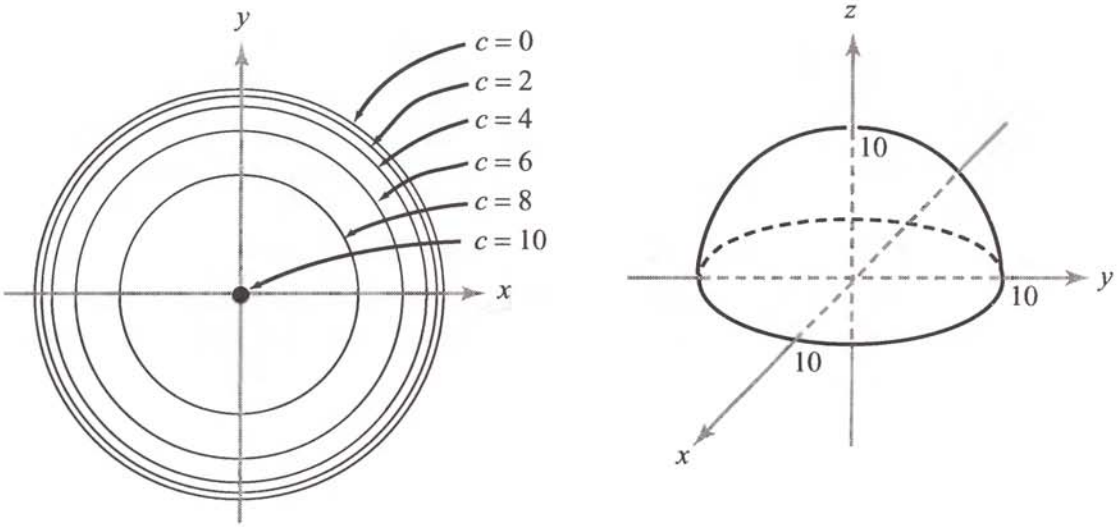

(b)




(c) $z = -xy$

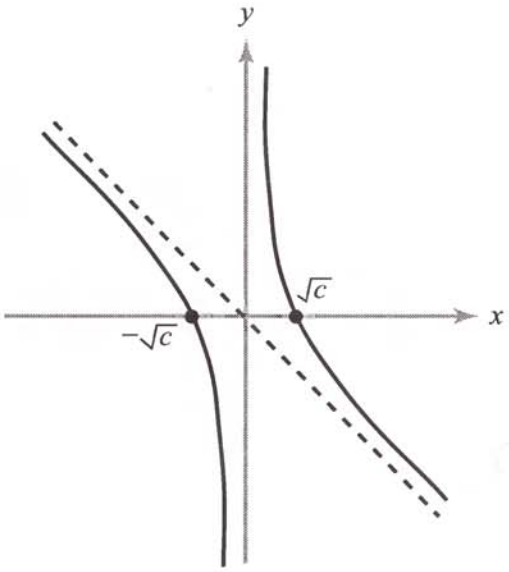
**3.** For Example 2, $z = r(\cos\theta + \sin\theta) + 2$, shape depends on $\theta$; for Example 3, $z = r^2$, shape is independent of $\theta$; for Example 4, $z = r^2(\cos^2\theta - \sin^2\theta)$, shape depends on $\theta$.

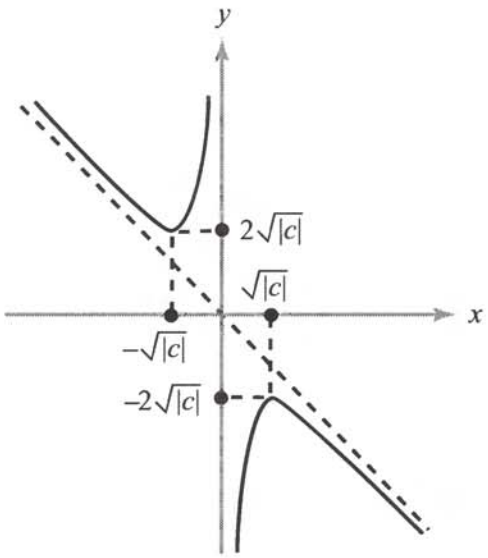
**5.** The level curves are circles $x^2 + y^2 = 100 - c^2$ when $c \leq 10$. The graph is the upper hemisphere of $x^2 + y^2 + z^2 = 100$.




**7.** The level curves are circles, and the graph is a paraboloid of revolution. See Example 3 of this section.

**9.** If $c = 0$, the level curve is the straight line $y = -x$ together with the line $x = 0$. If $c \neq 0$, then $y = -x + (c/x)$. The level curve is a hyperbola with the $y$ axis and the line $y = -x$ as asymptotes. The graph is a hyperbolic paraboloid. Sections along the line $y = ax$ are the parabolas $z = (1 + a)x^2$.



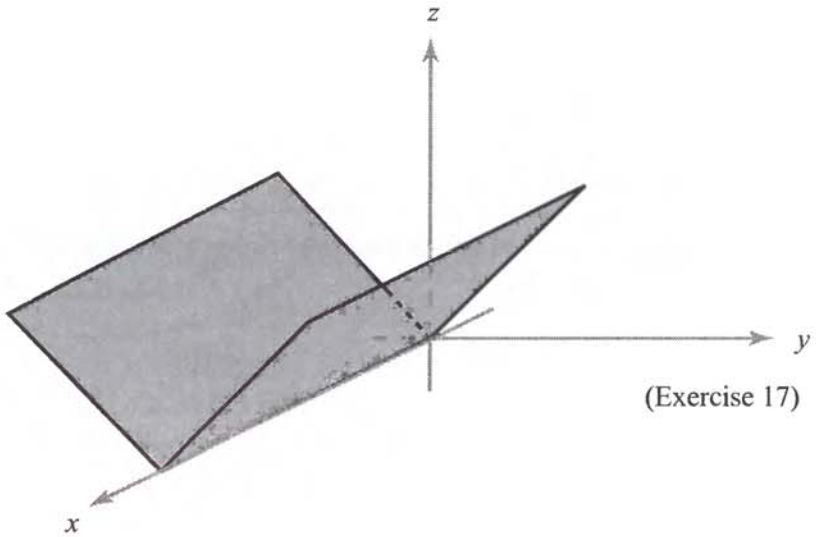




**11.** If $c > 0$, the level surface $f(x, y, z) = c$ is empty. If $c = 0$, the level surface is the point $(0, 0, 0)$. If $c < 0$, the level surface is the sphere of radius $\sqrt{-c}$ centered at $(0, 0, 0)$. A section of the graph determined by $z = a$ is given by $t = -x^2 - y^2 - a^2$, which is a paraboloid of revolution opening down in $xyt$ space.

**13.** If $c < 0$, the level surface is empty. If $c = 0$, the level surface is the $z$ axis. If $c > 0$, it is the right-circular cylinder $x^2 + y^2 = c$ of radius $\sqrt{c}$ whose axis is the $z$ axis. A section of the graph determined by $z = a$ is the paraboloid of revolution $t = x^2 + y^2$. A section determined by $x = b$ is a "trough" with parabolic cross section $t(y, z) = y^2 + b^2$.

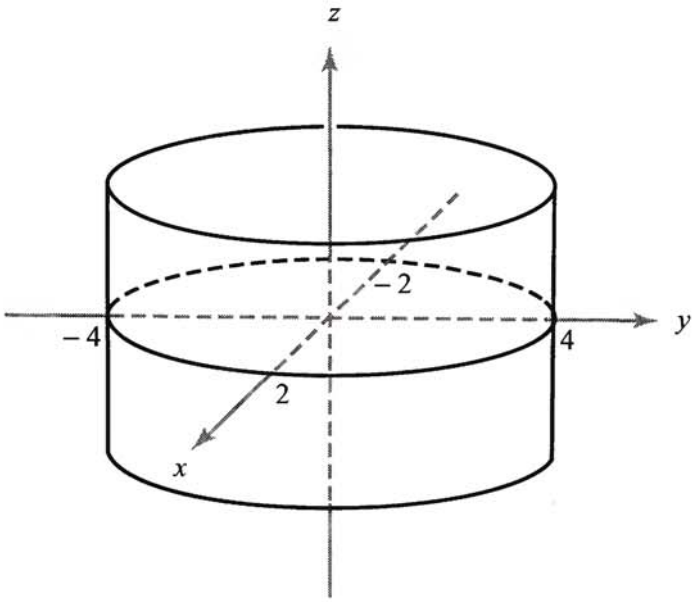
**15.** Setting $u = (x - z)/\sqrt{2}$ and $v = (x + z)/\sqrt{2}$ gives the $u$ and $v$ axes rotated 45° around the $y$ axis from the $x$ and $z$ axes. Because $f = vy\sqrt{2}$, the level surfaces $f = c$ are "cylinders" perpendicular to the $vy$ plane ($z = -x$) whose cross sections are the hyperbolas $vy = c/\sqrt{2}$, so the section $S_{x=a} \cap$ graph $f$ is the hyperbolic paraboloid $t = (z + a)y$ in $yzt$ space [see Exercise 1(c)]. The section $S_{y=b} \cap$ graph $f$ is the plane $t = bx + bz$ in $xzt$ space. The section $S_{z=b} \cap$ graph $f$ is the hyperbolic paraboloid $t = y(x + b)$ in $xyt$ space.

**17.** If $c < 0$, the level curve is empty. If $c = 0$, the level curve is the $x$ axis. If $c > 0$, it is the pair of parallel lines $|y| = c$. The sections of graph with $x$ constant are V-shaped curves $z = |y|$ in $yz$ space. The graph is shown in the accompanying figure.



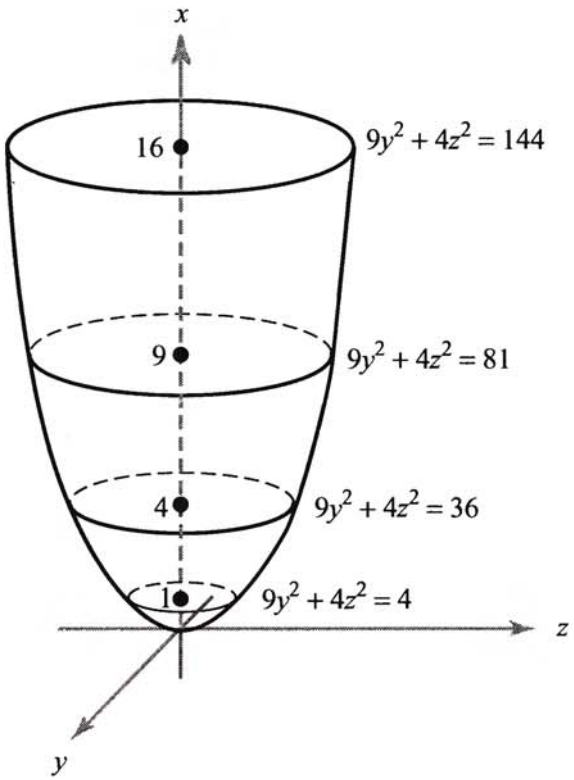

(Exercise 17)

**19.** The value of $z$ does not matter, so we get a "cylinder" of elliptic cross section parallel to the $z$ axis and intersecting the $xy$ plane in the ellipse $4x^2 + y^2 = 16$.

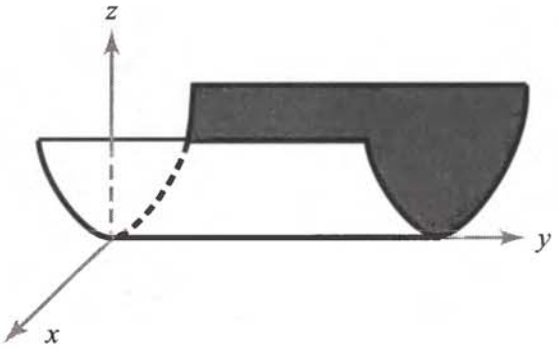



**21.** The value of $x$ does not matter, so we get a "cylinder" parallel to the $x$ axis of hyberbolic cross section intersecting the $yz$ plane in the hyperbola $z^2 - y^2 = 4$.

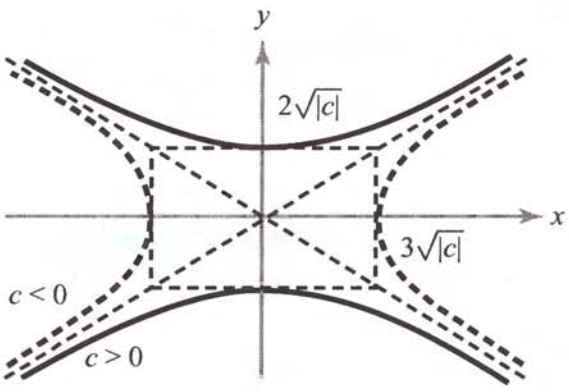
**23.** An elliptic paraboloid with axis along the $x$ axis.



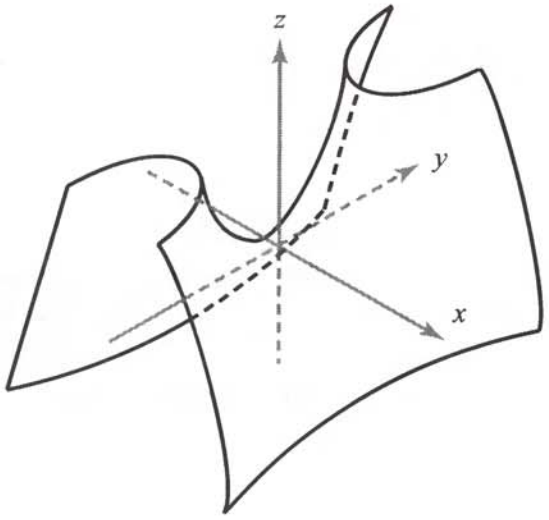

**25.** The value of $y$ does not matter, so we get a "cylinder" of parabolic cross section.



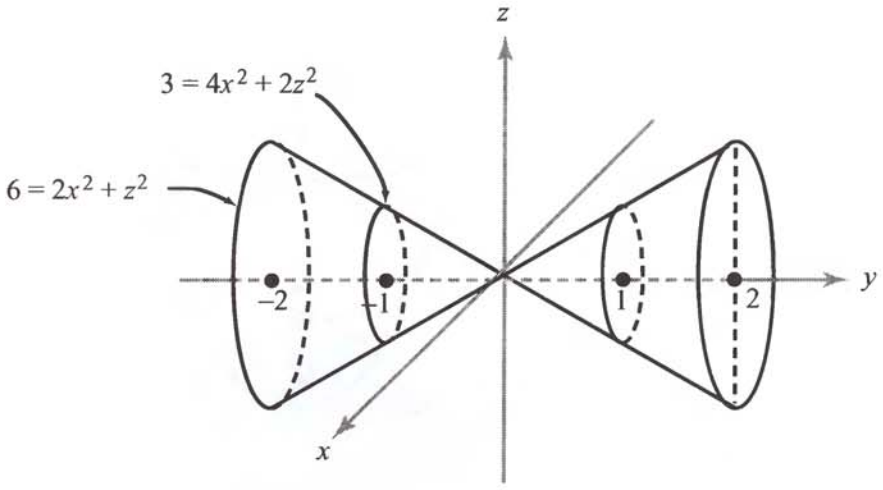

**27.** This is a saddle surface similar to that of Example 4, but the hyperbolas, which are level curves, no longer have perpendicular asymptotes.




Level curves

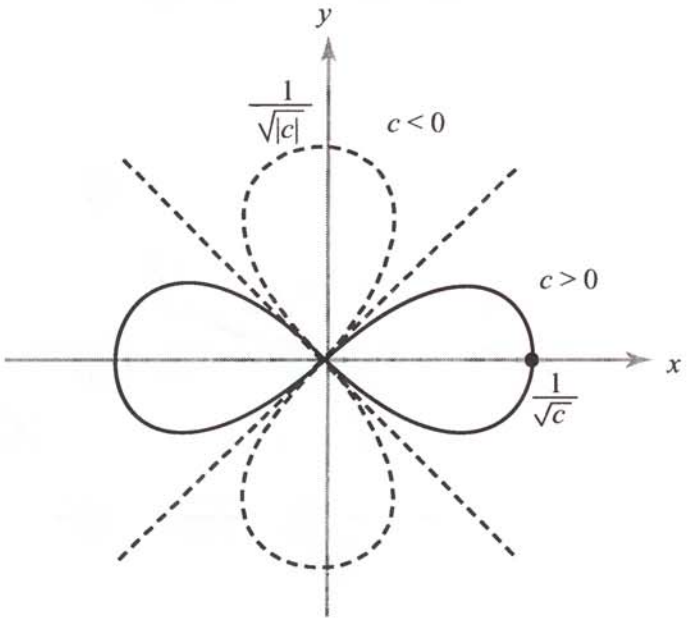



**29.** A double cone with axis along the $y$ axis and elliptical cross sections




**31.** Complete the square to get $(x + 2)^2 + (y - b/2)^2 + (z + \frac{9}{2})^2 = (b^2 + 4b + 97)/4$. This is an ellipsoid with center at $(-2, b/2, -\frac{9}{2})$ and axes parallel to the coordinate axes.

**33.** Level curves are described by $\cos 2\theta = cr^2$. If $c > 0$, then $-\pi/4 \le \theta \le \pi/4$ or $3\pi/4 \le \theta \le 5\pi/4$. If $c < 0$, then $\pi/4 \le \theta \le 3\pi/4$ or $5\pi/4 \le \theta \le 7\pi/4$. In either case you get a figure-eight shape, called a *lemniscate*, through the origin. (Such shapes were first studied by Jacques Bernoulli and are sometimes called Bernoulli's lemniscates.)




## Section 2.2

**1.** If $(x_0, y_0) \in A$, then $|x_0| < 1$ and $|y_0| < 1$. Let $r = \min(1 - |x_0|, 1 - |y_0|)$. Prove that $D_r(x_0, y_0) \subset A$ either analytically or by drawing a figure.

**3.** Let $r = \min(2 - \sqrt{x_0^2 + y_0^2}, \sqrt{x_0^2 + y_0^2} - \sqrt{2})$.

**5.** (a) 0 (b) $-1/2$ (c) 1

**7.** (a) 5 (b) 0 (c) $2x$

**9.** (a) 0 (b) $-1/2$ (c) 0

**11.** (a) Compose $f(x, y) = xy$ with $g(t) = (\sin t)/t$ for $t \neq 0$ and $g(0) = 1$.
(b) 0 (c) 0

**13.** (a) 1 (b) $\|\mathbf{x}_0\|$ (c) $(1, e)$
(d) Limit doesn't exist (look at the limits for $x = 0$ and $y = 0$ separately).

**15.** Use parts (ii) and (iii) of Theorem 4.

**17.** (a) Let the value of the function be 1 at (0, 0). (b) No.

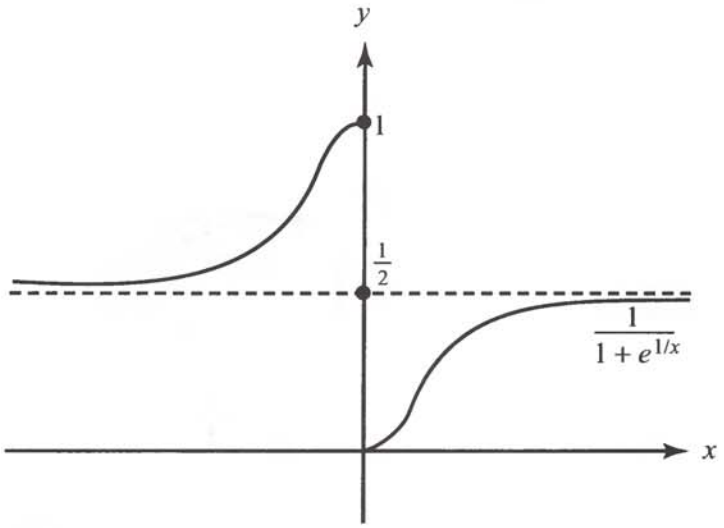
**19.** For $|x - 2| < \delta = \sqrt{\varepsilon + 4} - 2$, we have $|x^2 - 4| = |x - 2||x + 2| < \delta(\delta + 4) = \varepsilon$. By Theorem 3(iii), $\lim_{x \to 2} x^2 = (\lim_{x \to 2} x)^2 = 2^2 = 4$.

**21.** Let $r = \|\mathbf{x} - \mathbf{y}\|/2$. If $\|\mathbf{z} - \mathbf{y}\| \le r$, let $f(\mathbf{z}) = \|\mathbf{z} - \mathbf{y}\|/r$. If $\|\mathbf{z} - \mathbf{y}\| > r$, let $f(\mathbf{z}) = 1$.

**23.** (a) $\lim_{x\to b^+} f(x) = L$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $x > b$ and $0 < x - b < \delta$ imply $|f(x) - L| < \varepsilon$.

(b) $\lim_{x\to 0^-}(1/x) = -\infty$, $\lim_{t\to -\infty} e^t = 0$, and so $\lim_{x\to 0^-} e^{1/x} = 0$. Hence $\lim_{x\to 0^-} 1/(1 + e^{1/x}) = 1$. The other limit is 0.

(c)




**25.** If $\varepsilon > 0$ and $\mathbf{x}_0$ are given, let $\delta = (\varepsilon/K)^{1/\alpha}$. Then $\|f(\mathbf{x}) - f(\mathbf{x}_0)\| < K\delta^\alpha = \varepsilon$ whenever $\|\mathbf{x} - \mathbf{x}_0\| < \delta$. Notice that the choice of $\delta$ does not depend on $\mathbf{x}_0$. This means that $f$ is *uniformly continuous*.

**27.** (a) Choose $\delta < 1/500$. (b) Choose $\delta < 0.002$.

## Section 2.3

**1.** (a) $\partial f/\partial x = y$; $\partial f/\partial y = x$
(b) $\partial f/\partial x = ye^{xy}$; $\partial f/\partial y = xe^{xy}$
(c) $\partial f/\partial x = \cos x \cos y - x \sin x \cos y$; $\partial f/\partial y = -x \cos x \sin y$
(d) $\partial f/\partial x = 2x[1 + \log(x^2 + y^2)]$; $\partial f/\partial y = 2y[1 + \log(x^2 + y^2)]$; $(x, y) \neq (0, 0)$

**3.** (a) $\partial w/\partial x = (1 + 2x^2)\exp(x^2 + y^2)$; $\partial w/\partial y = 2xy \exp(x^2 + y^2)$
(b) $\partial w/\partial x = -4xy^2/(x^2 - y^2)^2$; $\partial w/\partial y = 4yx^2/(x^2 - y^2)^2$
(c) $\partial w/\partial x = ye^{xy}\log(x^2 + y^2) + 2xe^{xy}/(x^2 + y^2)$;
$\partial w/\partial y = xe^{xy}\log(x^2 + y^2) + 2ye^{xy}/(x^2 + y^2)$
(d) $\partial w/\partial x = 1/y$; $\partial w/\partial y = -x/y^2$
(e) $\partial w/\partial x = -y^2 e^{xy} \sin ye^{xy} \sin x + \cos ye^{xy} \cos x$;
$\partial w/\partial y = (xye^{xy} + e^{xy})(-\sin ye^{xy} \sin x)$

**5.** $z = 6x + 3y - 11$

**7.** (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 & e^z \\ 2xy & x^2 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} e^y & xe^y - \sin y \\ 1 & 0 \\ 1 & e^y \end{bmatrix}$ (d) $\begin{bmatrix} (y + xy^2)e^{xy} & (x + x^2 y)e^{xy} \\ \sin y & x \cos y \\ 5y^2 & 10xy \end{bmatrix}$

**9.** At $z = 1$.

**11.** Both are $xye^{xy}$.

**13.** (a) $\nabla f = (e^{-x^2-y^2-z^2}(-2x^2+1), -2xye^{-x^2-y^2-z^2}, -2xze^{-x^2-y^2-z^2})$
(b) $\nabla f = (x^2+y^2+z^2)^{-2}(yz(y^2+z^2-x^2), xz(x^2+z^2-y^2), xy(x^2+y^2-z^2))$
(c) $\nabla f = (z^2e^x \cos y, -z^2e^x \sin y, 2ze^x \cos y)$

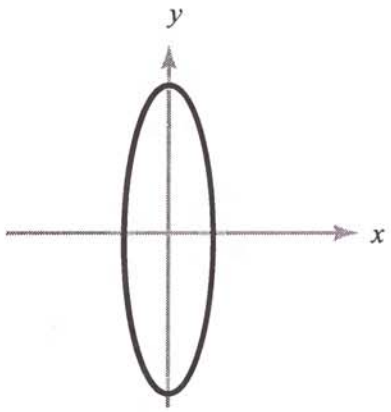
**15.** $2x + 6y - z = 5$

**17.** $-2\mathbf{k}$

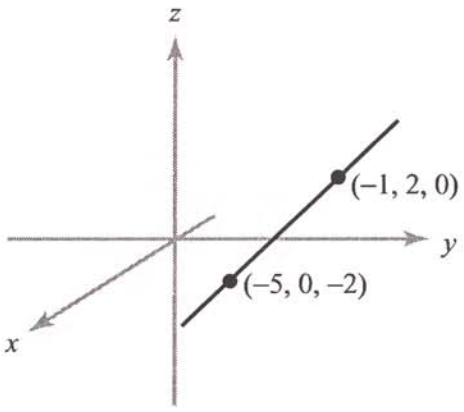
**19.** They are constant. Show that the derivative is the zero matrix.

## Section 2.4

**1.** This curve is the ellipse $(y^2/16) + x^2 = 1$:




**3.** This curve is the straight line through $(-1, 2, 0)$ with direction $(2, 1, 1)$:




**5.** $6\mathbf{i} + 6t\mathbf{j} + 3t^2\mathbf{k}$

**7.** $(-2 \cos t \sin t, 3 - 3t^2, 1)$

**9.** $\mathbf{c}'(t) = (e^t, -\sin t)$

**11.** $\mathbf{c}'(t) = (t \cos t + \sin t, 4)$

**13.** Horizontal when $t = (R/v)n\pi$, $n$ an integer; speed is zero if $n$ is even; speed is $2v$ if $n$ is odd.

**15.** $(\sin 3, \cos 3, 2) + (3 \cos 3, -3 \sin 3, 5)(t - 1)$

**17.** $(8, 8, 0)$

**19.** $(8, 0, 1)$

## Section 2.5

**1.** Use parts (i), (ii), and (iii) of Theorem 10. The derivative at $\mathbf{x}$ is $2(f(\mathbf{x})+1)\mathbf{D}f(\mathbf{x})$.

**3.** (a) $h(x,y) = f(x, u(x,y)) = f(p(x), u(x,y))$. We use $p$ here solely as notation: $p(x) = x$.

$$\text{Written out:} \quad \frac{\partial h}{\partial x} = \frac{\partial f}{\partial p}\frac{dp}{dx} + \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} = \frac{\partial f}{\partial p} + \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} \qquad \text{because} \qquad \frac{dp}{dx} = \frac{dx}{dx} = 1$$

JUSTIFICATION: Call $(p, u)$ the variables of $f$. To use the chain rule we must express $h$ as a composition of functions; that is, first find $g$ such that $h(x,y) = f(g(x,y))$. Let $g(x,y) = (p(x), u(x,y))$. Therefore, $\mathbf{D}h = (\mathbf{D}f)(\mathbf{D}g)$. Then

$$\begin{bmatrix} \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial p} & \frac{\partial f}{\partial u} \end{bmatrix} \begin{bmatrix} \frac{\partial g_1}{\partial x} & \frac{\partial g_1}{\partial y} \\ \frac{\partial g_2}{\partial x} & \frac{\partial g_2}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial p} & \frac{\partial f}{\partial u} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial f}{\partial p} + \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} & \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} \end{bmatrix},$$

and so $\dfrac{\partial h}{\partial x} = \dfrac{\partial f}{\partial p} + \dfrac{\partial f}{\partial u}\dfrac{\partial u}{\partial x}$. You may see $\dfrac{\partial h}{\partial x} = \dfrac{\partial f}{\partial x} + \dfrac{\partial f}{\partial u}\dfrac{\partial u}{\partial x}$ as an answer. This requires careful interpretation because of possible ambiguity about the meaning of $\partial f/\partial x$, which is why the name $p$ was used.

(b) $\dfrac{\partial h}{\partial x} = \dfrac{\partial f}{\partial x} + \dfrac{\partial f}{\partial u}\dfrac{du}{dx} + \dfrac{\partial f}{\partial v}\dfrac{dv}{dx}$ (c) $\dfrac{\partial h}{\partial x} = \dfrac{\partial f}{\partial u}\dfrac{\partial u}{\partial x} + \dfrac{\partial f}{\partial v}\dfrac{\partial v}{\partial x} + \dfrac{\partial f}{\partial w}\dfrac{dw}{dx}$

**5.** Compute each in two ways; the answers are

(a) $(f \circ \mathbf{c})'(t) = e^t(\cos t - \sin t)$
(b) $(f \circ \mathbf{c})'(t) = 15t^4 \exp(3t^5)$
(c) $(f \circ \mathbf{c})'(t) = (e^{2t} - e^{-2t})[1 + \log(e^{2t} + e^{-2t})]$
(d) $(f \circ \mathbf{c})'(t) = (1 + 4t^2)\exp(2t^2)$

**7.** Use Theorem 10(iii) and replace matrices by vectors.

**9.** $(f \circ g)(x,y) = (\tan(e^{x-y} - 1) - e^{x-y}, e^{2(x-y)} - (x-y)^2)$ and $\mathbf{D}(f \circ g)(1,1) = \begin{bmatrix} 0 & 0 \\ 2 & -2 \end{bmatrix}$.

**11.** $\frac{1}{2}\cos(1)\cos(\log\sqrt{2})$

**13.** $-2\cos t \sin t e^{\sin t} + \sin^4 t + \cos^3 t e^{\sin t} - 3\cos^2 t \sin^2 t$ for both (a) and (b).

**15.** $(2, 0)$

**17.** (a) $G(x, y(x)) = 0$ and so $\dfrac{\partial G}{\partial x} + \dfrac{\partial G}{\partial y}\dfrac{dy}{dx} = 0.$

(b) $$\begin{bmatrix} \dfrac{dy_1}{dx} \\ \dfrac{dy_2}{dx} \end{bmatrix} = -\begin{bmatrix} \dfrac{\partial G_1}{\partial y_1} & \dfrac{\partial G_1}{\partial y_2} \\ \dfrac{\partial G_2}{\partial y_1} & \dfrac{\partial G_2}{\partial y_2} \end{bmatrix}^{-1} \begin{bmatrix} \dfrac{\partial G_1}{\partial x} \\ \dfrac{\partial G_2}{\partial x} \end{bmatrix}$$ where $^{-1}$ means the inverse matrix.

The first component of this equation reads

$$\frac{dy_1}{dx} = \frac{-\dfrac{\partial G_1}{\partial x}\dfrac{\partial G_2}{\partial y_2} + \dfrac{\partial G_2}{\partial x}\dfrac{\partial G_1}{\partial y_2}}{\dfrac{\partial G_1}{\partial y_1}\dfrac{\partial G_2}{\partial y_2} - \dfrac{\partial G_2}{\partial y_1}\dfrac{\partial G_1}{\partial y_2}}.$$

(c) $$\frac{dy}{dx} = \frac{-2x}{3y^2 + e^y}$$

**19.** Apply the chain rule to $\partial G/\partial T$ where $G(t(T, P), p(T, P), V(T, P)) = P(V - b)e^{a/RVT} - RT$ is identically 0; $t(T, P) = T$; and $p(T, P) = P$.

**21.** Define $R_1(\mathbf{h}) = f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - [\mathbf{D}f(\mathbf{x}_0)]\mathbf{h}$.

**23.** Let $g_1$ and $g_2$ be $C^1$ functions from $\mathbb{R}^3$ to $\mathbb{R}$ such that $g_1(\mathbf{x}) = 1$ for $\|\mathbf{x}\| < \sqrt{2}/3$; $g_1(\mathbf{x}) = 0$ for $\|\mathbf{x}\| > 2\sqrt{2}/3$; $g_2(\mathbf{x}) = 1$ for $\|\mathbf{x} - (1, 1, 0)\| < \sqrt{2}/3$; and $g_2(\mathbf{x}) = 0$ for $\|\mathbf{x} - (1, 1, 0)\| > 2\sqrt{2}/3$. (See Exercise 22.) Let

$$h_1(\mathbf{x}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

and put $f(\mathbf{x}) = g_1(\mathbf{x})h_1(\mathbf{x}) + g_2(\mathbf{x})h_2(\mathbf{x})$.

**25.** Proof of rule (iii) follows:

$$\begin{aligned} &\frac{|h(\mathbf{x}) - h(\mathbf{x}_0) - [f(\mathbf{x}_0)\mathbf{D}g(\mathbf{x}_0) + g(\mathbf{x}_0)\mathbf{D}f(\mathbf{x}_0)](\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|} \\ &\qquad \le |f(\mathbf{x}_0)|\frac{|g(\mathbf{x}) - g(\mathbf{x}_0) - \mathbf{D}g(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|} \\ &\qquad\quad + |g(\mathbf{x}_0)|\frac{|f(\mathbf{x}) - f(\mathbf{x}_0) - \mathbf{D}f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|} \\ &\qquad\quad + \frac{|f(\mathbf{x}) - f(\mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|}\frac{|g(\mathbf{x}) - g(\mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|}\|\mathbf{x} - \mathbf{x}_0)\|. \end{aligned}$$

As $\mathbf{x} \to \mathbf{x}_0$, the first two terms go to 0 by the differentiability of $f$ and $g$. The third does so because $|f(\mathbf{x}) - f(\mathbf{x}_0)|/\|\mathbf{x} - \mathbf{x}_0\|$ and $|g(\mathbf{x}) - g(\mathbf{x}_0)|/\|\mathbf{x} - \mathbf{x}_0\|$ are bounded by a constant, say $M$, on some ball $D_r(\mathbf{x}_0)$. To see this, choose $r$ small enough that $[f(\mathbf{x}) - f(\mathbf{x}_0)]/\|\mathbf{x} - \mathbf{x}_0\|$ is within 1 of $\mathbf{D}f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)/\|\mathbf{x} - \mathbf{x}_0\|$ if $\|\mathbf{x} - \mathbf{x}_0\| < r$. Then we have

$|f(\mathbf{x}) - f(\mathbf{x}_0)|/\|\mathbf{x} - \mathbf{x}_0\| \leq 1 + |\mathbf{D}f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)|/\|\mathbf{x} - \mathbf{x}_0\| = 1 + |\nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0)|/\|\mathbf{x} - \mathbf{x}_0\| \leq 1 + \|\nabla f(\mathbf{x}_0)\|$ by the Cauchy–Schwarz inequality.

The proof of rule (iv) follows from rule (iii) and the special case of the quotient rule, with $f$ identically 1; that is, $\mathbf{D}(1/g)(\mathbf{x}_0) = [-1/g(\mathbf{x}_0)^2]\mathbf{D}g(\mathbf{x}_0)$. To obtain this answer, note that on some small ball $D_r(\mathbf{x}_0)$, $g(\mathbf{x}) > m > 0$. Use the triangle inequality and the Schwarz inequality to show that

$$\frac{\left|\dfrac{1}{g(\mathbf{x})} - \dfrac{1}{g(\mathbf{x})} + \dfrac{1}{g(\mathbf{x}_0)^2}\mathbf{D}g(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)\right|}{\|\mathbf{x} - \mathbf{x}_0\|}$$

$$\leq \frac{1}{|g(\mathbf{x})|}\frac{1}{|g(\mathbf{x}_0)|}\frac{|g(\mathbf{x}) - g(\mathbf{x}_0) - \mathbf{D}g(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|}$$

$$+ \frac{|g(\mathbf{x}) - g(\mathbf{x}_0)|}{|g(\mathbf{x})|g(\mathbf{x}_0)^2}\frac{|\mathbf{D}g(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|}$$

$$\leq \frac{1}{m^2}\frac{|g(\mathbf{x}) - g(\mathbf{x}_0) - \mathbf{D}g(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)|}{\|\mathbf{x} - \mathbf{x}_0\|} + \frac{\|\nabla g(\mathbf{x}_0)\|}{m^3}|g(\mathbf{x}) - g(\mathbf{x}_0)|.$$

These last two terms both go to 0, because $g$ is differentiable and continuous.

**27.** First find formula for $(\partial/\partial x)(F(x, x))$, using the chain rule. Let $F(x, z) = \int_0^x f(z, y)\,dy$ and use the fundamental theorem of calculus.

**29.** By Exercise 26 and Theorem 10(iii) (Exercise 25), each component of $k$ is differentiable and $\mathbf{D}k_i(\mathbf{x}_0) = f(\mathbf{x}_0)\mathbf{D}g_i(\mathbf{x}_0) + g_i(\mathbf{x}_0)\mathbf{D}f(\mathbf{x}_0)$. Because $[\mathbf{D}g_i(\mathbf{x}_0)]\mathbf{y}$ is the $i$th component of $[\mathbf{D}g(\mathbf{x}_0)]y$ and $[\mathbf{D}f(\mathbf{x}_0)]\mathbf{y}$ is the number $\nabla f(\mathbf{x}_0) \cdot \mathbf{y}$, we get $[\mathbf{D}k(\mathbf{x}_0)]\mathbf{y} = f(\mathbf{x}_0)[\mathbf{D}g(\mathbf{x}_0)]\mathbf{y} + [\mathbf{D}f(\mathbf{x}_0)]\mathbf{y}[g(\mathbf{x}_0)] = f(x_0)[\mathbf{D}g\,(x_0)]\mathbf{y} + [\nabla f(\mathbf{x}_0) \cdot \mathbf{y}]g(\mathbf{x}_0)$.

## Section 2.6

**1.** $\nabla f(1, 1, 2) \cdot \mathbf{v} = (4, 3, 4) \cdot (1/\sqrt{5}, 2/\sqrt{5}, 0) = 2\sqrt{5}$

**3.** (a) $17e^e/13$ (b) $e/\sqrt{3}$ (c) 0

**5.** (a) $z + 9x = 6y - 6$ (b) $z + y = \pi/2$ (c) $z = 1$

**7.** (a) $-\dfrac{1}{3\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k})$ (b) $2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ (c) $-\frac{2}{9}(\mathbf{i} + \mathbf{j} + \mathbf{k})$

**9.** $\mathbf{k}$

**11.** The graph of $f$ is the level surface $0 = F(x, y, z) = f(x, y) - z$. Therefore, the tangent plane is given by

$$0 = \nabla F(x_0, y_0, z_0) \cdot (x - x_0, y - y_0, z - z_0)$$

$$= \left(\frac{\partial f}{\partial x}(x_0, y_0), \frac{\partial f}{\partial y}(x_0, y_0), -1\right) \cdot (x - x_0, y - y_0, z - z_0).$$

Because $z_0 = f(x_0, y_0)$, this is $z = f(x_0, y_0) + (\partial f/\partial x)(x_0, y_0)(x - x_0) + (\partial f/\partial y)(x_0, y_0)(y - y_0)$.

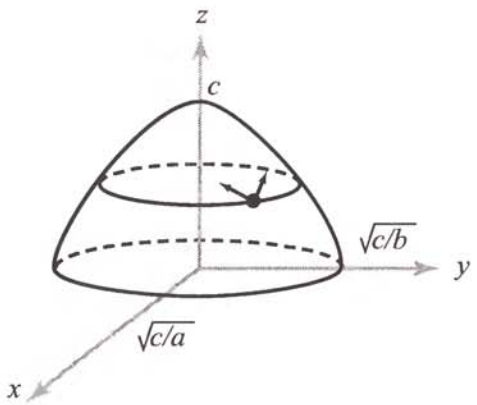
**13.** (a) $\nabla f = (z + y, z + x, x + y)$, $\mathbf{g}'(t) = (e^t, -\sin t, \cos t)$, $(f \circ \mathbf{g})'(1) = 2e \cos 1 + \cos^2 1 - \sin^2 1$
(b) $\nabla f = (yze^{xyz}, xze^{xyz}, xye^{xyz})$, $\mathbf{g}'(t) = (6, 6t, 3t^2)$, $(f \circ \mathbf{g})'(1) = 108e^{18}$
(c) $\nabla f = [1 + \log(x^2 + y^2 + z^2)](x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$, $\mathbf{g}' = (e^t, -e^{-t}, 1)$, $(f \circ \mathbf{g})'(1) = [1 + \log(e^2 + e^{-2} + 1)](e^2 - e^{-2} + 1)$

**15.** Let $f(x, y, z) = 1/r = (x^2 + y^2 + z^2)^{-1/2}$; $\mathbf{r} = (x, y, z)$. Then we calculate $\nabla f = -(x^2 + y^2 + z^2)^{-3/2}(x, y, z) = -(1/r^3)\mathbf{r}$.

**17.** $\nabla f = (g'(x), 0)$.

**19.** $\mathbf{D}f(0, 0, \ldots, 0) = [0, \ldots, 0]$

**21.** $\mathbf{d}_1 = [-(0.03 + 2by_1)/2a]\mathbf{i} + y_1\mathbf{j}$, $\mathbf{d}_2 = [-(0.03 + 2by_2)/2a]\mathbf{i} + y_2\mathbf{j}$, where $y_1$ and $y_2$ are the solutions of $(a^2 + b^2)y^2 + 0.03by + \left(\frac{0.03^2}{4} - a^2\right) = 0$.




**23.** $\nabla V = \dfrac{\lambda}{2\pi\varepsilon_0}\left[\left(\dfrac{x + x_0}{r_2^2} - \dfrac{x - x_0}{r_1^2}\right)\mathbf{i} + 2y\left(\dfrac{1}{r_2^2} - \dfrac{1}{r_1^2}\right)\mathbf{j}\right]$

**25.** Crosses at $(2, 2, 0)$, $\sqrt{5}/10$ seconds later.

## Review Exercises for Chapter 2

**1.** (a) Elliptic paraboloid.
(b) Let $y' = y + 3$ and write $z = xy'$. This is a (shifted) hyperbolic paraboloid.

**3.** (a) $\mathbf{D}f(x, y) = \begin{bmatrix} 2xy & x^2 \\ -ye^{-xy} & -xe^{-xy} \end{bmatrix}$

(b) $\mathbf{D}f(x) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

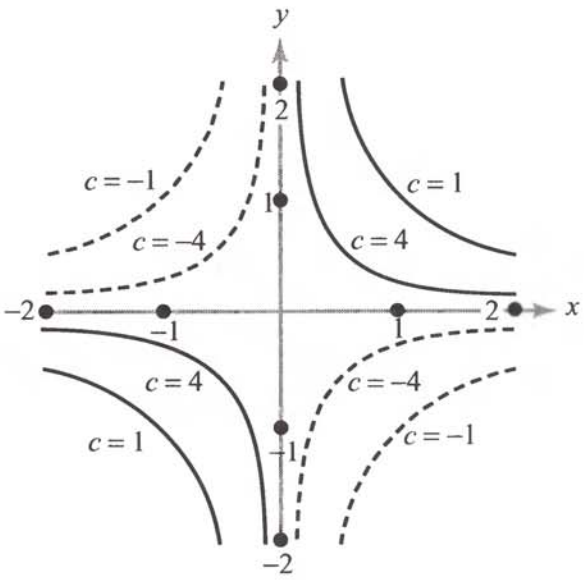
(c) $\mathbf{D}f(x, y, z)] = [e^x \quad e^y \quad e^z]$

(d) $\mathbf{D}f(x, y, z) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

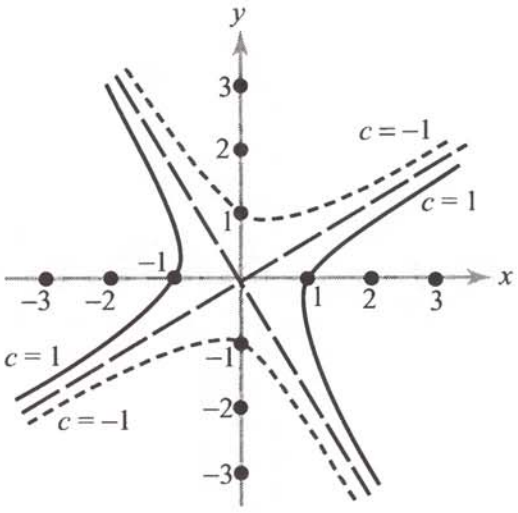
**5.** The plane tangent to a sphere at $(x_0, y_0, z_0)$ is normal to the line from the center to $(x_0, y_0, z_0)$.

**7.** (a) $z = x - y + 2$ (d) $10x + 6y - 4z = 6 - \pi$
(b) $z = 4x - 8y - 8$ (e) $2z = \sqrt{2}x + \sqrt{2}y$
(c) $x + y + z = -1$ (f) $x + 2y - z = 2$

**9.** (a) The level curves are hyperbolas $xy = 1/c$:




(b) $c = x^2 - xy - y^2 = \left(x - \dfrac{1+\sqrt{5}}{2}y\right)\left(x - \dfrac{1-\sqrt{5}}{2}y\right)$




**11.** (a) 0 (b) Limit does not exist.

**13.** $(1 + 2x^2)\exp(1 + x^2 + y^2)$

**15.** (a) The line $\mathbf{L}(t) = (x_0, y_0, f(x_0, y_0)) + t(a, b, c)$ lies in the plane $z = f(x_0, y_0)$ if $c = 0$ and is perpendicular to $\nabla f(x_0, y_0)$ if $a(\partial f/\partial x)(x_0, y_0) + b(\partial b/\partial y)(x_0, y_0) = 0$. On $\mathbf{L}$, we have

$$\begin{aligned} f(x_0, y_0) + \left[\frac{\partial f}{\partial x}(x_0, y_0)\right](x - x_0) + \left[\frac{\partial f}{\partial y}(x_0, y_0)\right](y - y_0) & \\ = f(x_0, y_0) + at\left[\frac{\partial f}{\partial x}(x_0, y_0)\right] + bt\left[\frac{\partial f}{\partial y}(x_0, y_0)\right] & \\ = f(x_0, y_0) = z & \end{aligned}$$

Therefore, $\mathbf{L}$ lies in the tangent plane. An upward unit normal to the tangent plane is $\mathbf{p} = (1 + \|\nabla f\|^{-1/2}(-(\partial f/\partial x)(x_0, y_0), -(\partial f/\partial y)(x_0, y_0), 1)$. Therefore, $\cos\theta = \mathbf{p} \cdot \mathbf{k} = (1 + \|\nabla f\|^2)^{-1/2}$, and $\tan\theta = \sin\theta/\cos\theta = \{\|\nabla f\|^2/(1 + \|\nabla f\|^2)\}^{1/2}/(1 + \|\nabla f\|^2)^{-1/2} = \|\nabla f\|$ as claimed.

(b) The tangent plane contains the horizontal line through $(1, 0, 2)$ perpendicular to $\nabla f(1, 0) = (5, 0)$, that is, parallel to the $y$ axis. It makes an angle of $\arctan(\|\nabla f(1, 0)\|) = \arctan 5 \approx 78.7^\circ$ with respect to the $xy$ plane.

**17.** $(1/\sqrt{2}, 1/\sqrt{2})$ or $(-1/\sqrt{2}, -1/\sqrt{2})$

**19.** A unit normal is $(\sqrt{2}/10)(3, 5, 4)$. The tangent plane is $3x + 5y + 4z = 18$.

**21.** $4\mathbf{i} + 16\mathbf{j}$

**23.** (a) Because $g$ is the composition $\lambda \mapsto \lambda x \mapsto f(\lambda x)$, the chain rule gives

$$g'(\lambda) = \mathbf{D}f(\lambda\mathbf{x}) \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

Thus,

$$g'(1) = \mathbf{D}f(\mathbf{x}) \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \nabla f(\mathbf{x}) \cdot \mathbf{x}$$

But also $g(\lambda) = \lambda^p f(\mathbf{x})$, so $g'(\lambda) = p\lambda^{p-1} f(\mathbf{x})$ and $g'(1) = pf(\mathbf{x})$.

(b) $p = 1$.

**25.** Differentiate directly using the chain rule, or use Exercise 23(a) with $p = 0$.

**27.** (a) If $(x, y) \neq (0, 0)$, then one calculates for (i) that $\partial f/\partial x = (y^3 - yx^2)/(x^2 + y^2)^2$ and $\partial f/\partial y = (x^3 - xy^2)/(x^2 + y^2)^2$. If $x = y = 0$, use the definition directly to find that both partial derivatives are 0. For (ii), if $(x, y) \neq (0, 0)$, then $\partial f/\partial x = 2xy^6/(x^2 + y^4)^2$ and $\partial f/\partial y = (2x^4y - 2x^2y^5)/(x^2 + y^4)^2$. The partials at the origin are zero.

(b) The function (i) is not continuous at (0, 0); the function (ii) is differentiable, but the derivative is not continuous.

**29.** (a) $\sqrt{2}\pi/8$ (b) $-\sin\sqrt{2}$ (c) $-2\sqrt{2}e^{-2}$

**31.** $(-4e^{-1}, 0)$

**33.** (a) See Theorem 11.

(b) $g(u) = (\sin 3u)^2 + \cos 8u$
$g'(u) = 6\sin 3u\cos 3u - 8\sin 8u$
$g'(0) = 0$

$\nabla f = (2x, 1)$
$\nabla f(\mathbf{h}(0)) = \nabla f(0, 1) = (0, 1)$
$\mathbf{h}'(u) = (3\cos 3u, -8\sin 8u)$
$g'(0) = \nabla f(\mathbf{h}(0)) \cdot \mathbf{h}'(0) = (0, 1) \cdot (3, 0) = 0$

**35.** $t = \sqrt{14}(-3 + \sqrt{359})/70 = (-3 + \sqrt{359})/5\sqrt{14}$

**37.** $\partial z/\partial x = 4(e^{-2x-2y+2xy})(1 + y)/(e^{-2x-2y} - e^{2xy})^2$
$\partial z/\partial y = 4(e^{-2y-2x+2xy})(1 + x)/(e^{-2x-2y} - e^{2xy})^2$

**39.** Notice that $y = x^2$, so that if $y$ is constant, $x$ cannot be a variable.

**41.** $[f'(t)g(t) + f(t)g'(t)]\exp[f(t)g(t)]$

**43.** $d[f(\mathbf{c}(t))]/dt = 2t/[(1 + t^2 + 2\cos^2 t)(2 - 2t^2 + t^4)]$
$-4t(t^2 - 1)\ln(1 + t^2 + 2\cos^2 t)/(2 - 2t^2 + t^4)^2$
$-4\cos t \sin t/[(1 + t^2 + 2\cos^2 t)(2 - 2t^2 + t^4)]$

**45.** Let $x = f(t)$, $y = t$, and use the chain rule to differentiate $u(x, y)$ with respect to $t$.

**47.** (a) $n = PV/RT$; $P = nRT/V$; $T = PV/nR$; $V = nRT/P$.

(b) $\partial V/\partial T = nR/P$; $\partial T/\partial P = V/nR$; $\partial P/\partial V = -nRT/V^2$. Multiply, remembering that $PV = nRT$.

**49.** (a) One can solve for any of the variables in terms of the other two.

(b) $\partial T/\partial P = (V - \beta)/R$;
$\partial P/\partial V = -RT/(V - \beta)^2 + 2\alpha/V^3$;
$\partial V/\partial T = R/[(V - \beta)(RT/(V - \beta)^2 - 2\alpha/V^3)]$

(c) Multiply and cancel factors.

**51.** (a) $(1/\sqrt{2}, 1/\sqrt{2})$

(b) The directional derivative is 0 in the direction

$$(x_0\mathbf{i} + y_0\mathbf{j})/\sqrt{x_0^2 + y_0^2}.$$

(c) The level curve through $(x_0, y_0)$ must be tangent to the line through (0, 0) and $(x_0, y_0)$. The level curves are lines or half-lines emanating from the origin.

**53.** $G(x, y) = x - y$

# Chapter 3

## Section 3.1

**1.** $\dfrac{\partial^2 f}{\partial x^2} = 24\dfrac{x^3y - xy^3}{(x^2+y^2)^4}$, $\dfrac{\partial^2 f}{\partial y^2} = 24\dfrac{-x^3y + xy^3}{(x^2+y^2)^4}$,

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{-6x^4 + 36x^2y^2 - 6y^4}{(x^2+y^2)^4}$$

**3.** $\dfrac{\partial^2 f}{\partial x^2} = -y^4\cos(xy^2)$, $\dfrac{\partial^2 f}{\partial y^2} = -2x\sin(xy^2) - 4x^2y^2\cos(xy^2)$,

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x} = -2y\sin(xy^2) - 2xy^3\cos(xy^2)$$

**5.** $\dfrac{\partial^2 f}{\partial x^2} = \dfrac{2(\cos^2 x + e^{-y})\cos 2x + 2\sin^2 2x}{(\cos^2 x + e^{-y})^3}$,

$$\frac{\partial^2 f}{\partial y^2} = \frac{e^{-y} - \cos^2 x}{e^y(\cos^2 x + e^{-y})^3}$$

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{2\sin 2x}{e^y(\cos^2 x + e^{-y})^3}$$

**7.** (a) $\partial^2 z/\partial x^2 = 6$, $\partial^2 z/\partial y^2 = 4$, $\partial^2 z/\partial x\,\partial y = \partial^2 z/\partial y\,\partial x = 0$ (b) $\partial^2 z/\partial x^2 = 0$, $\partial^2 z/\partial y^2 = 4x/3y^3$, $\partial^2 z/\partial x\,\partial y = \partial^2 z/\partial y\,\partial x = -2/3y^2$

**9.** $f_{xy} = 2x + 2y$, $f_{yz} = 2z$, $f_{zx} = 0$, $f_{xyz} = 0$

**11.** Because $f$ and $\partial f/\partial z$ are both of class $C^2$, we have

$$\frac{\partial^3 f}{\partial x\,\partial y\,\partial z} = \frac{\partial^2}{\partial x\,\partial y}\frac{\partial f}{\partial z} = \frac{\partial^2}{\partial y\,\partial x}\frac{\partial f}{\partial z} = \frac{\partial}{\partial y}\left(\frac{\partial^2 f}{\partial x\,\partial z}\right) = \frac{\partial}{\partial y}\left(\frac{\partial^2 f}{\partial z\,\partial x}\right) = \frac{\partial^3 f}{\partial y\,\partial z\,\partial x}.$$

**13.** $f_{xzw} = f_{zwx} = e^{xyz}[2xy\cos(xw) + x^2y^2z\cos(xw) - x^2yw\sin(xw)]$

**15.** (a) $\dfrac{\partial f}{\partial x} = \arctan\dfrac{x}{y} + \dfrac{xy}{x^2+y^2}$

$$\frac{\partial f}{\partial y} = \frac{-x^2}{x^2+y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{2y^3}{(x^2+y^2)^2}, \quad \frac{\partial^2 f}{\partial y^2} = \frac{2x^2y}{(x^2+y^2)^2}$$

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{-2xy^2}{(x^2+y^2)^2}$$

(b) $\dfrac{\partial f}{\partial x} = \dfrac{-x \sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}, \dfrac{\partial f}{\partial y} = \dfrac{-y \sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}$

$$\frac{\partial^2 f}{\partial x^2} = \frac{x^2 \sin\sqrt{x^2+y^2}}{(x^2+y^2)^{3/2}} - \frac{x^2\cos\sqrt{x^2+y^2}}{x^2+y^2} - \frac{\sin\sqrt{x^2+y^2}}{(x^2+y^2)^{1/2}}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{y^2 \sin\sqrt{x^2+y^2}}{(x^2+y^2)^{3/2}} - \frac{y^2\cos\sqrt{x^2+y^2}}{x^2+y^2} - \frac{\sin\sqrt{x^2+y^2}}{(x^2+y^2)^{1/2}}$$

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x} = xy\left[\frac{\sin\sqrt{x^2+y^2}}{(x^2+y^2)^{3/2}} - \frac{\cos\sqrt{x^2+y^2}}{x^2+y^2}\right]$$

(c) $\dfrac{\partial f}{\partial x} = -2x\exp(-x^2-y^2), \dfrac{\partial f}{\partial y} = -2y\exp(-x^2-y^2),$

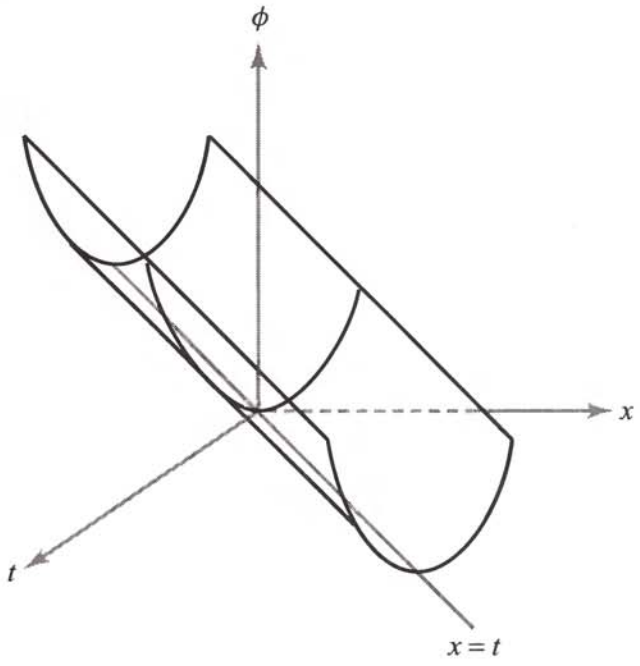
$$\frac{\partial^2 f}{\partial x^2} = (4x^2-2)\exp(-x^2-y^2), \frac{\partial^2 f}{\partial y^2} = (4y^2-2)\exp(-x^2-y^2),$$

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x} = 4xy\exp(-x^2-y^2)$$

**17.** $\dfrac{\partial^2 f}{\partial x^2}\left(\dfrac{dx}{dt}\right)^2 + 2\dfrac{\partial^2 f}{\partial x\,\partial y}\dfrac{dx}{dt}\dfrac{dy}{dt} + \dfrac{\partial^2 f}{\partial y^2}\left(\dfrac{dy}{dt}\right)^2 + \dfrac{\partial f}{\partial x}\dfrac{d^2x}{dt^2} + \dfrac{\partial f}{\partial y}\dfrac{d^2y}{dt^2},$
where $\mathbf{c}(t) = (x(t), y(t))$

**19.** Evaluate the derivatives $\partial^2 u/\partial x^2$ and $\partial^2 u/\partial y^2$ and add.

**21.** (a) Evaluate the derivatives and compare.
(b)




**23.** $V = -GmM/r = -GmM(x^2+y^2+z^2)^{-1/2}$. Check that
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = GmM(x^2+y^2+z^2)^{-3/2}[3 - 3(x^2+y^2+z^2)(x^2+y^2+z^2)^{-1}] = 0$$

## Section 3.2

**1.** $f(h_1, h_2) = h_1^2 + 2h_1h_2 + h_2^2$ $\quad [R_2(\mathbf{0}, \mathbf{h}) = 0$ in this case$]$

**3.** $f(h_1, h_2) = 1 + h_1 + h_2 + \dfrac{h_1^2}{2} + h_1h_2 + \dfrac{h_2^2}{2} + R_2(\mathbf{0}, \mathbf{h})$

**5.** $f(h_1, h_2) = 1 + h_1h_2 + R_2(\mathbf{0}, \mathbf{h})$

**7.** (a) Show that $|R_k(x, a)| \leq AB^{k+1}/(k+1)!$ for constants $A$, $B$, and $x$ in a fixed interval $[a, b]$. Prove that $R_k \to 0$ as $k \to \infty$. (Use convergence of the series $\sum c^k/k! = e^c$ and use Taylor's theorem.)

(b) The only possible trouble is at $x = 0$. Use L'Hôpital's rule to show that

$$\lim_{t\to\infty} p(t)e^t = \infty$$

for every polynomial $p(t)$. Using this, establish that $\lim_{x\to 0^+} p(x)e^{-1/x} = 0$ for every rational function $p(x)$, and conclude that $f^{(k)}(0) = 0$ for every $k$.

(c) $f\colon \mathbb{R}^n \to \mathbb{R}$ is analytic at $\mathbf{x}_0$ if the series

$$f(\mathbf{x}_0) + \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(\mathbf{x}_0) + \frac{1}{2}\sum_{i,j=1}^{n} h_ih_j \frac{\partial^2 f}{\partial x_i\, \partial x_j}(\mathbf{x}_0) + \cdots$$

$$+ \frac{1}{k!}\sum_{i_1,\ldots,i_k=1}^{n} h_{i_1}h_{i_2}\cdots h_{i_k} \frac{\partial^k f}{\partial x_{i_1}\cdots \partial x_{i_k}}(\mathbf{x}_0) + \cdots$$

converges to $f(\mathbf{x}_0 + \mathbf{h})$ for all $\mathbf{h} = (h_1, \ldots, h_n)$ in some sufficiently small disk $\|\mathbf{h}\| < \varepsilon$. The function $f$ is analytic if for every $R > 0$ there is a constant $M$ such that $|(\partial^k f/\partial x_{i_1}\cdots\partial x_{i_k})(\mathbf{x})| < M^k$ for each $k$th-order derivative at every $x$ satisfying $\|\mathbf{x}\| \leq R$.

(d) $f(x, y) = 1 + x + y + \frac{1}{2}(x^2 + 2xy + y^2) + \cdots + \dfrac{1}{k!}\displaystyle\sum_{j=0}^{k} \binom{k}{j} x^j y^{k-j} + \cdots$

## Section 3.3

**1.** $(0, 0)$; saddle point.

**3.** The critical points are on the line $y = -x$; they are local minima, because $f(x, y) = (x + y)^2 \geq 0$, equaling zero only when $x = -y$.

**5.** $(0, 0)$; saddle point.

**7.** $(-\frac{1}{4}, -\frac{1}{4})$; local minimum.

**9.** $(0, 0)$; local maximum. (The tests fail, but use the fact that $\cos z \leq 1$.)
$(\sqrt{\pi/2}, \sqrt{\pi/2})$, local minimum
$(0, \sqrt{\pi})$, local minimum.

**11.** No critical points.

**13.** (1, 1) is a local minimum.

**15.** $(0, n\pi)$; critical points, no local maxima or minima.

**17.** Minimum at (0, 0) and maxima at $(0, \pm 1)$ [and saddles at $(\pm 1, 0)$].

**19.** (a) $\partial f/\partial x$ and $\partial f/\partial y$ vanish at (0, 0).
(b) Show that $f(g(t)) = 0$ at $t = 0$ and that $f(g(t)) \geq 0$ if $|t| < |b|/3a^2$.
(c) $f$ is negative on the parabola $y = 2x^2$.

**21.** The critical points are on the line $y = x$ and they are local minima (see Exercise 3).

**23.** Minimize $S = 2xy + 2yz + 2xz$ with $z = V/xy$, $V$ the constant volume.

**25.** 40, 40, 40

**27.** The only critical point is (0, 0, 0). It is a minimum, because

$$f(x, y, z) \geq \frac{x^2 + y^2}{2} + z^2 + xy = \frac{1}{2}(x + y)^2 + z^2 \geq 0.$$

**29.** $(1, \frac{3}{2})$ is a saddle point; $(5, \frac{27}{2})$ is a local minimum.

**31.** $\frac{3}{2}$ is the absolute maximum and 0 is the absolute minimum.

**33.** $-2$ is the absolute minimum; 2 is the absolute maximum.

**35.** $(\frac{1}{2}, 4)$ is a local minimum.

**37.** If $u_n(x, y) = u(x, y) + (1/n)e^x$, then $\nabla^2 u_n = (1/n)e^x > 0$. Thus, $u_n$ is strictly subharmonic and can have its maximum only on $\partial D$, say, at $\mathbf{p}_n = (x_n, y_n)$. If $(x_0, y_0) \in D$, check that this implies $u(x_n, y_n) > u(x_0, y_0) - e/n$. Thus, there must be a point $\mathbf{q} = (x_\infty, y_\infty)$ on $\partial D$ such that arbitrarily close to $\mathbf{q}$ we can find an $(x_n, y_n)$ for $n$ as large as we like. Conclude from the continuity of $u$ that $u(x_\infty, y_\infty) \geq u(x_0, y_0)$.

**39.** Follow the methods of Exercise 37.

**41.** (a) If there were an $x_1$ with $f(x_1) < f(x_0)$, then the maximum of $f$ on the interval between $x_0$ and $x$ would be another critical point.

(b) Verify (i) by the second derivative test; for (ii), $f$ goes to $-\infty$ as $y \to \infty$ and $x = -y$.

## Section 3.4

**1.** Maximum at $\sqrt{\frac{2}{3}}(1, -1, 1)$, minimum at $\sqrt{\frac{2}{3}}(-1, 1, -1)$

**3.** Maximum at $(\sqrt{3}, 0)$, minimum at $(-\sqrt{3}, 0)$

**5.** Maximum at $(\frac{9}{\sqrt{70}}, \frac{4}{\sqrt{70}})$, minimum at $(-\frac{9}{\sqrt{70}}, -\frac{4}{\sqrt{70}})$

**7.** The minimum value 4 is attained at (0, 2). Use a geometric picture rather than Lagrange multipliers.

**9.** (0, 0, 2) is a minimum of $f$.

**11.** $\frac{3}{2}$ is the absolute maximum and 0 is the absolute minimum.

**13.** The diameter should equal the height, $20/\sqrt[3]{2\pi}$ cm.

**15.** Maximum value $\sqrt{3}$ at $(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$ and minimum value $-\sqrt{3}$ at $(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$.

**17.** Horizontal length is $\sqrt{qA/p}$, vertical length is $\sqrt{pA/q}$.

**19.** For Exercise 1, the bordered Hessians required are

$$|\bar{H}_2| = \begin{vmatrix} 0 & 2x & 2y \\ 2x & -2\lambda & 0 \\ 2y & 0 & -2\lambda \end{vmatrix} = 8\lambda(x^2 + y^2),$$

$$|\bar{H}_3| = \begin{vmatrix} 0 & 2x & 2y & 2z \\ 2x & -2\lambda & 0 & 0 \\ 2y & 0 & -2\lambda & 0 \\ 2z & 0 & 0 & -2\lambda \end{vmatrix} = -16\lambda(x^2 + y^2 + z^2).$$

At $\sqrt{\frac{2}{3}}(1, -1, 1)$ the Lagrange multiplier is $\lambda = \sqrt{6}/4 > 0$, indicating a maximum at $\sqrt{\frac{2}{3}}(1, -1, 1)$, and $\lambda = -\sqrt{6}/4 < 0$ indicates a minimum at $\sqrt{\frac{2}{3}}(-1, 1, -1)$. In Exercise 5, $|\bar{H}| = 24\lambda(4x^2 + 6y^2)$, and so $\lambda = \sqrt{70}/12 > 0$ indicates a maximum at $(9/\sqrt{70}, 4/\sqrt{70})$ and $\lambda = -\sqrt{70}/12 < 0$ indicates a minimum at $(-9/\sqrt{70}, -4/\sqrt{70})$.

**21.** 11,664 in$^3$

**23.** (a) $\nabla f(\mathbf{x}) = A\mathbf{x}$.

(b) $S$ is defined by the constraint function $g(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 - 1$. Because $\nabla g(\mathbf{x}) = 2\mathbf{x}$ is not $\mathbf{0}$, Theorem 9 applies. At an $\mathbf{x}$ where $f$ is extreme, there is a $\lambda/2$ such that $\nabla f(\mathbf{x}) = (\lambda/2)/\nabla g(\mathbf{x})$. That is, $A\mathbf{x} = \lambda\mathbf{x}$.

**25.** Minimum is $(-1/\sqrt{2}, 0)$ maximum is $(\frac{1}{4}, \pm\sqrt{7/8})$, local minimum at $(1/\sqrt{2}, 0)$.

**27.** No critical points; no maximum or minimum

**29.** $(-1, 0, 1)$

**31.** The point $(K, L) = (\alpha B/q, (1 - \alpha)B/p)$ optimizes the profit.

## Section 3.5

**1.** Use the special implicit function theorem with $n = 1$. (See Example 1.) Line (i) is given by $0 = (x - x_0, y - y_0) \cdot \nabla F(x_0, y_0) = (x - x_0)(\partial F/\partial x)(x_0, y_0) + (y - y_0)(\partial F/\partial y)(x_0, y_0)$. For line (ii), Theorem 11 gives $dy/dx = -(\partial F/\partial x)/(\partial F/\partial y)$, and so the lines agree and are given by

$$y = y_0 - \frac{(\partial F/\partial x)(x_0, y_0)}{(\partial F/\partial y)(x_0, y_0)}(x - x_0).$$

**3.** (a) If $x < -\frac{1}{4}$, we can solve for $y$ in terms of $x$ using the quadratic formula.
(b) $\partial F/\partial y = 2y + 1$ is nonzero for $\{y \mid y < -\frac{1}{2}\}$ and $\{y \mid > -\frac{1}{2}\}$. These regions correspond to the upper and lower halves of a horizontal parabola with vertex at $(-\frac{1}{4}, -\frac{1}{2})$ and to the choice of sign in the quadratic formula. The derivative $dy/dx = -3/(2y + 1)$ is negative on the top half of the parabola, positive on the bottom.

**5.** Let $F(x, y, z) = x^3z^2 - z^3yx; \partial F/\partial z = 2x^3z - 3z^2yx \neq 0$ at $(1, 1, 1)$. Near the origin, with $x = y \neq 0$, we get solutions $z = 0$ and $z = x$, and so there is no unique solution. At $(1, 1)$, $\partial z/\partial x = 2$ and $\partial z/\partial y = -1$.

**7.** With $F_1 = y + x + uv$ and $F_2 = uxy + v$, the determinant in the general implicit function theorem is

$$\begin{vmatrix} \partial F_1/\partial u & \partial F_1/\partial v \\ \partial F_2/\partial u & \partial F_2/\partial v \end{vmatrix} = v - uxy,$$

which is 0 at $(0, 0, 0, 0)$. Thus, the implicit function theorem does not apply. If we try directly, we find that $v = -uxy$, so $x + y = u^2xy$. For a particular choice of $(x, y)$ near $(0, 0)$, either there are no solutions for $(u, v)$ or else there are two.

**9.** No. $f(x, y) = (-1, 0)$ has infinitely many solutions, namely, $(x, y) = (0, y)$ for any $y$.

**11.** (a) $x_0^2 + y_0^2 \neq 0$.
(b) $f'(z) = -z(x + 2y)/(x^2 + y^2)$; $\quad g'(z) = z(y - 2x)/(x^2 + y^2)$.

**13.** Multiply and equate coefficients to get $a_0$, $a_1$, and $a_2$ as functions of $r_1$, $r_2$, and $r_3$. Then compute the Jacobian determinant $\partial(a_0, a_1, a_2)/\partial(r_1, r_2, r_3) = (r_3 - r_2)(r_1 - r_2)(r_1 - r_3)$. This is not zero if the roots are distinct. Thus, the inverse function theorem shows that the roots may be found as functions of the coefficients in some neighborhood of any point at which the roots are distinct. That is, if the roots $r_1, r_2, r_3$ of $x^3 + a_2x^2 + a_1x + a_0$ are all different, then there are neighborhoods $V$ of $(r_1, r_2, r_3)$ and $W$ of $(a_0, a_1, a_2)$ such that the roots in $V$ are smooth functions of the coefficients in $W$.

## Review Exercises for Chapter 3

**1.** (a) Saddle point.
(b) Saddle point for any $C$.

**3.** (a) 1 (b) $\sqrt{83}/6$

**5.** Use the second derivative test; $(0, 0)$ is a local maximum; $(-1, 0)$ is a saddle point; $(2, 0)$ is a local minumum.

**7.** Saddle points at $(n, 0)$, $n =$ integer.

**9.** Maximum $\approx 2.618$, minimum $\approx 0.382$.

**11.** Maximum 1, minimum $\cos 1$

**13.** $z = 1/4$

**15.** $(0, 0, \pm 1)$

**17.** If $b \geq 2$, the minimum distance is $2\sqrt{b-1}$; if $b \leq 2$, the minimum distance is $|b|$.

**19.** Not stable.

**21.** $f(-\frac{3}{2}, -\sqrt{3}/2) = 3\sqrt{3}/4$

**23.** $x = (20/3)\sqrt[3]{3}; y = 10\sqrt[3]{3}; z = 5\sqrt[3]{3}$

**25.** The determinant required in the general implicit function theorem is not zero, and so we can solve for $u$ and $v$; $(\partial u/\partial x)(2, -1) = 13/32$.

**27.** A new orthonormal basis may be found with respect to which the quadratic form given by the matrix

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

takes diagonal form. This change of basis defines new variables $\xi$ and $\eta$, which are linear functions of $x$ and $y$. Manipulations of linear algebra and the chain rule show that $Lv = \lambda(\partial^2 v/\partial \xi^2) + \mu(\partial^2 v/\partial \eta^2)$. The numbers $\lambda$ and $\mu$ are the eigenvalues of $A$ and are positive, because the quadratic form is positive-definite. At a maximum, $\partial v/\partial \xi = \partial v/\partial \eta = 0$. Moreover, $\partial^2 v/\partial \xi^2 \leq 0$ and $\partial^2 v/\partial \eta^2 \leq 0$, because if either were greater than 0, the cross section of the graph in that direction would have a minimum. Then $Lv \leq 0$, thus contradicting strict subharmonicity.

**29.** Reverse the inequalities in Exercises 27 and 28.

**31.** The equations for a critical point, $\partial s/\partial m = \partial s/\partial b = 0$, when solved for $m$ and $b$ give $m = (y_1 - y_2)/(x_1 - x_2)$ and $b = (y_2 x_1 - y_1 x_2)/(x_1 - x_2)$. The line $y = mx + b$ then goes through $(x_1, y_1)$ and $(x_2, y_2)$.

**33.** At a minimum of $s$, we have $0 = \partial s/\partial b = -2\sum_{i=1}^{n}(y_i - mx_i - b)$.

**35.** $y = \frac{9}{10}x + \frac{6}{5}$

# Chapter 4

## Section 4.1

**1.** $\mathbf{r}'(t) = -(\sin t)\mathbf{i} + 2(\cos 2t)\mathbf{j}$, $r'(0) = 2\mathbf{j}$, $\mathbf{a}(t) = -(\cos t)\mathbf{i} - 4(\sin 2t)\mathbf{j}$, $\mathbf{a}(0) = -\mathbf{i}$, $\mathbf{l}(t) = \mathbf{i} + 2t\mathbf{j}$

**3.** $\mathbf{r}'(t) = \sqrt{2}\mathbf{i} + e^t\mathbf{j} - e^{-t}\mathbf{k}$, $\mathbf{r}'(0) = \sqrt{2}\mathbf{i} + \mathbf{j} - \mathbf{k}$, $\mathbf{a}(t) = e^t\mathbf{j} + e^{-t}\mathbf{k}$, $\mathbf{a}(0) = \mathbf{j} + \mathbf{k}$, $\mathbf{l}(t) = \sqrt{2}t\mathbf{i} + (1 + t)\mathbf{j} + (1 - t)\mathbf{k}$

**5.** $(e^t - e^{-t}, \cos t - \sin t, -3t^2)$

**7.** $[-3t^2(2\sin t + \cos t) - t^3(2\cos t - \sin t)]\mathbf{i} + [3t^2(2e^t + e^{-t}) + t^3(2e^t - e^{-t})]\mathbf{j}$ $+ [e^t(\cos t - \sin t) - e^{-t}(-\sin t + \cos t)]\mathbf{k}$

**9.** $m(0, 6, 0)$

**11.** $-24\pi^2(\cos(2\pi t/5), \sin(2\pi t/5))/25$

**13.** $\dfrac{d}{dt}(\|\mathbf{v}\|^2) = \dfrac{d}{dt}(\mathbf{v}\cdot\mathbf{v}) = 2\mathbf{v}\cdot\dfrac{d\mathbf{v}}{dt} = 2\mathbf{v}\cdot\mathbf{a} = 0$

**15.** 6129 seconds

**17.** $\mathbf{c}(t) = \left(\dfrac{t^2}{2}, e^t - 6, \dfrac{t^3}{3} + 1\right)$

**19.** (a) $\mathbf{c}(t) = (t, e^t)$, $-\infty < t < \infty$. The image of this path is the graph $y = e^x$.
(b) $\mathbf{c}(t) = (\frac{1}{2}\cos t, \sin t)$, $0 \le t \le 2\pi$, an ellipse.
(c) $\mathbf{c}(t) = (at, bt, ct)$
(d) $\mathbf{c}(t) = (\frac{2}{3}\cos t, \frac{1}{2}\sin t)$, $0 \le t \le 2\pi$, an ellipse.

**21.** $\mathbf{c}(t) \times \mathbf{c}'(t)$ is normal to the plane of the orbit at time $t$. As in Exercise 20, its derivative is 0, and so the orbital plane is constant.

## Section 4.2

**1.** $2\sqrt{5}\pi$

**3.** $2(2\sqrt{2} - 1)$

**5.** $\dfrac{6 - \sqrt{3}}{\sqrt{2}} + \dfrac{1}{2}\log\left[\dfrac{2\sqrt{2} + 3}{\sqrt{2} + \sqrt{3}}\right]$

**7.** $2\pi(\sqrt{5} + \sqrt{2})$

**9.** $3 + \log 2$

**11.** (a) Because $\alpha$ is strictly increasing, it maps $[a, b]$ one-to-one onto $[\alpha(a), \alpha(b)]$. By definition, $\mathbf{v}$ is the image of $\mathbf{c}$ if and only if there is a $t$ in $[a, b]$ with $\mathbf{c}(t) = \mathbf{v}$. There is one

point $s$ in $[\alpha(a), \alpha(b)]$ with $s = \alpha(t)$, so $\mathbf{d}(s) = \mathbf{c}(t) = \mathbf{v}$. Therefore, the image of $\mathbf{c}$ is contained in that of $\mathbf{d}$. Use $\alpha^{-1}$ similarly for the opposite inclusion.

(b)
$$l_{\mathbf{d}} = \int_{\alpha(a)}^{\alpha(b)} \|\mathbf{d}'(s)\|\, ds = \int_{s=\alpha(a)}^{s=\alpha(b)} \|\mathbf{d}'(\alpha(t))\|\alpha'(t)\, dt$$
$$= \int_{t=a}^{t=b} \|\mathbf{d}'(\alpha(t))\alpha'(t)\| dt = \int_a^b \|\mathbf{c}'(t)\|\, dt = l_{\mathbf{c}}.$$

(c) Differentiate $\mathbf{d}$ using the chain rule.

**13.** (a) $l_{\mathbf{c}} = \int_a^b \|\mathbf{c}'(s)\|\, ds = \int_a^b ds = b - a$

(b) $\mathbf{T}(s) = \mathbf{c}'(s)/\|\mathbf{c}'(s)\| = \mathbf{c}'(s)$, so $\mathbf{T}'(s) = \mathbf{c}''(s)$. Then $k = \|\mathbf{T}'\| = \|\mathbf{c}''(s)\|$.

(c) Show that if $\mathbf{v}$ and $\mathbf{w}$ are in $\mathbb{R}^3$, $\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{w} - (\mathbf{v} \cdot \mathbf{w}/\|\mathbf{v}\|^2)\mathbf{v}\| \cdot \|\mathbf{v}\|$. Use this to show that if $\rho(t) = (x(t), y(t), z(t))$ is never $(0, 0, 0)$ and $\mathbf{f}(t) = \rho(t)/\|\rho(t)\|$, then

$$\frac{d\mathbf{f}}{dt} = \frac{1}{\|\rho(t)\|}\left[\rho'(t) - \frac{\rho(t) \cdot \rho'(t)}{\|\rho(t)\|^2}\rho(t)\right] \quad \text{and} \quad \frac{d\mathbf{f}}{dt} = \frac{\|\rho(t) \times \rho'(t)\|}{\|\rho(t)\|^2}.$$

With $\rho(t) = \mathbf{c}'(t)$, this gives

$$\mathbf{T}'(t) = \frac{\mathbf{c}''(t)}{\|\mathbf{c}'(t)\|} - \frac{\mathbf{c}'(t) \cdot \mathbf{c}''(t)}{\|\mathbf{c}'(t)\|^3}\mathbf{c}'(t) \quad \text{and} \quad \|\mathbf{T}'(t)\| = \frac{\|\mathbf{c}'(t) \times \mathbf{c}''(t)\|}{\|\mathbf{c}'(t)\|^2}.$$

If $s$ is the arc length of $\mathbf{c}$, $ds/dt = \|\mathbf{c}'(t)\|$, and therefore

$$\left\|\frac{d\mathbf{T}}{dt}\right\| = \left\|\frac{d\mathbf{T}}{ds}\ \frac{ds}{dt}\right\| = k\|\mathbf{c}'(t)\|.$$

Thus,

$$k = \frac{1}{\|\mathbf{c}'(t)\|}\frac{d\mathbf{T}}{dt} = \frac{\|\mathbf{c}'(t) \times \mathbf{c}''(t)\|}{\|\mathbf{c}'(t)\|^3}.$$

(This result is useful in Exercise 15.)

(d) $1/\sqrt{2}$

**15.** (a) Because $\mathbf{c}$ is parametrized by arc length, $\mathbf{T}(s) = \mathbf{c}'(s)$, and $\mathbf{N}(s) = \mathbf{c}''(s)/\|\mathbf{c}''(s)\|$. Use Exercise 13 to show that

$$\frac{d\mathbf{B}}{ds} = \left(\mathbf{c}'' \times \frac{\mathbf{c}''}{\|\mathbf{c}''\|}\right) + \mathbf{c}' \times \left(\frac{\mathbf{c}'''}{\|\mathbf{c}''\|} - \frac{\mathbf{c}'' \cdot \mathbf{c}'''}{\|\mathbf{c}''\|^3}\mathbf{c}''\right)$$

and

$$\tau = -\frac{d\mathbf{B}}{ds} \cdot \mathbf{N} = -\frac{(\mathbf{c}' \times \mathbf{c}''') \cdot \mathbf{c}''}{\|\mathbf{c}''\|^2} = \frac{(\mathbf{c}' \times \mathbf{c}'') \cdot \mathbf{c}'''}{\|\mathbf{c}''\|^2}.$$

(b) Obtain $\mathbf{T}'(t)$ and $\|\mathbf{T}'(t)\|$ as in Exercise 13. $\mathbf{B}$ is a unit vector in the direction of $\mathbf{c}' \times \mathbf{T}' = (\mathbf{c}' \times \mathbf{c}'')/\|\mathbf{c}'\|$, so $\mathbf{B} = (\mathbf{c}' \times \mathbf{c}'')/\|\mathbf{c}' \times \mathbf{c}''\|$. Use the solution of Exercise 13 with $\rho = \mathbf{c}' \times \mathbf{c}''$ to obtain

$$d\mathbf{B}/dt = (\mathbf{c}' \times \mathbf{c}''')/\|\mathbf{c}' \times \mathbf{c}''\| - \{[(\mathbf{c}' \times \mathbf{c}'') \cdot (\mathbf{c}' \times \mathbf{c}''')]/\|\mathbf{c}' \times \mathbf{c}''\|^3\}(\mathbf{c}' \times \mathbf{c}''),$$

and the values of $\mathbf{T}'$ and $\|\mathbf{T}'\|$ to get

$$\mathbf{N} = (\|\mathbf{c}'\|/\|\mathbf{c}' \times \mathbf{c}''\|)(\mathbf{c}'' - (\mathbf{c}' \times \mathbf{c}'')/\|\mathbf{c}'\|^2).$$

Finally, use the chain rule and the inner product of these to obtain

$$\tau = -\left[\frac{d\mathbf{B}}{ds}(s(t))\right] \cdot \mathbf{N}(s(t)) = -\frac{1}{|ds/dt|}\frac{d\mathbf{B}}{dt} \cdot \mathbf{N} = \frac{(\mathbf{c}' \times \mathbf{c}'') \cdot \mathbf{c}'''}{\|\mathbf{c}' \times \mathbf{c}''\|^2}.$$

(c) $\sqrt{2}/2$

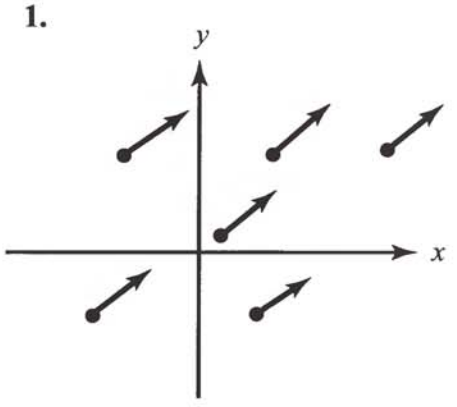
**17.** (a) $\mathbf{N}$ is defined as $\mathbf{T}'/\|\mathbf{T}'\|$, so $\mathbf{T}' = \|\mathbf{T}'\|\mathbf{N} = k\mathbf{N}$. Because $\mathbf{T} \cdot \mathbf{T}' = 0$, $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$ are an orthonormal basis for $\mathbb{R}^3$. Differentiating $\mathbf{B}(s) \cdot \mathbf{B}(s) = 1$ and $\mathbf{B}(s) \cdot \mathbf{T}(s) = 0$ shows that $\mathbf{B}' \cdot \mathbf{B} = 0$ and $\mathbf{B}' \cdot \mathbf{T} + \mathbf{B} \cdot \mathbf{T}' = 0$. But $\mathbf{T}' \cdot \mathbf{B} = \|\mathbf{T}'\|\mathbf{N} \cdot \mathbf{B} = 0$, so $\mathbf{B}' \cdot \mathbf{T} = 0$ also. Thus, $\mathbf{B}' = (\mathbf{B}' \cdot \mathbf{T})\mathbf{T} + (\mathbf{B}' \cdot \mathbf{N})\mathbf{N} + (\mathbf{B}' \cdot \mathbf{B})\mathbf{B} = (\mathbf{B}' \cdot \mathbf{N})\mathbf{N} = -\tau\mathbf{N}$. Also, $\mathbf{N}' \cdot \mathbf{N} = 0$, because $\mathbf{N} \cdot \mathbf{N} = 1$. Thus, $\mathbf{N}' = (\mathbf{N}' \cdot \mathbf{T})\mathbf{T} + (\mathbf{N}' \cdot \mathbf{B})\mathbf{B}$. But differentiating $\mathbf{N} \cdot \mathbf{T} = 0$ and $\mathbf{N} \cdot \mathbf{B} = 0$ gives $\mathbf{N}' \cdot \mathbf{T} = -\mathbf{N} \cdot \mathbf{T}' = -k$ and $\mathbf{N}' \cdot \mathbf{B} = -\mathbf{N} \cdot \mathbf{B}' = \tau$, and so the middle equation follows.

(b) $\omega = \tau\mathbf{T} + k\mathbf{B}$

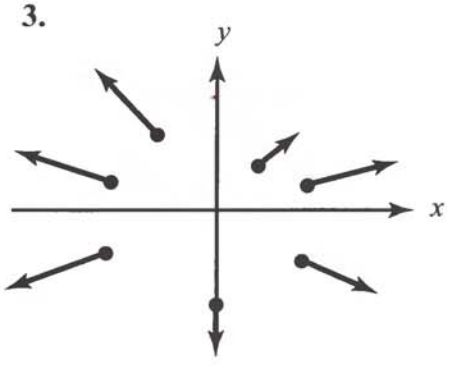
**19.** Follow the hint in the text.

## Section 4.3

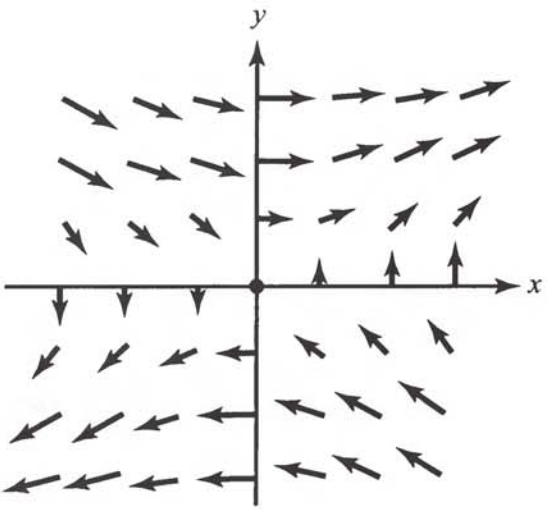
**1.**



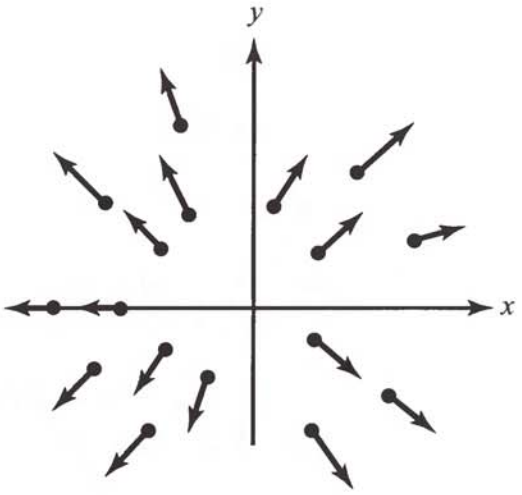

**3.**



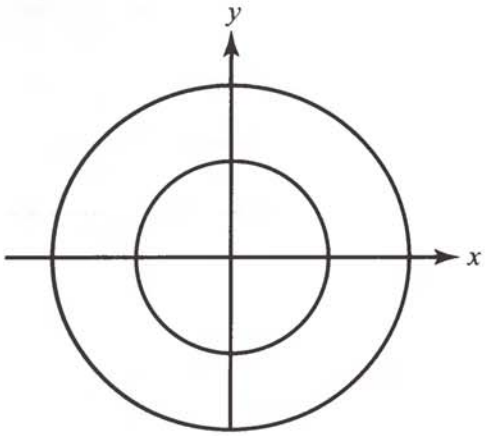
**5.** $\mathbf{F} = (2y, x)$:



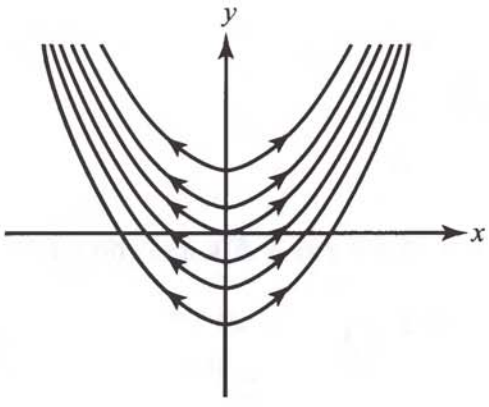

**7.**




**9.** The flow lines are concentric circles:




**11.** The flow lines for $t > 0$:




**13.** $\mathbf{c}'(t) = (2e^{2t}, 1/t, -1/t^2) = \mathbf{F}(\mathbf{c}(t))$

**15.** $\mathbf{c}'(t) = (\cos t, -\sin t, e^t) = \mathbf{F}(\mathbf{c}(t))$

**17.** Compare $\frac{1}{2}mv^2$ for the escape velocity $v_e = \sqrt{2gR_0}$ and the velocity in an orbit of radius $R_0$ given in Section 4.1. (Ignore the rotation of the earth.)

**19.** Use the fact that $-\nabla T$ is perpendicular to the surface $T =$ constant.

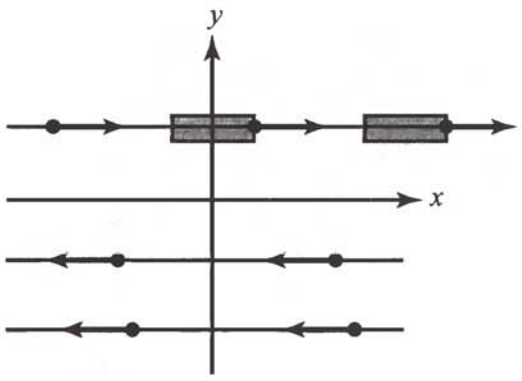
## Section 4.4

**1.** $ye^{xy} - xe^{xy} + ye^{yz}$

**3.** 3

**5.** div $\mathbf{V} > 0$ in the first and third quadrants,
div $\mathbf{V} < 0$ in the second and fourth quadrants.

**7.** $\nabla \cdot \mathbf{F} = 0$; if $\mathbf{F}$ represents a fluid, there is neither expansion nor compression; the area of a small rectangle remains the same.




**9.** $3x^2 - x^2 \cos(xy)$

**11.** $y \cos(xy) + x^2 \sin(x^2 y)$

**13.** $\mathbf{0}$

**15.** $(10y - 8z)\mathbf{i} + (6z - 10x)\mathbf{j} + (8x - 6y)\mathbf{k}$

**17.** $-\sin x$

**19.** $x$

**21.** $\nabla \times \nabla f = \mathbf{0}$

**23.** $\nabla \times \nabla f = \mathbf{0}$

**25.** $\nabla \times \mathbf{F} \neq \mathbf{0}$

**27.** Let $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ and compute both sides of the identity.

**29.** (a) $2xy\mathbf{i} + x^2\mathbf{j}$
(b) $(3y^2xz, 4xz - y^3z, 0)$
(c) $(-y^3zx^3, 2x^2y^4z, 2x^3z^2 - 2xy)$
(d) $4x^2yz^2 + x^2$

**31.** No.

**33.** Separate each expression into its real and imaginary parts and then treat the resulting quantity as a vector field on $\mathbb{R}^2$. Directly calculate its curl and divergence.

In (a), $\mathbf{F} = (x^2 - y^2)\mathbf{i} - 2xy\mathbf{j}$; in (b), $\mathbf{F} = (x^3 - 3xy^2)\mathbf{i} + (y^3 - 3x^2y)\mathbf{j}$; and in (c), $\mathbf{F} = (e^x \cos y)\mathbf{i} - (e^x \sin y)\mathbf{j}$. Show that $\nabla \cdot \mathbf{F} = 0$ and $\nabla \times \mathbf{F} = 0$ in each case.

## Review Exercises for Chapter 4

**1.** $\mathbf{v}(1) = (3, -e^{-1}, -\pi/2)$; $\mathbf{a}(1) = (6, e^{-1}, 0)$;
$s(1) = \sqrt{9 + e^{-2} + \frac{\pi^2}{4}}$; $\mathbf{l}(t) = (2, e^{-1}, 0) + (t-1)(3, -e^{-1}, -\pi/2)$

**3.** $\mathbf{v}(0) = (1, 1, 0)$; $\mathbf{a}(0) = (1, 0, -1)$; $s = \sqrt{2}$; $l(t) = (1, 0, 1) + t(1, 1, 0)$

**5.** Tangent vector: $\mathbf{v} = -(1/\sqrt{2})\mathbf{i} + (1/\sqrt{2})\mathbf{j} + \mathbf{k}$
Acceleration vector: $\mathbf{a} = -(1/\sqrt{2})(\mathbf{i} + \mathbf{j})$

**7.** $m(2, 0, -1)$

**9.** $\displaystyle\int_1^4 \sqrt{1 + \tfrac{4}{9}t^{-2/3} + \tfrac{4}{25}t^{-6/5}}\, dt$

**11.** (a) $\mathbf{v} = (-2t \sin(t^2), 2t \cos(t^2), 0)$; $s = 2t$

(b) $\left(\dfrac{1}{2}, -\dfrac{\sqrt{3}}{2}, 0\right)$

(c) $\sqrt{5\pi/3}$

(d) $\mathbf{v} = 2\sqrt{5\pi/3}(\sqrt{3}/2, 1/2, 0)$; $s = 2\sqrt{5\pi/3}$

(e) $\left(\dfrac{3}{2} + \dfrac{5\pi}{\sqrt{3}}\right) \Big/ \sqrt{5\pi}$

**13.** $x = 1 + t$, $y = -\dfrac{1}{2} + \dfrac{t}{2}$, $z = -\dfrac{2}{3} + \dfrac{t}{3}$

**15.** Compute $\mathbf{c}'(t)$ and check that it equals $\mathbf{F}(\mathbf{c}(t))$.

**17.** 9; $\mathbf{0}$

**19.** 3; $-\mathbf{i} - \mathbf{j} - \mathbf{k}$

**21.** 0; $-\mathbf{i} - \mathbf{j} - \mathbf{k}$

**23.** $\nabla f = (ye^{xy} - y \sin xy, xe^{xy} - x \sin xy, 0)$; verify that $\nabla \times \nabla f = 0$ in this case.

**25.** $\nabla f = (2xe^{x^2} + y^2 \sin xy^2, 2xy \sin xy^2, 0)$; check that $\nabla \times \nabla f = 0$ from this.

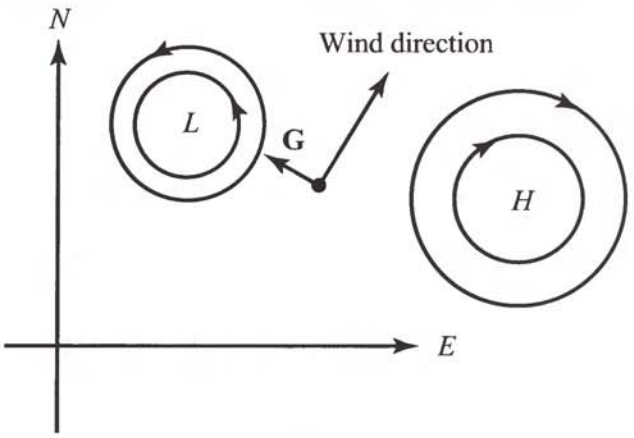
**27.** (a) $(yz^2, xz^2, 2xyz)$; (b) $(z - y, 0, -x)$
(c) $(2xyz^3 - 3xy^2z^2, 2x^2y^2z - y^2z^3, y^2z^3 - 2x^2yz^2)$

**29.** $\operatorname{div} \mathbf{F} = 0$; $\operatorname{curl} \mathbf{F} = (0, 0, 2(x^2 + y^2)f'(x^2 + y^2) + 2f(x^2 + y^2))$

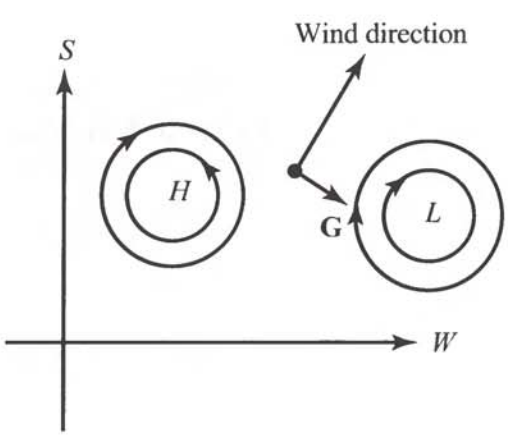
**31.** (a) A cone about $\mathbf{i}'$ making an angle of $\pi/3$ with $\mathbf{i}'$.
(b) $\nabla g = (3x^2, 5z, 5y + 2z)$

**33.** (a) $[\partial P/\partial x)^2 + (\partial P/\partial y)^2]^{1/2}$
(b) A small packet of air would obey $\mathbf{F} = m\mathbf{a}$.

(c)




(d)




**35.** (a) $\dfrac{\sqrt{R^2+\rho^2}}{\rho}(z_0 - z_1)$ (b) $\sqrt{\dfrac{2(R^2+\rho^2)z_0}{g\rho^2}}$

**37.** 680 miles per hour

# Chapter 5

## Section 5.1

**1.** (a) $\frac{13}{15}$ (b) $\pi + \frac{1}{2}$ (c) 1 (d) $\log 2 - \frac{1}{2}$

**3.** To show that the volumes of the two cylinders are equal, show that their area functions are equal.

**5.** $2r^3(\tan\theta)/3$

**7.** $\frac{26}{9}$

**9.** $(2/\pi)(e^2+1)$

**11.** $\frac{196}{15}$

## Section 5.2

**1.** (a) $\frac{7}{12}$ (b) $e-2$ (c) $\frac{1}{9}\sin 1$ (d) $2\ln 4 - 2$

**3.** 1/4

**5.** Use Fubini's theorem to write

$$\int\int_R [f(x)g(y)]\,dx\,dy = \int_c^d g(y)\left[\int_a^b f(x)\,dx\right]dy,$$

and notice that $\int_a^b f(x)\,dx$ is a constant and so may be pulled out.

**7.** 11/6

**9.** Because $\int_0^1 dy = \int_0^1 2y\,dy = 1$, we have $\int_0^1 [\int_0^1 f(x, y)\,dy]\,dx = 1$. In any partition of $R = [0, 1] \times [0, 1]$, each rectangle $R_{jk}$ contains points $\mathbf{c}_{jk}^{(1)}$ with $x$ rational and $\mathbf{c}_{jk}^{(2)}$ with $x$ irrational. If in the regular partition of order $n$ we choose $\mathbf{c}_{jk} = \mathbf{c}_{jk}^{(1)}$ in those rectangles with $0 \le y \le \frac{1}{2}$ and $\mathbf{c}_{jk} = \mathbf{c}_{jk}^{(2)}$ when $y > \frac{1}{2}$, the approximating sums are the same as those for

$$g(x, y) = \begin{cases} 1 & 0 \le y \le \frac{1}{2} \\ 2y & \frac{1}{2} < y < 1. \end{cases}$$

Because $g$ is integrable, the approximating sums must converge to $\int_R g\,dA = 7/8$. However, if we had picked all $\mathbf{c}_{ij} = \mathbf{c}_{jk}^{(1)}$, all approximating sums would have the value 1.

**11.** Fubini's theorem does not apply because the integrand is not continuous nor bounded at $(0, 0)$.

## Section 5.3

**1.** (a) 1/3, both (b) 5/2, both (c) $(e^2 - 1)/4$, both (d) 1/35, both

**3.** $A = \int_{-r}^{r} \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} dy\,dx = 2\int_{-r}^{r} \sqrt{r^2 - x^2}\,dx = r^2[\arcsin 1 - \arcsin(-1)] = \pi r^2.$

**5.** 28,000 ft$^3$

**7.** 0

**9.** $y$-simple; $\pi/2$

**11.** $50\,\pi$

**13.** $\pi/24$

**15.** Compute the integral with respect to $y$ first. Split that into integrals over $[-\phi(x), 0]$ and $[0, \phi(x)]$ and change variables in the first integral, or use symmetry.

**17.** Let $\{R_{ij}\}$ be a partition of a rectangle $R$ containing $D$ and let $f$ be 1 on $D$. Thus, $f^*$ is 1 on $D$ and 0 on $R \backslash D$. Let $\mathbf{c}_{jk} \in R \backslash D$ if $R_{ij}$ is not wholly contained in $D$. The approximating Riemann sum is the sum of the areas of those rectangles of the partition that are contained in $D$.

## Section 5.4

**1.** (a) 1/8 (b) $\pi/4$ (c) 17/12
(d) $G(b) - G(a)$, where $dG/dy = F(y, y) - F(a, y)$ and $\partial F/\partial x = f(x, y)$

**3.** Note that the maximum value of $f$ on $D$ is $e$ and the minimum value of $f$ on $D$ is $1/e$. Use the ideas in the proof of Theorem 4 to show that

$$\frac{1}{e} \le \frac{1}{4\pi^2} \iint f(x, y)\,dA \le e.$$

**5.** The smallest value of $f(x, y) = 1/(x^2 + y^2 + 1)$ on $D$ is $\frac{1}{6}$, at $(1, 2)$, and so

$$\iint_D f(x, y)\,dx\,dy \geq \frac{1}{6} \cdot \text{area } D = 1.$$

The largest value is 1, at $(0, 0)$, and so

$$\iint_D f(x, y)\,dx\,dy \leq 1 \cdot \text{area } D = 6.$$

**7.** $\frac{4}{3}\pi abc$

**9.** $\pi(20\sqrt{10} - 52)/3$

**11.** $\sqrt{3}/4$

**13.** $D$ looks like a slice of pie.

$$\int_0^1 \left[\int_0^x f(x, y)\,dy\right] dx + \int_1^{\sqrt{2}} \left[\int_0^{\sqrt{2-x^2}} f(x, y)\,dy\right] dx.$$

**15.** Use the chain rule and the fundamental theorem of calculus.

## Section 5.5

**1.** $1/3$

**3.** 10

**5.** $x^2 + y^2 \leq z \leq \sqrt{x^2 + y^2}, -\sqrt{1 - y^2} \leq x \leq \sqrt{1 - y^2}, -1 \leq y \leq 1$

**7.** $0 \leq z \leq \sqrt{1 - x^2 - y^2}, -\sqrt{1 - y^2} \leq x \leq \sqrt{1 - y^2}, -1 \leq y \leq 1$

**9.** $50\pi/\sqrt{6}$

**11.** $1/2$

**13.** 0

**15.** $a^5/20$

**17.** 0

**19.** $3/10$

**21.** $1/6$

**23.** $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{1} f(x, y, z)\,dz\,dy\,dx$

**25.** $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1+x^2}} \int_{0}^{\sqrt{4-x^2-y^2}} f(x, y, z)\,dz\,dy\,dx$

**27.** $\iint_D \int_0^{f(x,y)} dz\,dx\,dy = \iint_D f(x, y)\,dx\,dy$

**29.** Let $M_\epsilon$ and $m_\epsilon$ be the maximum and minimum of $f$ on $B_\epsilon$. Then we have the inequality $m_\epsilon \text{ vol}(B_\epsilon) \leq \iiint_{B_\epsilon} f\,dV \leq M_\epsilon \text{ vol}(B_\epsilon)$. Divide by $\text{vol}(B_\epsilon)$, let $\epsilon \to 0$ and use continuity of $f$.

## Review Exercises for Chapter 5

**1.** 81/2

**3.** $\frac{1}{4}e^2 - e + \frac{9}{4}$

**5.** 81/2

**7.** $\frac{1}{4}e^2 - e + \frac{9}{4}$

**9.** 7/60

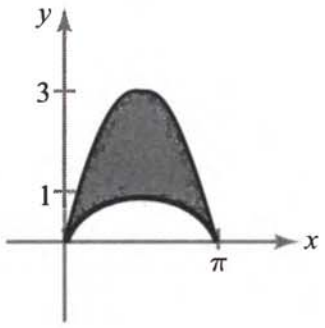
**11.** 1/2

**13.** In the notation of Figure 5.3.1,

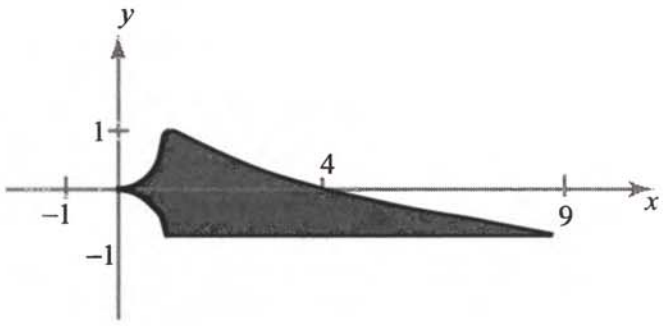
$$\iint_D dx\,dy = \int_a^b [\phi_2(x) - \phi_1(x)]\,dx.$$

**15.** (a) 0 (b) $\pi/24$ (c) 0

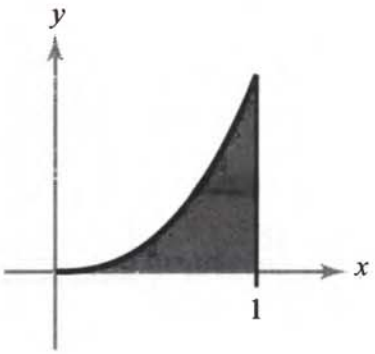
**17.** $y$-simple; $2\pi + \pi^2$



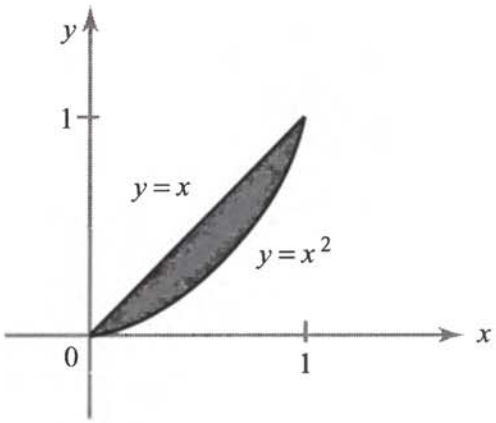

**19.** $x$-simple; 73/3




**21.** $y$-simple; 33/140



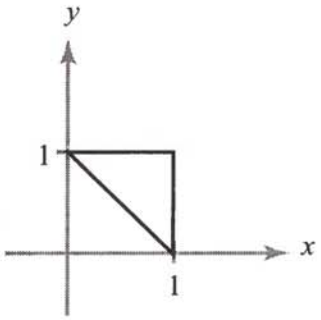
**23.** $y$-simple; 71/420.




**25.** 1/3

**27.** 19/3

**29.** 7/12



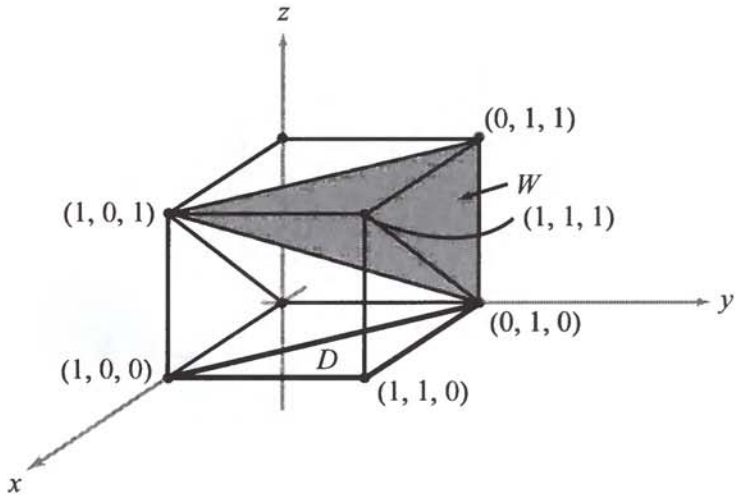

**31.** The function $f(x, y) = x^2 + y^2 + 1$ lies between 1 and $2^2 + 1 = 5$ on $D$, and so the integral lies between these values times $4\pi$, the area of $D$.

**33.** Interchange the order of integration (the reader should draw a sketch in the $(u, t)$ plane);

$$\int_0^x \int_0^t F(u)\,du\,dt = \int_0^x \int_u^x F(u)\,dt\,du = \int_0^x (x-u)F(u)\,du.$$

**35.** $\pi/12$

**37.** The region is the shaded region $W$ in the figure.



The integral in the order $dy\,dx\,dz$, for example, is

$$\int_0^1 \int_z^1 \int_{1-x}^1 f(x, y, z)\, dy\, dx\, dz.$$

# Chapter 6

## Section 6.1

**1.** $S =$ the unit disk minus its center.

**3.** $D = [0, 3] \times [0, 1]$; yes

**5.** The image is the triangle with vertices (0, 0), (0, 1), and (1, 1). $T$ is not one-to-one, but becomes so if we eliminate the portion $x^* = 0$.

**7.** $D$ is the set of $(x, y, z)$ with $x^2 + y^2 + z^2 \leq 1$ (the unit ball). $T$ is not one-to-one, but is one-to-one on $(0, 1] \times (0, \pi) \times (0, 2\pi]$.

**9.** Showing that $T$ is onto is equivalent in the $2 \times 2$ case to showing that the system $ax + by = e$, $cx + dy = f$ can always be solved for $x$ and $y$, where

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

When you do this by elimination or by Cramer's rule, the quantity by which you must divide is $\det A$. Thus, if $\det A \neq 0$, the equations can always be solved.

**11.** Because $\det A \neq 0$, $T$ maps $\mathbb{R}^2$ one-to-one onto $\mathbb{R}^2$. Let $T^{-1}$ be the inverse transformation. Show that $T^{-1}$ has matrix $A^{-1}$ and $\det(A^{-1}) = 1/\det A$, where $\det A \neq 0$. By Exercise 10, $P^* = T^{-1}(P)$ is a parallelogram.

## Section 6.2

**1.** $\pi(e - 1)$

**3.** $D$ is the region $0 \leq x \leq 4$, $\frac{1}{2}x + 3 \leq y \leq \frac{1}{2}x + 6$. (a) 140 (b) $-42$

**5.** $D^*$ is the region $0 \leq u \leq 1, 0 \leq v \leq 2$; $\frac{2}{3}(9 - 2\sqrt{2} - 3\sqrt{3})$.

**7.** $\pi$

**9.** $\dfrac{64\pi}{5}$

**11.** $3\pi/2$

**13.** $\dfrac{5\pi}{2}(e^4 - 1)$

**15.** $2a^2$

**17.** $\dfrac{21}{2}\left(e - \dfrac{1}{e}\right)$

**19.** $100\pi/3$

**21.** $2\pi[\sqrt{3} - 2\log(1+\sqrt{3}) + \log\sqrt{2}]$

**23.** $4\pi \log(a/b)$

**25.** $2\pi[(b^2+1)e^{-b^2} - (a^2+1)e^{-a^2}]$

**27.** 24 (use the change of variables $x = 3u - v + 1$, $y = 3u + v$).

**29.** (a) $\frac{4}{3}\pi abc$ (b) $\frac{4}{5}\pi abc$

**31.** (a) Check that if $T(u_1, v_1) = T(u_2, v_2)$, then $u_1 = u_2$ and $v_1 = v_2$.
(b) 160/3

**33.** $\frac{4}{9}a^{2/3}\iint_{D^*}[f((au^2)^{1/3}, (av^2)^{1/3})u^{-1/3}v^{-1/3}]\,du\,dv$

## Section 6.3

**1.** $[\pi^2 - \sin(\pi^2)]/\pi^3$

**3.** $\left(\frac{11}{18}, \frac{65}{126}\right)$

**5.** \$503.64

**7.** (a) $\delta$, where $\delta$ is the (constant) mass density. (b) 37/12

**9.** $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$

**11.** 1/4

**13.** Letting $\delta$ be density, the moment of inertia is $\delta\int_0^k\int_0^{2\pi}\int_0^{a\sec\phi}(\rho^4\sin^3\phi)\,d\rho\,d\theta\,d\phi$.

**15.** $(1.00 \times 10^8)m$

**17.** (a) The only plane of symmetry for the body of an automobile is the one dividing the left and right sides of the car.

(b) $\bar{z}\cdot\iiint_W \delta(x,y,z)\,dx\,dy\,dz$ is the $z$ coordinate of the center of mass times the mass of $W$. Rearrangement of the formula for $\bar{z}$ gives the first line of the equation. The next step is justified by the additivity property of integrals. By symmetry, we can replace $z$ by $-z$ and integrate in the region above the $xy$ plane. Finally, we can factor the minus sign outside the second integral, and because $\delta(x,y,z) = \delta(u,v,-w)$, we are subtracting the second integral from itself. Thus, the answer is 0.

(c) In part (b), we showed that $\bar{z}$ times the mass of $W$ is 0. Because the mass must be positive, $\bar{z}$ must be 0.

(d) By part (c), the center of mass must lie in both planes.

**19.** $V = -(4.71 \times 10^{19})Gm/R \approx -(3.04 \times 10^9)m/R$, where $m$ is the mass of a test particle at distance $R$ from the planet's center.

## Section 6.4

**1.** 4

**3.** 3/16

**5.** (a) $3\pi$ (b) $\lambda < 1$

**7.** Integration of $\iint e^{-xy}\,dx\,dy$ with respect to $x$ first and then $y$ gives log 2. Reversing the order gives the integral on the left side of the equality stated in the exercise.

**9.** Integrate over $[\varepsilon, 1] \times [\varepsilon, 1]$ and let $\varepsilon \to 0$ to show the improper integral exists and equals 2 log 2.

**11.** $\dfrac{2\pi}{9}[(1 + a^3)^{3/2} - a^{9/2} - 1]$

**13.** Use the fact that

$$\frac{\sin^2(x - y)}{\sqrt{1 - x^2 - y^2}} \leq \frac{1}{\sqrt{1 - x^2 - y^2}}.$$

**15.** Use the fact that $e^{x^2+y^2}/(x - y) \geq 1/(x - y)$ on the given region.

**17.** Each integral equals 1/4, and Theorem 3 (Fubini's theorem) does apply.

## Review Exercises for Chapter 6

**1.** (a) $T\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2u+v \\ 2v \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

(b) $\iint_P f(x, y)\,dx\,dy = 4\iint_S f(2u + v, 2v)\,du\,dv$

**3.** 3 (Use the change of variables $u = x^2 - y^2$, $v = xy$.)

**5.** $\dfrac{1}{3}\pi(4\sqrt{2} - \frac{7}{2})$

**7.** $(5\pi/2)\sqrt{15}$

**9.** $abc/6$

**11.** Cut with the planes $x + y + z = \sqrt[3]{k/n}$, $1 \leq k \leq n - 1$, $k$ an integer.

**13.** $(25 + 10\sqrt{5})\pi/3$

**15.** $(e - e^{-1})/4$ (Use the change of variables $u = y - x$, $v = y + x$.)

**17.** $(9.92 \times 10^6)\pi$ grams

**19.** (a) 32

(b) This occurs at the point of the unit sphere $x^2 + y^2 + z^2 = 1$ inscribed in the cube.

**21.** $(0, 0, 3a^{4/8})$

**23.** $4\pi \ln(a/b)$

**25.** $\pi/2$

**27.** (a) $9/2$ (b) $64\pi$

**29.** Work the integral with respect to $y$ first on the region $D_{\varepsilon,L} = \{(x, y) | \varepsilon \le x \le L, 0 \le y \le x\}$ to obtain $I_{\varepsilon,L} = \iint_{D_{\varepsilon,L}} f\,dx\,dy = \int_{\varepsilon}^{L} x^{-3/2}(1 - e^{-x})\,dx$. The integrand is positive, and so $I_{\varepsilon,L}$ increases as $\varepsilon \to 0$ and $L \to \infty$. Bound $1 - e^{-x}$ above by $x$ for $0 < x < 1$ and by 1 for $1 < x < \infty$ to see that $I_{\varepsilon,L}$ remains bounded and so must converge. The improper integral does exist.

**31.** (a) $1/6$ (b) $16\pi/3$

**33.** $2\pi$

# Chapter 7

## Section 7.1

**1.** $\int_{\mathbf{c}} f(x, y, z)\,ds = \int_I f(x(t), y(t), z(t))\|\mathbf{c}'(t)\|\,dt = \int_0^1 0 \cdot 1\,dt = 0.$

**3.** (a) 2 (b) $52\sqrt{14}$

**5.** $-\frac{1}{3}(1 + 1/e^2)^{3/2} + \frac{1}{3}(2^{3/2})$

**7.** (a) The path follows the straight line from (0, 0) to (1, 1) and back to (0, 0) in the $xy$ plane. Over the path, the graph of $f$ is a straight line from (0, 0, 0) to (1, 1, 1). The integral is the area of the resulting triangle covered twice and equals $\sqrt{2}$.

(b)
$$s(t) = \begin{cases} \sqrt{2}(1 - t^4) & \text{when} \quad -1 \le t \le 0 \\ \sqrt{2}(1 + t^4) & \text{when} \quad 0 < t \le 1. \end{cases}$$

The path is

$$\mathbf{c}(s) = \begin{cases} (1 - s/\sqrt{2})(1, 1) & \text{when} \quad 0 \le s \le \sqrt{2} \\ (s/(\sqrt{2} - 1))(1, 1) & \text{when} \quad \sqrt{2} \le s \le 2\sqrt{2} \end{cases}$$

and $\int_{\mathbf{c}} f\,ds = \sqrt{2}$.

**9.** $2a/\pi$

**11.** (a) $[2\sqrt{5} + \log(2 + \sqrt{5})]/4$ (b) $(5\sqrt{5} - 1)/[6\sqrt{5} + 3\log(2 + \sqrt{5})]$

**13.** The path is a unit circle centered at (0, 0, 0) in the plane $x + y + z = 0$ and so may be parametrized by $\mathbf{c}(\theta) = (\cos\theta)\mathbf{v} + (\sin\theta)\mathbf{w}$, where $\mathbf{v}$ and $\mathbf{w}$ are orthogonal unit vectors laying in that plane. For example, $\mathbf{v} = (1/\sqrt{2})(-1, 0, 1)$ and $\mathbf{w} = (1/\sqrt{6})(1, -2, 1)$ will do. The total mass is $2\pi/3$ grams.

**15.** Choosing either $\mathbf{c}(t) = (t^2, 1, 0)$ or $\mathbf{c}(t) = (1, t^2, 0)$, $0 \le t \le 1$ will do.

## Section 7.2

**1.** (a) 3/2 (b) 0 (c) 0 (d) 147

**3.** 9

**5.** By the Cauchy–Schwarz inequality, $|\mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)| \leq \|\mathbf{F}(\mathbf{c}(t))\| \, \|\mathbf{c}'(t)\|$ for every $t$. Thus,

$$\left| \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} \right| = \left| \int_a^b \mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)\, dt \right| \leq \int_a^b |\mathbf{F}(\mathbf{c}(t)) \cdot \mathbf{c}'(t)|\, dt$$

$$\leq \int_a^b \|\mathbf{F}(\mathbf{c}(t))\| \, \|\mathbf{c}'(t)\|\, dt \leq M \int_a^b \|\mathbf{c}'(t)\|\, dt = Ml.$$

**7.** $\frac{3}{4} - (n-1)/(n+1)$

**9.** 0

**11.** The length of $\mathbf{c}$.

**13.** If $\mathbf{c}'(t)$ is never 0, then the unit vector $\mathbf{T}(t) = \mathbf{c}'(t)/\|\mathbf{c}'(t)\|$ is a continuous function of $t$ and so is a smoothly turning tangent to the curve. The answer is no.

**15.** 7

**17.** Use the fact that $\mathbf{F}$ is a gradient to show that the work done is $\frac{1}{R_2} - \frac{1}{R_1}$, independent of the path.

**19.** (a) $\|\mathbf{c}'(x)\|$

(b) $f$ has a positive derivative; it is one-to-one and onto $[0, L]$ by the mean-value and intermediate-value theorems. It has a differentiable inverse by the inverse function theorem.

(c) $g'(s) = 1/\|\mathbf{c}'(x)\|$ where $s = f(x)$.

(d) By the chain rule, $\mathbf{b}'(s) = \mathbf{c}'(x) \cdot g'(s)$, which has unit length by part (c).

## Section 7.3

**1.** $z = 2(y-1) + 1$

**3.** $18(z-1) - 4(y+2) - (x-13) = 0$ or $18z - 4y - x - 13 = 0$.

**5.** The vector $\mathbf{n} = (\cos v \sin u, \sin v \sin u, \cos u) = (x, y, z)$. The surface is the unit sphere centered at the origin.

**7.** $\mathbf{n} = -(\sin v)\mathbf{i} - (\cos v)\mathbf{k}$; the surface is a cylinder.

**9.** (a) $x = x_0 + (y - y_0)(\partial h/\partial y)(y_0, z_0) + (z - z_0)(\partial h/\partial z)(y_0, z_0)$ describes the plane tangent to $x = h(y, z)$ at $(x_0, y_0, z_0)$, $x_0 = h(y_0, z_0)$.

(b) $y = y_0 + (x - x_0)(\partial k/\partial x)(x_0, z_0) + (z - z_0)(\partial k/\partial z)(x_0, z_0)$.

**11.** $z - 6x - 8y + 3 = 0$

**13.** (a) The surfaces is a helicoid. It looks like a spiral ramp winding around the $z$ axis. (See Figure 7.4.2.) It winds twice around, since $\theta$ goes up to $4\pi$.

(b) $\mathbf{n} = \pm(1/\sqrt{1+r^2})(\sin\theta, -\cos\theta, r)$.

(c) $y_0 x - x_0 y + (x_0^2 + y_0^2)z = (x_0^2 + y_0^2)z_0$.

(d) If $(x_0, y_0, z_0) = (r_0, \cos\theta_0, r_0 \sin\theta_0, \theta_0)$, then representing the line segment in the form $\{(r\cos\theta_0, r\sin\theta_0, \theta_0) | 0 \le r \le 1\}$ shows that the line lies in the surface. Representing the line as $\{(x_0, ty_0, z_0) | 0 \le t \le 1/(x_0^2 + y_0^2)\}$ and substituting into the results of part (c) shows that it lies in the tangent plane at $(x_0, y_0, z_0)$.

**15.** (a) Using cylindrical coordinates leads to the parametrization

$$\mathbf{\Phi}(z, \theta) = (\sqrt{25+z^2}\cos\theta, \sqrt{25+z^2}\sin\theta, z), \qquad -\infty < z < \infty, 0 \le \theta \le 2\pi$$

as one possible solution.

(b) $\mathbf{n} = (\sqrt{25+z^2}\cos\theta, \sqrt{25+z^2}\sin\theta, -z)/\sqrt{25+2z^2}$.

(c) $x_0 x + y_0 y = 25$.

(d) Substitute the coordinates along these lines into the defining equation of the surface and the result of part (c).

**17.** (a) $u \mapsto u$, $v \mapsto v$, $u \mapsto u^3$, and $v \mapsto v^3$ all map $\mathbb{R}$ onto $\mathbb{R}$.

(b) $\mathbf{T}_u \times \mathbf{T}_v = (0, 0, 1)$ for $\mathbf{\Phi}_1$, and this is never $\mathbf{0}$. For the surface $\mathbf{\Phi}_2$, $\mathbf{T}_u \times \mathbf{T}_v = 9u^2v^2(0, 0, 1)$, and this is $\mathbf{0}$ along the $u$ and $v$ axes.

(c) We want to show that any two parametrizations of a surface that are smooth near a point will give the same tangent plane there. Thus, suppose $\mathbf{\Phi}: D \subset \mathbb{R}^2 \to \mathbb{R}^3$ and $\mathbf{\Psi}: B \subset \mathbb{R}^2 \to \mathbb{R}^3$ are parametrized surfaces such that

$$\mathbf{\Phi}(u_0, v_0) = (x_0, y_0, z_0) = \mathbf{\Psi}(s_0, t_0) \tag{i}$$

and

$$\left(\mathbf{T}_u^{\Phi} \times \mathbf{T}_v^{\Phi}\right)\Big|_{(u_0, v_0)} \neq \mathbf{0} \qquad \text{and} \qquad \left(\mathbf{T}_s^{\Psi} \times \mathbf{T}_t^{\Psi}\right)\Big|_{(s_0, t_0)} \neq \mathbf{0}, \tag{ii}$$

so that $\mathbf{\Phi}$ and $\mathbf{\Psi}$ are smooth and one-to-one in neighborhoods of $(u_0, v_0)$ and $(s_0, t_0)$, which we may as well assume are $D$ and $B$. Suppose further that they "describe the same surface," that is, $\mathbf{\Phi}(D) = \mathbf{\Phi}(B)$. To see that they give the same tangent plane at $(x_0, y_0, z_0)$, show that they have parallel normal vectors. To do this, show that there is an open set $C$ with $(u_0, v_0) \in C \subset D$ and a differentiable map $f: C \to B$ such that $\mathbf{\Phi}(u, v) = \mathbf{\Psi}(f(u, v))$ for $(u, v) \in C$. Once you have done this, computation shows that the normal vectors are related by $\mathbf{T}_u^{\Phi} \times \mathbf{T}_v^{\Phi} = [\partial(s, t)/\partial(u, v)]\mathbf{T}_s^{\Psi} \times \mathbf{T}_t^{\Psi}$.

To see that there is such an $f$, notice that since $\mathbf{T}_s^{\Psi} \times \mathbf{T}_t^{\Psi} \neq \mathbf{0}$, at least one of the $2 \times 2$ determinants in the cross product is not zero. Assume, for example, that

$$\begin{vmatrix} \dfrac{\partial x}{\partial s} & \dfrac{\partial y}{\partial s} \\ \dfrac{\partial x}{\partial t} & \dfrac{\partial y}{\partial t} \end{vmatrix} \neq 0.$$

Now use the inverse function theorem to write $(s, t)$ as a differentiable function of $(x, y)$ in some neighborhood of $(x_0, y_0)$.

(d) No.

## Section 7.4

**1.** $4\pi$

**3.** $\frac{3}{2}\pi[\sqrt{2} + \log(1 + \sqrt{2})]$

**5.** $\frac{1}{3}\pi(6\sqrt{6} - 8)$

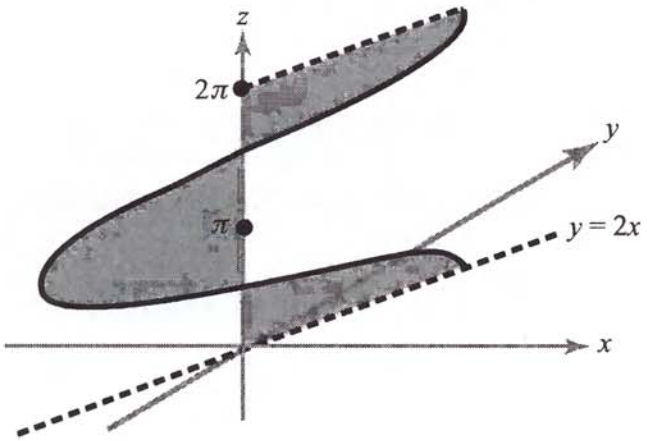
**7.** The integral for the volume converges, whereas that for the area diverges.

**9.** $A(E) = \int_0^{2\pi} \int_0^{\pi} \sqrt{a^2b^2 \sin^2\phi \cos^2\phi + b^2c^2 \sin^4\phi \cos^2\theta + a^2c^2 \sin^4\phi \sin^2\theta}\, d\phi\, d\theta$

**11.** $(\pi/6)(5\sqrt{5} - 1)$

**13.** $(\pi/2)\sqrt{6}$

**15.** $4\sqrt{5}$; for fixed $\theta$, $(x, y, z)$ moves along the horizontal line segment $y = 2x, z = \theta$ from the $z$ axis out to a radius of $\sqrt{5}|\cos\theta|$ into quadrant 1 if $\cos\theta > 0$ and into quadrant 3 if $\cos\theta < 0$.




**17.** $(\pi + 2)/(\pi - 2)$

**19.** $\pi(a + b)\sqrt{1 + m^2}(b - a)$

**21.** $\frac{4}{15}(9\sqrt{3} - 8\sqrt{2} + 1)$

**23.** With $f(x, y) = \sqrt{R^2 - x^2 - y^2}$, (4) becomes

$$A(S') = \iint_D \sqrt{\frac{x^2 + y^2}{R^2 - x^2 - y^2} + 1}\, dx\, dy$$

$$= \iint_D \frac{R}{\sqrt{R^2 - x^2 - y^2}}\, dx\, dy$$

where $D$ is the disk of radius $R$. Evaluate using polar coordinates, noting it is improper at the boundary, to get $2\pi R^2$.

## Section 7.5

**1.** $\dfrac{3\sqrt{2}+5}{24}$

**3.** $\pi a^3$

**5.** (a) $\sqrt{2}\pi R^2$ (b) $2\pi R^2$

**7.** $\dfrac{\pi}{4}\left(\dfrac{5\sqrt{5}}{3}+\dfrac{1}{15}\right)$

**9.** $16\pi R^3/3$

**11.** (a) The sphere looks the same from all three axes, so these three integrals should be the same quantity with different labels on the axes.
(b) $4\pi R^4/3$
(c) $4\pi R^4/3$

**13.** $(R/2, R/2, R/2)$

**15.** (a) Directly compute the vector cross product $\mathbf{T}_u \times \mathbf{T}_v$ and then calculate its length and compare your answer to the left-hand side.
(b) In this case, $F = 0$, so $A(s) = \iint_D \sqrt{EG}\, du\, dv$.
(c) $4\pi a^2$.

**17.** Let $a = \partial x/\partial u$, $b = \partial y/\partial u$, $c = \partial x/\partial v$, and $d = \partial y/\partial v$. The conditions (a) and (b) in Exercise 16 are then $a^2 + b^2 = c^2 + d^2$ and $ac + bd = 0$. Show that $a \neq 0$ and, by a normalization argument, show that you can assume $a = 1$. Now calculate further.

**19.** $2a^2$

## Section 7.6

**1.** $\pm 48\pi$ (the sign depends on orientation)

**3.** $4\pi$

**5.** $2\pi$ (or $-2\pi$, if you choose a different orientation)

**7.** $2\pi$

**9.** $12\pi/5$

**11.** With the usual spherical coordinate parametrization, $\mathbf{T}_\theta \times \mathbf{T}_\phi = -\sin\phi\, \mathbf{r}$ (see Example 1). Thus,

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint \mathbf{F}\cdot(\mathbf{T}_\phi \times \mathbf{T}_\theta)\, d\phi\, d\theta = \iint (\mathbf{F}\cdot\mathbf{r})\sin\phi\, d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} F_r \sin\phi\, d\phi\, d\theta$$

and

$$\iint_S f\,dS = \int_0^{2\pi}\int_0^{\pi} f\sin\phi\,d\phi\,d\theta.$$

**13.** For a cylinder of radius $R = 1$ and normal component $F_r$,

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \int_a^b\int_0^{2\pi} F_r\,d\theta\,dz.$$

**15.** $2\pi/3$

**17.** $\frac{2}{5}a^3bc\pi$

## Section 7.7

**1.** Apply formula (3) of this section and simplify; $H = 0$ and $K = -b^2/(u^2+b^2)^2$.

**3.** Apply formula (3) of this section and simplify.

**5.** $K = \dfrac{-4a^6b^6}{(a^4b^4 + 4b^4u^2 + 4a^4v^2)^2}$

**7.** Apply formula (3) of this section and simplify.

**9.** Apply formula (2) of this section and simplify; $K = -h''/[(1+(h')^2)^2h]$.

## Review Exercises for Chapter 7

**1.** (a) $3\sqrt{2}(1-e^{6\pi})/13$
(b) $-\pi\sqrt{2}/2$
(c) $(236, 158\sqrt{26} - 8)/35\cdot(25)^3$
(d) $8\sqrt{2}/189$

**3.** (a) $2/\pi + 1$ (b) $-1/2$

**5.** $2a^3$

**7.** (a) A sphere of radius 5 centered at (2, 3, 0); $\Phi(\theta,\phi) = (2+5\cos\theta\sin\phi,$ $3+5\sin\theta\sin\phi,\ 5\cos\phi)$; $0\le\theta\le 2\pi; 0\le\phi\le\pi$

(b) An ellipsoid with center at (2, 0, 0); $\Phi(\theta,\phi) = (2+(1/\sqrt{2})3\cos\theta\sin\phi,$ $3\sin\theta\sin\phi, 3\cos\phi)$; $0\le\theta\le 2\pi, 0\le\phi\le\pi$

(c) An elliptic hyperboloid of one sheet; $\Phi(\theta,z) = \left(\frac{1}{2}\sqrt{8+2z^2}\cos\theta,\right.$ $\left.\frac{1}{3}\sqrt{8+2z^2}\sin\theta, z\right)$; $0\le\theta\le 2\pi, -\infty < z < \infty$

**9.** $A(\Phi) = \frac{1}{2}\int_0^{2\pi}\sqrt{3\cos^2\theta+5}\,d\theta$; $\Phi$ describes the upper nappe of a cone with elliptical horizontal cross sections.

**11.** $11\sqrt{3}/6$

**13.** $\sqrt{2}/3$

**15.** $5\sqrt{5}/6$

**17.** (a) $(e^y \cos \pi z, xe^y \cos \pi z, -\pi x e^y \sin \pi z)$ (b) 0

**19.** $\frac{1}{2}(e^2 + 1)$

**21.** $\mathbf{n} = (1/\sqrt{5})(-1, 0, 2)$, $2z - x = 1$

**23.** 0

**25.** If $\mathbf{F} = \nabla\phi$, then $\nabla \times \mathbf{F} = \mathbf{0}$ (at least if $\phi$ is of class $C^2$; see Theorem 1, Section 3.4). Theorem 3 of Section 7.2 shows that $\int_{\mathbf{c}} \nabla\phi \cdot d\mathbf{s} = 0$ because $\mathbf{c}$ is a closed curve.

**27.** (a) $24\pi$ (b) $24\pi$ (c) $60\pi$

**29.** (a) $[\sqrt{R^2 + p^2}(z_0 - z_1)]/p$ (b) $\sqrt{2z_0(R^2 + p^2)/p^2 g}$

# Chapter 8

## Section 8.1

**1.** $-8$

**3.** (a) 0 (b) $-\pi R^2$ (c) 0 (d) $-\pi R^2$

**5.** $3\pi a^2$

**7.** $3\pi/2$

**9.** $3\pi(b^2 - a^2)/2$

**11.** (a) Both sides are $2\pi$. (b) 0

**13.** 0

**15.** $\pi ab$

**17.** A horizontal line segment divides the region into three regions of which Green's theorem applies; now use Exercise 8 or the technique in Figure 8.1.5.

**19.** $9\pi/8$

**21.** If $\varepsilon > 0$, there is a $\delta > 0$ such that $|u(\mathbf{q}) - u(\mathbf{p})| < \varepsilon$ whenever $\|\mathbf{p} - \mathbf{q}\| = \rho < \delta$. Parametrize $\partial B_\rho(\mathbf{p})$ by $\mathbf{q}(\theta) = \mathbf{p} + \rho(\cos\theta, \sin\theta)$. Then

$$|I(\rho) - 2\pi u(\mathbf{p})| \leq \int_0^{2\pi} |u(\mathbf{q}(\theta)) - u(\mathbf{p})|\, d\theta \leq 2\pi\varepsilon.$$

**23.** Parametrize $\partial B_\rho(\mathbf{p})$ as in Exercise 21. If $\mathbf{p} = (p_1, p_2)$, then $I(\rho) = \int_0^{2\pi} u(p_1 + \rho\cos\theta, p_2 + \rho\sin\theta)\,d\theta$. Differentiation under the integral sign gives

$$\frac{dI}{d\rho} = \int_0^{2\pi} \nabla u \cdot (\cos\theta, \sin\theta)\,d\theta = \int_0^{2\pi} \nabla u \cdot \mathbf{n}\,d\theta = \frac{1}{\rho}\int_{\partial B_\rho} \frac{\partial u}{\partial \mathbf{n}}\,ds = \frac{1}{\rho}\iint_{B_\rho} \nabla^2 u\,dA$$

(the last equality uses Exercise 22).

**25.** Using Exercise 24,

$$\begin{aligned}\iint_{B_R} u\,dA &= \int_0^R \int_0^{2\pi} u[\mathbf{p} + \rho(\cos\theta, \sin\theta)]\rho\,d\theta\,d\rho \\ &= \int_0^R \left(\int_{\partial B_\rho} u\,ds\right) d\rho = \int_0^R 2\pi\rho u(\mathbf{p})\,d\rho = \pi R^2 u(\mathbf{p}).\end{aligned}$$

**27.** Suppose $u$ is subharmonic. We establish the assertions corresponding to Exercises 26(a) and (b). The argument for superharmonic functions is similar, with inequalities reversed.

Suppose $\nabla^2 u \geq 0$ and $u(\mathbf{p}) \geq u(\mathbf{q})$ for all $\mathbf{q}$ in $B_R(\mathbf{p})$. By Exercise 23, $I'(\rho) \geq 0$ for $0 < \rho \leq R$, and so Exercise 24 shows that $2\pi u(\mathbf{p}) \leq I(\rho) \leq I(R)$ for $0 < \rho \leq R$. If $u(\mathbf{q}) < u(\mathbf{p})$ for some $\mathbf{q} = \mathbf{p} + \rho(\cos\theta_0, \sin\theta_0) \in B_R(\mathbf{p})$, then, by continuity, there is an arc $[\theta_0 - \delta, \theta_0 + \delta]$ on $\partial B_\rho(\mathbf{p})$ where $u < u(\mathbf{p}) - d$ for some $d > 0$. This would mean that

$$\begin{aligned}2\pi u(\mathbf{p}) \leq I(\rho) &= \frac{1}{\rho}\int_0^{2\pi} u[\mathbf{p} + \rho(\cos\theta, \sin\theta)]\rho\,d\theta \\ &\leq (2\pi - 2\delta)u(\mathbf{p}) + 2\delta[u(\mathbf{p}) - d] \leq 2\pi u(\mathbf{p}) - 2\delta d.\end{aligned}$$

This contradiction shows that we must have $u(\mathbf{q}) = u(\mathbf{p})$ for every $\mathbf{q}$ in $B_B(\mathbf{p})$.

If the maximum at $\mathbf{p}$ is absolute for $D$, the last paragraph shows that $u(\mathbf{x}) = u(\mathbf{p})$ for all $\mathbf{x}$ in some disk around $\mathbf{p}$. If $\mathbf{c}\colon [0, 1) \to D$ is a path from $\mathbf{p}$ to $\mathbf{q}$, then $u(\mathbf{c}(t)) = u(\mathbf{p})$ for all $t$ in some interval $[0, b)$. Let $b_0$ be the largest $b \in [0, 1]$ such that $u(\mathbf{c}(t)) = u(\mathbf{p})$ for all $t \in [0, b)$. (Strictly speaking, this requires the notion of the least upper bound from a good calculus text.) Because $u$ is continuous, $u(\mathbf{c}(b_0)) = u(\mathbf{p})$. If $b_0 \neq 1$, then the last paragraph would apply at $\mathbf{c}(b_0)$ and $u$ is constantly equal to $u(\mathbf{p})$ on a disk around $\mathbf{c}(b_0)$. In particular, there is a $\delta > 0$ such that $u(\mathbf{c}(t)) = u(\mathbf{c}(b_0)) = u(\mathbf{p})$ on $[0, b_0 + \delta)$. This contradicts the maximality of $b_0$, so we must have $b_0 = 1$. That is, $\mathbf{c}(\mathbf{q}) = \mathbf{c}(\mathbf{p})$. Because $\mathbf{q}$ was an arbitrary point in $D$, $u$ is constant on $D$.

**29.** Assume $\nabla^2 u_1 = 0$ and $\nabla^2 u_2 = 0$ are two solutions. Let $\phi = u_1 - u_2$. Then $\nabla^2\phi = 0$ and $\phi(x) = 0$ for all $x \in \partial D$. Consider the integral $\iint_D \phi\nabla^2\phi\,dA = -\iint_D \nabla\phi \cdot \nabla\phi\,dA$. Thus, $\iint_D \nabla\phi \cdot \nabla\phi\,dA = 0$, which implies that $\nabla\phi = \mathbf{0}$, and so $\phi$ is a constant function and hence must be identically zero.

## Section 8.2

**1.** $-2\pi$

**3.** Each integral in Stokes' theorem is zero.

**5.** 0

**7.** $-4\pi/\sqrt{3}$

**9.** 0

**11.** $\pm 2\pi$

**13.** Using Faraday's law, $\iint_S [\nabla \times \mathbf{E} + \partial \mathbf{H}/\partial t] \cdot d\mathbf{S} = 0$ for any surface $S$. If the integrand were a nonzero vector at some point, then by continuity the integral over some small disk centered at that point and lying perpendicular to that vector would be nonzero.

**15.** The orientations of $\partial S_1 = \partial S_2$ must agree.

**17.** Suppose $C$ is a closed loop on the surface drawn so that it divides the surface into two pieces $S_1$ and $S_2$. For the surface of a doughnut (torus) you must use two closed loops; can you see why? Then $C$ bounds both $S_1$ and $S_2$, but with positive orientation with respect to one and negative with respect to the other. Therefore,

$$\iint_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = \iint_{S_1} \nabla \times \mathbf{F} \cdot d\mathbf{S} + \iint_{S_2} \nabla \times \mathbf{F} \cdot d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{s} - \int_C \mathbf{F} \cdot d\mathbf{s} = 0.$$

**19.** (a) If $C = \partial S$, $\int_C \mathbf{v} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{v}) \cdot d\mathbf{S} = \iint_S \mathbf{0} \cdot d\mathbf{s} = 0$.

(b) $\int_C \mathbf{v} \cdot d\mathbf{s} = \int_a^b \mathbf{v} \cdot \mathbf{c}'(t) dt = \mathbf{v} \cdot \int_a^b \mathbf{c}'(t) dt = \mathbf{v} \cdot (\mathbf{c}(b) - \mathbf{c}(a))$, where $\mathbf{c}$: $[a, b] \to \mathbb{R}^3$ is a parametrization of $C$. (The vector integral is the vector whose components are the integrals of the component functions.) If $C$ is closed, the last expression is 0.

**21.** Both integrals give $\pi/4$.

**23.** (a) 0 (b) $\pi$ (c) $\pi$

**25.** $-20\pi$ (or $20\pi$ if the opposite orientation is used)

**27.** One possible answer: The Möbius curve $C$ is also the boundary of an *oriented* surface $\tilde{S}$; the equation in Faraday's law *is* valid for this new surface.

## Section 8.3

**1.** If $\mathbf{F} = \nabla f = \nabla g$ and $C$ is a curve from $\mathbf{v}$ to $\mathbf{w}$, then $(f - g)(\mathbf{w}) - (f - g)(\mathbf{v}) = \int_C \nabla(f - g) \cdot d\mathbf{s} = 0$ and so $f - g$ is constant.

**3.** $x^2yz - \cos x + C$

**5.** Yes, it is the gradient of $g(x, y) = F(x) + F(y)$, where $F'(x) = f(x)$.

**7.** No; $\nabla \times \mathbf{F} = (0, 0, -x) \neq \mathbf{0}$.

**9.** $e \sin 1 + \frac{1}{3}e^3 - \frac{1}{3}$

**11.** $3.5 \times 10^{29}$ ergs

**13.** (a) $f = x^2/2 + y^2/2 + C$ (c) $f = \frac{1}{3}x^3 + xy^2 + C$

(b) $\mathbf{F}$ is not a gradient field.

**15.** Use Theorem 7 in each case.

(a) $-3/2$ (b) $-1$ (c) $\cos(e^2) - \cos(1/e)/e$

**17.** (a) No. (b) $\left(\frac{1}{2}z^2, xy - z, x^2y\right)$ or $\left(\frac{1}{2}z^2 - 2xyz - \frac{1}{2}y^2, -x^2z - z, 0\right)$

**19.** $\frac{1}{3}(z^3\mathbf{i} + x^3\mathbf{j} + y^3\mathbf{k})$

**21.** $(-z\sin y + y\sin x, xz\cos y, 0)$ (Other answers are possible.)

**23.** (a) $\nabla \times \mathbf{F} = (0, 0, 2) \neq \mathbf{0}$

(b) Let $\mathbf{c}(t)$ be the path of an object in the fluid. Then $\mathbf{F}(\mathbf{c}(t)) = \mathbf{c}'(t)$. Let $\mathbf{c}(t) = (x(t), y(t), z(t))$. Then $x' = -y$, $y' = x$, and $z' = 0$, and so $z$ is constant and the motion is parallel to the $xy$ plane. Also, $x'' + x = 0$, $y'' + y = 0$. Thus, $x = A\cos t + B\sin t$ and $y = C\cos t + D\sin t$. Substituting these values in $x' = -y$, $y' = x$, we get $C = -B$, $D = A$, so that $x^2 + y^2 = A^2 + B^2$ and we have a circle.

(c) Counterclockwise

**25.** (a) $\mathbf{F} = -\dfrac{GmM}{(x^2 + y^2 + z^2)^{3/2}}(x, y, z)$;

$$\nabla \cdot \mathbf{F} = -GmM\left[\frac{x^2 + y^2 + z^2 - 3x^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{x^2 + y^2 + z^2 - 3y^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{x^2 + y^2 + z^2 - 3z^2}{(x^2 + y^2 + z^2)^{5/2}}\right]$$

$$= 0$$

(b) Let $S$ be the unit sphere, $S_1$ the upper hemisphere, $S_2$ the lower hemisphere, and $C$ the unit circle. If $\mathbf{F} = \nabla \times \mathbf{G}$, then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} + \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \int_C \mathbf{G} \cdot d\mathbf{s} - \int_C \mathbf{G} \cdot d\mathbf{s} = 0.$$

But $\iint_S \mathbf{F} \cdot d\mathbf{S} = -GmM \iint_S (\mathbf{r}/\|\mathbf{r}\|^3) \cdot \mathbf{n}\, dS = -4\pi GmM$, because $\|\mathbf{r}\| = 1$ and $\mathbf{r} = \mathbf{n}$ on $S$. Thus, $\mathbf{F} = \nabla \times \mathbf{G}$ is impossible. This does not contradict Theorem 8 because $\mathbf{F}$ is not smooth at the origin.

## Section 8.4

**1.** $4\pi$

**3.** 3

**5.** (a) 0 (b) 4/15 (c) $-4/15$

**7.** 6

**9.** 1

**11.** Apply the divergence theorem to $f\mathbf{F}$ using $\nabla \cdot (f\mathbf{F}) = \nabla f \cdot \mathbf{F} + f\nabla \cdot \mathbf{F}$.

**13.** If $\mathbf{F} = \mathbf{r}/r^2$, then $\nabla \cdot \mathbf{F} = 1/r^2$. If $(0, 0, 0) \notin \Omega$, the result follows from Gauss' theorem. If $(0, 0, 0) \in \Omega$, we compute the integral by deleting a small ball $B_\varepsilon = \{(x, y, z) | (x^2 + y^2 + z^2)^{1/2} < \varepsilon\}$ around the origin and then letting $\varepsilon \to 0$:

$$\begin{aligned}\iiint_\Omega \frac{1}{r^2} dV &= \lim_{\varepsilon \to 0} \iiint_{\Omega \setminus B_\varepsilon} \frac{1}{r^2} dV = \lim_{\varepsilon \to 0} \iint_{\partial(\Omega \setminus B_\varepsilon)} \frac{\mathbf{r} \cdot \mathbf{n}}{r^2} dS \\ &= \lim_{\varepsilon \to 0} \left( \iint_{\partial\Omega} \frac{\mathbf{r} \cdot \mathbf{n}}{r^2} dS - \iint_{\partial B_\varepsilon} \frac{\mathbf{r} \cdot \mathbf{n}}{r^2} dS \right) = \lim_{\varepsilon \to 0} \left( \iint_{\partial\Omega} \frac{\mathbf{r} \cdot \mathbf{n}}{r^2} dS - 4\pi\varepsilon \right) \\ &= \iint_{\partial\Omega} \frac{\mathbf{r} \cdot \mathbf{n}}{r^2} dS.\end{aligned}$$

The integral over $\partial B_\varepsilon$ is obtained from Theorem 10 (Gauss' law), because $r = \varepsilon$ everywhere on $B_\varepsilon$.

**15.** Use the vector identity for $\operatorname{div}(f\mathbf{F})$ and the divergence theorem for part (a). Use the vector identity $\nabla \cdot (f\nabla g - g\nabla f) = f\nabla^2 g - g\nabla^2 f$ for part (b).

**17.** (a) If $\phi(\mathbf{p}) = \iiint_W \rho(\mathbf{q})/(4\pi \|\mathbf{p} - \mathbf{q}\|)\, dV(\mathbf{q})$, then

$$\begin{aligned}\nabla\phi(\mathbf{p}) &= \iiint_W [\rho(\mathbf{q})/4\pi] \nabla_{\mathbf{p}} (1/\|\mathbf{p} - \mathbf{q}\|)\, dV(\mathbf{q}) \\ &= -\iiint_W [\rho(\mathbf{q})/4\pi][(\mathbf{p} - \mathbf{q})/\|\mathbf{p} - \mathbf{q}\|^3]\, dV(\mathbf{q}),\end{aligned}$$

where $\nabla_{\mathbf{p}}$ means the gradient with respect to the coordinates of $\mathbf{p}$ and the integral is the vector whose components are the three component integrals. If $\mathbf{p}$ varies in $V \cup \partial V$ and $\mathbf{n}$ is the outward unit normal to $\partial V$, we can take the inner product using these components and collect the pieces as

$$\nabla\phi(\mathbf{p}) \cdot \mathbf{n} = -\iiint_W \frac{\rho(\mathbf{q})}{4\pi} \frac{1}{\|\mathbf{p} - \mathbf{q}\|^3} (\mathbf{p} - \mathbf{q}) \cdot \mathbf{n}\, dV(\mathbf{q}).$$

Thus,

$$\iint_{\partial V} \nabla\phi(\mathbf{p}) \cdot \mathbf{n}\, dV(\mathbf{p}) = -\iint_{\partial V} \left( \iiint_W \frac{\rho(\mathbf{q})}{4\pi} \frac{1}{\|\mathbf{p} - \mathbf{q}\|^3} (\mathbf{p} - \mathbf{q}) \cdot \mathbf{n}\, d\mathbf{q} \right) dV(\mathbf{p}).$$

There are essentially five variables of integration here, three placing $\mathbf{q}$ in $W$ and two placing $\mathbf{p}$ on $\partial V$. Use Fubini's theorem to obtain

$$\iint_{\partial V} \nabla\phi \cdot \mathbf{n} \cdot d\mathbf{S} = -\iiint_W \frac{\rho(\mathbf{q})}{4\pi} \left[ \iint_{\partial V} \frac{(\mathbf{p} - \mathbf{q}) \cdot \mathbf{n}}{\|\mathbf{p} - \mathbf{q}\|^3} dS(\mathbf{p}) \right] dV(\mathbf{q}).$$

If $V$ is a symmetric elementary region, Theorem 10 says that the inner integral is $4\pi$ if $\mathbf{q} \in V$ and 0 if $\mathbf{q} \notin V$. Thus,

$$\iint_{\partial V} \nabla\phi \cdot \mathbf{n}\, dS = -\iiint_{W \cap V} \rho(\mathbf{q})\, dV(\mathbf{q}).$$

Because $\rho = 0$ outside $W$,

$$\iint_{\partial V} \nabla\phi \cdot \mathbf{n}\, dS = -\iiint_{V} \rho(\mathbf{q})\, dV(\mathbf{q}).$$

If $V$ is not a symmetric elementary region, subdivide it into a sum of such regions. The equation holds on each piece, and, upon adding them together, the boundary integrals along appropriately oriented interior boundaries cancel, leaving the desired result.

(b) By Theorem 9, $\iint_{\partial V} \nabla\phi \cdot d\mathbf{S} = \iiint_V \nabla^2\phi\, dV$, and so $\iiint_V \nabla^2\phi\, dV = -\iiint_V \rho\, dV$. Because both $\rho$ and $\nabla^2\phi$ are continuous and this holds for arbitrarily small regions, we must have $\nabla^2\phi = -\rho$.

**19.** If the charge $Q$ is spread evenly over the sphere $S$ of radius $R$ centered at the origin, the density of charge per unit area must be $Q/4\pi R^2$. If $\mathbf{p}$ is a point not on $S$ and $\mathbf{q} \in S$, then the contribution to the electric field at $\mathbf{p}$ due to charge near $\mathbf{q}$ is directed along the vector $\mathbf{p} - \mathbf{q}$. Because the charge is evenly distributed, the tangential component of this contribution will be canceled by that from a symmetric point on the other side of the sphere at the same distance from $\mathbf{p}$. (Draw the picture.) The total resulting field must be radial. Because $S$ looks the same from any point at a distance $\|\mathbf{p}\|$ from the origin, the field must depend only on radius and be of the form $\mathbf{E} = f(r)\mathbf{r}$.

If we look at the sphere $\Sigma$ of radius $\|\mathbf{p}\|$, we have

$$\begin{aligned}(\text{charge inside } \Sigma) &= \iint_\Sigma \mathbf{E} \cdot d\mathbf{S} = \iint_\Sigma f(\|\mathbf{p}\|)\mathbf{r} \cdot \mathbf{n}\, dS \\ &= f(\|\mathbf{p}\|)\|\mathbf{p}\| \text{ area } \Sigma = 4\pi \|\mathbf{p}\|^3 f(\|\mathbf{p}\|).\end{aligned}$$

If $\|\mathbf{p}\| < R$, there is no charge inside $\Sigma$; if $\|\mathbf{p}\| > R$, the charge inside $\Sigma$ is $Q$, and so

$$\mathbf{E}(\mathbf{p}) = \begin{cases} \dfrac{1}{4\pi}\dfrac{Q}{\|\mathbf{p}\|^3}\mathbf{p} & \text{if } \|\mathbf{p}\| > R \\ 0 & \text{if } \|\mathbf{p}\| < R. \end{cases}$$

**21.** By Theorem 10, $\iint_{\partial M} \mathbf{F} \cdot d\mathbf{S} = 4\pi$ for any surface enclosing the origin. But if $\mathbf{F}$ were the curl of some field, then the integral over such a closed surface would have to be 0.

**23.** If $S = \partial W$, then $\iint_S \mathbf{r} \cdot \mathbf{n}\, dS = \iiint_W \nabla \cdot \mathbf{r}\, dV = \iiint_W 3 dV = 3$ volume $(W)$. For the geometric explanation, assume $(0, 0, 0) \in W$ and consider the skew cone with its vertex at $(0, 0, 0)$ with base $\Delta S$ and altitude $\|\mathbf{r}\|$. Its volume is $\frac{1}{3}(\Delta S)(\mathbf{r} \cdot \mathbf{n})$.

## Section 8.5

**1.** Write the components of $\varphi$ as $\xi(\mathbf{x}, t)$, $\eta(\mathbf{x}, t)$, and $\zeta(\mathbf{x}, t)$. First, observe that by definition of $\varphi$,

$$\frac{\partial}{\partial t}\varphi(\mathbf{x}, t) = \mathbf{F}(\varphi(\mathbf{x}, t), t).$$

The determinant $J$ can be differentiated by recalling that the determinant of a matrix is multilinear in the columns (or rows). Thus, holding $\mathbf{x}$ fixed,

$$\frac{\partial}{\partial t}J = \begin{bmatrix} \frac{\partial}{\partial t}\frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} & \frac{\partial \zeta}{\partial x} \\ \frac{\partial}{\partial t}\frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} & \frac{\partial \zeta}{\partial y} \\ \frac{\partial}{\partial t}\frac{\partial \xi}{\partial z} & \frac{\partial \eta}{\partial z} & \frac{\partial \zeta}{\partial z} \end{bmatrix} + \begin{bmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial}{\partial t}\frac{\partial \eta}{\partial x} & \frac{\partial \zeta}{\partial x} \\ \frac{\partial \xi}{\partial y} & \frac{\partial}{\partial t}\frac{\partial \eta}{\partial y} & \frac{\partial \zeta}{\partial y} \\ \frac{\partial \xi}{\partial z} & \frac{\partial}{\partial t}\frac{\partial \eta}{\partial z} & \frac{\partial \zeta}{\partial z} \end{bmatrix} + \begin{bmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} & \frac{\partial}{\partial t}\frac{\partial \zeta}{\partial x} \\ \frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} & \frac{\partial}{\partial t}\frac{\partial \zeta}{\partial y} \\ \frac{\partial \xi}{\partial z} & \frac{\partial \eta}{\partial z} & \frac{\partial}{\partial t}\frac{\partial \zeta}{\partial z} \end{bmatrix}.$$

Now write

$$\frac{\partial}{\partial t}\frac{\partial \xi}{\partial x} = \frac{\partial}{\partial x}\frac{\partial \xi}{\partial t} = \frac{\partial}{\partial x}F_1(\varphi(\mathbf{x}, t), t),$$

$$\frac{\partial}{\partial t}\frac{\partial \xi}{\partial y} = \frac{\partial}{\partial y}\frac{\partial \xi}{\partial t} = \frac{\partial}{\partial y}F_2(\varphi(\mathbf{x}, t), t),$$

$$\frac{\partial}{\partial t}\frac{\partial \zeta}{\partial z} = \frac{\partial}{\partial z}\frac{\partial \zeta}{\partial t} = \frac{\partial}{\partial z}F_3(\varphi(\mathbf{x}, t), t).$$

The components $F_1$, $F_2$, and $F_3$ of $\mathbf{F}$ in this expression are functions of $x$, $y$, and $z$ through $\varphi(\mathbf{x}, t)$; therefore,

$$\frac{\partial}{\partial x}F_1(\varphi(\mathbf{x}, t), t) = \frac{\partial F_1}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial F_1}{\partial \eta}\frac{\partial \eta}{\partial x} + \frac{\partial F_1}{\partial \zeta}\frac{\partial \zeta}{\partial x},$$

$$\vdots$$

$$\frac{\partial}{\partial z}F_3(\varphi(\mathbf{x}, t), t) = \frac{\partial F_3}{\partial \xi}\frac{\partial \xi}{\partial z} + \frac{\partial F_3}{\partial \eta}\frac{\partial \eta}{\partial z} + \frac{\partial F_3}{\partial \zeta}\frac{\partial \zeta}{\partial z}.$$

When these are substituted into the previous expression for $\partial J/\partial t$, one gets for the respective terms

$$\frac{\partial F_1}{\partial x}J + \frac{\partial F_2}{\partial y}J + \frac{\partial F_3}{\partial z}J = (\operatorname{div}\mathbf{F})J.$$

**3.** HINTS: By the transport equation from Theorem 12, with $\mathbf{V}$ in place of $\mathbf{F}$,

$$\frac{d}{dt}\iiint_{W_t}\rho\,dx\,dy\,dz = \iiint_{W_t}\left(\frac{D\rho}{Dt} + \rho\operatorname{div}\mathbf{V}\right)dx\,dy\,dz.$$

Now use the fact that

$$\frac{D\rho}{Dt} + \rho\operatorname{div}\mathbf{V} = \operatorname{div}\mathbf{J} + \frac{\partial \rho}{\partial t},$$

where $\mathbf{J} = \rho\mathbf{V}$, as in the text.

**5.** If $v_i$ is the $i$th component of a vector $\mathbf{v}$, then by the transport equation (Exercise 2),

$$\begin{aligned}
\left[\frac{d}{dt}\iiint_{W_t} f\mathbf{F}\,dx\,dy\,dz\right]_i &= \frac{d}{dt}\iiint_{W_t}(f\mathbf{F})_i\,dx\,dy\,dz = \frac{d}{dt}\iiint_{W_t} fF_i\,dx\,dy\,dz \\
&= \iiint_{W_t}\left[\frac{D(fF_i)}{Dt} + (fF_i)\operatorname{div}\mathbf{F}\right]dx\,dy\,dz \\
&= \iiint_{W_t}\left[\frac{\partial}{\partial t}(fF_i) + \mathbf{D}_x(fF_i)\cdot\mathbf{F} + (f\mathbf{F}_i)\operatorname{div}\mathbf{F}\right]dx\,dy\,dz \\
&= \iiint_{W_t}\left[\frac{\partial}{\partial t}(f\mathbf{F}_i) + \nabla(fF_i)\cdot\mathbf{F} + (fF_i)\operatorname{div}\mathbf{F}\right]dx\,dy\,dz \\
&= \iiint_{W_t}\left\{\frac{\partial}{\partial t}(f\mathbf{F}_i) + [\mathbf{D}(f\mathbf{F})\mathbf{F}]_i + [(f\mathbf{F})\operatorname{div}\mathbf{F}]_i\right\}dx\,dy\,dz \\
&= \iiint_{W_t}\left[\frac{\partial}{\partial t}(f\mathbf{F}) + \mathbf{D}(f\mathbf{F})\mathbf{F} + (f\mathbf{F})\operatorname{div}\mathbf{F}\right]_i dx\,dy\,dz \\
&= \left[\iiint_{W_t}\frac{\partial}{\partial t}(f\mathbf{F}) + \mathbf{D}(f\mathbf{F})\mathbf{F} + (f\mathbf{F})\operatorname{div}\mathbf{F}\,dx\,dy\,dz\right]_i \\
&= \left[\iiint_{W_t}\left(\frac{\partial}{\partial t}(f\mathbf{F}) + (\mathbf{F}\cdot\nabla)(f\mathbf{F}) + (f\mathbf{F})\operatorname{div}\mathbf{F}\right)dx\,dy\,dz\right]_i.
\end{aligned}$$

**7.** (a) Because $\mathbf{V} = \nabla\phi$, $\nabla\times\mathbf{V} = \mathbf{0}$, and therefore $(\mathbf{V}\cdot\nabla)\mathbf{V} = \frac{1}{2}\nabla(\|\mathbf{V}\|^2)$, Euler's equation becomes

$$-\frac{\nabla p}{\rho} = \frac{d\mathbf{V}}{dt} + \frac{1}{2}\nabla(\|\mathbf{V}\|^2) = \nabla\left(\frac{d\phi}{dt} + \frac{1}{2}\|\mathbf{V}\|^2\right).$$

If $\mathbf{c}$ is a path from $P_1$ to $P_2$, then

$$\begin{aligned}
-\int_{\mathbf{c}}\frac{1}{\rho}\,dp &= -\int\frac{1}{\rho}\nabla p\cdot\mathbf{c}'(t)\,dt = \int_{\mathbf{c}}\nabla\left(\frac{d\phi}{dt} + \frac{1}{2}\|\mathbf{V}\|^2\right)\cdot\mathbf{c}'(t)\,dt \\
&= \left(\frac{d\phi}{dt} + \frac{1}{2}\|\mathbf{V}\|^2\right)\Bigg|_{P_1}^{P_2}.
\end{aligned}$$

(b) If $d\mathbf{V}/dt = \mathbf{0}$ and $\rho$ is constant, then $\frac{1}{2}\nabla(\|\mathbf{V}\|^2) = -(\nabla p)/\rho = -\nabla(p/\rho)$, and therefore $\nabla\left(\frac{1}{2}\|\mathbf{V}\|^2 + p/\rho\right) = \mathbf{0}$.

**9.** By Ampère's law, $\nabla\cdot\mathbf{J} = \nabla\cdot(\nabla\times\mathbf{H}) - \nabla\cdot(\partial\mathbf{E}/\partial t) = -\nabla\cdot(\partial\mathbf{E}/\partial t) = -(\partial/\partial t)(\nabla\cdot\mathbf{E})$. By Gauss' law this is $-\partial\rho/\partial t$. Thus, $\nabla\cdot\mathbf{J} + \partial\rho/\partial t = 0$.

## Section 8.6

**1.** (a) $(2xy^2 - yx^3)\,dx\,dy$
(b) $(x^2 + y^2)\,dx\,dy$
(c) $(x^2 + y^2 + z^2)\,dx\,dy\,dz$
(d) $(xy + x^2)\,dx\,dy\,dz$
(e) $dx\,dy\,dz$

**3.** (a) $2xy\,dx + (x^2 + 3y^2)\,dy$
(b) $-(x + y^2 \sin x)\,dx\,dy$
(c) $-(2x + y)\,dx\,dy$
(d) $dx\,dy\,dz$
(e) $2x\,dx\,dy\,dz$
(f) $2y\,dy\,dz - 2x\,dz\,dx$
(g) $-\dfrac{4xy}{(x^2 + y^2)^2}\,dx\,dy$
(h) $2xy\,dx\,dy\,dz$

**5.** (a)
$$\begin{aligned}\text{Form}_2(\alpha\mathbf{V}_1 + \mathbf{V}_2) &= \text{Form}_2(\alpha A_1 + A_2, \alpha B_1 + B_2, \alpha C_1 + C_2)\\ &= (\alpha A_1 + A_2)\,dy\,dz + (\alpha B_1 + B_2)\,dz\,dx\\ &\quad + (\alpha C_1 + C_2)\,dx\,dy\\ &= \alpha(A_1\,dy\,dz + B_1\,dz\,dx + C_1\,dx\,dy)\\ &\quad + (A_2\,dy\,dz + B_2\,dz\,dx + C_2\,dx\,dy)\\ &= \alpha\,\text{Form}_2(\mathbf{V}_1) + \text{Form}_2(\mathbf{V}_2)\end{aligned}$$

(b)
$$\begin{aligned}d\omega = &\left(\frac{\partial A}{\partial x}\,dx + \frac{\partial A}{\partial y}\,dy + \frac{\partial A}{\partial z}\,dz\right) \wedge dx + A(dx)^2\\ &+ \left(\frac{\partial B}{\partial x}\,dx + \frac{\partial B}{\partial y}\,dy + \frac{\partial B}{\partial z}\,dz\right) \wedge dy + B(dy)^2\\ &+ \left(\frac{\partial C}{\partial x}\,dx + \frac{\partial C}{\partial y}\,dy + \frac{\partial C}{\partial z}\,dz\right) \wedge dz + C(dz)^2.\end{aligned}$$

But $(dx)^2 = (dy)^2 = (dz)^2 = dx \wedge dx = dy \wedge dy = dz \wedge dz = 0$, $dy \wedge dx = -dx \wedge dy$, $dz \wedge dy = -dy \wedge dz$, and $dx \wedge dz = -dz \wedge dx$. Hence,

$$\begin{aligned}d\omega &= \left(\frac{\partial C}{\partial y} - \frac{\partial B}{\partial z}\right) dy\,dz + \left(\frac{\partial A}{\partial z} - \frac{\partial C}{\partial y}\right) dz\,dx + \left(\frac{\partial B}{\partial x} - \frac{\partial A}{\partial y}\right) dx\,dy\\ &= \text{Form}_2(\text{curl}\,\mathbf{V}).\end{aligned}$$

**7.** An oriented 1-manifold is a curve. Its boundary is a pair of points that may be considered a 0-manifold. Therefore, $\omega$ is a 0-form or function, and $\int_{\partial M} d\omega = \omega(b) - \omega(a)$ if the curve $M$ runs from $a$ to $b$. Furthermore, $d\omega$ is the 1-form $(\partial\omega/\partial x)\,dx + (\partial\omega/\partial y)\,dy$. Therefore, $\int_M d\omega$ is the line integral $\int_M (\partial\omega/\partial x)\,d\omega + (\partial\omega/\,dy)\,dy = \int_M \nabla\omega \cdot d\mathbf{s}$. Thus, we obtain Theorem 3 of Section 7.2, $\int_M \nabla\omega \cdot d\mathbf{s} = \omega(b) - \omega(a)$.

**9.** Put $\omega = F_1\,dx\,dy + F_2\,dy\,dz + F_3\,dz\,dx$. The integral becomes

$$\iint_{\partial T} \omega = \iiint_T d\omega = \iiint_T \left(\frac{\partial F_1}{\partial z} + \frac{\partial F_2}{\partial x} + \frac{\partial F_3}{\partial y}\right) dx\,dy\,dz.$$

(a) 0 (b) 40

**11.** Consider $\omega = x\,dy\,dz + y\,dz\,dx + z\,dx\,dy$. Compute that $d\omega = 3\,dx\,dy\,dz$, so that $\frac{1}{3}\iint_{\partial R} \omega = \frac{1}{3}\iiint_R d\omega = \iiint_R dx\,dy\,dz = v(R)$.

## Review Exercises for Chapter 8

**1.** (a) $2\pi a^2$ (b) 0

**3.** 0

**5.** (a) $f = x^4/4 - x^2y^3$ (b) $-1/4$

**7.** (a) Check that $\nabla \times \mathbf{F} = \mathbf{0}$ (b) $f = 3x^2y\cos z + C$ (c) 0

**9.** 23/6

**11.** No: $\nabla \times (\mathbf{a} \times \mathbf{r}) = 2\mathbf{a}$

**13.** (a) $\nabla f = 3ye^{z^2}\mathbf{i} + 3xe^{z^2}\mathbf{j} + 6xyze^{z^2}\mathbf{k}$ (b) 0 (c) Both sides are 0.

**15.** $8\pi/3$

**17.** $\pi a^2/4$

**19.** 21

**21.** (a) $\mathbf{G}$ is conservative; $\mathbf{F}$ is not.

(b) $\mathbf{G} = \nabla\phi$ if $\phi = (x^4/4) + (y^4/4) - \dfrac{3}{2}x^2y^2 + \dfrac{1}{2}z^2 + C$, where $C$ is any constant.

(c) $\displaystyle\int_\alpha \mathbf{F}\cdot d\mathbf{s} = 0; \int_\alpha \mathbf{G}\cdot d\mathbf{s} = -\frac{1}{2}; \int_\beta \mathbf{F}\cdot d\mathbf{s} = \frac{1}{3}; \int_\beta \mathbf{G}\cdot d\mathbf{s} = -\frac{1}{2}$

**23.** Use $(\nabla\cdot\mathbf{F})(x_0, y_0, z_0) = \operatorname{limit}_{\rho\to 0}\frac{1}{V(\Omega_\rho)}\iint_{\partial\Omega_\rho}\mathbf{F}\cdot\mathbf{n}\,dS$ from Section 8.4.

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**48.** $$\int \frac{x}{\sqrt{a+bx}}\,dx = \frac{2(bx-2a)\sqrt{a+bx}}{3b^2}$$

**49.** $$\int \frac{1}{x\sqrt{a+bx}}\,dx = \begin{cases} \dfrac{1}{\sqrt{a}}\log\left|\dfrac{\sqrt{a+bx}-\sqrt{a}}{\sqrt{a+bx}+\sqrt{a}}\right| & (a>0) \\ \dfrac{2}{\sqrt{-a}}\arctan\left|\sqrt{\dfrac{a+bx}{-a}}\right| & (a<0) \end{cases}$$

**50.** $$\int \frac{\sqrt{a^2-x^2}}{x}\,dx = \sqrt{a^2-x^2} - a\log\left|\frac{a+\sqrt{a^2-x^2}}{x}\right|$$

**51.** $$\int x\sqrt{a^2-x^2}\,dx = -\tfrac{1}{3}(a^2-x^2)^{3/2}$$

**52.** $$\int x^2\sqrt{a^2-x^2}\,dx = \frac{x}{8}(2x^2-a^2)\sqrt{a^2-x^2} + \frac{a^4}{8}\arcsin\frac{x}{a} \quad (a>0)$$

**53.** $$\int \frac{1}{x\sqrt{a^2-x^2}}\,dx = -\frac{1}{a}\log\left|\frac{a+\sqrt{a^2-x^2}}{x}\right|$$

**54.** $$\int \frac{x}{\sqrt{a^2-x^2}}\,dx = -\sqrt{a^2-x^2}$$

**55.** $$\int \frac{x^2}{\sqrt{a^2-x^2}}\,dx = -\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} \quad (a>0)$$

**56.** $$\int \frac{\sqrt{x^2+a^2}}{x}\,dx = \sqrt{x^2+a^2} - a\log\left|\frac{a+\sqrt{x^2+a^2}}{x}\right|$$

**57.** $$\int \frac{\sqrt{x^2-a^2}}{x}\,dx = \sqrt{x^2-a^2} - a\arccos\frac{a}{|x|} \quad (a>0)$$

**58.** $$\int \frac{x^2}{\sqrt{x^2+a^2}}\,dx = \frac{x\sqrt{x^2+a^2}}{2} - \frac{a^2}{2}\log\left(x+\sqrt{x^2+a^2}\right)$$

**59.** $$\int \frac{1}{x\sqrt{x^2+a^2}}\,dx = \frac{1}{a}\log\left|\frac{x}{a+\sqrt{x^2+a^2}}\right|$$

**60.** $$\int \frac{1}{x\sqrt{x^2-a^2}}\,dx = \frac{1}{a}\arccos\frac{a}{|x|} \quad (a>0)$$

**61.** $$\int \frac{1}{x^2\sqrt{x^2\pm a^2}}\,dx = \mp\frac{\sqrt{x^2\pm a^2}}{a^2x}$$

**62.** $$\int \frac{1}{\sqrt{x^2\pm a^2}}\,dx = \sqrt{x^2\pm a^2}$$

**63.** $$\int \frac{1}{ax^2+bx+c}\,dx = \begin{cases} \dfrac{1}{\sqrt{b^2-4ac}}\log\left|\dfrac{2ax+b-\sqrt{b^2-4ac}}{2ax+b+\sqrt{b^2-4ac}}\right| & (b^2>4ac) \\ \dfrac{2}{\sqrt{4ac-b^2}}\arctan\dfrac{2ax+b}{\sqrt{4ac-b^2}} & (b^2<4ac) \end{cases}$$

**64.** $$\int \frac{x}{ax^2+bx+c}\,dx = \frac{1}{2a}\log|ax^2+bx+c| - \frac{b}{2a}\int \frac{1}{ax^2+bx+c}\,dx$$

**65.** $$\int \frac{1}{\sqrt{ax^2+bx+c}}\,dx = \begin{cases} \dfrac{1}{\sqrt{a}}\log|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}| & (a>0) \\ \dfrac{1}{\sqrt{-a}}\arcsin\dfrac{-2ax-b}{\sqrt{b^2-4ac}} & (a<0) \end{cases}$$

**66.** $$\int \sqrt{ax^2+bx+c}\,dx = \frac{2ax+b}{4a}\sqrt{ax^2+bx+c} + \frac{4ac-b^2}{8a}\int \frac{1}{\sqrt{ax^2+b+c}}\,dx$$

**67.** $$\int \frac{x}{\sqrt{ax^2+bx+c}}\,dx = \frac{\sqrt{ax^2+bx+c}}{a} - \frac{b}{2a}\int \frac{1}{\sqrt{ax^2+bx+c}}\,dx$$

**68.** $$\int \frac{1}{x\sqrt{ax^2+bx+c}}\,dx = \begin{cases} \dfrac{-1}{\sqrt{c}}\log\left|\dfrac{2\sqrt{c}\sqrt{ax^2+bx+c}+bx+2c}{x}\right| & (c>0) \\ \dfrac{1}{\sqrt{-c}}\arcsin\dfrac{bx+2c}{|x|\sqrt{b^2-4ac}} & (c<0) \end{cases}$$

**69.** $$\int x^3\sqrt{x^2+a^2}\,dx = \left(\tfrac{1}{5}x^2 - \tfrac{2}{15}a^2\right)\sqrt{(a^2+x^2)^3}$$

**70.** $$\int \frac{\sqrt{x^2 \pm a^2}}{x^4}\,dx = \frac{\mp\sqrt{(x^2 \pm a^2)^3}}{3a^2x^3}$$

**71.** $$\int \sin ax \sin bx\,dx = \frac{\sin(a-b)x}{2(a-b)} - \frac{\sin(a+b)x}{2(a+b)} \quad (a^2 \neq b^2)$$

**72.** $$\int \sin ax \cos bx\,dx = -\frac{\cos(a-b)x}{2(a-b)} - \frac{\cos(a+b)x}{2(a+b)} \quad (a^2 \neq b^2)$$

**73.** $$\int \cos ax \cos bx\,dx = \frac{\sin(a-b)x}{2(a-b)} + \frac{\sin(a+b)x}{2(a+b)} \quad (a^2 \neq b^2)$$

**74.** $$\int \sec x \tan x\,dx = \sec x$$

**75.** $$\int \csc x \cot x\,dx = -\csc x$$

**76.** $$\int \cos^m x \sin^n x\,dx = \frac{\cos^{m-1}x \sin^{n+1}x}{m+n} + \frac{m-1}{m+n}\int \cos^{m-2}x \sin^n x\,dx$$
$$= -\frac{\sin^{n-1}x \cos^{m+1}x}{m+n} + \frac{n-1}{m+n}\int \cos^m x \sin^{n-2}x\,dx$$

**77.** $$\int x^n \sin ax\,dx = -\frac{1}{a}x^n \cos ax + \frac{n}{a}\int x^{n-1}\cos ax\,dx$$

**78.** $$\int x^n \cos ax\,dx = \frac{1}{a}x^n \sin ax - \frac{n}{a}\int x^{n-1}\sin ax\,dx$$

**79.** $$\int x^n e^{ax}\,dx = \frac{x^n e^{ax}}{a} - \frac{n}{a}\int x^{n-1}e^{ax}\,dx$$

**80.** $$\int x^n \log ax\,dx = x^{n+1}\left[\frac{\log ax}{n+1} - \frac{1}{(n+1)^2}\right]$$

**81.** $$\int x^n (\log ax)^m\,dx = \frac{x^{n+1}}{n+1}(\log ax)^m - \frac{m}{n+1}\int x^n (\log ax)^{m-1}\,dx$$

**82.** $$\int e^{ax}\sin bx\,dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}$$

**83.** $$\int e^{ax}\cos bx\,dx = \frac{e^{ax}(b\sin bx + a\cos bx)}{a^2+b^2}$$

**84.** $$\int \operatorname{sech} x \tanh x\,dx = -\operatorname{sech} x$$

**85.** $$\int \operatorname{csch} x \coth x\,dx = -\operatorname{csch} x$$

# SYMBOLS INDEX

SYMBOLS ARE LISTED IN ORDER OF THEIR APPEARANCE IN THE TEXT


| SYMBOL | NAME | PAGE |
|---|---|---|
| $\mathbb{R}$ | real numbers | xxv |
| $[a, b]$ | closed interval $\{x \mid a \leq x \leq b\}$ | xxv |
| $(a, b)$ | open interval $\{x \mid a < x < b\}$ | xxv |
| $[a, b)$ | half-open interval $\{x \mid a \leq x < b\}$ | xxv |
| $(a, b]$ | half-open interval $\{x \mid a < x \leq b\}$ | xxv |
| $\lvert a \rvert$ | absolute value of $a$ | xxvi |
| $\mathbb{Q}$ | rational numbers | xxvi |
| $\mathbb{R}^n$ | $n$-dimensional space | 3 |
| $\mathbf{i}, \mathbf{j}, \mathbf{k}$ | standard basis in $\mathbb{R}^3$ | 10 |
| $\lVert \mathbf{a} \rVert$ | norm of a vector $\mathbf{a}$ | 24 |
| $\mathbf{a} \cdot \mathbf{b}$ | inner product of the vectors $\mathbf{a}$ and $\mathbf{b}$ | 24 |
| $\mathbf{a} \times \mathbf{b}$ | cross product of the vectors $\mathbf{a}$ and $\mathbf{b}$ | 43 |
| $(r, \theta, z)$ | cylindrical coordinates | 66 |
| $(\rho, \theta, \phi)$ | spherical coordinates | 69 |
| $D_r(\mathbf{x}_0)$ | disk of radius $r$ about $\mathbf{x}_0$ | 107 |
| $\lim_{\mathbf{x} \to \mathbf{x}_0}$ | limit as $\mathbf{x}$ approaches $\mathbf{x}_0$ | 112 |
| $\lim_{x \to b^-}$ | left hand limit; $x \to b$ from below | 127 |
| $\frac{\partial f}{\partial x}$ | partial derivative of $f$ with respect to $x$ | 128 |
| $\mathrm{D}f(\mathbf{x}_0)$ | derivative of $f$ at the point $\mathbf{x}_0$ | 134 |
| $\nabla f$ | grad $f$, gradient of the function $f$ | 136 |
| $C^1$ | continuously differentiable | 138 |
| $\mathbf{c}$ | a path | 141 |
| $C^2$ | twice continuously differentiable | 182 |
| $Hf(\mathbf{x}_0)$ | Hessian of $f$ at the point $\mathbf{x}_0$ | 212 |
| $\nabla$ | del or nabla | 294 |
| $\nabla \cdot \mathbf{F}$ | div $\mathbf{F}$, divergence of $\mathbf{F}$ | 295 |
| $\nabla \times \mathbf{F}$ | curl $\mathbf{F}$, curl of $\mathbf{F}$ | 299 |
| $\nabla^2$ | Laplacian | 305 |
| $\iint_D f\, dA = \iint_D f(x, y)\, dx\, dy$ | double integral | 343 |
| $\iiint_W f\, dV = \iiint_W f(x, y, z)\, dx\, dy\, dz$ | triple integral | 357 |
| $J = \frac{\partial(x, y)}{\partial(u, v)}$ | Jacobian | 377 |
| $\mathbf{c}_{\mathrm{op}}$ | opposite path | 437 |
| $\int_C f\, ds$ | path integral | 442 |
| $\int_C \mathbf{F} \cdot d\mathbf{s}$ | line integral | 442 |
| $\iint_S f\, dS$ | scalar surface integral | 474 |
| $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n}\, dS$ | vector surface integral | 489 |

**About the cover:**

Isaac Newton, one of the greatest scientific and mathematical intellects of all time, created the notion of a vector through his conception of forces as being *vectorial*. Here you see him depicted in 1725 (with his natural hair) just two years before his death, leafing through the pages of his masterpiece, the *Philosopiae Naturalis Principia Mathematica*, arguably the most influential and profound scientific work ever written and the true starting point of vector calculus.

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